International Physics Olympiad

Training and Selection for the International Physics Olympiad

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Contents

I Theory **1**

1 Problems **2**
1.1 Mechanics . . . 2
1.2 Thermodynamics . . . 28
1.3 Electromagnetism . . . 34
1.4 Optics . . . 52
1.5 Atomic Physics . . . 58

2 Hints **64**
2.1 Mechanics . . . 64
2.2 Thermodynamics . . . 70
2.3 Electromagnetism . . . 71
2.4 Optics . . . 76
2.5 Atomic Physics . . . 77

3 Solutions **79**
3.1 Mechanics . . . 79
3.2 Thermodynamics . . . 213
3.3 Electromagnetism . . . 244
3.4 Optics . . . 321
3.5 Atomic Physics . . . 337

II Experiments **359**

4 Mechanics and Thermal Experiments **360**

5 Electromagnetic Experiments **415**

6 Optical Experiments **471**

III Appendices **513**

A The International Physics Olympiad **514**
A.1 The Syllabus of the International Physics Olympiad . . . 514
The 31st International Physics Olympiad . . . 515
The 32nd International Physics Olympiad . . . 537

The 33rd International Physics Olympiad 557
The 34th International Physics Olympiad 585

B Physical Data and Formulas **622**

Theory

Chapter 1

Problems

1.1 Mechanics

Problem 1.01. A small ball is thrown from the ground over a high wall of height h situated at a horizontal distance d. Find the minimum initial speed $v_{0\,\min}$ required and the corresponding angle of projection θ.

Problem 1.02. A small ball is thrown from the ground over two parallel walls, each of height h and separated by a distance d. Find the minimum initial speed required, the corresponding angle of projection, and the distance from the point of projection to the nearer wall.

Problem 1.03. A gun fires a shell outward from beneath a shelter. The shelter makes an angle α with the horizontal, and the gun position O is at a distance l from the edge P of the shelter, as shown in the figure. The muzzle speed of the shell is v_0. Find the greatest range of the shell.

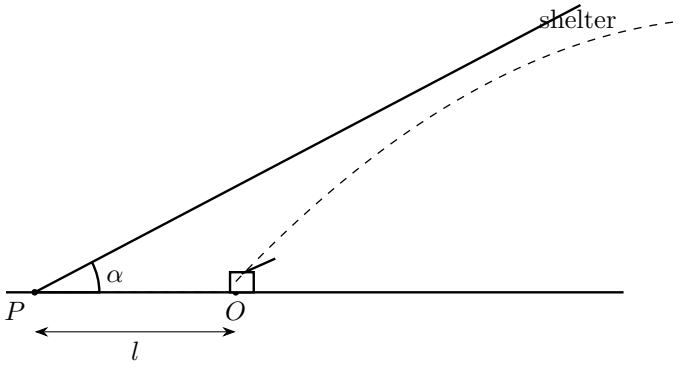


Fig. 1.03

Problem 1.04. A lamp hangs at a distance h_1 below the ceiling, and the lamp is at a height h_2 above the floor. The bulb suddenly bursts into many fragments, all of which fly off in every direction with the same initial speed v_0. The collisions of the fragments with the ceiling are elastic; the fragments do not rebound from the floor and do not strike the walls. Find the radius of the region in which the fragments land.

Problem 1.05. A person stands at a distance d from a high wall and plays a shooting game. Each bullet leaves the gun with the same initial speed v_0.

(1) To hit the same horizontal line at height h on the wall, find the locus of the aiming points on the wall.

(2) If the points of impact on the wall are to lie above this horizontal line, find the range of the aiming points.

Problem 1.06. An aircraft flies horizontally at a height h above the ground with speed v_0 and drops a bomb on a ground target.

(1) At what horizontal distance L from the target must the bomb be released in order that it hit the target?

(2) A gun is situated at a distance d from the target. At the instant the bomb is released the gun fires a shell. Find the minimum initial speed $v_{\min}$ of the shell and the corresponding angle of elevation, so that the shell can hit the bomb in the air (see the figure).

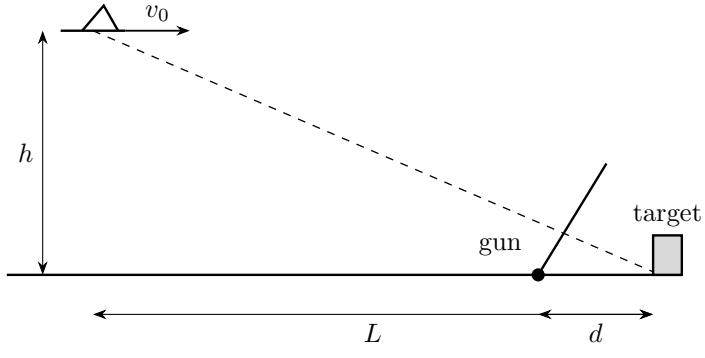


Fig. 1.06

Problem 1.07. A circle of radius R moves uniformly in the plane of the paper with speed v_1 to the right, while a straight line MN moves uniformly upward with speed v_2. When the angle between the line OP, joining the intersection point P of the line and the circle to the centre O, and the line MN is θ, find the velocity and acceleration of the point P describing the displacement of the intersection point.

Problem 1.08. A thin rod AB rests obliquely on a fixed semi-cylindrical surface of radius R. The end A, which touches the ground, moves along the horizontal plane with constant speed v, as shown. When the angle between the rod and the ground is θ, find:

(1) the velocity v_P of the point P of the rod that is in contact with the cylindrical surface;

(2) the velocity $v_{P'}$ with which the point of intersection P' of the rod and the cylindrical surface moves.

Problem 1.09. A hinge linkage is made of four thin rods of length $2l$ and four thin rods of length l; all the crossing points are connected by hinges, as shown in the figure. The hinge O_3 moves uniformly in the x direction with speed v. When each rod makes an angle θ with the x axis, find:

(1) the magnitudes of the velocity and acceleration of the hinges A_1 and B_3;

(2) the angular velocity and angular acceleration of the rod A_3B_2;

(3) the equations of the trajectories of all the hinges.

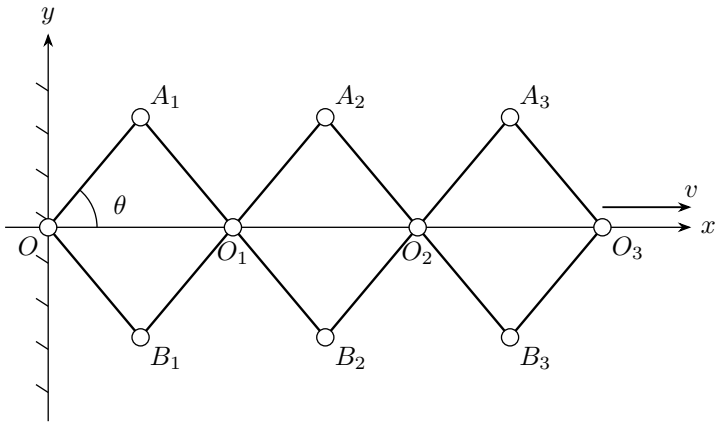


Fig. 1.09

Problem 1.10. A disc A of radius R rotates uniformly with angular velocity ω_1 about its fixed axis O_1. Another disc B of radius r rolls without slipping along the rim of A, turning uniformly about its own axis O_2 with angular velocity ω_2. Find the time required for B to make one complete circuit around A.

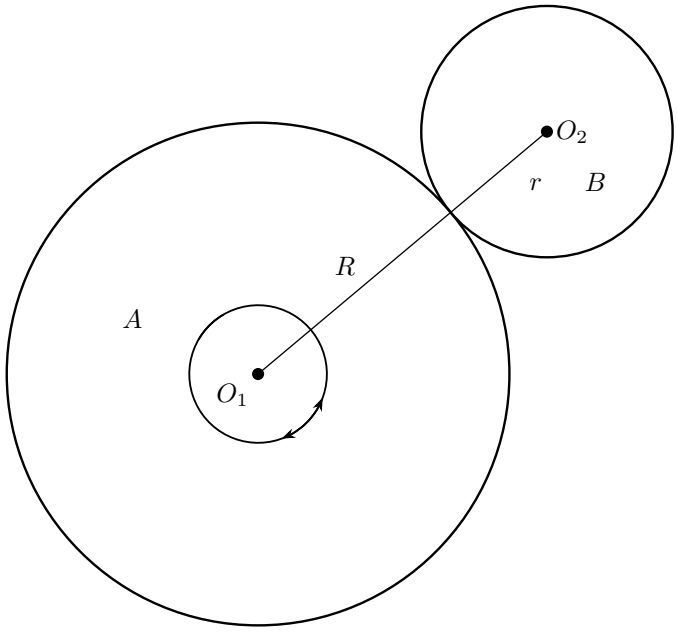


Fig. 1.10

Problem 1.11. A large steel ring of radius R rotates in its own plane, in the counterclockwise sense, with constant angular velocity ω about a point P on the ring. A small steel ring of radius $\frac{1}{3}R$ rolls without slipping along the inner side of the large ring in the same plane, in the direction shown in the figure. When the large ring turns once about P, the small ring turns exactly twice relative to the large ring. When the small ring has moved relative to the large ring to the position shown, find the accelerations of the points A and B of the small ring.

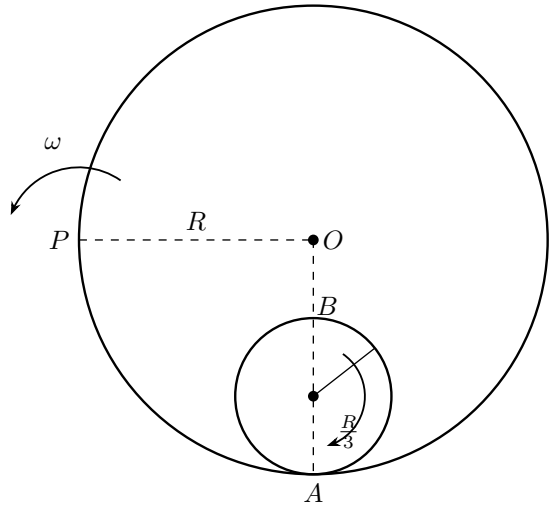


Fig. 1.11

Problem 1.12. Between the walls of two coaxial thin-walled cylinders A and B of radii R and $2R$, a small cylinder of radius exactly $\frac{1}{2}R$ is held. When A and B rotate uniformly in opposite directions with angular velocities ω_1 and ω_2, the small cylinder turns with them, there being no relative slipping at its points of contact with A and B, as shown in the figure. Find:

(1) the times required for the small cylinder to make one revolution relative to the ground and relative to cylinder B;

(2) the acceleration of the point C of the small cylinder that is in contact with B, at this instant, relative to the ground and relative to cylinder A.

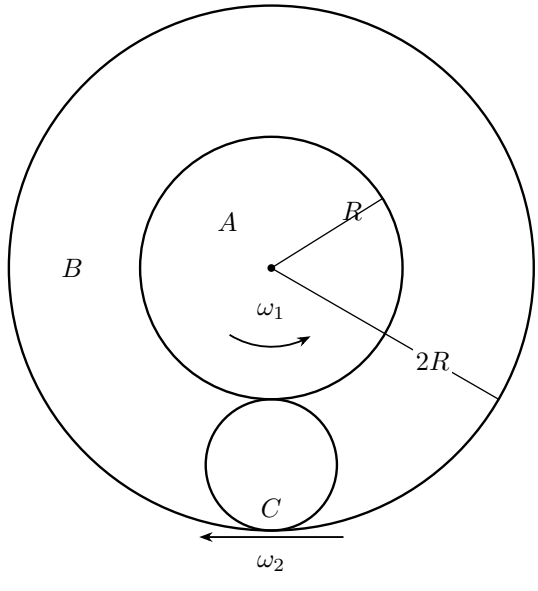


Fig. 1.12

Problem 1.13. Two particles A and B move uniformly in the same sense along concentric circles of radii a and b, their speeds being inversely proportional to the radii. Find the angle θ

between the lines joining the two particles to the centre at the instant when the direction of their relative velocity is parallel to the line joining them.

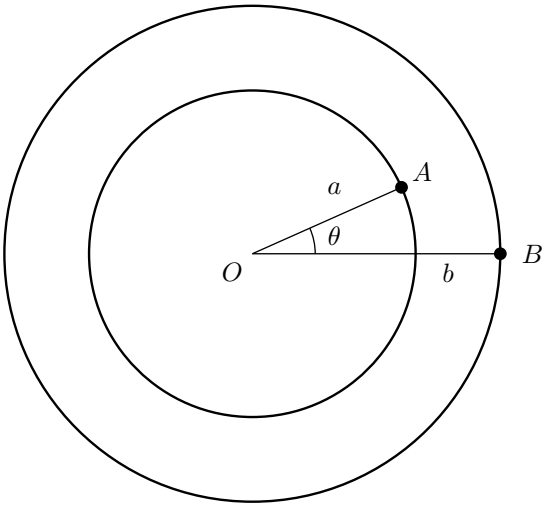


Fig. 1.13

Problem 1.14. A smooth inclined rod AB of length l makes a fixed angle α with the vertical axis through its end A, and rotates about this axis with constant angular velocity ω. A small ring of mass m is threaded on the rod and starts from A with an initial velocity v_0 relative to the rod, directed upward along the rod. Find:

(1) the minimum value of v_0 for the ring to reach the end B;

(2) the force exerted by the ring on the rod when the ring reaches B.

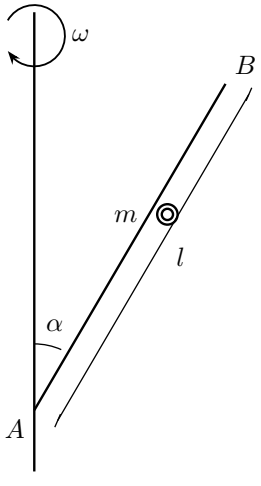


Fig. 1.14

Problem 1.15. A smooth thin rod is bent into the shape shown in the figure and fixed in the vertical plane (xOy), the equation of the curve being

$$y = \frac{1}{2a}x^2$$

where a is a positive constant. A small ring of mass m is threaded on the rod.

(1) If the ring is released from rest at $x = 2a$, find the force it exerts on the rod when it reaches $x = a$.

(2) If the rod rotates about the y axis with constant angular velocity ω_0, for what value of ω_0 can the ring remain at rest relative to the rod at any position on the rod?

(3) If the angular velocity of the rod about the y axis is taken to be $\frac{\sqrt{2}}{2}$ times the value found in (2), that is $\omega = \frac{\sqrt{2}}{2}\omega_0$, and the ring is released from rest relative to the rod at $x = 2a$, what is the force it exerts on the rod when it reaches $x = a$?

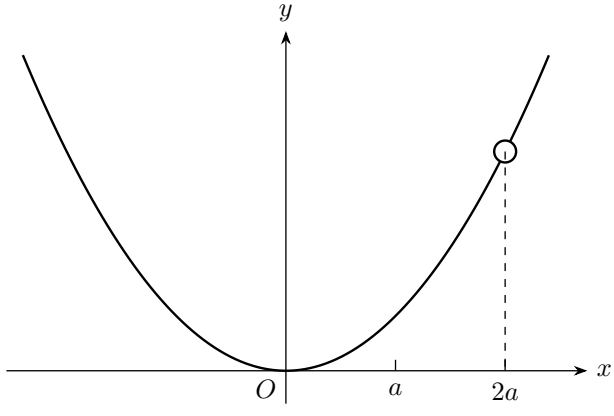


Fig. 1.15

Problem 1.16. A large horizontal disc of radius R rotates uniformly with angular velocity Ω about the vertical axis through its centre. A small disc of radius r and mass m is gently placed on the large disc from above. After the small disc has settled, the distance between the axes of the two discs is d ($d > r$, $R > d + r$). The coefficient of friction between the discs is μ.

(1) Find the angular velocity ω of the small disc in its steady rotation.

(2) What torque must be applied to the large disc to keep its angular speed unchanged?

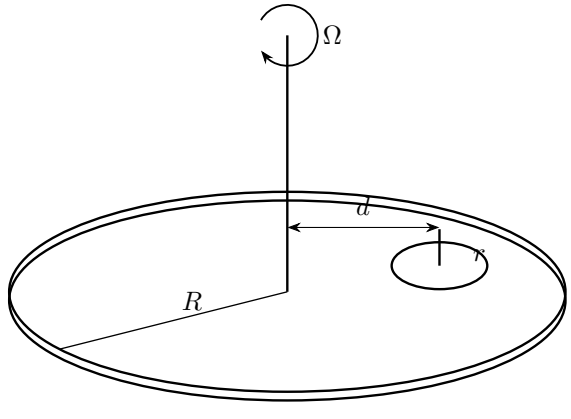


Fig. 1.16

Problem 1.17. Two identical small balls A and B, each of mass M and radius R, are suspended from the same point O by two strings each of length $l = 2R$. A small ball C of mass $m(= nM)$

and radius $r\left(=\frac{1}{2}R\right)$ is placed on the two balls. The surfaces of all three balls are smooth. Discuss the relation between the angle θ that the strings make with the vertical and n when the system is in equilibrium.

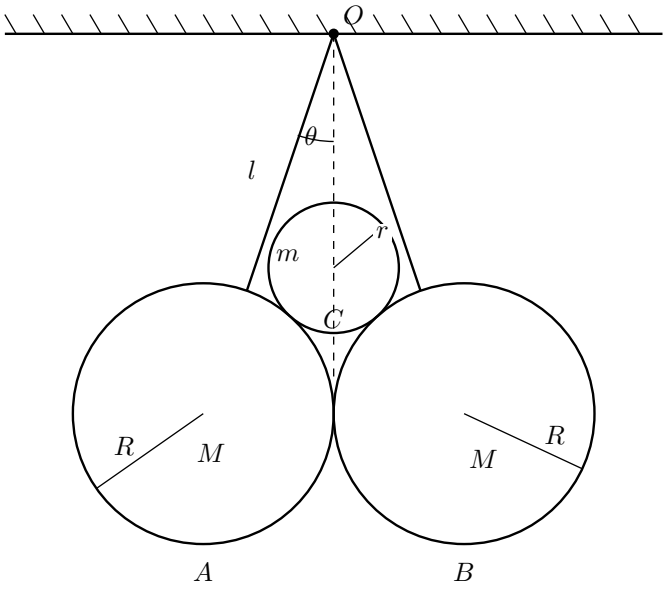


Fig. 1.17

Problem 1.18. A small ball of mass m is fixed to a large ring of mass M. The ring is hung on a rough nail, as shown in the figure. If the ring is to remain in equilibrium when hung on the nail at any point of the ring (except the point at which the small ball is fixed), what is the minimum coefficient of friction μ between the ring and the nail?

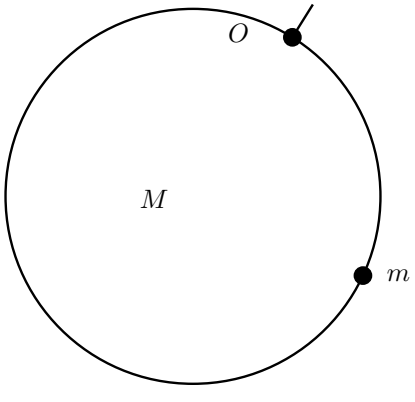


Fig. 1.18

Problem 1.19. Two small balls 1 and 2 of equal mass, connected by a light string, are placed on an inclined plane of angle θ. The coefficients of friction between the plane and balls 1 and 2 are μ_1 and μ_2 respectively, with $\mu_1 < \mu_2$ and $\tan\theta = \sqrt{\mu_1\mu_2}$. Find the maximum value of the angle α between the string and the line of greatest slope of the plane for the system to be in equilibrium.

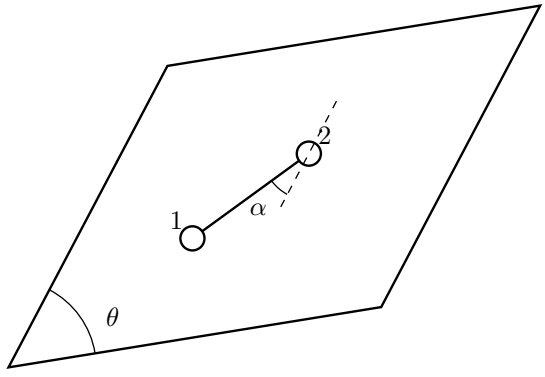


Fig. 1.19

Problem 1.20. A hinge structure is made of four uniform rods of length $2l$ and mass $2m$ and four uniform rods of length l and mass m. The structure has ten light smooth hinges. Hinge O_1 is suspended from a horizontal axis, and a string of length $\sqrt{2}\,l$ joins hinges O_3 and O_4. When the structure is in equilibrium, the angle between the two rods at every hinge is $90°$, as shown in the figure. Find the tension in the string in equilibrium.

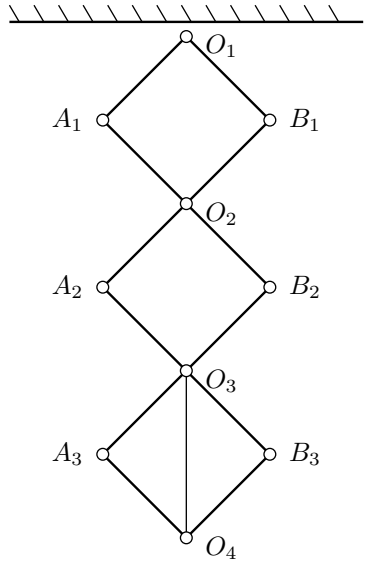


Fig. 1.20

Problem 1.21. At the foot of an inclined plane of angle θ, a vertical board of length l is connected by a smooth hinge. A sphere of radius R and mass m is held between the board and the plane, and a horizontal force F perpendicular to the board is applied at the upper end of the board so that both the sphere and the board remain in equilibrium, as shown in the figure. The coefficient of friction between the board and the sphere is μ_1, and that between the sphere and the plane is μ_2. The force F is now gradually increased. At what value of F is the equilibrium of the system destroyed?

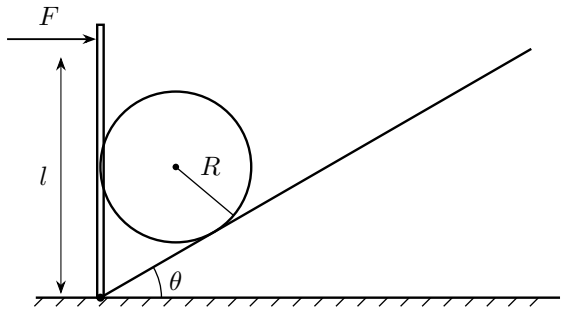


Fig. 1.21

Problem 1.22. Three identical uniform rods, each of mass m and length l, lean against one another as shown in the figure; the lines joining their points of contact with the ground form an equilateral triangle of side l. The coefficients of friction between the rods and the ground are equal.

(1) Find the magnitude and direction of the force acting on the upper end of rod OA.

(2) If a small ball of mass m is also fixed at the midpoint of rod OA, find again the magnitude and direction of the force acting on its upper end.

(3) Find the minimum coefficient of friction between the rods and the ground for the system to remain at rest.

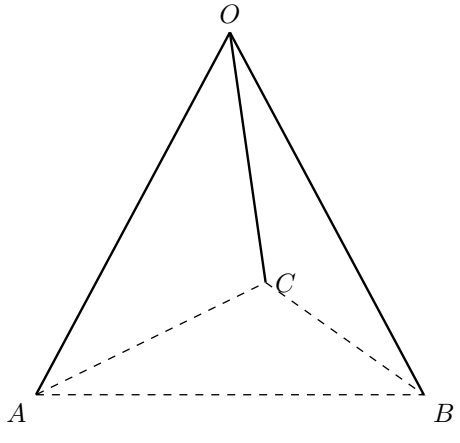


Fig. 1.22

Problem 1.23. A ring of radius R stands upright on the ground at point P. The line AP is perpendicular to the plane of the ring, and $\overline{AP} = \frac{3}{2}R$. One end of a thin rod is hinged at A, and the body of the rod rests on the ring at point B; the radius OB through B makes an angle $\theta = 60^\circ$ with OP, as shown in the figure. Find the minimum coefficient of friction between the rod and the ring for the rod to be in equilibrium in this position.

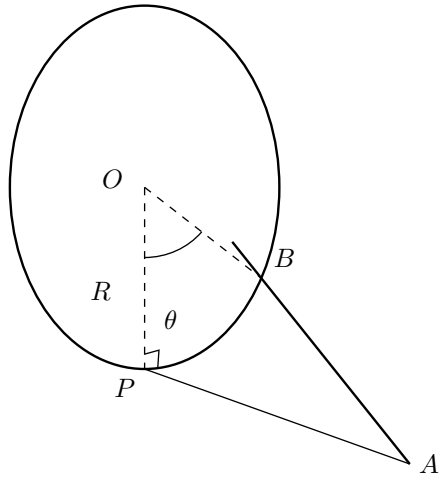


Fig. 1.23

Problem 1.24. A jet of water from a fountain holds a rubbish bin of mass M upside down in the air. The water leaves the fountain nozzle of area S_0 with speed v_0 and is directed vertically upward. After striking the inner bottom of the bin, half of the water is retained on the bottom and flows down along the inner wall, while the other half splashes vertically downward with its original speed, as shown in the figure. Find the height h at which the bin remains in the air. The density of water is ρ.

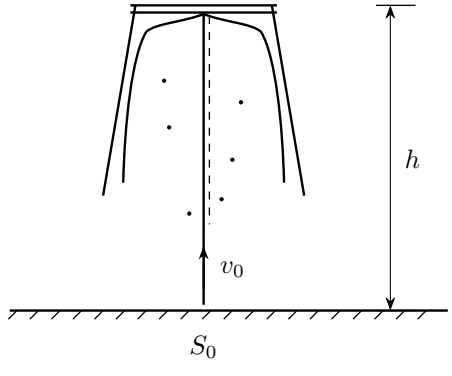


Fig. 1.24

Problem 1.25. A very small spherical raindrop condenses as a nucleus in a uniform, still cloud layer and falls from rest, capturing all the droplets it meets on its way, so that its volume increases continuously while it remains spherical. Neglecting the viscous drag on the drop, show that the drop tends to fall with uniform acceleration, and find this acceleration.

Problem 1.26. A block of mass m is connected to a soft rope of linear mass density λ. Initially the block is at the top of an inclined plane of angle θ, and the rope is coiled on the platform at the upper edge of the plane. The plane and the platform are both smooth. The block is released and slides down the plane, as shown in the figure. Find the speed of m after it has slid a distance x down the plane (x less than the length of the rope).

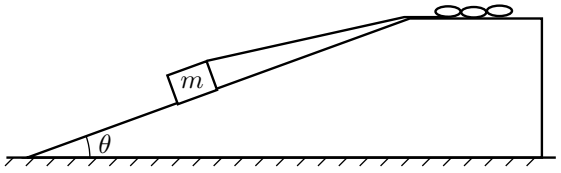


Fig. 1.26

Problem 1.27. A small ball of mass m lies on a table, and a vertical light spring of stiffness k is attached to it. Initially the spring is at its natural length; its upper end is then made to move vertically upward with constant speed v by an external force, as shown in the figure. Find the work done by the force acting on the upper end of the spring during the interval from the start of the motion of that end until the spring first reaches its maximum extension.

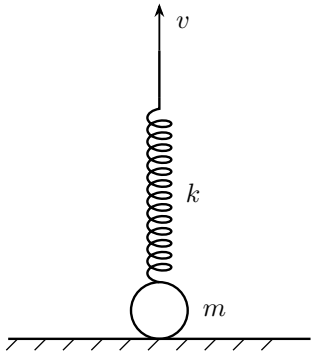


Fig. 1.27

Problem 1.28. Two small balls A and B, each of mass m, are connected by a light rod of length l and stand vertically on a smooth horizontal surface, as shown in the figure. A constant horizontal force F is applied to the lower ball B, so that B moves through a certain distance; at this moment the rod makes an angle θ with the horizontal. Find the force exerted by the horizontal surface on ball B at this moment.

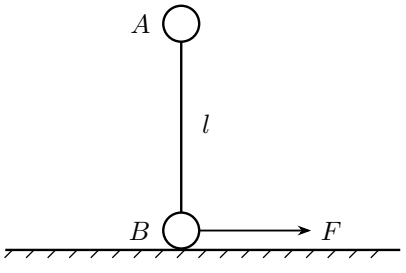


Fig. 1.28

Problem 1.29. A small ring A of mass m is threaded on a smooth fixed horizontal rod and is connected by a string of length l to a small ball B, also of mass m. The string is first held straight in the horizontal direction, and the system is then released from rest. Find:

(1) the angle θ between the string and the horizontal rod at which the speed of the ball is greatest, and this maximum speed;

(2) the tension in the string when the speed of the ball is greatest.

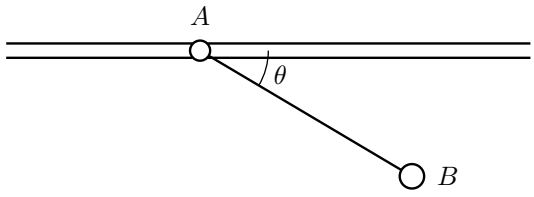


Fig. 1.29

Problem 1.30. A thin hollow tube is bent into a ring of radius R. The ring is fixed upright on a slider which rests on a smooth horizontal surface; the total mass of the ring and the slider is M. Inside the cavity of the tube there is a small ball of mass m, which can move along the cavity without friction, as shown in the figure. Initially the ball is at the highest point of the cavity, and both the ring and the ball are at rest. Under a small disturbance the ball slides down along the cavity from the right-hand side. When the ball has turned through an angle $\theta = \frac{\pi}{4}$ about the centre O of the ring, the leftward speed of the ring reaches its maximum. Find:

(1) the value of the ratio $\dfrac{M}{m}$ and the maximum leftward speed of the ring;

(2) the radius of curvature of the trajectory of the ball at this position.

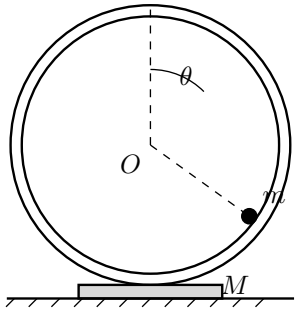


Fig. 1.30

Problem 1.31. A string passes over two small bearings a and b at the same height, the distance between them being $2l$. To the two ends of the string are attached bodies A and B, each of mass m, and to the midpoint of the part of the string between the two bearings is attached a body C of mass M, as shown in the figure. The system is released from rest, and body C moves vertically downward. Neglect the mass of the bearings and all friction.

(1) When the angle between the string joining C and the vertical is $\alpha = 60°$, the speed of C is greatest. Find the value of $\dfrac{M}{m}$ and the maximum speed of C;

(2) If $\dfrac{M}{m} = 1$, find the acceleration of body C when $\alpha = 60°$.

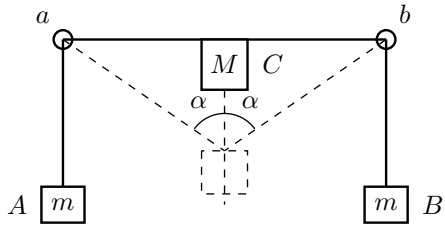


Fig. 1.31

Problem 1.32. Two blocks A and B of masses $M = 3.60\,\text{kg}$ and $m = 1.80\,\text{kg}$ respectively are connected to the two ends of an elastic string of natural length $l_0 = 0.300\,\text{m}$ and stiffness $k = 24.0\,\text{N/m}$, and are placed on a horizontal table, as shown in the figure. The coefficient of friction between either block and the table is $\mu = 0.300$. The two blocks are now pulled apart until the distance between them is $l = 1.200\,\text{m}$, and are released from rest. Find:

(1) the speeds v_A and v_B of the two blocks when they collide;

(2) the time elapsed from the release to the collision.

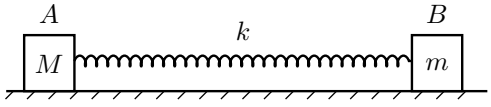


Fig. 1.32

Problem 1.33. Two supernovae of masses M and m respectively are separated by a distance d and move in their own circular orbits about their fixed centre of mass. In the explosion of the supernova, the star of mass M loses an amount ΔM of mass. Assume that the explosion is instantaneous and completely spherically symmetric, and neglect the direct action of the explosion debris on the star of mass m. What relation must ΔM satisfy in order that the remaining binary system should stay bound and not separate from each other?

Problem 1.34. A particle of mass m_1 moving with speed v_1 is captured by a stationary nucleus. As a result, a particle of mass m_2 is produced and emitted in a direction perpendicular to v_1, the mass of the remaining nucleus being m_3. In this process an amount Q of non-mechanical energy is converted into mechanical energy. Find the kinetic energy of the newly produced particle.

Problem 1.35. A block initially at rest explodes into three blocks 1, 2 and 3 of equal mass. The total kinetic energy released in the explosion has a fixed value Q, but the kinetic energies T_1, T_2, T_3 of the individual blocks can take many possible values. They can be represented by the lengths of the perpendiculars drawn from a point P inside an equilateral triangle of height Q to its three sides, as shown in the figure; however, not every set of kinetic energies corresponding to a point inside the triangle is physically allowed.

(1) Find the boundary of the region of physically allowed points;

(2) If instead a positronium at rest annihilates into three γ photons whose energies have the total value Q, find likewise the boundary of the region of physically allowed points.

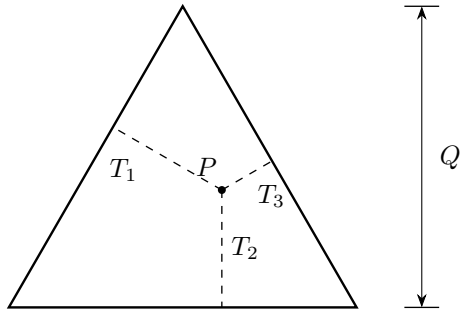


Fig. 1.35

Problem 1.36. On a smooth conical surface of semi-vertical angle $\theta = 60°$, a small ball of mass $m = 1.0\,\text{kg}$ is connected to a string which passes through a small hole at the apex of the cone,

and moves in uniform circular motion with speed $v_0 = \frac{\sqrt{15}}{4}\,\mathrm{m/s}$; the length of the string from the ball to the hole is $r_0 = 0.50\,\mathrm{m}$, as shown in the figure. The other end of the string is now pulled down slowly until the ball just leaves the conical surface. Find the work done by the pulling force during this process.

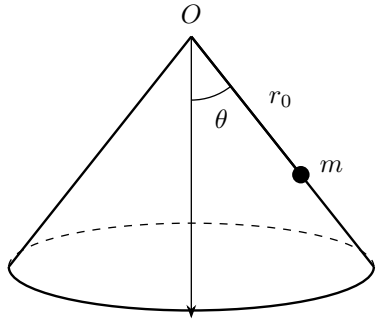


Fig. 1.36

Problem 1.37. On a smooth horizontal surface, two small balls A and B, each of mass m, are connected by a light spring of stiffness k, the natural length of the spring being a. Initially both balls are at rest and the spring is undeformed. A lead pellet, also of mass m, is shot with a horizontal velocity perpendicular to the line joining the two balls into one of the balls, B, and remains inside it, as shown in the figure. It is known that in the subsequent motion the maximum length of the spring is $2a$. Find the speed v_0 of the pellet before it entered the ball.

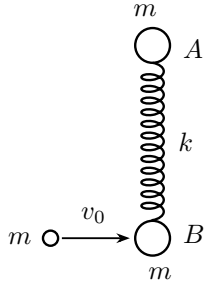


Fig. 1.37

Problem 1.38. Two comets of masses m_1 and m_2 move in their own parabolic orbits about the Sun, the two orbits being coplanar. When the two comets arrive at a distance R from the Sun, they collide at right angles and combine into a single body. Discuss the orbit of this combined body afterwards.

Problem 1.39. A lunar lander of mass m is attached to a space shuttle of mass $M(=2m)$; together they move in a circular orbit about the Moon, the radius of the orbit being 3 times the Moon's radius R_m. At a certain moment the shuttle ejects the lander in the direction opposite to the motion. Afterwards the lander still moves in the original direction and, along the elliptical orbit shown in the figure, descends to the Moon's surface. It stays on the Moon's surface for a period of time, and after completing its survey work it starts again quickly along the original elliptical orbit and returns to the point of separation, where it docks with the shuttle. Find the time for which the lander can stay on the Moon's surface. Given that the gravitational acceleration at the Moon's surface is $g_m = 1.62\,\mathrm{m/s^2}$ and the Moon's radius is $R_m = 1.74 \times 10^6\,\mathrm{m}$.

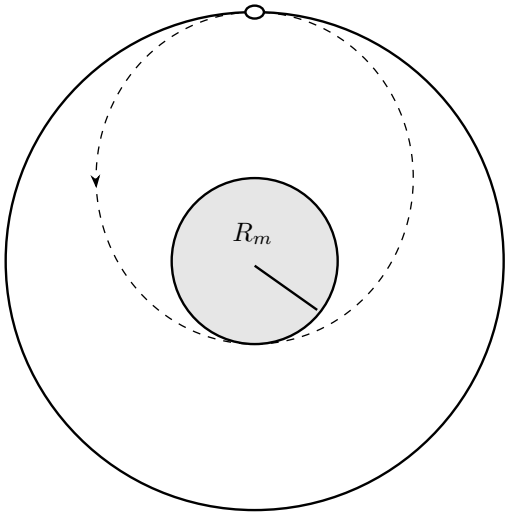


Fig. 1.39

Problem 1.40. An "interstellar bullet" is regarded as a dense gas cloud which passes like a ballistic particle through low-density interstellar gas clouds. Somewhere in the universe there is a spherical low-density uniform gas cloud of mass M and radius R; an "interstellar bullet" has a radius much smaller than R and a mass $m \ll M$. The angular momentum of the bullet about the centre of the gas cloud is

$$L = m\sqrt{\frac{GMR}{32}},$$

and the total energy of the bullet is

$$E = -\frac{5GMm}{4R}.$$

Neglect all non-gravitational forces.

(1) Is the motion of the bullet always inside the cloud, always outside the cloud, or sometimes inside and sometimes outside?

(2) Find the distances from the centre of the gas cloud of the turning points of the bullet's orbit;

(3) Discuss the shape of the bullet's orbit and find its period of motion.

Problem 1.41. To launch a spacecraft from the Earth to outside the solar system there are two schemes. Scheme I: launch it directly with a sufficiently large speed (greater than the escape speed from the solar system). Scheme II: make the spacecraft approach an outer planet of the solar system (for example Mars), let its direction of motion be changed under the action of that planet, and then escape from the solar system. Find:

(1) the minimum speed v_0 relative to the Earth required for launching from the ground in scheme I, and the direction of launch;

(2) the speed of the spacecraft when it crosses the orbit of Mars, if it is launched from the ground with speed v_0 relative to the Earth in the direction of part (1). Assume that when the spacecraft crosses the orbit of Mars it is very far from Mars;

(3) the speed of launch from the ground required for the spacecraft to enter the gravitational field of Mars and yet still be able to fly out of the solar system. Assume that the direction

of launch relative to the Earth is the same as in part (1), and that the spacecraft leaves the gravitational field of Mars along the tangent to the orbit of Mars;

(4) the maximum percentage of energy saved by scheme II compared with scheme I.

Assume that all the planets move in the same plane in the same direction about the Sun in circular orbits, and that air resistance and the effect of the Earth's rotation may be neglected.

Problem 1.42. Around a star of mass M, dust of mass density ρ is distributed uniformly and spherically symmetrically over a very large region. Within the dust region a planet of mass m moves in a circular orbit of radius r_0 about the star (neglect the drag of the dust on the planet). If the planet receives a small radial disturbance which makes it deviate slightly from the circular orbit, describe quantitatively the subsequent motion of the planet.

Problem 1.43. A particle of mass m moves in a circular orbit of radius r_0 under the action of a central attractive force, the magnitude of which can be written as

$$F = kr^n,$$

where r is the distance from the particle to the fixed centre of force, k is a positive constant and n is a constant which may be positive or negative (not necessarily an integer).

(1) For what values of n is this circular orbit stable against small radial disturbances?

(2) For a stable circular orbit, find the period of the radial oscillations of the particle under a small radial disturbance, in terms of r_0, m, k and n.

(3) For what values of n is the period of motion in the stable circular orbit an integral multiple of the period of the small radial oscillations?

Problem 1.44. The flyball governor of a steam engine consists of two balls, each of mass m, connected by four hinged arms, each of length l, to the upper and lower sleeves on a vertical shaft; the upper sleeve is fixed, while the lower sleeve, of mass M, can slide up and down the shaft without friction, as shown in the figure. The whole device rotates uniformly about the vertical shaft with constant angular velocity ω_0. Neglect the mass of the arms and the moment of inertia of the sleeve M.

(1) Find the distance between sleeve M and the upper sleeve in equilibrium;

(2) Find the frequency of small up-and-down oscillations of the sleeve M about the equilibrium position;

(3) If the shaft can rotate freely and the system rotates steadily with initial angular velocity ω_0, and the sleeve M is given a small up-and-down disturbance, is the frequency of its small oscillations the same as the value in (2)? If not, find this frequency.

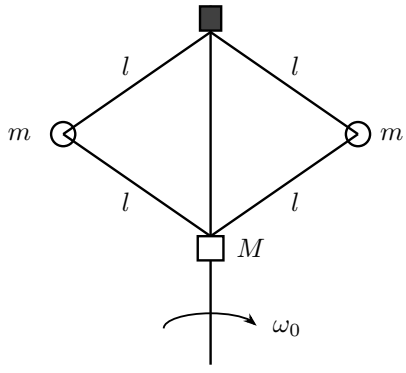


Fig. 1.44

Problem 1.45. A small ball of mass m is attached to one end of an elastic string of stiffness k. The string passes through a small hole O in a fixed vertical board, and its other end is attached to the ground directly below the hole. When the ball is at the hole, the string is just at its natural length, as shown in Fig. 1.45(1). The ball is now pulled along the board in the horizontal (x) direction through a distance a, and is set in motion with initial speed $v_0 = 2\sqrt{ga}$; the direction of v_0 is parallel to the board and makes an angle $\theta = 60°$ with the x direction, as shown in Fig. 1.45(2). Let $ka = mg$. Find:

(1) the time after which the ball reaches its highest position, and this height;

(2) the position and the velocity of the ball when it next passes through the x axis, and the time required.

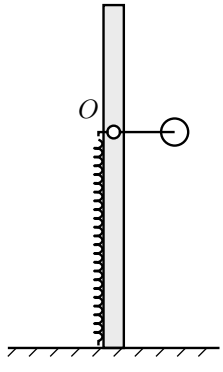


Fig. 1.45(1)

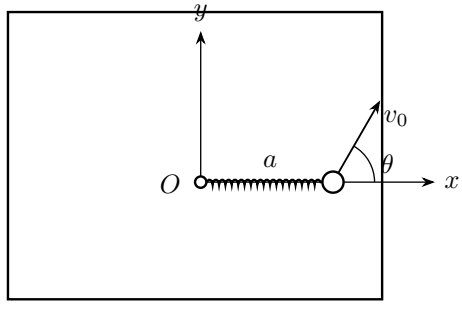


Fig. 1.45(2)

Problem 1.46. A long cylinder of mass M and length l, with its bottom upward and its mouth downward, is suspended from a spring of stiffness k. At the inner bottom of the cylinder there is a piston of mass m and negligible thickness; between the piston and the bottom of the cylinder a small spring of negligible natural length is compressed and locked, and the whole system is in equilibrium, as shown in the figure. The lock is now released, so that the cylinder and the piston acquire velocities in an extremely short time. It is known that the velocity acquired by the cylinder is $v_0 = \frac{l}{2}\sqrt{\frac{m}{k}}$, and that in the subsequent motion the friction between the piston and the cylinder is $f = \frac{1}{2}mg$; let $M = 3m$. Find:

(1) the lowest position which the cylinder can reach in the subsequent motion, and the time taken;

(2) the relation which the length l of the cylinder must satisfy in order that the cylinder can reach this lowest position.

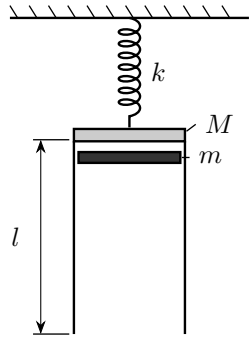


Fig. 1.46

Problem 1.47. On a smooth horizontal table, a small ball of mass m is connected to a fine string which passes through a small hole O in the table, and to the other end of the string a block of mass M is suspended. The ball moves on the table in uniform circular motion about the hole O with angular velocity ω_0, while the block below the table remains at rest, as shown in the figure. If the block is now given a small vertical disturbance, prove that it will oscillate up and down in simple harmonic motion, and find its period of oscillation.

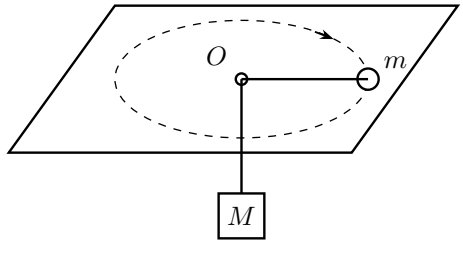


Fig. 1.47

Problem 1.48. A circle of radius R rolls without slipping along the straight line $y = 2R$. The trajectory of a point P on the circle is a cycloid. A fixed smooth steel wire has the same shape as this curve, and a small ring is threaded on it, as shown. Find the time required for the ring to slide from the highest point of the wire to the lowest point.

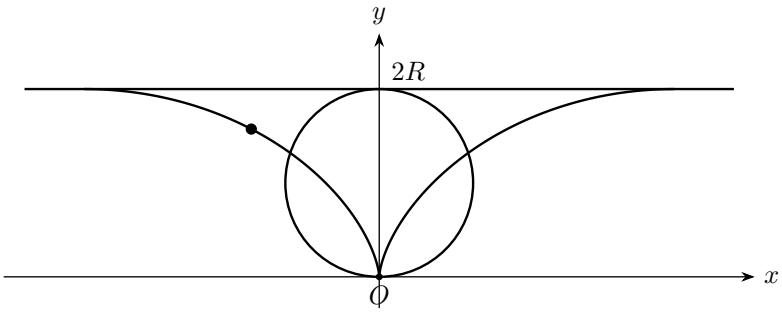


Fig. 1.48

Problem 1.49. When wind blows from a distant place, if the wind speed has a gradient in the vertical direction perpendicular to the ground, people occasionally hear the sound emitted by a distant source, and even with astonishing clarity. Suppose the distribution of the wind speed in the vertical (y) direction can be written as $v = v_0 + ky^2$, where k is a very small positive constant. A distant sound source ($x = 0$) emits a beam of sound; the angle between the emission direction and the vertical is θ_0, and the speed of sound at the ground ($y = 0$) is V_S. Assuming that throughout the region traversed by the sound wave $\dfrac{ky^2}{V_S} \ll 1$, find:

(1) the trajectory of the beam in space;

(2) the distance from the source to the points on the ground at which the sound can be heard clearly.

Problem 1.50. A string of linear density η and tension T lies along the x-axis, with a fixed end at $x = a$. A harmonic transverse wave propagates along the string in the x direction; the vibration excited at $x = -a$ is $y = A\cos(\omega t + \varphi_0)$. Find:

(1) the equation of the incident wave and the equation of the wave reflected from the fixed end;

(2) the expression for the resulting standing wave, and the positions of the nodes and antinodes;

(3) the total energy between two adjacent nodes (take the potential energy to be zero when every point of the string is at its equilibrium position).

Problem 1.51. Three blocks A, B, C are connected by two light springs, each of stiffness k, and rest on a smooth horizontal surface, as shown. The masses of A and C are both m, and the mass of B is M. Let x_1, x_2, x_3 denote the displacements of blocks A, B, C from their equilibrium positions shown in the figure. Block A is given an impulse in a very short time, so that it acquires an initial speed v_0 directed toward B. Find the displacements x_1, x_2, x_3 of the three blocks as functions of the time t thereafter.

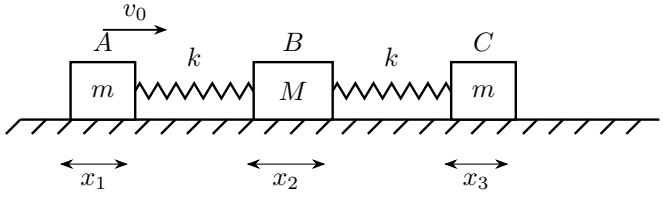


Fig. 1.51

Problem 1.52. A uniform rod of length l and mass m is suspended at its two ends by two light springs of stiffness k_1 and k_2 respectively. In equilibrium the rod is horizontal, as shown.

Assuming that the springs can move only in the vertical direction, the rod can perform small oscillations about its equilibrium position.

(1) For $k_1 = k_2$, find the eigenfrequencies of the normal modes and describe the corresponding normal-mode motions.

(2) For $k_1 \neq k_2$, find the eigenfrequencies of the normal modes.

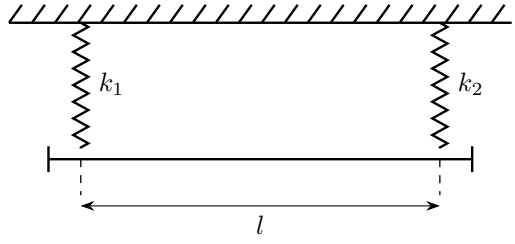


Fig. 1.52

Problem 1.53. On the ground there is a small hemispherical nozzle. Many small holes are distributed over the hemispherical surface of the nozzle. When water is sprayed, water issues from these holes in different directions with the same speed v_0. Compared with the area sprinkled by the water, the nozzle may be regarded as a point.

(1) The water sprayed from the nozzle traces out a solid figure in the air; describe this figure quantitatively.

(2) To make the amount of water received per unit area uniform within the region on the ground reached by the spray, how should the number density of the small holes on the hemispherical surface be distributed?

Problem 1.54. A planet of mass M is composed of an incompressible fluid of density ρ. The planet rotates slowly with constant angular velocity ω; owing to the rotation, its equator bulges slightly relative to its poles. The mean radius of the planet is R. To first order:

(1) find the pressure at a small depth h below the surface at polar angle θ;

(2) let the radii at the poles and at the equator be R_p and R_e respectively; find the value of $R_e - R_p$.

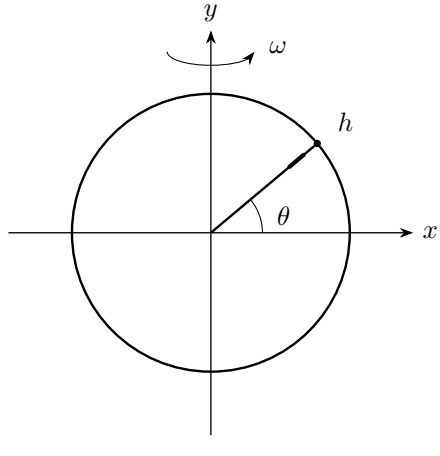


Fig. 1.54

Problem 1.55. Estimate the ratio of the height of the tide at the Earth's equator raised by the Sun to that raised by the Moon.

Problem 1.56. A satellite of mass m moves in a circular orbit of radius r around a planet of mass M, with orbital angular velocity ω, while the planet rotates about its spin axis through its centre of mass with angular velocity Ω; the two angular velocities have the same direction. The moment of inertia of the planet about its spin axis is I.

(1) In general ω and Ω are not equal. Owing to tidal friction, both the spin angular velocity of the planet and the orbital angular velocity of the satellite change. Find the relation between the changes of these two angular velocities.

(2) Tidal friction reduces the mechanical energy of the planet–satellite system. Under what condition can the system eventually reach a stable state, that is, a state in which the mechanical energy does not change?

Problem 1.57. A semi-cylindrical aircraft hangar of length l and radius R is exposed to the wind, the direction of the wind being perpendicular to its axis; the streamlines are as shown. The wind speed far away on the ground is v_0. The wind speed at any point around the hangar can be determined from the "velocity potential"

$$\phi = -v_0 \left(r + \frac{R^2}{r} \right) \cos\theta,$$

where r is the distance from a point B near the roof to the axis O of the semi-cylinder, and θ is the angle between OB and the x-axis. The relation between the velocity and the velocity potential is analogous to that between a conservative force and its potential energy. If the door A of the hangar is open, find the force exerted by the wind on the hangar. Take $l = 70\,\mathrm{m}$, $R = 10\,\mathrm{m}$, $v_0 = 20\,\mathrm{m/s}$, $\rho = 1.20\,\mathrm{kg/m^3}$, and neglect the pressure difference due to height. (Give two significant figures.)

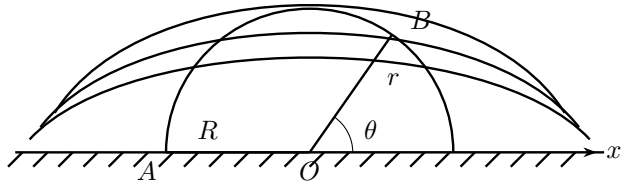


Fig. 1.57

Problem 1.58. Two uniform rods, each of length l and mass m, are joined at their upper ends by a light smooth hinge and stand upright on a smooth horizontal floor with the angle between them $\alpha = 60°$, as shown. They are released from rest. Find:

(1) the acceleration of the hinge immediately after release;

(2) the velocity and acceleration of the hinge when the angle between the rods becomes $\alpha' = 90°$.

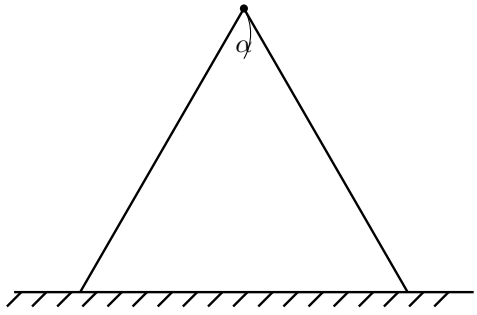


Fig. 1.58

Problem 1.59. A uniform rod of length l and mass m lies horizontally at the corner of a table, with a length $\frac{1}{3}l$ on the table and the remaining $\frac{2}{3}l$ projecting beyond the table edge, as shown. The rod is released from rest. Find the angle θ through which the rod has turned downward when it begins to slide relative to the table corner. The coefficient of friction between the rod and the table corner is μ.

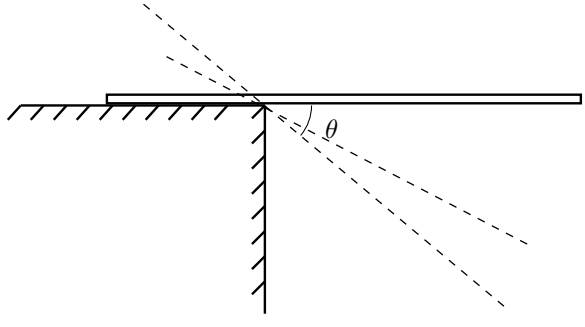


Fig. 1.59

Problem 1.60.

(1) A thin-walled cylindrical shell of radius R and mass m (its mass being concentrated on the cylindrical surface) is filled with water of mass μ. It rolls without slipping down the inclined face of a fixed wedge of inclination angle θ, as shown. Assuming that there is no friction between the water and the cylinder wall, find the acceleration of the centre of mass of the water-filled cylinder.

(2) Suppose the wedge is not fixed but is placed on a smooth horizontal surface and has mass M.

(i) Find the acceleration of the wedge while the water-filled cylinder rolls down.

(ii) If the cylinder starts from rest at a height h above the table top and rolls down, find the velocity of the wedge when it reaches the table top.

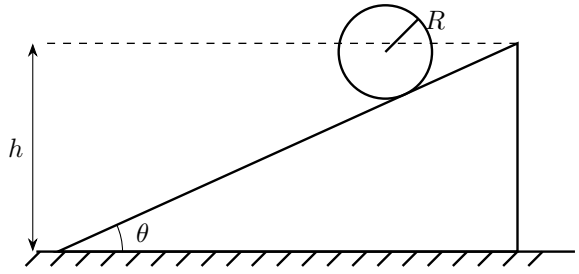


Fig. 1.60

Problem 1.61. A uniform rod AB of mass m and length $2a$ has its end A connected by a light smooth hinge to a horizontal light rod OA of length b. The rod AB can rotate freely about the hinge in a vertical plane. The two rods now rotate together about a vertical axis through O with constant angular velocity ω. Let θ be the angle between the rod AB and the vertical, as shown.

(1) Find the relation that the possible values of θ must satisfy when the rod AB is in equilibrium relative to the rod OA during the rotation.

(2) State how many possible equilibrium positions θ has between 0 and 2π, and discuss the stability of the equilibria.

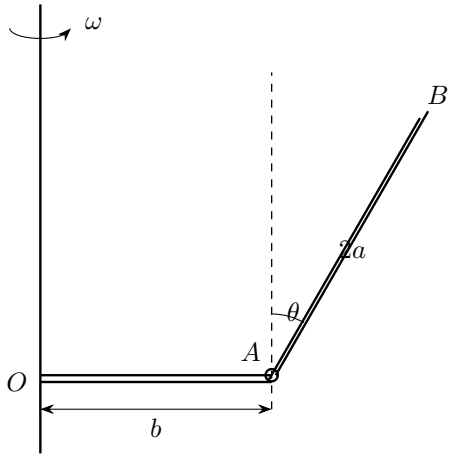


Fig. 1.61

Problem 1.62. A solid uniform sphere of radius R initially has its centre of mass at rest but rotates about a horizontal axis through the centre with angular velocity ω_0. It falls vertically to the ground from a position in which its lowest point is at a height h above the ground. The coefficient of restitution between the sphere and the ground is e, and the coefficient of friction between them is μ. Neglect air resistance and the deformation produced during the collision.

(1) Find the velocity of the centre of mass and the spin angular velocity of the sphere after the collision.

(2) Find the horizontal distance between the first and the second landing points.

(3) Draw the curve of the tangent of the angle θ between the velocity of the centre of mass after the collision and the ground against the initial angular velocity ω_0.

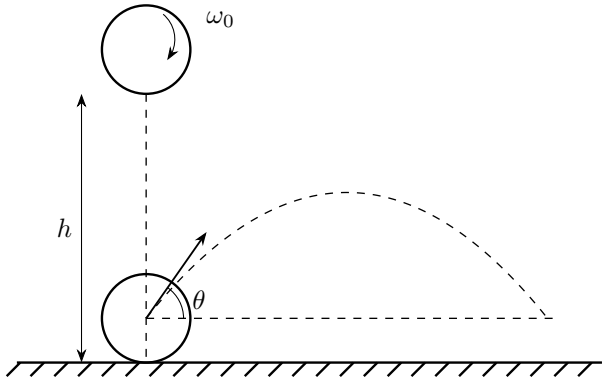


Fig. 1.62

Problem 1.63. A simple toy gyroscope consists of a uniform disc of mass m and radius R mounted perpendicularly and symmetrically at the midpoint of a light thin rod of length l and radius r ($\ll R$). Initially one end of the rod rests on the ground, the rod makes an angle θ with the vertical, and the gyroscope spins about the rod axis with a very large angular velocity ω, as shown. The friction between the rod and the horizontal ground is small, with coefficient μ. Neglect the nutation of the gyroscope, and assume that the fractional decrease of ω during one precession period is very small.

(1) Describe the motion of the gyroscope and find the precession angular velocity.

(2) After what time does the rod axis become vertical?

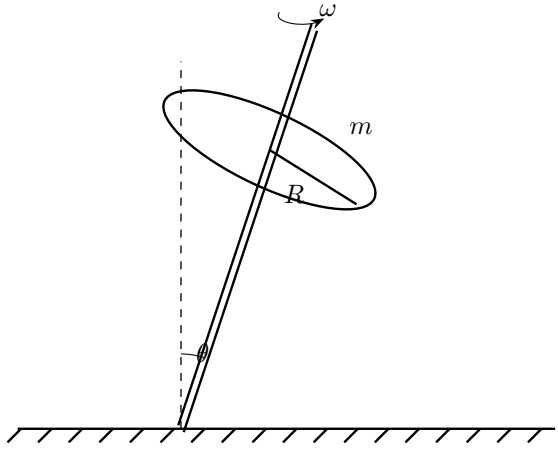


Fig. 1.63

Problem 1.64. A rocket ship flies past the Earth with speed $v = 0.8c$. Observers on the ship and on the Earth agree that this event occurs at 12:00.

(1) According to the clock on the ship, the ship passes a space station at 12:30. The space station is fixed relative to the Earth, and its clock is synchronised with the clocks on the Earth. When the ship passes the space station, what does the station's clock read?

(2) At ship time 12:30, that is, when it passes the space station, the ship sends a radio report signal back to the Earth. At what Earth time is this signal received?

(3) If, at the instant the Earth receives the signal, it immediately sends a reply signal, at what ship time is this reply signal received?

Problem 1.65. A thick glass plate moves to the right with speed v. At point A there is a flash lamp, and the light emitted by it passes through the thick glass and reaches point B, as shown. The distance between A and B is L, the thickness of the glass in its rest frame is D_0, and the refractive index of the glass is n. Find the time required for the light to travel from A to B.

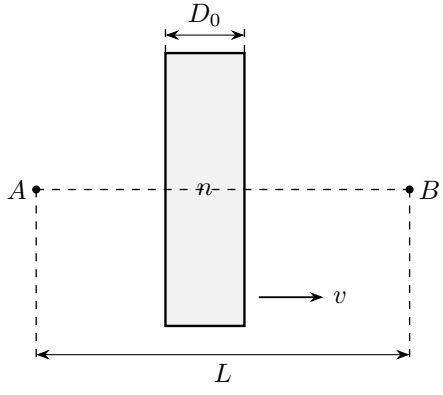


Fig. 1.65

Problem 1.66. In an inertial frame S, two spacecraft A and B are observed to fly parallel to each other in opposite directions along straight lines, the distance between their tracks being d, as shown. The speed of each spacecraft is $c/2$. When the two spacecraft reach their closest approach (shown by the dashed line in the figure), spacecraft A throws out a small packet with speed $\frac{3}{4}c$ (as observed in S).

(1) At what angle must the packet be thrown if it is to be received by spacecraft B? Assume that spacecraft A carries a coordinate system S' whose axes are parallel to those of S, and that S' moves relative to S in a direction parallel to the y-axis.

(2) As seen by the observer on spacecraft A, what is the speed of the packet?

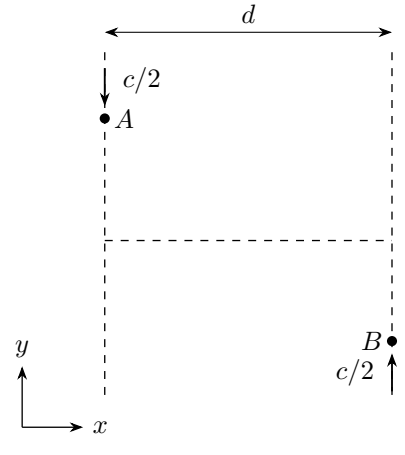


Fig. 1.66

Problem 1.67. Two straight rulers AB and $A'B'$ of equal proper length are at rest in frames S and S' respectively. Frame S' moves relative to S with speed v in the x direction, as shown. The clocks at rest at A and B are synchronised, and the clocks at rest at A' and B' are also synchronised. When A meets A', both clocks are set to zero; when B meets B', both clocks are also set to zero. Suppose that when A meets A', A emits a light signal, and it is known that when B' receives this signal, the clock at B' reads 1 time unit.

(1) When B receives this signal, what does the clock at B read, in time units?

(2) When B' receives this signal, it immediately sends a reply light signal. Find:

(i) the reading of the clock at A', in time units, when A' receives the reply signal;

(ii) the reading of the clock at A, in time units, when A receives the reply signal.

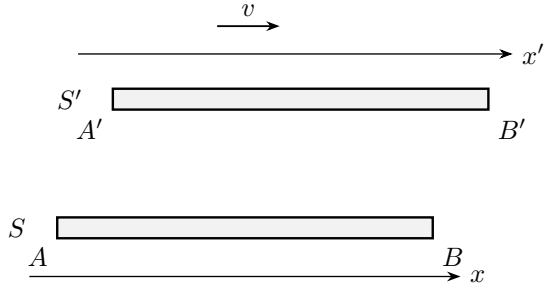


Fig. 1.67

Problem 1.68. At the end of a novel by Fred Hoyle, the hero flies with a high Lorentz factor in a direction perpendicular to the plane of the Galaxy. He says that he seems to be inside a "goldfish bowl" with a blue rim and a red bottom, flying toward the mouth of the bowl. Feynman bet 25 dollar bills that the light from the Galaxy would not look like that. It is known that his speed relative to the galactic frame is $v = 0.99c$, and that the angle between his flight direction and the line joining him to the edge of the Galaxy is observed to be $\phi = 45°$, as shown.

(1) Find the angle ϕ' between the direction of the light from the edge of the Galaxy and the flight direction, as seen by the traveller.

(2) Find the ratio ν'/ν of the frequency ν' of the light from the edge of the Galaxy received by the traveller to the frequency ν actually emitted.

(3) Calculate ϕ' and ν'/ν for various angles ϕ, and so decide who won the bet.

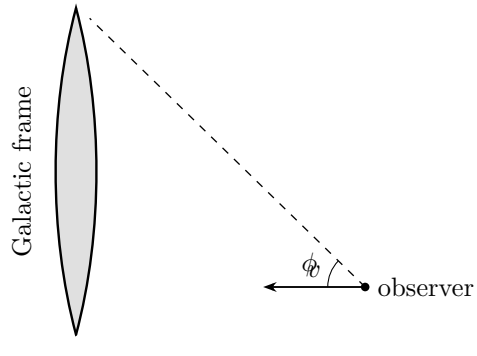


Fig. 1.68

Problem 1.69. When a light source approaches the Earth with speed v_1, person A, at rest in the frame S fixed to the Earth, sees the source emit green light of wavelength $\lambda_1 = 500\,\text{nm}$, while person B, in the frame S' moving relative to the Earth with speed v_2 in the same direction as v_1, sees the source emit red light of wavelength $\lambda_2 = 600\,\text{nm}$. When the source recedes from the Earth with the same speed v_1, A sees red light of wavelength λ_2. Find:

(1) the values of the speeds v_1 and v_2;

(2) the wavelength of the light seen by B when the source recedes from the Earth with speed v_1.

Problem 1.70. A particle decays into two γ-ray photons of energies E_1 and E_2 respectively, the angle between their directions being θ. Find the rest mass of the particle.

Problem 1.71. A π meson with momentum $200\,\text{GeV}/c$ decays into a μ meson and a neutrino ν, that is, $\pi \to \mu + \nu$. In the rest frame of the π meson, the lifetime of the π meson is $\tau_{\pi 0} = 2.60 \times 10^{-8}\,\text{s}$, its rest energy is $E_{0\pi} = 139.6\,\text{MeV}$, the rest energy of the μ meson is $E_{0\mu} = 105.7\,\text{MeV}$, and the rest mass of the neutrino is zero.

(1) Find the average distance travelled by the π meson before it decays.

(2) Find the maximum value of the angle between the direction of motion of the μ meson in the laboratory frame and the direction of motion of the π meson.

(3) Find the maximum and minimum momenta that the neutrino can have.

Problem 1.72. A photon rocket is to fly to the centre of the Galaxy, which is at a distance $R = 3.4 \times 10^4$ light years from the Earth. During the first half of the journey the rocket moves with acceleration $a' = 10\,\text{m/s}^2$ (relative to the rest frame of the rocket) in uniformly accelerated motion, and during the second half it decelerates uniformly with the same acceleration. When the rocket reaches its destination its rest mass is $M_0' = 1.0 \times 10^5\,\text{kg}$. What is the minimum power required of the rocket engine at the start of the launch?

1.2 Thermodynamics

Problem 2.01. A vacuum pump has a rotation rate of ω revolutions per minute and can withdraw c litres of gas per minute. The volume of a vessel is V. After what time can the pressure inside the vessel be reduced from p_0 to $10^{-2}p_0$?

Problem 2.02. A system consists of diatomic molecules and is at temperature T. The interaction potential energy between the atoms is

$$E_p = E_{p(0)} \left[\left(\frac{r_0}{r} \right)^{12} - 2 \left(\frac{r_0}{r} \right)^{6} \right].$$

Find the mean-square deviation of the distance r between the two atoms of a molecule.

Problem 2.03. Estimate the time required for an air molecule to cover a distance of 1 km under standard conditions. (Given: the mean molar mass of air is about 29 g, and the effective diameter of a molecule is 3×10^{-10} m.)

Problem 2.04. As shown in the figure, a vessel with a small hole contains a certain substance in which the gaseous and liquid phases coexist; the vapour molecules can escape through the small hole. Suppose that, when the two phases coexist in equilibrium, the temperature is T, the

latent heat of vaporisation is l, the molar mass of the substance is M, and the vapour may be regarded as an ideal gas. Find the relation describing how the relative intensity of the molecular beam issuing from the hole varies with temperature.

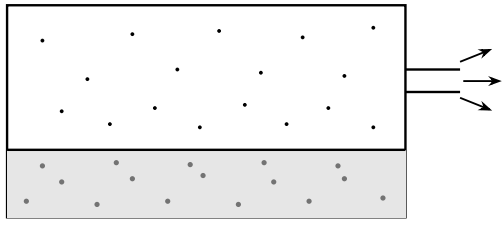


Fig. 2.04

Problem 2.05. It is known that one mole of a van der Waals gas undergoes a reversible cyclic process. From state 1 to state 2 the process is isothermal at temperature T_1, the volume changing from V_1 to V_2; from state 2 to state 3 the process is adiabatic, the volume changing from V_2 to V_3; from state 3 to state 4 the process is isothermal at temperature T_2; and from state 4 back to state 1 the process is adiabatic. Find:

(1) the heat absorbed by the system during the process 1–2;

(2) the work done by the system during the process 2–3;

(3) prove that the efficiency of the cycle is $\eta = 1 - T_2/T_1$. (Here T_1, T_2, V_1, V_2 and $C_{V,m}$ are all constants.)

Problem 2.06. One mole of a van der Waals gas is initially in the state (V_1, T_1). It undergoes a polytropic process in which its temperature increases to $T_2 = mT_1$, the polytropic index of the process being n; C_V is constant. Find:

(1) the heat absorbed by the system from the surroundings during the process;

(2) the work done by the system on the surroundings;

(3) the increase in entropy of the system.

Problem 2.07. Owing to convection in the atmosphere, air rises and descends. Since the atmospheric pressure varies with height, the volume of the gas changes as it rises or descends. Because the thermal conductivity of air is very small, the process of ascent or descent may be regarded as adiabatic. Find:

(1) the rate of change of atmospheric temperature with height, $\mathrm{d}T/\mathrm{d}z$;

(2) its value in the ordinary temperature range. (The molar mass of air is $M = 28.96$ g.)

Problem 2.08. A hollow shell of volume V is filled with thermal radiation whose energy per unit volume is $u = \sigma T^4$, where σ is the Stefan–Boltzmann constant and $\sigma > 0$; the radiation pressure is $p = \frac{1}{3}u$. Prove that the efficiency of a reversible Carnot cycle, in which thermal radiation is the working substance and which is in contact with two reservoirs at temperatures T_1 and T_2 $(< T_1)$, is

$$\eta = 1 - \frac{T_2}{T_1}.$$

Problem 2.09. As shown in the figure, a flat plate of area A, perpendicular to the x axis, moves in the direction opposite to the x axis with speed V. The surrounding rarefied gas is at

temperature T, its molecular number density is n, and the mass of a molecule is m. Using the Maxwell distribution, find the resistance experienced by the plate in its motion.

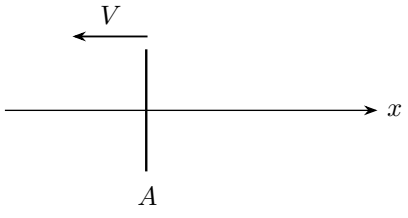


Fig. 2.09

Problem 2.10. The surface tension coefficient of water at one atmosphere and 4°C is $\sigma = 7.24 \times 10^{-2}\,\mathrm{N/m}$, and its isothermal compressibility is

$$\kappa = -\frac{1}{V}\left(\frac{\mathrm{d}V}{\mathrm{d}p}\right)_T = 4.75 \times 10^{-5}\,\mathrm{atm}^{-1}.$$

Find the density of a water drop of radius $r = 1.0 \times 10^{-6}$ cm at one atmosphere and 4°C.

Problem 2.11. A cylindrical vessel of height L and cross-sectional area A is filled with an ideal gas whose molecules have mass m. The temperature of the gas is T, in the room-temperature range. Find:

(1) the molar heat capacity at constant volume $C_{V,m}$ of the gas;

(2) $C_{V,m}$ for $T \to 0$ and for $T \to \infty$.

Problem 2.12. A certain amount of a liquid and its vapour are enclosed in a cylinder. The vapour may be regarded as an ideal gas, and the heat capacities of the liquid and of the vapour are both constant. Derive the Clausius equation for the saturated vapour pressure:

$$p = A\mathrm{e}^{-L(0)/kT} \cdot T^{(C_{gp}-C_{lp})/R},$$

where C_{gp} and C_{lp} are the constant-pressure heat capacities of the vapour and of the liquid respectively, and $L(0)$ is the latent heat of vaporisation at $T = 0$ K.

Problem 2.13. Two metals A and B of very large heat capacity are at temperatures T_1 and T_2 respectively, with $T_1 > T_2$. At a certain instant t they are brought into contact through a thin layer of a poor thermal conductor. The heat transmitted through the layer per unit time is $P_0\Delta T$, where ΔT is the temperature difference between the two metals. The whole system exchanges no heat with the surroundings. The heat capacities C_A and C_B of the metals do not vary with temperature. Find:

(1) the change in the entropy of metal A after a very short time Δt;

(2) the change in the total entropy of the two systems after a very short time Δt;

(3) the change in the entropy of the whole system after a sufficiently long time, when the two systems have reached thermal equilibrium.

Problem 2.14. A drunkard starts walking from the origin shown in the figure. He walks one step to the right with probability p_+, one step to the left with probability p_-, one step upward with probability q_+, and one step downward with probability q_-. Every step has length l. Find the probability that after N steps he is at the point P (whose x and y coordinates are integer multiples of l).

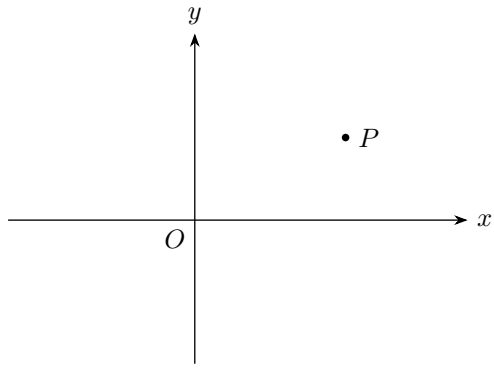


Fig. 2.14

Problem 2.15. Two balls of the same size and at the same temperature are both painted black, one of them being made of copper and the other of aluminium. They are suspended inside a cavity surrounded by melting ice. It is observed that the temperature of the aluminium ball falls from 13°C to 1°C in 10 minutes, whereas the copper ball takes 14.2 minutes to undergo the same temperature change. Given that the density of copper is $8.9 \times 10^3\,\mathrm{kg/m^3}$ and that of aluminium is $2.7 \times 10^3\,\mathrm{kg/m^3}$, calculate the ratio of the specific heat capacities of aluminium and copper.

Problem 2.16. Water at the freezing point (T_i, p_i) fills a vessel of invariable volume. The water is then cooled to T_f, whereupon the pressure increases to p_f. Find:

(1) the relation between the fraction y of the water (in all its states) that is in the solid state and the specific volumes of water in its several phases;

(2) given $t_i = 0°\mathrm{C}$, $p_i = 1.01 \times 10^5$ Pa, $t_f = -5°\mathrm{C}$, $p_f = 5.98 \times 10^7$ Pa; the coefficient of expansion at constant pressure of water over the temperature range involved is $\alpha = -67 \times 10^{-6}\,\mathrm{K^{-1}}$, and its isothermal compressibility is $k = 12.04 \times 10^{-11}\,\mathrm{Pa^{-1}}$; the specific volumes of water and of ice in the final state are $v_l = 1.0908 \times 10^{-3}\,\mathrm{m^3/kg}$ and $v_s = 1.1928 \times 10^{-3}\,\mathrm{m^3/kg}$ respectively. What is y?

Problem 2.17. A macroscopic ball of mass m' and radius r moves in vacuum with speed v_0. Ahead of it there are N moles of a rarefied ideal gas at temperature T with molecular number density n. At $t = 0$ the ball enters the gas. The gas consists of monatomic molecules of mass m, and the collisions of these atoms with the ball are elastic. The mean speed of the molecules of an ideal gas is known to be $\bar{v} = \sqrt{8RT/\pi M}$, and it is much greater than the initial speed of the ball; here M is the molar mass of the gas, $M = N_A m$. Assume that the number of collisions between atoms and the ball per unit time is $\lambda n \bar{v}$. Find:

(1) the change in the energy of the ball in a collision with one atom;

(2) the rate of decrease of the mean energy of the ball and its final mean energy;

(3) the change in the entropy of the system (the ball plus the ideal gas);

(4) $\lambda =$? (additional question)

Problem 2.18. One mole of a monatomic ideal gas undergoes the cyclic process shown in the figure. It is known that $p_2 = 2p_1$, $V_2 = 2V_1$ and $V_1 = 2V_3$. Find:

(1) the efficiency of the cycle;

(2) the highest temperature in the cycle;

(3) whether any of the three processes is a polytropic process.

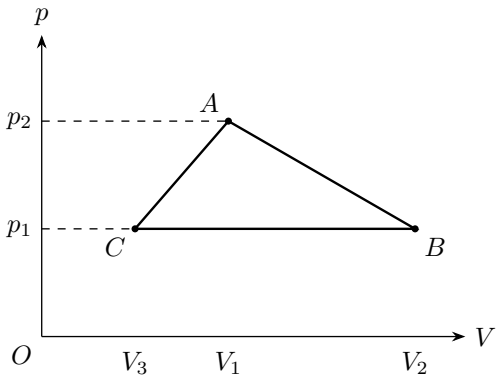


Fig. 2.18

Problem 2.19. As shown in the figure, a spherical vessel of volume V contains water vapour at an initial pressure p_0, and the temperature inside the sphere is maintained at T. A thin tube whose lower end has unit cross-sectional area is now inserted into liquid nitrogen. Assume that every molecule entering the tube condenses inside it. Find the time required for the pressure of the water vapour (which may be treated as an ideal gas) inside the sphere to fall to p, and the energy removed per unit time in the vicinity of that instant.

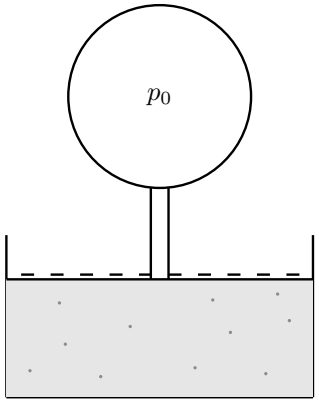


Fig. 2.19

Problem 2.20. It is known that a spherical soap film of radius R carries a charge Q, and that the surface tension coefficient of the soap solution is σ. Find the pressure difference between the inside and the outside of the film.

Problem 2.21. As shown in the figure, in case (a) the liquid in a capillary tube stands at a height h. In case (b) there is an inverted U-shaped capillary tube of height $h_1 < h$. Discuss the curvature of the liquid surface at the mouth of the U-tube when the distance from the mouth to the liquid surface is h_2.

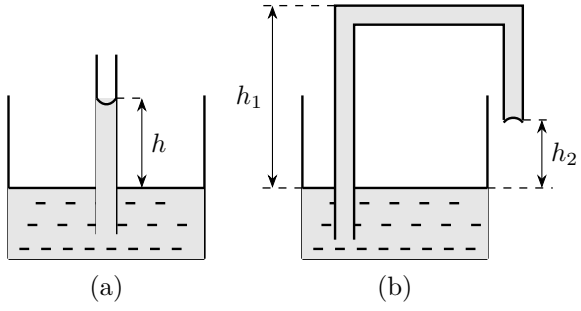


Fig. 2.21

Problem 2.22. The surface tension coefficient of a certain solution is $\sigma = a - bT$. In an isothermal process at temperature T, a small ball of radius R is divided into n small balls. Find:

(1) the work done on it by the surroundings;

(2) the heat absorbed from the surroundings;

(3) the increase in entropy of the system.

Problem 2.23. The temperature of the air is T_0, its molar mass is M_0, and the acceleration due to gravity is g. Find how the boiling point of water varies with height.

Problem 2.24. Suppose that a crystal lattice contains N atoms at lattice sites. When an atom leaves a lattice site and occupies an interstitial position, an interstitial atom and a vacancy are formed. Such a defect in a crystal is called a Frenkel defect.

(1) Suppose that the number of lattice sites and the number of interstitial positions are both N. Prove that when n vacancies and n interstitial atoms are formed in the lattice, the increase in entropy of the system is

$$\Delta s = 2k \ln \frac{N!}{n!\,(N-n)!}.$$

(2) Suppose that the energy difference between an atom at a lattice site and at an interstitial position is u, and that the temperature is T. Prove that the number of vacancies is

$$n = N\mathrm{e}^{-u/2kT}.$$

Problem 2.25. There is a one-dimensional chain consisting of N links ($N \gg 1$), each of length a, as shown in the figure; the distance between its two ends is x. Each link can rotate freely. Find:

(1) the entropy of the whole chain;

(2) the tension $f(x)$ that must be maintained when the chain is at temperature T and its ends are held a distance x apart.

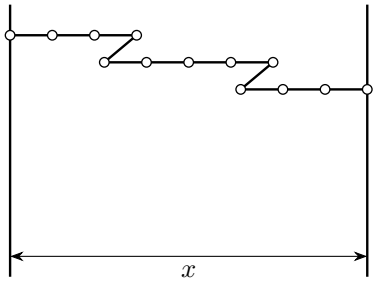


Fig. 2.25

Problem 2.26. A coin is tossed 400 times. What is the probability of obtaining 210 heads?

Problem 2.27. A uniform rod of mass m, length l and constant C_p has its two ends in contact with two heat reservoirs at temperatures T_1 and T_2 $(< T_1)$ until a steady state is reached; the two reservoirs are then removed and the rod is allowed to reach equilibrium. Find the increase in entropy of the rod between these two states.

Problem 2.28. N atoms are regularly arranged to form an ideal crystal. If an atom leaves a lattice site in the interior of the crystal and moves to the surface of the crystal, forming a new layer, a vacancy appears inside the crystal. Such a defect of the lattice is called a Schottky defect. Let N denote the number of atoms on the lattice, n the number of vacancies in the crystal, and u the energy required for one atom to move from the interior to the surface. Prove that, at temperature T, with $1 \ll n \ll N$ and neglecting the change in volume of the crystal, the following relations hold:

$$\frac{n}{N+n} = \mathrm{e}^{-u/kT} \qquad \text{or} \qquad n \approx N\mathrm{e}^{-u/kT},$$

where k is the Boltzmann constant.

1.3 Electromagnetism

Problem 3.01. Suppose that the force between electric charges were inversely proportional to the cube of the distance between them, all other properties being unchanged. Answer the following questions and give your reasons:

(1) Does Gauss's law still hold?

(2) Can the concept of electric potential still be introduced?

(3) Is there an electric field inside a uniformly positively charged spherical surface? What is its direction?

(4) In electrostatic equilibrium, is the charge on a conductor still distributed over its surface?

Problem 3.02.

(1) Two dipoles A and B have dipole moments p_1 and p_2 respectively, and the distance between their centres is r. The angle between p_1 and the vector drawn from A to B is θ, and the angle between p_2 and r is φ (so that the angle between p_1 and p_2 is $\varphi - \theta$), as shown in Fig. 1. Find the r-component F_r and the θ-component F_θ of the force exerted by p_1 on p_2.

(2) Four charges of equal magnitude and alternating sign are arranged as shown in Fig. 2 to form an electric quadrupole. Find the potential Φ and the field strength E produced by this charge system at the point $P(r, \theta)$ (find E_r and E_θ separately). (Assume $l \ll r$.)

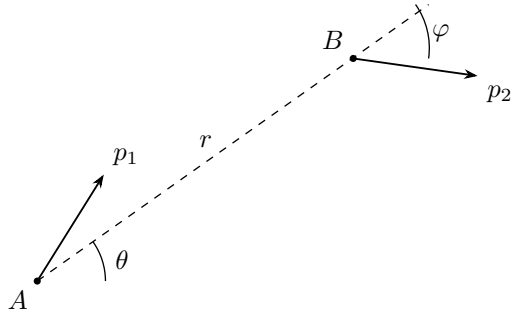


Fig. 3.02(a)

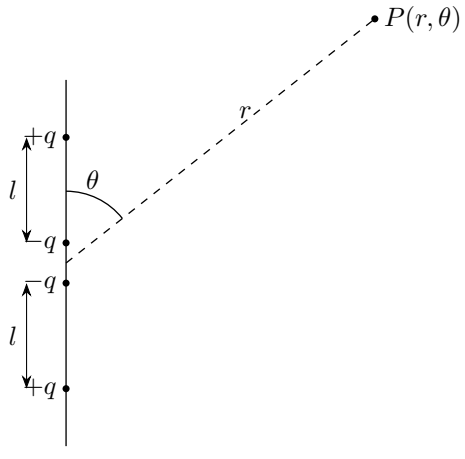


Fig. 3.02(b)

Problem 3.03. A point charge $+q$ and a point charge $-q'$ $(q' < q)$ are located at the points A and B on the x-axis respectively, the distance between A and B being l. A certain line of force issuing from $+q$ makes an angle α with the line AB.

(1) Find the angle β between the final tangent of this line of force and the x-axis.

(2) Find the position of the point C at which this line of force, or its final tangent, intersects the x-axis.

Problem 3.04. A uniformly charged ring of charge $+q$ and radius a and a uniformly charged ring of charge $-q$ and radius b are placed concentrically in the same plane, as shown. P is a distant point; its distance from the centre O of the rings is r, and OP makes an angle θ with the axis of the rings (taken as the z-axis), with $a < b \ll r$. Find the potential Φ at P, and the r-component E_r and the θ-component E_θ of the electric field strength at P, accurate to order a^2/r^2.

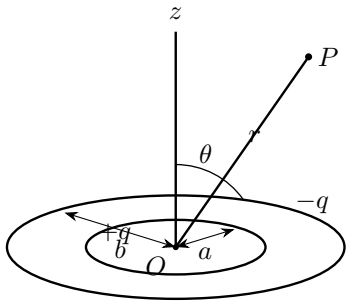


Fig. 3.04

Problem 3.05.

(1) Find the ratio of the potential at a point on the surface of a uniformly charged solid sphere to the potential at its centre.

(2) Find the ratio of the potential at a corner of a uniformly charged cube to the potential at its centre.

Problem 3.06. A spherical shell of radius a carries a uniform charge q. Outside the shell there is a volume charge distribution whose density depends only on the distance r from the centre of the shell. It is known that the magnitude of the electric field is the same at every point outside the shell. Find the volume charge density ρ as a function of r.

Problem 3.07. The space between two large planes separated by a distance l is filled with a plasma. Both the positive and the negative ions carry charge e, and both have number density n. The masses of the individual ions are m_+ and m_- respectively. Owing to a disturbance, the layers of positive and negative ions are displaced slightly in the direction perpendicular to the planes, and thereafter the two ion layers oscillate back and forth along this direction. Find the frequency of oscillation.

Problem 3.08.

(1) If the force between charges were not inversely proportional to the square of the distance but proportional to $r^{-(2+\delta)}$, everything else being the same as in Coulomb's law, find the potential at points inside and outside a uniformly charged spherical surface. Let the radius of the spherical surface be R and its surface charge density be σ.

(2) Building on Cavendish's work, Maxwell used the following experiment to test the value of δ: two thin conducting spherical shells of radii a and b ($a > b$) were placed concentrically and connected by a thin wire; the outer shell was charged to potential U, after which the power supply was disconnected and then the thin wire joining the two shells was removed; the outer shell was then earthed. The potential of the inner shell was measured to be no greater than u. Estimate the value of δ from this. Given $\delta \ll 1$, only first-order small quantities need be retained in the estimate, and earthing may be regarded as equivalent to connecting to infinity.

Problem 3.09. The figure shows a schematic diagram of Kelvin's water-drop generator. Water drops formed at the left and right nozzles of a T-shaped pipe pass respectively through the aluminium cylinders A_1 and A_2 and then drip into the aluminium cups B_1 and B_2. After the drops have fallen for some time, a potential difference of several kilovolts appears between the two initially uncharged aluminium cups. Analyse its principle. In the figure, the aluminium

cylinder A_1 is connected by a wire to the aluminium cup B_2, and the aluminium cylinder A_2 is connected by a wire to B_1.

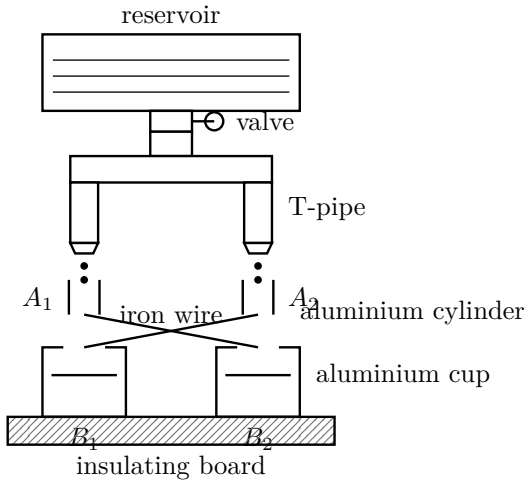


Fig. 3.09

Problem 3.10. There are two common types of microphone: the condenser type and the electret type. Each of them consists of a fixed plate and a movable plate, as shown in the figure. The area of both the fixed and the movable plates is S, and the separation between them is d. For the condenser microphone both plates are metal conductors. For the electret microphone the movable plate is a metal conductor and the fixed plate is a dielectric with one surface metallised, the thickness of the dielectric being d_1 and its relative permittivity ε_r; the non-metallised surface carries a constant charge acquired during manufacture, with surface charge density σ, and its metallised surface (which carries no external charge) serves as the electrode. Under the action of sound pressure the movable plate of either microphone vibrates harmonically about its equilibrium position with displacement $x_0 \sin \omega t$. Design the best circuit for each of the two microphones so as to convert this vibration into an output voltage signal. State the requirements on the parameters of the components used, and write the relation between the output voltage signal and the displacement.

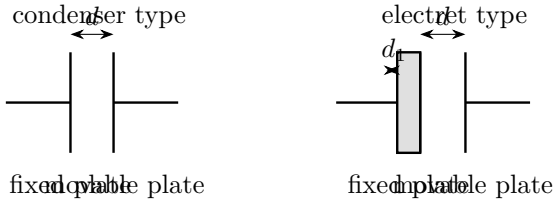


Fig. 3.10

Problem 3.11.

(1) Two semi-infinite conducting plates are placed at right angles to each other. A point charge q is moved from infinity to a point at a distance d from each of the two conducting plates, as shown in the figure. Find the work W_{in} done by the external force during this process.

(2) With the point charge q in this position, suppose the two plates change from conductors into insulators; the point charge q is then moved from this position back to infinity. Find the work W_{out} done by the external force during this process.

(3) What is the electrostatic energy of the system of charges on the insulating plates at this time?

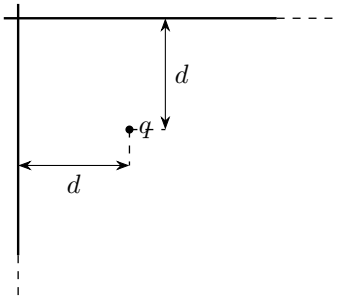


Fig. 3.11

Problem 3.12. A point charge Q is placed inside an electrically neutral conducting thin spherical shell of radius a, at a distance r from the centre. Find the electrostatic force acting on the point charge.

Problem 3.13. N uncharged conducting circular plates are mounted coaxially, parallel and equally spaced, and are numbered in order, as shown in the figure. The spacing between neighbouring plates is much smaller than the diameter of the plates. The negative terminal of a power supply of electromotive force U is connected to plate 1 and earthed, and the positive terminal is connected in turn to plate N, plate $N-1$, ..., plate k, ..., plate 3, plate 2. Find:

(1) the ratio of the absolute values of the charges carried by plate k and plate 1, $|q_k| : |q_1|$;

(2) the potential U_k of plate k.

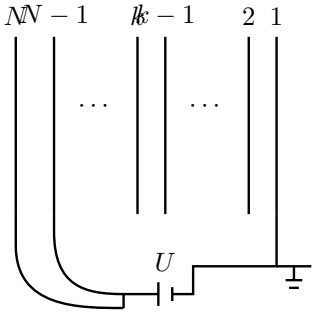


Fig. 3.13

Problem 3.14. A balloon of radius a is charged to potential V. The balloon leaks charge through the atmosphere, and at the same time a deflation valve is opened and adjusted so that its radius gradually decreases.

(1) Find the radius r of the balloon as a function of the time t if its potential is to remain constant. Let the resistivity of the atmosphere be ρ and its relative permittivity be 1.

(2) What is the Joule heat dissipated in the atmosphere during the whole leakage process? Is it equal to the initial electrostatic energy of the balloon? Why?

Problem 3.15. A capacitor is formed of two thin conducting spherical shells of radii a and b ($a < b$). There is a small hole in the outer shell, through which an insulated wire is led out to connect the inner shell with a third conductor of radius c; the distance r of this conductor from the capacitor is very large. The outer shell of the capacitor is earthed, and the two connected conductors are given a charge Q. Find an approximate expression for the electrostatic force acting on the third conductor.

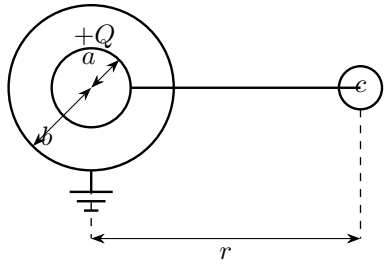


Fig. 3.15

Problem 3.16. Above a very large horizontal earthed conducting plate there is a uniform electric field E_0 directed vertically. A conducting hemisphere of mass density ρ and radius R is now placed flat on the conducting plate with its spherical surface facing upwards, as shown in the figure. Find the least value of E_0 required to lift the hemisphere off the plate.

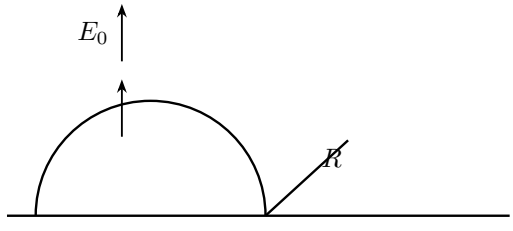


Fig. 3.16

Problem 3.17. A long cylindrical straight wire of radius r is placed horizontally with its axis at a distance d from the earth (which may be regarded as a conductor), as shown in the figure. Find the capacitance per unit length C^* between the wire and the earth, expressed as a function of d/r, in the following two cases:

(1) $r \ll d$;

(2) $r < d$, but of the same order of magnitude as d.

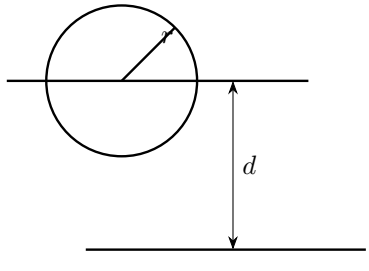


Fig. 3.17

Problem 3.18. A conducting spheroid of revolution has the surface equation

$$\frac{x^2}{a^2}+\frac{y^2}{a^2}+\frac{z^2}{b^2}=1.$$

(1) Find the capacitance of this conductor.

(2) Find the approximate expressions for the capacitance when $b \gg a$ and when $b \ll a$.

The following integral formulae may be used:

$$\int \frac{\mathrm{d}x}{\sqrt{1+x^2}} = \ln\left(x+\sqrt{1+x^2}\right), \qquad \int \frac{\mathrm{d}x}{1+x^2} = \arctan x,$$

$$\int \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \arcsin x, \qquad \int \frac{\mathrm{d}x}{1-x^2} = \frac{1}{2}\ln\frac{1+x}{1-x}.$$

Problem 3.19. On the axis of a grounded metal cylindrical surface of radius R there is a thin long wire of radius r_0 maintained at a high positive voltage V_0; this arrangement constitutes a cylindrical counter. As soon as ionising radiation appears, it triggers an "avalanche" near the surface of the wire, where the field is very strong. Suppose that at the surface of the wire N electrons are ionised from atoms, leaving N ions, which then begin to move away from the wire under the action of the electric field. Assume that the ion mobility (the ratio of the ion velocity to the electric field) is a constant w, and that the moving ions form a uniform thin cylindrical layer around the wire.

(1) Find the radial position r of the ions as a function of the time t, measured from the moment at which the ions are formed at the surface of the wire. In this part the avalanche and the change of the field caused by the ions are to be ignored.

(2) If the potential V_0 is kept constant, a charge Q must flow to the wire as the ions drift radially. Given that the total charge of the ions is Q_t, calculate the charge Q as a function of the time t.

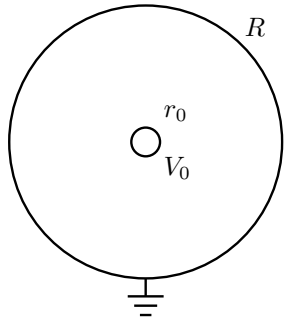


Fig. 3.19

Problem 3.20. A dielectric spherical shell has inner and outer radii a and b, the relative permittivity of the dielectric being ε_r. A point charge q is located at the centre of the shell. Find the work done by the external force in moving this point charge from the centre to infinity. (One may imagine a small hole made in the shell, and assume that this does not affect the polarisation of the dielectric.)

Problem 3.21. Dilute dielectric particles are distributed uniformly between the plates of a parallel-plate capacitor. Each particle may be regarded as a small sphere of radius a (a much

smaller than the dimensions of the capacitor), the relative permittivity of the dielectric being ε_r, and the number density of the particles being n. The space between the plates may then be regarded as filled with an equivalent uniform dielectric. Find the relative permittivity ε_r' of this equivalent dielectric.

Problem 3.22. A semi-infinite liquid dielectric of relative permittivity ε_r contains a point charge q at a depth h below its surface. Find the distribution of the polarisation charge on the surface of the liquid and the total amount of polarisation charge on the surface, and find the electrostatic force acting on the point charge.

Problem 3.23. An electrostatic precipitator consists of a vertical metal rod of radius a and a coaxial thin-walled cylinder of radius b; the height of the device is much greater than both a and b. A voltage U is applied between the rod and the cylinder.

(1) Find the electric field strength $E(r)$ at a distance r $(a < r < b)$ from the axis inside the cylinder.

(2) A dust particle may be regarded as a small dielectric sphere of radius c $(c \ll a, b)$ with relative permittivity ε_r. Suppose that the particle is inside the cylinder at a distance r $(a < r < b)$ from the axis. Estimate the magnitude and the direction of the electric force acting on the particle.

(3) Let the mass of the particle be m. Find the time required for the particle to travel from rest, in the direction of the electric force, from one electrode to the other (gravity and air resistance are neglected).

Problem 3.24. There is a uniform electric field E_0 in an infinite liquid dielectric of relative permittivity ε_r.

(1) A spherical cavity of radius R is hollowed out in the dielectric. Find the electric field inside the cavity.

(2) A conducting sphere of the same radius is placed in the spherical cavity and given a charge Q. Find the electric force acting on the conducting sphere.

Problem 3.25. Two point charges q and q' are placed at the centres of two small spherical cavities, both of radius a, in an infinite isotropic dielectric, the distance between the charges being r $(r \gg a)$. Find the electrostatic force acting on the charge q'. Let the relative permittivity of the dielectric be ε_r.

Problem 3.26. A point electric dipole of dipole moment p is located at the centre of a dielectric sphere of radius a and relative permittivity ε_r. Find the potential at points inside and outside the sphere. (Hint: the field outside the sphere is that of a dipole.)

Problem 3.27. Four metal pins are inserted vertically in a row in the horizontal surface of a very large piece of resistive material, the spacing between neighbouring pins being d, and the pins make good contact with the surface. A power supply is connected between the two outer pins, and a voltmeter between the two middle pins. Let the current flowing through the power supply be I and the reading of the voltmeter be U. Find the resistivity ρ of the material.

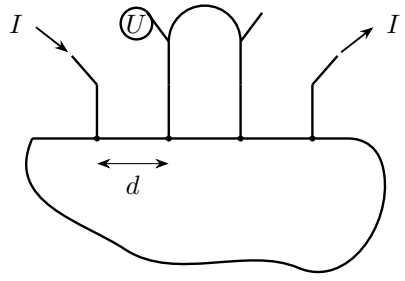


Fig. 3.27

Problem 3.28. Consider the infinite resistor network shown in the figure. Starting from a square of side L, joining the midpoints of its sides gives another square, and the process is repeated without end. The resistance of each side is proportional to its length. Find the equivalent resistance between two opposite vertices of the outermost square, expressing the result in terms of the resistance R of one side of that square.

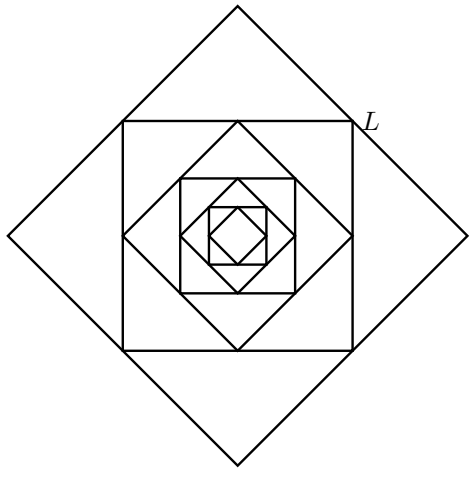


Fig. 3.28

Problem 3.29. Nine identical resistors, each of resistance $1\,\Omega$, are connected to form the edges of a triangular prism, as shown in the figure. A current enters the network at the vertex A and leaves it at the vertex E. Find the equivalent resistance R_{AE} between A and E.

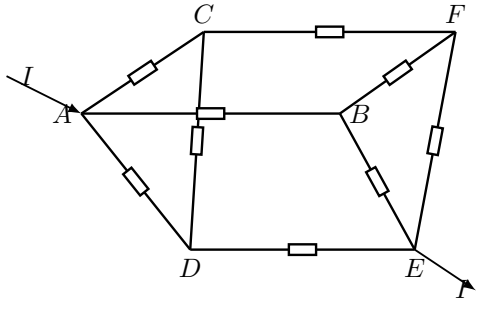


Fig. 3.29

Problem 3.30. The four arms of the Wheatstone bridge shown in the figure have resistances R_1, R_1, R_2 and $R_2(1-\delta)$, where $\delta \ll 1$. The galvanometer G is ideal, so that its resistance may be neglected. Find the ratio I_G/I of the current through the galvanometer to the total current, accurate to the first order in δ.

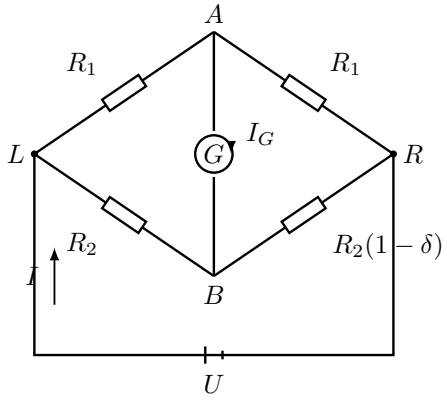


Fig. 3.30

Problem 3.31. The figure shows a ladder network of $n = 10$ cells. In the i-th cell a battery of electromotive force $V_i = (10 + i)\,\mathrm{V}$ and a resistor $2R$ are connected in series between the node A_i and the common return line B. Adjacent nodes A_i and A_{i+1} are connected by a resistor R, and the far end A_n is connected to the return line through a resistor $2R$. The output terminal A is the node A_1.

(1) Find the equivalent electromotive force and the internal resistance of the network between A and B.

(2) Which of the points $A_1, A_2, \ldots, A_{10}$ has the highest potential? Calculate this potential.

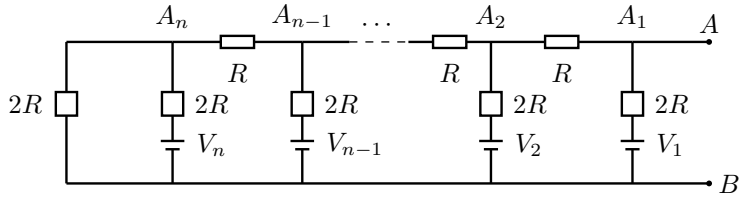


Fig. 3.31

Problem 3.32. A thin conducting ring of radius b and resistance R rotates about its axis with initial linear speed v_0. It is then brought to rest; the self-inductance of the ring is neglected.

(1) How can the sign of the charge carriers in the ring be determined from the direction of the current in it?

(2) During the deceleration a total charge q passes through a cross-section of the ring. Find the charge-to-mass ratio e/m of the carriers.

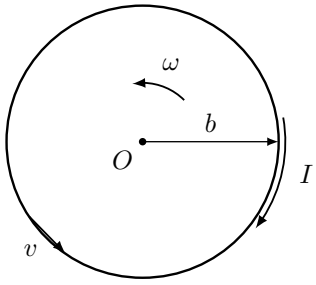


Fig. 3.32

Problem 3.33. A long straight wire of radius b and conductivity g carries a steady current whose density far from any disturbance is j_0. A spherical non-conducting bubble of radius a ($a \ll b$) is formed in the wire, its centre lying on the axis.

(1) Find the electric field E_{in} inside the bubble.

(2) Find the electric field at a point P in the wire at a distance r from the centre of the bubble ($r > a$), the radius OP making an angle θ with the direction of the current.

(3) Find the current I through the cross-section of the wire passing through the centre of the bubble, retaining the first non-vanishing correction due to the bubble.

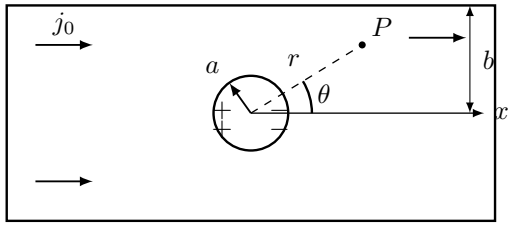


Fig. 3.33

Problem 3.34. The atmosphere surrounding the earth is ionised by cosmic rays and becomes a non-uniform conducting medium whose conductivity may be written as

$$\sigma = \sigma_0 + a(r - R)^2,$$

where r is the distance from the centre of the earth, $\sigma_0 = 3 \times 10^{-14}\,(\Omega \cdot \mathrm{m})^{-1}$ and $a = 0.5 \times 10^{-20}\,(\Omega \cdot \mathrm{m}^3)^{-1}$. In fine weather the electric field strength at the earth's surface is $E_0 = 100\,\mathrm{V/m}$, directed vertically downwards. The current in the atmosphere may be regarded as steady. Given that the earth's radius is $R = 6.4 \times 10^6\,\mathrm{m}$, calculate:

(1) the electric field $E(r)$ of the steady current in the atmosphere and the value of the total current I;

(2) the charge density distribution $\rho(r)$ in the atmosphere and the value of the surface charge density Σ at the earth's surface, the relative permittivity of the atmosphere being 1;

(3) the potential at the earth's surface and its numerical value;

(4) if the charge of the earth were not replenished, estimate how long it would take for the earth's charge to be neutralised by the current. In fact the total charge of the earth remains roughly constant. What natural phenomenon can keep the earth's charge from being neutralised by the current, so that it stays roughly constant?

The following integral formulae may be useful: putting $X = a + bx + cx^2$ and $q = 4ac - b^2$, one has

$$\int \frac{\mathrm{d}x}{X} = \frac{2}{\sqrt{q}} \arctan \frac{2cx + b}{\sqrt{q}}, \qquad \int \frac{\mathrm{d}x}{xX} = \frac{1}{2a} \ln \frac{x^2}{X} - \frac{b}{2a} \int \frac{\mathrm{d}x}{X}.$$

Problem 3.35. A d.c. supply transmits power to a distant load through two identical parallel thin wires (whose resistance is neglected) at a voltage of 33 kV. The capacitance per unit length between the two wires is $10\,\mu\mu$F/m. Find:

(1) the power transmitted when there is no net electromagnetic force acting between the two wires;

(2) the electric energy and the magnetic energy stored per unit length of the transmission line in this case.

Problem 3.36. A solenoid of length L and radius R has n turns per unit length and carries a current I.

(1) Find the magnetic flux density B at the centre of the axis of an end face of the solenoid.

(2) Suppose that the solenoid is semi-infinite. Find an approximate expression for the radial component B_ρ of the magnetic flux density at a point on the end face at a distance ρ_0 from the axis ($\rho_0 \ll R$).

Problem 3.37. The distance between the plates of a parallel-plate capacitor is d. The capacitor is initially uncharged. There is a uniform magnetic field of flux density B in space, its direction being parallel to the plates of the capacitor. When an electrically neutral liquid flows with velocity v between the two plates, what is the reading of a voltmeter connected between the plates of the capacitor? Let the relative permittivity of the liquid be ε_r.

Problem 3.38. A long dielectric cylinder of radius R and polarisability χ rotates about its axis with constant angular velocity ω. There is a uniform magnetic field of flux density B in space, directed along the axis. Find the distribution of the polarisation charge on the dielectric cylinder. (Edge effects are neglected.)

Problem 3.39. Two thin-walled metal mesh cylinders of radii a and b and length L ($a < b \ll L$) are placed coaxially, and the space between them is filled with a uniform medium of resistivity ρ. A voltage U is applied between the inner and the outer cylinder. Edge effects are neglected.

(1) Find the current I flowing through the inner and outer cylinders.

(2) If a uniform magnetic field of flux density B is applied along the axis of the cylinders, find the current I' then flowing through the inner and outer cylinders. Let the relative permeability of the medium be 1, the charge-to-mass ratio of the current carriers be e/m, and their number density be n. (The magnetic field produced by the current itself is neglected.)

Problem 3.40. A circular coil of radius b carries a current I. A small paramagnetic dielectric sphere of relative permeability μ_r and radius a ($a \ll b$) is placed at its centre.

(1) Find the magnetisation M of the small sphere (the sphere may be regarded as uniformly magnetised).

(2) The presence of the small sphere increases the self-inductance of the coil. Find this increase ΔL.

Problem 3.41. A cylindrical thin shell of length l and radius a ($l \gg a$) carries a uniform surface charge of density σ. The shell rotates about its axis with angular velocity $\omega = kt$, where k is a constant. Edge effects are neglected. Find:

(1) the magnetic field B at points inside the cylindrical shell;

(2) the electric field E at points inside the cylindrical shell;

(3) the total electric energy W_E and the total magnetic energy W_m inside the cylindrical shell;

(4) the Poynting vector inside the cylindrical shell, and its relation to the energy of the electromagnetic field inside the shell.

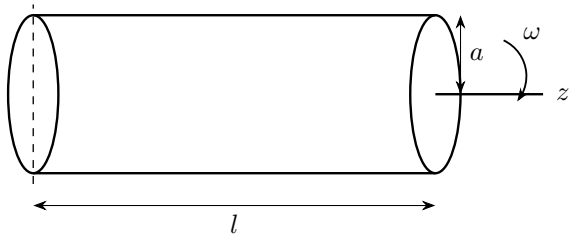


Fig. 3.41

Problem 3.42. The axis of a long rod of radius R coincides with the z-axis, and the relative permeability of the rod is μ_r. There is a uniform external magnetic field of flux density B_0 in the positive x direction in space. Find the magnetic flux density B_1 inside the rod and B_2 outside the rod at an arbitrary point (the latter expressed in terms of its r and θ components).

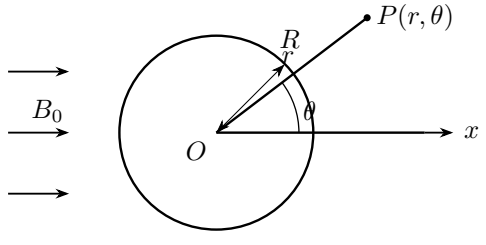


Fig. 3.42

Problem 3.43. Motion of electrons in a magnetron. Two coaxial metal cylinders of radii a and b ($a < b$) are placed in a uniform magnetic field whose direction is parallel to the axis. A radial electric field directed towards the inner cylinder exists between the two cylinders, the voltage between them being U. Electrons emitted without initial velocity from the surface of the inner cylinder are accelerated by the electric field and fly towards the outer cylinder, while the magnetic field deflects the direction of their motion and may even make the electrons return to the inner cylinder. Find the magnetic flux density B that just prevents the electrons from reaching the outer cylinder.

Problem 3.44. A charged particle of charge $+q$ and mass m moves in a circle of radius R in the horizontal symmetry plane of a cyclotron. In this plane the magnetic field is along the vertical direction (taken as the z direction), with flux density $B = B_z$. At the location of the particle, however, the flux density begins to decrease gradually in the radial direction (the direction of increasing r).

(1) Find the cyclotron frequency of the particle.

(2) Show that, when the particle suffers a small disturbance in the vertical direction, it performs simple harmonic motion in the vertical direction.

(3) Show that the angular frequency ω of this oscillation is proportional to the cyclotron frequency ω_c of the particle, that is,

$$\omega = \alpha\omega_c,$$

and find the coefficient α.

Problem 3.45. A particle of mass m and charge $+q$ moves in a spiral in an axisymmetric non-uniform magnetic field, circling about the axis of the field (taken as the z-axis). It passes gradually from a region of flux density B_1 into a region of stronger magnetic field; the radius of gyration slowly decreases, and the particle is finally reflected (a magnetic mirror). Suppose that in the region B_1 the particle has speed v_0 and its velocity makes an angle φ_0 with the z-axis, and that the flux density at the strongest point of the field is B_2. Find the least value φ_{0c} of the angle φ_0 for which the particle can be reflected. (Reasonable approximations may be made.)

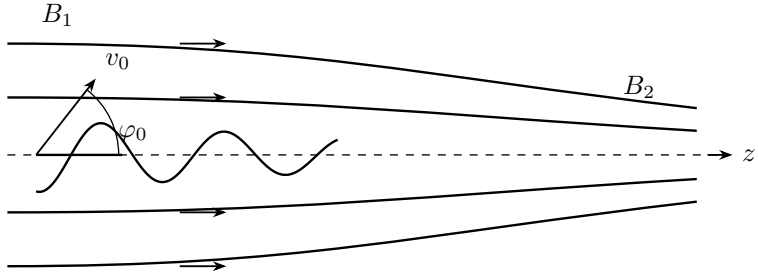


Fig. 3.45

Problem 3.46. A rectangular coil of mass m and width b, of sufficient length, moves in the plane of the paper along its length (the x direction) with initial speed v_0 into a uniform magnetic field of flux density B perpendicular to the paper and directed inwards; the edge of the field is parallel to one side of the coil, as shown. Take the moment at which the right side of the coil just enters the field as the origin of time, and let its coordinate at that moment be $x = 0$. Gravity is neglected. Find the subsequent motion of the coil (that is, x as a function of t) in the following two cases:

(1) the resistance of the coil is R and its self-inductance is negligible;

(2) the self-inductance of the coil is L and its resistance is negligible.

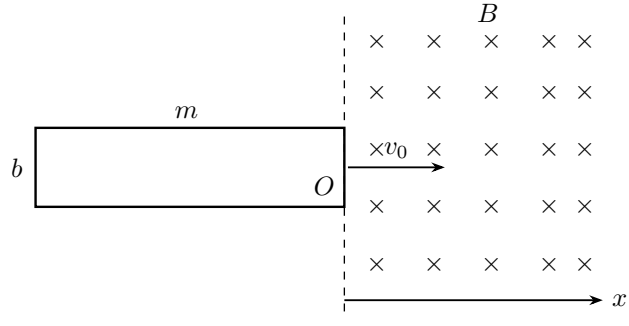


Fig. 3.46

Problem 3.47. A circular ring made of gold wire stands upright on a horizontal table, resting on a point of its rim. There is a uniform magnetic field in space, directed vertically. The ring is disturbed and starts to fall from a position making an angle of 0.1 rad with the vertical. It is known that the magnetic flux density is $B = 10^4\,\mathrm{Gs}$, the radius of the ring is $r = 1\,\mathrm{cm}$, the radius of the cross-section of the gold wire is $r_1 = 1\,\mathrm{mm}$, the conductivity of gold is $\sigma = 4.25 \times 10^7\,(\Omega \cdot \mathrm{m})^{-1}$, and its density is $\rho = 19.3\,\mathrm{g/cm^3}$. Assume that the point of support does not slip while the ring falls.

(1) Is the gravitational potential energy released during the fall of the ring mainly used to increase the kinetic energy of the ring, or is it converted into Joule heat? Explain your reasoning (an order-of-magnitude estimate is sufficient).

(2) Neglecting secondary factors, calculate the time taken for the ring to fall over.

Integral formula:

$$\int \frac{\mathrm{d}x}{\sin x} = \ln \tan \frac{x}{2}.$$

Problem 3.48. A small cylindrical permanent magnet is placed vertically. At a point 1 m directly above the centre of the magnet, the magnetic flux density is B_0. A small superconducting circular coil is brought from far away to a position directly above the magnet, coaxial with it, as shown. The radius of the coil is a, its mass is m, and its self-inductance is L; the coil can move only up and down.

(1) Find the equilibrium height z_0 of the coil above the centre of the magnet. It is given that $a \ll z_0$.

(2) Find the period of the small vertical oscillations of the coil after a small disturbance (express the answer in terms of z_0).

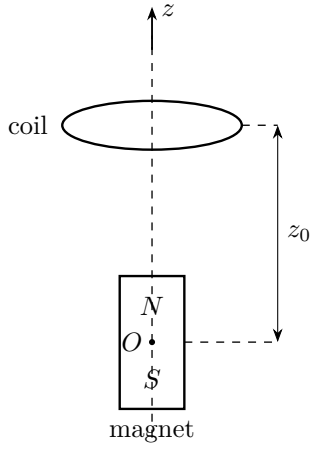


Fig. 3.48

Problem 3.49. There is an experiment demonstrating the magnetic effect of eddy currents. A cylindrical magnet is released from rest at the mouth of a long vertical metal tube, as shown. After a short acceleration, the magnet falls slowly at a certain constant speed. If the magnet is replaced by an iron block of the same shape, the block accelerates downwards just as in free fall. (The fall of the magnet or of the iron block in the tube can be observed through holes drilled horizontally in the wall of the tube, as shown.) Give a quantitative explanation of this phenomenon.

It is known that the diameter and the height of the magnet are both 1.5 cm, its density is $7.8\,\mathrm{g/cm^3}$, and the magnetic flux density at the centre of its end face is 0.3 T. The internal diameter of the metal (copper) tube is 1.7 cm, its external diameter is 2.7 cm, and its resistivity is $1.67 \times 10^{-8}\,\Omega \cdot \mathrm{m}$. Estimate from these data the steady speed of the magnet.

Integral formula:

$$\int \sin^n x \,\mathrm{d}x = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \,\mathrm{d}x.$$

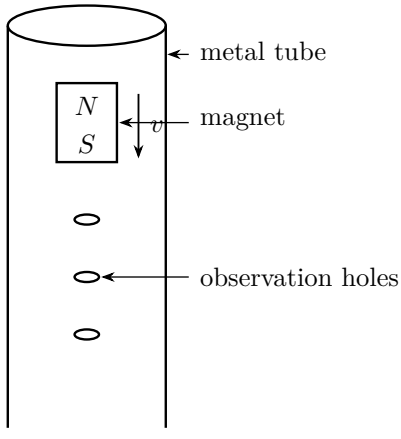


Fig. 3.49

Problem 3.50. An inductive electrical appliance consumes a power $P = 100\,\mathrm{W}$ from a supply of voltage $U = 220\,\mathrm{V}$ and frequency $f = 50\,\mathrm{Hz}$, its power factor being $\cos\varphi = 0.6$. What capacitance must be connected in parallel with it in order to raise the power factor to 1?

Problem 3.51. The edges of a regular tetrahedron form an RLC circuit, as shown. Two opposite edges are resistors R, two other opposite edges are capacitors C, and the remaining two opposite edges are inductors L. An alternating voltage of amplitude V_0 is applied across the two ends of one resistor. If the frequency of the alternating voltage is $\omega = 1/\sqrt{LC}$ and the resistance is $R = \sqrt{L/C}$, find the total current flowing through the circuit.

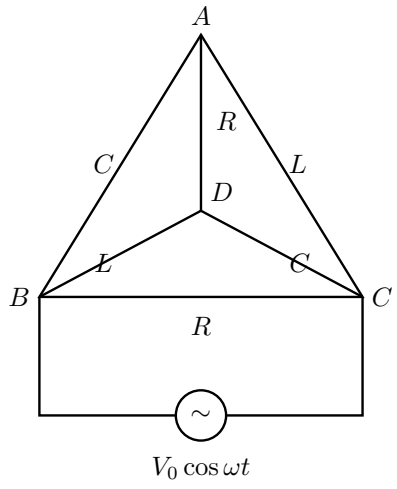


Fig. 3.51

Problem 3.52. A semi-infinite network is made of capacitors C and inductors L, as shown. The network starts at the terminals A and B on the left and extends infinitely to the right.

(1) Find the complex impedance between the terminals A and B.

(2) An alternating voltage $v_0 \cos \omega t$ is applied between A and B, causing a current to flow in. Calculate the average power input to the network.

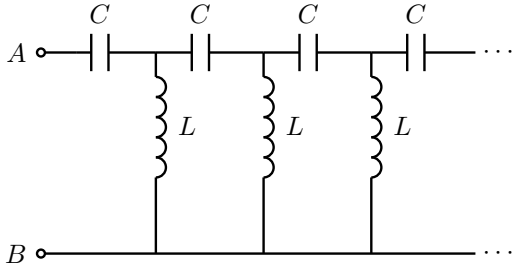


Fig. 3.52

Problem 3.53. A parallel-plate capacitor is formed of two circular metal plates of radius a separated by a distance d. The capacitor is supplied by long resistance-free wires along the axis of the two plates, and the voltage between the plates varies with time as

$$V(t) = V_0 \sin \omega t.$$

Assume that

$$d \ll a \ll c/\omega,$$

so that edge effects can be neglected and retardation effects can also be neglected (that is, the quasi-steady condition is satisfied).

(1) Find the electric field and the magnetic field inside the capacitor as functions of time.

(2) Find the electromagnetic field energy flux density and the energy flux entering the capacitor through its edge.

(3) Is this energy flux equal to the time rate of increase of the electromagnetic field energy inside the capacitor? Explain.

Problem 3.54. A circuit is formed of two resistors (R_1 and R_2), a capacitor C, and an alternating voltage source V, connected as shown.

(1) When $V(t) = V_0 \cos \omega t$, find the amplitude and the phase of the voltage drop $V_1(t)$ across R_1.

(2) When $V(t)$ is a pulse of very large amplitude U_0 and very short duration Δt, that is,

$$V(t) = \begin{cases} U_0, & 0 < t < \Delta t, \\ 0, & t < 0, \quad t > \Delta t, \end{cases}$$

and $U_0 \Delta t = A$ is a constant, find how the voltage drop across R_1 varies with time.

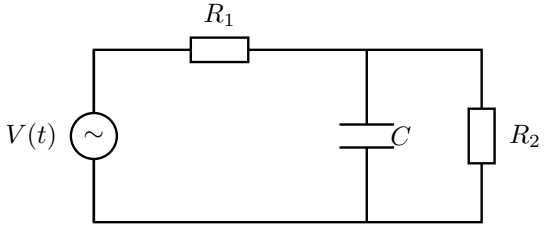


Fig. 3.54

Problem 3.55. As shown in the circuit, capacitor C_1 initially carries a charge Q_0, and capacitor C_2 is initially uncharged. When the switch K is closed, C_1 discharges through the inductor L and the resistor R. The resistance R is adjusted to a value R_0 such that the discharge process is as fast as possible without oscillation.

(1) Find R_0.

(2) In such a discharge process, after what time from the closing of K is the current through the resistor R_0 a maximum? Find this maximum current I_m.

Integral formula:

$$\int x\mathrm{e}^x \, \mathrm{d}x = x\mathrm{e}^x - \mathrm{e}^x.$$

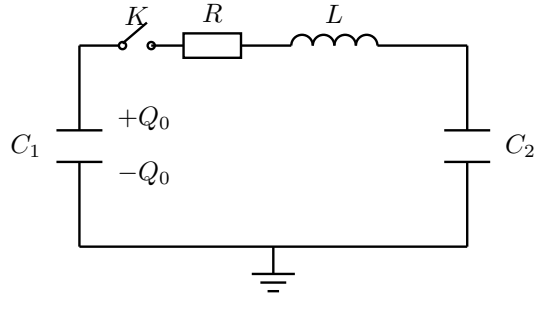


Fig. 3.55

Problem 3.56.

(1) Figure 1 shows an LC filter circuit. An alternating signal of frequency ω is applied to the input. The ratio of its output voltage U_{out} to the input voltage U_{in} is called the propagation factor and is denoted by α:

$$\alpha = U_{\text{out}}/U_{\text{in}}.$$

1.4 Optics

Problem 4.01. Light is incident on the principal section of a prism of apex angle A at an angle of incidence i, and emerges after two refractions, as shown in the figure. The refractive index of the prism material is n.

(1) Find the angle of deviation δ between the emergent ray and the incident ray.

(2) Under what condition is the angle of deviation δ a minimum? Find this minimum angle of deviation δ_m.

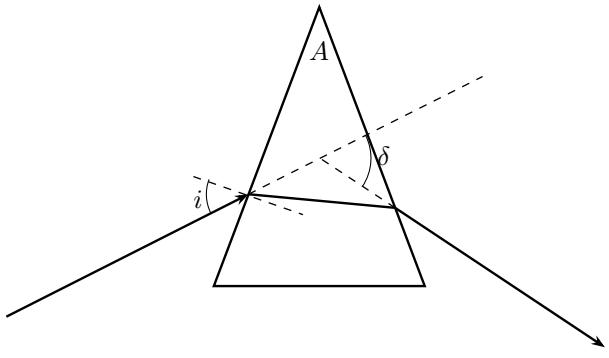


Fig. 4.01

Problem 4.02.

(1) An Amici prism is made by cementing together two right-angle prisms and one isosceles prism, as shown in the figure. The refractive indices of the two prism materials are n_1 and n_2 respectively. By choosing the apex angle a of the isosceles prism appropriately, a ray incident parallel to the base in the principal section can emerge without any deviation, thus forming a direct-vision prism. Find a.

(2) Suppose the right-angle prisms and the isosceles prism are made of crown glass and flint glass respectively. Calculate the value of a for the F line ($\lambda = 4861$Å), and also calculate the angle of dispersion between the C line ($\lambda = 6563$Å) and the G line ($\lambda = 4341$Å). The refractive indices of crown glass and flint glass for the D ($\lambda = 5893$Å), F, C and G lines are shown in the table below.

Material	n_D	$n_F - n_C$	$n_F - n_D$	$n_G - n_F$
Crown glass	1.5100	0.00805	0.00565	0.00451
Flint glass	1.7550	0.02743	0.01975	0.01730

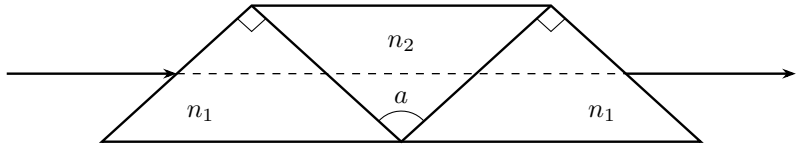


Fig. 4.02

Problem 4.03. A surface of revolution separating two homogeneous media of refractive indices n and n' has the following property: light emitted from a fixed point P on the axis of revolution, after refraction at the surface, converges without aberration to another point P' on the same axis. This surface is called a Cartesian oval. Taking a plane through the axis to cut this surface, find the equation of the curve of intersection. Investigate under what conditions this curve becomes a conic.

Problem 4.04. According to Descartes' theory, the rainbow is formed by the refraction and reflection of sunlight in water drops. The angular radius of the rainbow can be determined from the requirement that the light emerging after one internal reflection and refraction has maximum intensity. Treating the water drop as a sphere of refractive index n, find the angular radius α of the rainbow. Given that the refractive indices of water for red and violet light are $n_{\text{red}} = 1.329$ and $n_{\text{violet}} = 1.343$ respectively, find the values of the angular radii α_{red} and α_{violet} of the rainbow for red and violet light.

Problem 4.05. A person stands on a long straight runway. Owing to the temperature difference, the refractive index of the air above the runway varies linearly with height, namely $n = n_0(1+\alpha y)$, where $|\alpha| = 1.5 \times 10^{-6}\,\text{m}^{-1}$. As a result, he cannot see clearly the runway at a distance d away ($y = 0$). If his eyes are at a height of 1.7 m above the ground, find d. Also, how does the temperature vary with height?

Problem 4.06. Astronomical refraction is the change in the azimuth α of a star (the angle between the direction of the star and the normal to the ground) caused by refraction in the atmosphere. Let the apparent azimuth of the star be α_0 and the refractive index at the ground be n_0.

(1) Neglecting the curvature of the Earth's surface, prove that the astronomical refraction is independent of the law according to which the refractive index of the air varies with height and depends only on the values of n_0 and α_0. Find an approximate expression for the astronomical refraction $\alpha - \alpha_0$ (this formula is valid for $\alpha < 75°$).

(2) If the curvature of the Earth's surface is taken into account, the astronomical refraction does depend on the law of variation of the refractive index of the air with height. Suppose the air is a uniform transparent layer of refractive index n and height h; find an expression for the astronomical refraction $\alpha - \alpha_0$.

(3) Let $n = 1.00028$, $h = 4.0\,\text{km}$, the Earth's radius $R = 6370\,\text{km}$ and $\alpha_0 = 90°$; find the value of the astronomical refraction $\alpha - \alpha_0$.

Problem 4.07. A plano-convex lens of diameter $2a$ and radius of curvature R of its convex surface is placed horizontally and illuminated by a vertically upward laser beam, as shown in the figure. What should be the power P and the intensity I of the laser so that the lens can hover in the air without falling? The refractive index of the lens material is n and its density is ρ. Assume that the laser beam fills the lens and that all rays are paraxial.

Let $n = 1.50$, $R = 5\,\text{mm}$, $a = 1\,\text{mm}$, $\rho = 2.5\,\text{g} \cdot \text{cm}^{-3}$ and $g = 9.8\,\text{m} \cdot \text{s}^{-2}$; find the numerical values of P and I.

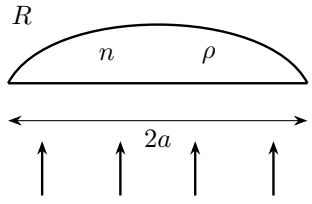


Fig. 4.07

Problem 4.08. A thin biconvex lens of refractive index n and radii of curvature R_1 and R_2 is placed between media of refractive indices n_1 and n_2, as shown in the figure. Find the focal length of the lens and the object-image formula.

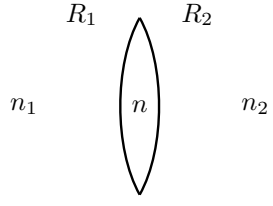


Fig. 4.08

Problem 4.09. Lenses L_1 and L_2 of focal lengths f and $-f$ respectively are placed coaxially, separated by a distance f. An opaque screen P with a small hole of diameter d is placed exactly midway between them, the centre of the hole lying on the axis. A screen Q is placed at a distance f to the right of L_2. In order that the diameter of the light spot formed on Q by a point source S on the principal axis to the left of L_1 be not greater than $0.4d$, find the range of positions of S (expressed in terms of the distance u from L_1).

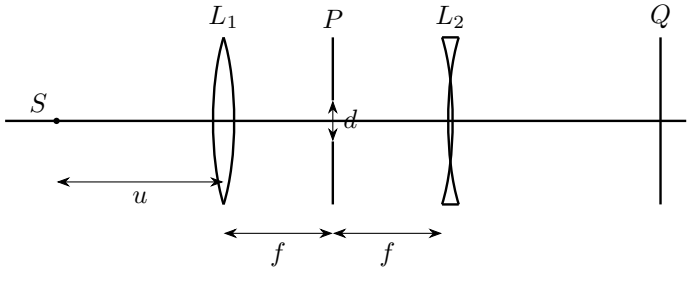


Fig. 4.09

Problem 4.10. An object O is placed in front of the focal point of a convex lens L and is imaged by the lens together with a concave mirror M placed behind the lens, as shown in the figure.

(1) Prove that, with the positions of the object and the lens fixed, there exist two positions of the concave mirror for which the magnification of the final image of the object is $+1$.

(2) Let $u = 10\,\text{cm}$, the radius of the concave mirror $R = 7\,\text{cm}$, and the distance between the two positions of the concave mirror for which the magnification is $+1$ be $l = 2\,\text{cm}$; find the focal length f of the lens.

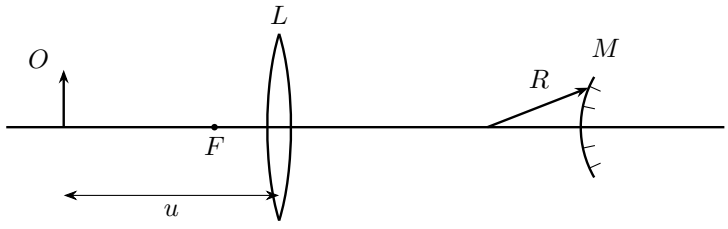


Fig. 4.10

Problem 4.11.

(1) A transparent sphere of radius R has refractive index n. P is a point on the principal axis, located at a distance R/n to the left of the centre C of the sphere, as shown in the figure. Prove that any ray emitted from P to the right (not necessarily a paraxial ray), after refraction at the spherical surface, converges to a single point. Find the position of this point Q. P and Q are called aplanatic points.

(2) The concept of aplanatic points is often used in microscope objectives in order to increase the aperture angle of the light entering the microscope. Suppose that the objective of a microscope is a hemisphere of refractive index n_1 and radius R_1, whose plane base and the object are both immersed in oil of refractive index equal to that of the objective material, and that the object is located at an aplanatic point.

(i) Design the radii R_2 and R_3 of the two spherical surfaces of the second lens of the objective and its configuration, so that its object and image are also aplanatic points. It is known that the refractive index of this lens material is n_2, the distance between the front spherical surface of this lens and the rear spherical surface of the first lens is d_1, and the central thickness of the lens is d_2.

(ii) Estimate the aperture angle β of the light emerging from the second lens.

(iii) Find the magnification k after imaging through the two lenses.

(iv) Let $n_1 = 1.5$, $n_2 = 1.6$, $R_1 = 3\,\text{mm}$, $d_1 = 2.0\,\text{mm}$ and $d_2 = 1.5\,\text{mm}$; find the numerical values of R_2, R_3, β and k.

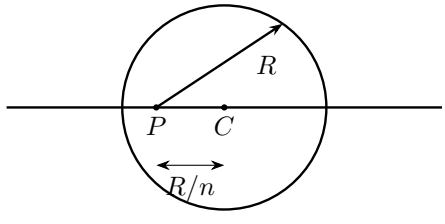


Fig. 4.11

Problem 4.12.

(1) A Fresnel biprism has refractive index n and refracting edge angle α ($\alpha \ll 1$). A monochromatic slit S (wavelength λ) lying on the axis of symmetry is at a distance a from the prism, and the observation screen is at a distance b from the prism. Find an expression for the distance x from the centre of the interference pattern to the m-th bright fringe.

(2) If the slit source S is displaced slightly upward by a distance h in the direction perpendicular to the principal axis, in which direction will the zero-order fringe move? What is the distance y by which it moves?

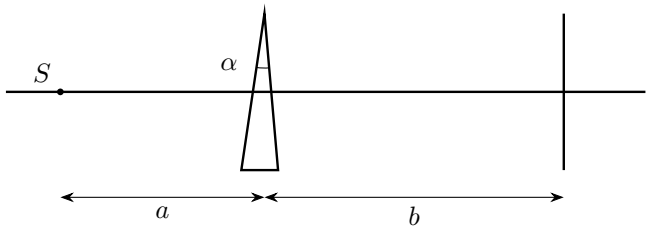


Fig. 4.12

Problem 4.13. The figure shows a schematic of an arrangement for observing interference fringes with a Billet split lens. A lens of focal length f is cut into two equal halves L_1 and L_2, which are displaced relative to each other along the principal axis by a distance f. A monochromatic point source S of wavelength λ is placed on the optical axis to the left of L_1, at a distance f from L_1. A screen is placed at a distance $3f$ to the right of L_2 to observe the interference fringes.

(1) Describe the shape of the fringes. Is the centre of the pattern dark or bright?

(2) Calculate the distance r of the m-th bright fringe from the axis.

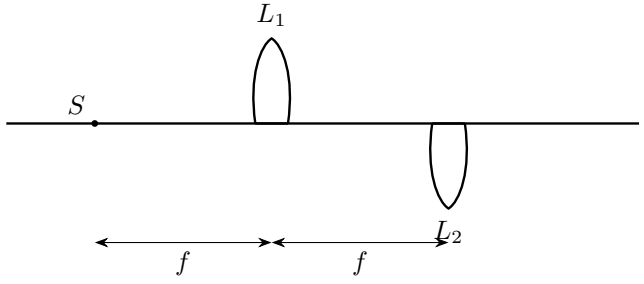


Fig. 4.13

Problem 4.14.

(1) A plane electromagnetic wave is incident normally on the interface between two media labelled 1 and 2. The amplitude reflection and transmission coefficients for a wave incident from medium 1 into medium 2 are r and t, and those for a wave incident from medium 2 into medium 1 are r' and t'. Using the superposition principle and time-reversal invariance, derive the Stokes relations

$$r^2 + tt' = 1, \qquad r = -r'.$$

(2) A plane electromagnetic wave is incident from vacuum normally on a thin film of thickness d covering a semi-infinite dielectric. Suppose the relative permeability of the two media is $\mu_r = 1$, and the refractive indices of the film and of the substrate are n_1 and n_2. Express, in terms of n_1, n_2 and the vacuum wavelength λ, the expression for the wave reflected back into vacuum. Under what condition is the reflected wave zero?

Problem 4.15. A monochromatic slit source S of wavelength λ is placed in the focal plane of lens L_1. The lens splits the light into two parallel beams, which after passing through two narrow tubes filled with liquid are focused by lens L_2 onto the focal plane S', where interference fringes are observed, as shown in the figure. The liquid in both tubes has refractive index n; the liquid in the upper tube moves with velocity u, while the liquid in the lower tube is at rest. Both tubes have length L. Find the minimum value of u for which destructive interference occurs at S'.

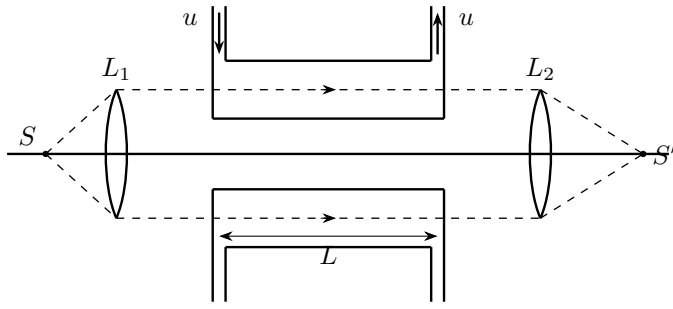


Fig. 4.15

Problem 4.16. Consider a reflection grating whose grooves are equally spaced with spacing d, but whose reflectivity alternates: $(1+\alpha)$, $(1-\alpha)$, $(1+\alpha)$, $(1-\alpha)$, and so on, where α is a constant much smaller than 1. Discuss the angular distribution of the diffracted intensity. Under what circumstances is the effect of α most significant?

Problem 4.17. A sailor comes out of a tavern drunk and staggers along many steps, each step having the same length L, but the direction of each step being random. After walking N steps over a fairly long time, on average how far is he from the tavern?

Problem 4.18. The aperture of a lens is square, with side D, and a monochromatic point source is placed on the principal axis of the lens. On the plane passing through its geometrical image point and perpendicular to the principal axis, find the intensity distribution near the geometrical image point produced by diffraction at the edge of the aperture.

Problem 4.19. A beam of monochromatic partially elliptically polarised light propagates along the z direction and passes through a linear polariser. When the transmission axis of the polariser is along the x direction, the transmitted intensity is maximum, with value $1.5I_0$. When the transmission axis is along the y direction, the transmitted intensity is minimum, with value I_0.

(1) When the transmission axis makes an angle θ with the x axis, what is the transmitted intensity? Does your answer depend on the unpolarised part of the beam?

(2) Let the original beam first pass through a quarter-wave plate and then through a linear polariser, the axis of the quarter-wave plate being along the x axis or the y axis. It is now found that when the transmission axis of the polariser makes an angle of 30° with the x axis the transmitted intensity is maximum. Find this maximum intensity. Also find the percentage of the unpolarised component in the incident light.

Problem 4.20. In order to observe monochromatic light from the sun, the French astronomer B. Lyot invented a birefringent filter consisting of a set of birefringent crystals (C). Starting from the first crystal (thickness d), the thickness of each subsequent crystal is twice that of the preceding one. Polarising films (P) are placed between the crystals and at both ends (see Fig. 1). The optic axes of all the crystals are parallel to each other and perpendicular to the direction of propagation of the light. The transmission axes of the polarising films are also all parallel, but make an angle of 45° with the direction of the optic axes (see Fig. 2). Only certain wavelength bands can pass through the filter. For a filter consisting of s crystals, calculate the transmission coefficient as a function of the wavelength λ, and find the bandwidth $\Delta\lambda$ of the light that can pass through the filter.

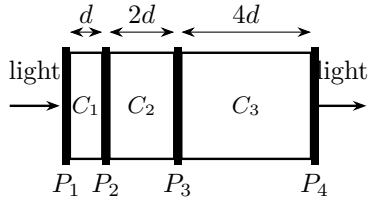


Fig. 4.20(a)

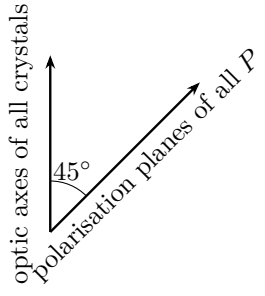


Fig. 4.20(b)

1.5 Atomic Physics

Problem 5.01. A clean metal surface is irradiated with ultraviolet light of wavelength $\lambda = 200\,\text{nm}$, and the stopping voltage of the photoelectrons is measured to be 2.6 V. What is the work function of this metal? If ultraviolet light of wavelength $\lambda = 300\,\text{nm}$ is used instead, what is the stopping voltage? What happens if visible light is used? ($hc = 1240\,\text{eV}\cdot\text{nm}$)

Problem 5.02. In order to decide whether a particle of very great penetrating power is a γ photon, scientists carried out the following work: (a) they measured the attenuation of this "photon" in lead and estimated its energy to be about 15 MeV; (b) they bombarded paraffin (CH_4) with a beam of these particles, producing a large number of protons, the maximum recoil energy of the protons being 5.7 MeV. From these two experimental results, decide by calculation whether or not this particle is a γ photon. (Electron rest energy $m_e c^2 = 0.511\,\text{MeV}$, $hc = 12\,400\,\text{eV}\cdot\text{Å}$)

Problem 5.03. (1) Find the Compton shift $\Delta\lambda$ in the following two cases, the photon being incident on a free electron initially at rest: (a) if after scattering the electron moves along the direction of incidence of the photon; (b) if the photon still moves along its original direction of incidence.

(2) A photon collides with a high-energy relativistic electron; the photon is scattered through an angle of 60°, while the electron comes to rest. (a) Find the wavelength difference $\Delta\lambda = \lambda - \lambda'$ between the incident photon (wavelength λ) and the outgoing photon (wavelength λ'). (b) If the energy of the incident photon is equal to the rest energy of the electron, find the energy of the scattered photon and the kinetic energy of the electron before the collision.

Problem 5.04. SLAC (Stanford Linear Accelerator Center) once used the following scheme to produce high-energy monochromatic photons: a low-energy photon beam from a laser (in the visible region) collides head-on with a 20 GeV electron beam.

(1) Let θ be the angle between the scattered photon beam and the incident electron beam. Find the energy of the scattered photon as a function of θ.

(2) Give an approximate value for the maximum energy of the scattered photon (in GeV).

Problem 5.05. The track of a 1 MeV electron in a cloud chamber is a string of small water droplets, the linear size of a droplet being about 10^{-5} m. Will the track of the electron differ appreciably from a straight line? Why? (Electron rest energy $m_e c^2 = 0.511\,\text{MeV}$, $hc = 12\,400\,\text{eV}\cdot\text{Å}$)

Problem 5.06. An ion has nuclear charge number z and only one electron of rest mass m outside the nucleus. Using the uncertainty relation $\Delta x\,\Delta p \gtrsim \hbar$, estimate the ground-state energy of this hydrogen-like ion in the following two cases (1) and (2).

(1) In the non-relativistic case, when the electron is at a distance r from the nucleus and has momentum p, the energy of the ion is

$$E = \frac{p^2}{2m} - \frac{ze^2}{4\pi\varepsilon_0 r}.$$

(2) In the relativistic case, the energy of the ion is

$$E = \sqrt{p^2c^2 + m^2c^4} - \frac{ze^2}{4\pi\varepsilon_0 r}.$$

(3) What is the connection between the results of (1) and (2)?

Problem 5.07. Calculate:

(1) the ratio of the circumference of the n-th orbit of the hydrogen atom to the de Broglie wavelength of the electron moving in that orbit;

(2) the de Broglie wavelength of the electron moving in the $n = 1$ orbit of the hydrogen atom.

Problem 5.08. Because the hydrogen gas contained helium gas, the spectral lines of the helium ion (He^+) were mixed into the photographed spectrum of atomic hydrogen. Show how, without measuring the wavelengths of the spectral lines, one can decide directly from the photograph which lines belong to the hydrogen spectrum and which to the He^+ spectrum, and briefly explain the basis of your judgement.

Problem 5.09. The μ^- meson carries a charge $-e$ and has mass $m_\mu = 106\,\text{MeV}/c^2$; the π^+ meson carries a charge $+e$ and has mass $m_\pi = 140\,\text{MeV}/c^2$. If in the "exotic atom" formed by these two mesons both particles may be regarded as point particles, find:

(1) the lowest energy of the atom;

(2) the distance between the two mesons in the ground state;

(3) the speed of the μ^- meson in the ground state, neglecting the overall motion of the atom.

Problem 5.10. A μ^- meson is captured by a lead nucleus ($^{208}_{82}$Pb, $z = 82$), forming a lead muonic atom. Apart from its different mass, the μ^- meson is the same thing as an electron; the mass of the μ^- meson is known to be 207 times the electron mass. Assume that the nuclear charge is uniformly distributed within a sphere of radius $R = 7.7\,\text{fm}$ and that the μ^- meson can move inside this sphere. Using Bohr's theory, find:

(1) the radii of the first two Bohr orbits of the μ^- meson;

(2) the energies of the first two levels of the lead muonic atom;

(3) the wavelength of the first line of the Lyman series of the lead muonic atom;

(4) how the result of part (3) changes if the lead nucleus is treated as a point charge.

Problem 5.11. The sodium atom ($z = 11$) has 11 electrons outside the nucleus, of which 10 are rather tightly bound to the nucleus and form the atomic core, while the outermost valence electron is rather loosely bound to the core. In the spectrum emitted when the valence electron of the sodium atom makes a transition, a spectral series similar to the Lyman series of hydrogen is described by the formula

$$\frac{1}{\lambda} = 0.378R - \frac{R}{m^2},$$

where R is the Rydberg constant and m is an integer.

(1) The formula holds only within a certain range; explain, in terms of atomic structure, the conditions under which the formula holds.

(2) Use this formula to calculate the ionisation energy of sodium (that is, the energy required to ionise the valence electron).

Problem 5.12. Assume that Bohr's theory of the hydrogen atom applies to each of the following parts, and neglect the motion of the helium nucleus.

(1) Find the ground-state energy of the helium ion (He^+) and the radius of the first Bohr orbit of the electron.

(2) The mass of the μ^- meson is about 200 times the electron mass; apart from the mass, it is identical with the electron in every respect. Suppose a helium nucleus captures a μ^- meson and forms a hydrogen-like ion. Find the ground-state energy of this system and the radius of the first Bohr orbit of the μ^- meson.

(3) Suppose the atom formed when one electron of the helium atom is replaced by a μ^- meson is in its ground state. (a) Estimate the radius of the first Bohr orbit of the electron; (b) estimate the energy of this system; (c) to which chemical element is the chemical behaviour of this atom most similar, and why?

(Numerical results in this problem need only two significant figures; energies are to be given in eV and radii in nm.)

Problem 5.13. Hydrogen atoms are bombarded with particles and thereby excited. In each of the following cases, what is the minimum kinetic energy that the particle must have in order that the hydrogen atom emit the H_α line?

(1) An electron collides with a hydrogen atom at room temperature.

(2) A proton bombards a ground-state hydrogen atom initially at rest.

(3) A proton and a ground-state hydrogen atom approach each other with equal speeds and collide head-on.

Problem 5.14. The helium atom is the simplest many-electron atom.

(1) If there were no interaction between the electrons, what would be the ground-state energy of the helium atom?

(2) Using a rough model, estimate the ground-state energy of the helium atom when the Coulomb interaction between the electrons is taken into account.

(3) If one electron of the helium atom is excited to the $4f$ state while the other electron remains in the $1s$ state, what is the energy of the helium atom then?

Problem 5.15. The natural radioactive uranium series starts from $^{238}_{92}\mathrm{U}$, the uranium isotope with the longest half-life, and after a series of α and β decays finally reaches the stable nucleus $^{206}_{82}\mathrm{Pb}$. The relative ratio of the numbers of $^{238}_{92}\mathrm{U}$ and $^{206}_{82}\mathrm{Pb}$ nuclei measured at present in uranium ore is

$$\frac{N(^{238}_{92}\mathrm{U})}{N(^{206}_{82}\mathrm{Pb})} = 2.785.$$

Estimate the age of the Earth. (Given: the half-life of $^{238}_{92}\mathrm{U}$ is $T_{1/2} = 4.468 \times 10^9$ years, and among its daughter nuclei $^{234}_{92}\mathrm{U}$ has the longest half-life, $T_{1/2} = 2.45 \times 10^5$ years.)

Problem 5.16. It is known that the concentration ratio $^{14}\mathrm{C}/^{12}\mathrm{C}$ in the atmosphere is $\eta = 1.2 \times 10^{-12}$, and that the half-life of $^{14}\mathrm{C}$ is $T_{1/2} = 5730$ years. Since a living organism must necessarily exchange carbon with the atmosphere through respiration or photosynthesis, the content ratio of $^{14}\mathrm{C}$ to $^{12}\mathrm{C}$ in a living body is the same as in the atmosphere.

(1) Calculate the specific activity of the carbon in the atmosphere, that is, the number of disintegrations per minute of one gram of carbon.

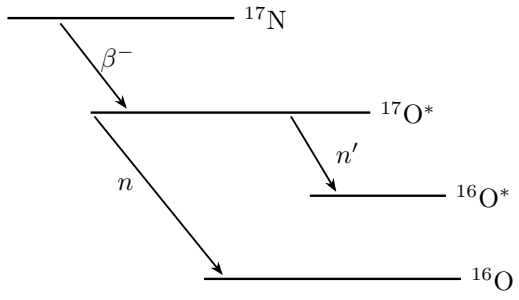
(2) The $^{14}\mathrm{C}$ content of an ancient biological sample is 1/1.831 of that of a living biological sample. Estimate the age of this ancient sample.

(3) The relative statistical error of a radioactive count N is defined as $\delta = 1/\sqrt{N}$. In order that the statistical error of the measurement be $\delta = 1\%$, how many grams of the above ancient sample are required so that the count over one hour satisfies this requirement on δ? Also estimate the precision, plus or minus how many years, of the age determined at this δ. (Assume that the efficiency of the detector is 100%.)

Problem 5.17. (1) Write down the expression for β^- decay and the formula for its decay energy (the decay-energy formula is required to be expressed in terms of atomic masses).

(2) As the figure shows, $^{17}\mathrm{N}$ decays by β^- emission, with a maximum electron kinetic energy of 3.72 MeV, to an excited state of $^{17}\mathrm{O}$ ($^{17}\mathrm{O}^*$), and from this excited state it decays further by emitting a neutron to the ground state or to an excited state of $^{16}\mathrm{O}$. It is known that $^{16}\mathrm{O}$ has several excited states ($^{16}\mathrm{O}^*$) at 3.05, 6.13, 6.90 MeV, etc. (the ground-state energy of $^{16}\mathrm{O}$ is taken as 0). State how many possible values the kinetic energy of the emitted neutron can take, and find the corresponding values. The recoil of the $^{16}\mathrm{O}$ nucleus when the neutron is emitted must be taken into account in the calculation.

(Known: atomic masses $M(^{17}\mathrm{N}) = 17.008\,449\,\mathrm{u}$, $M(^{16}\mathrm{O}) = 15.994\,915\,\mathrm{u}$, neutron mass $m_n = 1.008\,665\,\mathrm{u}$, $1\,\mathrm{u}\,c^2 = 931.5\,\mathrm{MeV}$, where u is the atomic mass unit.)



In the figure ^{16}O* denotes an excited state; in reality it is not a single level. n' denotes the neutron emitted in the decay to this excited state, and its kinetic energy differs from that of the neutron n emitted in the decay to the ground state.

Fig. 5.17

Problem 5.18.

(1) Write down the semi-empirical formula for the binding energy of a nucleus in the liquid-drop model, and state the physical meaning of each term and which terms come from quantum effects.

(2) The atomic masses of a pair of mirror nuclei (that is, two nuclei with the same mass number A but with the proton and neutron numbers interchanged) — boron and carbon — are known to be $M(^{11}\mathrm{B}) = 11.009\,31\,\mathrm{u}$ and $M(^{11}\mathrm{C}) = 11.011\,43\,\mathrm{u}$. Use the semi-empirical binding-energy formula to find the value of the constant r_0 in the corresponding nuclear-radius formula $R = r_0 A^{1/3}$.

(Known: hydrogen atom mass $M(^1\mathrm{H}) = 1.007\,825\,\mathrm{u}$, neutron mass $m_n = 1.008\,665\,\mathrm{u}$, combination constant $e^2/4\pi\varepsilon_0 = 1.44\,\mathrm{MeV}\cdot\mathrm{fm}$, $1\,\mathrm{u}\,c^2 = 931.5\,\mathrm{MeV}$.)

Problem 5.19. In the Rutherford elastic-scattering experiment, the scattering of the incident particle and the recoil of the target nucleus occur together (see the figure). In the figure, θ_L and φ_L denote respectively the scattering angle and the recoil angle of the target nucleus in the laboratory system, and θ_C and φ_C denote the corresponding angles in the centre-of-mass system. Assume that the incident particle is an alpha particle ($^4_2\mathrm{He}$) of kinetic energy 2 MeV bombarding a hydrogen nucleus ($^1_1\mathrm{H}$) assumed to be at rest in the target, and carry out the following calculations:

(1) What is the kinetic energy of the recoiling hydrogen nucleus in the direction $\varphi_L = 30^\circ$ in the laboratory system?

(2) What is the Rutherford differential scattering cross-section of the recoiling hydrogen nucleus at $\varphi_L = 30^\circ$ in the laboratory system?

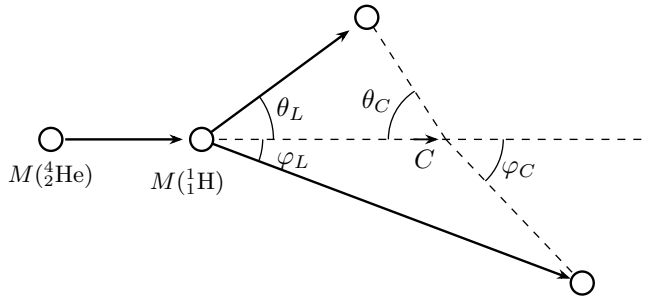
Given: in the centre-of-mass system, the Rutherford differential scattering cross-section for the outgoing particle is

$$\sigma_C(\theta_C) = \left(\frac{a}{4}\right)^2 \frac{1}{\sin^4(\theta_C/2)},$$

where the Coulomb scattering factor is

$$a = \frac{z_1 z_2 e^2}{4\pi\varepsilon_0 E_C},$$

z_1 and z_2 being respectively the charge numbers of the incident particle and of the target nucleus, and E_C the centre-of-mass energy of the incident particle. The combination constant is $e^2/(4\pi\varepsilon_0) = 1.44\,\mathrm{eV}\cdot\mathrm{nm}$.



Elastic scattering of an alpha particle ($^4_2$He) by a hydrogen nucleus; C is the centre of mass of the two nuclei.

Fig. 5.19

Chapter 2

Hints

2.1 Mechanics

Hint 1.01. First find the initial speed v_0 required for the ball to reach the top of the wall as a function of the angle of projection θ; then find the minimum of v_0.

Hint 1.02. Treat the motion of the ball above the tops of the walls as an independent projectile motion; then relate the motion above the wall tops to the motion below them.

Hint 1.03. Discuss the condition for the trajectory of the shell to be tangent to the shelter.

Hint 1.04. Discuss separately the two cases in which the farthest-flying fragment after the lamp bursts does, and does not, strike the ceiling.

Hint 1.05. For the motion of the bullet, set up a three-dimensional coordinate system with the point of firing as origin; to find the locus of the aiming points, set up another two-dimensional coordinate system on the wall.

Hint 1.06. Note the velocity of the shell relative to the bomb, and discuss the relation between the minimum relative speed required for the shell to hit the bomb in the air and the speed with which the bomb is released.

Hint 1.07. The acceleration of P relative to the paper is equal to its acceleration relative to the straight line.

Hint 1.08. The angular speed of the rod relative to the ground is equal to its angular speed relative to A (that is, relative to the frame moving together with A).

Hint 1.09. The accelerations of hinges B_3 and A_1 are equal in magnitude, and the angular displacements of the rods are also equal.

Hint 1.10. First find the relative velocity of the centre of disc B in the rotating frame of disc A; then use the relative-motion formula to find its absolute velocity.

Hint 1.11. In the rotating frame of the large steel ring, find the relative velocities and relative accelerations of points A and B on the small ring; the absolute accelerations of these two points then follow from the relative-motion formula.

Hint 1.12. Correctly finding the velocity and acceleration of the axis of the small cylinder relative to the ground, relative to cylinder A and relative to cylinder B is the key to this problem.

Hint 1.13. Draw the vector diagrams of the velocities and of the relative velocities of A and B,

and the diagram of the positions of A, B and the centre O; then solve by geometry.

Hint 1.14. In the frame rotating with the rod, analyse whether the small ring has a position at which the resultant force on it vanishes; if it has, the ring need only reach this position in order to reach end B of the rod.

Hint 1.15. The radius of curvature of the parabolic bent rod at a point can be found from the normal equation of a projectile motion whose trajectory has the same shape. Moreover, in the frame rotating with the rod, besides gravity, the normal reaction and the centrifugal force of inertia, the ring is also acted on by the Coriolis force because it is moving.

Hint 1.16. Analyse the motion, relative to the points of contact with the large disc, of two small segments on any diameter of the small disc that are symmetric about its centre; note that the friction forces on these two segments are directed opposite to this relative motion. Discuss the effect of the couple of these friction forces on the rotation of the small disc, and the angular speed of the small disc for which this frictional torque vanishes.

Hint 1.17. The three-ball system has two possible equilibrium configurations: the two lower balls separated, and the two lower balls close together.

Hint 1.18. When the large ring hangs on the nail in any position, the angle between the force exerted on it by the nail and the normal at that point is less than the angle of friction.

Hint 1.19. When α is the greatest value for which the system can be in equilibrium, there are two possible cases: in the first, ball 1 has already reached the critical state between sliding and not sliding relative to the inclined plane, while ball 2 has not yet reached this state; in the second, ball 2 has already reached this critical state, while ball 1 has not. Discuss the conditions for these two cases and the corresponding conclusions.

Hint 1.20. From the equilibrium conditions find the forces at the hinges at the ends of each rod, and hence the tension in the string; alternatively, use the principle of virtual work.

Hint 1.21. When the sphere is in the critical state between equilibrium and non-equilibrium, discuss which of the two contact points, that of the sphere with the board and that of the sphere with the inclined plane, has reached the critical state of relative sliding. Find the conditions for the two possible cases separately, and the corresponding value of F follows.

Hint 1.22. After the small ball is fixed at the midpoint of rod OA, the symmetry of the system is destroyed. For the system to remain at rest, the minimum coefficient of friction sought must ensure not only that rod OA does not slide relative to the ground, but also that the other two rods do not slide relative to the ground.

Hint 1.23. The direction of the normal reaction N of the ring on the rod is perpendicular both to the rod and to the ring; the direction of the friction force f acting on the rod is perpendicular both to the rod and to the direction of the normal reaction N acting on it.

Hint 1.24. Note the continuity equation for the water, that is, the mass of water issuing from the fountain nozzle per second equals the mass of water rushing up to the bottom of the bucket.

Hint 1.25. By integrating twice the equation obtained from the differential form of the momentum theorem, find how the radius r of the raindrop varies with time t; then find the velocity and the acceleration. The relations of r and t can also be found directly by solving the differential

equation.

Hint 1.26. By combining the equation obtained from the differential form of the momentum theorem into an exact differential and then integrating, find the velocity.

Hint 1.27. After the ball leaves the table, change to the frame that moves upward with speed v together with the upper end of the spring, and discuss the motion of the ball in this frame.

Hint 1.28. Note the motion of the centre of mass of the system; the force exerted by the light rod on ball B can be found by analysing the motion of ball A in the frame of ball B.

Hint 1.29. The speed of the ball is greatest when the resultant force along the tangent to its trajectory is zero.

Hint 1.30. When the ball slides to the right along the tube and ceases to interact with the tube, the ring has its greatest speed to the left; the radius of curvature of the trajectory at this point can be found by analysing the normal motion of the ball.

Hint 1.31. In the plane polar coordinate system with a certain bearing a (or b) as pole, when the speed of body c is greatest its acceleration is zero, hence its radial acceleration is also zero; also analyse the accelerations of bodies A and B at this instant.

Hint 1.32. In the centre-of-mass frame the two blocks vibrate while connected to a fixed point (the centre of mass) by an elastic rope; since the rope can only be stretched and not compressed, once it has returned to its natural length it exerts no force on the blocks.

Hint 1.33. Although after the supernova explosion the speed of the remaining star is unchanged, the centre of mass of the system formed by the remaining star and the other, unexploded star is no longer at rest. Moreover, in the centre-of-mass frame the mechanical energy of the system must be less than or equal to zero if the binary is not to separate.

Hint 1.34. Use conservation of momentum and the work-energy theorem.

Hint 1.35. Set up a plane polar coordinate system with the orthocentre of the equilateral triangle as pole, and in it discuss the boundary of the region of physically allowed points; when discussing the γ photon, use the relativistic relation between the energy and momentum of a photon, namely $T = cP$.

Hint 1.36. The angular momentum of the ball about the vertical axis through the small hole (the symmetry axis of the conical surface) is conserved.

Hint 1.37. In the centre-of-mass frame the mechanical energy and the angular momentum of the system are both conserved; and when the spring reaches its maximum extension the ball has only a transverse velocity and no radial velocity.

Hint 1.38. The mechanical energy of a comet moving around the Sun on a parabolic orbit is zero; if the mechanical energy of the system is less than zero, the orbit is an ellipse or a circle.

Hint 1.39. The motion of the space shuttle (including the lunar module) around the Moon is similar to the motion of a planet around the Sun and can be solved by means of Kepler's laws.

Hint 1.40. The motion of the bullet can be analysed by means of a potential energy curve or an effective potential energy curve.

Hint 1.41. In the first scheme the motion of the spacecraft can be divided into two stages: the first stage is to escape the Earth's gravitational field, during which the Sun's gravitation may be regarded as unchanged; the second stage is then to escape from the solar system. In the second scheme the motion of the spacecraft can be divided into three stages: the first stage is the same as in the first scheme; the second stage is that, after escaping the Earth's gravitation, the spacecraft flies from the Earth's orbit to the orbit of Mars; the third stage is that, under the action of Mars's gravitation, it changes its direction of motion, leaves Mars's gravitational field tangentially from another point of the Martian orbit, and, with the aid of the orbital speed of Mars, continues to escape from the solar system.

Hint 1.42. After a planet receives a small radial perturbation it executes small radial oscillations; comparing the period of the small radial oscillation with the period of the original circular motion helps to describe the subsequent motion of the planet.

Hint 1.43. If the particle executes small radial oscillations, its orbit will be stable.

Hint 1.44. Whether the axis of rotation turns with constant angular velocity or is free to turn, the mechanical energy of the system in the rotating frame (including the centrifugal potential energy) is conserved; from conservation of mechanical energy the equation of oscillation of the ball is obtained.

Hint 1.45. The ball executes simple harmonic oscillations along the x direction and along the y direction with the same frequency but different amplitudes and different initial phases.

Hint 1.46. Compare which of the following two cases makes the lowest position reached by the cylinder lower: (1) before the cylinder descends to its lowest position the piston always remains inside the cylinder; (2) at some point before the cylinder descends to its lowest position the piston has already left the cylinder.

Hint 1.47. The angular momentum of the ball about the small hole O is conserved; note that the rate of change of the radial velocity of the ball ($\dot{r}$) is the acceleration of the up-and-down motion of the heavy object.

Hint 1.48. Find the tangential acceleration along the trajectory when the small ring moves along this steel wire, and compare this tangential acceleration with the acceleration of a particle executing simple harmonic motion along a straight line.

Hint 1.49. Because the wind speed has a gradient in the vertical direction perpendicular to the ground, when sound waves emitted from the ground travel between layers of different wind speeds, a refraction phenomenon similar to that occurring when light propagates in layers of different media appears, which changes their direction of propagation; after a long distance they bend back to the ground.

Hint 1.50. A wave reflected at a fixed end suffers a half-wave loss, that is, the oscillation excited by the reflected wave at the fixed end always has a phase difference of π from that excited by the incident wave at the fixed end.

Hint 1.51. In the centre-of-mass frame the system has two normal modes: mode 1, B is at rest while A and C move towards each other; mode 2, A and C move in the same direction while B moves in the opposite direction.

Hint 1.52. This system has two normal modes: mode 1 is the symmetric mode, in which the

rod remains horizontal and executes small oscillations along the vertical direction; mode 2 is the antisymmetric mode, in which the rod executes small angular oscillations about its centre of mass.

Hint 1.53. The trajectory of water ejected at any elevation angle θ is a quadratic equation in u $(= \tan\theta)$; for this equation to have a real solution its discriminant must be greater than or equal to zero, and hence the equation of the boundary curve of the region that the water jet can reach in space is obtained. The surface of revolution formed by rotating this curve about the axis of symmetry is the surface of the solid figure.

Hint 1.54. At depth h, the difference between the radial components of the gravitational force and of the centrifugal force of inertia acting on a small mass element is balanced by the force produced by the pressure difference between the two cross-sections along the radial direction. Moreover, if the centrifugal potential energy is introduced in the rotating frame, the total potential energy on the surface of the star is the same everywhere, and the value of $R_e - R_p$ is thereby easily found.

Hint 1.55. The surface of the Earth is an equipotential surface; this includes the Earth's gravitational potential, the Moon's gravitational potential, and the centrifugal potential in the centre-of-mass frame of the Moon–Earth system (the rotating frame).

Hint 1.56. From conservation of the angular momentum of the system formed by the planet and the satellite about the planet's centre, the relation between the variations of ω, Ω and r can be found; when the system reaches a steady state (a state in which the mechanical energy does not change), $\dfrac{\mathrm{d}E}{\mathrm{d}\omega} = 0$ and $\dfrac{\mathrm{d}^2E}{\mathrm{d}\omega^2} > 0$.

Hint 1.57. Bernoulli's equation gives the pressure at the points near the canopy, and the force on the canopy then follows from the pressure difference between the points inside and outside it.

Hint 1.58. At the instant of release, in the frame of the centre of mass of one rod, the rod turns about its centre of mass; relative to the ground the upper end of the rod moves vertically and its lower end moves horizontally; the various velocities then follow from the relative-motion relations.

Hint 1.59. Before the rod begins to slide relative to the corner of the table, the centre of mass of the rod moves in a circle; writing the tangential and normal equations of motion of the centre of mass gives the angle at which sliding begins.

Hint 1.60. In discussing the rotation of the cylinder, this cylinder differs from a solid cylinder of mass $m + M$, because the water inside does not take part in the rotation, so its moment of inertia is only that of the thin-walled cylinder. If the wedge-shaped block is not fixed, the force exerted by the cylinder on the block must be found by solving the motion of the cylinder in the frame of the block.

Hint 1.61. In the rotating frame the potential energy of the rod consists of the gravitational potential energy and the centrifugal potential energy; when the rod is in equilibrium the first derivative of the potential energy with respect to θ is zero, and the sign of the second derivative of the potential energy with respect to θ determines the stability of this equilibrium position.

Hint 1.62. The collision of a rotating sphere with the ground has two possible cases: in the first, the point of the sphere in contact with the ground keeps slipping during the whole collision (although the time is very short); in the second, relative slipping exists only during the first

part of the collision and has ceased before the collision ends. In these two cases the impulse of the friction force is different, which leads to different horizontal velocities of the ball after the collision.

Hint 1.63. A rapidly spinning top has a large spin angular momentum; when its axis tilts, the torque about the centre of mass produced by the friction on the side of the point of contact between the axis and the ground makes the spin axis of the top gradually approach the vertical axis.

Hint 1.64. In the frame of the spacecraft the observer sees the Earth receding with speed v, so the speed of the radio signal emitted by the spacecraft relative to the Earth is $c - v$, and the time from the emission of the signal by the spacecraft to its reception on the Earth is equal to the time from the immediate retransmission by the Earth to the reception of this signal by the spacecraft.

Hint 1.65. The speed of light in the frame at rest relative to the glass is $\frac{c}{n}$; the speed of light in the moving glass in the ground frame can be found from the relativistic velocity-addition formula.

Hint 1.66. First find the x- and y-components of the velocity of the small packet thrown out in frame S; then, using the relativistic velocity transformation formula, find the x'- and y'-components of the velocity of the small packet in frame S' of spacecraft A.

Hint 1.67. To the observer in frame S', B and B' meet earlier than A and A'; having found this time difference, one can find, from the condition that the reading of clock B' is one unit of time when it receives the signal emitted by A, the relation between the proper length l_0 and v; while to the observer in frame S, A and A' meet earlier than B and B', and from the above relation between l_0 and v the reading of clock B when it receives the signal can be found.

Hint 1.68. The relativistic velocity transformation formula gives the angle between the direction from which light arrives and the direction of his flight, as seen by the pilot, for light coming from the edge of the Galaxy and from different parts of it, so as to determine the approximate shape of the Galaxy as he sees it; moreover, from the Doppler effect of light find the critical angle for redshift and blueshift, so as to determine who won the bet.

Hint 1.69. First find the proper wavelength of the light emitted by the source.

Hint 1.70. The decay of the particle is a process in which momentum and energy are conserved.

Hint 1.71. After finding the momentum and energy of the μ meson in the centre-of-mass frame, the momentum of the μ meson in the laboratory frame can be found by means of the relativistic momentum transformation formula.

Hint 1.72. The flight of the rocket from the Earth to the Galaxy is an accelerated process, so the rocket frame S' is not an inertial frame. If at time t a coordinate system is set up in S' which moves uniformly relative to the Earth with the velocity v of the rocket at time t during an extremely short interval, then during this extremely short interval the frame S' may be regarded as inertial, so the motion of the rocket can be discussed by means of special relativity, and the whole journey of the rocket is a combination of these infinitely short processes.

2.2 Thermodynamics

Hint 2.01. Use the equation of state of an ideal gas.

Hint 2.02. Use the equipartition theorem to find the mean potential energy of the molecular vibration.

Hint 2.03. Using the theoretical results of Brownian motion and the random-walk model, find the relation between the mean square value of the distance x of a molecule from the origin and the time.

Hint 2.04. The number of collisions of molecules with unit area of the container per unit time is just the intensity of the molecular beam when the molecules leak out through the small hole; Clapeyron's equation.

Hint 2.05. For a van der Waals gas the internal energy is $U = C_V T - \dfrac{a}{V} + U_0$; the equation of an adiabatic process is $T(V-b)^{R/C_{V,m}} = \text{const.}$

Hint 2.06. The polytropic process equation of one mole of a van der Waals gas: $\left(P + \dfrac{a}{V^2}\right)(V - b)^n = \text{const.}$

Hint 2.07. Regard the atmosphere as an ideal gas and use the adiabatic process equation of an ideal gas to find the relation between the temperature T and the height z.

Hint 2.08. The equation of state and the internal energy of a photon system are respectively $P = \dfrac{1}{3}\sigma T^4$, $U = V\sigma T^4$.

Hint 2.09. Use Maxwell's distribution law to find the number of molecules colliding with the plate in time $\mathrm{d}t$; note that the numbers of molecules colliding with the two faces of the plate are different.

Hint 2.10. Omitted.

Hint 2.11. $\lim_{T\to 0} \dfrac{1}{T^2}\mathrm{e}^{-1/T} = 0$.

Hint 2.12. When the temperature range of variation is rather large, the latent heat of phase change L depends on the temperature T.

Hint 2.13. Omitted.

Hint 2.14. This is a two-dimensional random-walk problem, usually discussed in mathematics by means of the multinomial distribution formula.

Hint 2.15. In a cavity at temperature T_0, a small black ball (regarded as a black body) at temperature T transfers heat to the cavity wall at the rate $\dfrac{\mathrm{d}Q}{\mathrm{d}t} = A\sigma(T^4 - T_0^4)$.

Hint 2.16. Omitted.

Hint 2.17. Omitted.

Hint 2.18. Omitted.

Hint 2.19. When the ideal gas in the container is in equilibrium, the mean translational kinetic energy of a molecule striking the wall is $\bar{\varepsilon}_{k\mathrm{tr}} = 2kT$.

Hint 2.20. When the volume of the liquid is constant, the work done by the surroundings on the liquid in an isothermal process is equal to the increase of the free energy of the liquid: $\mathrm{d}W = \mathrm{d}F = \sigma\,\mathrm{d}A$.

Hint 2.21. At the interface where a liquid touches a solid, the contact angle made by the liquid surface is determined by the properties of the liquid and of the solid, and is independent of the position of the liquid surface.

Hint 2.22. The surface tension coefficient σ of a given liquid is a function of the temperature T only, and is independent of the area and the pressure of the liquid surface.

Hint 2.23. Omitted.

Hint 2.24. The relation between the entropy S of a system and the thermodynamic probability W^T is called the Boltzmann relation, or Boltzmann's hypothesis: $S = k \ln W^T$.

Hint 2.25. A one-dimensional chain means a chain that can be folded in one dimension.

Hint 2.26. Omitted.

Hint 2.27. Omitted.

Hint 2.28. Omitted.

2.3 Electromagnetism

Hint 3.01. (1) Consider the field of a point charge, taking a sphere centred on the point charge as a Gaussian surface. (2) Analyse the conservative nature of this electric force. (3) Consider the electric field at any point inside a uniformly charged spherical shell: take this point as vertex and construct a small cone, and examine the field produced at this point by the two surface elements of the shell cut out by the cone. (4) Take the conducting sphere as an example and use the result of (3).

Hint 3.02. (1) Use the energy method. (2) Regard the field of a quadrupole as the superposition of the fields of two oppositely directed dipoles, and consider the potential first.

Hint 3.03. (1) Relate the lines of force to the electric flux. (2) Examine the relation between the electric field (a vector) at a distant point and the two charges, noting that the distances from this point to the two charges are almost the same.

Hint 3.04. First take the approximation for a small quantity, then integrate.

Hint 3.05. (1) Omitted. (2) Divide the cube into eight equal small cubes; the centre of the original cube is then a vertex of the eight small cubes.

Hint 3.06. Use Gauss's law, taking as Gaussian surfaces concentric spheres of radii r and $r + \mathrm{d}r$ respectively.

Hint 3.07. Consider the force exerted by the positive ion layer on the negative ion layer and

prove that this force is proportional to the small displacement. Replace the mass of the negative ion layer by the reduced mass to compensate for the motion of the positive ions.

Hint 3.08. (1) Note that the results of the integration inside and outside the spherical surface have different forms. (2) Since $\delta \ll 1$, when the result of (1) is applied to the inner and outer spherical shells, take the small-quantity approximation.

Hint 3.09. Note the asymmetry of the charge on the two aluminium cylinders caused by accidental factors, and then analyse the positive feedback of this asymmetry.

Hint 3.10. To make the microphone give a voltage output, a condenser microphone must have a power supply, whereas an electret one needs none. In order that the motion of the diaphragm (which changes the capacitance) be reflected to the greatest extent in the output voltage, the charge on the diaphragm should be kept constant as far as possible.

Hint 3.11. (1) When a charge is moved, the image charge moves correspondingly, but the external force does work only on the charge being moved. (2) At this time the image charge no longer moves. (3) Consider the effects of (1) and (2) together.

Hint 3.12. Use the method of images; to find the position and magnitude of the image charge it suffices to consider special points on the axis of symmetry.

Hint 3.13. Regard these parallel conducting plates as capacitors connected in series.

Hint 3.14. (1) According to Ohm's law, write the differential equation for the relation between the radius and the time. (2) Note that external forces take part in doing work in the process.

Hint 3.15. Determine the charge on the third conducting sphere from the parallel connection of the capacitances, then determine the induced charge on the outer surface of the outer shell from electrostatic equilibrium; the third conductor is acted on by this charge.

Hint 3.16. First consider the induced charge distribution on an electrically neutral conducting sphere in a uniform external field (this can be obtained from two uniformly charged spheres of opposite sign displaced slightly in the external field, which makes the total field in the overlapping part vanish); the charge distribution on its upper half is the same as that on the conducting hemisphere in the problem, and the force on the hemisphere can then be found.

Hint 3.17. (1) At this time the charge may be regarded as uniformly distributed on the cylindrical surface, and the field in space is the field excited jointly by this charged cylindrical surface and its image. (2) Examine the equipotential surfaces of two parallel oppositely charged straight lines and prove that they are a family of non-concentric cylindrical surfaces; the problem is then solvable.

Hint 3.18. (1) Imagine the charge on the uniformly charged spherical surface to be frozen to the sphere. When the sphere is deformed along the z axis into an ellipsoid of revolution, use the scaling principle to prove that the field inside is still everywhere zero, and hence obtain the charge distribution on the conducting ellipsoid of revolution. Find the potential at the centre by integration, and the capacitance follows. Note that when $b > a$ and when $b < a$ the results of the integration have different forms. (2) Find the small-quantity approximation separately.

Hint 3.19. (1) From the field inside the column, write the differential equation satisfied by the motion of the ions. (2) The charge Q flowing to the conducting wire serves to cancel the reversed

field formed by the ion layer.

Hint 3.20. Use the energy method, and regard the system as three spherical capacitors in series (the point charges being regarded as distributed on small conducting spheres), one of which is filled with a dielectric. When the point charge is moved to infinity, the three capacitors may still be regarded as in series, but all are then vacuum capacitors.

Hint 3.21. Suppose an electric field exists between the plates; find the polarisation intensity of each dielectric sphere under the action of the external field (neglecting the influence of the other dielectric spheres), and hence the electric moment of each dielectric particle, and then the equivalent polarisation intensity and the equivalent permittivity of the dielectric made of sparse particles.

Hint 3.22. First, since polarisation charges of opposite sign gather around the point charge, the behaviour of the point charge q is equivalent to that of a point charge of magnitude q/ε_r. Then use the boundary conditions at the surface of the dielectric to find the distribution of the surface polarisation charge; the behaviour of the surface polarisation charge is equivalent to that of an "image charge".

Hint 3.23. (1) Use Gauss's law. (2) Use the method of the solution to Problem 3.21 to find the polarisation intensity and the dipole moment of the spherical dust particle, and hence the force on the dust. (3) Write the equation of motion of the dust and integrate.

Hint 3.24. (1) The spherical cavity is a uniform field, which can be found from the boundary conditions. (2) Use the boundary conditions to find the charge distribution on the surface of the conducting sphere and on the surface of the dielectric cavity, and use the energy method to find the force on the conducting sphere.

Hint 3.25. The cavity in which q' is located lies in the field excited by q and by the polarisation charge on the surface of the cavity in which it is located (this field may be regarded as uniform at the cavity of q'); use the result of part (1) of Problem 3.24.

Hint 3.26. The distribution of the polarisation charge on the surface of the dielectric sphere is the same as that of a uniformly polarised dielectric sphere.

Hint 3.27. The current distribution in the resistive material may be regarded as the superposition of the current distribution when only the current flows in and that when only the current flows out. From the differential form of Ohm's law it follows that the electric field in the material is equivalent to the field of a pair of point charges of opposite sign.

Hint 3.28. Using similarity and scaling relations, the equipotential points and the superposition principle, simplify the original circuit to the simple circuit shown in the figure.

Hint 3.29. This problem can be solved by Kirchhoff's method, by the superposition of currents, or by the star–delta transformation.

Hint 3.30. Use Thévenin's theorem and the small-quantity approximation.

Hint 3.31. (1) Use Thévenin's theorem repeatedly from left to right. (2) Looking to the left from A_i gives an equivalent source, and looking to the right from A_{i-1} also gives an equivalent source; the original circuit is thus simplified to two equivalent sources and the resistance R between A_i and A_{i-1} connected in series.

Hint 3.32. (1) This can be decided by whether the directions of the current in the ring and of the velocity of the ring are the same. (2) When the ring decelerates in its rotation, the current carriers are acted on by an inertial force. This inertial force may be regarded as a force of non-electrostatic origin, from which an equivalent electromotive force is obtained; using Ohm's law and integrating the current over time, the specific charge can be found.

Hint 3.33. (1) Charge will accumulate on the surface of the bubble; the surface charge distribution can be found from the boundary conditions for the current and for the electric field. The field inside the bubble is the superposition of the original field in the conductor and the field produced by the surface charge (a uniform field). (2) The field at any point P inside the conductor is the superposition of the original field and the field produced by the surface charge (the field of a dipole).

Hint 3.34. (1) Regard the current as steady; the current density depends only on the radius, and the electric field follows from Ohm's law. (2) Use Gauss's law. (3) The line integral of the electric field along a path gives the potential of the Earth's surface. (4) Write the differential equation for the relation between the charge and the time.

Hint 3.35. (1) Use Gauss's law and Ampère's circuital law to find the electric and magnetic fields; from the statement of the problem (the electric and magnetic forces cancel) find the power. (2) The self-inductance L^* per unit length of the transmission line can be found from the capacitance C^* per unit length ($L^*C^* = \varepsilon_0\mu_0$).

Hint 3.36. (1) The field can be obtained by superposing (reducing to an integral) the magnetic fields produced on the axis by circular coils. (2) It can be found from the variation of the magnetic field along the axis by using Gauss's law for the magnetic field.

Hint 3.37. The dielectric is polarised under the joint action of the Lorentz force and the electric field produced by its own polarisation charges.

Hint 3.38. Same as Hint 3.37. Use Gauss's law to find the charge distribution.

Hint 3.39. (2) In producing a current the Lorentz force is equivalent to the electric force; the current intensity between the inner and outer cylinders is determined by the radial component of the current density, while the radial component of the Lorentz force is determined by the transverse component of the drift velocity of the carriers, and the transverse component of the Lorentz force is in turn determined by the radial component of the drift velocity; a relation between the current density, the radial momentum and the electric and magnetic fields can thus be established.

Hint 3.40. (1) The dielectric sphere will be uniformly magnetised in the magnetic field at the centre of the current-carrying coil. (2) The magnetised dielectric sphere may be regarded as a magnetic dipole; it will increase the magnetic flux through the coil.

Hint 3.41. (1) The motion of charge produces a current. (2) A changing magnetic field produces an electric field.

Hint 3.42. The long rod will be uniformly magnetised along the z direction; the magnetisation current in the rod produces a uniform magnetic field inside the rod, and outside the rod it is equivalent to the field produced by two opposite currents.

Hint 3.43. Use the angular momentum theorem and conservation of energy. The torque acting

on the electron comes from the Lorentz force.

Hint 3.44. (1) The centripetal force for the circular motion of the particle is the Lorentz force. (2) Since the magnetic flux density decreases along the radial direction, the lines of magnetic induction must be barrel-shaped; this makes the magnetic flux density have a radial component at $z > 0$ and at $z < 0$, so that when the particle is slightly displaced in the vertical direction it experiences a restoring force.

Hint 3.45. The particle moves in a helix about the z axis; since the magnetic field gradually increases along the z direction, a radial component of the magnetic field appears. The Lorentz force produced by this radial component points in the $-z$ direction, so that the particle is finally reflected. Using Newton's law and conservation of energy, establish the equation relating the rate of change of v_z with z to the rate of change of B_z with z.

Hint 3.46. In the two cases (1) and (2), write the equation of motion of the coil and the circuit equation separately.

Hint 3.47. (1) Take the time of falling over as a parameter and find the ratio of the kinetic energy to the Joule heat. The time of falling over can be estimated from the case without a magnetic field. (2) The time taken for the ring to fall over is found by balancing the moment of the gravitational force against the moment of the Ampere force.

Hint 3.48. (1) An induced current will appear in the superconducting coil so as to keep the magnetic flux through the coil zero. This current, acting in the radial magnetic field, will be acted on by an upward vertical force that balances gravity, and the radial component of the magnetic field can be found from the rate of change of the z component of the magnetic field along the z direction. (2) From the condition that the magnetic flux through the coil is zero, find the variation of the induced current in the coil with position, then write the equation of motion of the coil; note the small-quantity approximation.

Hint 3.49. Reduce the steel magnet to a magnetic dipole and the metal tube to a series of coaxial circular coils; use conservation of energy.

Hint 3.50. Use the phasor method.

Hint 3.51. Write Kirchhoff's equations (using the complex method).

Hint 3.52. Use the complex method together with the boundary condition at infinity.

Hint 3.53. Use the quasi-stationary condition, and note that the displacement current is the same as the conduction current in exciting a magnetic field.

Hint 3.54. (2) When the pulse is applied, the capacitor may be regarded as a short circuit; it is charged by the current U_0/R_1 and then discharges through the parallel combination of R_1 and R_2.

Hint 3.55. (1) Adjust R to the critically damped state. (2) When the current is maximum, $\mathrm{d}I/\mathrm{d}t = 0$; use the final-state condition.

Hint 3.56. (1) Express the propagation factor in complex form, then use the condition $|a| < 1$ to find ω_c. (2) Use the boundary condition at infinity. (3) Examine the propagation factor between two adjacent sections, $a_n = U_{n+1}/U_n$ (n the section number), and set $|a_n| = 1$. (4)

Under the condition $\omega \ll \omega_c$, find the phase difference δ between adjacent sections; then the speed of propagation of the signal along the network is $v = \Delta r/(\delta/\omega)$.

Hint 3.57. Use the star–delta transformation of complex impedances.

Hint 3.58. (1) Superpose the incident and reflected waves, and use $E = 0$ inside an ideal conductor and the boundary conditions for the electric and magnetic fields. (2) Apply Ampère's circuital law at the surface of the conductor to find the surface current, then use Ampère's law to find the force on the surface of the conductor.

2.4 Optics

Hint 4.01. (1) Express the deviation angle δ as $\delta = i + r' - A$ (r' the angle of emergence), and use the law of refraction to express δ as a function of the angle of incidence i. (2) Set $\mathrm{d}\delta/\mathrm{d}i = 0$; the result then follows.

Hint 4.02. (1) Use the symmetry of the optical path and the law of refraction. (2) Calculate successively with a calculator.

Hint 4.03. Take the xy plane as the cross-section and the x axis as the axis of rotation; using the equality of the optical paths of an object and its image, write the equation of the cross-section curve, then simplify and discuss.

Hint 4.04. First use the law of refraction and geometrical relations to express the complement α of the deviation angle δ as a function of the angle of incidence i and the angle of refraction r (see the figure); $\mathrm{d}\alpha/\mathrm{d}i = 0$ determines α, because in that direction the light is most concentrated; α is the angular radius of the rainbow.

Hint 4.05. Use the law of refraction and the small-angle approximation ($\alpha \ll 1$, the ray is bent only slightly), then integrate.

Hint 4.06. (1) Use the law of refraction and the small-angle approximation [$(\alpha - \alpha_0)$ is a small angle]. (2) Omitted.

Hint 4.07. Regard the laser beam as a stream of photons; the change of momentum caused by the deflection of the photons by the lens immediately produces a force on the lens.

Hint 4.08. Regard the thin convex lens as a combination of two spherical surfaces.

Hint 4.09. The diameter of the light spot on screen Q is determined by the position of the image point and by the angular aperture of the emergent beam, and the angular aperture of the emergent beam is determined by the size of the hole in screen P; more precisely, it is determined by the size of the image of the hole formed by lens L_2.

Hint 4.10. (1) From the object–image correspondence, when the concave mirror is at the position of the image of object O formed by the lens, the final image will coincide with the original object. Another position of the concave mirror can be found from the requirement that a ray emitted from O parallel to the optical axis retraces its path after passing through the lens. Of course, the two positions of the concave mirror can also be found by writing the equation from the condition that the magnification is $+1$. (2) Omitted.

Hint 4.11. (1) Use the law of refraction and the sine rule for triangles; alternatively, first find the position of the image by paraxial rays. (2) 1) From the position of the aplanatic point obtained in part (1) and the conditions given in the problem, R_2 and R_3 can be found (the second lens being concave–convex). 2), 3), 4) Omitted.

Hint 4.12. (1) Use the small-angle approximation to find the image of S formed by the upper and lower prisms (note that the angles of incidence and refraction of the ray travelling from S to the prisms are also small). (2) Find the optical path difference of the two rays passing through the upper and lower prisms after S has been displaced upward.

Hint 4.13. (1) Note the phase jump that occurs when light passes through a focal point. (2) Omitted.

Hint 4.14. (1) Use the reversibility of time (that is, the reversibility of the optical path) and the superposition principle. (2) Superpose the complex amplitudes of the reflected waves and use the result of part (1).

Hint 4.15. Use the relativistic velocity-addition formula.

Hint 4.16. Sum separately the complex amplitudes of the terms with coefficient 1 and with coefficient α in the reflectivity $(1 \pm \alpha)$.

Hint 4.17. Regard each step of the sailor as a vector; the problem reduces to finding the magnitude of the sum (resultant) of a large number of vectors of equal magnitude whose directions vary at random.

Hint 4.18. Regard the light disturbance at any point near the geometrical image point on the observation screen as the superposition (integral) of the light disturbances produced at that point by all the surface elements of the geometrical wavefront formed by the square aperture.

Hint 4.19. (1) The incident light may be regarded as the superposition of unpolarised light and elliptically polarised light; the x and y axes are then the principal axes of the ellipse, and the phase difference between the components of the vibration of the elliptically polarised light along these two axes is $90°$. (2) Elliptically polarised light becomes linearly polarised after passing through a quarter-wave plate.

Hint 4.20. In a birefringent crystal, light polarised parallel and perpendicular to the optic axis travels with different speeds; after passing through a crystal of a certain thickness, these two polarised components acquire a phase difference proportional to the thickness of the crystal, and when they pass through the analyser behind, they again combine into linearly polarised light.

2.5 Atomic Physics

Hint 5.01. Use Einstein's photoelectric equation.

Hint 5.02. If the particle is a γ photon, the phenomenon in (b) is the Compton effect of the proton; comparing the experimental results of (a) and (b), one sees that only by abandoning conservation of energy and momentum can the hypothesis that the particle is a γ ray be supported.

Hint 5.03. The Compton effect and the inverse Compton effect.

Hint 5.04. Conservation of momentum and energy, and the relativistic expressions for the momentum–energy relation.

Hint 5.05. Estimate by means of the uncertainty relation. Since the kinetic energy of the electron is comparable to its rest energy, the relativistic momentum–energy relation should be used for the electron.

Hint 5.06. Use the uncertainty relation $\Delta x\,\Delta p \geq \hbar$ to eliminate r or p from the energy expression.

Hint 5.07. Use the quantisation condition for the orbital angular momentum of the electron.

Hint 5.08. Hydrogen-like ion spectra; take the finite nuclear mass correction into account.

Hint 5.09. Take the finite nuclear mass correction into account.

Hint 5.10. The influence of the finite size of the nucleus on the energy levels and spectrum of the exotic atom.

Hint 5.11. The atomic core of the sodium atom may be regarded as an effective nucleus; the valence electron is subject to different effective nuclear charges in different states.

Hint 5.12. The answers to (1) and (2) are helpful for the solution of (3).

Hint 5.13. The Franck–Hertz experiment.

Hint 5.14. Use the Bohr model of the hydrogen atom to estimate the energies of some states of the helium atom (a two-electron atom).

Hint 5.15. This problem mainly uses the fact that the uranium series reaches secular equilibrium after a certain time, that is, the activities of the parent and of all the daughters are equal, and the assumption that the age of formation of the uranium ore is approximately the age of the Earth.

Hint 5.16. The age of ancient biological samples can be determined from the exponential law of variation of activity with time.

Hint 5.17. The key is to write down the energy requirement for the excited state $^{17}\mathrm{O}^{*}$ to emit a neutron and reach $^{16}\mathrm{O}$ (which may be in the ground state or in an excited state).

Hint 5.18. Using the Coulomb energy formula directly related to the same mass number A and radius R, $B_c = \dfrac{3}{5}\dfrac{z(z-1)e^2}{4\pi\varepsilon_0 R}$, the corresponding nuclear radius constant can be found (note that the nuclear radius constants for light and heavy nuclei actually differ somewhat).

Hint 5.19. Use the characteristics of elastic collisions in the centre-of-mass frame and the conversion relation between the differential cross-section in the centre-of-mass frame and that in the laboratory frame.

Chapter 3

Solutions

3.1 Mechanics

Solution 1.01. Method 1. Take the point of projection as the origin of the coordinate system, and take the horizontal and vertical directions as those of the x- and y-axes respectively. The trajectory corresponding to the ball being thrown over the wall with the minimum initial speed must be the parabola tangent to the top of the wall. When the ball reaches the top of the wall,

$$d = v_0 \cos\theta\, t, \tag{1}$$

$$h = v_0 \sin\theta\, t - \frac{1}{2} g t^2. \tag{2}$$

Eliminating t from (1) and (2), we obtain

$$h = d\tan\theta - \frac{g d^2}{2 v_0^2 \cos^2\theta}. \tag{3}$$

Equation (3) can be written as

$$v_0^2 = \frac{g d^2}{2\cos\theta\,(d\sin\theta - h\cos\theta)}. \tag{4}$$

From (4) it is seen that to find the minimum of v_0^2 we need only find the maximum of the denominator on the right-hand side. This denominator can be transformed into

$$2\cos\theta\,(d\sin\theta - h\cos\theta) = 2\cos\theta\sqrt{d^2+h^2}\,\sin(\theta-\alpha) = \sqrt{d^2+h^2}\,[\sin(2\theta-\alpha) - \sin\alpha]\,,$$

where α is the angle between the line joining the point of projection to the top of the wall and the horizontal, as shown in the figure. From the above it is seen that when θ satisfies $2\theta - \alpha = \pi/2$, i.e. $\theta = \pi/4 + \alpha/2$, the denominator of (4) is maximum, that is, the value of v_0^2 is minimum. The minimum value of v_0^2 is

$$v_{0\,\min}^2 = \frac{g d^2}{\sqrt{d^2+h^2}\,(1-\sin\alpha)} = \frac{g d^2}{\sqrt{d^2+h^2} - h} = g\left(\sqrt{d^2+h^2} + h\right),$$

i.e. when the angle of projection is $\theta = \dfrac{\pi}{4} + \dfrac{1}{2}\arctan\dfrac{h}{d}$, the initial speed with which the ball is thrown over the wall is minimum, its minimum value being $v_{0\,\min} = \sqrt{g\left(\sqrt{d^2+h^2}+h\right)}$.

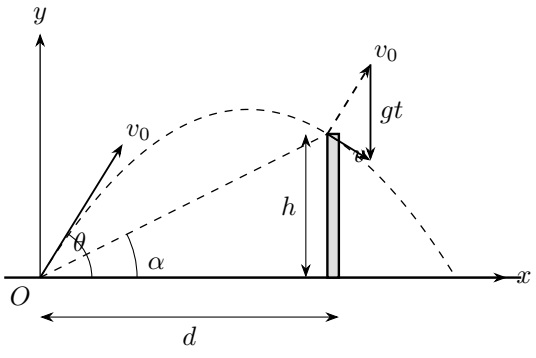


Fig. 1.01

Method 2. When the ball is thrown to the top of the wall, the relation between the velocity $\vec{v}$ of the ball and the initial velocity $\vec{v}_0$ is

$$\vec{v} = \vec{v}_0 + \vec{g}t.$$

A vector triangle satisfying this equation can be drawn, as shown in the figure. The area of this triangle is

$$S = \frac{1}{2}gtv_0\cos\theta = \frac{1}{2}gd. \tag{5}$$

The above shows that however large the initial speed v_0 and however large the angle of projection θ with which the ball is thrown, as long as the ball passes tangent to the top of the wall, the area of the velocity vector triangle at the point of tangency is constant. Since $v = \sqrt{v_0^2 - 2gh}$, it is seen that the minimum v_0 corresponds to the minimum v. From geometry, only when the directions of $\vec{v}$ and $\vec{v}_0$ are mutually perpendicular are the required v_0 and v minimum; at this instant

$$S = \frac{1}{2}v_0 v = \frac{1}{2}v_0\sqrt{v_0^2 - 2gh}. \tag{6}$$

From (5) and (6) we obtain

$$v_0\sqrt{v_0^2 - 2gh} = gd,$$

i.e.

$$(v_0^2)^2 - 2gh(v_0^2) - g^2d^2 = 0,$$

which can be solved to give

$$v_{0\,\min}^2 = g\left(\sqrt{d^2 + h^2} + h\right).$$

From the velocity vector triangle,

$$v_0\cos\theta = v\sin\theta = \sqrt{v_0^2 - 2gh}\,\sin\theta.$$

Substituting $v_0^2 = g\left(\sqrt{d^2 + h^2} + h\right)$ into the above, we obtain

$$\tan\theta = \frac{h + \sqrt{d^2 + h^2}}{d},$$

that is,

$$\theta = \frac{\pi}{4} + \frac{\alpha}{2} = \frac{\pi}{4} + \frac{1}{2}\arctan\frac{h}{d}.$$

Comment. What is the shape of the parabola corresponding to throwing the ball over the wall with the minimum initial speed? Is its highest point at the top of the wall, or in front of the wall, or behind it? From the second method of this problem it is known that at the top of the wall the velocity $\vec{v}$ of the ball should be perpendicular to $\vec{v}_0$; hence the highest point must be in front of the wall, and not in either of the other two positions.

The second method combines the physical problem with a geometric one and analyses the problem from another point of view, thereby making the solution more concise.

Solution 1.02. The parabola corresponding to throwing the ball over the two walls A and B with the minimum initial speed must be tangent to the top of wall A. Let the velocity of the ball when it reaches the top of wall A be v, making an angle α with the horizontal. This is equivalent to the ball being projected from the top of wall A with speed v and angle of projection α. As long as it can be thrown to the top of wall B, that is, pass tangent to the top of wall B, it can be thrown over the two walls.

From the range formula for projectile motion,

$$\frac{v^2 \sin 2\alpha}{g} = d. \tag{1}$$

From this it is seen that when $\alpha = \pi/4$ the required v is minimum:

$$v_{\min} = \sqrt{gd}. \tag{2}$$

From (2) and the conservation of mechanical energy, the minimum initial speed $v_{0\,\min}$ required to throw the ball over the two walls is

$$v_{0\,\min} = \sqrt{v_{\min}^2 + 2gh} = \sqrt{g(d+2h)}. \tag{3}$$

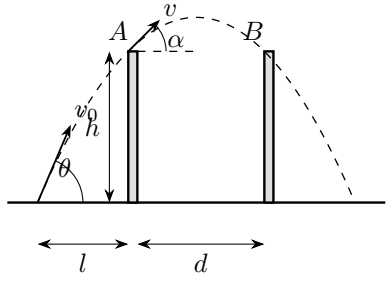


Fig. 1.02

Let the distance from the point of projection to wall A be l and the angle of projection be θ; then

$$v_0 \cos\theta = v \cos\alpha,$$

hence

$$\cos\theta = \frac{v}{v_0} \cos\alpha. \tag{4}$$

Substituting $v_0 = v_{0\,\min} = \sqrt{g(d+2h)}$, $v = v_{\min} = \sqrt{gd}$ and $\alpha = \pi/4$ into (4), we obtain

$$\theta = \arccos \sqrt{\frac{d}{2(d+2h)}}.$$

In addition,

$$v_0 \cos\theta\, t = l, \tag{5}$$

$$v_0 \sin\theta - gt = v \sin\alpha. \tag{6}$$

From (5) and (6), eliminating t and substituting (2), (3), (4) and $\alpha = \pi/4$, we obtain

$$l = \frac{1}{2}\left(\sqrt{d(d+4h)} - d\right).$$

Comment. This problem looks very similar to Problem 1.01, but because the point of projection is not restricted, the method of solution must be completely different. The key to solving this problem is to discuss separately the motion of the ball between the two wall tops and its motion on the two sides of the walls, and then to solve the problem from the connection between them.

Solution 1.03. Take point O as the origin of the coordinate system, take the direction parallel to the shelter as the direction of the x-axis, and the direction perpendicular to the shelter as the direction of the y-axis, as shown in the figure. In this coordinate system the initial velocity and the acceleration of the shell in the y direction are respectively

$$v_{0y} = v_0 \sin(\theta - \alpha), \qquad a_y = -g\cos\alpha.$$

If the trajectory of the shell is tangent to the shelter, then the y-coordinate of the point of tangency A is

$$y_A = h = l\sin\alpha = \frac{v_{0y}^2}{-2a_y} = \frac{v_0^2 \sin^2(\theta-\alpha)}{2g\cos\alpha},$$

that is,

$$v_0^2 \sin^2(\theta - \alpha) = gl \sin 2\alpha. \tag{1}$$

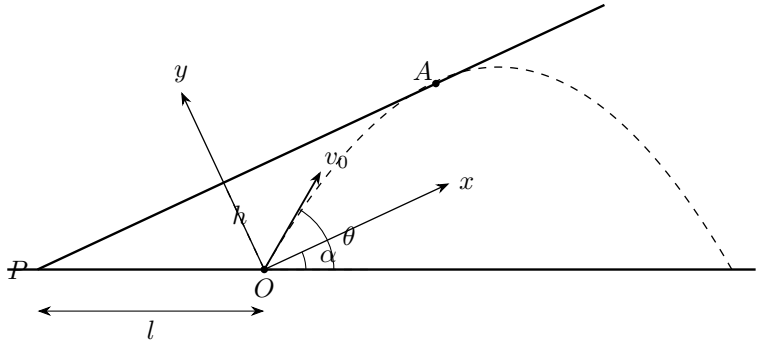


Fig. 1.03

From (1) the following discussion can be made.

(1) If $v_0 < \sqrt{gl\sin 2\alpha}$, the trajectory of the shell cannot be tangent to the shelter, so it is sufficient to take $\theta = \pi/4$, and the shell can then obtain the maximum range

$$L_{\max} = \frac{v_0^2}{g}.$$

(2) If $v_0 \geqslant \sqrt{gl\sin 2\alpha}$, it is known from (1) that when the firing angle of the shell takes the following value, its trajectory is tangent to the shelter:

$$\theta = \alpha + \arcsin\frac{\sqrt{gl\sin 2\alpha}}{v_0}. \tag{2}$$

(i) If the θ given by (2) satisfies $\theta \geqslant \pi/4$, i.e. $v_0 \leqslant \sqrt{\dfrac{2gl\sin 2\alpha}{1-\sin 2\alpha}}$, then it is sufficient to take $\theta = \pi/4$, and the shell can obtain the maximum range

$$L_{\max} = \frac{v_0^2}{g}.$$

(ii) If $\theta < \pi/4$, i.e. $v_0 > \sqrt{\dfrac{2gl\sin 2\alpha}{1-\sin 2\alpha}}$, then the firing angle of the shell is taken as

$$\theta = \alpha + \arcsin\frac{\sqrt{gl\sin 2\alpha}}{v_0},$$

and the shell can obtain the maximum range

$$L_{\max} = \frac{v_0^2 \sin 2\left(\alpha + \arcsin\dfrac{\sqrt{gl\sin 2\alpha}}{v_0}\right)}{g}.$$

Comment. This problem asks for the maximum range obtained by a shell fired with a fixed initial speed. Without any restriction, and without considering the influence of air resistance, one need only fire the shell at an elevation angle $\theta = \pi/4$ to obtain the maximum range. Because of the existence of the shelter in this problem, it is extremely necessary to discuss whether the shell will collide with the shelter during its flight, and to find the corresponding conditions. Moreover, when finding the condition for the trajectory of the shell to be tangent to the shelter, choosing the two directions parallel and perpendicular to this slope as the directions of the x- and y-axes of the coordinate system makes the solution more concise.

Solution 1.04. Take the position of the lamp as the origin of the coordinate system, with the coordinate axes as shown in the figure. If there were no restriction by the ceiling, the fragments of the burst lamp would fly out with initial speed v_0 and elevation angle θ, and the distance from their landing point to the point O_2 on the ground directly below the lamp is denoted by R.

From the formulas of projectile motion,

$$v_0\cos\theta\, t = R, \tag{1}$$

$$v_0\sin\theta\, t - \frac{1}{2}gt^2 = -h_2. \tag{2}$$

Squaring both sides of (1) and (2) respectively and then adding the two equations, we obtain

$$R^2 + h_2^2 = v_0^2t^2 - gv_0\sin\theta\, t^3 + \frac{1}{4}g^2t^4. \tag{3}$$

Substituting (2) into (3), we obtain

$$\begin{aligned} R^2 &= (v_0^2 + gh_2)t^2 - \frac{1}{4}g^2t^4 - h_2^2 \\ &= \left(\frac{v_0^2}{g} + h_2\right)^2 - h_2^2 - \left[\frac{1}{2}gt^2 - \left(\frac{v_0^2}{g} + h_2\right)\right]^2. \end{aligned} \tag{4}$$

From (4) it is seen that when $t = \dfrac{1}{g}\sqrt{2(v_0^2 + gh_2)}$, R is maximum, and its value is

$$R_{\max} = \frac{v_0}{g}\sqrt{v_0^2 + 2gh_2}. \tag{5}$$

Substituting (5) and $t = \dfrac{1}{g}\sqrt{2(v_0^2 + gh_2)}$ into (1), we obtain the angle of projection θ_0 corresponding to $R_{\max}$:

$$\cos\theta_0 = \sqrt{\frac{v_0^2 + 2gh_2}{2(v_0^2 + gh_2)}}. \tag{6}$$

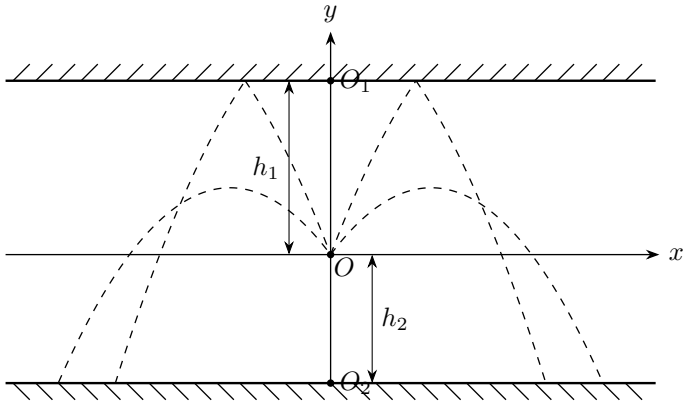


Fig. 1.04

From (6) the maximum height to which the farthest-flying fragment can rise is

$$h = \frac{v_0^2 \sin^2\theta_0}{2g} = \frac{v_0^2(1 - \cos^2\theta_0)}{2g} = \frac{v_0^4}{4g(v_0^2 + gh_2)}. \tag{7}$$

Discussion:

(1) If $h_1 \geqslant h$, i.e. the farthest-flying fragment does not strike the ceiling during its flight, then the radius r of the region in which the fragments land is just $R_{\max}$:

$$r = R_{\max} = \frac{v_0}{g}\sqrt{v_0^2 + 2gh_2}.$$

The condition for this case can also be expressed as

$$v_0^2 \leqslant 2g\left(h_1 + \sqrt{h_1(h_1 + h_2)}\right).$$

(2) If $h_1 < h$, i.e. $v_0^2 > 2g\left(h_1 + \sqrt{h_1(h_1 + h_2)}\right)$, then the farthest-flying fragment will strike the ceiling during its flight and thereby change its trajectory. For this case it can be proved that among all the fragments, the fragment whose trajectory is tangent to the ceiling can fly farthest. The proof is as follows.

(i) From the law of conservation of mechanical energy, whether a fragment is tangent to the ceiling or collides with it, its speed v on reaching the ceiling is the same. After an elastic collision with the ceiling, the fragment moves as a projectile thrown obliquely downward with the original speed v, whereas the fragment tangent to the ceiling is equivalent to being projected horizontally with the same speed v at the point of tangency. From the knowledge of projectile motion it is known that, at the same height, a body projected horizontally can fly farther than a body projected obliquely downward in any direction.

(ii) Compared with the various fragments that collide with the ceiling, the fragment tangent to the ceiling reaches the ceiling at a position farther from O_1, the point on the ceiling directly above the lamp.

From the above two points it is known that the fragment tangent to the ceiling will fly farthest.

Let the angle of projection of the fragment tangent to the ceiling at the moment it flies out be θ'; then

$$\frac{v_0^2 \sin^2 \theta'}{2g} = h_1,$$

i.e. $v_0 \sin \theta' = \sqrt{2gh_1}$. Substituting this into (2), we obtain

$$t^2 - 2\sqrt{\frac{2h_1}{g}}\, t - \frac{2h_2}{g} = 0.$$

The solution of this equation is

$$t = \sqrt{\frac{2h_1}{g}} \pm \sqrt{\frac{2(h_1 + h_2)}{g}}.$$

Discarding the solution with the "$-$" sign and substituting the solution with the "$+$" sign into (1), we obtain the radius of the region in which the fragments land in this case:

$$r = \frac{1}{g}\sqrt{v_0^2 - 2gh_1}\left[\sqrt{2gh_1} + \sqrt{2g(h_1 + h_2)}\right].$$

Solution 1.05. (1) The coordinate system O-xyz is as shown in the figure, with the origin O at the point from which the bullet is fired, the wall being parallel to the xOz plane and at distance d from O.

Let the trajectory of a certain bullet lie in the yOz plane, its angle of elevation be θ, and the aiming point on the wall be M; the bullet hits point P on the wall at height h above the ground. For this bullet,

$$v_0 \sin\theta\, t - \frac{1}{2}gt^2 = h, \tag{1}$$

$$v_0 \cos\theta\, t = d. \tag{2}$$

From (1) and (2), eliminating t, we obtain

$$h = d\tan\theta - \frac{gd^2}{2v_0^2 \cos^2\theta},$$

i.e.

$$\tan\theta = \frac{v_0^2 \pm \sqrt{v_0^4 - 2ghv_0^2 - g^2d^2}}{gd}. \tag{3}$$

Through O draw another coordinate axis y', as shown in the figure. This axis intersects the wall at K; take $KO' = l$, then $KO = \sqrt{d^2 + l^2}$. Suppose that the trajectory of another bullet lies in the $y'Oz$ plane, its angle of elevation is θ', the aiming point on the wall is N, and the bullet hits point Q on the wall at the same height h above the ground. For this bullet a relation similar to (3) is obtained:

$$\tan\theta' = \frac{v_0^2 \pm \sqrt{v_0^4 - 2ghv_0^2 - g^2(d^2 + l^2)}}{g\sqrt{d^2 + l^2}}. \tag{4}$$

Through O' take another plane coordinate system $O'x'z'$, with $O'K$ as the x'-axis and $O'M$ as the z'-axis. In this coordinate system the coordinates of point N are denoted by (x', z'). From the geometric relations shown in the figure,

$$x' = l, \qquad z' = \sqrt{d^2 + l^2}\,\tan\theta'.$$

Substituting (4), we obtain

$$z' = \frac{v_0^2}{g} \pm \frac{\sqrt{v_0^4 - 2ghv_0^2 - g^2d^2 - g^2x'^2}}{g}.$$

From this it follows that

$$\left(z' - \frac{v_0^2}{g}\right)^2 + x'^2 = \left(\frac{\sqrt{v_0^4 - 2ghv_0^2 - g^2d^2}}{g}\right)^2. \tag{5}$$

Equation (5) is the equation, in the $O'x'z'$ coordinate system, of a circle whose centre is at the point $\left(0, \dfrac{v_0^2}{g}\right)$ and whose radius is $R = \dfrac{1}{g}\sqrt{v_0^4 - 2ghv_0^2 - g^2d^2}$; this circle is precisely the locus of the aiming points for the bullet to hit the horizontal line at height h on the wall.

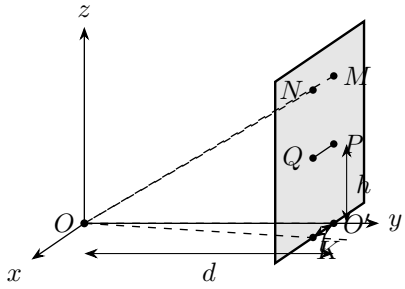


Fig. 1.05

(2) If the bullet is to hit a point on the wall above the horizontal line of height h, it is only necessary to substitute h' $(> h)$ for h in (5). It is known that the equation then becomes a circle of radius R' $(< R)$. Hence the aiming points lie inside the circle shown in (5).

Comment. This problem involves three-dimensional space; choosing a suitable coordinate system is the key to solving it.

Solution 1.06. (1) After the bomb is released it moves as a horizontally projected body, with the speed v_0 of the aeroplane as its initial speed; hence

$$h = \frac{1}{2}gt^2 \tag{1}$$

$$L = v_0 t \tag{2}$$

Combining (1) and (2), we obtain the horizontal distance L between the point at which the bomb is released and the target:

$$L = v_0\sqrt{\frac{2h}{g}}.$$

(2) Take the instant at which the bomb is released as $t = 0$. At the time t the velocity $\vec{v}_1$ of the bomb is

$$\vec{v}_1 = \vec{v}_0 + \vec{g}t \tag{3}$$

and at this instant the velocity $\vec{v}_2$ of a shell fired from the gun on the ground with initial speed $\vec{v}$ is

$$\vec{v}_2 = \vec{v} + \vec{g}t. \tag{4}$$

Take the bomb as the frame of reference; in this frame the relative velocity $\vec{v}'$ of the shell is

$$\vec{v}' = \vec{v}_2 - \vec{v}_1 = \vec{v} - \vec{v}_0. \tag{5}$$

The vector relation (5) is shown in Fig. 1.06(a).

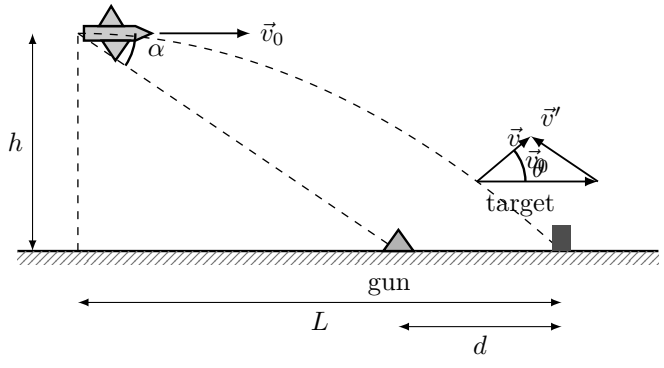


Fig. 1.06(a)

Suppose that after a time t_1 the shell hits the bomb in the air; then

$$v't_1 = \sqrt{(L-d)^2 + h^2}. \tag{6}$$

Since the point at which the shell hits the bomb must be in the air, while the longest time for which the bomb can fly in the air is $t = \sqrt{\dfrac{2h}{g}}$, it follows that necessarily

$$t_1 \le t = \sqrt{\frac{2h}{g}}. \tag{7}$$

Combining (6) and (7) gives

$$v' \ge \sqrt{(L-d)^2 + h^2}\,\sqrt{\frac{g}{2h}}. \tag{8}$$

From (8) it is seen that the minimum relative speed required for the shell to hit the bomb in the air is

$$v'_m = \sqrt{(L-d)^2 + h^2}\,\sqrt{\frac{g}{2h}}.$$

To find the minimum initial speed of the shell relative to the ground, the following discussion may be made.

(i) If $v_0 \cos\alpha < v'_m = \sqrt{(L-d)^2+h^2}\sqrt{\dfrac{g}{2h}}$, where $\cos\alpha = \dfrac{L-d}{\sqrt{(L-d)^2+h^2}}$, this condition can be expressed as

$$v_0 < \frac{d^2+h^2}{d}\sqrt{\frac{g}{2h}}.$$

Then from the vector triangle of Fig. 1.06(a) it is seen that for the value of v to be a minimum the value of v' must be as small as possible; hence we take v'_m and obtain

$$\begin{aligned} v_{\min}^2 &= v_m'^2 + v_0^2 - 2v'_m v_0 \cos\alpha \\ &= (L-d)^2\frac{g}{2h} + \frac{1}{2}gh + v_0^2 - 2v_0(L-d)\sqrt{\frac{g}{2h}} \end{aligned}$$

Substituting $v_0 = L\sqrt{\dfrac{g}{2h}}$ into the above expression, we obtain the minimum launching speed of the shell in this case:

$$v_{\min} = \sqrt{\frac{g}{2h}(d^2+h^2)}.$$

From the geometrical relations shown in the figure, the launching angle θ of the shell corresponding to $v_{\min}$ satisfies

$$v_{\min}\sin\theta = v'_m \sin\alpha.$$

Hence

$$\sin\theta = \frac{v'_m}{v_{\min}}\sin\alpha = \frac{h}{\sqrt{d^2+h^2}},$$

or, equivalently,

$$\tan\theta = \frac{h}{d}, \qquad \text{i.e.} \qquad \theta = \arctan\frac{h}{d}.$$

(ii) If $v_0\cos\alpha \geq v'_m$, the condition for this case can also be expressed as

$$v_0 \geq \frac{d^2+h^2}{d}\sqrt{\frac{g}{2h}}.$$

The velocity vector triangle for this case is shown in Fig. 1.06(a). From the figure it is seen that we need only make $v \perp v'$ for v to attain a minimum; note that in this case $v' > v'_m$. Hence the minimum initial speed $v_{\min}$ is

$$v_{\min} = v_0\sin\alpha = \frac{v_0 h}{\sqrt{(L-d)^2+h^2}} = \frac{v_0 h}{\sqrt{\left(v_0\sqrt{\dfrac{2h}{g}} - d\right)^2 + h^2}}.$$

The launching angle θ corresponding to this minimum initial speed is

$$\theta = \frac{\pi}{2} - \alpha, \qquad \text{or} \qquad \theta = \arctan\left(\frac{v_0\sqrt{\dfrac{2h}{g}} - d}{h}\right).$$

The discussion above is all for the case $d < L$, i.e. the case shown in Fig. 1.06(a). If $d \geq L$, the velocity vector triangle is as shown in Fig. 1.06(b). From the geometrical relations shown in the

figure it is seen that for v to be a minimum, v' should be as small as possible, so we take $v' = v'_m$, and the law of cosines then gives

$$v_{\min} = \sqrt{\frac{g}{2h}(h^2 + d^2)}.$$

This value is the same as in case (i), and the corresponding launching angle is also the same as in case (i), that is,

$$\theta = \arctan\left(\frac{h}{d}\right).$$

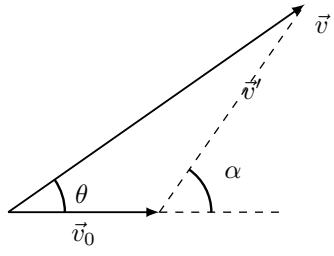


Fig. 1.06(b)

Solution 1.07. First find the velocity v_P with which the position of point P changes.

Take the circle as the frame of reference. In this frame the velocity of the straight line in the direction perpendicular to the line is v_2, and the direction of the velocity v'_P of point P (its relative velocity) is along the tangent to the circle. As a point on the straight line, P must satisfy

$$v'_P \cos\theta = v_2 \tag{1}$$

From the relative motion formula, the velocity v_P with which the position of point P changes relative to the plane of the paper (its absolute velocity) is

$$\vec{v}_P = \vec{v}'_P + \vec{v}_1 \tag{2}$$

The vector relation is shown in Fig. 1.07(a). From the geometrical relations shown in the figure we obtain

$$\begin{aligned} v_P &= \sqrt{(v_1 - v'_P \sin\theta)^2 + v_2^2} \\ &= \sqrt{(v_1 - v_2 \tan\theta)^2 + v_2^2} \end{aligned}$$

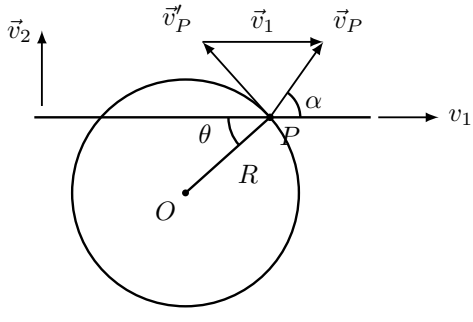


Fig. 1.07(a)

The angle α between the direction of v_P and the straight line is

$$\cot\alpha = \frac{v_1 - v_2\tan\theta}{v_2} = \frac{v_1}{v_2} - \tan\theta,$$

that is,

$$\alpha = \operatorname{arccot}\left(\frac{v_1}{v_2} - \tan\theta\right).$$

Next find the acceleration with which the position of point P changes.

First take the circle as the frame of reference. In this frame, as a point on the circle, P moves in a circle, and indeed in a circle with varying speed, so its relative acceleration $\vec{a}'_P$ can be expressed as

$$\vec{a}'_P = \vec{a}'_n + \vec{a}'_t.$$

In the above expression $\vec{a}'_n$ and $\vec{a}'_t$ are the normal and the tangential acceleration respectively. The value of the normal acceleration is

$$a'_n = \frac{v'^2_P}{R} = \frac{v_2^2}{R\cos^2\theta} \tag{3}$$

Since the circle moves uniformly relative to the plane of the paper, from the theory of relative motion it is known that the acceleration $\vec{a}_P$ of point P relative to the plane of the paper is equal to $\vec{a}'_P$, that is,

$$\vec{a}_P = \vec{a}'_P = \vec{a}'_n + \vec{a}'_t \tag{4}$$

Next take the straight line as the frame of reference. In this frame, as a point on the straight line, P moves along a straight line, so the direction of its relative acceleration $\vec{a}''_P$ is along this straight line. Since the straight line also moves uniformly relative to the plane of the paper, the theory of relative motion likewise gives

$$\vec{a}_P = \vec{a}''_P \tag{5}$$

that is, relative to the plane of the paper the direction of the acceleration $\vec{a}_P$ of point P is along the direction of the straight line. Fig. 1.07(b) shows the vector relation in (4); the value of a_P can then be found:

$$a_P = \frac{a'_n}{\cos\theta} \tag{6}$$

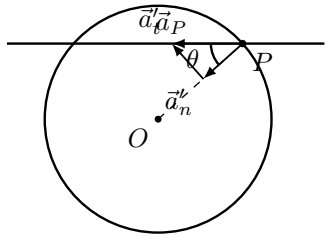


Fig. 1.07(b)

Substituting (3) into (6) gives

$$a_P = \frac{v_2^2}{R\cos^3\theta}.$$

Comment. The acceleration of the point P in this problem can also be found by first differentiating v'_P with respect to time to obtain a'_t, and then using $a_P = a'_P = \sqrt{a'^2_n + a'^2_t}$ to find a_P. The preceding

method takes the circle and then the straight line as frames of reference in turn and finds by analysis that the direction of $\vec{a}_P$ is along the straight line. Hence one need only find the value of a'_n to obtain the value of a_P. This method of solution avoids differentiation and is therefore more convenient.

Solution 1.08. (1) Take the point of contact A of the rod with the ground as the frame of reference. In this frame the rod rotates about A, so the direction of the velocity $\vec{v}'_P$ (the relative velocity) of the point P of the rod that is in contact with the cylinder is perpendicular to the rod, that is, along the radius of the cylinder towards its centre O. Let the angular velocity with which the rod rotates about A at this instant be ω; then

$$v'_P = \omega l \tag{1}$$

where l is the length of the rod from the point P to the point A. Owing to the constraint by the fixed semi-cylindrical body, the absolute velocity $\vec{v}_P$ of the point P of the rod can have no component perpendicular to the rod and directed obliquely downwards (i.e. towards the point O), and of course it can have no component perpendicular to the rod and directed obliquely upwards either, since otherwise the rod would necessarily leave the cylindrical surface; hence at this instant the direction of $\vec{v}_P$ must be along the rod towards the point A. From the relative motion formula we obtain

$$\vec{v}_P = \vec{v}'_P + \vec{v}.$$

Fig. 1.08(a) shows the vector relation of the above expression. From the geometrical relations shown in the figure we obtain

$$v_P = v\cos\theta \tag{2}$$

and at the same time

$$v'_P = v\sin\theta. \tag{3}$$

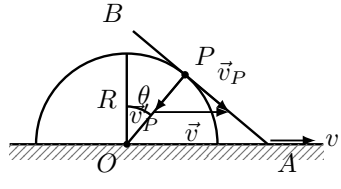


Fig. 1.08(a)

(2) Combining (1) and (3), and noting that $l = R\cot\theta$, we obtain

$$v\sin\theta = l\omega = R\cot\theta\,\omega,$$

and hence

$$\omega = \frac{v\sin^2\theta}{R\cos\theta}. \tag{4}$$

This ω is the angular velocity with which the rod rotates about A in the frame of reference attached to A. Since this frame translates, ω is also the angular velocity of the rod relative to the frame of reference of the ground. From Fig. 1.08(b) it is seen that when the rod turns through an angle $\Delta\theta$, the point of intersection P' of the rod and the cylindrical surface also turns through an angle $\Delta\theta$ in its circular motion about O, that is, the angular velocity of this circular motion is also ω. Hence

$$v_{P'} = \omega R. \tag{5}$$

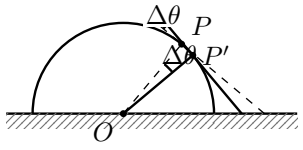


Fig. 1.08(b)

Substituting (4) into (5) gives

$$v_{P'} = \frac{v \sin^2 \theta}{\cos \theta}.$$

Solution 1.09. (1) When hinge O_3 moves along the x-axis with speed v, the velocities of hinges O_1 and O_2 are respectively

$$v_{O_1} = \frac{1}{3}v, \qquad v_{O_2} = \frac{2}{3}v,$$

and the components along the x direction of the velocities of hinges A_1 and B_3 are respectively

$$v_{A_1,x} = \frac{1}{6}v, \qquad v_{B_3,x} = \frac{5}{6}v.$$

Since point A_1 moves in a circle about O, the direction of v_{A_1} is perpendicular to the rod OA_1. From the geometric relations it is known that the angle between the direction of v_{A_1} and the x-axis is $\dfrac{\pi}{2} - \theta$. Hence

$$v_{A_1} = \frac{v_{A_1,x}}{\cos\left(\dfrac{\pi}{2} - \theta\right)} = \frac{v}{6 \sin \theta}. \tag{1}$$

At the same time the value of the component of the velocity of point A_1 in the y direction can be found:

$$v_{A_1,y} = v_{A_1} \cos \theta = \frac{v}{6} \cot \theta.$$

Since the value of the component in the y direction of the velocity of point B_3 is the same as that of point A_1, i.e.

$$v_{B_3,y} = v_{A_1,y} = \frac{v}{6} \cot \theta,$$

we therefore obtain

$$v_{B_3} = \sqrt{v_{B_3,x}^2 + v_{B_3,y}^2} = \frac{v}{6}\sqrt{25 + \cot^2 \theta}. \tag{2}$$

In this problem, except for hinge O, all the other hinges move uniformly along the x direction. Therefore the accelerations of A_1 and B_3 are along the y-axis; let their values be a_{A_1} and a_{B_3} respectively. The normal acceleration of hinge A_1 in its circular motion is

$$a_{A_1,n} = \frac{v_{A_1}^2}{l} = \frac{v^2}{36l \sin^2 \theta}, \tag{3}$$

hence

$$a_{A_1} = \frac{a_{A_1,n}}{\sin \theta} = \frac{v^2}{36l \sin^3 \theta}. \tag{4}$$

The value of the acceleration of hinge B_3 is the same as that of A_1, i.e.

$$a_{B_3} = a_{A_1} = \frac{v^2}{36l \sin^3 \theta}.$$

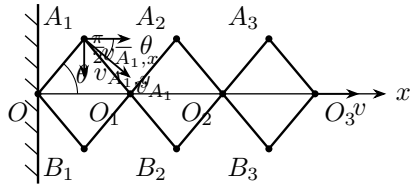


Fig. 1.09

(2) Since the angular speeds and the angular accelerations of all the rods are the same, it is only necessary to find the angular speed ω and the angular acceleration β of rod OA_1:

$$\omega = \frac{v_{A_1}}{l} = \frac{v}{6l\sin\theta}. \tag{5}$$

From (3) and (4) the tangential acceleration of A_1 can be found:

$$a_{A_1,t} = \sqrt{a_{A_1}^2 - a_{A_1,n}^2} = \frac{v^2\cos\theta}{36l\sin^3\theta}.$$

From the tangential acceleration, β can be found:

$$\beta = \frac{a_{A_1,t}}{l} = \frac{v^2\cos\theta}{36l^2\sin^3\theta}. \tag{6}$$

Equations (5) and (6) are the expressions for the angular speed and the angular acceleration of rod A_3B_2.

(3) Hinges O_1, O_2 and O_3 move along the x-axis, and their trajectory equations can be expressed as

$$y = 0.$$

Hinges A_1, A_2, A_3, B_1, B_2, B_3 can be denoted by the symbols A_n, B_n $(n = 1, 2, 3)$, and their position coordinates are

$$x_n = (2n-1)l\cos\theta,$$
$$y_n = \pm l\sin\theta.$$

Hence the trajectory equations are

$$\frac{x^2}{(2n-1)^2l^2} + \frac{y^2}{l^2} = 1 \qquad (n = 1, 2, 3),$$

which are equations of ellipses.

Solution 1.10. Take the disc A as the frame of reference. In this uniformly rotating frame the disc B rolls without slipping along the rim of A, and the angular velocity with which B turns about its own centre O_2 is

$$\omega_2' = \omega_2 - \omega_1.$$

Hence the relative velocity v_{O_2}' of the point O_2 is

$$v_{O_2}' = r\omega_2' = r(\omega_2 - \omega_1).$$

From the theory of relative motion it is known that the absolute velocity v_{O_2} of the point O_2 is related to the relative velocity v'_{O_2} by

$$\vec{v}_{O_2} = \vec{v}'_{O_2} + \vec{\omega}_1 \times \vec{r}_{O_1O_2}, \tag{1}$$

where $\vec{r}_{O_1O_2}$ is the vector from O_1 to O_2; hence

$$\begin{aligned} v_{O_2} &= r(\omega_2 - \omega_1) + (R + r)\omega_1 \\ &= R\omega_1 + r\omega_2. \end{aligned}$$

From the absolute velocity of the point O_2 the time t required for the centre O_2 of the disc B to make one complete circuit about the centre O_1 of the disc A is found to be

$$t = \frac{2\pi(R + r)}{v_{O_2}} = \frac{2\pi(R + r)}{R\omega_1 + r\omega_2}.$$

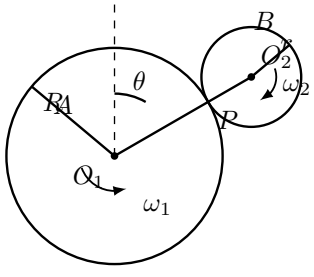


Fig. 1.10

Comment. This problem concerns relative motion in a frame of reference that rotates uniformly. Equation (1) is the velocity transformation formula, in which $\vec{\omega}_1 \times \vec{r}_{O_1O_2}$ is the transport velocity. This expression shows that the transport velocity depends on the position of the particle. This is different from the velocity transformation for relative motion in a frame of reference that translates, where the transport velocity is independent of the position of the particle.

Solution 1.11. Take the large steel ring as the frame of reference. In this uniformly rotating frame, let the angular velocity of the small steel ring be ω' when it rolls without slipping along the inner side of the large ring; then the relative velocity of its centre C is

$$v'_C = \frac{R}{3}\omega'. \tag{1}$$

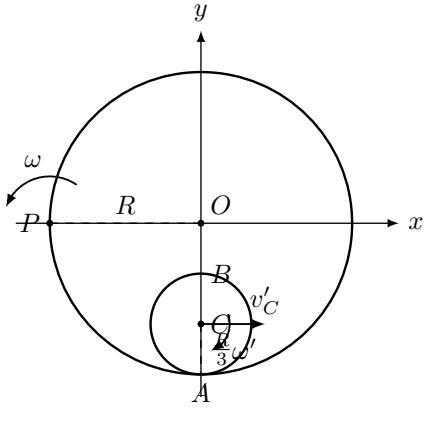


Fig. 1.11

According to the conditions given in the problem: during the time t' in which the small steel ring turns once relative to the large steel ring, the large steel ring turns through half a revolution about the point P; hence

$$t' = \frac{2\pi \times \frac{2}{3}R}{v'_C} = \frac{4\pi}{\omega'} = \frac{1}{2} \cdot \frac{2\pi}{\omega}.$$

From the above expression we obtain

$$\omega' = 4\omega \tag{2}$$

Substituting (2) into (1) gives

$$v'_C = \frac{4}{3}\omega R. \tag{3}$$

In order to find the relative accelerations of the points A and B of the small steel ring, we may take the large steel ring as a frame of reference at rest and take the centre C of the small steel ring as a moving (instantaneously translating) frame of reference; in this frame the points A and B both move uniformly in circles, and the value of their relative acceleration is

$$a''_A = a''_B = \omega'^2 \frac{R}{3} = \frac{16}{3}\omega^2 R.$$

The transport acceleration is the acceleration $\vec{a}'_C$ of the point C relative to the large steel ring:

$$a'_C = \frac{v'^2_C}{\frac{2}{3}R} = \frac{8}{3}\omega^2 R.$$

Hence the accelerations of the points A and B relative to the large steel ring are respectively

$$\vec{a}'_A = \vec{a}''_A + \vec{a}'_C = (a''_A + a'_C)\hat{\jmath} = 8\omega^2 R\,\hat{\jmath},$$

$$\vec{a}'_B = \vec{a}''_B + \vec{a}'_C = (a'_C - a''_B)\hat{\jmath} = -\frac{8}{3}\omega^2 R\,\hat{\jmath}.$$

Having found the accelerations of the points A and B relative to the large steel ring, we can use the relative motion formula to find the absolute accelerations of the points A and B respectively:

$$\begin{aligned}
\vec{a}_A &= \vec{a}'_A + \vec{\omega} \times (\vec{\omega} \times \vec{r}_A) \\
&= \vec{a}'_A + \vec{\omega} \times [\vec{\omega} \times (R\hat{\imath} - R\hat{\jmath})] \\
&= 8\omega^2 R\,\hat{\jmath} + \omega^2 R\,\hat{\jmath} - \omega^2 R\,\hat{\imath} \\
&= -\omega^2 R\,\hat{\imath} + 9\omega^2 R\,\hat{\jmath} \\
&= \omega^2 R(-\hat{\imath} + 9\hat{\jmath})
\end{aligned} \tag{4}$$

$$\vec{a}_B = \vec{a}'_B + \vec{\omega} \times (\vec{\omega} \times \vec{r}_B) + 2\vec{\omega} \times \vec{v}'_B \tag{5}$$

In the above expression $\vec{v}'_B$ is the velocity of the point B relative to the large steel ring:

$$\vec{v}'_B = 2v'_C = \frac{8}{3}\omega R\,\hat{\imath}. \tag{6}$$

Substituting (6) into (5) gives

$$\begin{aligned}\vec{a}_B &= \vec{a}'_B + \vec{\omega} \times \left[\vec{\omega} \times \left(R\hat{\imath} - \frac{R}{3}\hat{\jmath}\right)\right] + 2\vec{\omega} \times \frac{8}{3}\omega R\,\hat{\imath} \\ &= -\frac{8}{3}\omega^2 R\,\hat{\jmath} + \frac{\omega^2 R}{3}\hat{\jmath} - \omega^2 R\,\hat{\imath} + \frac{16}{3}\omega^2 R\,\hat{\jmath} \\ &= \omega^2 R(-\hat{\imath} + 3\hat{\jmath}).\end{aligned}$$

Comment. Equations (4) and (5) are respectively the relations between the absolute accelerations of the points A and B and their relative accelerations in the uniformly rotating frame of reference; here $\vec{\omega} \times (\vec{\omega} \times \vec{r}_A)$ and $\vec{\omega} \times (\vec{\omega} \times \vec{r}_B)$ are the transport accelerations. Unlike the point A, which is at rest in the rotating frame of reference, the point B moves in this frame with velocity $\vec{v}'_B$. Therefore, compared with (4), equation (5) contains one more term $2\vec{\omega} \times \vec{v}'_B$, and this term is the Coriolis acceleration.

Solution 1.12. (1) Let the velocity of the centre O of the small cylinder relative to the ground be v_0, and its angular speed of rotation be ω. Since there is no relative slipping between the small cylinder and cylinders A and B at the points of contact, take O as the moving (translating) frame of reference. From the relative-motion velocity formula the absolute velocities v_C and v_D of points C and D on the small cylinder can be obtained, and v_D and v_C are precisely the velocities of the points of contact of the walls of cylinders A and B with the small cylinder, so

$$v_C = v_0 + \frac{R}{2}\omega = 2R\omega_2, \tag{1}$$

$$v_D = -v_0 + \frac{R}{2}\omega = R\omega_1. \tag{2}$$

From (1) and (2) we can solve

$$\omega = \omega_1 + 2\omega_2, \tag{3}$$

$$v_0 = \frac{R}{2}\left(2\omega_2 - \omega_1\right). \tag{4}$$

Since the centre O of the small cylinder moves in a circle of radius $\dfrac{3}{2}R$ about the axis O' of cylinders A and B, the time t required for it to make one circuit about O' relative to the ground is

$$t = \frac{2\pi \times \dfrac{3}{2}R}{v_0} = \frac{6\pi}{2\omega_2 - \omega_1}.$$

Take cylinder B as the moving (uniformly rotating) frame of reference; let the velocity of point O in this frame be v'_0. From the relative-motion velocity formula,

$$v'_0 = v_0 - \omega_2 \times r'_0, \tag{5}$$

where r'_0 is the vector from O' to O, $|r'_0| = \dfrac{3}{2}R$. Substituting into (5), we obtain

$$\begin{aligned}v'_0 &= \frac{R}{2}\left(2\omega_2 - \omega_1\right) - \omega_2 \times \frac{3}{2}R \\ &= -\frac{\omega_1 + \omega_2}{2}R.\end{aligned} \tag{6}$$

Hence the time t' for the small cylinder to make one circuit relative to cylinder B is

$$t' = \frac{2\pi \times \frac{3}{2}R}{|v_0'|} = \frac{6\pi}{\omega_1 + \omega_2}.$$

(2) First find the acceleration a_C of point C of the small cylinder relative to the ground.

Take O as the moving (translating) frame of reference. Since O moves uniformly in a circle about O', the value of the entrainment acceleration a_0 is

$$a_0 = \frac{v_0^2}{\frac{3}{2}R} = \frac{R}{6}\left(2\omega_2 - \omega_1\right)^2, \tag{7}$$

and since point C moves uniformly in a circle relative to O, the relative acceleration is

$$a_C' = \omega^2 \cdot \frac{R}{2} = \frac{R}{2}\left(\omega_1 + 2\omega_2\right)^2. \tag{8}$$

From the relative-motion formula,

$$a_C = a_C' + a_0.$$

From the above and (7), (8), we obtain

$$\begin{aligned} a_C = a_C' + a_0 &= \frac{R}{2}\left(\omega_1 + 2\omega_2\right)^2 + \frac{R}{6}\left(2\omega_2 - \omega_1\right)^2 \\ &= \frac{2}{3}R\left[\left(\omega_1 + \omega_2\right)^2 + 3\omega_2^2\right]. \end{aligned} \tag{9}$$

The direction of a_C is from C towards O'.

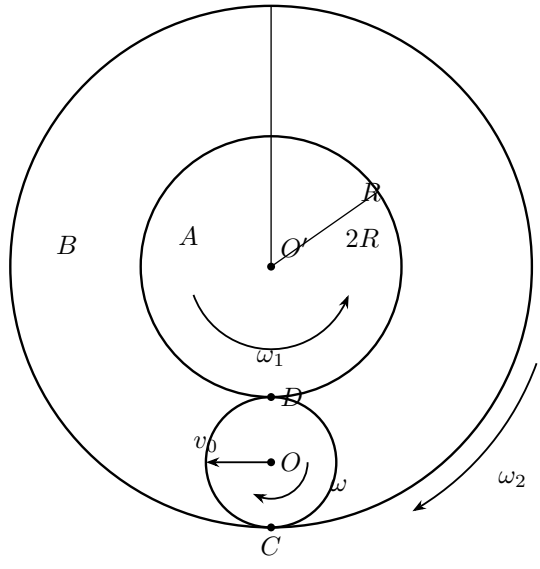


Fig. 1.12

Next find the acceleration a''_C of point C relative to cylinder A.

Method 1.

Take cylinder A as the moving (rotating) frame of reference; in this frame let the relative velocity of point O be v''_0; then from the relative-motion velocity formula

$$v''_0 = v_0 - \omega_1 \times r'.$$

Hence

$$\begin{aligned} v''_0 &= \frac{R}{2}(2\omega_2 - \omega_1) - \left(-\omega_1 \times \frac{3}{2}R\right) \\ &= (\omega_1 + \omega_2) R. \end{aligned} \tag{10}$$

In this frame the angular speed of rotation of cylinder B is $\omega_1 + \omega_2$. Let the angular speed of the small cylinder about O be ω'; then the relative velocity v''_C of point C is

$$v''_C = (\omega_1 + \omega_2) \cdot 2R = v''_0 + \omega' \frac{R}{2}. \tag{11}$$

Substituting (10) into (11), we can solve

$$\omega' = 2(\omega_1 + \omega_2). \tag{12}$$

Take cylinder A as a fixed frame and take point O as the moving (translating) frame; the relative acceleration of point C is $\omega' \times (\omega' \times r''_C)$, so its absolute acceleration a''_C relative to cylinder A is

$$a''_C = a''_0 + \omega' \times (\omega' \times r''_C). \tag{13}$$

In (13), r''_C is the vector from O to C, $|r''_C| = \dfrac{R}{2}$, and $a''_0 = \dfrac{v''^2_0}{\dfrac{3}{2}R} = \dfrac{2}{3}(\omega_1 + \omega_2)^2 R$. Substituting this and (12) into (13), we obtain

$$\begin{aligned} a''_C &= \frac{2}{3}(\omega_1 + \omega_2)^2 R + 4(\omega_1 + \omega_2)^2 \times \frac{R}{2} \\ &= \frac{8}{3}(\omega_1 + \omega_2)^2 R. \end{aligned}$$

The direction of a''_C is from C towards O'.

Method 2.

Take cylinder A as the moving (rotating) frame of reference, and find a''_C directly from the relative-acceleration formula for a rotating frame of reference:

$$a''_C = a_C - \omega_1 \times (\omega_1 \times r'_C) - 2\omega_1 \times v''_C. \tag{14}$$

In (14), the value of v''_C can be found by substituting (10) and (12) into (11):

$$\begin{aligned} v''_C &= v''_0 + \omega' \frac{R}{2} = (\omega_1 + \omega_2) R + 2(\omega_1 + \omega_2) \times \frac{R}{2} \\ &= 2(\omega_1 + \omega_2) R. \end{aligned} \tag{15}$$

Substituting (9) and (15) into (14), we obtain

$$\begin{aligned} a''_C &= \frac{2}{3}R\left[(\omega_1+\omega_2)^2+3\omega_2^2\right]-\omega_1^2\times 2R+2\omega_1\times 2\left(\omega_1+\omega_2\right)R \\ &= \frac{8}{3}\left(\omega_1+\omega_2\right)^2 R. \end{aligned}$$

Solution 1.13. When the direction of the relative velocity $\vec{v}' = \vec{v}_A - \vec{v}_B$ of the two particles A and B is parallel to the line joining the two particles, the vector relation between the velocities $\vec{v}_A$, $\vec{v}_B$ of A and B and $\vec{v}'$ is as shown in the figure. Applying the sine rule to $\triangle AOB$, we obtain

$$\frac{a}{\sin\phi} = \frac{b}{\sin(\theta+\phi)}.$$

The above expression may also be written as

$$b\sin\phi = a\sin(\theta+\phi). \tag{1}$$

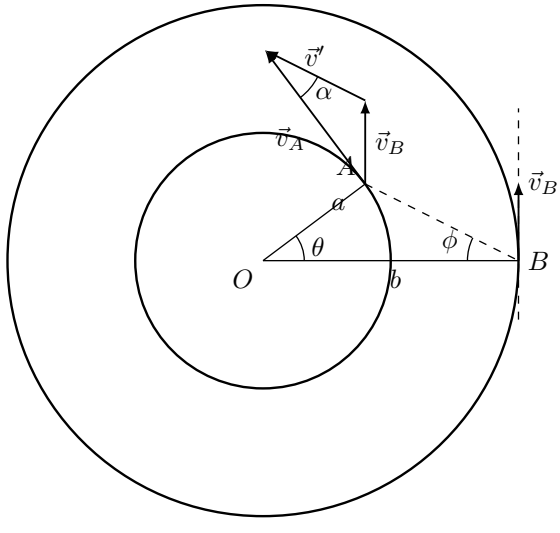


Fig. 1.13

Applying the sine rule to the vector triangle formed by $\vec{v}_A$, $\vec{v}_B$ and $\vec{v}'$, we obtain

$$\frac{v_A}{\sin[\pi-(\theta+\alpha)]} = \frac{v_B}{\sin\alpha}. \tag{2}$$

Since at this instant the vector line of $\vec{v}'$ is parallel to the line joining A and B, we have

$$\pi-(\theta+\alpha) = \pi-\left(\frac{\pi}{2}-\phi\right),$$

hence

$$\alpha = \frac{\pi}{2}-(\theta+\phi). \tag{3}$$

Substituting (3) into (2), we obtain

$$\frac{v_A}{\cos\phi} = \frac{v_B}{\cos(\theta+\phi)}. \tag{4}$$

From the condition given in the problem, that the speeds of the two particles are inversely proportional to the radii, we have

$$\frac{v_A}{v_B} = \frac{b}{a}. \tag{5}$$

Substituting (5) into (4), we obtain

$$a\cos\phi = b\cos(\theta + \phi). \tag{6}$$

Adding the two sides of (1) and (6), we obtain

$$b\sin\phi + a\cos\phi = a\sin(\theta + \phi) + b\cos(\theta + \phi).$$

Multiplying both sides of the above expression by $\dfrac{1}{\sqrt{a^2 + b^2}}$ and setting $\tan\beta = \dfrac{b}{a}$, the above expression becomes

$$\cos(\beta - \phi) = \sin(\theta + \phi + \beta). \tag{7}$$

From (7) we obtain

$$\theta + \phi + \beta = \frac{\pi}{2} - \beta + \phi,$$

that is,

$$\theta = \frac{\pi}{2} - 2\beta.$$

Therefore

$$\cos\theta = \sin 2\beta = \frac{2ab}{a^2 + b^2}.$$

Hence θ is

$$\theta = \arccos\frac{2ab}{a^2 + b^2}.$$

Solution 1.14. Let the distance of the small ring from the end A of the rod be x, as shown in the figure. In the rotating reference frame that moves together with the rod, among the external forces acting on the small ring two have components along the rod: the weight mg and the centrifugal force of inertia $f_c = m\omega^2 x\sin\alpha$. Suppose that when the small ring moves to the position $x = x_0$, the resultant of the forces acting on it along the rod is zero, that is,

$$mg\cos\alpha = m\omega^2 x_0 \sin^2\alpha.$$

Solving, we obtain

$$x_0 = \frac{g\cos\alpha}{\omega^2\sin^2\alpha}. \tag{1}$$

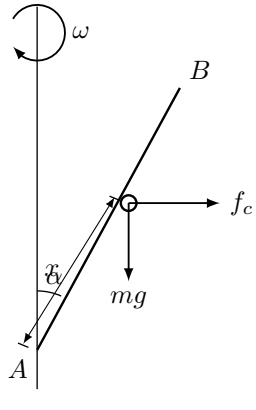


Fig. 1.14

(1) If $x_0 < l$, that is, $\omega > \sqrt{\dfrac{g\cos\alpha}{l\sin^2\alpha}}$, then in this case, when the small ring moves in the region $x < x_0$, the resultant external force along the rod is directed towards A, while when it moves in the region $x > x_0$, the resultant external force along the rod is directed towards B. It is seen that the small ring need only reach the point $x = x_0$, and it can then be accelerated to the end B. In the rotating reference frame, the equation of motion of the small ring is

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = m\omega^2 x\sin^2\alpha - mg\cos\alpha. \tag{2}$$

Equation (2) can be written as

$$\frac{\mathrm{d}v}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = \omega^2\sin^2\alpha\, x - g\cos\alpha,$$

that is,

$$v\,\mathrm{d}v = \left(\omega^2\sin^2\alpha\, x - g\cos\alpha\right)\mathrm{d}x.$$

Integrating both sides between definite limits, we obtain

$$\int_{v_0}^{0} v\,\mathrm{d}v = \int_0^{x_0}\left(\omega^2\sin^2\alpha\, x - g\cos\alpha\right)\mathrm{d}x. \tag{3}$$

Carrying out the integration, we obtain

$$-\frac{1}{2}v_0^2 = \frac{1}{2}\omega^2\sin^2\alpha\, x_0^2 - g\cos\alpha\, x_0. \tag{4}$$

Substituting (1) into (4), we obtain

$$v_0 = \sqrt{2g\cos\alpha\,\frac{g\cos\alpha}{\omega^2\sin^2\alpha} - \omega^2\sin^2\alpha\,\frac{g^2\cos^2\alpha}{\omega^4\sin^4\alpha}} = \frac{g}{\omega}\cot\alpha.$$

Therefore it is only necessary that $v_0 \geq \dfrac{g}{\omega}\cot\alpha$, and the small ring can move to the end B.

(2) If $x_0 \geq l$, that is, $\omega \leq \sqrt{\dfrac{g\cos\alpha}{l\sin^2\alpha}}$, then during the motion of the small ring from A to B the direction of the resultant force along the rod is always towards the end A. For this case the integral in (3) should be written as

$$\int_{v_0}^{0} v\,\mathrm{d}v = \int_0^{l}\left(\omega^2\sin^2\alpha\, x - g\cos\alpha\right)\mathrm{d}x. \tag{5}$$

Integrating (5), we obtain

$$-\frac{1}{2}v_0^2 = \frac{1}{2}\omega^2\sin^2\alpha\, l^2 - g\cos\alpha\, l,$$

$$\therefore \quad v_0 = \sqrt{2gl\cos\alpha - \omega^2\sin^2\alpha\, l^2}.$$

That is, provided $v_0 \geq \sqrt{2gl\cos\alpha - \omega^2\sin^2\alpha\, l^2}$, the small ring can move to the end B.

Comment. In the rotating reference frame, two of the external forces acting on the small ring have components along the rod: the first, the weight, is a constant force; the second, the centrifugal force of inertia, is a variable force. As the position of the small ring rises, this force increases accordingly.

Hence the resultant of the forces acting on the small ring along the rod gradually decreases from a comparatively large value directed towards the end A to zero, and then its direction is towards the end B and it gradually increases. In this problem it is only necessary to analyse whether or not there is a point on the rod at which the direction of this resultant changes, and the upper limit of the integral on the right-hand side of the equals sign in (3) or (5) can then be determined.

Solution 1.15. (1) From the equation of the shape of the bent rod, $y = \dfrac{1}{2a}x^2$, it follows that the values of y of the small ring at $x = 2a$ and at $x = a$ are respectively

$$y(2a) = 2a, \qquad y(a) = \frac{1}{2}a.$$

The small ring is released from rest at $x = 2a$. Let the speed with which it arrives at $x = a$ be v; then from the law of conservation of mechanical energy we obtain

$$\frac{1}{2}mv^2 = mg\left[y(2a) - y(a)\right],$$

$$v = \sqrt{2g\left[y(2a) - y(a)\right]} = \sqrt{3ga}. \tag{1}$$

In what follows we find the radius of curvature of the rod at $x = a$.

Method 1. The curve of the shape of the rod is compared with the parabolic trajectory of a body projected horizontally. That is, a body is projected from point O with initial speed v_0 (along the x direction), and the direction of the gravitational acceleration g of this body is taken to be along the y direction; then the equations of motion of this body are

$$x = v_0 t \tag{2}$$

$$y = \frac{1}{2}gt^2 \tag{3}$$

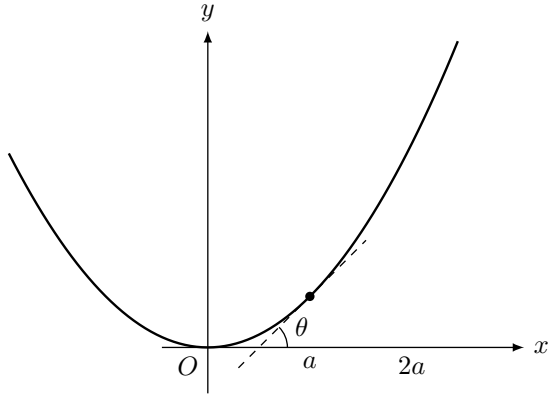


Fig. 1.15(1)

Eliminating t from (2) and (3), we obtain

$$y = \frac{g}{2v_0^2}x^2. \tag{4}$$

Comparing (4) with the equation of the shape of the bent rod, $y = \dfrac{1}{2a}x^2$, we obtain

$$a = \frac{v_0^2}{g}. \tag{5}$$

At $x = a$, $x = a = v_0 t$, so $t = \dfrac{a}{v_0}$; substituting (5) into this gives $t = \dfrac{v_0}{g}$. Hence at this instant

$$v_y = gt = v_0 = v_x.$$

It is seen that here the tangent to the trajectory makes an angle $\theta = \dfrac{\pi}{4}$ with the x-axis, so the normal acceleration here is

$$a_n = g\cos\theta = \frac{\sqrt{2}}{2}g.$$

From this the radius of curvature $\rho(a)$ here can be found:

$$\rho(a) = \frac{v^2}{a_n} = \frac{v_x^2 + v_y^2}{a_n} = \frac{2v_0^2}{\frac{\sqrt{2}}{2}g} = 2\sqrt{2}\,\frac{v_0^2}{g} = 2\sqrt{2}\,a.$$

Method 2. Use the formula $\rho = \left|\dfrac{\left[1 + (y')^2\right]^{3/2}}{y''}\right|$ to solve the problem.

$y' = \dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x}{a}$, $y'' = \dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \dfrac{1}{a}$. Substituting these two expressions into the formula, we obtain

$$\rho = \left|\frac{\left[1 + \left(\frac{x}{a}\right)^2\right]^{3/2}}{\frac{1}{a}}\right|_{x=a} = 2\sqrt{2}\,a.$$

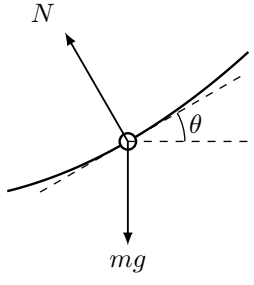


Fig. 1.15(2)

When the small ring moves to $x = a$, the forces acting on it being as shown in Fig. 2, its equation of motion in the normal direction is

$$N - mg\cos\theta = m\frac{v^2}{\rho}. \tag{6}$$

Substituting (1) into (6), we can solve for

$$N = mg\cos\theta + m\frac{v^2}{\rho} = \frac{5}{4}\sqrt{2}\,mg.$$

(2) In the rotating reference frame that rotates together with the rod, the forces acting on the ring, which is at rest relative to the rod, are as shown in Fig. 3. Hence

$$mg\sin\theta = f_c\cos\theta = m\omega_0^2 x\cos\theta. \tag{7}$$

where the slope of the tangent to the track is

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tan\theta = \frac{x}{a}. \tag{8}$$

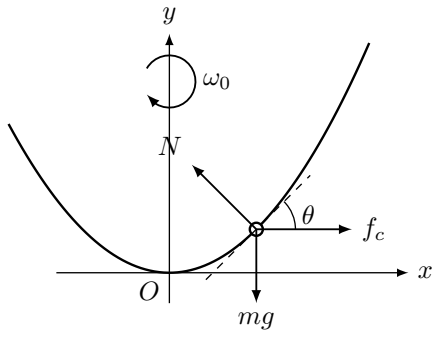


Fig. 1.15(3)

Substituting (8) into (7), we obtain

$$\omega_0 = \sqrt{\frac{g}{a}}.$$

That is, the bent rod need only rotate about the y-axis with angular velocity $\omega_0 = \sqrt{\dfrac{g}{a}}$, and the small ring can then remain at rest relative to the rod at any position on the rod.

(3) If $\omega = \dfrac{\sqrt{2}}{2}\omega_0 = \sqrt{\dfrac{g}{2a}}$, then in the reference frame rotating together with the rod, during the process in which the small ring slides from $x = 2a$ to $x = a$ the work done by gravity is

$$W_g = mg\,[y(2a) - y(a)] = mg\left(2a - \frac{a}{2}\right) = \frac{3}{2}mga.$$

In addition, the work done by the centrifugal force of inertia $f = m\omega^2 x$ can be found by analogy with the elastic force $f = kx$, and is

$$W_{f_c} = \frac{1}{2}m\omega^2\left[a^2 - (2a)^2\right] = -\frac{3}{2}m\omega^2 a^2.$$

From the work-energy theorem we obtain

$$\frac{1}{2}mv'^2 = W_g + W_{f_c} = \frac{3}{2}mga - \frac{3}{2}m\omega^2 a^2,$$

that is,

$$v'^2 = 3ga - 3\,\frac{g}{2a}a^2 = \frac{3}{2}ga. \tag{9}$$

At $x = a$, besides the weight mg, the centrifugal force of inertia $f_c = m\omega^2 x$ and the supporting force N, the small ring, since it has a velocity v' along the rod, is also acted upon by the Coriolis force $f_C = 2mv'\omega\cos\theta$. The direction of this force is that of $\vec{v}' \times \vec{\omega}$, that is, it is perpendicular to the plane of the page and points inwards (it is not marked in Fig. 4). The supporting force N exerted by the rod on the small ring should have components in two directions, that is,

$$\vec{N} = \vec{N}_1 + \vec{N}_2. \tag{10}$$

Here N_1 lies in the plane of the page, and its value is determined by the equation of motion of the small ring in the normal direction within the plane of the page,

$$N_1 - mg\cos\theta - m\omega^2 a\sin\theta = m\frac{v'^2}{\rho}. \tag{11}$$

Substituting (9) and $\rho = 2\sqrt{2}\,a$ into (11), we obtain

$$\begin{aligned} N_1 &= \frac{\sqrt{2}}{2}mg + m\times\frac{g}{2a}\times a\times\frac{\sqrt{2}}{2} + m\frac{\frac{3}{2}ga}{2\sqrt{2}\,a} \\ &= \frac{9}{8}\sqrt{2}\,mg. \end{aligned} \tag{12}$$

And N_2 in (10) is equal in magnitude and opposite in direction to the Coriolis force, that is, its direction is perpendicular to the plane of the page and points outwards, and its value is

$$\begin{aligned} N_2 &= 2mv'\omega\cos\theta \\ &= 2m\sqrt{\frac{3}{2}ga}\times\sqrt{\frac{g}{2a}}\times\frac{\sqrt{2}}{2} = \frac{\sqrt{6}}{2}mg. \end{aligned} \tag{13}$$

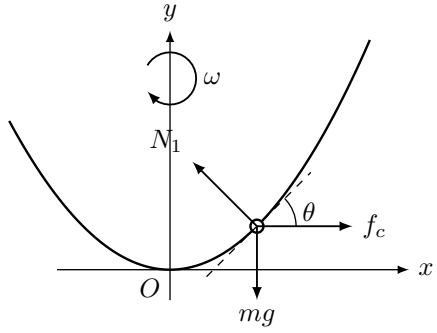


Fig. 1.15(4)

From (12) and (13) the value of the supporting force N of the rod on the small ring can be obtained; this value is also the value of the force N' exerted by the small ring on the rod,

$$N' = N = \sqrt{N_1^2 + N_2^2} = \frac{\sqrt{258}}{8}mg \approx 2.01mg.$$

Comment. In the rotating reference frame, apart from the weight, the centrifugal force of inertia and the supporting force, the small ring, since it has a velocity v', is also acted upon by the Coriolis force, and this force can be expressed as $\vec{f}_C = 2m\vec{v}'\times\vec{\omega}$. The Coriolis force is very easily overlooked, and it should be noted that the direction of this force is perpendicular both to $\vec{v}'$ and to $\vec{\omega}$, that is, it is perpendicular to the plane (the plane of the page) formed by these two vectors. Since the small ring has no motion in this direction, the Coriolis force is balanced by the component N_2 of the supporting force, while of the supporting force only the value of the other component N_1 within the plane of the page has to be found. According to the Pythagorean theorem, the value of the resultant N of N_1 and N_2 can then be found.

Solution 1.16. (1) On the small disc take an arbitrary diameter, and on this diameter take two small segments that are symmetric with respect to the centre O' of the disc; the centres of these two small segments are A_1 and A_2, as shown in Fig. 2.

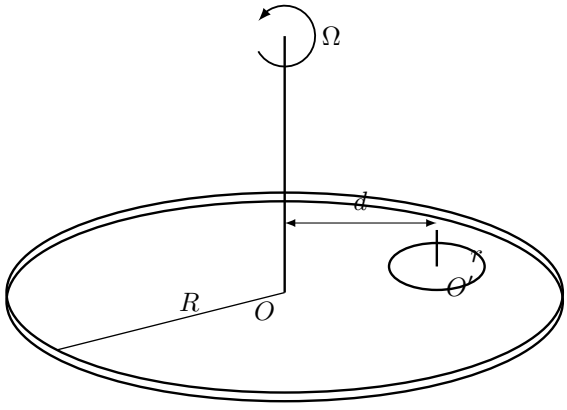


Fig. 1.16(1)

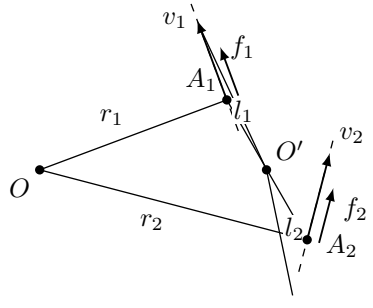


Fig. 1.16(2)

When the small disc has just been put down, the velocities of the two points on the large disc that are in contact with the points A_1 and A_2 of the small disc are respectively v_1 and v_2, their directions being respectively perpendicular to r_1 and r_2. At this time the directions of the friction forces f_1 and f_2 acting at the points A_1 and A_2 are respectively along the directions of v_1 and v_2. From Fig. 2 it is seen that the lever arm l_1 of f_1 with respect to O' is smaller than the lever arm l_2 of f_2 with respect to O', so the resultant torque of f_1 and f_2 with respect to O' makes the small disc rotate with acceleration about O' in the same sense of rotation as the large disc.

Suppose that at some instant the angular velocity of the rotation of the small disc is $\omega(< \Omega)$; the velocities of the points A_1 and A_2 on the small disc at this instant are respectively $u_1 = \vec{\omega} \times \vec{r}_1'$ and $u_2 = \vec{\omega} \times \vec{r}_2'$, while the velocities of the two points on the large disc that are in contact with A_1 and A_2 are respectively $v_1 = \vec{\Omega} \times \vec{r}_1$ and $v_2 = \vec{\Omega} \times \vec{r}_2$. Hence the relative velocities at the two points of contact are respectively

$$v_1' = v_1 - u_1, \qquad v_2' = v_2 - u_2.$$

Since $v_1 - v_2 = \vec{\Omega} \times (\vec{r}_1 - \vec{r}_2)$, while $\vec{r}_1 - \vec{r}_2 = \vec{r}_1' - \vec{r}_2'$, that is, its direction is the direction of the line joining A_1 and A_2. From this it is known that the direction of $v_1 - v_2$ is perpendicular to the line joining A_1 and A_2, and since the directions of u_1 and u_2 are respectively perpendicular to r_1' and r_2', that is, both are perpendicular to the line joining A_1 and A_2, the directions of u_1 and u_2 lie along the vector line of $v_1 - v_2$.

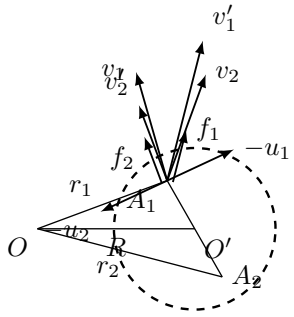


Fig. 1.16(3)

The vector diagram of $v_2' = v_2 - u_2$ at the point A_2 is drawn over at the point A_1 for convenience of comparison; then the directions of the friction forces f_1 and f_2 are respectively along the directions of their own relative velocities v_1' and v_2', as shown in Fig. 3. Since

$$u_1 - u_2 = \vec{\omega} \times (\vec{r}_1' - \vec{r}_2') = \vec{\omega} \times (\vec{r}_1 - \vec{r}_2),$$

$$v_1 - v_2 = \vec{\Omega} \times (\vec{r}_1 - \vec{r}_2),$$

and $\omega < \Omega$, from the above two expressions it is known that

$$|u_1 - u_2| < |v_1 - v_2|.$$

Therefore there is an angle between f_1 and f_2, that is, the resultant torque of f_1 and f_2 with respect to the point O' is still not zero, so that the small disc continues to rotate with acceleration.

When $\omega = \Omega$, we have

$$|u_1 - u_2| = |v_1 - v_2|.$$

At this time the directions of f_1 and f_2 are both along the direction f, since

$$v_1' = v_1 - u_1 = \vec{\Omega} \times (\vec{r}_1 - \vec{r}_1') = \vec{\Omega} \times \vec{R},$$

$$v_2' = v_2 - u_2 = \vec{\Omega} \times (\vec{r}_2 - \vec{r}_2') = \vec{\Omega} \times \vec{R}.$$

It is seen that $f_1 = f_2 = f$, and the direction is perpendicular to $\vec{R}$; therefore the resultant torque of f_1 and f_2 with respect to O' is zero. The rotation of the small disc then becomes steady, and the steady angular velocity is

$$\omega = \Omega.$$

(2) When $\omega = \Omega$, the directions of the friction forces acting on any two symmetric small segments on any diameter of the small disc are all perpendicular to $\vec{R}$, that is, perpendicular to the direction of the line OO'; hence the total friction force acting on the small disc is $f = \mu mg$, and its equivalent point of application is O'. From Newton's second law it is known that the torque of the friction force acting on the large disc with respect to the point O is

$$M' = \mu mgd.$$

Therefore, in order to maintain the rotational speed of the large disc unchanged, the external torque that must be applied is $M = \mu mgd$.

Comment. In the solution of this problem, the vector diagram of $v_2' = v_2 - u_2$ at the point A_2 is drawn at the point A_1 and combined with the vector diagram of $v_1' = v_1 - u_1$ at the point A_1 into

a single geometrical figure. From this it can be seen very clearly that, in the case $\omega < \Omega$, because $|u_1 - u_2| < |v_1 - v_2|$ the directions of the friction forces f_1 and f_2 are different, and the correct conclusion is thereby obtained that under the action of this friction torque the small disc continues to rotate with acceleration until $\omega = \Omega$.

Solution 1.17. Since the weights acting on the two balls A and B pass through their respective centres, while the surfaces of the three balls are all smooth, the force N exerted by ball C on balls A and B also passes through the centres of the two balls. Therefore, in equilibrium the tension T of the string also passes through the centres of the two balls.

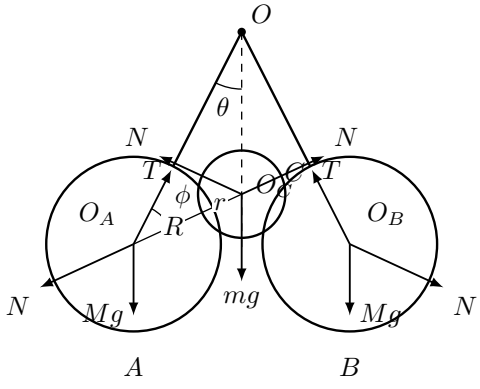


Fig. 1.17

In the figure it is known that $OO_A = 3R$, $O_AO_C = \dfrac{3}{2}R$; from the sine rule we obtain

$$\frac{\frac{3}{2}R}{\sin\theta} = \frac{3R}{\sin(\theta+\phi)},$$

that is,

$$\sin(\theta+\phi) = 2\sin\theta. \tag{1}$$

From the equilibrium of the forces acting on ball C we obtain

$$2N\cos(\theta+\phi) = mg = nMg. \tag{2}$$

From the equilibrium of the forces acting on ball A (or B) in the horizontal direction and in the vertical direction, the following two expressions are obtained respectively:

$$T\sin\theta = N\sin(\theta+\phi) \tag{3}$$

$$T\cos\theta - N\cos(\theta+\phi) - Mg = 0 \tag{4}$$

From (1), (2), (3) and (4) we can solve and obtain

$$\sin\theta = \frac{1}{4}\sqrt{\frac{4+4n-3n^2}{1+n}}. \tag{5}$$

When balls A and B are in contact with each other, θ has a minimum value θ_0; from the geometrical relation we obtain

$$\sin\theta \geq \sin\theta_0 = \frac{1}{3}. \tag{6}$$

From (5) and (6) we obtain

$$\frac{1}{4}\sqrt{\frac{4+4n-3n^2}{1+n}} \geq \frac{1}{3}.$$

Solving, we obtain

$$n \leq \frac{10+8\sqrt{10}}{27} \approx 1.3.$$

Therefore the following analysis can be made:

(1) If $n > 1.3$, the system cannot be in equilibrium.

(2) If $n = 1.3$, then when the system is in equilibrium $\theta = \theta_0 = \arcsin\frac{1}{3}$, that is, balls A and B lean against each other, but there is no interaction between the two balls.

(3) If $n < 1.3$, the system has two possible states of equilibrium:

(i) $\theta = \theta_0 = \arcsin\frac{1}{3}$, that is, balls A and B are pressed tightly against each other, and there is an interaction between the two balls; the magnitude of this force F is related to n: the larger n is, the smaller F is, and when $n = 1.3$, $F = 0$.

(ii) $\theta = \arcsin\left[\frac{1}{4}\sqrt{\frac{4+4n-3n^2}{1+n}}\right]$, that is, balls A and B are separated; the smaller n is, the farther apart they are. If $n \to 0$, then $\theta \to \arcsin\frac{1}{2}$.

Solution 1.18. Let the radius of the large ring be R. The nail is located at the point P, and the small ball is fixed at the point A of the large ring, as shown in the figure. The large ring is acted upon by the supporting force N and the friction force f of the nail. The direction of the resultant F of these two forces must be vertically upward. The point B is the intersection of the vertical line through P with OA. Let $OB = r$.

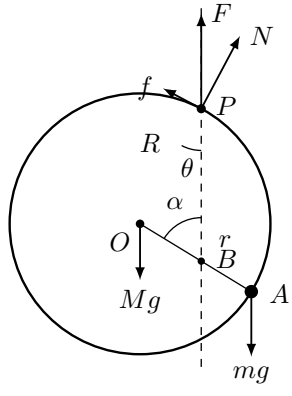


Fig. 1.18

For $\triangle OPB$, from the sine rule we obtain

$$\frac{R}{\sin\alpha} = \frac{r}{\sin\theta},$$

that is,

$$\sin\theta = \frac{r}{R}\sin\alpha.$$

It is seen that when $\alpha = \dfrac{\pi}{2}$, the value of θ is a maximum, and its maximum value θ_m is

$$\theta_m = \arcsin \frac{r}{R}. \tag{1}$$

When the large ring is in equilibrium, the moments of the external forces acting on it with respect to the point P balance, and we obtain

$$Mgr \sin \alpha = mg(R - r) \sin \alpha,$$

that is,

$$\frac{r}{R} = \frac{m}{M + m}. \tag{2}$$

Substituting (2) into (1), we obtain

$$\theta_m = \arcsin \frac{m}{M + m}. \tag{3}$$

Since the large ring does not slip relative to the nail, the angle θ must be smaller than or equal to the angle of friction $\phi_{\max}(= \arctan \mu)$, that is,

$$\theta \leq \phi_{\max}. \tag{4}$$

Expression (3) represents the maximum value of θ for equilibrium when different points of the large ring are hung on the nail; as long as this angle is smaller than or equal to the angle of friction, the equilibrium of the large ring hung on the nail at any position can be ensured. In addition, from $\tan \phi_{\max} = \mu$ we obtain $\sin \phi_{\max} = \dfrac{\mu}{\sqrt{\mu^2 + 1}}$; hence we obtain

$$\sin \theta_m = \frac{m}{M + m} \leq \sin \phi_m = \frac{\mu}{\sqrt{\mu^2 + 1}}.$$

From the above expression we can solve and obtain

$$\mu \geq \frac{m}{\sqrt{M(M + 2m)}}.$$

Solution 1.19. (1) Let α be the maximum value α_m at which the system can be in equilibrium. At this time the tendency of relative motion between ball 1 and the inclined plane is greatest, that is, the friction force acting on ball 1 is the maximum static friction force, $f_1 = \mu_1 mg \cos \theta$, while the tendency of relative motion between ball 2 and the inclined plane has not reached its maximum, that is, f_2 is an ordinary static friction force. Since the direction of the motion of ball 1 relative to the inclined plane is perpendicular to the string, it is seen that f_1 is also perpendicular to the string. Let the tension in the string be T, and let the component of the weight of each of the two balls along the inclined plane be F, that is, $F = mg \sin \theta$. Fig. 2 shows the vector relations of the external forces acting on the two balls; drawing the two vector diagrams together will be helpful for comparing the magnitudes and directions of the various vectors. From the geometrical relations shown in the figure we obtain

$$T = F \cos \alpha_m = mg \sin \theta \cos \alpha_m \tag{1}$$

$$f_1 = \mu_1 mg \cos \theta = F \sin \alpha_m = mg \sin \theta \sin \alpha_m \tag{2}$$

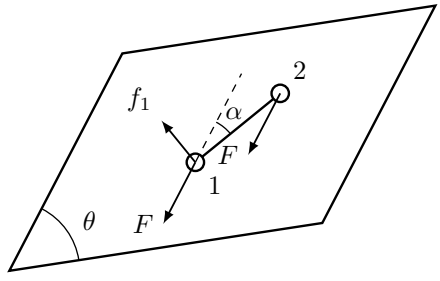


Fig. 1.19(1)

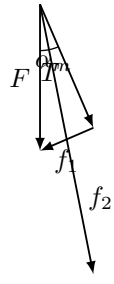


Fig. 1.19(2)

From (2) we can solve and obtain

$$\sin\alpha_m = \mu_1 \cot\theta = \sqrt{\frac{\mu_1}{\mu_2}}.$$

At the same time, we also have

$$\begin{aligned} f_2^2 &= F^2 + T^2 + 2FT\cos\alpha_m \\ &= (mg\sin\theta)^2 + (mg\sin\theta\cos\alpha_m)^2 + 2(mg\sin\theta)(mg\sin\theta\cos\alpha_m)\cos\alpha_m \\ &= m^2g^2\sin^2\theta\left(1+3\cos^2\alpha_m\right). \end{aligned} \tag{3}$$

Since f_2 is an ordinary static friction force, that is, $f_2 \le \mu_2 mg\cos\theta$; substituting this relation into (3), we obtain

$$f_2^2 = m^2g^2\sin^2\theta\left(1+3\cos^2\alpha_m\right) \le \left(\mu_2 mg\cos\theta\right)^2,$$

that is,

$$\tan^2\theta\left(1+3\cos^2\alpha_m\right) \le \mu_2^2. \tag{4}$$

Substituting $\tan\theta = \sqrt{\mu_1\mu_2}$ and $\cos\alpha_m = \sqrt{\dfrac{\mu_2-\mu_1}{\mu_2}}$ into (4), we obtain

$$\mu_1\mu_2\left[1+\frac{3(\mu_2-\mu_1)}{\mu_2}\right] \le \mu_2^2. \tag{5}$$

Let $x = \mu_1/\mu_2$; then (5) becomes

$$3x^2 - 4x + 1 \ge 0. \tag{6}$$

Solving this inequality, we obtain two sets of solutions:

$x_1 \geq 1$, which does not agree with the given condition $\mu_1 < \mu_2$, and is discarded;

$$x_2 \leq \frac{1}{3}, \quad \text{that is, } \mu_2 \geq 3\mu_1.$$

From the above analysis it is known that if $\mu_2 \geq 3\mu_1$, then the maximum value α_m of α satisfies

$$\sin\alpha_m = \sqrt{\frac{\mu_1}{\mu_2}}.$$

(2) If $\mu_1 < \mu_2 < 3\mu_1$, then when α takes the maximum value α_m at which the system can be in equilibrium, the tendency of relative motion between ball 2 and the inclined plane reaches its maximum, that is, the friction force acting on ball 2 is the maximum static friction force, so that $f_2 = \mu_2 mg\cos\theta$, while f_1 is an ordinary static friction force, that is, $f_1 \leq \mu_1 mg\cos\theta$. In this case the vector relations of the external forces acting on the two balls are as shown in Fig. 3. From the geometrical relations shown in the figure we obtain

$$f_1^2 = T^2 + (mg\sin\theta)^2 - 2Tmg\sin\theta\cos\alpha_m \leq (\mu_1 mg\cos\theta)^2 \tag{7}$$

$$f_2^2 = T^2 + (mg\sin\theta)^2 + 2Tmg\sin\theta\cos\alpha_m = (\mu_2 mg\cos\theta)^2 \tag{8}$$

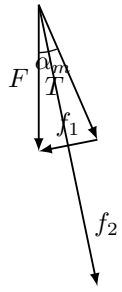


Fig. 1.19(3)

Adding the two sides of (7) and (8), we obtain

$$T \leq \sqrt{\left(\frac{\mu_1^2+\mu_2^2}{2}\right)(mg\cos\theta)^2 - (mg\sin\theta)^2}. \tag{9}$$

Subtracting the two sides of (7) and (8), we obtain

$$4Tmg\sin\theta\cos\alpha_m \geq (\mu_2^2 - \mu_1^2)(mg\cos\theta)^2. \tag{10}$$

Substituting (9) into (10), we obtain

$$\cos\alpha_m \geq \frac{\sqrt{2}(\mu_1+\mu_2)}{4\sqrt{\mu_1\mu_2}}.$$

Comment. The given condition of this problem is $\mu_2 > \mu_1$, and for this reason it is very easy to conclude that, when α takes the value α_m, it is the friction force f_1 acting on ball 1 that reaches the maximum static friction force while the friction force f_2 acting on ball 2 does not reach the maximum static friction force; such a conclusion would lead to the omission of the discussion of case (2) in the above

analysis. From the above analysis it is known that when $\mu_2 \geq 3\mu_1$ it is indeed true that when $\alpha = \alpha_m$, $f_1 = \mu_1 mg\cos\theta$ and $f_2 \leq \mu_2 mg\cos\theta$; but if $\mu_1 < \mu_2 < 3\mu_1$, then $f_2 = \mu_2 mg\cos\theta$ while $f_1 \leq \mu_1 mg\cos\theta$. By calculation the value of α_m or its range of values can be found in the two cases separately.

Solution 1.20. *Method 1.* Let the components along the horizontal and vertical directions of the force acting on the end A_1 of rod O_1A_1 be F_{1x} and F_{1y} respectively, their directions being as shown in Fig. 1. From symmetry it is known that F_{1y} should be equal to half of the weight of the rods below the hinges A_1 and B_1. Since the mass of the rod between any two hinges is m, we have

$$F_{1y} = 5mg. \tag{1}$$

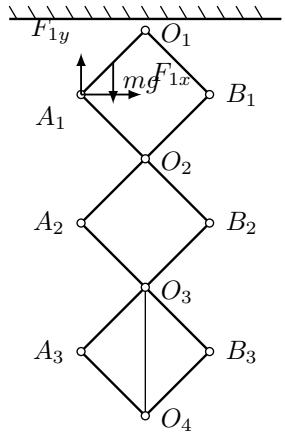


Fig. 1.20(1)

From the equilibrium of the moments of the external forces acting on rod O_1A_1 with respect to the point O_1, we obtain

$$F_{1x}l\sin\theta = F_{1y}l\cos\theta + mg\,\frac{l}{2}\cos\theta. \tag{2}$$

Substituting $\theta = 45^\circ$ and (1) into (2), we obtain

$$F_{1x} = \frac{11}{2}mg. \tag{3}$$

The external forces acting on rod A_1B_2 that have moments with respect to the point O_2 are as shown in Fig. 2, where F_{2x} and F_{2y} are the components in the horizontal and vertical directions of the force acting on the end B_2. Similarly, we have

$$F_{2y} = 3mg. \tag{4}$$

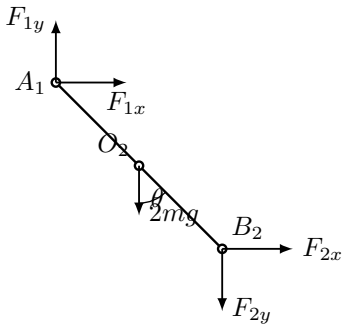


Fig. 1.20(2)

From the equilibrium of the moments of the external forces acting on rod A_1B_2 with respect to the point O_2, we obtain

$$F_{1y}l\cos\theta + F_{1x}l\sin\theta + F_{2y}l\cos\theta = F_{2x}l\sin\theta. \tag{5}$$

Substituting (1), (3) and (4) into (5), we can solve and obtain

$$F_{2x} = \frac{27}{2}mg. \tag{6}$$

Taking rod B_2A_3 and rod A_3O_4 as a single whole, the external forces acting on it that have moments with respect to the point O_3 are as shown in Fig. 3. In the figure, N is the external force acting on the end O_4; from the symmetry of the whole hinge structure it is known that N must be horizontal. Hence, from the equilibrium of the moments of this whole with respect to the point O_3, we obtain

$$F_{2x}l\sin\theta + F_{2y}l\cos\theta + mg\,\frac{l}{2}\cos\theta = N\sqrt{2}\,l. \tag{7}$$

Substituting (4) and (6) into (7), we can solve and obtain

$$N = \frac{17}{2}mg. \tag{8}$$

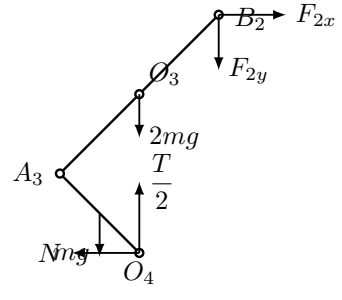


Fig. 1.20(3)

From the equilibrium of the moments of rod A_3O_4 with respect to the point A_3, we obtain

$$mg\,\frac{l}{2}\cos\theta + Nl\sin\theta = \frac{T}{2}l\cos\theta. \tag{9}$$

In (9), T is the tension in the string; again because of the symmetry of the whole hinge structure, the force acting on the end O_4 of rod A_3O_4 should be $\frac{T}{2}$. Substituting (8) into (9), we can solve and obtain

$$T = 18mg.$$

Method 2. This problem can also be solved by means of the principle of virtual work.

Let the point O_2 undergo a very small virtual displacement δx upwards; then the points O_3 and O_4 undergo the corresponding virtual displacements $2\delta x$ and $3\delta x$, and the virtual work done by the tension in the string is

$$\delta W = T \cdot 3\delta x - T \cdot 2\delta x = T\delta x. \tag{10}$$

At this time the gravitational potential energy of the whole hinge structure increases by δE_p:

$$\begin{aligned} \delta E_p &= 4mg\,\frac{\delta x}{2} + 4mg\left(\delta x + \frac{\delta x}{2}\right) + 4mg\left(2\delta x + \frac{\delta x}{2}\right) \\ &= 18mg\,\delta x. \end{aligned} \tag{11}$$

According to the principle of virtual work, $\delta W = \delta E_p$; substituting (10) and (11) into it, we immediately obtain

$$T = 18mg.$$

Comment. The principle of virtual work used in Method 2 of this problem may be described roughly as follows.

When a mechanical system is in equilibrium, then for any particle P_i of the system the forces acting on it are the active force $\vec{F}_i$ and the passive force (that is, the constraint reaction) $\vec{N}_i$, and we have

$$\vec{F}_i + \vec{N}_i = 0 \qquad (i = 1, 2, \cdots, N).$$

If the particle P_i undergoes a virtual displacement (that is, any displacement consistent with the constraints) $\delta\vec{r}_i$, then

$$\vec{F}_i \cdot \delta\vec{r}_i + \vec{N}_i \cdot \delta\vec{r}_i = 0 \qquad (i = 1, 2, \cdots, N).$$

Adding the equations for all the particles of the system, we obtain

$$\sum_{i=1}^{N} \vec{F}_i \cdot \delta\vec{r}_i + \sum_{i=1}^{N} \vec{N}_i \cdot \delta\vec{r}_i = 0.$$

For a system with ideal constraints (that is, $\sum\limits_{i=1}^{N} \vec{N}_i \cdot \delta\vec{r}_i = 0$) in equilibrium, the equilibrium condition is

$$\delta W = \sum \vec{F}_i \cdot \delta\vec{r}_i = 0,$$

that is, the equilibrium condition of a mechanical system with ideal constraints is that, for any virtual displacement, the sum of the virtual work done by all the active forces is equal to zero.

For the mechanical system of this problem — the hinge structure — when it is in equilibrium, the active forces acting on it are gravity and the tension of the string, and the total virtual work done by them during a virtual displacement is zero. Here the virtual work done by gravity is equal to the change of its gravitational potential energy. Hence the conclusion is obtained that the virtual work done by the tension of the string is equal to the change of the gravitational potential energy.

Solution 1.21. When the sphere is in equilibrium, the external forces acting on it are as shown in the figure. From the equilibrium of the moments of these external forces about the centre C of the sphere we obtain

$$f_1 = f_2 = f. \tag{1}$$

From the vanishing of the resultant of the external forces in the horizontal and in the vertical direction we obtain respectively

$$N_1 - N_2 \sin\theta - f_2 \cos\theta = 0, \tag{2}$$

$$N_2 \cos\theta - mg - f_1 - f_2 \sin\theta = 0. \tag{3}$$

Solving (1), (2) and (3) simultaneously, we obtain

$$N_1 = \frac{1+\sin\theta}{\cos\theta} f + mg\tan\theta. \tag{4}$$

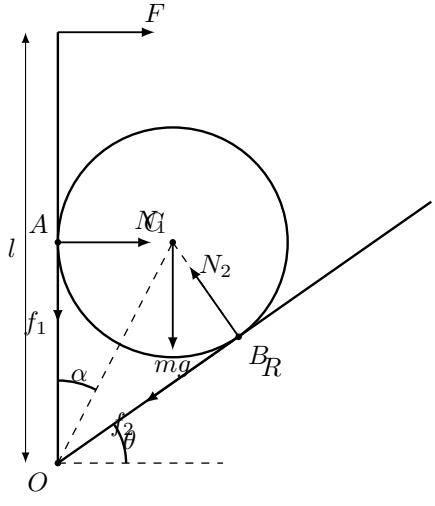


Fig. 1.21

Also, from (1) and (3) we obtain

$$N_2 = \frac{1+\sin\theta}{\cos\theta} f + \frac{mg}{\cos\theta}. \tag{5}$$

(i) If $N_1\mu_1 \geq N_2\mu_2$: when F increases until the equilibrium of the system is just broken, relative sliding begins to occur at the contact point B between the sphere and the inclined plane, while at the contact point A between the sphere and the board the maximum tendency to relative sliding has not yet been reached; that is, at this moment $f = N_2\mu_2$. Substituting this relation into (4) and (5), we obtain

$$N_2 = \frac{mg}{\cos\theta - \mu_2(1+\sin\theta)}, \tag{6}$$

$$N_1 = \frac{(\sin\theta + \mu_2\cos\theta)mg}{\cos\theta - \mu_2(1+\sin\theta)}. \tag{7}$$

Substituting (6) and (7) into $N_1\mu_1 \geq N_2\mu_2$, we obtain

$$\frac{(\sin\theta + \mu_2\cos\theta)mg\mu_1}{\cos\theta - \mu_2(1+\sin\theta)} \geq \frac{mg\mu_2}{\cos\theta - \mu_2(1+\sin\theta)},$$

from which the condition for this case follows:

$$\mu_1 \geq \frac{\mu_2}{\sin\theta + \mu_2\cos\theta}. \tag{8}$$

(ii) If $N_1\mu_1 < N_2\mu_2$, then when the equilibrium of the system is just broken it is at the contact point A between the sphere and the board that relative sliding occurs, while at the contact point B the maximum tendency to relative sliding has not yet been reached; at this moment $f = N_1\mu_1$. Substituting this relation into (4) and (5), we obtain

$$N_1 = \frac{mg\sin\theta}{\cos\theta - \mu_1(1+\sin\theta)}, \tag{9}$$

$$N_2 = \frac{1-\mu_1\cos\theta}{\cos\theta - \mu_1(1+\sin\theta)}\, mg. \tag{10}$$

Substituting (9) and (10) into $N_1\mu_1 < N_2\mu_2$, we obtain

$$\frac{mg\sin\theta\,\mu_1}{\cos\theta - \mu_1(1+\sin\theta)} < \frac{(1-\mu_1\cos\theta)\mu_2\, mg}{\cos\theta - \mu_1(1+\sin\theta)},$$

from which the condition for this case follows:

$$\mu_1 < \frac{\mu_2}{\sin\theta + \mu_2\cos\theta}. \tag{11}$$

From the equilibrium of the moments of the external forces acting on the board about point O we obtain

$$Fl = N_1 \cdot AO. \tag{12}$$

From the geometrical relations shown in the figure we obtain

$$AO = R\cot\alpha = R\cot\left(\frac{\pi}{4} - \frac{\theta}{2}\right). \tag{13}$$

Substituting (13) into (12) gives

$$F = \frac{R}{l}\cot\left(\frac{\pi}{4} - \frac{\theta}{2}\right) N_1. \tag{14}$$

It is only necessary to substitute (7) and (9) respectively into (14) to obtain the value of F in the two cases. The results may therefore be summarized as follows.

(i) If $\mu_1 \geq \dfrac{\mu_2}{\sin\theta + \mu_2\cos\theta}$, then

$$F = \frac{R(\sin\theta + \mu_2\cos\theta)}{l\,[\cos\theta - \mu_2(1+\sin\theta)]}\, mg\cot\left(\frac{\pi}{4} - \frac{\theta}{2}\right).$$

(ii) If $\mu_1 < \dfrac{\mu_2}{\sin\theta + \mu_2\cos\theta}$, then

$$F = \frac{R}{l}\cdot\frac{\sin\theta}{\cos\theta - \mu_1(1+\sin\theta)}\, mg\cot\left(\frac{\pi}{4} - \frac{\theta}{2}\right).$$

Comment. In this problem the friction coefficients μ_1 and μ_2 at the two contact points A and B are different, so when F increases until the equilibrium of the system is just broken, which of the contact points begins to slide relatively is determined by the relative magnitudes of μ_1 and μ_2. Therefore two cases must be discussed in the solution, and the condition and the corresponding value of N_1 must be found separately for each case.

Solution 1.22. (1) The upper ends of the three rods lean against one another, as shown in Fig. 1. By symmetry, the resultant F of the forces exerted by the other two rods on the upper end of any one rod (for instance rod OA) must be in the horizontal direction, as shown in Fig. 2. In Fig. 1, D is the midpoint of BC, and

$$AD = OD = \frac{\sqrt{3}}{2}l,$$

$$\cos\alpha = \frac{\sqrt{3}}{3}. \tag{1}$$

From the equilibrium of the moments of the external forces acting on rod OA about point A we obtain

$$Fl\sin\alpha - mg\,\frac{l}{2}\cos\alpha = 0. \tag{2}$$

Substituting (1) into (2), we obtain

$$F = \frac{\sqrt{2}}{4}mg.$$

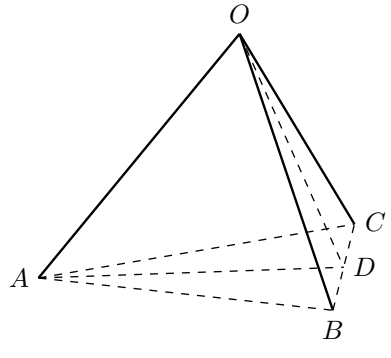


Fig. 1.22(a)

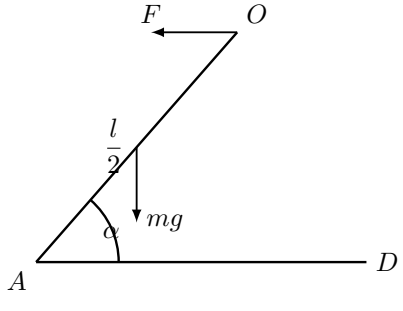


Fig. 1.22(b)

(2) After a small ball of mass m has been fixed at the midpoint of rod OA, the forces on the three rods all change, and the configuration is no longer symmetric; however, rods OB and OC are in the same situation. The force at the upper end of these two rods may be regarded as, besides the original force F, an additional force T_B and T_C respectively, with $T_B = T_C$. Since these two rods are still in equilibrium, T_B and T_C must be along the directions of their respective rods; hence the resultant of these two forces is along OD, and its reaction T acts on the upper end of rod OA, as shown in Fig. 3. From the vanishing of the total moment of T and of the weight of the small ball about point A, we obtain

$$Tl\sin\alpha - mg\,\frac{l}{2}\cos\alpha = 0.$$

Solving, we obtain

$$T = \frac{\sqrt{2}}{4}mg.$$

From the vector relation of F and T shown in Fig. 3 we obtain the force F_A acting on the upper end of rod OA as

$$\begin{aligned} F_A &= 2F\cos\left(\frac{\pi}{2} - \alpha\right) = 2F\sin\alpha = 2 \times \frac{\sqrt{2}}{4}mg\sqrt{\frac{2}{3}} \\ &= \frac{\sqrt{3}}{3}mg. \end{aligned}$$

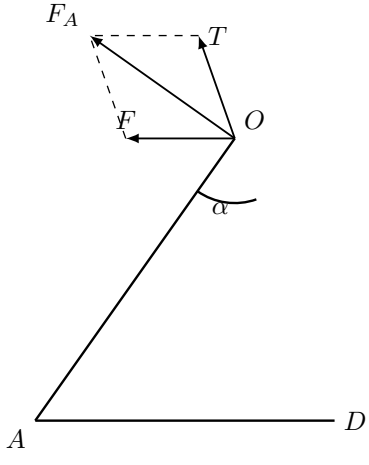


Fig. 1.22(c)

(3) From the vanishing of the resultant of the forces acting on rod OA in the vertical and in the horizontal direction we obtain respectively

$$N_A = 2mg - T\sin(\pi - 2\alpha), \tag{1}$$
$$f_A = F + T\cos(\pi - 2\alpha). \tag{2}$$

Substituting $T = F = \frac{\sqrt{2}}{4}mg$ into (1) and (2), we obtain

$$N_A = \frac{5}{3}mg, \tag{3}$$

$$f_A = \frac{\sqrt{2}}{3}mg. \tag{4}$$

Substituting (3) and (4) into $f_A \leq N_A\mu_A$, we obtain

$$\mu_A \geq \frac{f_A}{N_A} = \frac{\sqrt{2}}{5}. \tag{5}$$

The forces acting on rod OB are shown in Fig. 4. From the vanishing of the resultant of the forces on this rod in the vertical and in the horizontal direction we obtain respectively

$$N_B = mg + T_B \sin\alpha, \tag{6}$$

$$f_B = F + T_B \cos\alpha. \tag{7}$$

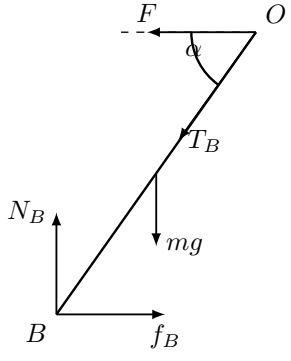


Fig. 1.22(d)

From the vector relations shown in Fig. 5, the relation between T_B, T_C and T is

$$T = 2T_B \cos 30°,$$

that is,

$$T_B = \frac{1}{\sqrt{3}}T = \frac{1}{\sqrt{3}} \times \frac{\sqrt{2}}{4}mg = \frac{\sqrt{6}}{12}mg. \tag{8}$$

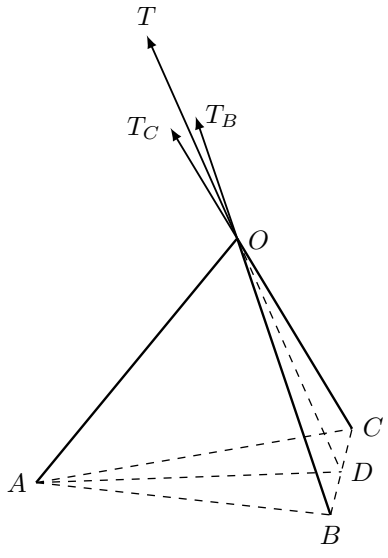


Fig. 1.22(e)

Substituting (8) into (6) and (7), we obtain

$$N_B = mg + \frac{\sqrt{6}}{12}mg \times \sqrt{\frac{2}{3}} = \frac{7}{6}mg, \tag{9}$$

$$f_B = \frac{\sqrt{2}}{4}mg + \frac{\sqrt{6}}{12}mg \times \frac{1}{\sqrt{3}} = \frac{\sqrt{2}}{3}mg. \tag{10}$$

Substituting (9) and (10) into $f_B \le \mu_B N_B$, we obtain

$$\mu_B \ge \frac{f_B}{N_B} = \frac{2\sqrt{2}}{7}. \tag{11}$$

Since rods B and C are in exactly the same situation as regards the forces, the minimum coefficient of friction required for the equilibrium of rod C is equal to that of rod B. Comparing (5) with (11), we obtain that the coefficient of friction between the rods and the ground must satisfy

$$\mu \ge \frac{2\sqrt{2}}{7}.$$

Comment. In the solution of part (3) it is very easy to omit the discussion and calculation of the minimum coefficient of friction required for the equilibrium of rod OB or OC, and to suppose mistakenly that, since the small ball is fixed at the midpoint of rod OA, the system is certainly in equilibrium as long as rod OA can remain in equilibrium. That leads to the erroneous conclusion that it is sufficient to satisfy $\mu \ge \frac{\sqrt{2}}{5}$.

Solution 1.23. Through the centre O of the ring establish a rectangular coordinate system: the x-axis is along the direction PA, the yOz plane is the plane of the ring, the y-axis is along the horizontal direction, and the z-axis is along the vertical direction, as shown in the figure.

Since the direction of the supporting force $\vec{N}$ exerted by the ring on the rod is perpendicular both to the rod and to the ring, its component in the plane of the ring must be along the radial

direction; denote it by N_1. Its other component N_2 is along the negative direction of the x-axis. Moreover, the direction of N $(= N_1 + N_2)$ is perpendicular to the rod, hence the resultant of the component of N_1 in the plane PAB and of N_2 is perpendicular to the rod; that is,

$$N_1 \cos\theta \sin\alpha = N_2 \cos\alpha. \tag{1}$$

From the geometrical relations shown in the figure,

$$\tan\alpha = \frac{2}{3}. \tag{2}$$

From (1) and (2) we obtain

$$N_1 = 3N_2. \tag{3}$$

The direction of the friction force $\vec{f}$ acting on the rod is determined by the following analysis. First, the rod has a tendency to slide downwards relative to point B in the plane of the ring (the yOz plane), so the component of $\vec{f}$ in this plane must be along the upward tangent at point B; denote it by f_1. Secondly, the ring has a tendency to move in the direction BA relative to the rod, so the other component f_2 of $\vec{f}$ must be along the direction BA, as shown in the figure. Moreover, the direction of f $(= f_1 + f_2)$ is perpendicular both to the rod and to the direction of N. From this it follows that the component of f_1 along the rod must be equal and opposite to f_2, so that the two cancel each other; that is,

$$f_1 \sin\theta \sin\alpha = f_2. \tag{4}$$

Substituting $\sin\alpha = \dfrac{2}{\sqrt{13}}$ and $\sin\theta = \dfrac{\sqrt{3}}{2}$ into (4), we obtain

$$f_1 = \sqrt{\frac{13}{3}}\, f_2. \tag{5}$$

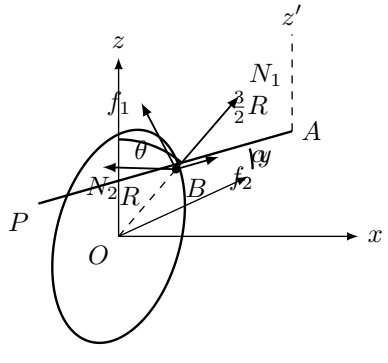


Fig. 1.23

Through point A establish a z' axis parallel to the z-axis, as shown in the figure. From the equilibrium of the moments of the external forces acting on the rod about the z' axis (since the weight of the rod is along the vertical direction, its moment about the z' axis is zero), we obtain

$$N_1 \sin\theta \times \frac{3}{2}R + N_2 R \sin\theta = f_1 \cos\theta \times \frac{3}{2}R. \tag{6}$$

Substituting (3) into (6), we obtain

$$N_2 = \frac{\sqrt{3}}{11} f_1. \tag{7}$$

Substituting (3) into $N = \sqrt{N_1^2 + N_2^2}$, we obtain

$$N = \sqrt{10}\, N_2. \tag{8}$$

From the above analysis the value of the friction force f is obtained as

$$\begin{aligned} f &= \sqrt{(f_1 \sin\theta \cos\alpha)^2 + (f_1 \cos\theta)^2} \\ &= \sqrt{\frac{10}{13}}\, f_1. \end{aligned} \tag{9}$$

Combining (7), (8) and (9), we obtain

$$f = \frac{11}{\sqrt{39}} N \le N\mu,$$

from which we obtain

$$\mu \ge \frac{11}{\sqrt{39}}.$$

Comment. This problem asks for the minimum coefficient of friction required for the rod to be in equilibrium at the position $\theta = 60^\circ$. For this purpose the relation between the supporting force N and the friction force f exerted by the ring on the rod at this position must first be found, and the correct analysis and judgement of the directions of N and f is the key to obtaining this relation: since the direction of the supporting force between two contact surfaces is perpendicular to both surfaces, the direction of N is perpendicular both to the rod and to the ring; and the direction of f must be perpendicular to N, while, because the rod has a tendency to rotate about point A, the direction of f is also perpendicular to the rod. Once the directions of N and f are known, the relations between their components can be found, and the relation between these two forces is obtained by requiring the resultant moment of N and f about the vertical axis through A to vanish.

Solution 1.24. The speed v with which the water reaches the inner bottom of the garbage can after spraying out from the fountain nozzle is

$$v = \sqrt{v_0^2 - 2gh}. \tag{1}$$

From the continuity of the water stream it follows that the mass $\dfrac{\Delta m}{\Delta t}$ of water that rushes onto the inner bottom of the can per second should be equal to the mass of water sprayed from the fountain nozzle per second; that is,

$$\frac{\Delta m}{\Delta t} = \rho v_0 S_0. \tag{2}$$

In (2), ρ is the density of water. Half of the water that rushes onto the inner bottom of the can is adsorbed on the inner bottom and flows down along the wall of the can, while the other half splashes vertically downwards with its original speed; hence the change of momentum per second of the water rushing onto the inner bottom is

$$\frac{\Delta p}{\Delta t} = \frac{1}{2}\frac{\Delta m}{\Delta t} \cdot v + \frac{1}{2}\frac{\Delta m}{\Delta t} \cdot 2v. \tag{3}$$

Substituting (2) into (3), we obtain

$$\frac{\Delta p}{\Delta t} = \frac{3}{2}\rho S_0 v_0 v. \tag{4}$$

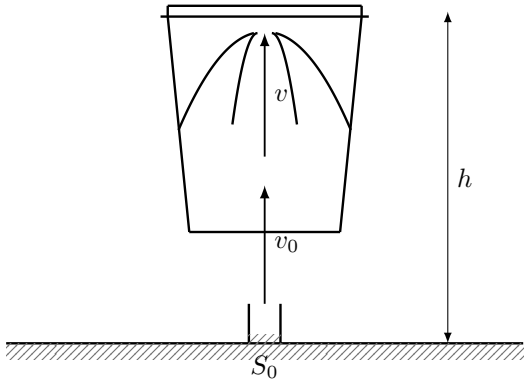


Fig. 1.24

Solution 1.25. Let the density of the raindrop be ρ. Take the instant at which the raindrop begins to fall as $t = 0$. At time t the mass of the raindrop is $m(t)$, its velocity $v(t)$, its radius $r(t)$ and its volume $V(t) = \frac{4}{3}\pi r^3$. On its way down the raindrop adsorbs all the mist that it encounters, so that its volume and mass increase continually; hence

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{\mathrm{d}(\rho V)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\rho\frac{4}{3}\pi r^3\right) = 4\pi r^2\rho\frac{\mathrm{d}r}{\mathrm{d}t} = \rho_0\pi r^2 v,$$

where ρ_0 is the density of the mist. From the above equation we obtain

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{\rho_0}{4\rho}v. \tag{1}$$

Since the viscous drag and the buoyant force acting on the raindrop during its fall are neglected, the only external force is gravity; hence, from the momentum theorem,

$$\frac{\mathrm{d}(mv)}{\mathrm{d}t} = mg. \tag{2}$$

Substituting $m = \rho V$ into (2), we have

$$\frac{\mathrm{d}(Vv)}{\mathrm{d}t} = Vg.$$

The above equation can also be written as

$$\frac{\mathrm{d}(Vv)}{\mathrm{d}V}\frac{\mathrm{d}V}{\mathrm{d}t} = Vg. \tag{3}$$

From (1) we have $\frac{\mathrm{d}V}{\mathrm{d}t} = 4\pi r^2\frac{\mathrm{d}r}{\mathrm{d}t} = \pi r^2\frac{\rho_0}{\rho}v$; substituting this into (3), we obtain

$$\frac{\mathrm{d}(Vv)}{\mathrm{d}V}\cdot\frac{\pi r^2\rho_0}{\rho}v = Vg. \tag{4}$$

Substituting $r = \sqrt[3]{\frac{3V}{4\pi}}$ into (4), we obtain

$$(Vv)\,\mathrm{d}(Vv) = \sqrt[3]{\frac{16\rho g}{9\pi\rho_0}}\,V^{4/3}\,\mathrm{d}V.$$

Integrating both sides of the above equation,

$$\int_0^{Vv} (Vv)\,\mathrm{d}(Vv) = \sqrt[3]{\frac{16\rho g}{9\pi\rho_0}}\int_0^{V} V^{4/3}\,\mathrm{d}V,$$

we obtain

$$\frac{1}{2}(Vv)^2 = \frac{3}{7}\sqrt[3]{\frac{16}{9\pi}\frac{\rho g}{\rho_0}}\,V^{7/3}.$$

Hence

$$v^2 = \frac{4}{7}\sqrt[3]{\frac{6}{\pi}\frac{\rho g}{\rho_0}}\,V^{1/3} = \frac{8\rho g}{7\rho_0}r,$$

that is,

$$v = \sqrt{\frac{8\rho g}{7\rho_0}r}. \tag{5}$$

Substituting (5) into (1), we obtain

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{\rho_0}{4\rho}\sqrt{\frac{8\rho g}{7\rho_0}r} = \sqrt{\frac{\rho_0 g}{14\rho}r}.$$

Multiplying both sides of the above equation by $\frac{\mathrm{d}t}{\sqrt{r}}$ and then integrating both sides,

$$\int_0^r \frac{\mathrm{d}r}{\sqrt{r}} = \sqrt{\frac{\rho_0 g}{14\rho}}\int_0^t \mathrm{d}t,$$

we obtain

$$2\sqrt{r} = \sqrt{\frac{\rho_0 g}{14\rho}}\,t,$$

that is,

$$r = \frac{\rho_0 g}{56\rho}t^2. \tag{6}$$

Substituting (6) into (5), we obtain

$$v = \sqrt{\frac{8\rho g}{7\rho_0}\times\frac{\rho_0 g}{56\rho}t^2} = \frac{1}{7}gt. \tag{7}$$

From (7) we immediately obtain

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{1}{7}g.$$

One method of solving this problem has been given above; another method is as follows. From (2) we obtain

$$3v\frac{\mathrm{d}r}{\mathrm{d}t} + r\frac{\mathrm{d}v}{\mathrm{d}t} = rg. \tag{8}$$

Substituting (1) into (8), we obtain

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} + \frac{3}{r}\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 = \frac{\rho_0}{4\rho} g. \tag{9}$$

The particular solution of this non-homogeneous differential equation is

$$r = bt^2. \tag{10}$$

Substituting (10) into (9), we can solve for b:

$$b = \frac{1}{56}\frac{\rho_0}{\rho} g. \tag{11}$$

From (1), (10) and (11) we obtain

$$v = \frac{4\rho}{\rho_0}\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{4\rho}{\rho_0} \times 2bt = \frac{1}{7} gt. \tag{12}$$

Hence

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{1}{7} g.$$

Solution 1.26. Take the instant at which the block begins to slide down the inclined plane as $t = 0$. At time t let the velocity of the block be $v(t)$, and let the total mass of the block together with that part of the rope which moves with the block at velocity v (this part having length x) be $M(t)$. Then, from the momentum theorem,

$$\frac{\mathrm{d}(Mv)}{\mathrm{d}t} = Mg \sin\theta.$$

The above equation can be rewritten as

$$\frac{\mathrm{d}(Mv)}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = Mg \sin\theta.$$

Substituting $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ into the above equation and multiplying both sides by $M\,\mathrm{d}x$, and noting that $M = m + \lambda x$, we obtain

$$(Mv)\,\mathrm{d}(Mv) = (m + \lambda x)^2 g \sin\theta\,\mathrm{d}x.$$

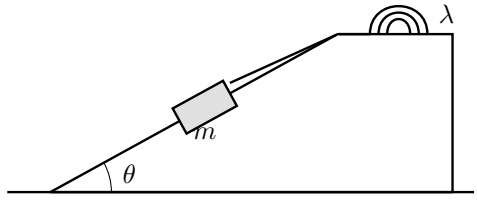


Fig. 1.26

Integrating both sides of the above equation,

$$\int_0^{Mv} (Mv)\,\mathrm{d}(Mv) = \int_0^x (m + \lambda x)^2 g \sin\theta\,\mathrm{d}x,$$

we obtain

$$\frac{1}{2}(Mv)^2 = \frac{g}{3\lambda}\sin\theta\left[(m+\lambda x)^3 - m^3\right].$$

From this we can solve

$$v = \sqrt{\frac{2g\sin\theta\left[(m+\lambda x)^3 - m^3\right]}{3\lambda(m+\lambda x)^2}}.$$

Solution 1.27. When the extension of the spring is x_0, that is, when the spring force is equal to the weight of the small ball, the ball leaves the table. We have

$$x_0 = \frac{mg}{k}. \tag{1}$$

After this, take the upper end of the spring, which moves uniformly upward at speed v_0, as the frame of reference. In this inertial frame the ball begins to move downward with initial speed v_0 and does so until it stops; at this instant the extension of the spring reaches its maximum for the first time. Let this maximum extension be x. From the conservation of mechanical energy of the system,

$$\frac{1}{2}mv_0^2 - mgx_0 + \frac{1}{2}kx_0^2 = \frac{1}{2}kx^2 - mgx. \tag{2}$$

The zero points of both the gravitational potential energy and the elastic potential energy of the system are taken at the position of the undeformed spring. Substituting (1) into (2), we obtain

$$kx^2 - 2mgx + \frac{(mg)^2}{k} - mv_0^2 = 0. \tag{3}$$

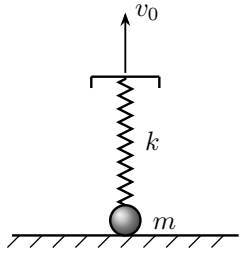


Fig. 1.27

Solving this quadratic equation, a reasonable solution is

$$x = \frac{mg}{k} + v_0\sqrt{\frac{m}{k}}. \tag{4}$$

The time elapsed while the extension of the spring changes from x_0 to x is equal to one quarter of the period of the simple harmonic motion of this oscillator, that is,

$$t = \frac{T}{4} = \frac{\pi}{2}\sqrt{\frac{m}{k}}. \tag{5}$$

From (5) the height h of the ball above the table at this instant is found to be

$$h = v_0 t - (x - x_0) = \left(\frac{\pi}{2} - 1\right)v_0\sqrt{\frac{m}{k}}. \tag{6}$$

From the work-energy theorem for the system, the work W done by the force acting on the upper end of the spring during the whole process is

$$W = \frac{1}{2}mv_0^2 + mgh + \frac{1}{2}kx^2. \tag{7}$$

Substituting (4) and (6) into (7), we obtain

$$W = mv_0^2 + \frac{\pi}{2}mgv_0\sqrt{\frac{m}{k}} + \frac{1}{2}\frac{m^2g^2}{k}.$$

Comment. The key to the solution of this problem is that, after the ball leaves the table, one changes to the frame of reference moving together with the upper end of the spring. This is an inertial frame; by the principle of relativity of mechanics, the laws of mechanics have the same form in every inertial frame. In this frame the force acting on the upper end of the spring does no work, so after the ball leaves the table the mechanical energy of the system is conserved; equation (2) is thus obtained, and hence the maximum extension x of the spring. Moreover, in this frame the ball performs simple harmonic motion, so the time t from the ball leaving the table until the spring first reaches its maximum extension is easily found. In the table frame, on the other hand, the mechanical energy of the system is not conserved, and the ball does not perform simple harmonic motion either, so x and t are not easily found.

Solution 1.28. Suppose that under the action of the force F, after a time t, ball B has moved a distance s, and at this moment the rod makes an angle θ with the horizontal, as shown in Fig. 1.28(a). Let the velocity of ball B be v, and let the velocity of ball A relative to ball B be v'. From the work-energy principle we have

$$Fs = \frac{1}{2}mv^2 + \frac{1}{2}m\left[(v - v'\sin\theta)^2 + (v'\cos\theta)^2\right] - mgl(1 - \sin\theta) \tag{1}$$

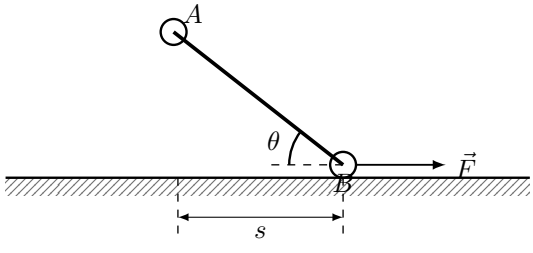


Fig. 1.28(a)

The acceleration a_C of the centre of mass of the system along the horizontal direction is

$$a_C = \frac{F}{2m} \tag{2}$$

The displacement x_C of the centre of mass along the horizontal direction during this process is

$$x_C = \frac{1}{2}a_C t^2 \tag{3}$$

From the geometric relation shown in Fig. 1.28(a), the displacement x_C of the centre of mass along the horizontal direction is

$$x_C = s - \frac{l}{2}\cos\theta \tag{4}$$

From (2), (3) and (4) we obtain

$$t = \sqrt{\frac{2m}{F}(2s - l\cos\theta)} \tag{5}$$

From the momentum theorem,

$$Ft = mv + m(v - v'\sin\theta) \tag{6}$$

Substituting (5) into (6) gives

$$v = \frac{1}{2}v'\sin\theta + \sqrt{\frac{F}{2m}(2s - l\cos\theta)} \tag{7}$$

Substituting (7) into (1), we obtain

$$v'^2 = \frac{2Fl\cos\theta + 4mgl(1 - \sin\theta)}{m(1 + \cos^2\theta)} \tag{8}$$

Let the force exerted on ball B by the rod be T, directed along the rod, and let the normal force exerted on it by the ground be N; the forces acting on ball B are then as shown in Fig. 1.28(b). Accordingly, the equations of motion of ball B along the horizontal and vertical directions are respectively

$$F + T\cos\theta = ma \tag{9}$$

$$N - T\sin\theta - mg = 0 \tag{10}$$

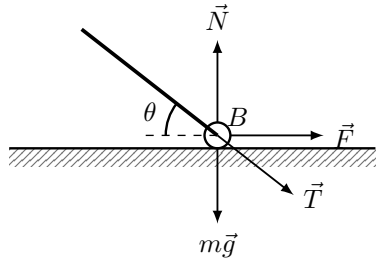


Fig. 1.28(b)

Take ball B as the reference frame; then the forces acting on ball A are as shown in Fig. 1.28(c). In the figure the inertial force is $f_i = ma$, where a is the acceleration of ball B at this moment. Hence the normal equation of motion of ball A, which moves in a circle about B in this frame, is

$$mg\sin\theta - T - ma\cos\theta = m\frac{v'^2}{l} \tag{11}$$

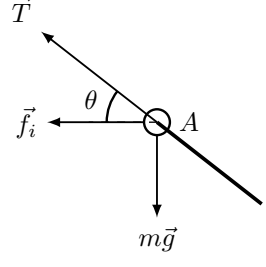


Fig. 1.28(c)

Substituting (8) and (9) into (11) gives

$$T = \frac{mg(6\sin\theta - \sin^3\theta - 4) - F\cos\theta(\cos^2\theta + 3)}{(1+\cos^2\theta)^2} \tag{12}$$

Substituting (12) into (10), we obtain the force N exerted by the horizontal surface on ball B at this moment:

$$\begin{aligned} N &= mg + T\sin\theta \\ &= mg\left[1 + \frac{\sin\theta(6\sin\theta - \sin^3\theta - 4) - \dfrac{F}{mg}\sin\theta\cos\theta(\cos^2\theta + 3)}{(1+\cos^2\theta)^2}\right]. \end{aligned}$$

Comment. The system in this problem consists of two small balls connected by the two ends of a light rod. Whatever the motion of such a system, the force exerted by the rod on a ball at any instant must be directed along the rod; otherwise its reaction, acting on the two ends of the light rod, would subject the light rod to a torque. This contradicts the conclusion, obtained from the law of rotation, that the resultant torque acting on a rigid body of zero moment of inertia is zero. Moreover, in order to find the force exerted by the rod on ball B, one may discuss the motion of ball A in the frame of ball B (a non-inertial frame). The velocity of ball A relative to ball B can then be found by applying, in the inertial frame, such principles as the law of motion of the centre of mass, the work-energy principle and the momentum theorem.

Solution 1.29. Suppose that when the angle between the string and the rod is θ, the velocity of the small ring A is v_A, and the velocity of the small ball B relative to the small ring is v'. Then the velocity v_B of ball B relative to the ground is given by the formula for relative motion, namely

$$v_B = v' + v_A \tag{1}$$

The vector relation of (1) is shown in Fig. 1.29(a). From the figure we obtain

$$v_{Bx} = v_A - v'\sin\theta \tag{2}$$

$$v_B = \sqrt{(v_A - v'\sin\theta)^2 + (v'\cos\theta)^2} \tag{3}$$

From the conservation of horizontal momentum and the conservation of mechanical energy of the system we obtain respectively

$$mv_A + mv_{Bx} = 0 \tag{4}$$

$$\frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 - mgl\sin\theta = 0 \tag{5}$$

where the zero of gravitational potential energy is taken at the point from which ball B is released. Substituting (2) and (3) into (4) and (5) respectively gives

$$mv_A + m(v_A - v'\sin\theta) = 0 \tag{6}$$

$$\frac{1}{2}mv_A^2 + \frac{1}{2}m\left[(v_A - v'\sin\theta)^2 + (v'\cos\theta)^2\right] - mgl\sin\theta = 0 \tag{7}$$

Solving (6) and (7) simultaneously, we obtain

$$v'^2 = \frac{4gl\sin\theta}{1+\cos^2\theta} \tag{8}$$

$$v_A = \frac{1}{2}\sin\theta\sqrt{\frac{4gl\sin\theta}{1+\cos^2\theta}} \tag{9}$$

Let the angle between v_B and v' be α. From the geometric relation shown in Fig. 1.29(a), and using the sine rule, we have

$$\frac{v_A}{\sin\alpha} = \frac{v'}{\sin\left(\frac{\pi}{2}+\theta-\alpha\right)} \tag{10}$$

Substituting (8) and (9) into (10), we obtain

$$\frac{v'}{v_A} = \frac{\cos(\theta-\alpha)}{\sin\alpha} = \frac{\cos\theta\cos\alpha+\sin\theta\sin\alpha}{\sin\alpha} = \frac{2}{\sin\theta}.$$

Hence we can solve for

$$\cot\alpha = \frac{1+\cos^2\theta}{\sin\theta\cos\theta},$$

that is,

$$\begin{cases} \sin\alpha = \dfrac{\sin\theta\cos\theta}{\sqrt{1+3\cos^2\theta}} \\ \cos\alpha = \dfrac{1+\cos^2\theta}{\sqrt{1+3\cos^2\theta}} \end{cases} \tag{11}$$

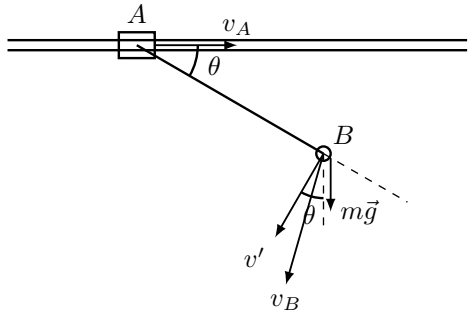


Fig. 1.29(a)

Let the tension in the string at this moment be T, and let the acceleration of the small ring A be a. Then

$$T\cos\theta = ma \tag{12}$$

Take the small ring A as the reference frame; in this non-inertial frame the forces acting on ball B are as shown in Fig. 1.29(b). In the figure $f_i = ma$. From the figure, the normal equation of motion of ball B, which moves in a circle, is

$$T + ma\cos\theta - mg\sin\theta = m\frac{v'^2}{l} \tag{13}$$

Substituting (8) into (13) gives

$$T = \frac{5+\cos^2\theta}{1+\cos^2\theta}mg\sin\theta - ma\cos\theta \tag{14}$$

Solving (12) and (14) simultaneously, we obtain

$$T = \frac{5+\cos^2\theta}{(1+\cos^2\theta)^2}mg\sin\theta \tag{15}$$

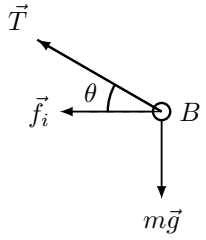


Fig. 1.29(b)

(1) The direction of the velocity v_B of ball B relative to the ground is just the direction of the tangent to its trajectory at that point. When the speed of the ball reaches its maximum, the resultant force acting on the ball along this tangential direction must be zero. Hence

$$mg\cos(\theta-\alpha)=T\sin\alpha \tag{16}$$

Substituting (11) and (15) into (16), we obtain a quadratic equation in $\cos^2\theta$:

$$3\cos^4\theta+8\cos^2\theta-3=0 \tag{17}$$

The solution of this equation is

$$\cos^2\theta=\frac{1}{3},$$

that is,

$$\begin{cases}\cos\theta=\dfrac{\sqrt{3}}{3}\\[2ex] \sin\theta=\sqrt{\dfrac{2}{3}}\end{cases} \tag{18}$$

Substituting (8) and (9) into (3), and then substituting (18), we obtain the maximum speed $v_{B\max}$ of ball B:

$$\begin{aligned} v_{B\max}&=\sqrt{(v_A-v'\sin\theta)^2+(v'\cos\theta)^2}\\ &=\sqrt{\frac{gl(4-3\sin^2\theta)\sin\theta}{1+\cos^2\theta}}=\sqrt{\frac{\sqrt{6}}{2}gl}. \end{aligned}$$

(2) Substituting (18) into (15), we obtain the tension in the string when ball B has its maximum speed:

$$T=\frac{5+\cos^2\theta}{(1+\cos^2\theta)^2}mg\sin\theta=\sqrt{6}mg.$$

The above is one method of solving this problem; of course, the maximum value of v_B can also be found by the method of differentiation.

Solution 1.30. (1) Suppose that when the small ball has turned through an angle θ relative to the ring, the speed of the ring is V and the speed of the small ball relative to the ring is v'; the speed of the small ball relative to the ground is v. The relation between these three velocity vectors is given by the formula for relative motion, namely

$$v=v'+V \tag{1}$$

From the geometric relation shown in Fig. 1.30 we obtain

$$v_x = v' \cos\theta - V \tag{2}$$

$$v = \sqrt{(v' \cos\theta - V)^2 + (v' \sin\theta)^2} \tag{3}$$

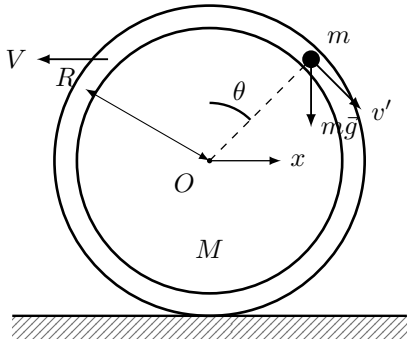


Fig. 1.30

From the conservation of momentum of the system along the horizontal (x) direction, and using (2), we obtain

$$m(v' \cos\theta - V) - MV = 0 \tag{4}$$

Take the centre of the ring as the zero of gravitational potential energy; from the conservation of mechanical energy of the system, and using (3), we obtain

$$mgR = mgR\cos\theta + \frac{1}{2}MV^2 + \frac{1}{2}m\left[(v' \cos\theta - V)^2 + (v' \sin\theta)^2\right] \tag{5}$$

From (4) and (5) we can solve for

$$v' = \sqrt{\frac{2(M+m)gR(1-\cos\theta)}{M + m\sin^2\theta}} \tag{6}$$

$$V = \sqrt{\frac{2m^2 gR\cos^2\theta(1-\cos\theta)}{(M+m)(M+m\sin^2\theta)}} \tag{7}$$

When the leftward speed of the ring reaches its maximum, the small ball separates from the tube of the ring, that is, there is no longer any interaction between the two; at this moment the acceleration of the ring is zero. Now take the ring as the reference frame; at this moment this frame is an inertial frame. In this frame the normal equation of motion of the small ball, which moves in a circle along the tube, is

$$mg\cos\theta = m\frac{v'^2}{R} \tag{8}$$

Substituting (6) into (8) gives

$$\frac{M}{m} = \frac{2(1-\cos\theta) - \sin^2\theta\cos\theta}{3\cos\theta - 2} \tag{9}$$

Substituting $\theta = 45^\circ$ into (9), we obtain

$$\frac{M}{m} = \sqrt{2} + \frac{1}{2} \approx 1.914 \tag{10}$$

Substituting (10) and $\theta = 45^\circ$ into (7), we obtain the maximum leftward speed $V_{\max}$ of the ring:

$$V_{\max} = \sqrt{(17\sqrt{2} - 24)gR} \approx 0.2\sqrt{gR}$$

(2) Substituting (6) and (7) into (3), we obtain the absolute speed v of the small ball:

$$v = \sqrt{\frac{2gR(1-\cos\theta)\left[M^2 + m(2M+m)\sin^2\theta\right]}{(M+m)(M+m\sin^2\theta)}} \tag{11}$$

Let the angle between v and the vertical direction be α. Then from the geometric relation shown in Fig. 1.30, and from (6) and (7), we obtain

$$\tan\alpha = \frac{v'\cos\theta - V}{v'\sin\theta} = \cot\theta - \frac{V}{v'\sin\theta} = \frac{M}{M+m}\cot\theta,$$

that is,

$$\sin\alpha = \frac{M\cos\theta}{\sqrt{M^2 + m(m+2M)\sin^2\theta}} \tag{12}$$

Since at this moment there is no interaction force between the small ball and the ring, the acceleration of the small ball is equal to the acceleration of gravity, that is, $a = g$; the component of the acceleration along the normal to the trajectory is

$$a_n = a\sin\alpha = g\sin\alpha \tag{13}$$

The radius of curvature ρ of the trajectory of the small ball at this point is

$$\rho = \frac{v^2}{a_n} = \frac{v^2}{g\sin\alpha} \tag{14}$$

Substituting (11) and (12) into (14), we obtain

$$\rho = \frac{2(1-\cos\theta)\left[M^2 + m(2M+m)\sin^2\theta\right]^{\frac{3}{2}}}{M\cos\theta(M+m)(M+m\sin^2\theta)}R \tag{15}$$

Substituting $\theta = 45^\circ$ and (10) into (15), we obtain

$$\rho = \frac{8}{7}\left(46\sqrt{2} - 65\right)\left(\frac{13}{4} + 2\sqrt{2}\right)^{\frac{3}{2}} R \approx 0.92R.$$

Comment. The radius of curvature of the trajectory of the small ball at a certain point can be found in this problem either by first finding the equation of the trajectory of the small ball and then calculating the radius of curvature, or, as in the solution above, by first finding the normal acceleration a_n of the small ball at this point and then using the formula $a_n = \dfrac{v^2}{\rho}$ to find the radius of curvature ρ. Since the moment when the leftward speed of the ring reaches its maximum is precisely the moment when the small ball and the ring separate from each other, that is, when there is no interaction force, the acceleration of the small ball at this moment is equal to the acceleration of gravity, while the normal to the trajectory at this point is the direction perpendicular to the tangent, i.e. the direction perpendicular to the velocity of the small ball at this moment. Hence, by using the conservation of momentum of the system, the conservation of energy and the principle of relative motion to find this velocity and its direction, the radius of curvature of the trajectory at this point can be obtained.

Solution 1.31. (1) From the moment of release until the string connected to C makes an angle α with the vertical line, the height H through which object C descends is

$$H = l\cot\alpha \tag{1}$$

while objects A and B rise by h:

$$h = \frac{l}{\sin\alpha} - l \tag{2}$$

Let the speeds of A (or B) and C at this moment be v and V respectively; from the law of conservation of mechanical energy of the system we have

$$MgH - 2mgh = \frac{1}{2}MV^2 + 2\times\frac{1}{2}mv^2 \tag{3}$$

and the constraint relation between v and V is

$$v = V\cos\alpha \tag{4}$$

Solving (1), (2), (3) and (4) simultaneously, we obtain

$$V = \sqrt{\frac{M\cos\alpha + 2m(\sin\alpha - 1)}{(M + 2m\cos^2\alpha)\sin\alpha}\cdot 2gl} \tag{5}$$

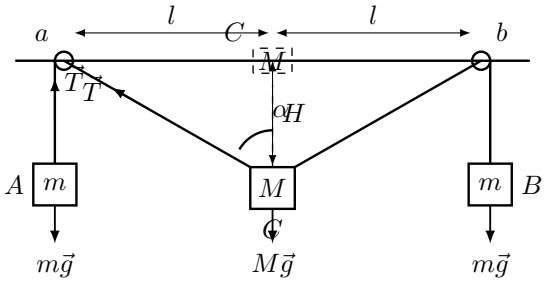


Fig. 1.31

When the speed of object C reaches its maximum, its acceleration is zero. Hence, from the force analysis of object C, we obtain

$$2T\cos\alpha = Mg \tag{6}$$

At this moment the equation of motion of object A (or B) is

$$T - mg = ma \tag{7}$$

In a plane polar coordinate system with a (or b) as the pole, since the acceleration of object C is zero, its radial acceleration is zero. That is,

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 \tag{8}$$

In (8), $\ddot{r}$ is just the a of (7), while $r = \dfrac{l}{\sin\alpha}$ and $\dot{\theta} = \dfrac{V\sin\alpha}{l/\sin\alpha} = \dfrac{V\sin^2\alpha}{l}$. Substituting these relations into (8), we obtain

$$a = \frac{V^2\sin^3\alpha}{l} \tag{9}$$

Solving (5), (6), (7) and (9) simultaneously, and substituting $\alpha = 60^\circ$, we obtain

$$4\left(\frac{M}{m}\right)^2 - 5\left(\frac{M}{m}\right) + (10 - 6\sqrt{3}) = 0 \tag{10}$$

Solving this quadratic equation in $\dfrac{M}{m}$, we obtain

$$\frac{M}{m} = 1.324 \tag{11}$$

Substituting (11) and $\alpha = 60^\circ$ into (5), we obtain the maximum speed of object C:

$$V_{\max} = 0.706\sqrt{gl}.$$

(2) If $\dfrac{M}{m} = 1$ is taken, then (5) becomes

$$V = \sqrt{\frac{\cos\alpha + 2(\sin\alpha - 1)}{(1 + 2\cos^2\alpha)\sin\alpha} \cdot 2gl} \tag{12}$$

The radial equation of motion of object C is

$$T + T\cos 2\alpha - mg\cos\alpha = m\left[\frac{(V\sin\alpha)^2}{l/\sin\alpha} - a\right] \tag{13}$$

The equation of motion of object A (or B) is

$$T - mg = ma \tag{14}$$

Let the acceleration of object C be a_C, directed along the vertical direction; the equation of motion of object C along this direction is

$$2T\cos\alpha - mg = ma_C \tag{15}$$

Solving (12), (13), (14) and (15) simultaneously, and substituting $\alpha = 60^\circ$, we obtain

$$a_C = \left(\frac{2}{\sqrt{3}} - 1\right) g \approx 0.155g.$$

The direction of this acceleration is vertically upward.

Comment. A key point in solving this problem is that the moment when the speed of object C reaches its maximum is not the moment when the speeds of objects A and B reach their maxima. In the plane polar coordinate system with the bearing a (or b) as the pole, when the speed of object C reaches its maximum both its radial acceleration and its transverse acceleration are zero, and from the vanishing of the radial acceleration it follows that $\ddot{r} = r\dot{\theta}^2$, where $\ddot{r}$ is precisely the acceleration of the motion of A and B at this moment. This shows that, as long as the angular velocity $\dot{\theta}$ of the rotation of the string connected to object C is not zero at this moment, the acceleration of A and B is not zero, that is, the speeds of A and B are not at their maxima at this moment. This point is very easily overlooked.

Solution 1.32. (1) After the two bodies are released, let the acceleration of the centre of mass of the system be a_C. Then

$$-Mg\mu + mg\mu = (M+m)a_C,$$
$$a_C = -\frac{M-m}{M+m}g\mu = -1\,\mathrm{m/s^2}.$$

Let the distances from the centre of mass to A and B at this moment be l_A and l_B; then

$$\begin{cases} l_A + l_B = l \\ Ml_A = ml_B \end{cases}$$

From these two equations we obtain

$$l_A = \frac{m}{M+m}l = \frac{1}{3}l,$$
$$l_B = \frac{M}{M+m}l = \frac{2}{3}l.$$

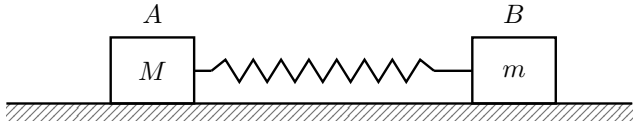


Fig. 1.32(a)

When the string is at its natural length l_0, by the same method we can find the lengths l_{0A} and l_{0B} of the two segments of the elastic string from A and B to the centre of mass:

$$l_{0A} = \frac{m}{M+m}l_0 = \frac{1}{3}l_0,$$
$$l_{0B} = \frac{M}{M+m}l_0 = \frac{2}{3}l_0.$$

The stiffness coefficients k_A and k_B of these two segments of elastic string are respectively

$$k_A = \frac{l_0}{l_{0A}}k = \frac{M+m}{m}k = 3k,$$
$$k_B = \frac{l_0}{l_{0B}}k = \frac{M+m}{M}k = \frac{3}{2}k.$$

Hence the angular frequency of the simple harmonic motion performed by the two bodies in the centre-of-mass frame is

$$\omega_A = \omega_B = \omega = \sqrt{\frac{k(M+m)}{Mm}} = \sqrt{20}\,\mathrm{s^{-1}}.$$

Let the distance between the equilibrium position O' of block B and the position O at which the string is at its natural length be b_B; the forces acting on B are as shown in Fig. 1.32(b), where the inertial force is $f_{iB} = ma_C$. Then

$$k_B b_B = f_{iB} + f_B = m|a_C| + mg\mu,$$

$$b_B = \frac{m(g\mu + |a_C|)}{k_B} = 0.2\,\text{m}.$$

Hence the amplitude A_B of the simple harmonic motion performed by B is

$$\begin{aligned} A_B &= (l_B - l_{0B}) - b_B = \frac{2}{3}(l - l_0) - b_B \\ &= 0.4\,\text{m}. \end{aligned}$$

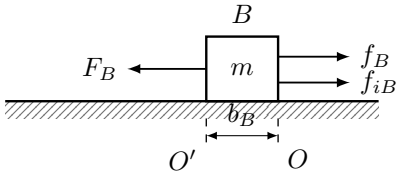


Fig. 1.32(b)

When the elastic string returns to its natural length, B reaches point O, that is, the point $\frac{1}{2}A_B$ to the left of O'. Let the time from the release of B to its arrival at point O be t_1; from the amplitude vector diagram shown in Fig. 1.32(c) we obtain

$$\omega t_1 = \frac{2\pi}{3},$$

that is,

$$t_1 = \frac{2\pi}{3\omega} = \frac{\sqrt{5}}{15}\pi\,\text{s}.$$

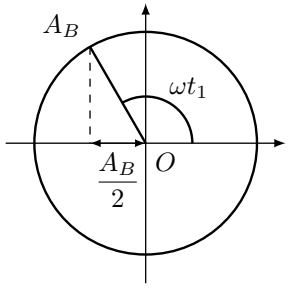


Fig. 1.32(c)

At this moment the velocity v_B of B is

$$v'_B = -A_B\omega \sin\frac{2\pi}{3} = -\frac{2}{5}\sqrt{15}\,\text{m/s}.$$

From the properties of the centre-of-mass frame it is known that block A, also after the time t_1 from release, has the segment of the string between A and the centre of mass restored to its natural length. At this moment the velocity v'_A of A is

$$v'_A = -\frac{m}{M}v'_B = \frac{1}{2}\cdot\frac{2}{5}\sqrt{15} = \frac{1}{5}\sqrt{15}\,\text{m/s}.$$

At this moment the velocity v_C of the centre of mass is

$$v_C = a_C t_1 = -\frac{\sqrt{5}}{15}\pi\,\text{m/s}.$$

Hence at this moment the velocities of A and B relative to the ground are respectively

$$v_A = v'_A + v_C = \frac{1}{5}\sqrt{15} - \frac{\sqrt{5}}{15}\pi = \frac{\sqrt{5}}{5}\left(\sqrt{3} - \frac{\pi}{3}\right) \text{ m/s},$$

$$v_B = v'_B + v_C = -\frac{2}{5}\sqrt{15} - \frac{\sqrt{5}}{15}\pi = -\frac{\sqrt{5}}{5}\left(2\sqrt{3} + \frac{\pi}{3}\right) \text{ m/s}.$$

After this both bodies move with uniform deceleration; the acceleration a is

$$a = -g\mu = -3\,\text{m/s}^2.$$

When the elastic string is restored to its natural length, the velocity v_A of block A is the smaller one; when it comes to rest after a time t', the displacement it has covered is Δl_A, and we have

$$t' = \frac{v_A}{|a|} = \frac{\sqrt{5}}{15}\left(\sqrt{3} - \frac{\pi}{3}\right) \text{ s},$$

$$\begin{aligned}\Delta l_A &= \frac{1}{2}|a|t'^2 = \frac{1}{2} \times 3 \times \frac{5}{15^2}\left(\sqrt{3} - \frac{\pi}{3}\right)^2 \\ &\approx 0.016\,\text{m}.\end{aligned}$$

The displacement Δl_B covered by block B during this process is

$$\begin{aligned}\Delta l_B &= v_B t' + \frac{1}{2}at'^2 = -\frac{\sqrt{5}}{5}\left(2\sqrt{3} + \frac{\pi}{3}\right) \times \frac{\sqrt{5}}{15}\left(\sqrt{3} - \frac{\pi}{3}\right) - \frac{3}{2}\frac{5}{15^2}\left(\sqrt{3} - \frac{\pi}{3}\right)^2 \\ &= -\frac{1}{30}\left(9 - \frac{\pi^2}{3}\right) \approx -0.190\,\text{m}.\end{aligned}$$

Obviously, the distance covered by the two bodies A and B during this process is

$$\Delta l_A - \Delta l_B \approx 0.206\,\text{m} < l_0 = 0.30\,\text{m}.$$

It is thus seen that when A has stopped the two bodies have not yet collided, and B must cover a distance $\Delta l'_B$ before it collides with A:

$$\Delta l'_B = l_0 - \Delta l_A = 0.284\,\text{m}.$$

Let the value of the velocity of B when the two bodies collide be V_B; then

$$V_B^2 - v_B^2 = 2a\Delta l'_B,$$

$$\begin{aligned}\therefore \quad V_B &= \sqrt{v_B^2 + 2a\Delta l'_B} = \sqrt{\frac{1}{5}\left(2\sqrt{3} + \frac{\pi}{3}\right)^2 - 2 \times 3 \times 0.284} \\ &\approx 1.54\,\text{m/s}.\end{aligned}$$

(2) Let the time from the restoration of the elastic string to its natural length until the two bodies collide be t_2; then

$$\Delta l'_B = |v_B|t_2 + \frac{1}{2}at_2^2.$$

Substituting the data into the above expression gives

$$0.284 = \frac{\sqrt{5}}{5}\left(2\sqrt{3} + \frac{\pi}{3}\right) t_2 - \frac{3}{2}t_2^2,$$

that is,

$$1.5t_2^2 - 2.02t_2 + 0.284 = 0.$$

We obtain

$$t_2 = \frac{1}{3}(2.02 - 1.54) = 0.16\,\mathrm{s}.$$

The other solution of the equation, $t_2' = \frac{1}{3}(2.02 + 1.54)$, is discarded. Hence the total time from the release of the two bodies until their collision is

$$t = t_1 + t_2 = \frac{\sqrt{5}}{15}\pi + 0.16 = 0.63\,\mathrm{s}.$$

Solution 1.33. Let the distances of the two stars M and m from the centre of mass before the explosion be r_M and r_m respectively. Then

$$r_M + r_m = d \tag{1}$$

$$Mr_M = mr_m \tag{2}$$

From (1) and (2) we obtain

$$r_M = \frac{m}{M+m}\,d \tag{3}$$

$$r_m = \frac{M}{M+m}\,d \tag{4}$$

The two stars revolve about the centre of mass with the same angular velocity. Let it be ω; then

$$\frac{GMm}{d^2} = M\omega^2 r_M,$$

that is,

$$Gm = r_M\omega^2 d^2. \tag{5}$$

Since the explosion of the star of mass M is instantaneous and spherically symmetric, the velocity of the remaining part of this star does not change at that instant. In this way, for the system formed by the remaining part of the star and the star of mass m, the centre of mass has velocity v_C given by

$$v_C = \frac{m\omega r_m - (M - \Delta M)\omega r_M}{M - \Delta M + m} = \frac{\Delta M\, r_M\omega}{M - \Delta M + m}.$$

The kinetic energy E_{kC} of the centre of mass is

$$E_{kC} = \frac{1}{2}(M - \Delta M + m)v_C^2 = \frac{\Delta M^2 r_M^2\omega^2}{2(M - \Delta M + m)}. \tag{6}$$

The total kinetic energy E_k of the system at this time is

$$E_k = \frac{1}{2}(M - \Delta M)(r_M\omega)^2 + \frac{1}{2}m(r_m\omega)^2. \tag{7}$$

The potential energy U of the system at this time is

$$U = -\frac{G(M-\Delta M)m}{d}. \tag{8}$$

Taking the centre of mass as the reference frame, the mechanical energy E' of the system in the centre-of-mass frame is

$$E' = E_k - E_{kC} + U. \tag{9}$$

Substituting (6), (7) and (8) into (9), we obtain

$$E' = \frac{1}{2}(M-\Delta M)(r_M\omega)^2 + \frac{1}{2}m(r_m\omega)^2 - \frac{\Delta M^2 r_M^2\omega^2}{2(M-\Delta M+m)} - \frac{G(M-\Delta M)m}{d}.$$

Substituting (5) into the above expression, we obtain

$$E' = -\frac{1}{2}r_M\omega^2\left(Md - 2\Delta Md + \Delta M r_M + \frac{(\Delta M)^2 r_M}{M-\Delta M+m}\right). \tag{10}$$

If the mechanical energy of the system in the centre-of-mass frame is less than zero, i.e. $E' < 0$, then the binary star will be bound and will not separate from each other; hence

$$Md - 2\Delta Md + \Delta M r_M + \frac{(\Delta M)^2 r_M}{M-\Delta M+m} > 0. \tag{11}$$

Substituting (3) into (11), we obtain

$$\Delta M < \frac{1}{2}(M+m).$$

Comment. Before the supernova explosion the centre of mass of the system is at rest. Although in the explosion the mass ΔM lost by the star of mass M flies apart spherically symmetrically, so that the instantaneous velocity of the remaining part of this star does not change, the system formed by the remainder of the star M after the explosion and the star m is not the original system. The centre of mass of this new system is not at rest, but begins to move uniformly with velocity v_C; this point is easily overlooked. In addition, in solving this problem use is made of König's theorem, i.e. that the total kinetic energy of a system equals the sum of the kinetic energy of the centre of mass and the kinetic energy of the system in the centre-of-mass frame.

Solution 1.34. Let the remaining nucleus m_3 be emitted with velocity v_3, and let the angle between the direction of v_3 and the direction of v_1 be θ, as shown in the figure. Take the direction of v_1 as the x direction and the direction of v_2 as the y direction. From the conservation of momentum of the system in the x direction and in the y direction we obtain respectively

$$m_1 v_1 = m_3 v_3 \cos\theta \tag{1}$$
$$m_2 v_2 = m_3 v_3 \sin\theta \tag{2}$$

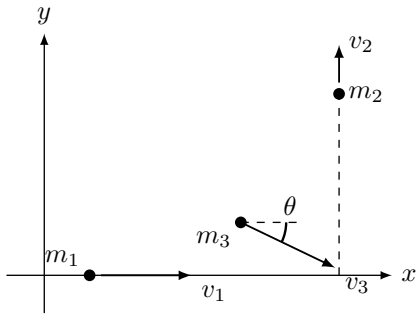


Fig. 1.34

Squaring both sides of (1) and (2) respectively and then adding the two equations, we obtain

$$m_1^2 v_1^2 + m_2^2 v_2^2 = m_3^2 v_3^2.$$

The above expression may also be rewritten as

$$\frac{1}{2} m_3 v_3^2 = \frac{1}{2} \frac{m_1^2}{m_3} v_1^2 + \frac{1}{2} \frac{m_2^2}{m_3} v_2^2. \tag{3}$$

Since in this nuclear reaction a quantity of non-mechanical energy of magnitude Q is converted into mechanical energy, the energy relation of this reaction is

$$\frac{1}{2} m_1 v_1^2 + Q = \frac{1}{2} m_2 v_2^2 + \frac{1}{2} m_3 v_3^2. \tag{4}$$

Substituting (3) into (4), we obtain

$$\frac{1}{2} m_1 v_1^2 + Q = \frac{1}{2} m_1 v_1^2 \frac{m_1}{m_3} + \frac{1}{2} m_2 v_2^2 \left(1 + \frac{m_2}{m_3}\right). \tag{5}$$

From (5) we immediately obtain the kinetic energy E_k of the particle m_2 as

$$E_k = \frac{1}{2} m_2 v_2^2 = \frac{m_3}{m_2 + m_3} \left[Q + \frac{1}{2} m_1 v_1^2 \left(\frac{m_3 - m_1}{m_3} \right) \right].$$

Solution 1.35. (1) Let the kinetic energies of the three blocks 1, 2 and 3 after the explosion be T_1, T_2 and T_3 respectively, and let their momenta be p_1, p_2 and p_3 respectively. The total kinetic energy of the three blocks is known to be Q; hence

$$T_1 + T_2 + T_3 = Q. \tag{1}$$

From the conservation of momentum of the system we obtain

$$\vec{p}_1 + \vec{p}_2 + \vec{p}_3 = 0. \tag{2}$$

The vector relation expressed by (2) is shown in Fig. 1.35(a). From the figure we obtain

$$p_1^2 = p_2^2 + p_3^2 - 2 p_2 p_3 \cos\theta_{23},$$

that is,

$$\cos\theta_{23} = \frac{p_2^2 + p_3^2 - p_1^2}{2 p_2 p_3}. \tag{3}$$

Since $|\cos\theta_{23}| \leq 1$, equation (3) may be written as

$$\left(\frac{p_2^2 + p_3^2 - p_1^2}{2p_2p_3}\right)^2 \leq 1. \tag{4}$$

Substituting the relation between momentum and kinetic energy $T = p^2/2m$ into (4), we obtain

$$\frac{(T_2 + T_3 - T_1)^2}{4T_2T_3} \leq 1,$$

that is,

$$(T_2 + T_3 - T_1)^2 \leq 4T_2T_3. \tag{5}$$

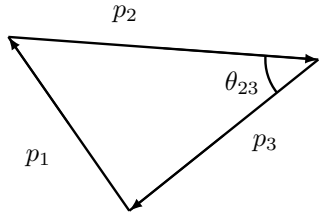


Fig. 1.35(a)

Any point inside the equilateral triangle shown in Fig. 1.35(b) satisfies (1); the points that can also satisfy (2), i.e. that satisfy (5), are the physically admissible points. Take an arbitrary point P inside this triangle, and denote its position by the polar coordinates (r, θ), taking the centroid O of the triangle as the pole of the polar coordinates and OA as the polar axis. Since $OA = \frac{2}{3}Q$, from the figure we obtain

$$T_1 = \frac{Q}{3} + r\cos\theta \tag{6}$$

$$T_2 = \frac{Q}{3} - r\cos(60^\circ + \theta) \tag{7}$$

$$T_3 = \frac{Q}{3} - r\cos(60^\circ - \theta) \tag{8}$$

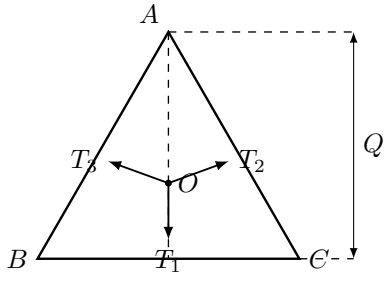


Fig. 1.35(b)

Substituting (6), (7) and (8) into (5), we obtain

$$\left(\frac{Q}{3} - 2r\cos\theta\right)^2 \leq \frac{4}{9}Q^2 - \frac{4}{3}Qr\cos\theta + 4r^2\cos^2\theta - 3r^2. \tag{9}$$

From (9) we obtain

$$r \leq \frac{Q}{3}.$$

Thus the boundary of the region of physically admissible points inside the equilateral triangle is the circumference of the inscribed circle. The admissible region is the region inside this circle, i.e. the shaded region shown in Fig. 1.35(c).

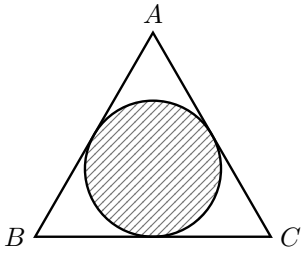


Fig. 1.35(c)

(2) For the case in which an electron-positron pair decays into three γ photons, equations (1), (2), (3) and (4) above all apply; only the speed of a photon is c, and the relation between its energy and its momentum must be the relativistic one. Since a photon has no rest mass, its kinetic energy T is just the energy E of the photon; hence

$$T = E = cp. \tag{10}$$

Thus (4) becomes

$$\left| \frac{T_2^2 + T_3^2 - T_1^2}{2T_2T_3} \right| \leq 1,$$

that is,

$$T_2^2 + T_3^2 - T_1^2 \leq 2T_2T_3,$$

or

$$(T_2 - T_3)^2 \leq T_1^2. \tag{11}$$

Equation (11) has two solutions:

$$T_2 \leq T_1 + T_3 \tag{12}$$
$$T_3 \leq T_1 + T_2 \tag{13}$$

Similarly we obtain

$$T_1 \leq T_2 + T_3. \tag{14}$$

Any one of (12), (13) and (14), for example (12), may also be written as

$$2T_2 \leq T_1 + T_2 + T_3 = Q,$$

so that

$$T_2 \leq \frac{Q}{2}. \tag{15}$$

From (13) and (14) we likewise obtain

$$T_3 \leq \frac{Q}{2} \tag{16}$$

$$T_1 \leq \frac{Q}{2} \tag{17}$$

Equations (15), (16) and (17) are the conditions that a physically admissible point must satisfy; inside the equilateral triangle the region of physically admissible points is the interior of the triangle joining the midpoints of the sides together with its three sides, i.e. the shaded region shown in Fig. 1.35(d).

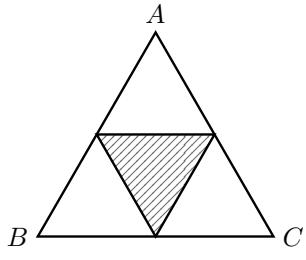


Fig. 1.35(d)

Solution 1.36. Let the length of the string from the ball to the hole at the moment when the ball leaves the conical surface be r, and let the speed of the ball be v. Since none of the external forces acting on the ball has a vertical component of moment about point O, the angular momentum of the ball about the axis of symmetry of the cone (i.e. the vertical axis through the small hole O) is conserved; hence

$$mv_0 r_0 \sin\alpha = mvr \sin\alpha.$$

That is,

$$v = \frac{r_0}{r} v_0. \tag{1}$$

When the ball leaves the conical surface, the forces acting on it are as shown in the figure. Hence

$$T\cos\alpha = mg \tag{2}$$

$$T\sin\alpha = m\frac{v^2}{r\sin\alpha} \tag{3}$$

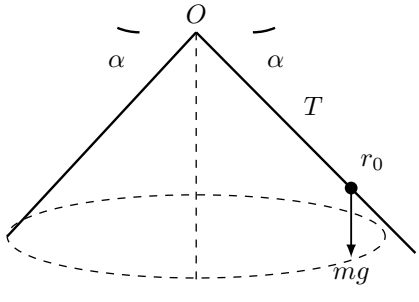


Fig. 1.36

From (2) and (3) we obtain

$$v = \sin\alpha\sqrt{\frac{gr}{\cos\alpha}}. \tag{4}$$

Comparing (1) and (4), we obtain

$$\begin{aligned} r &= \sqrt[3]{\frac{v_0^2 r_0^2 \cos\alpha}{g\sin^2\alpha}} = \sqrt[3]{\frac{\frac{15}{16}\times\frac{1}{4}\times\frac{1}{2}}{9.8\times\frac{3}{4}}} \\ &= 0.25\,\mathrm{m}, \end{aligned} \tag{5}$$

$$\begin{aligned} v &= v_0 r_0 \sqrt[3]{\frac{g\sin^2\alpha}{v_0^2 r_0^2\cos\alpha}} = \frac{\sqrt{15}}{4}\times\frac{1}{2}\times\sqrt[3]{\frac{9.8\times\frac{3}{4}}{\frac{15}{16}\times\frac{1}{4}\times\frac{1}{2}}} \\ &= 1.92\,\mathrm{m/s}. \end{aligned} \tag{6}$$

From the principle of work and energy we can find the work W done by the force pulling the string during this process:

$$W = \frac{1}{2}m(v^2 - v_0^2) + mg(r_0 - r)\cos\alpha. \tag{7}$$

Substituting (5), (6) and the data into (7), we obtain

$$W = 2.61\,\mathrm{J}.$$

Solution 1.37. After the lead bullet is shot into the small ball B and remains inside it, the spring begins to rotate about the centre of mass with an initial angular velocity ω_0, while the two small balls vibrate along the direction of the spring relative to the centre of mass, and the centre of mass C of the system always moves in a straight line with uniform velocity. Let the distance between the balls A, B and the centre of mass C be l_A and l_B respectively when the length of the spring is l; then

$$ml_A = 2ml_B \tag{1}$$

$$l_A + l_B = l \tag{2}$$

From (1) and (2) we obtain

$$l_A = \frac{2}{3}l$$

$$l_B = \frac{1}{3}l$$

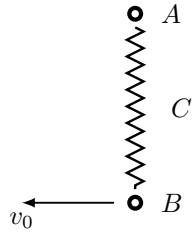


Fig. 1.37

It is thus seen that when the spring is at its natural length a and when its length has become $2a$, the distances of the balls A and B from the centre of mass are $\frac{2}{3}a$, $\frac{1}{3}a$ and $\frac{4}{3}a$, $\frac{2}{3}a$ respectively. Before and after the lead bullet is shot into the small ball B, the angular momentum of the system about the spatial point C at which the centre of mass is located at that instant is conserved; hence

$$mv_0 \cdot \frac{1}{3}a = \left[2m\left(\frac{a}{3}\right)^2 + m\left(\frac{2a}{3}\right)^2\right]\omega_0. \tag{3}$$

From (3) we obtain

$$\omega_0 = \frac{v_0}{2a}. \tag{4}$$

Take the centre of mass as the reference frame. In the centre-of-mass frame the angular momentum of the system about the centre of mass is conserved and the mechanical energy is conserved. Let the angular velocity of rotation be ω when the length of the spring reaches its maximum value $2a$; then respectively

$$mv_0 \cdot \frac{1}{3}a = \left[2m\left(\frac{2a}{3}\right)^2 + m\left(\frac{4a}{3}\right)^2\right]\omega, \tag{5}$$

$$\frac{1}{2}\left[2m\left(\frac{a}{3}\right)^2 + m\left(\frac{2a}{3}\right)^2\right]\omega_0^2 = \frac{1}{2}\left[2m\left(\frac{2a}{3}\right)^2 + m\left(\frac{4a}{3}\right)^2\right]\omega^2 + \frac{1}{2}ka^2. \tag{6}$$

Solving (4), (5) and (6) simultaneously, we obtain

$$v_0 = 2a\sqrt{\frac{2k}{m}}.$$

Comment. This problem discusses the conservation of angular momentum and the conservation of mechanical energy of a system in the centre-of-mass frame. Since the centre of mass of the system always moves in a straight line with uniform velocity, the centre-of-mass frame is an inertial frame, and by the principle of relativity of mechanics its laws of mechanics are the same as in any inertial frame. However, the centre-of-mass frame is a special reference frame: even if the centre-of-mass frame were a non-inertial frame, in solving problems by means of the principle of work and energy, the theorem of angular momentum and so on, one need not take the action of inertial forces into account, i.e. it is equivalent to an ordinary inertial frame.

Solution 1.38. The orbits of the two comets are both parabolas; hence the mechanical energy of the system formed by each comet together with the Sun is zero. Let the speeds of the two comets just before their collision, when they have moved to a distance R from the Sun, be v_1 and v_2 respectively; then

$$\frac{1}{2}m_1v_1^2 - \frac{GM_sm_1}{R} = 0 \tag{1}$$

$$\frac{1}{2}m_2v_2^2 - \frac{GM_sm_2}{R} = 0 \tag{2}$$

From (1) and (2) we obtain

$$v_1 = v_2 = \sqrt{\frac{2GM_s}{R}}. \tag{3}$$

The momentum of the system formed by the two comets is conserved before and after their collision, that is,

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2)v. \tag{4}$$

In (4), v is the velocity with which the two comets combine into a single composite celestial body after the collision. Since $v_1 \perp v_2$, as shown in the figure, we have

$$(m_1 v_1)^2 + (m_2 v_2)^2 = (m_1 + m_2)^2 v^2.$$

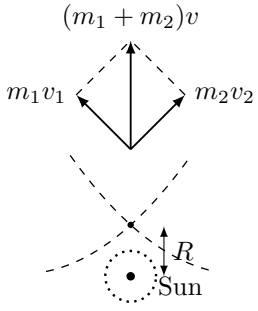


Fig. 1.38

we immediately obtain

$$v^2 = \frac{m_1^2 + m_2^2}{(m_1 + m_2)^2} \frac{2GM_s}{R}. \tag{5}$$

Since the collision is a perfectly inelastic one, mechanical energy is lost in the collision. Therefore the mechanical energy of the system formed by the composite celestial body and the Sun after the collision must be less than zero. Let the mass of the composite body be $M(= m_1 + m_2)$; then

$$E = \frac{1}{2}Mv^2 - \frac{GM_s M}{R} < 0.$$

From this it is seen that the orbit of the composite celestial body must be an elliptical orbit with the Sun as focus. Since the semi-major axis a of an elliptical orbit is determined by the energy, whereas the semi-minor axis b is determined jointly by the energy and the angular momentum, and since from the given conditions of this problem it is impossible to determine the angle between the velocity of the composite body after the collision of the two comets and the radius vector at that place, i.e. it is impossible to determine the angular momentum, the semi-minor axis b of the elliptical orbit cannot be determined. Since the semi-major axis a is determined by the energy E alone, in order to find a we may as well assume that the velocity of the composite celestial body after the collision is perpendicular to the radius vector at that place, i.e. that the point of collision is the perihelion (or aphelion) of the elliptical orbit. Let the radius vector and the velocity of this composite body be r and v' when it moves to the aphelion (or perihelion); then from the conservation of angular momentum of the system about the centre of the Sun and from the conservation of mechanical energy we obtain respectively

$$Mv'r = MvR \tag{6}$$

$$\frac{1}{2}Mv'^2 - \frac{GM_s M}{r} = \frac{1}{2}Mv^2 - \frac{GM_s M}{R} \tag{7}$$

From (5), (6) and (7) we obtain

$$r = \frac{m_1^2 + m_2^2}{2m_1 m_2}R.$$

From this the semi-major axis a of the elliptical orbit is found to be

$$a = \frac{1}{2}(r + R) = \frac{(m_1 + m_2)^2}{4m_1 m_2} R.$$

That is, the orbit of the composite celestial body after the collision of the two comets is an elliptical orbit with the Sun as focus and semi-major axis $\dfrac{(m_1 + m_2)^2}{4m_1 m_2} R$.

The semi-major axis a may also be found from the following formula:

$$v^2 = k^2 \left(\frac{2}{r} - \frac{1}{a} \right), \tag{8}$$

where $k^2 = GM_s$ and $r = R$. Substituting (5) into (8), we immediately obtain

$$a = \frac{2k^2 r}{k^2 - v^2 r} = \frac{(m_1 + m_2)^2}{4m_1 m_2} R.$$

Solution 1.39. Let the speed with which the space shuttle together with the lunar module revolves about the Moon in a circular orbit be v_0; then

$$\frac{GM_m(M + m)}{(3R_m)^2} = (M + m)\frac{v_0^2}{3R_m}. \tag{1}$$

In (1), M_m is the mass of the Moon. From (1) we obtain

$$v_0 = \sqrt{\frac{GM_m}{3R_m}}. \tag{2}$$

Its period T_0 of revolution about the Moon is

$$T_0 = \frac{2\pi(3R_m)}{v_0} = 6\pi R_m \sqrt{\frac{3R_m}{GM_m}}. \tag{3}$$

Since the gravitational acceleration at the surface of the Moon g_m is

$$g_m = \frac{GM_m}{R_m^2}, \tag{4}$$

substituting (4) into (3) and inserting the data, we obtain

$$\begin{aligned} T_0 &= 6\pi R_m \sqrt{\frac{3}{g_m R_m}} = 6 \times 3.14 \times 1.74 \times 10^6 \times \sqrt{\frac{3}{1.62 \times 1.74 \times 10^6}} \\ &= 33812\,\mathrm{s} \approx 9.4\,\mathrm{h}. \end{aligned} \tag{5}$$

Let the velocities of the lunar module and the space shuttle immediately after their separation be v_1 and v_2 respectively; from the conservation of momentum,

$$(M + m)v_0 = mv_1 + Mv_2. \tag{6}$$

After the lunar module and the space shuttle have separated, the two move along their respective elliptical orbits, as shown in the figure. Let the speed of the lunar module when it reaches the

surface of the Moon be v_1'; then from the conservation of angular momentum about the centre of the Moon and the conservation of mechanical energy we obtain respectively

$$3R_m m v_1 = R_m m v_1' \tag{7}$$

$$\frac{1}{2}mv_1^2 - \frac{GM_m m}{3R_m} = \frac{1}{2}mv_1'^2 - \frac{GM_m m}{R_m}. \tag{8}$$

Solving (7) and (8) simultaneously, we obtain

$$v_1 = \sqrt{\frac{GM_m}{6R_m}} = \frac{\sqrt{2}}{2}v_0. \tag{9}$$

Substituting (9) into (6), we obtain

$$v_2 = v_0 + \frac{m}{M}\left(1 - \frac{\sqrt{2}}{2}\right)v_0. \tag{10}$$

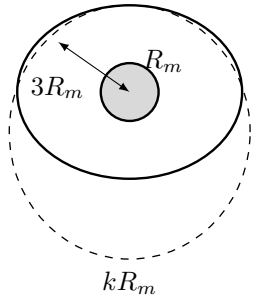


Fig. 1.39

Let the speed of the space shuttle be v_2' when it moves to its farthest point from the Moon, and let the distance of this point from the centre of the Moon be kR_m. Likewise, from the conservation of angular momentum of the space shuttle about the centre of the Moon and the conservation of mechanical energy we obtain respectively

$$3R_m M v_2 = kR_m M v_2' \tag{11}$$

$$\frac{1}{2}Mv_2^2 - \frac{GM_m M}{3R_m} = \frac{1}{2}Mv_2'^2 - \frac{GM_m M}{kR_m}. \tag{12}$$

Solving (11) and (12) simultaneously, we obtain

$$k = \frac{3}{2\left(\dfrac{v_0}{v_2}\right)^2 - 1}. \tag{13}$$

Substituting (10) into (13) and inserting $\dfrac{M}{m} = 2$, we obtain

$$k \approx 5.75.$$

Thus the semi-major axis a_M of the elliptical orbit of the space shuttle is

$$a_M = \frac{1}{2}(k+3)R_m \approx 4.38R_m,$$

while the semi-major axis a_m of the elliptical orbit of the lunar module is

$$a_m = \frac{1}{2}(3+1)R_m = 2R_m.$$

According to Kepler's third law, the ratios of the period T_m of the lunar module and the period T_M of the space shuttle to T_0 are found respectively to be

$$\frac{T_m}{T_0} = \left(\frac{a_m}{3R_m}\right)^{\frac{3}{2}} = \left(\frac{2}{3}\right)^{\frac{3}{2}} = 0.54,$$

$$\frac{T_M}{T_0} = \left(\frac{a_M}{3R_m}\right)^{\frac{3}{2}} = \left(\frac{4.38}{3}\right)^{\frac{3}{2}} = 1.76.$$

That is,

$$T_m = 0.54T_0 \tag{14}$$

$$T_M = 1.76T_0 \tag{15}$$

In order that the lunar module may return along its original elliptical orbit to the point of separation and dock with the space shuttle, the time Δt during which it may stay on the Moon is

$$\Delta t = (n+1)T_M - T_m \qquad (n = 0, 1, 2, \dots). \tag{16}$$

Substituting (14), (15) and (5) into (16), we obtain

$$\Delta t = (1.76n + 1.22) \times 9.4\,\mathrm{h} \qquad (n = 0, 1, 2, \dots).$$

From this expression it is seen that the shortest time during which the lunar module may stay on the surface of the Moon is

$$\Delta t_{\min} = 11.5\,\mathrm{h}.$$

Solution 1.40. (1) The gravitational force exerted on the bullet by the gas cloud is

$$\vec{F}(r) = \begin{cases} -\dfrac{GMmr}{R^3}\,\hat{r}, & (r \le R) \\ -\dfrac{GMm}{r^2}\,\hat{r}, & (r > R) \end{cases} \tag{1}$$

If the gravitational potential energy is taken to be zero at infinity, the gravitational potential energy is found from the following integral:

$$U(r) = \int_r^\infty \vec{F}(r) \cdot \mathrm{d}\vec{l} = \int_r^\infty F(r)\mathrm{d}r. \tag{2}$$

In the region $r \le R$,

$$\begin{aligned} U(r) &= \int_r^R \left(-\frac{GMmr}{R^3}\right)\mathrm{d}r + \int_R^\infty \left(-\frac{GMm}{r^2}\right)\mathrm{d}r \\ &= -\frac{GMm}{2R^3}\left(3R^2 - r^2\right). \end{aligned} \tag{3}$$

In the region $r > R$,

$$U(r) = \int_R^\infty \left(-\frac{GMm}{r^2}\right)\mathrm{d}r = -\frac{GMm}{r}. \tag{4}$$

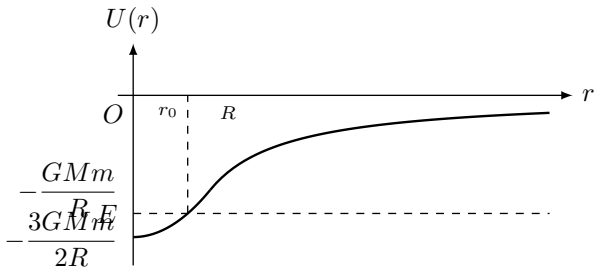


Fig. 1.40(a)

Fig. 1.40(a) is the potential-energy curve. From the statement of the problem, the total energy of the bullet is $E = -\dfrac{5GMm}{4R}$, and the region in which the bullet can appear must satisfy $E \geq U$. From the figure it is seen that the bullet can appear only in the region $r \leq r_0$, and $r_0 < R$; that is to say, the bullet is always inside the cloud.

(2) The effective potential energy $U_{\text{eff}}(r)$ of the bullet is

$$U_{\text{eff}}(r) = U(r) + \frac{L^2}{2mr^2}. \tag{5}$$

In (5), L is the angular momentum of the bullet about the centre of the gas cloud, $L = m\sqrt{\dfrac{GMR}{32}}$. Substituting this expression together with (3) into (5), we obtain

$$U_{\text{eff}}(r) = -\frac{GMm}{2R^3}\left(3R^2 - r^2\right) + \frac{GMmR}{64r^2}. \tag{6}$$

The mechanical energy E of the bullet is

$$E = \frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r). \tag{7}$$

At the turning points of the orbit, $\dot{r} = 0$, hence

$$E = -\frac{5GMm}{4R} = U_{\text{eff}}(r) = -\frac{GMm\left(3R^2 - r^2\right)}{2R^3} + \frac{GMmR}{64r^2}. \tag{8}$$

From (8) we obtain the equation

$$32\left(\frac{r}{R}\right)^4 - 16\left(\frac{r}{R}\right)^2 + 1 = 0. \tag{9}$$

The solutions of this quadratic equation are

$$\left(\frac{r}{R}\right)^2 = \frac{2 \pm \sqrt{2}}{8}, \qquad \text{that is,} \qquad r_1 = \sqrt{\frac{2+\sqrt{2}}{8}}\,R, \quad r_2 = \sqrt{\frac{2-\sqrt{2}}{8}}\,R.$$

At a turning point of the orbit the distance of the bullet from the centre of the gas cloud is $r_1 = \sqrt{\dfrac{2+\sqrt{2}}{8}}\,R$ or $r_2 = \sqrt{\dfrac{2-\sqrt{2}}{8}}\,R$.

(3) From the discussion and calculation of part (2) it is known that equation (9) has two solutions r_1 and r_2, i.e. in the effective-potential-energy curve of the bullet shown in Fig. 1.40(b)

the mechanical energy $E = -\dfrac{5GMm}{4R}$ of the bullet has two points of intersection with the effective potential-energy curve. Therefore the motion of the bullet is an elliptical orbit, whose semi-major axis a and semi-minor axis b are respectively

$$a = r_1 = \frac{1}{2}\sqrt{1 + \frac{\sqrt{2}}{2}}\, R,$$

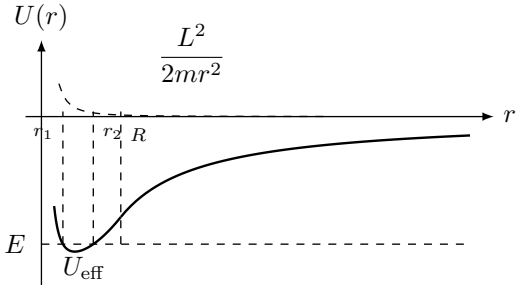


Fig. 1.40(b)

and

$$b = r_2 = \frac{1}{2}\sqrt{1 - \frac{\sqrt{2}}{2}}\, R.$$

The shape is roughly as shown in Fig. 1.40(c). It is seen that the orbit has four turning points in all. Since the area S of the ellipse is

$$S = \pi ab = \frac{\sqrt{2}}{8}\pi R^2,$$

and since the angular momentum of the bullet about the centre of the gas cloud is conserved, its areal velocity k is constant:

$$k = \frac{\mathrm{d}S}{\mathrm{d}t} = \frac{L}{2m} = \frac{1}{2}\sqrt{\frac{GMR}{32}}.$$

Therefore the period T of the motion of the bullet in its orbit is

$$T = \frac{S}{k} = \frac{2\pi R^{3/2}}{\sqrt{GM}}.$$

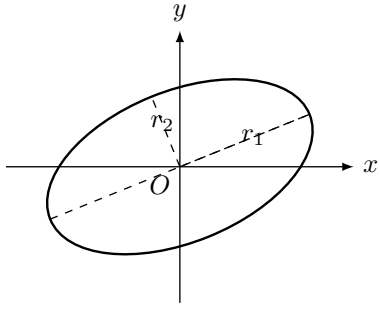


Fig. 1.40(c)

Comment. When a particle moves in a one-dimensional potential field, the potential-energy curve is often used to discuss its motion; and when a particle moves in two dimensions (i.e. with the two variables r and θ) in the gravitational field of a fixed centre of force, the effective potential energy $U_{\text{eff}}(r)$ is often introduced in order to discuss conveniently the range of variation of its radius vector r. The effective potential energy consists of two parts: one is the gravitational potential energy $U(r)$, the other is the transverse kinetic energy $\frac{1}{2}m(r\dot{\theta})^2$ contained in its mechanical energy. Since the angular momentum of the particle about the fixed centre of force is conserved, i.e. $L = mr^2\dot{\theta} = C$, the $\dot{\theta}$ in the expression for the transverse kinetic energy may be replaced by $\dfrac{L}{mr^2}$; hence

$$U_{\text{eff}}(r) = U(r) + \frac{L^2}{2mr^2}.$$

It is seen that the effective potential energy is a function of r alone, and the mechanical energy E of the system is also a function of r alone, namely

$$E = \frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r).$$

This is analogous to the motion of a particle in one dimension (a single variable r); once the effective potential-energy curve is drawn, the range of the radial motion of the particle can be discussed very conveniently.

Solution 1.41. (1) According to scheme 1, the process in which a spaceship launched from the ground escapes the solar system directly may be divided into two stages. The first stage is the stage of first escaping the Earth's gravitational field. In order that the launch speed be the smallest, the Earth's orbital velocity should be used as far as possible in the solar system; hence the spaceship should be launched in the direction of the Earth's orbital velocity. When the spaceship has reached a place sufficiently far from the Earth, that is, when it has already escaped the Earth's gravitational field, it may be considered that the spaceship is still approximately in the Earth's orbit around the Sun, and at this time the spaceship still has a certain speed relative to the Earth; during this stage the gravitational potential energy due to the Sun may be regarded as unchanged. The second stage is the process in which the spaceship, after escaping the Earth's gravity, continues to escape from the Sun until it flies to an infinite distance from the Sun.

Taking the Earth as the frame of reference, let the speed of the spaceship relative to the Earth after it has escaped the Earth's gravity be v_{a1}; then from the conservation of mechanical energy we have

$$\frac{1}{2}mv_a^2 - \frac{GM_em}{R_e} = \frac{1}{2}mv_{a1}^2 \tag{1}$$

In (1), M_e and R_e are respectively the mass and the radius of the Earth. Now take the Sun as the frame of reference. By the statement of the problem the Earth moves in a circular orbit around the Sun; let the orbital speed of the Earth be v_e, the mass of the Sun be M_s, and the radius of the Earth's orbit be r_{se}; then

$$\frac{GM_sM_e}{r_{se}^2} = M_e\frac{v_e^2}{r_{se}},$$

that is,

$$v_e = \sqrt{\frac{GM_s}{r_{se}}} \tag{2}$$

In the frame of the Sun, the velocity v_{a2} with which the spaceship leaves the Earth's gravitational field is

$$v_{a2} = v_{a1} + v_e \tag{3}$$

For the spaceship on the Earth's orbit to continue to escape from the Sun, the mechanical energy of the system formed by it and the Sun must be greater than or equal to zero, that is,

$$\frac{1}{2}mv_{a2}^2 - \frac{GM_s m}{r_{se}} \geq 0.$$

The minimum v_{a2} corresponds to taking the equality sign in the above expression; hence

$$v_{a2} = \sqrt{\frac{2GM_s}{r_{se}}} = \sqrt{2}\, v_e \tag{4}$$

Substituting (4) into (3), we obtain

$$v_{a1} = (\sqrt{2} - 1)v_e = 12.3 \text{ (km/s)} \tag{5}$$

Substituting (5) into (1), we obtain

$$v_a = \sqrt{v_{a1}^2 + 2G\frac{M_e}{R_e}} = 16.4 \text{ (km/s)}.$$

(2) In the frame of the Earth, let the launch speed of the spaceship at the Earth's surface be v_b, and let its speed when it leaves the Earth's gravity be v_{b1}; then from the conservation of mechanical energy,

$$\frac{1}{2}mv_b^2 - \frac{GM_e m}{R_e} = \frac{1}{2}mv_{b1}^2 \tag{6}$$

In the frame of the Sun, the velocity v_{b2} of the spaceship at this time is

$$v_{b2} = v_{b1} + v_e \tag{7}$$

Thereafter, while the spaceship moves from the Earth's orbit to the orbit of Mars, it is acted on by the Sun's gravitation alone. When the spaceship reaches the orbit of Mars, let its velocity be v $(= v_r + v_\theta)$, as shown in the figure. Let the radius of the orbit of Mars be r_{sm}; then from the conservation of the angular momentum of the spaceship about the centre of the Sun we obtain

$$mv_{b2}r_{se} = mv_\theta r_{sm} \tag{8}$$

From the conservation of mechanical energy,

$$\frac{1}{2}mv_{b2}^2 - \frac{GM_s m}{r_{se}} = \frac{1}{2}m\left(v_r^2 + v_\theta^2\right) - \frac{GM_s m}{r_{sm}} \tag{9}$$

Substituting (6), (7) and (8) into (9), and using $v_e = \sqrt{GM_s/r_{se}}$ and $g = GM_e/R_e^2$, we obtain

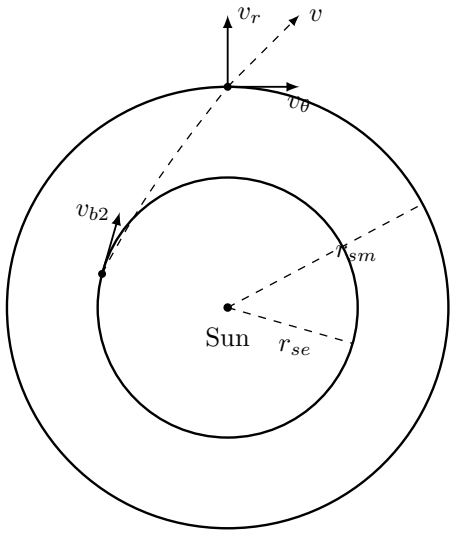


Fig. 1.41(a)

$$v_r^2 = v_{b2}^2 - v_\theta^2 - \frac{2GM_s}{r_{se}} + \frac{2GM_s}{r_{sm}}$$
$$= (v_{b1} + v_e)^2 \left[1 - \left(\frac{r_{se}}{r_{sm}}\right)^2\right] - 2v_e^2 \left(1 - \frac{r_{se}}{r_{sm}}\right)$$

Hence

$$v_r = \left[\left(\sqrt{v_b^2 - 2R_e g} + v_e\right)^2 \left(1 - \frac{r_{se}^2}{r_{sm}^2}\right) - 2v_e^2 \left(1 - \frac{r_{se}}{r_{sm}}\right)\right]^{\frac{1}{2}} \tag{10}$$

$$v_\theta = \frac{r_{se}}{r_{sm}} \left(\sqrt{v_b^2 - 2R_e g} + v_e\right) \tag{11}$$

(3) According to scheme 2, the motion of the spaceship may be divided into three stages. The first stage is the same as the first stage of scheme 1, being the stage in which the spaceship escapes the Earth's gravity. The second stage is the stage in which, after escaping the Earth's gravity, the spaceship moves from the Earth's orbit to the orbit of Mars, while the third stage is the stage in which the spaceship enters the gravitational field of Mars somewhere on the orbit of Mars, changes its direction of motion under the action of Mars's gravity, and leaves the gravitational field of Mars at another point of the orbit of Mars along the tangential direction of that orbit, so that it can make use of the orbital velocity of Mars and continue to escape from the solar system with a greater speed relative to the Sun.

The discussion of the first and second stages is exactly the same as in part (2). When the spaceship flies to a certain point of the orbit of Mars, its speed relative to the Sun is v and its speed relative to Mars is v'; from the formula for relative motion we have

$$v' = v - v_m \tag{12}$$

In (12), v_m is the orbital velocity of Mars; (12) may also be expressed as

$$v'^2 = (v_\theta - v_m)^2 + v_r^2 \tag{13}$$

The orbital velocity v_m of Mars is obtained from

$$\frac{GM_sM_m}{r_{sm}^2} = M_m\frac{v_m^2}{r_{sm}}$$
$$v_m = \sqrt{\frac{GM_s}{r_{sm}}} \tag{14}$$

By the statement of the problem, when the spaceship leaves the gravitational field of Mars at another point of the orbit of Mars, the direction of its velocity is the tangential direction of the orbit, as shown in the figure. Taking Mars as the frame of reference, it is known from the conservation of mechanical energy that the speed with which it leaves the gravitational field of Mars is still v', that is, Hence, in the frame of the Sun, the velocity v_{b3} of the spaceship is

$$v_{b3} = v' + v_m \tag{15}$$

At the orbit of Mars the spaceship escapes from the solar system under the action of the Sun's gravitation alone, so the mechanical energy of the system formed by the spaceship and the Sun must be at least zero; hence

$$\frac{1}{2}mv_{b3}^2 - \frac{GM_sm}{r_{sm}} = 0,$$

that is,

$$v_{b3} = \sqrt{\frac{2GM_s}{r_{sm}}} = \sqrt{2}\,v_m \tag{16}$$

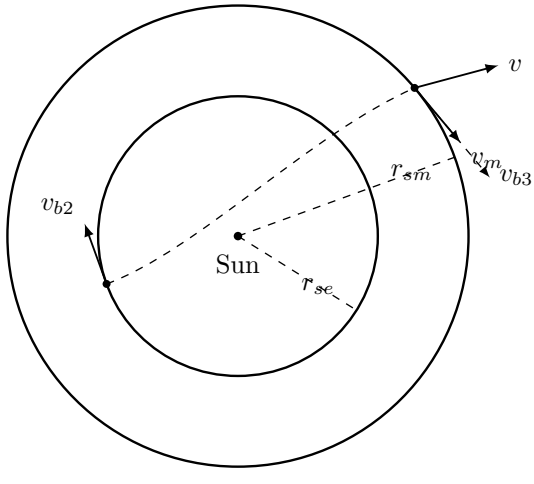


Fig. 1.41(b)

Solution 1.42.

Note. The page on which this solution begins is missing from the scan; the text below is its continuation. Equations (1)–(5) are not reproduced.

where $\delta(t) \ll r_0$. Adopting plane polar coordinates, the radial equation of motion of the planet is

$$-\frac{GMm}{r^2} - kmr = m\left(\ddot{r} - r\dot{\theta}^2\right) \tag{6}$$

Since the angular momentum of the planet about the centre of the star, $L(= mr^2\dot{\theta})$, is conserved,

$$\dot{\theta} = \frac{L}{mr^2} \tag{7}$$

Substituting (5) and (7) into (6), we obtain

$$-\frac{GM}{(r_0+\delta)^2} - k(r_0+\delta) = \ddot{\delta} - \frac{L^2}{m^2(r_0+\delta)^3}.$$

Since $\delta \ll r_0$, the term $(\delta/r_0)^2$ and the higher-order terms in the above expression may all be neglected, and the above expression becomes

$$\ddot{\delta} = -\frac{GM}{r_0^2}\left(1 - \frac{2\delta}{r_0}\right) - kr_0\left(1 + \frac{\delta}{r_0}\right) + \frac{L^2}{m^2 r_0^3}\left(1 - \frac{3\delta}{r_0}\right) \tag{8}$$

Substituting (4) into (8), we obtain

$$\ddot{\delta} + \left(\frac{L^2}{m^2 r_0^4} + 3k\right)\delta = 0 \tag{9}$$

This is the equation of motion of simple harmonic motion, and the angular frequency of the simple harmonic motion is

$$\omega_r = \sqrt{\frac{L^2}{m^2 r_0^4} + 3k} \tag{10}$$

When the planet moves in a circle of radius r_0 about the star, its angular velocity is $\omega_0 = \dfrac{L}{mr_0^2}$; if there is no cosmic dust in space, that is, if $\rho = 0$ and hence $k = 0$, then $\omega_r = \omega_0$, which shows that when the planet moves once around the star, its radial vibration in r also goes through exactly one period. In this process the value of r has two maxima and two minima, so the orbit of the planet is a stable ellipse. When cosmic dust is present, that is, when $\rho \neq 0$ and hence $k \neq 0$, it is known from (10) that $\omega_r > \omega_0$. This shows that the orbit of the planet is no longer a strictly elliptical one; but since k is very small, ω_r is only slightly larger than ω_0, so the shape of the orbit is still roughly an ellipse, only the ellipse does not close: the radial vibration of the planet is ahead of the circular motion by one phase, and the ellipse will precess, as shown in the figure. That is, if the planet moves anticlockwise about the star, the direction of precession will be clockwise. Let the angular velocity of precession be ω_p; then the time required for the axis of the ellipse to turn through one revolution is T_p, where

$$T_p = \frac{2\pi}{\omega_p} \tag{11}$$

and when the ellipse has precessed through one revolution, the phase difference between the radial vibration of the planet and its circular motion is 2π, that is,

$$(\omega_r - \omega_0)T_p = 2\pi \tag{12}$$

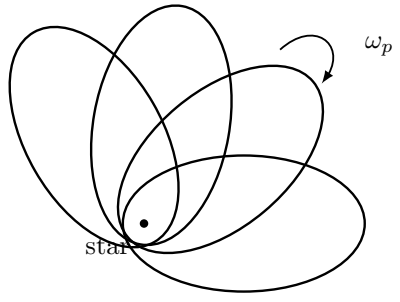


Fig. 1.42

From (11) and (12) we obtain

$$\begin{aligned}\omega_p = \omega_r - \omega_0 &= \sqrt{\frac{L^2}{m^2 r_0^4} + 3k} - \frac{L}{mr_0^2} \\ &= \frac{L}{mr_0^2}\left[\left(1 + \frac{3km^2 r_0^4}{L^2}\right)^{\frac{1}{2}} - 1\right] \\ &\approx \frac{L}{mr_0^2} \cdot \frac{3km^2 r_0^4}{2L^2} = \frac{3kmr_0^2}{2L}\end{aligned} \tag{13}$$

Substituting $k = \dfrac{4}{3}\pi G\rho$ and $L = mr_0^2\omega_0 = m\sqrt{GMr_0}$ into (13) gives

$$\omega_p = 2\pi\rho\sqrt{\frac{Gr_0^3}{M}}.$$

Comment. The motion of a planet about a star is a two-dimensional problem, and in plane polar coordinates the two variables r and θ are used to describe its motion. Using the conservation of the angular momentum of the planet about the centre of the star, $\dot{\theta}$ in its radial equation of motion is replaced by $\dfrac{L}{mr^2}$, so that the equation becomes a one-dimensional equation of motion with the single variable r. From the equation it is known that the small radial vibration of the planet is a simple harmonic vibration; comparing the angular frequency of the radial vibration with the angular frequency of the orbital motion, the conclusion that the elliptical orbit of the planet precesses can be obtained, and both the direction of the precession and the angular frequency of precession can be found.

Solution 1.43. (1) Under the action of the central attractive force $F = kr^N$ the particle moves in a circle of radius r_0; let its speed of motion be v_0; then

$$kr_0^N = \frac{mv_0^2}{r_0},$$

that is,

$$v_0 = \sqrt{\frac{kr_0^{N+1}}{m}} \tag{1}$$

When it suffers a small radial disturbance its motion will deviate slightly from the circular orbit, and its position vector r may be written as

$$r = r_0 + \delta \tag{2}$$

In (2), $\delta \ll r_0$. Adopting plane polar coordinates, its radial equation of motion is

$$-kr^N = m\left(\ddot{r} - r\dot{\theta}^2\right) \tag{3}$$

Since the force acting on the particle is a central force, the angular momentum L of the particle about the centre of force is conserved, L being

$$L = mr^2\dot{\theta} = mv_0 r_0 \tag{4}$$

that is,

$$\dot{\theta} = \frac{L}{mr^2} \tag{5}$$

Substituting (2) and (5) into (3), we obtain

$$-k(r_0+\delta)^N = m\left[\ddot{\delta} - \frac{L^2}{m^2(r_0+\delta)^3}\right].$$

Since $\delta \ll r_0$, neglecting $\left(\frac{\delta}{r_0}\right)^2$ and higher-order terms in the above expression, we obtain

$$\ddot{\delta} = -\frac{k}{m}r_0^N + \frac{L^2}{m^2 r_0^3} - \delta\left(\frac{k}{m}Nr_0^{N-1} + \frac{3L^2}{m^2 r_0^4}\right) \tag{6}$$

Substituting (1) into (4), we obtain

$$L = mv_0 r_0 = mr_0\sqrt{\frac{kr_0^{N+1}}{m}}.$$

That is, we have the equality

$$\frac{L^2}{m^2 r_0^3} - \frac{k}{m}r_0^N = 0 \tag{7}$$

Substituting (7) into (6), (6) becomes

$$\ddot{\delta} + \left(\frac{k}{m}Nr_0^{N-1} + \frac{3L^2}{m^2 r_0^4}\right)\delta = 0 \tag{8}$$

From (8) it is known that if the small radial disturbance is to be stable, then we must have

$$\frac{k}{m}Nr_0^{N-1} + \frac{3L^2}{m^2 r_0^4} > 0 \tag{9}$$

Substituting (4) into (9), we obtain

$$N > \frac{-\frac{3v_0^2}{r_0^2}}{\frac{k}{m}r_0^{N-1}} = -\frac{3mv_0^2}{kr_0^{N+1}} \tag{10}$$

Substituting (1) into (10), we find

$$N > -3.$$

(2) Equation (8) is the equation of motion of the small radial vibration of the particle, and from this equation it is known that the angular frequency ω of this vibration is

$$\omega = \sqrt{\frac{k}{m} N r_0^{N-1} + \frac{3L^2}{m^2 r_0^4}} \tag{11}$$

Substituting $L = mr_0 v_0 = mr_0 \sqrt{\frac{kr_0^{N+1}}{m}}$ into (11), we obtain

$$\omega = \sqrt{(N+3)\frac{k}{m} r_0^{N-1}} \tag{12}$$

Hence the period T is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{(N+3)kr_0^{N-1}}} \tag{12}$$

(3) The period T_0 of the motion of the particle in a circular orbit of radius r_0 is

$$T_0 = \frac{2\pi r_0}{v_0} = 2\pi \sqrt{\frac{m}{kr_0^{N-1}}} \tag{13}$$

If the period of the stable circular motion is to be an integral multiple of the period of the small radial vibration, then we have

$$T_0 = nT \qquad (n = 1, 2, 3, \cdots) \tag{14}$$

Substituting (12) and (13) into (14), we obtain

$$2\pi \sqrt{\frac{m}{kr_0^{N-1}}} = n \cdot 2\pi \sqrt{\frac{m}{(N+3)kr_0^{N-1}}}.$$

From the above expression we obtain

$$N = n^2 - 3 \qquad (n = 1, 2, 3, \cdots),$$

that is, when $N = -2, 1, 6, \cdots$, the period of the stable circular motion is $1, 2, 3, \cdots$ times the period of the small radial vibration.

Solution 1.44. (1) In the rotating frame which rotates about the axis with angular velocity ω_0, the forces acting on one of the balls m are as shown in the figure. The inertial centrifugal force is $f_c = m\omega_0^2 l \sin\theta$; hence

$$(T_1 - T_2)\cos\theta = mg \tag{1}$$

$$(T_1 + T_2)\sin\theta = f_c = m\omega_0^2 l \sin\theta \tag{2}$$

From (1) and (2) we obtain

$$T_2 = \frac{1}{2} m \left(\omega_0^2 l - \frac{mg}{\cos\theta}\right) \tag{3}$$

From the equilibrium of the forces on sleeve M we obtain

$$2T_2 \cos\theta = Mg \tag{4}$$

Substituting (3) into (4), we obtain

$$\cos\theta = \frac{(M+m)g}{m\omega_0^2 l} \tag{5}$$

Hence, when sleeve M is in equilibrium, the distance L between the upper and lower sleeves is

$$L = 2l\cos\theta = \frac{2(M+m)g}{m\omega_0^2}.$$

Of course, $\theta = 0$ should also be a possible equilibrium state of the system; in this case $L = 2l$.

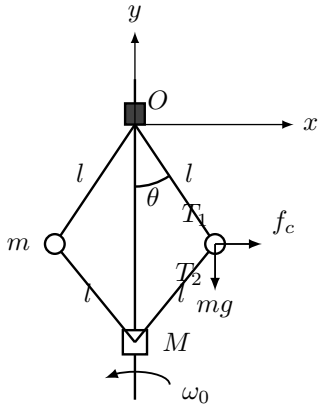


Fig. 1.44(b)

(2) *Method 1.* Set up the coordinate system shown in the figure in the rotating frame; then the coordinates x, y of the ball m and the coordinate Y of sleeve M are respectively

$$x = l\sin\theta, \qquad y = -l\cos\theta,$$

$$Y = -2l\cos\theta.$$

Hence

$$\dot{x} = l\cos\theta\,\dot{\theta}, \qquad \dot{y} = l\sin\theta\,\dot{\theta}, \qquad \dot{Y} = 2l\sin\theta\,\dot{\theta},$$

$$\ddot{x} = -l\sin\theta\,\dot{\theta}^2 + l\cos\theta\,\ddot{\theta},$$

$$\ddot{y} = l\cos\theta\,\dot{\theta}^2 + l\sin\theta\,\ddot{\theta},$$

$$\ddot{Y} = 2l\cos\theta\,\dot{\theta}^2 + 2l\sin\theta\,\ddot{\theta}.$$

Then the equations of motion of the ball m and of sleeve M are respectively

$$m\omega_0^2 l\sin\theta - (T_1 + T_2)\sin\theta = m\ddot{x} \tag{6}$$

$$(T_1 - T_2)\cos\theta - mg = m\ddot{y} \tag{7}$$

$$2T_2\cos\theta - Mg = M\ddot{Y} \tag{8}$$

From (6) and (7) we obtain

$$2T_2\cos\theta = m\cot\theta\left(\omega_0^2 l\sin\theta - \ddot{x}\right) - m(\ddot{y} + g) \tag{9}$$

From (8) and (9) we obtain

$$M(\ddot{Y}+g)=m\cot\theta\left(\omega_0^2 l\sin\theta-\ddot{x}\right)-m(\ddot{y}+g).$$

Substituting the expressions for $\ddot{x}$, $\ddot{y}$ and $\ddot{Y}$ into the above expression, we obtain

$$2Ml\cos\theta\,\dot{\theta}^2+\frac{(2M\sin^2\theta+m)l\ddot{\theta}}{\sin\theta}=m\omega_0^2 l\cos\theta-(M+m)g.$$

Let the system be in equilibrium at $\theta=\theta_0$; then after a small disturbance let $\theta=\theta_0+\theta'$, with $\theta'\ll\theta_0$, so that $\dot{\theta}=\dot{\theta}'$ and $\ddot{\theta}=\ddot{\theta}'$; since $\dot{\theta}^2=\dot{\theta}'^2$ is a second-order small quantity and is neglected, the above expression becomes

$$(2M\sin^2\theta+m)l\ddot{\theta}'=m\omega_0^2 l\cos\theta\sin\theta-(M+m)g\sin\theta.$$

Since $\sin\theta=\sin(\theta_0+\theta')\approx\sin\theta_0+\cos\theta_0\cdot\theta'$ and $\cos\theta=\cos(\theta_0+\theta')\approx\cos\theta_0-\sin\theta_0\cdot\theta'$, substitution into the above expression and neglect of second-order and higher-order small quantities give

$$(2M\sin^2\theta_0+m)l\ddot{\theta}'-\left[m\omega_0^2 l\cos 2\theta_0-(M+m)g\cos\theta_0\right]\theta'=0.$$

Substituting (5) into the above expression, we obtain

$$(2M\sin^2\theta_0+m)l\ddot{\theta}'+ml\omega_0^2\sin^2\theta_0\,\theta'=0.$$

Hence the frequency of the small oscillation is

$$\nu=\frac{\omega_0}{2\pi}\sqrt{\frac{m\sin^2\theta_0}{2M\sin^2\theta_0+m}}.$$

Method 2. Introduce the centrifugal potential energy in the rotating frame (the conservative force being the inertial centrifugal force). Take the axis of rotation as the zero of potential energy; when the ball is at a distance x from the axis, the centrifugal potential energy U_{cf} is

$$U_{\mathrm{cf}}=\int_x^0 m\omega_0^2 x\,\mathrm{d}x=-\frac{1}{2}m\omega_0^2 x^2.$$

Take the upper sleeve as the zero of gravitational potential energy; then when the hinged arms make an angle θ with the axis, the potential energy U of the system is

$$U=U_{\mathrm{cf}}+U_{\mathrm{g}}=-2\times\frac{1}{2}m\omega_0^2 l^2\sin^2\theta-2(m+M)gl\cos\theta \tag{10}$$

Let the speed of the ball be $v(=l\dot{\theta})$ and the speed of the sleeve be V; the constraint relation between them is

$$V\cos\theta=v\cos\left(\frac{\pi}{2}-2\theta\right).$$

That is,

$$V=2v\sin\theta.$$

The kinetic energy of the system is

$$\begin{aligned}E_k&=2\times\frac{1}{2}mv^2+\frac{1}{2}MV^2\\&=mv^2+2M\sin^2\theta\,v^2\end{aligned} \tag{11}$$

From (10) and (11) the mechanical energy of the system is

$$E = E_k + U = (2M\sin^2\theta + m)l^2\dot{\theta}^2 - m\omega_0^2 l^2\sin^2\theta - 2(M+m)gl\cos\theta \tag{12}$$

Since the mechanical energy of the system is conserved,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = 4M\sin\theta\cos\theta\, l^2\dot{\theta}^3 + (2M\sin^2\theta + m)l^2\cdot 2\dot{\theta}\ddot{\theta} - 2m\omega_0^2 l^2\sin\theta\cos\theta\,\dot{\theta} + 2(M+m)gl\sin\theta\,\dot{\theta} = 0.$$

The ball executes small oscillations about the equilibrium position. Let $\theta = \theta_0 + \theta'$, $\theta' \ll \theta_0$; then $\dot{\theta} = \dot{\theta}'$ and $\ddot{\theta} = \ddot{\theta}'$, $\sin\theta \approx \sin\theta_0 + \cos\theta_0\cdot\theta'$ and $\cos\theta \approx \cos\theta_0 - \sin\theta_0\cdot\theta'$. Neglecting second-order and higher-order small quantities, the above expression becomes

$$(2M\sin^2\theta + m)l\ddot{\theta}' - m\omega_0^2 l\,(\sin\theta_0\cos\theta_0 + \cos 2\theta_0\,\theta') + (M+m)g\,(\sin\theta_0 + \cos\theta_0\cdot\theta') = 0.$$

Substituting (5) into the above expression, we obtain

$$(2M\sin^2\theta + m)l\ddot{\theta}' + m\omega_0^2 l\sin^2\theta_0\,\theta' = 0.$$

From this the frequency of the small oscillation is obtained as

$$\nu = \frac{\omega_0}{2\pi}\sqrt{\frac{m\sin^2\theta_0}{2M\sin^2\theta_0 + m}}.$$

(3) If the axis can rotate freely, then when the system suffers a small disturbance the angular velocity of rotation ω will change; let $\omega = \dot{\phi}$. Since the system is not acted on by an external torque about the axis, the angular momentum of the system about the axis is conserved; let this angular momentum be L,

$$L = 2ml^2\sin^2\theta\,\dot{\phi} \tag{13}$$

Taking a point at an infinite distance from the axis as the zero of centrifugal potential energy, the centrifugal potential energy of a ball is

$$U_{\mathrm{cf}} = \int_x^\infty m\dot{\phi}^2 x\,\mathrm{d}x = \int_x^\infty m\frac{L^2}{4m^2x^3}\,\mathrm{d}x = \frac{L^2}{8mx^2}.$$

Hence the mechanical energy of the system is

$$E = (2M\sin^2\theta + m)l^2\dot{\theta}^2 + \frac{2\times L^2}{8ml^2\sin^2\theta} - 2(M+m)gl\cos\theta.$$

Since the mechanical energy of the system is conserved,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = 4M\sin\theta\cos\theta\, l^2\dot{\theta}^3 + 2(2M\sin^2\theta + m)l^2\dot{\theta}\ddot{\theta} - \frac{L^2\cos\theta}{2ml^2\sin^3\theta}\dot{\theta} + 2(M+m)gl\sin\theta\,\dot{\theta} = 0.$$

Let $\theta = \theta_0 + \theta'$, $\theta' \ll \theta_0$, and $\dot{\theta} = \dot{\theta}'$, $\ddot{\theta} = \ddot{\theta}'$; substituting into the above expression and neglecting second-order and higher-order small quantities, the above expression becomes

$$(2M\sin^2\theta + m)l\ddot{\theta}' - \frac{L^2\cos\theta_0}{2ml^2\sin^3\theta_0}\left(1 - \frac{1+2\cos^2\theta_0}{\sin\theta_0\cos\theta_0}\theta'\right) + 2(M+m)g\,(\sin\theta_0 + \cos\theta_0\theta') = 0.$$

Substituting $L = 2ml^2 \sin^2\theta_0\,\omega_0$ and $\cos\theta_0 = \dfrac{(M+m)g}{m\omega_0^2 l}$ into the above expression, we obtain

$$(2M\sin^2\theta_0 + m)l\ddot{\theta}' + m\omega_0^2\left(1 + 3\cos^2\theta_0\right)\theta' = 0.$$

From this the frequency of the small oscillation in this case is obtained as

$$\nu = \frac{\omega_0}{2\pi}\sqrt{\frac{m\left(1+3\cos^2\theta_0\right)}{2M\sin^2\theta_0 + m}}.$$

Comment. In a rotating frame the inertial centrifugal force acting on a particle is a conservative force, and it is customary to introduce a centrifugal potential energy. If the centrifugal potential energy is treated as one of the potential energies of the system, just like the potential energies of other real conservative internal forces, then the question of the mechanical energy of the system may be discussed in the same way as in any inertial frame; in this problem, for instance, since no external force does work and no non-conservative internal force does work, the mechanical energy of the system is conserved. It must be pointed out here that in a rotating frame the particle is acted on not only by the inertial centrifugal force but also by the Coriolis force; however, since the Coriolis force is always perpendicular to the direction of the particle's velocity, this force does no work, and hence the presence of the Coriolis force does not affect the discussion of the mechanical-energy question of the system.

Solution 1.45. (1) When the string has turned through an angle ϕ, let the elongation of the string be l, as shown in the figure; then the equations of motion of the ball in the x and y directions are respectively

$$-kl\cos\phi = -kx = m\ddot{x} \tag{1}$$

$$-kl\sin\phi - mg = -ky - mg = m\ddot{y} \tag{2}$$

Let $y' = y + \dfrac{mg}{k} = y + a$; then $\ddot{y}' = \ddot{y}$, and substitution into (2) gives

$$-ky' = m\ddot{y}' \tag{3}$$

From (1) and (3) the angular frequencies of the oscillations in the x and y directions are obtained as

$$\omega_x = \omega_y = \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{a}}.$$

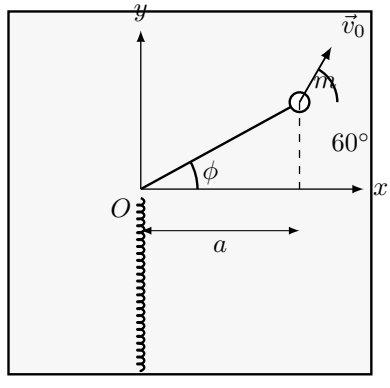


Fig. 1.45(b)

From the initial conditions $x_0 = a$, $\dot{x}_0 = v_0 \cos\theta = \frac{1}{2}v_0$, $y_0' = a$, $\dot{y}_0' = v_0 \sin\theta = \frac{\sqrt{3}}{2}v_0$, the amplitudes and initial phases of the oscillations in the x and y directions are obtained as

$$A_x = \sqrt{x_0^2 + \left(\frac{\dot{x}_0}{\omega_x}\right)^2} = \sqrt{a^2 + \frac{ga}{g} \cdot a} = \sqrt{2}\, a,$$

$$A_y = \sqrt{y_0'^2 + \left(\frac{\dot{y}_0'}{\omega_y}\right)^2} = \sqrt{a^2 + \frac{ga \cdot 3a}{g}} = 2a,$$

$$\tan\phi_x = \frac{-\dot{x}_0}{\omega_x x_0} = \frac{-\frac{1}{2} \times 2\sqrt{ga}}{\sqrt{\frac{g}{a}} \cdot a} = -1.$$

$$\therefore\ \phi_x = -\frac{\pi}{4},$$

$$\tan\phi_y = \frac{-\dot{y}_0'}{\omega_y y_0'} = \frac{-\frac{\sqrt{3}}{2} \times 2\sqrt{ga}}{\sqrt{\frac{g}{a}} \cdot a} = -\sqrt{3},$$

$$\therefore\ \phi_y = -\frac{\pi}{3}.$$

Hence the equations of the oscillations in the x and y directions are respectively

$$x = \sqrt{2}\, a \cos\left(\sqrt{\frac{g}{a}}\, t - \frac{\pi}{4}\right) \tag{1}$$

$$y' = 2a \cos\left(\sqrt{\frac{g}{a}}\, t - \frac{\pi}{3}\right) \tag{2}$$

When the ball reaches its highest position, we have

$$y' = A_y = 2a = y + a,$$

$$\therefore\ y = a.$$

Let the time elapsed be t_1; then from (2) we obtain

$$y' = 2a \cos\left(\sqrt{\frac{g}{a}}\, t_1 - \frac{\pi}{3}\right) = 2a,$$

that is,

$$\sqrt{\frac{g}{a}}\, t_1 - \frac{\pi}{3} = 0,$$

$$\therefore\ t_1 = \frac{\pi}{3}\sqrt{\frac{a}{g}}.$$

(2) When the ball passes through the x-axis again, $y = 0$, that is, $y' = a$. Let the time elapsed be t_2; then

$$y' = 2a \cos\left(\sqrt{\frac{g}{a}}\, t_2 - \frac{\pi}{3}\right) = a,$$

that is,

$$\cos\left(\sqrt{\frac{g}{a}}\,t_2 - \frac{\pi}{3}\right) = \frac{1}{2}.$$

$$\therefore\ \sqrt{\frac{g}{a}}\,t_2 - \frac{\pi}{3} = \frac{\pi}{3},$$

and hence

$$t_2 = \frac{2\pi}{3}\sqrt{\frac{a}{g}}.$$

The x-coordinate at this time is

$$\begin{aligned} x &= \sqrt{2}\,a\cos\left(\sqrt{\frac{g}{a}}\,t_2 - \frac{\pi}{4}\right) \\ &= \sqrt{2}\,a\cos\left(\sqrt{\frac{g}{a}} \times \frac{2\pi}{3}\sqrt{\frac{a}{g}} - \frac{\pi}{4}\right) \\ &= \sqrt{2}\,a\cos\frac{5}{12}\pi = 0.366a \end{aligned} \tag{3.1}$$

At this time the velocities v_x and v_y of the ball in the x and y directions are respectively

$$\begin{aligned} v_x &= -A_x\omega_x\sin(\omega_x t + \phi_x) \\ &= -\sqrt{2ga}\sin\frac{5}{12}\pi = -1.366\sqrt{ga}, \end{aligned} \tag{3.2}$$

$$\begin{aligned} v_y &= -A_y\omega_y\sin(\omega_y t + \phi_y) \\ &= -2\sqrt{ga}\sin\frac{\pi}{3} = -\sqrt{3ga} = -1.73\sqrt{ga}. \end{aligned} \tag{3.3}$$

Let the angle between the direction of the ball's velocity at this time and the x-axis be α; then

$$\tan\alpha = \frac{v_y}{v_x} = 1.266,$$

$$\therefore\ \alpha = 51.7^\circ.$$

The magnitude v of the velocity is

$$v = \sqrt{v_x^2 + v_y^2} = 2.2\sqrt{ga}.$$

Comment. This is a two-dimensional oscillation problem, that is, the motion of the ball is regarded as the superposition of two oscillations of the same frequency whose directions are mutually perpendicular but whose initial phases and amplitudes are all different. It is only necessary to use the initial conditions to find the initial phases and amplitudes of the oscillations in the two directions, and so to obtain the equations of these two oscillations; then, from the connection between the oscillations in the two directions, the required results can be found.

Solution 1.46. (1) After the locking of the small spring is released, the cylinder acquires an upward initial speed V_0. From the conservation of momentum, the downward initial speed v_0 acquired by the piston is found to be

$$v_0 = \frac{MV_0}{m} = 3V_0 = \frac{3}{2}g\sqrt{\frac{m}{k}}.$$

Afterwards the piston moves downwards with uniform acceleration a, and the equation of motion is

$$mg - f = ma.$$

Since $f = \frac{1}{2}mg$, its acceleration a is

$$a = \frac{1}{2}g.$$

Take the positive direction of the x-axis to be vertically downwards, and take the origin of the coordinate axis at the equilibrium position of the cylinder after the locking of the small spring is released and before the piston leaves the cylinder, as shown in Fig. 1.46(a). Take the instant at which the lock is released as $t = 0$. At the instant $t = 0$ the position of the cylinder is $x_0 = \dfrac{mg - f}{k}$, its speed is $V_0 = -\dfrac{g}{2}\sqrt{\dfrac{m}{k}}$, and the angular frequency of the cylinder's oscillation is $\omega = \sqrt{\dfrac{k}{M}}$. Hence its amplitude and initial phase are respectively

$$A = \sqrt{x_0^2 + \left(\frac{V_0}{\omega}\right)^2} = \sqrt{\left(\frac{mg}{2k}\right)^2 + \left(\frac{g}{2}\sqrt{\frac{m}{k}} \cdot \sqrt{\frac{M}{k}}\right)^2} = \frac{mg}{k}.$$

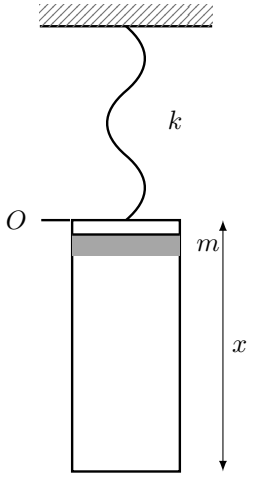


Fig. 1.46(a)

$$\tan\phi = \frac{-V_0}{x_0\omega} = \frac{\dfrac{g}{2}\sqrt{\dfrac{m}{k}}}{\dfrac{mg}{2k}\sqrt{\dfrac{k}{M}}} = \sqrt{3},$$

hence

$$\phi = \frac{\pi}{3}.$$

Therefore, before the piston leaves the cylinder, the motion of the cylinder can be expressed as

$$x = \frac{mg}{k}\cos\left(\sqrt{\frac{k}{M}}\,t + \frac{\pi}{3}\right) \tag{1}$$

In this case the lowest position that the cylinder can reach is $x = A = \dfrac{mg}{k}$.

Suppose that at some instant before the cylinder reaches the above lowest position the piston has already left the cylinder. Can the cylinder afterwards move to a position lower than the above lowest position? Let the piston leave the cylinder when the cylinder has moved to the position x' ($< A$), and let its speed at this instant be V'. From the conservation of mechanical energy,

$$\frac{1}{2}kx'^2 + \frac{1}{2}MV'^2 = \frac{1}{2}kA^2. \tag{2}$$

Since the piston has left the cylinder, the equilibrium position of the cylinder has changed: compared with the situation before the separation it has risen by $x_0' = \dfrac{f}{k} = \dfrac{mg}{2k}$. If the new equilibrium position is taken as the origin of the new coordinate axis x', and the amplitude of the subsequent oscillation of the cylinder is denoted by A', then again from the conservation of mechanical energy,

$$\frac{1}{2}k\left(x' + x_0'\right)^2 + \frac{1}{2}MV'^2 = \frac{1}{2}kA'^2. \tag{3}$$

Rewriting (2) and (3) gives

$$\left(x'\right)^2 + \left(\sqrt{\frac{M}{k}}V'\right)^2 = A^2, \tag{4}$$

$$\left(x' + x_0'\right)^2 + \left(\sqrt{\frac{M}{k}}V'\right)^2 = A'^2. \tag{5}$$

The relation among the magnitudes of the quantities in (4) and (5) is similar to the relation of the Pythagorean theorem in a right triangle, and can be represented by the geometrical relation shown in Fig. 1.46(b). From the figure it is known that

$$A + x_0' > A'. \tag{6}$$

Equation (6) shows that, if the piston leaves the cylinder at any position before the cylinder has moved to the position $x = A$, then although the amplitude A' of the subsequent motion is greater than A, its equilibrium position has risen by x_0', so its lowest position is still higher than the position $x = A$. Therefore the lowest position that the cylinder can reach is $x = A = \dfrac{mg}{k}$, and the elapsed time t can be found from (1):

$$x = \frac{mg}{k}\cos\left(\sqrt{\frac{k}{M}}\,t + \frac{\pi}{3}\right) = \frac{mg}{k}.$$

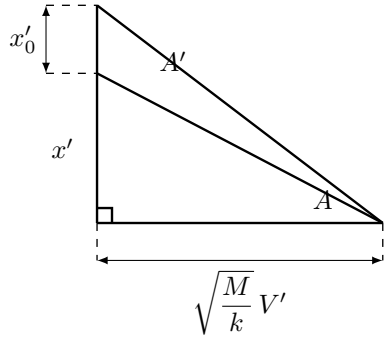


Fig. 1.46(b)

That is,

$$\sqrt{\frac{k}{M}}\,t+\frac{\pi}{3}=2\pi,$$

and hence

$$t=\frac{5\pi}{3}\sqrt{\frac{M}{k}}.$$

(2) For the cylinder to be able to reach the above lowest position, the piston must not leave the cylinder before the cylinder reaches this position; hence the length l of the cylinder must satisfy

$$\begin{aligned} l &\geq v_0 t+\frac{1}{2}at^2-(A-x_0)\\ &=\frac{3}{2}g\sqrt{\frac{m}{k}}\times\frac{5\pi}{3}\sqrt{\frac{M}{k}}+\frac{1}{4}g\left(\frac{5\pi}{3}\sqrt{\frac{M}{k}}\right)^2-\frac{mg}{2k}\\ &=33.6\,\frac{mg}{k}. \end{aligned}$$

Solution 1.47. When the system is in a steady state, let the distance from the small ball moving on the table top to the hole O be r_0; then

$$m\omega_0^2 r_0=Mg.$$

Hence

$$r_0=\frac{Mg}{m\omega_0^2}. \tag{1}$$

After the system is slightly disturbed, the motion of the small ball on the table top can be described in a polar coordinate system with origin O, and its radial equation of motion is

$$-T=m\left(\ddot{r}-r\dot{\theta}^2\right). \tag{2}$$

In the above expression T is the tension in the string. The equation of motion of the block M is

$$T-Mg=Ma. \tag{3}$$

The quantity a in (3) should be equal to $\ddot{r}$ in (2), that is,

$$a=\ddot{r}. \tag{4}$$

From (2), (3) and (4) we obtain

$$(M+m)\ddot{r}-mr\dot{\theta}^2+Mg=0. \tag{5}$$

Since the angular momentum L of the small ball m about the point O is conserved, we have

$$L=m\omega_0 r_0^2=m\dot{\theta}r^2.$$

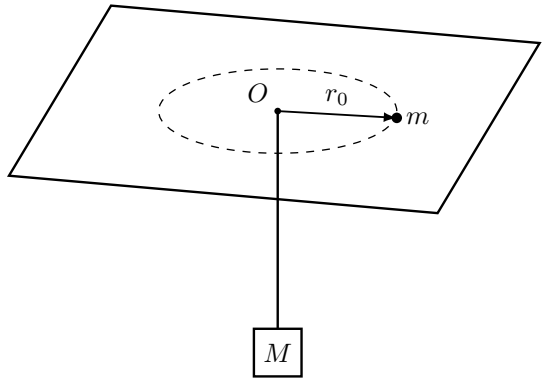


Fig. 1.47

That is,

$$\dot{\theta} = \frac{L}{mr^2}. \tag{6}$$

Substituting (6) into (5) gives

$$(M+m)\ddot{r} - \frac{L^2}{mr^3} + Mg = 0. \tag{7}$$

Let $r = r_0 + \delta$, where $\delta \ll r_0$; substituting this into (7) gives

$$(M+m)\ddot{\delta} - \frac{L^2}{mr_0^3}\left(1 + \frac{\delta}{r_0}\right)^{-3} + Mg = 0.$$

Since $\dfrac{\delta}{r_0} \ll 1$, we take the approximation in the above expression and neglect small quantities of second and higher order. Substituting (1) and $L = m\omega_0 r_0^2$, we obtain

$$(M+m)\ddot{\delta} + \frac{3L^2}{mr_0^4}\delta = 0,$$

or

$$\ddot{\delta} + \frac{3m\omega_0^2}{M+m}\delta = 0. \tag{8}$$

From (8) the period of the small oscillation of the block is obtained at once:

$$T = \frac{2\pi}{\omega_0}\sqrt{\frac{M+m}{3m}}.$$

This problem may also be solved by a method similar to Method 2 of Problem 1.44: by taking the first derivative of the mechanical energy of the system the equation of oscillation is obtained, and hence the period of oscillation is found. This is not repeated here; the reader is asked to work it out independently.

Solution 1.48. Suppose that the circle rolls uniformly without slipping along a straight line; then the coordinates of the point P on it at time t can be expressed as

$$x = R(\omega t + \sin \omega t) \tag{1}$$

$$y = R(1 - \cos \omega t) \tag{2}$$

where ω is the angular speed of rotation about the centre of the circle when the circle rolls without slipping.

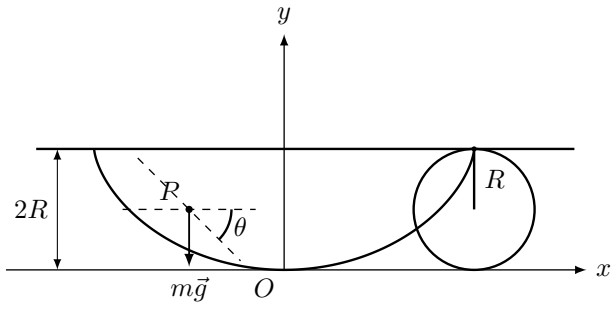


Fig. 1.48

Differentiating both sides of (1) and (2) gives

$$\mathrm{d}x = R\omega\,(1+\cos\omega t)\,\mathrm{d}t \tag{3}$$

$$\mathrm{d}y = R\omega\sin\omega t\,\mathrm{d}t \tag{4}$$

From (3) and (4) the element of arc length $\mathrm{d}s$ traversed by the point P along the cycloid during the time $\mathrm{d}t$ is

$$\begin{aligned}\mathrm{d}s &= \sqrt{(\mathrm{d}x)^2+(\mathrm{d}y)^2} = R\omega\sqrt{2+2\cos\omega t}\,\mathrm{d}t \\ &= R\omega\sqrt{4\cos^2\left(\frac{\omega t}{2}\right)}\,\mathrm{d}t = 2R\omega\cos\frac{\omega t}{2}\,\mathrm{d}t.\end{aligned} \tag{5}$$

Integrating both sides of (5),

$$\int_0^s \mathrm{d}s = \int_0^t 2R\omega\cos\frac{\omega t}{2}\,\mathrm{d}t,$$

we obtain

$$s = 4R\sin\frac{\omega t}{2}. \tag{6}$$

When a small ring is threaded on a smooth steel wire of the same shape as this cycloid and slides along the wire, its tangential acceleration $\ddot{s}$ is

$$\ddot{s} = -g\sin\theta. \tag{7}$$

Since the slope of the tangent is

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tan\theta,$$

we have

$$\sin\theta = \frac{\mathrm{d}y}{\sqrt{(\mathrm{d}x)^2+(\mathrm{d}y)^2}} = \frac{\mathrm{d}y}{\mathrm{d}s}. \tag{8}$$

Substituting (8) into (7) gives

$$\ddot{s} = -g\frac{\mathrm{d}y}{\mathrm{d}s}. \tag{9}$$

Substituting (4) and (5) into (9) gives

$$\ddot{s} = -g\,\frac{\sin\omega t}{2\cos\dfrac{\omega t}{2}} = -g\sin\frac{\omega t}{2}. \tag{10}$$

Substituting (6) into (10) gives

$$\ddot{s} + \frac{g}{4R}s = 0. \tag{11}$$

Equation (11) shows that the small ring performs a motion similar to simple harmonic motion along this steel wire, and its period is

$$T = 2\pi\sqrt{\frac{4R}{g}} = 4\pi\sqrt{\frac{R}{g}}.$$

Hence the time required for the small ring to slide from the highest point of the steel wire to the lowest point is

$$t = \frac{1}{4}T = \pi\sqrt{\frac{R}{g}}.$$

Solution 1.49. (1) The rectangular coordinate system is chosen as shown in the figure. The source of sound is located at the origin O of the coordinate system. The wind that blows over the ground is divided along the vertical (y) direction into many thin layers, and the wind speed within each thin layer $\mathrm{d}y$ is uniform. The sound wave is refracted successively at the interfaces of the thin layers. Suppose that a sound wave is incident on an interface at an angle of incidence θ in the layer where the wind speed is v, and enters the layer where the wind speed is $v + \mathrm{d}v$ at an angle of refraction $\theta + \mathrm{d}\theta$; then from the law of refraction,

$$\frac{\sin(\theta + \mathrm{d}\theta)}{\sin\theta} = \frac{V_s + (v + \mathrm{d}v)\sin(\theta + \mathrm{d}\theta)}{V_s + v\sin\theta}.$$

Expanding the above expression, neglecting small quantities of higher order and rearranging, we obtain

$$\frac{\mathrm{d}v}{V_s} = \frac{\mathrm{d}\,(\sin\theta)}{\sin^2\theta}.$$

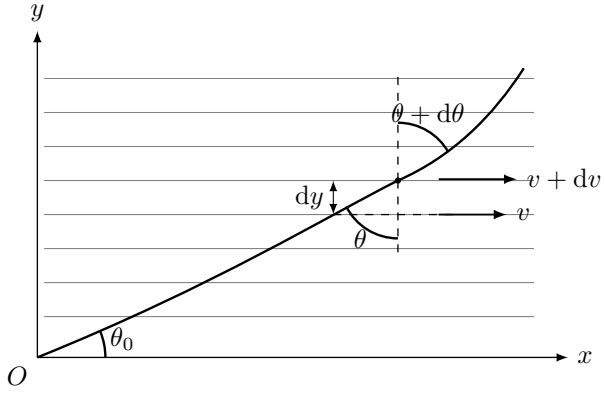


Fig. 1.49

Substituting $v = v_0 + ky^2$ into the above expression and integrating both sides,

$$\int_0^y \frac{2ky}{V_s}\,\mathrm{d}y = \int_{\theta_0}^{\theta} \frac{\mathrm{d}\,(\sin\theta)}{\sin^2\theta},$$

we obtain

$$\frac{k}{V_s}y^2 = \frac{1}{\sin\theta_0} - \frac{1}{\sin\theta}. \tag{1}$$

Since

$$\sin\theta = \frac{\mathrm{d}x}{\sqrt{(\mathrm{d}x)^2 + (\mathrm{d}y)^2}} = \frac{1}{\sqrt{1 + \left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2}},$$

substitution of this relation into (1) gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \pm\sqrt{\frac{1}{\sin^2\theta_0}\left(1 - \frac{k\sin\theta_0}{V_s}y^2\right)^2 - 1}.$$

By the condition given in the problem, $\dfrac{ky^2}{V_s} \ll 1$, that is, $\dfrac{k\sin\theta_0}{V_s}y^2 \ll 1$. Hence, expanding the above expression and neglecting small quantities of higher order, we obtain

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \pm\sqrt{\frac{1}{\sin^2\theta_0}\left(1 - \frac{2k\sin\theta_0}{V_s}y^2\right) - 1} \\ &= \pm\sqrt{\cot^2\theta_0 - \frac{2k}{V_s\sin\theta_0}y^2},\end{aligned}$$

or

$$\frac{\mathrm{d}y}{\sqrt{\cot^2\theta_0 - \dfrac{2k}{V_s\sin\theta_0}y^2}} = \pm\mathrm{d}x.$$

Integrating both sides of the above expression gives

$$\sqrt{\frac{V_s\sin\theta_0}{2k}}\arcsin\left(\sqrt{\frac{2k\sin\theta_0}{V_s}}\cdot\frac{y}{\cos\theta_0}\right) = \pm x + C.$$

From the boundary conditions $x_0 = 0$, $y_0 = 0$ the constant of integration is found to be $C = 0$; dropping the "$-$" sign in the above expression, we obtain the trajectory along which the sound wave propagates:

$$y = \cos\theta_0\sqrt{\frac{V_s}{2k\sin\theta_0}}\sin\left(\sqrt{\frac{2k}{V_s\sin\theta_0}}\,x\right). \tag{2}$$

(2) When the sound wave reaches the ground, $y = 0$, that is,

$$y = \cos\theta_0\sqrt{\frac{V_s}{2k\sin\theta_0}}\sin\left(\sqrt{\frac{2k}{V_s\sin\theta_0}}\,x\right) = 0.$$

Hence

$$\sqrt{\frac{2k}{V_s\sin\theta_0}}\,x = 0 \quad \text{or} \quad \pi.$$

The value 0 corresponds to the position of the source of sound; therefore the distance x from the source to the position at which the sound can be heard clearly is

$$x = \pi\sqrt{\frac{V_s \sin\theta_0}{2k}}.$$

Since k is very small, the value of x is comparatively large, which shows that the sound can be transmitted rather far.

Solution 1.50. (1) Let the equation of the incident wave be

$$y_1 = A\cos(\omega t - kx + \phi_1).$$

In the above expression $k = \dfrac{2\pi}{\lambda}$ is the wave number. The equation of the vibration excited by this wave at $x = -a$ is

$$y_1(x = -a) = A\cos(\omega t + ka + \phi_1).$$

According to the problem this vibration is given by $y = A\cos(\omega t + \phi_0)$; hence

$$ka + \phi_1 = \phi_0,$$

and we obtain

$$\phi_1 = \phi_0 - ka.$$

Thus

$$y_1 = A\cos(\omega t - kx + \phi_0 - ka).$$

Let the equation of the reflected wave be

$$y_2 = B\cos(\omega t + kx + \phi_2).$$

Since $x = a$ is a fixed end, the reflection there is total, that is,

$$B = A.$$

At this fixed end the reflected wave suffers a half-wave loss; hence

$$ka + \phi_2 = -ka + \phi_0 - ka + \pi,$$

and we obtain

$$\phi_2 = \phi_0 - 3ka + \pi.$$

Thus

$$y_2 = A\cos(\omega t + kx + \phi_0 - 3ka + \pi).$$

(2) The incident wave and the reflected wave superpose to form a standing wave:

$$\begin{aligned} y_{\text{st}} &= y_1 + y_2 = 2A\cos\left(kx - ka + \frac{\pi}{2}\right)\cos\left(\omega t + \phi_0 - 2ka + \frac{\pi}{2}\right) \\ &= 2A\sin k(x-a)\sin(\omega t + \phi_0 - 2ka). \end{aligned}$$

From the standing-wave equation shown above, the positions of the nodes are obtained as follows:

$$k(x-a) = n\pi \qquad (n = 0, -1, -2, \cdots),$$

that is,

$$x = a + \frac{n\pi}{k} \qquad (n = 0, -1, -2, \cdots).$$

Since the wave number k is

$$k = \frac{2\pi}{\lambda} = \frac{\omega}{v} = \omega\sqrt{\frac{\eta}{T}},$$

the positions of the nodes are

$$x = a + \frac{n\pi}{\omega}\sqrt{\frac{T}{\eta}} \qquad (n = 0, -1, -2, \cdots),$$

while the positions of the antinodes are

$$k(x - a) = \left(n + \frac{1}{2}\right)\pi \qquad (n = -1, -2, \cdots),$$

that is,

$$x = a + \frac{(2n+1)\pi}{2\omega}\sqrt{\frac{T}{\eta}} \qquad (n = -1, -2, \cdots).$$

(3) Take an arbitrary small segment $\mathrm{d}x$ of the standing wave; the kinetic energy of this small segment is

$$\begin{aligned}\mathrm{d}E_k &= \frac{1}{2}\eta\,\mathrm{d}x\left(\frac{\partial y}{\partial t}\right)^2 \\ &= 2\eta A^2\omega^2 \sin^2 k(x-a)\cos^2(\omega t + \phi_0 - 2ka)\,\mathrm{d}x.\end{aligned}$$

From the properties of a standing wave it is known that the total mechanical energy between two adjacent nodes is conserved, and that all points between the two nodes are in phase, that is, they pass through the equilibrium position simultaneously; at that instant the kinetic energy of every mass element reaches its maximum, while the potential energy is zero. The kinetic energy of the segment $\mathrm{d}x$ at that instant is

$$\mathrm{d}E_{k\max} = 2\eta A^2\omega^2 \sin^2 k(x-a)\,\mathrm{d}x.$$

Therefore the total energy between two adjacent nodes is

$$\begin{aligned}E &= \int \mathrm{d}E_{k\max} = 2\eta A^2\omega^2 \int_{a-\frac{\pi}{k}}^{a} \sin^2 k(x-a)\,\mathrm{d}x \\ &= 2\eta A^2\omega^2 \cdot \frac{\pi}{2k} = \eta A^2\omega^2\frac{\pi}{k} \\ &= \pi\eta A^2\omega\sqrt{\frac{T}{\eta}}.\end{aligned}$$

Solution 1.51. The oscillating system shown in the figure has two normal modes: in mode 1, B does not move, while A and C move towards each other. The eigenfrequency of this mode is $\omega_1 = \sqrt{\dfrac{k}{m}}$. In mode 2, A and C move in the same direction, while B moves in the opposite direction. The eigenfrequency of this mode is $\omega_2 = \sqrt{\dfrac{k(M+2m)}{Mm}}$. A general motion is a superposition of the oscillations of these two normal modes.

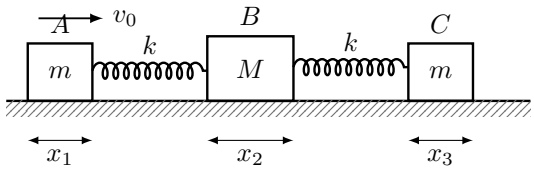


Fig. 1.51

After A is given the initial speed v_0, the centre of mass of the system begins to move uniformly, and the speed of the centre of mass v_C is

$$v_C = \frac{mv_0}{M+2m}.$$

In the centre-of-mass frame the displacements of the three bodies can be expressed respectively as

$$x_1' = A_1 \sin \omega_1 t + A_2 \sin \omega_2 t + x_{10}' \tag{1}$$
$$x_2' = -B_2 \sin \omega_2 t + x_{20}' \tag{2}$$
$$x_3' = -A_1 \sin \omega_1 t + A_2 \sin \omega_2 t + x_{30}' \tag{3}$$

Since $x_1' = x_2' = x_3' = 0$ at $t = 0$, we have $x_{10}' = x_{20}' = x_{30}' = 0$ in the above three expressions. Differentiating the above three expressions with respect to time t gives

$$\dot{x}_1' = A_1\omega_1 \cos \omega_1 t + A_2\omega_2 \cos \omega_2 t \tag{4}$$
$$\dot{x}_2' = -B_2\omega_2 \cos \omega_2 t \tag{5}$$
$$\dot{x}_3' = -A_1\omega_1 \cos \omega_1 t + A_2\omega_2 \cos \omega_2 t \tag{6}$$

At $t = 0$ we have

$$\dot{x}_1' = v_0 - v_C = v_0 - \frac{mv_0}{M+2m} = \frac{M+m}{M+2m}v_0, \tag{7}$$
$$\dot{x}_2' = -v_C = -\frac{mv_0}{M+2m}, \tag{8}$$
$$\dot{x}_3' = -v_C = -\frac{mv_0}{M+2m}. \tag{9}$$

Substituting (7), (8) and (9) respectively into (4), (5) and (6), we obtain

$$A_1 = \frac{v_0}{2\omega_1},$$
$$A_2 = \frac{Mv_0}{2\omega_2(M+2m)},$$
$$B_2 = \frac{mv_0}{\omega_2(M+2m)}.$$

Substituting the above relations into (1), (2) and (3), and using the principle of relative motion, we can obtain the relations for the variation with time of the displacements of the three bodies relative to the laboratory frame of reference.

Solution 1.52.

Note. The first page of this solution (page 124 of the original) is missing from the available copy. The translation therefore begins with equation (7), the first line preserved on the following page.

$$-k_1A_1 - k_2A_2 = \frac{m}{2}\left(-A_1\omega^2 - A_2\omega^2\right), \tag{7}$$

$$-\frac{l}{2}\left(k_2A_2 - k_1A_1\right) = \frac{I_0}{l}\left(-A_2\omega^2 + A_1\omega^2\right). \tag{8}$$

The necessary and sufficient condition for the system of equations (7) and (8) to have a non-zero solution is

$$\begin{vmatrix} k_1 - \dfrac{m\omega^2}{2} & k_2 - \dfrac{m\omega^2}{2} \\ \dfrac{I_0\omega^2}{l} - \dfrac{lk_1}{2} & \dfrac{lk_2}{2} - \dfrac{I_0\omega^2}{l} \end{vmatrix} = 0.$$

Hence

$$m^2\omega^4 - 4m(k_1 + k_2)\omega^2 + 12k_1k_2 = 0.$$

Solving, we obtain

$$\omega^2 = \frac{2(k_1 + k_2) \pm 2\sqrt{k_1^2 + k_2^2 - 3k_1k_2}}{m}.$$

The values of ω corresponding to the "+" and "−" signs in the above expression are the eigenfrequencies of the two normal modes of this system.

Solution 1.53. (1) Take the nozzle as the origin of coordinates, take some horizontal direction as the direction of the x-axis and the vertical upward direction as the direction of the y-axis. The water sprayed from the nozzle is symmetric about the y-axis, so it suffices to consider, in one cross-section (the xOy plane), the jets of water sprayed by the nozzle in all directions, to find the envelope of the parabolas formed by these jets, and then to rotate this envelope about the y-axis through one complete revolution; the solid figure is thereby obtained.

In the xOy plane, suppose that the jet of water from a certain hole is ejected at an elevation angle θ; then the equations of motion in the x and y directions are

$$x = v_0 \cos\theta\, t \tag{1}$$

$$y = v_0 \sin\theta\, t - \frac{1}{2}gt^2 \tag{2}$$

Combining (1) and (2) and eliminating t, we obtain

$$y = \tan\theta\, x - \frac{g}{2v_0^2\cos^2\theta}x^2. \tag{3}$$

Equation (3) is the trajectory equation of the jet. Setting $u = \tan\theta$, equation (3) can be written as

$$\frac{gx^2}{2v_0^2}u^2 - xu + \left(y + \frac{gx^2}{2v_0^2}\right) = 0. \tag{4}$$

If x and y are regarded as fixed values, then (4) is a quadratic equation in u. If this equation is to have real solutions, its discriminant must be greater than or equal to zero, that is,

$$x^2 - 4 \cdot \frac{gx^2}{2v_0^2}\left(y + \frac{gx^2}{2v_0^2}\right) \geq 0.$$

Solving, we obtain

$$y \leq \frac{v_0^2}{2g} - \frac{g}{2v_0^2}x^2. \tag{5}$$

The inequality expressed by (5) divides the xOy plane into two parts by means of a parabola. Water can reach the various points below this parabola, but cannot reach the points on and above it; this parabola is the required envelope, as shown by the dashed line in Fig. 1.53(a). The equation of this parabola is

$$y = \frac{v_0^2}{2g} - \frac{g}{2v_0^2}x^2. \tag{6}$$

Therefore the surface of the solid figure traced out by the water in the air is the paraboloid of revolution formed by rotating the parabola given by (6) about the y-axis through one complete revolution; the height of the solid figure is $\dfrac{v_0^2}{2g}$, and the radius of the region wetted on the ground is $R = \dfrac{v_0^2}{g}$.

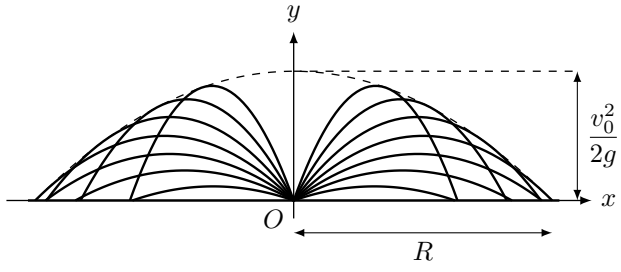


Fig. 1.53(a)

(2) Let the radius of the spherical surface of the small nozzle be r, as shown in Fig. 1.53(b). The range of the water ejected at the angle θ in the figure is

$$R = \frac{v_0^2 \sin 2\theta}{g}.$$

Differentiating both sides of the above expression, we obtain

$$\mathrm{d}R = \frac{2v_0^2 \cos 2\theta}{g}\mathrm{d}\theta.$$

Therefore the water ejected from the small holes on the narrow annular band (the dashed band in the figure) on the spherical surface of the nozzle, which makes an angle from θ to $\theta + \mathrm{d}\theta$ with the vertical axis, lands, at the points where it splashes down, on an annular band of radius R and width $\mathrm{d}R$, whose area $\mathrm{d}S$ is

$$\mathrm{d}S = 2\pi R\,\mathrm{d}R = \frac{2\pi v_0^4 \sin 4\theta}{g^2}\mathrm{d}\theta.$$

Let the volume of water received per unit area per second within the region of splash-down points be k; then the volume of water received per second on the above annular band of splash-down points is

$$\mathrm{d}Q = k\,\frac{2\pi v_0^4 \sin 4\theta}{g^2}\mathrm{d}\theta.$$

The area $\mathrm{d}S'$ of the narrow annular band $\theta \to \theta + \mathrm{d}\theta$ on the spherical surface of the nozzle is

$$\mathrm{d}S' = 2\pi r^2 \sin\theta\,\mathrm{d}\theta.$$

Let the area of a small hole be S_0, and let the distribution of the number of small holes per unit area on the spherical surface be $N(\theta)$; then the volume of water sprayed per second from this narrow annular band, $\mathrm{d}Q'$, is

$$\mathrm{d}Q' = N(\theta)S_0 v_0 \,\mathrm{d}S' = 2\pi r^2 \sin\theta \, N(\theta)S_0 v_0 \,\mathrm{d}\theta.$$

From $\mathrm{d}Q = \mathrm{d}Q'$ we obtain

$$k\,\frac{2\pi v_0^4 \sin 4\theta}{g^2}\mathrm{d}\theta = 2\pi r^2 \sin\theta \, N(\theta)S_0 v_0 \,\mathrm{d}\theta.$$

From this expression $N(\theta)$ is obtained as

$$N(\theta) = \frac{k v_0^3 \sin 4\theta}{r^2 g^2 S_0 \sin\theta} = C\,\frac{\sin 4\theta}{\sin\theta}.$$

In the above expression the constant C is a quantity related to k, v_0, r, g, S_0 and other such quantities.

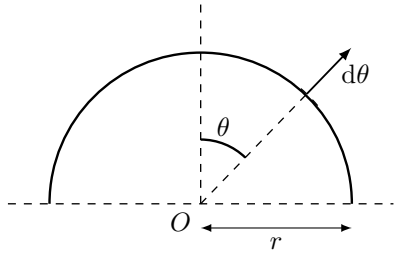


Fig. 1.53(b)

Solution 1.54. (1) Take a small volume element of cross-section $\mathrm{d}S$ and depth $\mathrm{d}r$ at a depth h near the surface at polar angle θ. Its volume is $\mathrm{d}V = \mathrm{d}S \cdot \mathrm{d}r$ and its mass is $\mathrm{d}m = \rho\,\mathrm{d}V$. In the rotating reference frame that rotates together with the planet at angular velocity ω, the inertial centrifugal force $\mathrm{d}\vec{f_i}$ acting on this small volume element is

$$\mathrm{d}\vec{f_i} = \mathrm{d}m\,\omega^2 r\cos\theta = \rho\omega^2 r\cos\theta\,\mathrm{d}V,$$

and its direction is shown in Fig. 1.54(a). The universal gravitational force $\mathrm{d}\vec{F}$ acting on the small volume element is

$$\mathrm{d}F = \frac{G\,\frac{4}{3}\pi r^3 \rho\,\mathrm{d}m}{r^2} = \frac{4}{3}\pi G\rho^2 r\,\mathrm{d}V,$$

its direction being along the radial direction r. The difference between the radial component of $\mathrm{d}\vec{f_i}$ and $\mathrm{d}F$ is balanced by the force produced at the two cross-sections of the small volume element, separated by $\mathrm{d}r$, by the pressure difference $\mathrm{d}p(r)$, that is,

$$-\mathrm{d}p(r)\,\mathrm{d}S = \frac{4}{3}\pi G\rho^2 r\,\mathrm{d}V - \rho\omega^2 r\cos^2\theta\,\mathrm{d}V.$$

Since $\rho = \dfrac{M}{\frac{4}{3}\pi R^3}$, the above expression can also be written as

$$-\mathrm{d}p(r) = \left(\frac{GM\rho}{R^3} - \rho\omega^2\cos^2\theta\right) r\,\mathrm{d}r.$$

Integrating both sides of the above expression as definite integrals, and since $p = 0$ at $r = R$, we have

$$-\int_{p(r)}^{0} \mathrm{d}p(r) = \int_{r}^{R} \left(\frac{GM\rho}{R^3} - \rho\omega^2 \cos^2\theta \right) r \, \mathrm{d}r.$$

Integrating, we obtain

$$p(r) = \frac{1}{2} \left(\frac{GM\rho}{R^3} - \rho\omega^2 \cos^2\theta \right) \left(R^2 - r^2 \right). \tag{1}$$

Substituting $r = R - h$ into the above expression, and since $h/R \ll 1$, expanding the above expression and neglecting quantities of higher order in h/R, we obtain

$$\begin{aligned} p(h) &= \frac{1}{2}\rho \left(\frac{GM}{R^3} - \omega^2 \cos^2\theta \right) \left[R^2 - (R-h)^2 \right] \\ &= \frac{1}{2}\rho \left(\frac{GM}{R^3} - \omega^2 \cos^2\theta \right) R^2 \left[1 - \left(1 - \frac{h}{R} \right)^2 \right] \\ &\approx \left(\frac{GM}{R^2} - \omega^2 \cos^2\theta \, R \right) \rho h. \end{aligned} \tag{2}$$

It is seen that within the range $h \ll R$ the pressure is approximately proportional to the depth.

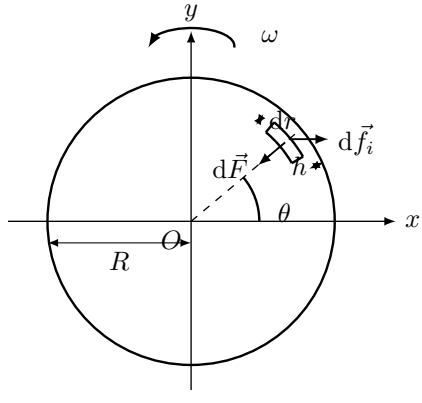


Fig. 1.54(a)

(2) We introduce the inertial centrifugal potential energy in the rotating reference frame (hereafter called the centrifugal potential energy). Taking the axis of rotation as the zero point of this potential energy, the centrifugal potential energy per unit volume at the equatorial surface of this planet (i.e. the centrifugal potential) is

$$U_{\mathrm{cent}}(\theta = 0) = -\int_{0}^{R_e} \rho\omega^2 r \, \mathrm{d}r = -\frac{1}{2}\rho\omega^2 R_e^2.$$

The gravitational potential and the total potential there are respectively

$$U_{\mathrm{grav}}^{(\theta=0)} = -\frac{\rho GM}{R_e},$$

$$U_{\mathrm{tot}}^{(\theta=0)} = U_{\mathrm{cent}} + U_{\mathrm{grav}} = -\frac{1}{2}\rho\omega^2 R_e^2 - \frac{\rho GM}{R_e}.$$

At the polar surface of the planet, on the other hand, the centrifugal potential is zero, so its total potential is

$$U_{\text{tot}}\left(\theta = \frac{\pi}{2}\right) = -\frac{\rho GM}{R_p}.$$

The surface of the planet is an equipotential surface everywhere; hence from

$$U_{\text{tot}}(\theta = 0) = U_{\text{tot}}\left(\theta = \frac{\pi}{2}\right)$$

it follows that

$$-\frac{1}{2}\rho\omega^2 R_e^2 - \frac{\rho GM}{R_e} = -\frac{\rho GM}{R_p},$$

that is,

$$R_e - R_p = \frac{\omega^2 R_e^2 R_p}{2GM} \approx \frac{\omega^2 R^4}{2GM}. \tag{3}$$

If the mass and the radius of this planet are both similar to those of the Earth, and its rotation is also the same as that of the Earth, that is, $R = 6.4 \times 10^6\,\text{m}$, $M = 6 \times 10^{24}\,\text{kg}$ and $\omega = \dfrac{2\pi}{T} = 7.3 \times 10^{-5}\,\text{s}^{-1}$, then substituting these data into (3) gives the estimate

$$R_e - R_p \approx 11\,\text{km}.$$

This sub-question can also be solved by the following method. Take a small volume element $\mathrm{d}V = R_p\,\mathrm{d}\theta \cdot \mathrm{d}S$ at polar angle $\theta \to \theta + \mathrm{d}\theta$ and at a distance R_p from the centre O, as shown in Fig. 1.54(b). The transverse component of the inertial centrifugal force acting on this volume is balanced by the force caused by the pressure difference $\mathrm{d}p(\theta)$ on the two transverse cross-sections $\mathrm{d}S$ of the small volume element, that is,

$$-\mathrm{d}p(\theta)\,\mathrm{d}S = \rho R_p\,\mathrm{d}\theta\,\mathrm{d}S\,\omega^2 R_p \cos\theta \sin\theta.$$

Integrating both sides of the above expression as definite integrals, and since $p = 0$ at $\theta = \dfrac{\pi}{2}$, we have

$$-\int_{p(\theta=0)}^{0} \mathrm{d}p(\theta) = \int_0^{\frac{\pi}{2}} \rho\omega^2 R_p^2 \sin\theta\cos\theta\,\mathrm{d}\theta.$$

Integrating, we obtain

$$p(\theta = 0) = \frac{1}{2}\rho\omega^2 R_p^2. \tag{4}$$

From equation (1) of sub-question (1) it is known that the pressure at $\theta = 0$, $r = R_p$ is

$$p(\theta = 0,\, h) = \frac{1}{2}\left(\frac{GM}{R_e^3} - \omega^2\right)\rho\left(R_e^2 - R_p^2\right). \tag{5}$$

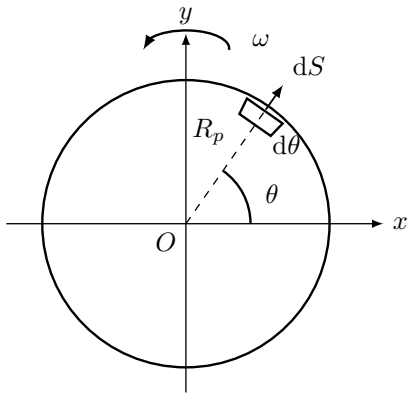


Fig. 1.54(b)

The pressures p in (4) and (5) must be equal, hence

$$\frac{1}{2}\omega^2 R_p^2 = \frac{1}{2}\left(\frac{GM}{R_e^3} - \omega^2\right)\left(R_e^2 - R_p^2\right).$$

That is,

$$R_e - R_p = \frac{\omega^2 R_e^5}{GM\left(R_e + R_p\right)} \approx \frac{\omega^2 R_e^4}{2GM} \approx \frac{\omega^2 R^4}{2GM}. \tag{6}$$

Solution 1.55. First consider the tide caused by the Moon. Let the mass of the Moon be m, the mass of the Earth be M_e, the radius of the solid part of the Earth be r_0, and the distance from the Moon to the centre of the Earth be r_{me}. The distance from the centre of mass C of the Moon–Earth system to the centre of the Earth is r_{Ce}. In the cross-section shown in the figure, the position of a point P under consideration on the water surface is expressed by polar coordinates (r, θ), and the distance from C to P is denoted by η. Suppose that the Moon–Earth system rotates about an axis through the centre of mass C perpendicular to the plane of the figure, with angular velocity ω, and that the Earth's axis of rotation is also perpendicular to the plane of the figure. This setup makes it convenient to calculate the relation between the tidal variation at the equator and θ over a 24-hour period. In the above setup, r_{Ce} is

$$r_{Ce} = \frac{m}{M_e + m} r_{me}. \tag{1}$$

In addition,

$$\frac{GM_e m}{r_{me}^2} = M_e \omega^2 r_{Ce}. \tag{2}$$

From (1) and (2) we obtain

$$\omega = \sqrt{\frac{G\left(M_e + m\right)}{r_{me}^3}}. \tag{3}$$

Moreover, in the above setup η is

$$\eta = \left(r_{Ce}^2 + r^2 + 2 r r_{Ce} \cos\theta\right)^{\frac{1}{2}}. \tag{4}$$

The potential energy per unit mass of water at the point $P(r, \theta)$ on the Earth's surface, that is, the potential V, is composed of three potentials. The first is the Earth's gravitational potential

V_e:

$$V_e = -\frac{GM_e}{r}. \tag{5}$$

The second is the Moon's gravitational potential V_m:

$$V_m = \frac{-Gm}{(r_{me}^2 + r^2 + 2rr_{me}\cos\theta)^{\frac{1}{2}}}. \tag{6}$$

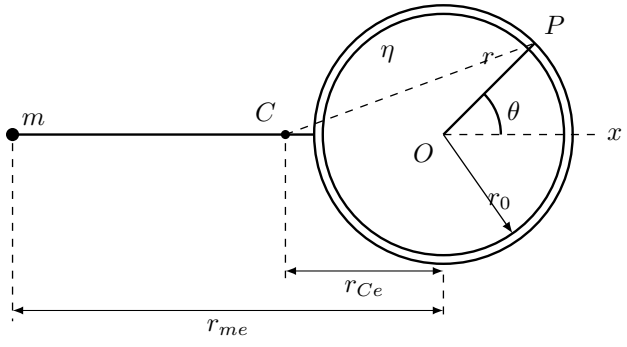


Fig. 1.55

Since $\dfrac{r}{r_{me}} \ll 1$, we may expand (6) in powers of $\dfrac{r}{r_{me}}$ and neglect terms of higher order than $\left(\dfrac{r}{r_{me}}\right)^2$, obtaining

$$\begin{aligned} V_m &= \frac{-Gm}{r_{me}}\left[1 + \left(\frac{r}{r_{me}}\right)^2 + 2\left(\frac{r}{r_{me}}\right)\cos\theta\right]^{-\frac{1}{2}} \\ &\approx \frac{-Gm}{r_{me}}\left[1 - \left(\frac{r}{r_{me}}\right)\cos\theta + \frac{r^2}{2r_{me}^2}\left(3\cos^2\theta - 1\right)\right]. \end{aligned} \tag{7}$$

The third is the centrifugal potential V_{cent} in the reference frame of the centre of mass of the Moon–Earth system:

$$\begin{aligned} V_{\text{cent}} &= -\frac{1}{2}\omega^2\eta^2 = -\frac{1}{2}\frac{G\left(M_e + m\right)}{r_{me}^3}\left(r_{Ce}^2 + r^2 + 2r_{Ce}r\cos\theta\right) \\ &= -\frac{Gm}{r_{me}}\left[\frac{m}{2\left(M_e + m\right)} + \frac{M_e + m}{2m}\left(\frac{r}{r_{me}}\right)^2 + \left(\frac{r}{r_{me}}\right)\cos\theta\right]. \end{aligned} \tag{8}$$

Hence V is

$$V = V_e + V_m + V_{\text{cent}}. \tag{9}$$

Substituting (5), (7) and (8) into (9), and noting that the surface of the Earth is an equipotential surface, we obtain

$$-\frac{GM_e}{r} - \frac{Gm}{r_{me}}\left[\frac{2M_e + 3m}{2\left(M_e + m\right)} + \frac{M_e + 3m\cos^2\theta}{2m}\left(\frac{r}{r_{me}}\right)^2\right] = C. \tag{10}$$

Substituting $r = r_0 + h$ into (10) and making reasonable approximations, we obtain

$$\frac{GM_e}{r_0^2}h - \frac{3Gmr_0^2}{2r_{me}^3}\cos^2\theta = C.$$

That is,

$$h = \frac{3mr_0^4\cos^2\theta}{2M_e r_{me}^3} + C. \tag{11}$$

At $\theta = \dfrac{\pi}{2}$ the height of the water surface is a minimum h_0, so the C in (11) is $C = h_0$; while at $\theta = 0$ the tide is a maximum, which we take as $h_{\max}$. Hence the tidal variation $(\Delta h)_m$ at the equator during a 24-hour period is

$$(\Delta h)_m = h_{\max} - h_0 = \frac{3mr_0^4}{2M_e r_{me}^3}. \tag{12}$$

Similar to the calculation of the tide caused by the Moon, the tidal variation $(\Delta h)_s$ caused by the Sun is

$$(\Delta h)_s = \frac{3M_s r_0^4}{2M_e r_{se}^3}. \tag{13}$$

In (13) M_s is the mass of the Sun and r_{se} is the distance from the Sun to the Earth; hence from (12) and (13) we obtain

$$\frac{(\Delta h)_s}{(\Delta h)_m} = \frac{M_s r_{me}^3}{m r_{se}^3} = \frac{2.0\times10^{30}\times\left(3.8\times10^8\right)^3}{7.3\times10^{22}\times\left(1.5\times10^{11}\right)^3} \approx 0.44.$$

Solution 1.56. Take the planet and the satellite as the system; the angular momentum L of this system about the centre of the planet is

$$L = I\Omega + mr^2\omega. \tag{1}$$

Owing to the action of tidal friction, both the angular velocity of rotation Ω of the planet and the angular velocity ω of the satellite's circular orbital motion about the planet will change, and the change in ω will in turn lead to a change in the orbital radius r. The relations among these changes can be obtained by differentiating (1):

$$\mathrm{d}L = I\,\mathrm{d}\Omega + mr^2\,\mathrm{d}\omega + 2mr\omega\,\mathrm{d}r. \tag{2}$$

Since the satellite moves in a circle about the planet, we have

$$G\frac{Mm}{r^2} = m\omega^2 r.$$

That is,

$$\omega^2 = \frac{GM}{r^3}. \tag{3}$$

Differentiating both sides of (3), we obtain

$$2\omega\,\mathrm{d}\omega = -\frac{3GM}{r^4}\mathrm{d}r = -\frac{3\omega^2}{r}\mathrm{d}r.$$

Hence the relation between $\mathrm{d}\omega$ and $\mathrm{d}r$ is

$$\mathrm{d}r = -\frac{2r}{3\omega}\mathrm{d}\omega. \tag{4}$$

Substituting (4) into (2), we obtain

$$\mathrm{d}L = I\,\mathrm{d}\Omega - \frac{1}{3}mr^2\,\mathrm{d}\omega. \tag{5}$$

Since the angular momentum of the system about the centre of the planet is conserved, that is, $\mathrm{d}L = 0$, from (5) we obtain the relation between $\mathrm{d}\Omega$ and $\mathrm{d}\omega$:

$$\mathrm{d}\Omega = \frac{mr^2}{3I}\mathrm{d}\omega. \tag{6}$$

(2) The mechanical energy E of the system is

$$E = \frac{1}{2}I\Omega^2 + \frac{1}{2}mr^2\omega^2 - G\frac{Mm}{r}. \tag{7}$$

Since tidal friction causes a decrease in the mechanical energy of the system, differentiating (7) gives

$$\mathrm{d}E = I\Omega\,\mathrm{d}\Omega + mr^2\omega\,\mathrm{d}\omega + mr\omega^2\,\mathrm{d}r + \frac{GMm}{r^2}\mathrm{d}r.$$

Substituting (3) into the above expression, we obtain

$$\mathrm{d}E = I\Omega\,\mathrm{d}\Omega + mr^2\omega\,\mathrm{d}\omega + 2mr\omega^2\,\mathrm{d}r.$$

Substituting (4) and (6) into the above expression, we obtain

$$\frac{\mathrm{d}E}{\mathrm{d}\omega} = \frac{1}{3}mr^2\left(\Omega - \omega\right). \tag{8}$$

When the system finally reaches a stable state, that is, a state in which the energy does not change, $\dfrac{\mathrm{d}E}{\mathrm{d}\omega} = 0$; hence from (8) we obtain

$$\Omega = \omega.$$

From this it is seen that when the orbital angular velocity ω of the satellite is the same as the angular velocity of rotation Ω of the planet, the mechanical energy of the system no longer changes; at this time there is no longer any phenomenon of the tide rising and falling on the planet, and tidal friction no longer exists. For the system to be in a stable state, the mechanical energy at this time should take a minimum value, that is,

$$\frac{\mathrm{d}^2E}{\mathrm{d}\omega^2} > 0.$$

Using (8), we obtain

$$\frac{\mathrm{d}^2E}{\mathrm{d}\omega^2} = \frac{1}{3}mr^2\left(\frac{\mathrm{d}\Omega}{\mathrm{d}\omega} - 1\right) + \frac{2}{3}mr\left(\Omega - \omega\right)\frac{\mathrm{d}r}{\mathrm{d}\omega} > 0.$$

Substituting (6) and $\Omega = \omega$ into the above expression, we obtain

$$\frac{mr^2}{I} > 3.$$

This is the condition that the system should satisfy.

Solution 1.57. The radial component v_r and the transverse component v_θ of the wind velocity $\vec{v}$ at point B near the hangar are respectively

$$v_r = -\left.\frac{\partial \phi}{\partial r}\right|_{r=R} = v_0 \left(1 - \frac{R^2}{r^2}\right) \cos\theta \Big|_{r=R} = 0,$$

$$v_\theta = -\frac{1}{r}\left.\frac{\partial \phi}{\partial \theta}\right|_{r=R} = -v_0 \left(1 + \frac{R^2}{r^2}\right) \sin\theta \Big|_{r=R} = -2v_0 \sin\theta.$$

It is known from the statement of the problem that the wind speed far away is v_0 and the pressure there is p_0; hence from Bernoulli's equation the pressure $p(\theta)$ at point B is

$$p(\theta) + \frac{1}{2}\rho v_\theta^2 = p_0 + \frac{1}{2}\rho v_0^2.$$

That is,

$$\begin{aligned} p(\theta) &= p_0 + \frac{1}{2}\rho \left(v_0^2 - v_\theta^2\right) \\ &= p_0 + \frac{1}{2}\rho v_0^2 \left(1 - 4\sin^2\theta\right). \end{aligned} \tag{3.4}$$

The pressure inside the hangar, that is, the pressure p_A at the door A of the hangar at $\theta = \pi$, is

$$p_A = p_0 + \frac{1}{2}\rho v_0^2.$$

Hence the force F exerted by the wind on the hangar is

$$\begin{aligned} F &= \int_0^\pi [p_A - p(\theta)] \sin\theta \, \mathrm{d}s \\ &= \frac{1}{2}\rho v_0^2 LR \int_0^\pi 4\sin^2\theta \cdot \sin\theta \, \mathrm{d}\theta \\ &= 2\rho v_0^2 LR \left(\frac{1}{3}\cos^3\theta - \cos\theta\right)\Big|_0^\pi \\ &= \frac{8}{3}\rho v_0^2 LR \approx 9.0 \times 10^5 \, \mathrm{N}. \end{aligned} \tag{3.5}$$

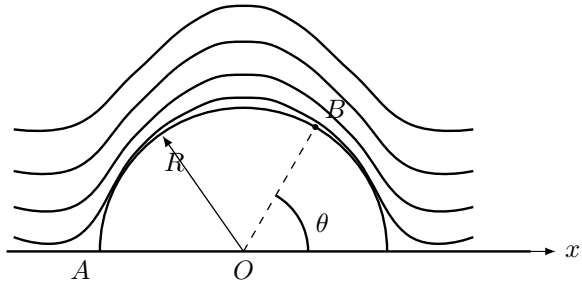


Fig. 1.57

Solution 1.58. (1) At the moment of release the angle between the two rods and the ground is $\theta = \alpha = 60°$, and the forces acting on one of the rods, AB, are shown in Fig. 1.58(a). Let the horizontal (x) and vertical (y) components of the acceleration of the centre of mass C of this rod at this moment be a_{Cx} and a_{Cy} respectively; then the equations of motion of the centre of mass of this rod along the x and y directions are respectively

$$F = ma_{Cx} \tag{1}$$

$$mg - N = ma_{Cy} \tag{2}$$

The equation of rotation of this rod about its centre of mass C is

$$\frac{1}{2}lN\cos\theta - F \cdot \frac{l}{2}\sin\theta = \frac{1}{12}ml^2\ddot{\theta}. \tag{3}$$

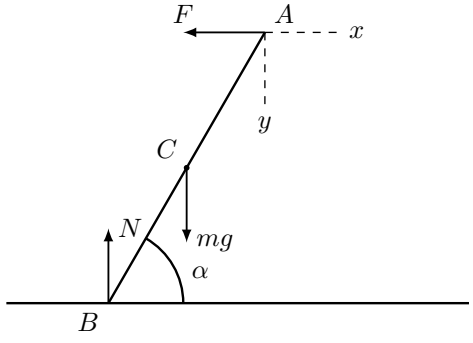


Fig. 1.58(a)

From the symmetry of the system it is known that the acceleration of the hinge A has no horizontal component, that is, the acceleration of A is along the y direction. Hence, from the formula for relative motion, we obtain

$$a_{Ax} = a_{Cx} - \frac{1}{2}l\ddot{\theta}\sin\theta = a_{Cx} - \frac{\sqrt{3}}{4}l\ddot{\theta} = 0 \tag{4}$$

$$a_{Ay} = a_{Cy} + \frac{1}{2}l\ddot{\theta}\cos\theta = a_{Cy} + \frac{1}{4}l\ddot{\theta} \tag{5}$$

From (4) we obtain

$$a_{Cx} = \frac{\sqrt{3}}{4}l\ddot{\theta}. \tag{6}$$

The acceleration of the end B of the rod is along the x direction, that is, it has no y component; hence

$$a_{By} = a_{Cy} - \frac{1}{2}l\ddot{\theta}\cos\theta = a_{Cy} - \frac{1}{4}l\ddot{\theta} = 0,$$

and we obtain

$$a_{Cy} = \frac{1}{4}l\ddot{\theta}. \tag{7}$$

Substituting (6) into (1), we obtain

$$F = \frac{\sqrt{3}}{4}ml\ddot{\theta}. \tag{8}$$

Substituting (7) into (2), we obtain

$$N = mg - \frac{1}{4}ml\ddot{\theta}. \tag{9}$$

Substituting (8), (9) and $\theta = 60^\circ$ into (3), we obtain

$$\frac{1}{2}\left(mg - \frac{1}{4}ml\ddot{\theta}\right) - \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{4}ml\ddot{\theta} = \frac{1}{6}ml\ddot{\theta}.$$

That is,

$$l\ddot{\theta} = \frac{3}{4}g. \tag{10}$$

From (5), (7) and (10) we obtain

$$a_A = a_{Ay} = \frac{3}{8}g.$$

(2) When the angle between the two rods is $\alpha' = 90^\circ$, the angle between rod AB and the ground is $\theta' = 45^\circ$. At this moment the rod may be regarded as making pure rotation about the instantaneous axis O, as shown in Fig. 2. In the figure the distance between point O and the centre of mass C of the rod is $\frac{1}{2}l$.

Solution 1.59.

Note. The first page of this solution (page 134 of the original) is missing from the available copy. The translation therefore begins with equation (6), the first line preserved on the following page.

Since the rod now begins to slide relative to the corner of the table, f is the kinetic friction force, so

$$f = \mu N. \tag{6}$$

From (3) we obtain

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = \sqrt{\frac{3g}{l}} \cdot \frac{\cos\theta}{2\sqrt{\sin\theta}} \cdot \omega = \frac{3g}{2l}\cos\theta. \tag{7}$$

Substituting (7) into (4), we obtain

$$N = \frac{3}{4}mg\cos\theta. \tag{8}$$

Substituting (8) into (6), we obtain

$$f = \frac{3}{4}mg\mu\cos\theta. \tag{9}$$

Substituting (3) and (9) into (5), we obtain

$$\tan\theta = \frac{\mu}{2}.$$

Comment. In solving this problem, attention should be paid to the fact that when the rod begins to slide relative to the corner of the table, the resultant force on the centre of mass of the rod in the direction along the rod is not zero, that is, it is not $f = mg\sin\theta$; it should be equal to the centripetal force of the centre of mass in its circular motion. The equation of motion of the centre of mass in this direction is (5). This is because, from the start of the motion until the rod begins to slide off the corner of the table, the centre of mass always moves in a circle.

Solution 1.60. (1) The water-filled cylinder rolls without slipping on the inclined plane, so besides gravity it is also acted on by a static friction force f; hence the equation of motion of its centre of mass is

$$(m+\mu)g\sin\theta - f = (m+\mu)a. \tag{1}$$

The equation of rotation of the cylinder about its own axis is

$$fR = I\beta. \tag{2}$$

In (2), I is the moment of inertia of the water-filled cylinder about its own axis. Since, while the cylinder rolls, the water inside it does not rotate together with the cylinder but moves translationally, we have

$$I = mR^2. \tag{3}$$

In (2), β is the angular acceleration of the cylinder; since the cylinder rolls without slipping,

$$\beta = \frac{a}{R}. \tag{4}$$

From (1), (2), (3) and (4) we obtain

$$a = \frac{m+\mu}{2m+\mu}g\sin\theta.$$

(2) (i) Suppose that when the cylinder rolls down the inclined plane of the wedge-shaped block, the acceleration of the block is w. Take the block as the reference frame. In this frame the forces acting on the cylinder are as shown in the figure, where the inertial force is $f_i = (m+\mu)w$; let the acceleration of the centre of mass of the cylinder relative to the block be a', and let its angular acceleration of rotation be β. The equation of motion of its centre of mass is

$$(m+\mu)g\sin\theta + (m+\mu)w\cos\theta - f = (m+\mu)a'. \tag{5}$$

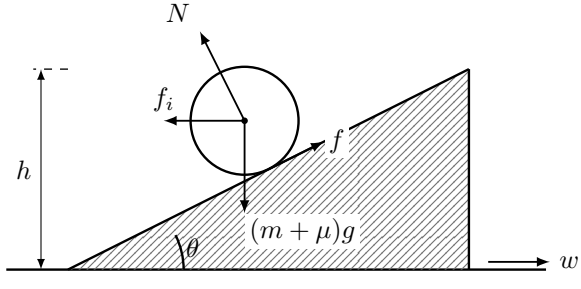


Fig. 1.60

The equation of rotation of the cylinder about its centre-of-mass axis is

$$fR = I\beta. \tag{6}$$

The relation between a' and β is the same as in (4), namely

$$a' = \beta R. \tag{7}$$

In the direction perpendicular to the inclined plane we have

$$N - (m+\mu)g\cos\theta + (m+\mu)w\sin\theta = 0, \tag{8}$$

and the equation of motion of the block is

$$N\sin\theta - f\cos\theta = Mw. \tag{9}$$

From (3), (6) and (7) we obtain $f = ma'$; substituting this into (5), we obtain

$$(m+\mu)g\sin\theta + (m+\mu)w\cos\theta = (2m+\mu)a'. \tag{10}$$

Eliminating N from (8) and (9), and using $f = ma'$, we obtain

$$(m+\mu)g\sin\theta\cos\theta - (m+\mu)w\sin^2\theta - ma'\cos\theta = Mw. \tag{11}$$

Eliminating a' from (10) and (11), we obtain the acceleration of the block

$$w = \frac{(m+\mu)^2 g\sin\theta\cos\theta}{(M+m+\mu)(2m+\mu) - (m+\mu)^2\cos^2\theta}. \tag{12}$$

Substituting (12) into (10), we obtain the acceleration a' of the cylinder relative to the block:

$$a' = \frac{(M+m+\mu)(m+\mu)g\sin\theta}{m(m+\mu) + (m+\mu)^2\sin^2\theta + M(2m+\mu)}. \tag{13}$$

(ii) Let t be the time taken by the water-filled cylinder to roll from a height h above the table top down to the table top; then

$$\frac{h}{\sin\theta} = \frac{1}{2}a't^2,$$

that is,

$$t = \sqrt{\frac{2h}{a'\sin\theta}}. \tag{14}$$

Using (12), (13) and (14), the velocity u of the wedge-shaped block at this moment is obtained as

$$u = wt = \sqrt{\frac{2gh(m+\mu)^3\cos^2\theta}{(M+m+\mu)\left[(M+m+\mu)(2m+\mu) - (m+\mu)^2\cos^2\theta\right]}}.$$

Solution 1.61. (1) In the rotating reference frame, the potential energy of rod AB is the sum of the gravitational potential energy and the centrifugal potential energy caused by the inertial centrifugal force, in which the gravitational potential energy is $mga\cos\theta$ (taking point A as the zero of gravitational potential energy), while the centrifugal potential energy can be found by the following calculation.

Take an element $\mathrm{d}l$ on rod AB at a distance l from end A, as shown in Fig. 1.61(a). Let the distance of the element from the axis of rotation be ξ. If the axis of rotation is taken as the zero of centrifugal potential energy, then the centrifugal potential energy of this element is

$$\mathrm{d}U_{\mathrm{cent}} = -\int_0^{b+l\sin\theta} \frac{m}{2a}\,\mathrm{d}l\,\omega^2\xi\,\mathrm{d}\xi = -\frac{m}{4a}\omega^2(b+l\sin\theta)^2\,\mathrm{d}l.$$

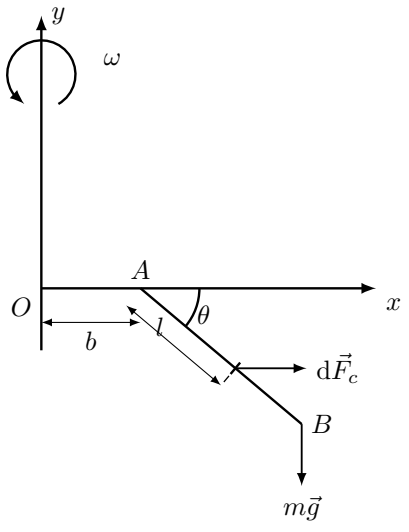


Fig. 1.61(a)

Therefore the centrifugal potential energy of rod AB is

$$\begin{aligned} U_{\text{cent}} &= \int \mathrm{d}U_{\text{cent}} = \int_0^{2a} -\frac{m}{4a}\omega^2(b + l\sin\theta)^2\,\mathrm{d}l \\ &= -\frac{m\omega^2}{12a\sin\theta}\left[(b + 2a\sin\theta)^3 - b^3\right] \\ &= -\frac{1}{2}m\omega^2\left[\frac{1}{3}a^2\sin^2\theta + (b + a\sin\theta)^2\right] \end{aligned} \tag{1}$$

Therefore the potential energy of rod AB is

$$U = U_{\text{cent}} + U_{\text{grav}} = -\frac{1}{2}m\omega^2\left[\frac{1}{3}a^2\sin^2\theta + (b + a\sin\theta)^2\right] + mga\cos\theta \tag{2}$$

When the rod is in equilibrium the condition $\dfrac{\mathrm{d}U}{\mathrm{d}\theta} = 0$ is satisfied, so

$$\frac{\mathrm{d}U}{\mathrm{d}\theta} = -m\omega^2 a\cos\theta\left(b + \frac{4}{3}a\sin\theta\right) - mga\sin\theta = 0,$$

that is,

$$\tan\theta = -\frac{a}{g}\omega^2\left(\frac{b}{a} + \frac{4}{3}\sin\theta\right) \tag{3}$$

Equation (3) is the relation satisfied by the possible values of θ when rod AB is in equilibrium.

(2) Each of the terms on the left and right sides of (3) is set equal to a function, that is, we set $y_1 = \tan\theta$ and $y_2 = -\dfrac{a}{g}\omega^2\left(\dfrac{b}{a} + \dfrac{4}{3}\sin\theta\right)$; the curves of the two functions are shown in Fig. 1.61(b). From the figure it is seen that θ has one equilibrium position in each of quadrants II and IV, while whether there is an equilibrium position in quadrant III depends on the relation between a and b: there may be two equilibrium positions, or there may be one equilibrium

position, or there may be no equilibrium position. This is because the curve y_1 is fixed, whereas the shape of the curve y_2 is determined by the constants $-\dfrac{b\omega^2}{g}$ and $\dfrac{4a\omega^2}{3g}$, that is, the curve y_2 oscillates about $y=-\dfrac{b\omega^2}{g}$. When the value of ω changes, $y=-\dfrac{b\omega^2}{g}$ changes correspondingly, and its amplitude of oscillation $\dfrac{4a\omega^2}{3g}$ changes accordingly as well, so that the two curves y_1 and y_2 in the third quadrant either have two points of intersection, or are tangent at one point, or have no point of intersection; Fig. 1.61(b) shows the case of no point of intersection, that is, the case in which there is no equilibrium position.

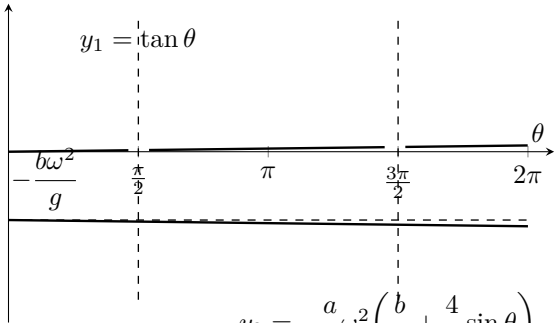


Fig. 1.61(b)

The stability of an equilibrium position depends on whether the value of $\dfrac{\mathrm{d}^2U}{\mathrm{d}\theta^2}$ at that equilibrium position is positive or negative, and $\dfrac{\mathrm{d}^2U}{\mathrm{d}\theta^2}$ is

$$\frac{\mathrm{d}^2U}{\mathrm{d}\theta^2}=-\frac{4}{3}ma^2\omega^2\left(\cos^2\theta-\sin^2\theta\right)+mab\omega^2\sin\theta-mga\cos\theta \tag{4}$$

Substituting (3) into the above expression gives

$$\frac{\mathrm{d}^2U}{\mathrm{d}\theta^2}=\frac{ma\cos^2\theta}{\sin\theta}\left(b\omega^2-g\tan^3\theta\right) \tag{5}$$

Discussion:

(i) For the equilibrium point of θ in quadrant II, that is, in the interval $\dfrac{\pi}{2}<\theta<\pi$, we have $\sin\theta>0$ and $\tan\theta<0$; therefore from (5) we obtain

$$\left.\frac{\mathrm{d}^2U}{\mathrm{d}\theta^2}\right|_{\mathrm{II}}>0,$$

that is, the equilibrium property of this equilibrium position is stable equilibrium.

(ii) For the equilibrium point of θ in quadrant IV, that is, in the interval $\dfrac{3\pi}{2}<\theta<2\pi$, we have $\sin\theta<0$ and $\tan\theta<0$; therefore from (5) we obtain

$$\left.\frac{\mathrm{d}^2U}{\mathrm{d}\theta^2}\right|_{\mathrm{IV}}<0,$$

that is, the equilibrium property of this equilibrium position is unstable equilibrium.

(iii) For the equilibrium point of θ in quadrant III, that is, in the interval $\pi < \theta < \dfrac{3\pi}{2}$, we have $\sin\theta < 0$ and $\tan\theta > 0$, and (4) can also be expressed as

$$\begin{aligned}\frac{\mathrm{d}^2U}{\mathrm{d}\theta^2} &= \frac{ma\cos^2\theta}{\sin\theta}\left[\frac{4}{3}a\omega^2\sin\theta\cdot\tan^2\theta + b\omega^2\left(\tan^2\theta+1\right)\right] \\ &= \frac{ma\omega^2}{\sin\theta}\left(\frac{4}{3}a\sin^2\theta + b\right) \end{aligned} \tag{6}$$

From (6) it is seen that when $\dfrac{4}{3}a\sin^2\theta + b < 0$, that is, $b < -\dfrac{4}{3}a\sin^2\theta$, we have $\dfrac{\mathrm{d}^2U}{\mathrm{d}\theta^2} > 0$, so the equilibrium property of this equilibrium position is stable equilibrium; and when $b > -\dfrac{4}{3}a\sin^2\theta$, we have $\dfrac{\mathrm{d}^2U}{\mathrm{d}\theta^2} < 0$, so the equilibrium property of this equilibrium position is unstable equilibrium.

In this problem the centrifugal potential energy is the negative of the work done by the inertial centrifugal force during the displacement of the rod from the zero of potential energy (that is, the position of the axis of rotation) to the equilibrium position. In fact, this work is just the kinetic energy of this rod rotating about the axis of rotation in the inertial frame, that is,

$$U_{\text{cent}} = -\frac{1}{2}I\omega^2 \tag{7}$$

where I is the moment of inertia of the rod about the axis of rotation. I is

$$I = \frac{1}{3}ma^2\sin^2\theta + m\left(b + a\sin\theta\right)^2 \tag{8}$$

Substituting (8) into (7) gives

$$U_{\text{cent}} = -\frac{1}{2}m\omega^2\left[\frac{1}{3}a^2\sin^2\theta + \left(b + a\sin\theta\right)^2\right].$$

This expression is completely the same as (1).

Solution 1.62. (1) Before the ball strikes the ground, the velocity of its centre of mass is

$$v_0 = \sqrt{2gh} \tag{1}$$

Before the collision the angular velocity of the ball is ω_0. Let the angular velocity after the collision be ω and the velocity of the centre of mass after the collision be v, as shown in Fig. 1.62(a). The two components of v are v_x and v_y. For the process of the collision between the ball and the ground there are the following two possible cases.

Case 1. During the time Δt of the collision between the ball and the ground, there is always relative sliding between the points of contact of the ball and the ground, so the friction between the two is always sliding friction, that is, $f = \mu N$, where N is the supporting force of the ground. According to the momentum theorem, in the x and y directions respectively,

$$\int_0^{\Delta t} f\,\mathrm{d}t = \int_0^{\Delta t} N\mu\,\mathrm{d}t = mv_x \tag{2}$$

$$\int_0^{\Delta t} N\,\mathrm{d}t - mg\Delta t = mv_y - m(-v_0) = m\left(v_y + v_0\right) \tag{3}$$

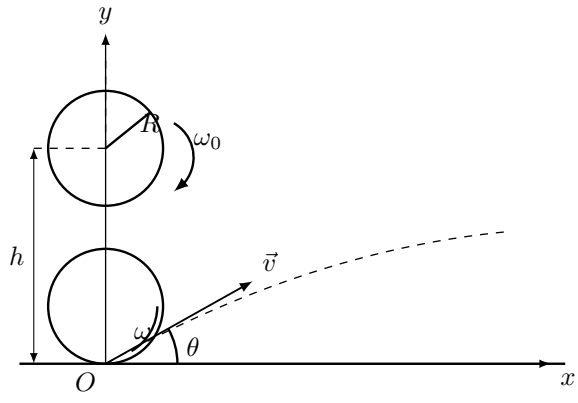


Fig. 1.62(a)

From the definition of the coefficient of restitution e,

$$e = \frac{v_y}{v_0} \tag{4}$$

From (2), (3) and (4) we obtain

$$v_x = \mu(1+e)v_0 \tag{5}$$

Hence the velocity v of the centre of mass of the ball after the collision is

$$v = \sqrt{v_x^2 + v_y^2} = v_0\sqrt{\mu^2(1+e)^2 + e^2} \tag{6}$$

The angle θ between the direction of v and the x-axis is given by

$$\tan\theta = \frac{v_y}{v_x} = \frac{e}{\mu(1+e)} \tag{7}$$

In the collision the rotational speed of the ball becomes smaller because of the action of the friction torque; from the angular momentum theorem,

$$\int_0^{\Delta t} -fR\,\mathrm{d}t = -N\mu R\Delta t = I\,(\omega - \omega_0) \tag{8}$$

From (2), (5) and (8) we obtain

$$\omega = \omega_0 - \frac{\mu mR(1+e)v_0}{I}$$

The moment of inertia of the ball about the axis through its centre of mass is $I = \dfrac{2}{5}mR^2$; substituting this and (1) into the above expression gives

$$\omega = \omega_0 - \frac{5\mu(1+e)}{2R}\sqrt{2gh} \tag{9}$$

This case is the one in which, while the ball is in contact with the ground, there is always relative sliding between the two; the requirement for this is

$$\omega R \geq v_x.$$

Substituting (5) and (9) into the above expression gives

$$\omega_0 \geq \frac{7\mu(1+e)}{2R}\sqrt{2gh} = \omega_{0c},$$

and this is the initial condition corresponding to this case.

Case 2. During the time Δt of the collision between the ball and the ground, there is relative sliding between the points of contact of the two only during the first part of the time Δt_1, while during the latter part of the time there is no longer relative sliding, that is, the ball has already reached pure rolling before the collision ends, and the friction will become zero. Similarly, applying the momentum theorem in the x direction,

$$\int_0^{\Delta t_1} f \,\mathrm{d}t = \int_0^{\Delta t_1} \mu N \,\mathrm{d}t = m v_x \tag{10}$$

From the angular momentum theorem,

$$\int_0^{\Delta t_1} -fR \,\mathrm{d}t = \int_0^{\Delta t_1} -\mu N R \,\mathrm{d}t = I\left(\omega - \omega_0\right) \tag{11}$$

Since at the end of this stage the ball has already reached pure rolling, we have

$$\omega = \frac{v_x}{R} \tag{12}$$

From (10), (11) and (12) we obtain

$$v_x = \frac{IR}{I + mR^2}\omega_0 = \frac{2}{7}\omega_0 R,$$

while v_y is still given by (4); hence the velocity of the centre of mass after the collision is

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{\left(\frac{2}{7}\omega_0 R\right)^2 + \left(ev_0\right)^2} \tag{13}$$

The angle θ between the direction of v and the x-axis satisfies

$$\tan\theta = \frac{v_y}{v_x} = \frac{7ev_0}{2\omega_0 R} \tag{14}$$

The angular velocity of the ball after the collision is

$$\omega = \frac{v_x}{R} = \frac{2}{7}\omega_0.$$

The initial condition corresponding to this case is

$$\omega_0 < \frac{7\mu(1+e)}{2R}\sqrt{2gh} = \omega_{0c}.$$

(2) After the collision between the ball and the ground the centre of the ball performs projectile motion. From the range formula for projectile motion, the horizontal distance s between the first and the second point of landing of the ball is

$$s = \frac{2v^2 \sin\theta\cos\theta}{g} \tag{15}$$

For case 1, from (7) we obtain

$$\sin\theta = \frac{e}{\sqrt{e^2+\mu^2(1+e)^2}}, \qquad \cos\theta = \frac{\mu(1+e)}{\sqrt{e^2+\mu^2(1+e)^2}}.$$

Substituting (6) and the above two expressions into (15) gives

$$\begin{aligned} s &= \frac{2}{g}\left[e^2+\mu^2(1+e)^2\right] v_0^2 \left[\frac{\mu e(1+e)}{e^2+\mu^2(1+e)^2}\right] \\ &= 4\mu(1+e)h. \end{aligned}$$

For case 2, from (14) we obtain

$$\sin\theta = \frac{ev_0}{\sqrt{\left(\frac{2}{7}\omega_0 R\right)^2+(ev_0)^2}}, \qquad \cos\theta = \frac{\frac{2}{7}\omega_0 R}{\sqrt{\left(\frac{2}{7}\omega_0 R\right)^2+(ev_0)^2}}.$$

Substituting (13) and the above two expressions into (15) gives

$$\begin{aligned} s &= \frac{2}{g}\left[\left(\frac{2}{7}\omega_0 R\right)^2+(ev_0)^2\right]\left[\frac{\frac{2}{7}\omega_0 R e v_0}{\left(\frac{2}{7}\omega_0 R\right)^2+(ev_0)^2}\right] \\ &= \frac{4}{7}e\omega_0 R\sqrt{\frac{2h}{g}}. \end{aligned}$$

(3) Equations (7) and (14) respectively give the relation of $\tan\theta$ with the initial angular velocity ω_0 in case 1 and in case 2. For case 1, that is, when $\omega_0 \geq \omega_{0c}$, $\tan\theta$ is independent of ω_0 and is the constant $\dfrac{e}{\mu(1+e)}$; for case 2, that is, when $\omega_0 < \omega_{0c}$, $\tan\theta$ is inversely proportional to ω_0. Hence the curve of $\tan\theta$ against ω_0 is as shown in Fig. 1.62(b).

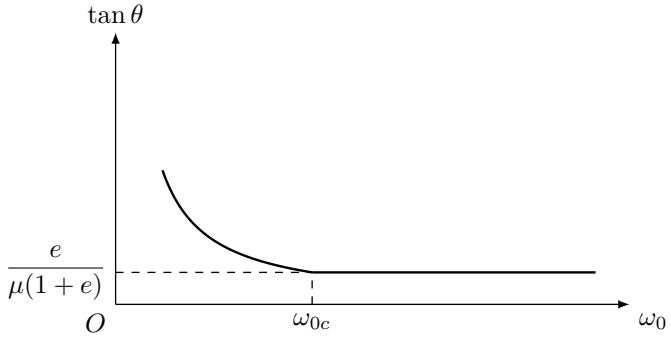


Fig. 1.62(b)

Solution 1.63. (1) When the top spins rapidly about its axis of spin, the top has a very large spin angular momentum L along the direction of the axis of spin, as shown in Fig. 1.63(a).

The torque τ_g of the weight acting on the top about the point of support O gives the angular momentum L, within the time $\mathrm{d}t$, an increment $\mathrm{d}L$ in the direction of τ_g. This means that the axis of spin of the top will precess about the vertical axis through O. In addition, the axis of spin of the top in Fig. 1.63(a) inclines to the left, and the left-hand part of the lower end of the axis is in contact with the ground; because of the precession, the direction of the friction f acting on the end of the axis is perpendicular to the plane of the figure and inwards, so the direction of the friction torque τ_f about the centre of mass C is as shown in Fig. 1.63(b). Therefore, within the time $\mathrm{d}t$ the angular momentum has an increment $\mathrm{d}L$ along the direction of τ_f. This causes the axis of spin of the top to perform another kind of precession, that is, while precessing about the vertical axis it also moves gradually towards the vertical axis, and finally becomes vertical.

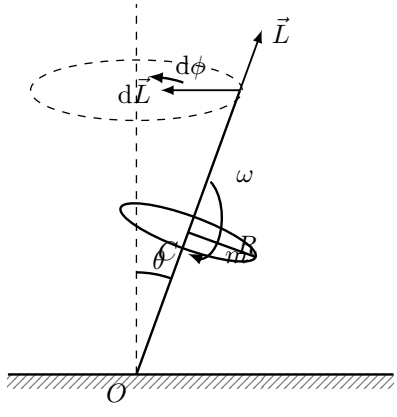


Fig. 1.63(a)

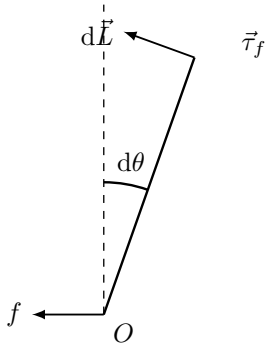


Fig. 1.63(b)

The torque τ_g of the weight acting on the top about point O is

$$\tau_g = mg\frac{l}{2}\sin\theta \tag{1}$$

The increment $\mathrm{d}L$ of the angular momentum within the time $\mathrm{d}t$ is

$$\mathrm{d}L = L\sin\theta\,\mathrm{d}\phi \tag{2}$$

From the angular momentum theorem,

$$\tau_g\,\mathrm{d}t = \mathrm{d}L \tag{3}$$

From (1), (2) and (3) we obtain the angular velocity Ω of the precession of the top about the vertical axis:

$$\Omega = \frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{mgl}{2L} = \frac{mgl}{2I\omega} \tag{4}$$

where I is the moment of inertia of the top about its axis of spin. From (4) it is seen that Ω is inversely proportional to ω; as ω becomes smaller and smaller, the angular velocity of precession Ω becomes larger and larger.

(2) The friction is $f = \mu mg$, and the torque τ_f of the friction about the centre of mass C is

$$\tau_f = \mu mg\frac{l}{2} \tag{5}$$

The increment $\mathrm{d}L$ of the angular momentum within the time $\mathrm{d}t$ is

$$\mathrm{d}L = -L\,\mathrm{d}\theta \tag{6}$$

From the angular momentum theorem,

$$\tau_f\,\mathrm{d}t = \mathrm{d}L \tag{7}$$

From (5), (6) and (7) we obtain the angular velocity Ω' of the precession of the axis of spin of the top towards the vertical axis:

$$\Omega' = \frac{\mathrm{d}\theta}{\mathrm{d}t} = -\frac{\mu mgl}{2I\omega} \tag{8}$$

Hence the time t that elapses from the moment when the axis of spin makes an angle θ with the vertical line until it turns to the vertical position is

$$t = -\frac{\theta}{\Omega'} = \frac{2I\omega\theta}{\mu mgl}.$$

Solution 1.64. (1) In the reference frame S' of the spaceship, the time that elapses from the spaceship's flying past the Earth to its flying past the space station is $\Delta t_1' = 30 \times 60 = 1800\,\mathrm{s}$. Since this is the proper time, the time Δt_1 measured in the space-station–Earth reference frame S is

$$\Delta t_1 = \frac{\Delta t_1'}{\sqrt{1 - v^2/c^2}} = \frac{1800}{\sqrt{1 - (0.8)^2}} = 3000\ (\mathrm{s}),$$

that is, 50 minutes; hence the reading of the clock on the space station is 12:50.

(2) In frame S the distance between the space station and the Earth is

$$l = 0.8c \times 50 \times 60 = 2400c.$$

The time Δt_2 that elapses from the moment the spaceship sends out the report signal until the Earth receives this signal is

$$\Delta t_2 = \frac{l}{c} = 2400\ (\mathrm{s}),$$

that is, 40 minutes; hence the clock on the Earth at this moment indicates 1:30 p.m.

(3) In frame S' the distance between the space station and the Earth is

$$l' = 0.8c \times 30 \times 60 = 1440c.$$

Let the time that elapses in frame S' from the moment the spaceship sends out the report signal until it receives the signal sent back by the Earth be $\Delta t'$; then

$$(c-v)\frac{\Delta t'}{2} = l'$$

we obtain

$$\Delta t' = \frac{2l'}{c-v} = \frac{2l'}{0.2c} = \frac{10l'}{c} = 14400 \text{ (s)},$$

that is, 4 hours; hence when the signal sent back by the Earth is received, the clock on the spaceship indicates 4:30 p.m.

This part of the problem can also be solved by the following method.

In frame S, when the Earth receives the report signal sent out by the spaceship, the distance between the spaceship and the Earth is $L = l + v\Delta t_2$. Let the time that elapses from the moment the Earth sends out the reply signal until the spaceship receives this signal be Δt_3; then

$$(c-v)\Delta t_3 = L = l + v\Delta t_2,$$

so

$$\Delta t_3 = \frac{l + v\Delta t_2}{c-v} = 21600 \text{ (s)}.$$

Hence the total time Δt that elapses from the moment the spaceship sends out the report signal until it receives the reply signal sent back by the Earth is

$$\Delta t = \Delta t_2 + \Delta t_3 = 24000 \text{ (s)}.$$

Since in frame S' the time $\Delta t'$ is the proper time and Δt is the dilated time, $\Delta t'$ is

$$\Delta t' = \Delta t\sqrt{1 - v^2/c^2} = 24000 \times 0.6 = 14400 \text{ (s)}.$$

Solution 1.65. Take the ground as the reference frame S and the moving thick glass as the reference frame S', the frame S' moving with velocity v in the x direction relative to S. In frame S' the speed of propagation of light in the glass is $u'_x = \dfrac{c}{n}$; from the velocity transformation formula, the speed of this propagation u_x in frame S is

$$u_x = \frac{u'_x + v}{1 + \dfrac{u'_x v}{c^2}} = \frac{\dfrac{c}{n} + v}{1 + \dfrac{v}{cn}} = \frac{c^2 + cnv}{cn + v} \tag{1}$$

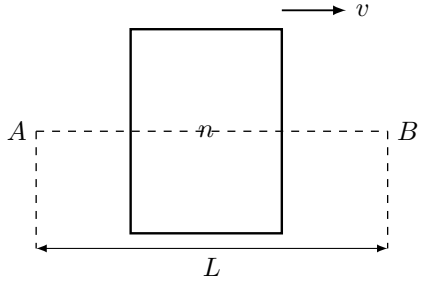


Fig. 1.65

In frame S' the thickness of the glass is D_0; then the thickness D in frame S is

$$D = D_0\sqrt{1-\beta^2} \tag{2}$$

As seen by an observer in frame S, let the time required for the light to propagate through the thick glass be Δt_1; during this time the glass has moved a distance $v\Delta t_1$ in the x direction, hence

$$D + v\Delta t_1 = u_x \Delta t_1 \tag{3}$$

Substituting (1) and (2) into (3) gives

$$\Delta t_1 = \frac{D_0\sqrt{1-\beta^2}}{u_x - v} = \frac{D_0(cn+v)}{c^2\sqrt{1-\beta^2}} \tag{4}$$

Let the time that the light takes to propagate through the space outside the glass be Δt_2; then

$$c\Delta t_2 = L - u_x \Delta t_1 \tag{5}$$

Substituting (1) and (4) into (5) gives

$$\Delta t_2 = \frac{L}{c} - \frac{D_0(c+nv)}{c^2\sqrt{1-\beta^2}} \tag{6}$$

Hence the total time Δt required for the light to propagate from point A to point B is

$$\Delta t = \Delta t_1 + \Delta t_2 \tag{7}$$

Substituting (4) and (6) into (7) gives

$$\begin{aligned}\Delta t &= \frac{D_0(cn+v)}{c^2\sqrt{1-\beta^2}} + \frac{L}{c} - \frac{D_0(c+nv)}{c^2\sqrt{1-\beta^2}} \\ &= \frac{L}{c} + \frac{D(n-1)(c-v)}{c^2\sqrt{1-\beta^2}}.\end{aligned}$$

Equation (4) in this problem can also be obtained by the following method.

In frame S' the speed of propagation of light in the glass is $u'_x = \dfrac{c}{n}$. Hence the time $\Delta t'_1$ taken by the light to propagate through the thick glass is

$$\Delta t'_1 = \frac{D_0}{u'_x} = \frac{nD_0}{c}.$$

From the Lorentz transformation, the time Δt_1 measured in frame S for the light to propagate through the glass can be found:

$$\begin{aligned}\Delta t_1 &= \frac{\Delta t'_1 + \dfrac{v}{c^2}\Delta x'}{\sqrt{1-\beta^2}} = \frac{\dfrac{nD_0}{c} + \dfrac{v}{c^2}D_0}{\sqrt{1-\beta^2}} \\ &= \frac{D_0(cn+v)}{c^2\sqrt{1-\beta^2}}.\end{aligned}$$

This is (4).

Solution 1.66. (1) As shown in Fig. 1.66(a), in the inertial frame S spaceship B flies along the y direction with speed $\frac{c}{2}$; therefore, in order that the packet thrown from spaceship A can be received by spaceship B, the y component u_y of its velocity u must be the same as the flying speed of spaceship B, that is,

$$u_y = \frac{c}{2}.$$

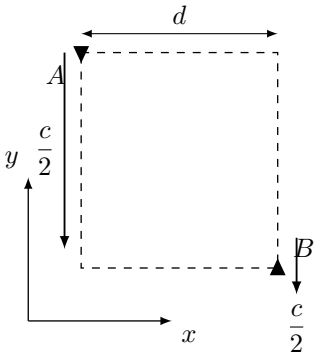


Fig. 1.66(a)

In frame S the speed with which the packet is thrown is $u = \frac{3}{4}c$, so the x component u_x of the velocity is

$$u_x = \sqrt{u^2 - u_y^2} = \frac{\sqrt{5}}{4}c.$$

The coordinate system S', which is at rest relative to spaceship A, moves relative to frame S with speed $\frac{c}{2}$ along the negative y direction, that is, $v = -\frac{c}{2}$; from the velocity transformation formula, the two velocity components u'_y and u'_x of the packet at the moment it is thrown, as measured in frame S', are respectively

$$u'_y = \frac{u_y - v}{1 - \frac{u_x v}{c^2}} = \frac{\frac{c}{2} + \frac{c}{2}}{1 + \frac{1}{4}} = \frac{4}{5}c,$$

$$u'_x = \frac{u_x\sqrt{1-\beta^2}}{1 - \frac{u_y v}{c^2}} = \frac{\frac{\sqrt{5}}{4}c\sqrt{1 - \frac{1}{4}}}{1 + \frac{1}{4}} = \frac{\sqrt{15}}{10}c.$$

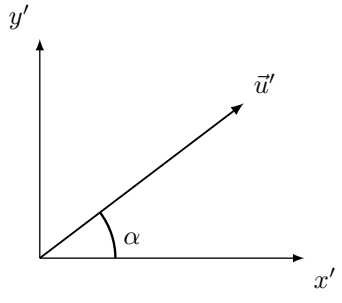


Fig. 1.66(b)

As shown in Fig. 1.66(b), therefore, as seen by an observer in frame S' (that is, the reference frame of spaceship A), the angle α between the direction of the velocity of the packet at the moment it is thrown and the x' axis is given by

$$\tan\alpha = \frac{u'_y}{u'_x} = \frac{8}{\sqrt{15}},$$

that is,

$$\alpha = \arctan\frac{8}{\sqrt{15}} \approx 64^\circ.$$

(2) In frame S' the speed with which the packet is thrown is u':

$$u' = \sqrt{u'^2_x + u'^2_y} = \frac{\sqrt{79}}{10}c \approx 0.89c.$$

Solution 1.67. (1) Let the proper length of the two rulers AB and $A'B'$ be l_0. The observer in the S' frame measures the length of ruler AB to be $\sqrt{1-\beta^2}\,l_0$; in his view, B' meets B first, and A' meets A later. The time difference between these two events is

$$\Delta t' = \frac{l_0 - \sqrt{1-\beta^2}\,l_0}{v}.$$

When A meets A', the light signal emitted by A' takes a time l_0/c to reach B' in the S' frame. Hence when B' receives this light signal the reading $t_{B'}$ of the clock B' is

$$t_{B'} = \Delta t' + \frac{l_0}{c} = \frac{l_0}{c} + \frac{l_0 - \sqrt{1-\beta^2}\,l_0}{v}. \tag{1}$$

From the statement of the problem,

$$t_{B'} = 1. \tag{2}$$

From (1) and (2) we obtain

$$\frac{l_0}{c} = \frac{\beta}{\beta + \left(1 - \sqrt{1-\beta^2}\right)}. \tag{3}$$

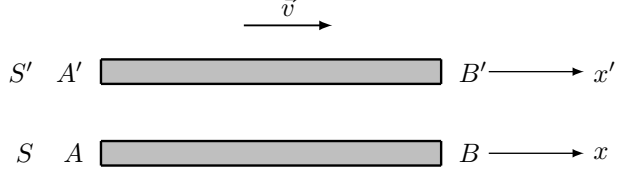


Fig. 1.67

The observer in the S frame measures the length of ruler $A'B'$ to be $\sqrt{1-\beta^2}\,l_0$; in his view, A meets A' first, and B meets B' later. The time difference between these two events is

$$\Delta t = \frac{l_0 - \sqrt{1-\beta^2}\,l_0}{v}.$$

When A meets A', the light signal emitted by A' takes a time l_0/c to reach B in the S frame, hence when B receives this light signal the reading t_B of the clock B is

$$t_B = \frac{l_0}{c} - \Delta t = \frac{l_0}{c} - \frac{l_0 - \sqrt{1-\beta^2}\, l_0}{v}$$
$$= \frac{l_0}{c}\left(1 - \frac{1-\sqrt{1-\beta^2}}{\beta}\right). \tag{4}$$

Substituting (3) into (4) gives

$$t_B = \frac{\beta}{\beta + \left(1-\sqrt{1-\beta^2}\right)} \cdot \frac{\beta - \left(1-\sqrt{1-\beta^2}\right)}{\beta}$$
$$= \sqrt{\frac{1-\beta}{1+\beta}} \qquad \text{(time units)}.$$

(2) (a) In the view of the observer in the S' frame, after B' has received the signal emitted by A' it immediately emits a reply signal; this signal takes a time l_0/c to reach A', hence when A' receives this reply signal the reading $t_{A'}$ of the clock A' is

$$t_{A'} = \frac{2l_0}{c}. \tag{5}$$

Substituting (3) into (5) gives

$$t_{A'} = \frac{2\beta}{\beta + \left(1-\sqrt{1-\beta^2}\right)} = 1 + \sqrt{\frac{1-\beta}{1+\beta}} \qquad \text{(time units)}.$$

(b) Let the reading of clock A in the S frame be t_{A1} at the moment when A' receives the reply signal. Since $t_{A'}$ is a proper time, t_{A1} is

$$t_{A1} = \frac{t_{A'}}{\sqrt{1-\beta^2}} = \frac{2l_0}{c\sqrt{1-\beta^2}}.$$

At this moment the distance between A' and A is

$$\Delta l_{AA'} = v t_{A1} = \frac{2vl_0}{c\sqrt{1-\beta^2}}.$$

The light signal then takes a further time $\dfrac{\Delta l_{AA'}}{c}$ to reach A, hence when A receives this reply signal the reading t_A of clock A is

$$t_A = t_{A1} + \frac{\Delta l_{AA'}}{c} = \frac{2(1+\beta)l_0}{c\sqrt{1-\beta^2}}. \tag{6}$$

Substituting (3) into (6) gives

$$t_A = 1 + \sqrt{\frac{1+\beta}{1-\beta}} \qquad \text{(time units)}.$$

Solution 1.68. (1) Take the Galaxy as the S frame and the traveller as the S' frame; the S' frame moves with velocity v along the x direction relative to the S frame, as shown in Fig. 1.68(a). In the S frame the velocity of the light coming from the edge of the Galaxy makes an angle $\varphi = 45°$ with the x-axis. Hence the components of the light velocity along the x and y directions are respectively $u_x = c\cos\varphi$ and $u_y = c\sin\varphi$; from the velocity transformation formulas the two components of the light velocity in the S' frame are respectively

$$u'_x = \frac{u_x - v}{1 - \dfrac{u_x v}{c^2}} = \frac{c\cos\varphi - v}{1 - \dfrac{v}{c}\cos\varphi},$$

$$u'_y = \frac{u_y\sqrt{1-\beta^2}}{1 - \dfrac{u_x v}{c^2}} = \frac{c\sin\varphi\sqrt{1-\beta^2}}{1 - \dfrac{v}{c}\cos\varphi}.$$

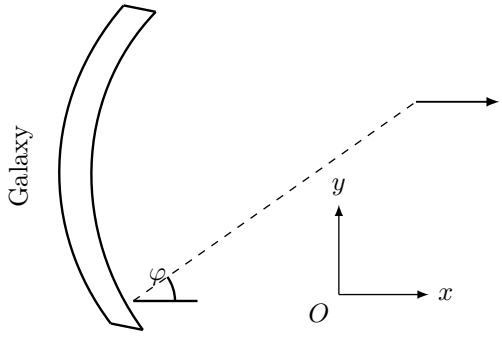


Fig. 1.68(a)

Let the angle between this light velocity direction and the x'-axis in the S' frame be φ'; then

$$\tan\varphi' = \frac{u'_y}{u'_x} = \frac{\sqrt{1-\beta^2}}{1 - \frac{\beta}{\cos\varphi}} \cdot \tan\varphi.$$

Substituting $\beta = \dfrac{v}{c} = 0.99$ and $\varphi = 45°$ into the above expression gives

$$\tan\varphi' = \frac{\sqrt{1-(0.99)^2}}{1 - 0.99 \times \sqrt{2}} = -0.3529,$$

that is, $\varphi' = 160.58°$. Fig. 1.68(b) shows the shape of the Galaxy as it appears to the traveller: just as described by the hero of the novel, it looks like a "goldfish bowl", and he is flying towards the mouth of the bowl.

(2) According to the formula for the Doppler effect of light, the frequency ν' of the light from the edge of the Galaxy received by the traveller is

$$\nu' = \frac{\nu\sqrt{1-\beta^2}}{1 - \frac{v}{c}\cos\theta}.$$

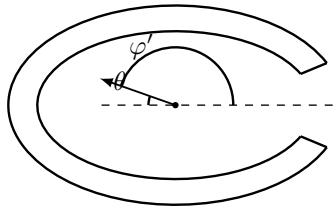


Fig. 1.68(b)

In the above expression ν is the frequency of the light actually emitted at the edge of the Galaxy. θ and φ' are supplementary to each other, as shown in Fig. 1.68(b), that is, $\cos\theta = 0.9431$. Substituting this and $\beta = 0.99$ into the above expression gives

$$\frac{\nu'}{\nu} = \frac{\sqrt{1-(0.99)^2}}{1-0.99\times 0.9431} = 2.127.$$

From this ratio it is seen that the light from the edge of the Galaxy observed by the traveller is blue-shifted.

(3) Take $\varphi' = 0$, that is, $\theta = \pi$, $\cos\theta = -1$. Then

$$\frac{\nu'}{\nu} = \frac{\sqrt{1-(0.99)^2}}{1+0.99} = 0.0709.$$

This is the light from the centre of the Galaxy, which appears red-shifted to the traveller.

If $\nu' = \nu$ is taken, then

$$\cos\theta = \frac{1-\sqrt{1-\beta^2}}{\beta} = \frac{1-\sqrt{1-(0.99)^2}}{0.99} = 0.8676,$$

that is, $\theta = 29.8°$. At $\varphi' = -150.18°$ there is a critical angle: if θ is greater than this angle the light is red-shifted, whereas if θ is smaller than this angle the light is blue-shifted. This means that the closer to the centre of the Galaxy the light comes from, the more marked is its red shift, and the closer to the edge of the Galaxy the light comes from, the more marked is its blue shift.

When the traveller is at the centre of the Galaxy and flying outwards, for the light coming from the edge of the Galaxy one has $\varphi = \dfrac{\pi}{2}$, $\varphi' = \arctan\left(\dfrac{\sqrt{1-\beta^2}}{-\beta}\right) = 171.89°$, that is, $\theta = 8.11°$; hence

$$\frac{\nu'}{\nu} = \frac{\sqrt{1-(0.99)^2}}{1-0.99\times 0.99} = 7.089.$$

From this ratio it is seen that the closer the traveller is to the centre of the Galaxy, the greater is the degree of blue shift of the light from the edge of the Galaxy.

In summary, what the space hero says in the novel written by Hall — namely his description of flying inside a "goldfish bowl" with a blue rim and a red body, towards the mouth of the bowl — is basically correct, and Feynman lost his bet.

Solution 1.69. (1) Let the proper wavelength of the light emitted by the source be λ_0. When the source approaches the Earth with velocity v_1, the formula for the Doppler effect of light gives the wavelength λ_1 of the light received by the person A in the S frame, which is at rest relative to the Earth, as

$$\lambda_1 = \lambda_0\sqrt{\frac{c-v_1}{c+v_1}}. \tag{1}$$

Another inertial frame S' moves relative to the Earth with velocity v_2 along the same straight line as v_1. Let the velocity of the source observed in the S' frame be v_1'; then the velocity transformation formula gives

$$v_1' = \frac{v_1 - v_2}{1 - \dfrac{v_1 v_2}{c^2}}. \tag{2}$$

Let the wavelength of the light emitted by the source as seen by the observer B in the S' frame be λ_2. By the formula for the Doppler effect of light, λ_2 is

$$\lambda_2 = \lambda_0 \sqrt{\frac{c - v_1'}{c + v_1'}}. \tag{3}$$

When the source recedes from the Earth with the same speed v_1, the wavelength of the light seen by A in the S frame is λ_2, hence

$$\lambda_2 = \lambda_0 \sqrt{\frac{c + v_1}{c - v_1}}. \tag{4}$$

From (1) and (4) we obtain

$$\lambda_0 = \sqrt{\lambda_1 \lambda_2} = 547.7 \text{ (nm)}. \tag{5}$$

Substituting (5) into (1) gives

$$\left(\frac{500}{547.7}\right)^2 = \frac{c - v_1}{c + v_1}.$$

Solving, we obtain

$$v_1 = \frac{1}{11} c. \tag{6}$$

From (3) and (4) we obtain

$$v_1' = -v_1 = -\frac{1}{11} c. \tag{7}$$

Substituting (6) and (7) into (2) gives

$$v_2 = 0.18c.$$

v_2 being positive means that the S' frame moves in the direction approaching the Earth.

(2) When the source recedes from the Earth with speed v_1, the speed v_1'' of the source as observed by B in the S' frame is

$$v_1'' = \frac{v_1 - v_2}{1 - \dfrac{v_1 v_2}{c^2}} = \frac{-\frac{1}{11}c - 0.18c}{1 + \dfrac{0.18}{11}} = -0.267c.$$

Hence, by the formula for the Doppler effect of light, the wavelength λ_1' of the light seen by B is

$$\lambda_1' = \lambda_0 \sqrt{\frac{c - v_1''}{c + v_1''}} = \lambda_0 \sqrt{\frac{1 + 0.267}{1 - 0.267}} = 720.1 \text{ (nm)}.$$

Solution 1.70. Let the rest mass of this particle be m_0, its velocity before decay be v, and let the directions of the two γ rays, whose energies after the decay are E_1 and E_2 respectively, make angles θ_1 and θ_2 respectively with the direction of motion of the particle (the x direction), as

shown in Fig. 1.70. From the conservation of momentum of the system in the x and y directions we obtain respectively

$$\frac{m_0 v}{\sqrt{1-\beta^2}} = \frac{E_1}{c}\cos\theta_1 + \frac{E_2}{c}\cos\theta_2, \tag{1}$$

$$0 = \frac{E_1}{c}\sin\theta_1 - \frac{E_2}{c}\sin\theta_2. \tag{2}$$

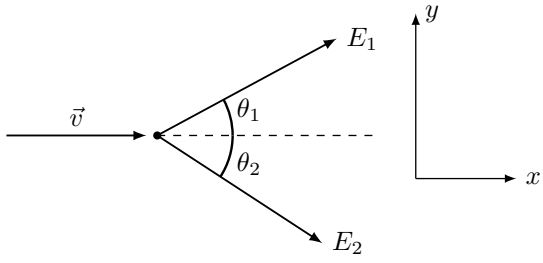


Fig. 1.70

From the statement of the problem,

$$\theta_1 + \theta_2 = \theta. \tag{3}$$

From the conservation of energy of the system before and after the decay of the particle,

$$\frac{m_0 c^2}{\sqrt{1-\beta^2}} = E_1 + E_2. \tag{4}$$

Squaring both sides of (1) and (2) separately and then adding the two equations, we obtain

$$\frac{m_0^2 v^2}{1 - v^2/c^2} = \frac{E_1^2}{c^2} + \frac{E_2^2}{c^2} + 2\frac{E_1 E_2}{c^2}\left(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2\right).$$

Substituting (3) into the above expression gives

$$\frac{m_0^2 c^4}{1 - v^2/c^2}\cdot\frac{v^2}{c^4} = \frac{1}{c^2}\left(E_1^2 + E_2^2 + 2E_1E_2\cos\theta\right).$$

Substituting (4) into the above expression gives

$$(E_1 + E_2)^2\cdot\frac{v^2}{c^2} = (E_1 + E_2)^2 - 2E_1E_2(1-\cos\theta),$$

that is,

$$\sqrt{1 - v^2/c^2} = \frac{\sqrt{2E_1E_2(1-\cos\theta)}}{E_1 + E_2}.$$

Substituting this expression into (4), we obtain the rest mass m_0 of this particle:

$$m_0 = \frac{1}{c^2}\sqrt{2E_1E_2(1-\cos\theta)}.$$

Solution 1.71. (1) Let the velocity of the π meson be v, its rest mass be $m_0 = \dfrac{E_{0\pi}}{c^2} = 139.6\ \text{MeV}/c^2$, and its momentum P_π be

$$P_\pi = \frac{m_0 v}{\sqrt{1-\beta^2}} = \frac{m_0 c\beta}{\sqrt{1-\beta^2}} = 2\times 10^5\ (\text{MeV}/c),$$

that is,

$$\frac{\beta}{\sqrt{1-\beta^2}} = \frac{2\times 10^5}{139.6}.$$

We obtain

$$\beta = 0.999\,999\,756.$$

In the laboratory frame the lifetime τ_π of the π meson is

$$\tau_\pi = \frac{\tau_{\pi 0}}{\sqrt{1-\beta^2}}.$$

Hence the distance l travelled by the π meson before it decays is

$$\begin{aligned} l = \tau_\pi v &= \tau_{\pi 0}\cdot c\cdot \frac{\beta}{\sqrt{1-\beta^2}} \\ &= 2.6\times 10^{-8}\times 3\times 10^8 \times \frac{2\times 10^5}{139.6} \\ &= 11\,175\ (\text{m}). \end{aligned}$$

(2) Take the laboratory frame as the S frame and the centre-of-mass frame of the system as the S' frame. In the S' frame the total energy of the system is just the rest energy of the π meson, $E' = 139.6$ MeV. Let the energy of the μ meson be E'_μ and the energy of the neutrino be E'_ν; from the conservation of energy,

$$E' = E'_\mu + E'_\nu. \tag{1}$$

In the centre-of-mass frame S', let the momentum of the μ meson be p'; from the properties of the centre-of-mass frame, the momentum of the neutrino is $-p'$. For these two particles respectively we have

$$E'^2_\mu = E^2_{\mu 0} + p'^2 c^2, \tag{2}$$

$$E'_\nu = p'c. \tag{3}$$

Substituting (2) and (3) into (1) and solving, we obtain

$$\begin{aligned} p' &= \frac{E'^2 - E^2_{\mu 0}}{2E'c} = \frac{(139.6)^2 - (105.7)^2}{2\times(139.6)}\ (\text{MeV}/c) \\ &= 29.783\,9\ (\text{MeV}/c). \end{aligned}$$

Substituting the above expression into (2) gives

$$E'_\mu = \sqrt{(105.7)^2 + (29.8)^2} = 109.816\ (\text{MeV}).$$

Take the direction of motion of the π meson as the positive direction of the x-axis, and let θ' denote the angle between p' and the x-axis. From the relativistic momentum transformation

formulas, the components of the momentum of the μ meson along the x and y directions in the S frame are respectively

$$p_{\mu x} = \frac{p'_x + \frac{v}{c^2}E'_\mu}{\sqrt{1-\beta^2}} = \frac{p'\cos\theta' + \frac{v}{c^2}E'_\mu}{\sqrt{1-\beta^2}}, \tag{4}$$

$$p_{\mu y} = p'_y = p'\sin\theta'. \tag{5}$$

Let the angle between the momentum of the μ meson and the x-axis in the S frame be θ; then from (4) and (5) we obtain

$$\tan\theta = \frac{p_{\mu y}}{p_{\mu x}} = \frac{p'\sin\theta'\sqrt{1-\beta^2}}{p'\cos\theta' + \frac{v}{c^2}E'_\mu}. \tag{6}$$

Differentiating both sides of (6) with respect to θ', when θ takes its maximum value we have $\frac{\mathrm{d}\theta}{\mathrm{d}\theta'} = 0$, and hence

$$\cos\theta' = -\frac{p'^2c^2}{E'_\mu v} = -0.271\,2,$$

that is,

$$\theta' = 105.736^\circ.$$

Substituting this value of θ' into (6), the tangent of the maximum value $\theta_{\max}$ of θ is found to be

$$\tan\theta_{\max} = 0.000\,196\,6,$$

that is,

$$\theta_{\max} = 0.011\,267^\circ.$$

(3) The maximum and minimum momenta of the neutrino in the laboratory frame S occur respectively when its momentum in the centre-of-mass frame S' is along the positive and the negative direction of the x-axis. From the momentum transformation formulas for relative motion, the maximum momentum $p_{\nu\,\max}$ and the minimum momentum $p_{\nu\,\min}$ of the neutrino are respectively

$$p_{\nu\,\max} = \frac{p' + \frac{v}{c^2}E'_\nu}{\sqrt{1-\beta^2}} = \frac{p'(1+\beta)}{\sqrt{1-\beta^2}} = 85\,358.5\ (\mathrm{MeV}/c),$$

$$p_{\nu\,\min} = \frac{-p' + \frac{v}{c^2}E'_\nu}{\sqrt{1-\beta^2}} = -\frac{p'(1-\beta)}{\sqrt{1-\beta^2}} = -0.010\,413\,7\ (\mathrm{MeV}/c).$$

Solution 1.72. During its flight from the Earth to the Galaxy the rocket is always in accelerated motion, so if the rocket is taken as the S' frame of reference, this frame is a non-inertial frame. If the special theory of relativity is to be used to solve this kind of problem, one may set up in succession a series of coordinate systems in the S' frame, the coordinate system at any instant t moving uniformly and rectilinearly relative to the Earth with the velocity v of the rocket at that instant during an extremely short interval of time. Hence during this short interval S' may be regarded as an inertial frame, and the special theory of relativity can then be used; the whole accelerating process of the rocket is thus a combination of this series of infinitely short processes of motion.

In the S' frame the rocket moves with acceleration a'. In the time $\mathrm{d}t'$ the increment of the momentum of the rocket is $\mathrm{d}p'$; according to the laws of dynamics,

$$\frac{\mathrm{d}p'}{\mathrm{d}t'} = Ma', \tag{1}$$

where M is approximately the rest mass of the rocket at that instant. From the conservation of momentum it is known that the increment of the momentum of the rocket should be equal to the momentum of the photons radiated in the time $\mathrm{d}t'$; at the same time as the rocket radiates photons, the energy E of the rocket (and of the fuel in it) changes, so according to the mass-energy formula $E = Mc^2$,

$$\mathrm{d}p' = -\frac{1}{c}\mathrm{d}E = -c\,\mathrm{d}M. \tag{2}$$

From (1) and (2) we obtain

$$Ma' = -c\frac{\mathrm{d}M}{\mathrm{d}t'}.$$

The power N of the rocket engine at any moment is

$$N = -\frac{\mathrm{d}E}{\mathrm{d}t'} = -c^2\frac{\mathrm{d}M}{\mathrm{d}t'} = cMa'. \tag{3}$$

Equation (3) shows that the power of the rocket engine depends on the mass of the rocket at that moment; hence one need only find the initial mass M_0 of the rocket in order to find the required initial power. Let the rest mass of the rocket at half the journey (that is, at the end of the accelerating process) be M_h, its velocity be v, and let the total energy of the photons radiated during this half of the journey be E_r; then from the conservation of energy and the conservation of momentum we obtain respectively

$$M_0c^2 = \frac{M_hc^2}{\sqrt{1 - v^2/c^2}} + E_r, \tag{4}$$

$$\frac{M_hv}{\sqrt{1 - v^2/c^2}} = \frac{E_r}{c}. \tag{5}$$

From (4) and (5) we obtain

$$\frac{M_h}{M_0} = \sqrt{\frac{c - v}{c + v}}. \tag{6}$$

The second half of the journey may be treated in the same way, giving

$$\frac{M_0'}{M_h} = \sqrt{\frac{c - v}{c + v}}. \tag{7}$$

From (6) and (7) we obtain

$$M_0 = \frac{c + v}{c - v}M_0'. \tag{8}$$

Equation (8) shows that, in order to find the initial mass M_0 of the rocket, one must first find the velocity v of the rocket at half the journey. For this purpose, let the velocity of the rocket at time t in the terrestrial frame S be u; after a time $\mathrm{d}t$ the increment of u is $\mathrm{d}u$, while the corresponding increment in the S' frame is $\mathrm{d}u'$. Then the velocity transformation formula gives

$$u + \mathrm{d}u = \frac{u + \mathrm{d}u'}{1 + \dfrac{u}{c^2}\mathrm{d}u'},$$

that is,

$$(u + \mathrm{d}u)\left(1 + \frac{u}{c^2}\mathrm{d}u'\right) = u + \mathrm{d}u'.$$

Neglecting infinitesimals of higher order in the above expression, we obtain

$$\mathrm{d}u' = \frac{1}{1 - u^2/c^2}\mathrm{d}u = \frac{1}{1 - \beta^2}\mathrm{d}u. \tag{9}$$

Since $\mathrm{d}t'$ is a proper time, time dilation gives

$$\mathrm{d}t = \frac{\mathrm{d}t'}{\sqrt{1 - \beta^2}}. \tag{10}$$

From (9) and (10) we obtain

$$a' = \frac{\mathrm{d}u'}{\mathrm{d}t'} = \frac{1}{(1 - \beta^2)^{3/2}}\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{1}{(1 - \beta^2)^{3/2}}a,$$

where $a = \dfrac{\mathrm{d}u}{\mathrm{d}t}$ is the acceleration of the rocket in the S frame. Since a' is constant, a is not constant; hence the above expression may be rewritten as

$$\frac{\mathrm{d}u}{\left(1 - \dfrac{u^2}{c^2}\right)^{3/2}} = a'\mathrm{d}t.$$

Integrating the above expression and using the initial condition $u = 0$ at $t = 0$, we obtain

$$\frac{u}{\sqrt{1 - u^2/c^2}} = a't,$$

that is,

$$u = \frac{a't}{\sqrt{1 + \dfrac{a'^2}{c^2}t^2}}. \tag{11}$$

Since $u = \dfrac{\mathrm{d}x}{\mathrm{d}t}$, (11) may be written as

$$\mathrm{d}x = \frac{a't}{\sqrt{1 + \dfrac{a'^2}{c^2}t^2}}\mathrm{d}t.$$

Integrating the above expression and using the initial condition $x = 0$ at $t = 0$, we obtain

$$x = \frac{c^2}{a'}\left[\sqrt{1 + \frac{a'^2}{c^2}t^2} - 1\right].$$

When the rocket has flown half the journey, $x = \dfrac{R}{2}$; substituting this into the above expression, the terrestrial time t required to complete the half journey is found to be

$$t = \frac{R}{2c}\sqrt{1 + \frac{4c^2}{a'R}}. \tag{12}$$

Substituting (12) into (11), the velocity v of the rocket at half the journey is obtained as

$$v = \frac{c}{\sqrt{1+\dfrac{c^2}{a'^2t^2}}} = c\left[1+\frac{4c^4}{a'^2R^2}\left(1+\frac{4c^2}{a'R}\right)^{-1}\right]^{-1/2}.$$

Since $\dfrac{c^2}{a'R} \ll 1$, neglecting quantities of higher order in the above expression gives

$$v \approx c\left(1-\frac{2c^4}{a'^2R^2}\right). \tag{13}$$

Substituting (13) into (8), the rest mass M_0 of the rocket at take-off is

$$M_0 = \frac{c+v}{c-v}M_0' = \left(\frac{a'^2R^2}{c^4}-1\right)M_0' \approx \frac{a'^2R^2}{c^4}M_0'. \tag{14}$$

Substituting (14) into (3), the power N required by the engine at the moment the rocket starts is

$$N = cM_0a' = \frac{a'^3R^2}{c^3}M_0' = 3.8\times10^{24}\ (\mathrm{J/s}).$$

3.2 Thermodynamics

Solution 2.01. *Method 1.* Since the rotational speed of the pump is ω revolutions per minute and the volume of gas extracted per minute is c litres, the volume of gas extracted in one revolution of the pump is

$$v = \frac{c}{\omega}.$$

Let the pressure in the vessel after the pump has extracted the gas once be p_1; then

$$p_1(V+v) = p_0V,$$

that is,

$$p_1 = \left(\frac{V}{V+v}\right)p_0 = \left(\frac{V}{V+c/\omega}\right)p_0.$$

Let the pressure in the vessel after n strokes be p_n; then from the above analysis,

$$p_n = \left(\frac{V}{V+c/\omega}\right)^n p_0, \qquad n = \frac{\ln\dfrac{p_n}{p_0}}{\ln\left(\dfrac{V}{V+c/\omega}\right)}.$$

The time taken is

$$t = \frac{n}{\omega} = \frac{\ln\dfrac{p_0}{p_n}}{\omega\ln\left(1+\dfrac{c}{V\omega}\right)}.$$

When $p_n = 10^{-2}p_0$,

$$t = \frac{2\ln 10}{\omega\ln\left(1+\dfrac{c}{\omega V}\right)}.$$

If $\dfrac{c}{\omega V} \ll 1$, then

$$t = \frac{V}{c} \ln \frac{p_0}{p_n} = 2\,\frac{V}{c} \ln 10 \approx 4.606\,\frac{V}{c}\ \text{(min)}.$$

Method 2. Suppose that in time $\mathrm{d}t$ the pump extracts a volume of gas $\mathrm{d}v = c\,\mathrm{d}t$, and the pressure in the vessel is $p + \mathrm{d}p$; then

$$(p + \mathrm{d}p)(V + c\,\mathrm{d}t) = pV,$$

that is,

$$\frac{\mathrm{d}p}{p} = -\frac{c}{V}\,\mathrm{d}t.$$

Integrating both sides gives

$$t = \frac{V}{c} \ln \frac{p_0}{p} = \frac{V}{c} \ln \frac{p_0}{10^{-2} p_0} = \frac{2V}{c} \ln 10 = 4.606\,\frac{V}{c}\ \text{(min)}.$$

Solution 2.02. Since the interaction potential energy E_p of the two atoms in a diatomic molecule is

$$E_p = E_p(0) \left[\left(\frac{r_0}{r} \right)^{12} - \left(\frac{r_0}{r} \right)^{6} \right],$$

the interaction force $F(r)$ between the two atoms is

$$F(r) = -\frac{\partial E_p}{\partial r} = 12\,\frac{E_p(0)}{r_0} \left[\left(\frac{r_0}{r} \right)^{13} - \left(\frac{r_0}{r} \right)^{7} \right].$$

If, when the temperature of the system is T, the vibrational energy of the molecule is not large, the two atoms in the molecule will perform simple harmonic motion about the equilibrium position. Expanding $F(r)$ in a Taylor series about $r = r_0$ and keeping the term linear in $(r - r_0)$, we obtain

$$F(r) = -\frac{72 E_p(0)}{r_0^2}(r - r_0) = -Kx.$$

Since, when the system is in equilibrium, the mean kinetic energy and the mean potential energy of the molecular vibration are both $\dfrac{1}{2}kT$,

$$\frac{1}{2} K \overline{x^2} = \frac{1}{2} K\, \overline{(r - r_0)^2} = \frac{1}{2} kT.$$

Because the molecule performs simple harmonic motion, $\bar{r} = r_0$, and hence the mean square deviation of r is

$$\overline{(r - \bar{r})^2} = \frac{k}{K} T = \frac{k}{72 E_p(0)} \cdot r_0^2.$$

Solution 2.03. According to the theory of Brownian motion, or the random-walk model, let the distance of the molecule from the origin be x; then

$$\overline{x^2} = 2Dt = \frac{l^2}{\tau} t = \overline{v_x} \cdot \lambda \cdot t,$$

that is,

$$t = \frac{\overline{x^2}}{\overline{v_x} \lambda}.$$

Because

$$\frac{1}{2}m\overline{v_x^2} = \frac{1}{2}kT, \qquad \lambda = \frac{1}{\sqrt{2}\,n\sigma}, \qquad n = \frac{p}{kT}, \qquad \sigma = \pi d^2,$$

we have

$$\overline{v_x} \approx \sqrt{\frac{kT}{m}},$$

and hence

$$t = \frac{\overline{x^2}\sqrt{2}\,\pi d^2 p}{kT}\left(\frac{m}{kT}\right)^{1/2} = \frac{\overline{x^2}\sqrt{2}\,\pi d^2 p}{kT}\left(\frac{M}{RT}\right)^{1/2}.$$

Substituting $x = 1\times10^3$, $d = 3\times10^{-10}$, $M = 29\times10^{-3}$, $p = 1.013\times10^5$, $T = 273$, $k = 1.38\times10^{-23}$, $R = 8.31$ into the above expression gives

$$t = 3.6 \times 10^{10}\ \text{(s)}.$$

Solution 2.04. From the concept, in the kinetic theory of molecules, of the number of molecules striking unit area of the wall in unit time, it is known that the intensity I of the molecular beam flowing out of the small hole is

$$I = \frac{1}{4}n\overline{v} = \frac{1}{4}n\sqrt{\frac{8kT}{\pi m}}.$$

Because

$$p = nkT,$$

we have

$$I = \frac{1}{4}\frac{p}{kT}\sqrt{\frac{8kT}{\pi m}} = \frac{p}{\sqrt{2\pi mkT}}.$$

Differentiating both sides of the above expression gives

$$\frac{1}{I}\frac{\mathrm{d}I}{\mathrm{d}T} = \frac{1}{p}\frac{\mathrm{d}p}{\mathrm{d}T} - \frac{1}{2T}.$$

Because

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{l}{T(v_g - v_L)} = \frac{l}{v_g T} = \frac{lMp}{RT^2},$$

where $v_g \gg v_L$ and the equation of state of an ideal gas have been used, we have

$$\frac{1}{I}\frac{\mathrm{d}I}{\mathrm{d}T} = \frac{Ml}{RT^2} - \frac{1}{2T} = \frac{1}{T}\left(\frac{Ml}{RT} - \frac{1}{2}\right).$$

Solution 2.05. (1) The change of the system from the state (V_1, T_1) to the state (V_2, T_1) is an isothermal process. Since

$$\mathrm{d}Q_1 = T\,\mathrm{d}S_1 = \mathrm{d}U + p\,\mathrm{d}V, \qquad U = C_{Vm}T - \frac{a}{V} + U_0, \qquad \left(p + \frac{a}{V^2}\right)(V - b) = RT,$$

we have

$$\mathrm{d}U = C_{Vm}\,\mathrm{d}T + \frac{a}{V^2}\,\mathrm{d}V,$$

and hence

$$\mathrm{d}Q_1 = C_{Vm}\,\mathrm{d}T + \left(\frac{a}{V^2} + p\right)\mathrm{d}V = C_{Vm}\,\mathrm{d}T + \frac{RT}{V - b}\,\mathrm{d}V.$$

Integrating the above expression gives

$$Q_1 = RT_1 \ln \frac{V_2 - b}{V_1 - b}.$$

(2) Since the system passes adiabatically from the state (V_2, T_1) to the state (V_3, T_2), the work W done by the system on the outside is

$$W = -\Delta U = C_{Vm}(T_2 - T_1) + \left(\frac{a}{V_3} - \frac{a}{V_2}\right).$$

Because the equation of an adiabatic process of one mole of a van der Waals gas is

$$\left(p + \frac{a}{V^2}\right)(V - b)^{1+\frac{R}{C_{Vm}}} = \text{constant} = A, \qquad \text{or} \quad T(V - b)^{R/C_{Vm}} = \text{constant} = B,$$

we obtain from this

$$V_3 = b + \left(\frac{T_1}{T_2}\right)^{C_{Vm}/R} \cdot (V_2 - b).$$

Substituting this into the expression for W gives the work done by the system on the outside in the adiabatic process.

Alternatively, W may also be found as follows:

$$W = \int_{V_2}^{V_3} p\,\mathrm{d}V = \int_{V_2}^{V_3} \left[\frac{A}{(V - b)^n} - \frac{a}{V^2}\right] \mathrm{d}V = \frac{A}{1 - n}\left[(V_3 - b)^{1-n} - (V_2 - b)^{1-n}\right] + a\left(\frac{1}{V_3} - \frac{1}{V_2}\right),$$

where $n = 1 + R/C_{Vm}$. Because

$$A = \left(p_3 + \frac{a}{V_3^2}\right)(V_3 - b)^n = \left(p_2 + \frac{a}{V_2^2}\right)(V_2 - b)^n,$$

we have

$$W = \frac{R}{1 - n}(T_2 - T_1) + a\left(\frac{1}{V_3} - \frac{1}{V_2}\right) = C_{Vm}(T_2 - T_1) + a\left(\frac{1}{V_3} - \frac{1}{V_2}\right),$$

which is the same as the result obtained above.

(3) The change of the system from the state (V_3, T_2) to the state (V_4, T_2) is an isothermal process, and the heat Q_2 absorbed is

$$Q_2 = RT_2 \ln \frac{V_4 - b}{V_3 - b}.$$

Hence the efficiency of the system in one cyclic process is

$$\eta = 1 - \frac{|Q_2|}{Q_1} = 1 - \frac{RT_2 \ln \dfrac{V_3 - b}{V_4 - b}}{RT_1 \ln \dfrac{V_2 - b}{V_1 - b}}.$$

Because

$$\left(p_3+\frac{a}{V_3^2}\right)(V_3-b)^n=\left(p_2+\frac{a}{V_2^2}\right)(V_2-b)^n,$$
$$\left(p_4+\frac{a}{V_4^2}\right)(V_4-b)^n=\left(p_1+\frac{a}{V_1^2}\right)(V_1-b)^n,$$
$$\left(p+\frac{a}{V^2}\right)(V-b)=RT,$$

we have

$$\frac{V_3-b}{V_4-b}=\frac{V_2-b}{V_1-b}.$$

Substituting the above result into η gives

$$\eta=1-\frac{T_2}{T_1}.$$

This is Carnot's theorem.

Solution 2.06. (1) Since the equation of a polytropic process of one mole of a van der Waals gas is

$$\left(p+\frac{a}{v^2}\right)(v-b)^n=\text{constant},$$

where

$$n=1-\frac{R}{C-C_V},$$

and here C and C_V are respectively the molar heat capacity and the molar heat capacity at constant volume, we have

$$C=C_V+\frac{R}{1-n}.$$

Hence

$$Q=\int_{T_1}^{T_2}C\,\mathrm{d}T=\int_{T_1}^{T_2}\left(C_V+\frac{R}{1-n}\right)\mathrm{d}T=\left(C_V+\frac{R}{1-n}\right)(T_2-T_1)=(m-1)\left(C_V+\frac{R}{1-n}\right)T_1.$$

(2) Let the final state of the process be (T_2,V_2). Then, since

$$\left(p+\frac{a}{v^2}\right)(v-b)^n=\text{constant},\qquad\text{and}\qquad\left(p+\frac{a}{v^2}\right)(v-b)=RT,$$

substituting the initial state (p_1,v_1,T_1) and the final state (p_2,v_2,T_2) into the above two expressions gives

$$\left(p_1+\frac{a}{v_1^2}\right)(v_1-b)^n=\left(p_2+\frac{a}{v_2^2}\right)(v_2-b)^n,$$
$$\left(p_1+\frac{a}{v_1^2}\right)(v_1-b)=RT_1,\qquad\left(p_2+\frac{a}{v_2^2}\right)(v_2-b)=RT_2.$$

Therefore

$$\frac{v_2-b}{v_1-b}=\left(\frac{T_2}{T_1}\right)^{\frac{1}{1-n}}=m^{\frac{1}{1-n}},$$

that is,

$$v_2=b+(v_1-b)\,m^{\frac{1}{1-n}}.$$

Since

$$\left(p+\frac{a}{v^2}\right)(v-b)^n = \text{constant}, \qquad \left(p+\frac{a}{v^2}\right)(v-b) = RT,$$

we have

$$RT(v-b)^{n-1} = \text{constant}.$$

Hence

$$\left(p+\frac{a}{v^2}\right)(v-b)^{n-1} = RT_1(v_1-b)^{n-1},$$

and we obtain

$$p = \frac{RT_1(v_1-b)^{n-1}}{(v-b)^n} - \frac{a}{v^2}.$$

Thus the work done by the system on the outside is

$$\begin{aligned} W' &= \int_{v_1}^{v_2} p\,\mathrm{d}v = \int_{v_1}^{v_2} \left[\frac{RT_1(v_1-b)^{n-1}}{(v-b)^n} - \frac{a}{v^2}\right]\mathrm{d}v \\ &= \frac{RT_1}{1-n}\left[\left(\frac{v_1-b}{v_2-b}\right)^{n-1} - 1\right] + a\left(\frac{1}{v_2}-\frac{1}{v_1}\right) \\ &= \frac{RT_1}{1-n}(m-1) + a\left[\frac{1}{b+(v_1-b)\,m^{\frac{1}{1-n}}} - \frac{1}{v_1}\right]. \end{aligned}$$

(3)

$$\Delta S = \int \frac{\mathrm{d}Q}{T} = \int_{T_1}^{T_2} \frac{C\,\mathrm{d}T}{T} = C\ln\frac{T_2}{T_1} = \left(C_V + \frac{R}{1-n}\right)\ln m.$$

Solution 2.07. As shown in Fig. 2.07, consider a thin layer of air of area A and thickness $\mathrm{d}z$ at height z. It is in mechanical equilibrium, so that

$$(p+\mathrm{d}p)\cdot A = pA - M'g,$$

$$M' = n\cdot m\cdot A\,\mathrm{d}z,$$

where p is the gas pressure, M' the mass of this thin layer of air, n the number density of gas molecules at that place, and g the acceleration due to gravity. From the above two equations we obtain

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -nmg = -\frac{p}{kT}mg \tag{1}$$

Suppose that 1 mol of gas rises by $\mathrm{d}z$, its temperature changing from T to $T+\mathrm{d}T$, and that the process is adiabatic; when the gas is regarded as an ideal gas, we have

$$T^{\gamma}p^{1-\gamma} = \text{constant}.$$

Differentiating the above expression with respect to p gives

$$\frac{\mathrm{d}T}{\mathrm{d}p} = \frac{\gamma-1}{\gamma}\frac{T}{p}. \tag{2}$$

From (1) and (2) we obtain the rate of change of the atmospheric temperature with height:

$$\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{\mathrm{d}T}{\mathrm{d}p}\cdot\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{\gamma-1}{\gamma}\frac{m}{k}g = -\frac{\gamma-1}{\gamma}\frac{M}{R}g,$$

where $M = N_A \cdot m$ is the molar mass of air.

At ordinary temperatures, $C_p = \frac{7}{2}R$, $C_V = \frac{5}{2}R$, $\gamma = C_p/C_V = \frac{7}{5}$, $M = 28.96\,\mathrm{g}\cdot\mathrm{mol}^{-1}$, $R = 8.31\,\mathrm{J}\cdot\mathrm{mol}^{-1}\mathrm{K}^{-1}$. Substituting these data into the above expression gives

$$\frac{\mathrm{d}T}{\mathrm{d}z} = -\frac{\left(\frac{7}{5}-1\right)}{7/5}\cdot\frac{28.96\times10^{-3}\times10}{8.31} = -10\times10^{-3}\ (\mathrm{K/m}).$$

The above expression shows that, under the conditions given in the problem, the temperature of the air falls by 10 degrees for every 1 kilometre of ascent.

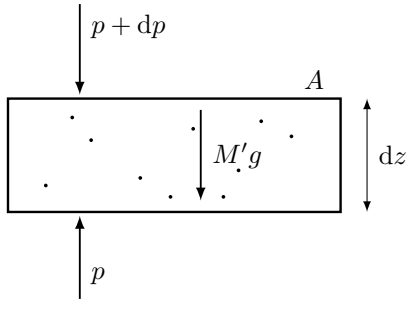


Fig. 2.07

Solution 2.08. Imagine that thermal radiation is taken as the working substance and is made to perform the Carnot cycle shown in Fig. 2.08. Because

$$p = \frac{1}{3}u = \frac{1}{3}\sigma T^4,$$

the isothermal process of this system is just an isobaric process.

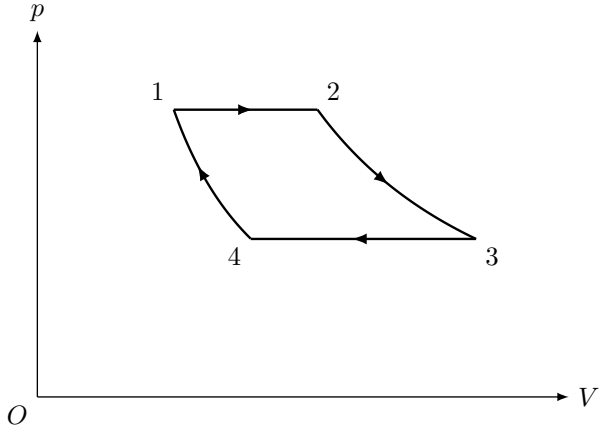


Fig. 2.08

Since the internal energy U of the system is

$$U = V\cdot u = V\sigma T^4,$$

we have

$$\mathrm{d}Q = \mathrm{d}U + p\,\mathrm{d}V = \mathrm{d}(V\sigma T^4) + \frac{1}{3}\sigma T^4\,\mathrm{d}V = \frac{4}{3}\sigma T^4\,\mathrm{d}V + 4\sigma T^3 V\,\mathrm{d}T.$$

From the above expression it is seen that, for the isothermal process $1 \to 2$, the heat absorbed by the system is

$$Q_1 = \frac{4}{3}\sigma T_1^4(V_2 - V_1),$$

and for the isothermal process $3 \to 4$, the heat absorbed by the system is

$$Q_2 = \frac{4}{3}\sigma T_2^4(V_4 - V_3).$$

For an adiabatic process $\mathrm{d}Q = 0$, so

$$\frac{4}{3}\sigma T^4\,\mathrm{d}V + 4\sigma T^3 V\,\mathrm{d}T = 0,$$

that is,

$$\frac{\mathrm{d}V}{V} = -3\frac{\mathrm{d}T}{T}, \qquad \ln VT^3 = \text{constant},$$

so

$$VT^3 = \text{constant}.$$

Since both $2 \to 3$ and $4 \to 1$ are adiabatic processes,

$$V_1T_1^3 = V_4T_2^3, \qquad V_2T_1^3 = V_3T_2^3.$$

Hence

$$\frac{V_1}{V_2} = \frac{V_4}{V_3}, \qquad \frac{V_3 - V_4}{V_2 - V_1} = \left(\frac{T_1}{T_2}\right)^3.$$

Therefore

$$\eta = 1 - \frac{|Q_2|}{Q_1} = 1 - \frac{T_2^4(V_3 - V_4)}{T_1^4(V_2 - V_1)} = 1 - \frac{T_2}{T_1}.$$

Solution 2.09. Because the gas is dilute and the motion of the plate is comparatively slow, it may be considered that the motion of the plate does not affect the distribution function of the molecules of the gas with respect to velocity. As shown in Fig. 2.09, suppose that the collision between a molecule and the plate is elastic. Then, relative to the reference frame at rest with the plate, a molecule of velocity v collides with the plate with velocity $v_x + V$, its component in the x direction is $-(v_x + V)$, and its component in the x direction relative to the ground frame is $-(v_x + 2V)$. Hence, with respect to the ground frame, the increment of the velocity of this molecule in the x direction is

$$-(v_x + 2V) - v_x = -2(v_x + V),$$

and the corresponding momentum acquired by the plate in the x direction is

$$2m(v_x + V).$$

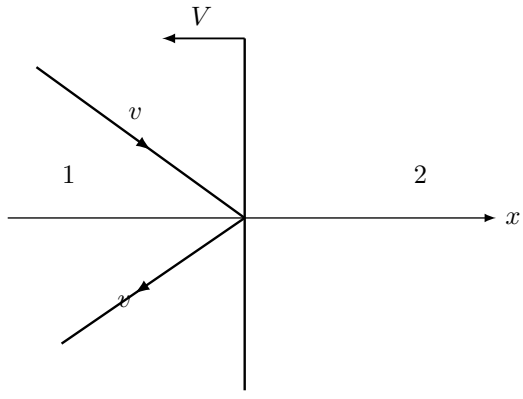


Fig. 2.09

In the time $\mathrm{d}t$, among the molecules of velocity v, those that can collide with the part of the plate of area $\mathrm{d}A$ must lie in the volume whose height is $(v_x + V) \cdot \mathrm{d}t$ and whose base is $\mathrm{d}A$; their number is

$$\mathrm{d}N = n \cdot (v_x + V) f(v)\, \mathrm{d}v_x\, \mathrm{d}v_y\, \mathrm{d}v_z\, \mathrm{d}A.$$

Therefore, in the time $\mathrm{d}t$, the momentum acquired by the plate A in the x direction is

$$\mathrm{d}k_1 = 2mn(v_x + V)^2 \cdot A f(v)\, \mathrm{d}v\, \mathrm{d}t.$$

The force f_1 exerted on the plate by the gas 1 on the left is

$$\begin{aligned} f_1 = \frac{\mathrm{d}k_1}{\mathrm{d}t} &= \int_{-V}^{\infty} \iint_{-\infty}^{\infty} 2mnA(v_x + V)^2 f(v_x) f(v_y) f(v_z)\, \mathrm{d}v_x\, \mathrm{d}v_y\, \mathrm{d}v_z \\ &= \int_{-V}^{\infty} 2mnA(v_x + V)^2 f(v_x)\, \mathrm{d}v_x. \end{aligned}$$

Similarly, the force f_2 exerted on the plate A by the gas 2 on the right along the x direction is

$$f_2 = \int_{V}^{\infty} 2mnA(v_x - V)^2 f(v_x)\, \mathrm{d}v_x.$$

The resultant force f on the plate along the x direction is

$$f = f_1 - f_2 = \int_{-V}^{\infty} 2mnA(v_x + V)^2 f(v_x)\, \mathrm{d}v_x - \int_{V}^{\infty} 2mnA(v_x - V)^2 f(v_x)\, \mathrm{d}v_x.$$

Because the plate moves slowly, V is a small quantity, so the above expression may be written approximately as

$$\begin{aligned} f &= \int_{0}^{\infty} 2mnA(v_x + V)^2 f(v_x)\, \mathrm{d}v_x - \int_{0}^{\infty} 2mnA(v_x - V)^2 f(v_x)\, \mathrm{d}v_x \\ &= 2mnA \int_{0}^{\infty} 4 \cdot V v_x f(v_x)\, \mathrm{d}v_x = 8mnAV \left(\frac{kT}{2\pi m}\right)^{1/2} = 2mnA\overline{v}\, V. \end{aligned}$$

This shows that the resistance experienced by a plate moving slowly in air is proportional to the velocity V of the plate, to the cross-sectional area A of the plate, to the number density n of the molecules of the air, and to the mean speed of the molecules of the air.

Solution 2.10. Because the isothermal compressibility κ_T of the system is

$$\kappa_T = -\frac{1}{V}\left(\frac{\mathrm{d}V}{\mathrm{d}p}\right)_T = 4.75 \times 10^{-5}\ (\mathrm{atm}^{-1}),$$

for an isothermal process with κ constant we have

$$\frac{\mathrm{d}V}{V} = -\kappa\,\mathrm{d}p, \qquad \ln\frac{V}{V_0} = -\kappa(p - p_0),$$

that is,

$$V = V_0 \mathrm{e}^{-\kappa(p-p_0)}.$$

Let the density of water at a flat liquid surface be ρ_0, and let the density of the same mass M of water, in a small droplet at the same temperature and under the same atmospheric pressure, be ρ; then

$$M = V\rho = V_0\rho_0,$$

that is,

$$V = V_0\rho_0/\rho.$$

Hence the density of the droplet is

$$\rho = \rho_0 \mathrm{e}^{\kappa(p-p_0)}.$$

If the small droplet is spherical, the pressure inside the droplet is

$$p = p_0 + \frac{2\alpha}{r},$$

where p_0 is the atmospheric pressure and r the radius of the droplet. Hence the density of the droplet is

$$\rho = \rho_0 \mathrm{e}^{2\kappa\alpha/r}.$$

Because

$$\frac{2\kappa\alpha}{r} = 2 \times 4.75 \times 10^{-5} \times (1.013 \times 10^5)^{-1} \times \frac{7.24 \times 10^{-2}}{1.0 \times 10^{-8}} = 6.79 \times 10^{-3}$$

is a small quantity, we have

$$\rho = \rho_0\left(1 + \frac{2\kappa\alpha}{r}\right) = \rho_0(1 + 6.79 \times 10^{-3}) = 1.01 \times 10^3\ \ \mathrm{kg \cdot m^{-3}}.$$

Solution 2.11. (1) Suppose that the gas in the container is in a state of equilibrium, that the temperature of the system is T and that the number of molecules is N. Because of the influence of gravity, the number density of molecules n depends on the height z, that is,

$$n = n_0 \mathrm{e}^{-\frac{mg}{kT}z},$$

where n_0 is the number density of molecules at $z = 0$, i.e. at the bottom of the container, k is Boltzmann's constant, m is the mass of a molecule, and g is the acceleration due to gravity. Hence

$$\mathrm{d}n = -\frac{n_0 mg}{kT}\mathrm{e}^{-\frac{mg}{kT}z}\mathrm{d}z.$$

Therefore

$$N = A\int \mathrm{d}n = n_0 A\int_0^L \mathrm{d}\big(\mathrm{e}^{-\frac{mg}{kT}z}\big) = n_0 A\big(\mathrm{e}^{-\frac{mgL}{kT}} - 1\big).$$

The internal energy of the system is

$$U = N\bar{\epsilon} + N\bar{\epsilon}_p,$$

where $\bar{\epsilon}_p$ is the average gravitational potential energy of a molecule and $\bar{\epsilon}$ is the average energy apart from the gravitational potential energy (such as the sum of the translational, rotational and vibrational energies).

$$\begin{aligned}\bar{\epsilon}_p &= \frac{A}{N}\int mgz\cdot \mathrm{d}n = \frac{An_0}{N}\left(mgz\,\mathrm{e}^{-\frac{mg}{kT}z}\Big|_0^L - mg\int_0^L \mathrm{e}^{-\frac{mg}{kT}z}\mathrm{d}z\right)\\ &= \frac{n_0 A}{N}\Big[mgL\,\mathrm{e}^{-\frac{mgL}{kT}} + kT\big(\mathrm{e}^{-\frac{mgL}{kT}} - 1\big)\Big] = \frac{mgL\,\mathrm{e}^{-\frac{mgL}{kT}}}{\big(\mathrm{e}^{-\frac{mgL}{kT}} - 1\big)} + kT.\end{aligned}$$

Since

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = N\frac{\partial \bar{\epsilon}}{\partial T} + N\frac{\partial \bar{\epsilon}_p}{\partial T},$$

therefore

$$\begin{aligned}C_{Vm} &= N_A\frac{\partial \bar{\epsilon}}{\partial T} + N_A\frac{\partial \bar{\epsilon}_p}{\partial T} = C'_{Vm} + N_A mgL\frac{\partial}{\partial T}\left(\frac{\mathrm{e}^{-\frac{mgL}{kT}}}{\mathrm{e}^{-\frac{mgL}{kT}} - 1}\right) + N_A k\\ &= C'_{Vm} - R\left(\frac{mgL}{kT}\right)^2\frac{\mathrm{e}^{-\frac{mgL}{kT}}}{\big(\mathrm{e}^{-\frac{mgL}{kT}} - 1\big)^2} + R,\end{aligned}$$

where C'_{Vm} is the molar heat capacity of one mole of ideal gas when gravity is not considered.

(2) When $T \to 0$, the molecules of the ideal gas possess only translational kinetic energy, i.e. the degrees of freedom of rotation and vibration are frozen, so $C'_{Vm} = \frac{3}{2}R$; and $\mathrm{e}^{-\frac{mgL}{kT}} \to 0$, therefore

$$\lim_{T\to 0} C_{Vm} = \frac{3}{2}R + R = \frac{5}{2}R.$$

(2) When $T \to \infty$,

$$C_{Vm} = \begin{cases}\dfrac{3}{2}R & \text{monatomic gas,}\\[2ex] \dfrac{7}{2}R & \text{diatomic gas.}\end{cases}$$

Since

$$\lim_{T\to\infty} \mathrm{e}^{-\frac{mgL}{kT}} = \mathrm{e}^{-0} = 1,$$

that is, $\dfrac{mgL}{kT}$ is a small quantity, therefore

$$\lim_{T\to\infty}\big(\mathrm{e}^{-\frac{mgL}{kT}} - 1\big)^2 = \left(\frac{mgL}{kT}\right)^2.$$

Thus

$$C_{Vm} = C'_{Vm} - R + R = C'_{Vm} = \begin{cases} \dfrac{3}{2}R & \text{monatomic gas,} \\ \dfrac{7}{2}R & \text{diatomic gas.} \end{cases}$$

Solution 2.12. Construct a small Carnot cycle as shown in the figure; then its efficiency is

$$\eta = \frac{W'}{Q} = \frac{\Delta T}{T},$$

$$W' = \Delta p\,(V_g - V_L),$$

$$Q = L,$$

where V_g and V_L are the volumes of the gas and of the liquid when the system is at temperature T and pressure $p + \Delta p$, and L is the heat absorbed when the liquid is entirely converted into gas at the same temperature, that is, the latent heat of phase change. Since the process AB is an isothermal and isobaric process,

$$Q = H(g) - H(L),$$

where $H(g)$ and $H(L)$ are the enthalpies of the gas and of the liquid at the beginning and at the end of the process respectively. Since

$$H_g(T) - H_g(0) = \int_0^T C_{gp}\mathrm{d}T = C_{gp} \cdot T,$$

similarly

$$H_L(T) = C_{Lp} \cdot T + H_L(0),$$

where C_{gp} and C_{Lp} are the heat capacities at constant pressure of the gas and of the liquid respectively. Therefore

$$L = (C_{gp} - C_{Lp}) \cdot T + [H_g(0) - H_L(0)] = (C_{gp} - C_{Lp}) \cdot T + L(0).$$

Substituting W' and $L = Q$ into the expression for η gives

$$\frac{(V_g - V_L)\mathrm{d}p}{(C_{gp} - C_{Lp}) \cdot T + L(0)} = \frac{\mathrm{d}T}{T}.$$

Since

$$V_g \gg V_L, \qquad V_g = \frac{\nu RT}{p},$$

where ν is the number of moles of the gas. Substituting V_g into the above expression gives

$$\mathrm{d}p = \big[(C_{gp} - C_{Lp}) \cdot T + L(0)\big]\frac{p\,\mathrm{d}T}{RT^2}.$$

Integrating the above expression gives

$$p = A\mathrm{e}^{-L(0)/RT} \cdot T^{(C_{gp} - C_{Lp})/R}.$$

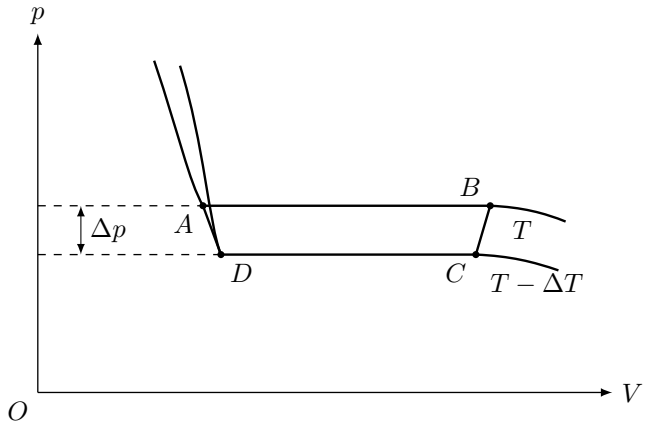


Fig. 2.12

Solution 2.13.

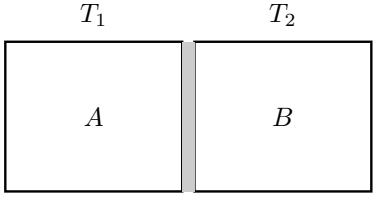


Fig. 2.13

(1) As shown in the figure, A and B are two pieces of metal of very large heat capacity, and between them is a substance of very poor thermal conductivity inserted at time t. Since the heat it conducts per unit time is $P_Q\Delta T$, the heat ΔQ_A lost by metal A in the time Δt is

$$\Delta Q_A = -P_Q\Delta T\cdot\Delta t.$$

But since the heat capacity of metal A is very large, the loss of this heat does not change its temperature; hence after the time Δt the increment of the entropy of metal A, ΔS_A, is

$$\Delta S_A = \frac{\Delta Q_A}{T_1} = -\frac{P_Q\Delta T\Delta t}{T_1} = -\frac{P_Q(T_1-T_2)\Delta t}{T_1}.$$

(2) Similarly, after the time Δt the increment of the entropy of metal B is

$$\Delta S_B = \frac{\Delta Q_B}{T_2} = \frac{P_Q\Delta T\Delta t}{T_2} = \frac{P_Q(T_1-T_2)\Delta t}{T_2}.$$

After Δt the total entropy change of the two systems is

$$\Delta S = \Delta S_A + \Delta S_B = P_Q(T_1-T_2)\Delta t\left(\frac{1}{T_2}-\frac{1}{T_1}\right) = \frac{P_Q\Delta t\cdot(T_1-T_2)^2}{T_1T_2}.$$

Since $(T_1-T_2)^2>0$, therefore

$$\Delta S>0.$$

(3) Suppose that after a sufficiently long time the whole system reaches thermal equilibrium at temperature T, and that the isolating substance is very thin and absorbs no heat; then

$$C_A(T_1 - T) = C_B(T - T_2).$$

Hence the common temperature T finally reached by metals A and B is

$$T = \frac{C_A T_1 + C_B T_2}{C_A + C_B}.$$

Let the temperature of metal A at a certain instant be T_1'; when the temperature changes by $\mathrm{d}T_1'$, the change of its entropy is

$$\mathrm{d}S_A = \frac{C_A \mathrm{d}T_1'}{T_1'}.$$

Hence the increment of the entropy of A while its temperature changes from T_1 to T is

$$\Delta S_A = C_A \int_{T_1}^{T} \frac{\mathrm{d}T_1'}{T_1'} = C_A \ln \frac{T}{T_1} = C_A \ln \left[\frac{C_A T_1 + C_B T_2}{T_1(C_A + C_B)} \right].$$

Similarly, the increment of the entropy of metal B is

$$\Delta S_B = C_B \int_{T_2}^{T} \frac{\mathrm{d}T_2'}{T_2'} = C_B \ln \frac{T}{T_2} = C_B \ln \left[\frac{C_A T_1 + C_B T_2}{T_2(C_A + C_B)} \right].$$

When equilibrium is reached, the increment of the entropy of the system, ΔS, is

$$\Delta S = \Delta S_A + \Delta S_B = \ln \left[\left(\frac{C_A T_1 + C_B T_2}{C_A + C_B} \right)^{(C_A + C_B)} \Big/ T_1^{C_A} \cdot T_2^{C_B} \right].$$

Solution 2.14.

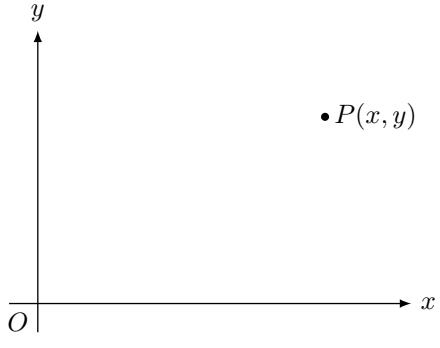


Fig. 2.14

As shown in the figure, suppose that among the N steps the drunkard takes in reaching the point P, he walks n steps to the right, n' steps to the left, m steps upwards and m' steps downwards; then

$$n - n' = \frac{x}{l}; \qquad m - m' = \frac{y}{l}; \qquad n + n' + m + m' = N.$$

From the above three equations we obtain

$$\begin{cases} n = \dfrac{1}{2}\left[N + \left(\dfrac{x}{l} - \dfrac{y}{l}\right)\right] - m', \\ n' = \dfrac{1}{2}\left[N - \left(\dfrac{x}{l} + \dfrac{y}{l}\right)\right] - m', \\ m = \dfrac{y}{l} + m'. \end{cases}$$

Hence the probability that the drunkard, after N steps, is at the point $P(x, y)$ is

$$W(P) = \sum_{m'=0}^{\frac{1}{2}\left[N-\left(\frac{x}{l}+\frac{y}{l}\right)\right]} \frac{N!}{n!\, n'!\, m!\, m'!}\, p_+^n p_-^{n'} q_+^m q_-^{m'},$$

where p_+, p_-, q_+ and q_- denote respectively the probabilities that the drunkard walks one step to the right, to the left, upwards and downwards, and satisfy the following relation:

$$p_+ + p_- + q_+ + q_- = 1.$$

Since n, n', m and m' are all positive integers, when $\dfrac{x}{l} + \dfrac{y}{l}$ is even N must be even, and when $\dfrac{x}{l} + \dfrac{y}{l}$ is odd N must be odd. The summation in the expression for $W(P)$ extends from $m' = 0$ to the value of m' for which $n' = 0$.

For example, let $x = 3l$, $y = 2l$, $N = 11$. Then the following 4 kinds of walk exist: $(m' = 0,\ m = 2,\ n' = 3,\ n = 6)$; $(m' = 1,\ m = 3,\ n' = 2,\ n = 5)$; $(m' = 2,\ m = 4,\ n' = 1,\ n = 4)$; $(m' = 3,\ m = 5,\ n' = 0,\ n = 3)$. Substituting these sets of numbers into the expression for $W(P)$ and summing, one obtains the probability that the drunkard reaches the point P within 11 steps.

Again, if $p_+ = p_- = q_+ = q_- = \dfrac{1}{4}$, $x = 0$, $y = 0$, $N = 2N'$, then

$$n = n', \qquad m = m', \qquad 2n + 2m = 2N', \qquad n + m = N',$$

hence the probability that the drunkard returns to the origin after $2N'$ steps is

$$W(O) = \sum_m \frac{N!}{(n!)^2 (m!)^2} \cdot \left(\frac{1}{4}\right)^N = \sum_n \frac{N!}{(n!)^2 \left[\left(\dfrac{N}{2} - n\right)!\right]^2} \cdot \left(\frac{1}{4}\right)^N.$$

Solution 2.15. *Method 1.* Let the radius of the copper ball be r, its density ρ_1 and its specific heat c_1, and let the heat it gives out through its surface per unit time be κ_1; then the heat Q_1 it gives out in the time Δt_1 is

$$Q_1 = \kappa_1 \Delta t_1 = M_1 \cdot c_1 \Delta T_1, \tag{1}$$

where M_1 is the mass of the copper ball, that is,

$$M_1 = \frac{4}{3}\pi r^3 \rho_1.$$

Similarly, for the aluminium ball, the heat Q_2 it gives out in the time Δt_2 is

$$Q_2 = \kappa_2 \Delta t_2 = M_2 c_2 \Delta T_2. \tag{2}$$

Since the two small balls have the same radius r, their surfaces are both blackened, and the conditions outside the spherical surfaces are also completely the same, therefore

$$\kappa_1 = \kappa_2, \qquad \text{and} \qquad M_2 = \frac{4}{3}\pi r^3 \rho_2.$$

Hence from (1) and (2) we obtain

$$\frac{c_1}{c_2} = \frac{\Delta t_1}{\Delta t_2} \cdot \frac{M_2}{M_1} = \frac{\Delta t_1}{\Delta t_2} \cdot \frac{\rho_2}{\rho_1} = \frac{14.2 \times 2.7}{10 \times 8.9} = 0.43.$$

Method 2. Since the blackened small ball is placed in a cavity at temperature T_0, the rate at which heat is transferred between the ball and the walls of the cavity is

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = A\sigma(T^4 - T_0^4),$$

where A is the surface area of the small ball, σ is the Stefan–Boltzmann constant, whose experimental value is 5.6703×10^{-8} W/(m$^2 \cdot$ K^4), and T is the temperature of the small ball. After the ball has lost heat $\mathrm{d}Q$, its temperature falls by $\mathrm{d}T$, so

$$\mathrm{d}Q = Mc\,\mathrm{d}T.$$

Substituting into the above expression gives

$$Mc\frac{\mathrm{d}T}{\mathrm{d}t} = A\sigma(T^4 - T_0^4), \qquad \text{or} \qquad \frac{\mathrm{d}T}{(T^4 - T_0^4)} = \frac{A\sigma}{Mc}\mathrm{d}t.$$

For the copper ball,

$$\int_{276.15}^{274.15} \frac{\mathrm{d}T}{(T^4 - T_0^4)} = \frac{A_1\sigma}{M_1 c_1}\Delta t_1.$$

For the aluminium ball,

$$\int_{276.15}^{274.15} \frac{\mathrm{d}T}{(T^4 - T_0^4)} = \frac{A_2\sigma}{M_2 c_2}\Delta t_2.$$

From the above two expressions it is known that

$$\frac{A_1\sigma}{M_1 c_1}\Delta t_1 = \frac{A_2\sigma}{M_2 c_2}\Delta t_2.$$

Since $A_1 = A_2$, therefore

$$\frac{c_1}{c_2} = \frac{M_2}{M_1} \cdot \frac{\Delta t_1}{\Delta t_2} = 0.43.$$

Solution 2.16. (1) Let the mass of the solid water (ice) be m_s and the mass of the liquid water be m_l; then the fraction y of the water that is solid is

$$y = \frac{m_s}{m_s + m_l}.$$

Let the total mass of the system be m, and let the specific volume of water at temperature T_f and pressure p_f be v_l and that of ice be v_s; further let

$$V_2 = mv_l, \qquad V_1 = mv_s, \qquad V_1 + V_2 = m(v_l + v_s),$$

which are respectively the volumes of the system when it is entirely water or entirely ice at temperature T_f and pressure p_f. Then

$$V_2 - V_1 = m(v_l - v_s), \qquad V = m_s v_s + m_l v_l,$$

where V is the volume in the state (T_f, p_f), that is, the volume in the initial state (T_i, p_i). Substituting into the expression for y gives

$$y = \frac{m_s(v_l - v_s)}{V_2 - V_1} = \frac{m_s v_l - (V - m_l v_l)}{V_2 - V_1} = \frac{m v_l - V}{V_2 - V_1} = \frac{V_2 - V}{V_2 - V_1} = \frac{V_2 - V}{m(v_l - v_s)}.$$

Since

$$\mathrm{d}V = \left(\frac{\partial V}{\partial T}\right)_p \mathrm{d}T + \left(\frac{\partial V}{\partial p}\right)_T \mathrm{d}p, \qquad \alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p, \qquad \kappa = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T,$$

therefore

$$\mathrm{d}V = V(\alpha\,\mathrm{d}T - \kappa\,\mathrm{d}p).$$

Since α and κ are constants during the process $(T_i, p_i) \to (T_f, p_f)$, integration of the above expression gives

$$\int_V^{V_2} \frac{\mathrm{d}V}{V} = \alpha\Delta T - \kappa\Delta p,$$

that is,

$$\ln\frac{V_2}{V} = \alpha\Delta T - \kappa\Delta p, \qquad \text{so} \qquad V = V_2\exp(-\alpha\Delta T + \kappa\Delta p).$$

Substituting into the expression for y gives

$$y = \frac{V_2}{m(v_l - v_s)}\big[1 - \exp(-\alpha\Delta T + \kappa\Delta p)\big] = \frac{v_l}{(v_l - v_s)}\big[1 - \exp(-\alpha\Delta T + \kappa\Delta p)\big].$$

Since $\alpha\Delta T$ and $\kappa\Delta p$ are small quantities, the above expression becomes

$$y = \frac{v_l}{v_l - v_s}(\alpha\Delta T - \kappa\Delta p).$$

(2) Substituting $v_l = 1.0908 \times 10^{-3}$, $v_s = 1.1928 \times 10^{-3}$, $\alpha = -67 \times 10^{-6}$, $\kappa = 12.04 \times 10^{-11}$, $\Delta T = (T_f - T_i) = -5$ and $\Delta p = p_f - p_i = 5.98 \times 10^7 - 1.01 \times 10^5 = 5.97 \times 10^7$ into the expression for y gives

$$\begin{aligned} y &= \frac{1.0908 \times 10^{-3}}{(1.0908 - 1.1928) \times 10^{-3}}\big[-67 \times 10^{-6} \times (-5) - 12.04 \times 10^{-11} \times 5.97 \times 10^7\big] \\ &= \frac{1.0908 \times 10^{-3}}{-1.02 \times 10^{-4}} \times (-6.853 \times 10^{-3}) = 7.33 \times 10^{-2} = 7.33\%. \end{aligned}$$

Solution 2.17. (1) As shown in Fig. (1) of this problem, an atom of mass m collides perpendicularly with the surface element $\mathrm{d}s$ on the surface of the small ball with speed $\bar{v}$. Since the small ball moves with speed v_0 in the direction opposite to the x-axis, and the collision is elastic, the change of the momentum of the atom along the direction $\vec{n}$ (the normal to the surface element $\mathrm{d}s$) after the collision is

$$m(\bar{v} + v_0\cos\theta) - [-m(\bar{v} + v_0\cos\theta)] = 2m(\bar{v} + v_0\cos\theta).$$

Hence the momentum Δp gained by the small ball in the direction $\vec{n}$ is

$$\Delta p = -2m(\bar{v} + v_0\cos\theta). \tag{1}$$

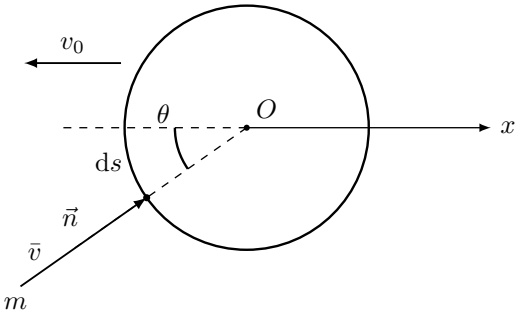


Fig. 2.17(1)

The increment of the energy of the small ball ΔE is

$$\Delta E = \frac{(p+\Delta p)^2}{2m'} - \frac{p^2}{2m'},$$

where $p = m'v_0$, therefore

$$\begin{aligned}\Delta E &= \frac{1}{2m'}\left[p^2 + 2p\cdot\Delta p + (\Delta p)^2 - p^2\right] \\ &= \frac{1}{2m'}\left[-2m'v_0\cdot 2m(\bar{v}+v_0\cos\theta)\cos\theta + 4m^2(\bar{v}+v_0\cos\theta)^2\right] \\ &= -2mv_0(\bar{v}+v_0\cos\theta)\cos\theta + \frac{2m}{m'}(\bar{v}+v_0\cos\theta)^2.\end{aligned}$$

(2) When the small ball enters a rarefied ideal gas, it is bombarded by the surrounding atoms from all directions. It is known (as given in the problem) that the number of collisions between the atoms and the small ball per unit time is $\lambda n\bar{v}$; since $v_0 \ll \bar{v}$, it may be considered that the motion of the small ball does not affect the equilibrium state in which the ideal gas is, and hence the number of collisions per unit time with unit area is $\dfrac{\lambda n\bar{v}}{4\pi r^2}$. From part (1) of the solution of Problem 2.17 it is known that the number of molecules colliding with the surface element $\mathrm{d}s$ of the small ball in the time $\mathrm{d}t$ is

$$\frac{\lambda n\bar{v}}{4\pi r^2}\,\mathrm{d}s\cdot\mathrm{d}t.$$

Since the increment of the momentum gained by each molecule in the direction $\vec{n}$ is $2m(\bar{v}+v_0\cos\theta)$, the increment of the momentum gained by the small ball in the time $\mathrm{d}t$ through $\mathrm{d}s$ (along the direction opposite to $\vec{n}$) is

$$\mathrm{d}k = 2m(\bar{v}+v_0\cos\theta)\cdot\frac{\lambda n\bar{v}}{4\pi r^2}\,\mathrm{d}s\,\mathrm{d}t.$$

Then the normal force $\mathrm{d}f$ acting on $\mathrm{d}s$ is

$$\mathrm{d}f = \frac{\mathrm{d}k}{\mathrm{d}t} = \frac{1}{2}\,\frac{m\lambda n\bar{v}}{\pi r^2}(\bar{v}+v_0\cos\theta)\,\mathrm{d}s.$$

It is also seen from the figure that the components of this force $\mathrm{d}f$ in the y and z directions give a resultant force of zero, and only the component in the x direction survives. The resultant force

in the x direction is

$$f = \int \mathrm{d}f \cos\theta = \frac{1}{2}\,\frac{m\lambda n\bar{v}}{\pi r^2}\int_0^{\pi}(\bar{v}+v_0\cos\theta)\cos\theta\, r^2\sin\theta\,\mathrm{d}\theta\int_0^{2\pi}\mathrm{d}\varphi$$
$$= -m\lambda n\bar{v}\cdot v_0\cdot\frac{1}{3}\cos^3\theta\Big|_0^{\pi} = \frac{2}{3}m\lambda n\bar{v}v_0.$$

After the small ball is acted on by the resisting force, its velocity v will change; from the above expression,

$$f = m'\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{2}{3}m\lambda n\bar{v}\,v,$$
$$\frac{\mathrm{d}v}{v} = -\frac{2}{3}\,\frac{m\lambda n\bar{v}}{m'}\cdot\mathrm{d}t.$$

Integrating the above expression gives

$$\ln\frac{v}{v_0} = -\frac{2}{3}\,\frac{m\lambda n\bar{v}}{m'}t, \qquad v = v_0\mathrm{e}^{-\frac{2}{3}\frac{m\lambda n\bar{v}}{m'}t}.$$

The energy E of the small ball is

$$E = \frac{1}{2}m'v^2 = \frac{1}{2}m'v_0^2\mathrm{e}^{-\frac{4}{3}\frac{m\lambda n\bar{v}}{m'}t}.$$

The rate of change of the energy E, $\dfrac{\mathrm{d}E}{\mathrm{d}t}$, is

$$\frac{\mathrm{d}E}{\mathrm{d}t} = -\frac{2}{3}m\lambda n\bar{v}v_0^2\mathrm{e}^{-\frac{4}{3}\frac{m\lambda n\bar{v}}{m'}t}.$$

When $t\to\infty$, the velocity of the directed motion of the small ball is 0, and the energy of the directed motion is $E=0$. At this time the small ball, under the action of the surrounding gas molecules, performs random motion, that is, Brownian motion. From the theory of Brownian motion it is known that the average translational kinetic energy of the small ball is equal to the average translational kinetic energy $\bar{\varepsilon}$ of the molecules:

$$\bar{\varepsilon} = \frac{3}{2}kT',$$

where T' is the temperature finally reached by the ideal gas after absorbing the heat Q produced by the frictional resistance suffered by the small ball. Since Q is equal to the work done by the resisting force f, that is, to the decrease of the kinetic energy of the directed motion of the small ball, therefore

$$Q = \frac{1}{2}m'v_0^2.$$

When the gas absorbs heat, the average translational kinetic energy of every molecule increases. Let the temperature of the system at $t=\infty$ be T'; then

$$\frac{3}{2}kT' = \frac{3}{2}kT + \frac{Q}{NN_A} = \frac{3}{2}kT + \frac{1}{2}\frac{m'v_0^2}{NN_A},$$

$$T' = T + \frac{1}{3}\frac{m'v_0^2}{NR},$$

where $R = N_A \cdot k$ is the gas constant and N is the number of moles of the ideal gas.

(3) Let the temperature of the system at a certain instant be T''; then from the above expression it is known that

$$\frac{3}{2}k\,\mathrm{d}T'' = \frac{\mathrm{d}Q}{NN_A}, \qquad \mathrm{d}Q = \frac{3}{2}NR\,\mathrm{d}T''.$$

When the system formed by the ideal gas and the small ball is isolated from the surroundings and absorbs heat $\mathrm{d}Q$, its entropy increment $\mathrm{d}S$ is

$$\mathrm{d}S = \frac{\mathrm{d}Q}{T''} = \frac{3}{2}NR\,\frac{\mathrm{d}T''}{T''}.$$

Hence when $t \to \infty$ the entropy increment of the system, ΔS, is

$$\begin{aligned}\Delta S &= \int \mathrm{d}S = \int_T^{T'} \frac{3}{2}NR\,\frac{\mathrm{d}T''}{T''} = \frac{3}{2}NR\ln\frac{T'}{T} = \frac{3}{2}NR\ln\left(1+\frac{1}{3}\frac{m'v_0^2}{NRT}\right)\\ &= \frac{3}{2}NR\ln\left(1+\frac{8}{3}\frac{m'v_0^2}{\pi NM\bar{v}^2}\right) \approx \frac{4km'v_0^2}{\pi m(\bar{v})^2}.\end{aligned}$$

In the above approximation use has been made of $m' \sim NM$ and $(\bar{v})^2 \gg v_0^2$.

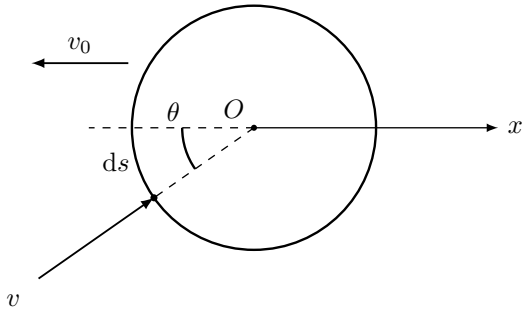


Fig. 2.17(2)

(4) As shown in Fig. (2), the molecules of velocity $\vec{v}$ that collide with the surface element $\mathrm{d}s$ in the time $\mathrm{d}t$ must be located in a cylinder of height $(v_x + v_0\cos\theta)\mathrm{d}t$ and base $\mathrm{d}S$. The number of molecules of velocity $\vec{v}$ in this volume is $nf(v)\mathrm{d}v$, so the number $\mathrm{d}\Gamma$ of molecules of velocity $\vec{v}$ colliding with $\mathrm{d}s$ per unit time is

$$\begin{aligned}\mathrm{d}\Gamma &= (v_x + v_0\cos\theta)nf(v)\mathrm{d}v\,\mathrm{d}S\\ &= n(v_x + v_0\cos\theta)f(v)\mathrm{d}v\,r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi\end{aligned} \tag{2}$$

The number Γ of molecules colliding with the small ball per unit time is

$$\Gamma = \int \mathrm{d}\Gamma = \int_0^{2\pi}\mathrm{d}\varphi\int_0^{\pi}\mathrm{d}\theta\int_{-v_0\cos\theta}^{\infty}\mathrm{d}v_x\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\mathrm{d}v_y\,\mathrm{d}v_z\cdot nr^2(v_x + v_0\cos\theta)\sin\theta.$$

Since $v_0\cos\theta$ is a small quantity, the above expression becomes

$$\Gamma = 4\pi nr^2\int_0^{\infty} v_x f(v_x)\mathrm{d}v_x = 4\pi nr^2\cdot\left(\frac{m}{2\pi kT}\right)^{1/2}\frac{kT}{m} = n\pi r^2\bar{v} = \lambda n\bar{v}.$$

Hence $\lambda = \pi r^2$, which is the collision cross-section between the small ball and the molecules.

Solution 2.18. (1) As shown in the figure, the efficiency of the cycle is

$$\eta = \frac{W'}{Q_1}.$$

Here W' is the work done by the system on the exterior, that is, the area of $\triangle ABC$ in the figure, namely

$$W' = \frac{1}{2}(p_2 - p_1)(V_2 - V_3) = \frac{1}{2}\,p_1\left(2V_1 - \frac{1}{2}V_1\right) = \frac{3}{4}\,p_1 V_1.$$

The heat Q_1 absorbed by the system during the cycle is equal to the sum of the heats absorbed in the processes AB, BC and CA. Since BC is an isobaric compression, the exterior does work on the system, the temperature falls and the internal energy decreases, so this is a heat-releasing process. In the process CA the temperature of the system increases monotonically, i.e. the internal energy increases monotonically, while the system does work on the exterior, so it must absorb heat from the exterior. From the given conditions it is known that $T_A = T_B$, so in the process AB the temperature does not vary monotonically; hence in this process the system absorbs heat in a certain range and releases heat in another range. For this reason we first find the relation between the heat absorbed by the system and the process. Since

$$\mathrm{d}Q = C_{V,m}\mathrm{d}T + p\,\mathrm{d}V = \frac{3}{2}R\,\mathrm{d}T + p\,\mathrm{d}V,$$

$$pV = RT, \qquad p\,\mathrm{d}V + V\,\mathrm{d}p = R\,\mathrm{d}T,$$

therefore

$$\mathrm{d}Q = \frac{C_{V,m}}{R}\left(p\,\mathrm{d}V + V\,\mathrm{d}p\right).$$

Since the equation of the process AB is

$$p = a + bV,$$

and since

$$p_2 = a + bV_1, \qquad p_1 = a + bV_2,$$

we have

$$b = \frac{p_2 - p_1}{V_1 - V_2} = -\frac{p_2 - p_1}{V_2 - V_1} = -\frac{p_1}{V_1}; \qquad a = 3p_1.$$

Thus

$$p = 3p_1 - \frac{p_1}{V_1}V = p_1\left(3 - \frac{V}{V_1}\right),$$

$$\mathrm{d}p = -\frac{p_1}{V_1}\,\mathrm{d}V.$$

Substituting into the expression for $\mathrm{d}Q$, we obtain

$$\mathrm{d}Q = \frac{C_{V,m}}{R}\left[p_1\left(3 - \frac{V}{V_1}\right)\mathrm{d}V - \frac{p_1}{V_1}V\,\mathrm{d}V\right] = \frac{C_{V,m}}{R}\,p_1\left(3 - 2\frac{V}{V_1}\right)\mathrm{d}V.$$

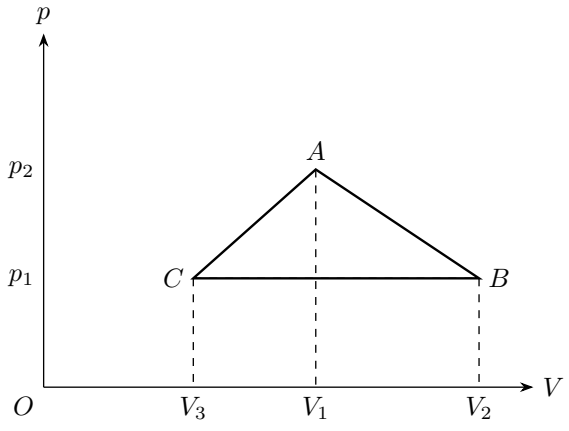


Fig. 2.18

It is seen from the above expression that when $\left(3-2\frac{V}{V_1}\right)>0$, i.e. when $V<\frac{3}{2}V_1$, the volume of the system increases and the system absorbs heat; and when $V>\frac{3}{2}V_1$, it releases heat. Hence the heat absorbed by the system in the process AB is

$$Q_{AB}=\int \mathrm{d}Q=\frac{C_{V,m}}{R}\,p_1\int_{V_1}^{\frac{3}{2}V_1}\left(3-2\frac{V}{V_1}\right)\mathrm{d}V=\frac{C_{V,m}}{R}\,p_1\left[3V-\frac{V^2}{V_1}\right]_{V_1}^{\frac{3}{2}V_1}=\frac{C_{V,m}}{4R}\,p_1V_1.$$

For the process CA, the heat absorbed is

$$\begin{aligned}Q_{CA}&=\Delta U+W'=C_{V,m}\left(T_A-T_C\right)+\frac{1}{2}\left(p_2-p_1\right)\left(V_1-V_3\right)\\&=C_{V,m}\left(\frac{p_2V_1}{R}-\frac{p_1V_3}{R}\right)+\frac{1}{2}\,p_1\cdot\frac{V_1}{2}\\&=\frac{C_{V,m}}{R}\left(2p_1V_1-\frac{1}{2}p_1V_1\right)=\frac{3}{2}C_{V,m}p_1V_1.\end{aligned}$$

Therefore

$$Q=Q_{AB}+Q_{CA}=\frac{C_{V,m}}{4R}\,p_1V_1+\frac{3C_{V,m}}{2R}\,p_1V_1=\frac{7}{4}\frac{C_{V,m}}{R}\,p_1V_1,$$

$$\eta=\frac{W'}{Q}=\frac{3}{7}\frac{C_{V,m}}{R}=\frac{9}{14}=64.3\%,$$

where use has been made of $C_{V,m}=\frac{3}{2}R$.

(2) Since the process BC is one in which the temperature falls continuously, its highest temperature is T_B, while the process CA is one in which the temperature increases monotonically, its highest temperature being T_A, and $T_A=T_B$. Hence the highest temperature in the cycle may occur within the process AB. For this reason we first find the relation between the temperature and the volume in the process AB.

Since

$$\begin{cases}p=p_1\left(3-\dfrac{V}{V_1}\right),\\pV=RT,\end{cases}$$

therefore

$$T = \frac{p_1}{R}\left(3 - \frac{V}{V_1}\right)V,$$
$$\frac{\mathrm{d}T}{\mathrm{d}V} = \frac{p_1}{R}\left(3 - 2\frac{V}{V_1}\right) = 0, \qquad V = \frac{3}{2}V_1,$$
$$\frac{\mathrm{d}^2T}{\mathrm{d}V^2} = -\frac{2p_1}{RV_1} < 0.$$

Hence

$$T_{\max} = \frac{9}{4}\frac{p_1V_1}{R}.$$

(3) Since the equation of the process AB is $p = p_1\left(3 - \dfrac{V}{V_1}\right)$, which cannot be written in the form $pV^n = \text{const}$, the process AB is not a polytropic process.

Since BC is an isobaric process, it is a polytropic process.

Since CA is a straight line, it can be written as $p = c + dV$. Substituting the p and V of points C and A into this equation gives $p_1 = c + \dfrac{1}{2}dV_1$, $p_2 = c + dV_1$. From these, $d = \dfrac{2p_1}{V_1}$, $c = 0$. Therefore the process equation of the process CA is

$$p = \frac{2p_1}{V_1}V, \qquad \text{i.e.} \quad pV^{-1} = \text{const}.$$

The polytropic index is $n = -1$, hence the process CA is a polytropic process.

Solution 2.19. (1) Since all the water vapour that enters the narrow tube condenses into liquid, the number $\mathrm{d}N$ of molecules that enter the narrow tube and condense into water during the time $\mathrm{d}t$ is just the number by which the water vapour decreases during the same time:

$$\mathrm{d}N = -\frac{1}{4}n\bar{v}\cdot\mathrm{d}t,$$

where n is the molecular number density of the water vapour and $\bar{v}$ its mean speed. From this expression it is seen that the decrease, during $\mathrm{d}t$, of the water-vapour molecules per unit volume in the container is

$$\mathrm{d}n = \mathrm{d}\left(\frac{N}{V}\right) = -\frac{1}{4}\frac{n}{V}\bar{v}\,\mathrm{d}t.$$

Since $p = nkT$, therefore

$$\mathrm{d}p = -\frac{1}{4}\frac{p}{V}\bar{v}\,\mathrm{d}t, \qquad \frac{\mathrm{d}p}{p} = -\frac{1}{4}\frac{\bar{v}}{V}\,\mathrm{d}t.$$

Integrating the above expression, the pressure p and the molecular number density n in the container at time t are respectively

$$p = p_0\mathrm{e}^{-\frac{\bar{v}}{4V}t}, \qquad n = n_0\mathrm{e}^{-\frac{\bar{v}}{4V}t},$$

where p_0 and n_0 are the pressure and the molecular number density of the water vapour at $t = 0$.

(2) Since the mean energy of each molecule is

$$\bar{\varepsilon} = \bar{\varepsilon}_{\text{trans}} + \bar{\varepsilon}_{\text{rot}} + \bar{\varepsilon}_{\text{vib}},$$

for a molecule striking the wall (i.e. entering the narrow tube) the mean translational kinetic energy $\bar{\varepsilon}_{\text{trans}}$ is

$$\bar{\varepsilon}_{\text{trans}} = 2kT.$$

Since a water molecule has 3 rotational degrees of freedom and 3 vibrational degrees of freedom,

$$\bar{\varepsilon}_{\text{rot}} = \frac{3}{2}kT, \qquad \bar{\varepsilon}_{\text{vib}} = \frac{1}{2} \times 6kT = 3kT.$$

Therefore

$$\bar{\varepsilon} = 2kT + \frac{3}{2}kT + 3kT = 6\frac{1}{2}kT.$$

Since the number of collisions with the wall per unit time at time t, i.e. the number entering the narrow tube, is

$$\mathrm{d}N = \frac{1}{4}n\bar{v} = \frac{1}{4}n_0\bar{v}\mathrm{e}^{-\frac{\bar{v}}{4V}t},$$

when these molecules condense into liquid, the water vapour loses energy

$$\mathrm{d}E = \bar{\varepsilon}\,\mathrm{d}N = \frac{13}{8}n_0\bar{v}kT\mathrm{e}^{-\frac{\bar{v}}{4V}t}.$$

When the temperature of the water vapour is room temperature, its vibrational degrees of freedom are frozen, so

$$\bar{\varepsilon} = \frac{7}{2}kT.$$

Therefore

$$\mathrm{d}E = \bar{\varepsilon}\,\mathrm{d}N = \frac{7}{8}n_0\bar{v}kT\mathrm{e}^{-\frac{\bar{v}}{4V}t}.$$

Comment. This problem involves two basic pieces of knowledge, which are used directly in the solution. The first is that the number of molecules striking the wall per unit time is $\mathrm{d}\Gamma = \frac{1}{4}n\bar{v}$; this can be derived directly from Maxwell's distribution law. The second is that the mean translational kinetic energy of the molecules striking the wall is $\bar{\varepsilon}_{\text{trans}} = 2kT$; this can be obtained from the distribution function of the speeds of the molecules striking the wall.

Solution 2.20. Let the spherical soap film have radius R and charge Q; then the potential ϕ at the surface of the film is

$$\phi = \frac{Q}{4\pi\varepsilon_0 R}.$$

Suppose that the soap film undergoes an isothermal process in which its radius changes from R to $R + \mathrm{d}R$; then the work $\mathrm{d}W$ done by the exterior on the film is

$$\mathrm{d}W = \mathrm{d}W_1 + \mathrm{d}W_2 = (p_i - p_0) \cdot \mathrm{d}V - Q\,\mathrm{d}\phi,$$

where $\mathrm{d}W_1 = (p_i - p_0)\mathrm{d}V$ is the work done by the gases inside and outside, and $\mathrm{d}W_2 = -Q\,\mathrm{d}\phi$ is the work done by the electric field.

Since in an isothermal process the work done by the exterior on the liquid (here the soap liquid film) is equal to the increment of the free energy of the liquid, i.e. $\mathrm{d}W = \mathrm{d}F = \sigma\,\mathrm{d}A$, therefore

$$(p_i - p_0)\mathrm{d}V - Q\,\mathrm{d}\phi = \sigma\,\mathrm{d}A.$$

Because

$$V = \frac{4}{3}\pi R^3, \qquad \mathrm{d}V = 4\pi R^2\,\mathrm{d}R, \qquad \mathrm{d}\phi = -\frac{Q}{4\pi\varepsilon_0 R^2} \cdot \mathrm{d}R,$$

$$A = 2 \times 4\pi R^2, \qquad \mathrm{d}A = 16\pi R\,\mathrm{d}R,$$

therefore

$$(p_i - p_0) \cdot 4\pi R^2\,\mathrm{d}R + \frac{Q^2}{4\pi\varepsilon_0 R^2}\,\mathrm{d}R = 16\pi\sigma R\,\mathrm{d}R,$$

i.e.

$$\Delta p = p_i - p_0 = \frac{4\sigma}{R} - \frac{Q^2}{\varepsilon_0\,(4\pi R^2)^2}.$$

Comment. The key to this problem is $\mathrm{d}W = \sigma\,\mathrm{d}A$. Its meaning is: for a liquid of constant volume or for a surface of zero volume, in a reversible isothermal process the work $\mathrm{d}W$ done by the exterior on the system is equal to the increment $\mathrm{d}F$ of the free energy, and $F = \sigma A$, where A is the surface area of the liquid and σ the surface-tension coefficient. Since σ is a function of the temperature only and is independent of A, in an isothermal process $\mathrm{d}F = \sigma\,\mathrm{d}A$.

Solution 2.21. As shown in the figure, let a point A on the liquid surface at the mouth of the U-tube and a point B inside the tube lie in the same horizontal plane, and let points C and D lie in the same horizontal plane. Then

$$p_A = p_B = p_C - \rho g h_2 = p_D - \rho g h_2.$$

Therefore

$$\Delta p = p_A - p_0 = -\rho g h_2.$$

This shows that the pressure difference between the inside and the outside of the liquid surface at the tube mouth is $-\rho g h_2$. When h_2 is positive, that is, when point A lies above the horizontal liquid surface, $h_2 > 0$, and then

$$\Delta p < 0.$$

At this time the pressure at point A inside the liquid surface is smaller than the atmospheric pressure outside the surface, so the liquid surface must be a concave surface.

When $h_2 = 0$, $\Delta p = 0$, i.e. $p_A = p_0$; at this time the liquid surface must be a flat surface.

When $h_2 < 0$, that is, when the tube mouth A lies below the horizontal liquid surface, $\Delta p > 0$. Hence at this time the liquid surface must be a convex surface. As h_2 continues to decrease, i.e. as the mouth of the U-tube is lowered further below the liquid surface, the surface bulges further downwards and the liquid drop is elongated. When the weight of the liquid drop exceeds the upward component of the surface tension, the drop falls; when the tube mouth is lowered still further, the liquid flows out continuously from the tube mouth.

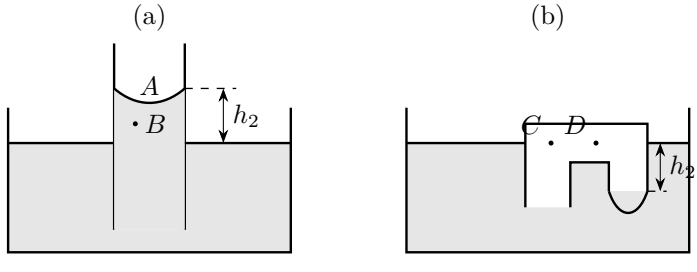


Fig. 2.21

Comment. When discussing surface-tension problems one must bear in mind that the contact angle between the liquid surface and the wall of the vessel is determined only by the properties of the liquid

and of the solid with which it is in contact. Hence, in the discussion of this problem, that is, while h_2 changes from greater than zero to less than zero, the contact angle between the liquid and the capillary tube does not change. When $h_2 = 0$ the liquid surface is flat, which does not mean that the liquid surface is perpendicular to the tube wall, but represents the situation in which the liquid surface is in contact with the wall of the tube mouth of a certain thickness. The contact angle between the liquid surface and the tube wall is the same as in the case $h_2 > 0$ with a concave surface. When $h_2 < 0$, the liquid surface comes into contact with the outer wall of the tube mouth and forms an upward surface-tension pull acting on the liquid drop; at this time the liquid surface is a convex surface.

Solution 2.22. (1) Since $V = \frac{4}{3}\pi R^3 = n \cdot \frac{4}{3}\pi r^3$, therefore

$$r = n^{-1/3} \cdot R \qquad \text{or} \qquad n = \left(\frac{R}{r}\right)^3 .$$

The work W done by the exterior in the isothermal process is

$$W = \sigma \cdot \Delta A = \sigma\left(n \cdot 4\pi r^2 - 4\pi R^2\right) = \sigma\left(\frac{R^3}{r^3}\, 4\pi r^2 - 4\pi R^2\right) = 4\pi R^2 \sigma\left(\frac{R}{r} - 1\right)$$
$$= 4\pi R^2 (a - bT)\left(\frac{R}{r} - 1\right).$$

(2)

$$\Delta Q = T\Delta S = -T\frac{\mathrm{d}\sigma}{\mathrm{d}T}\Delta A = 4\pi R^2 bT\left(\frac{R}{r} - 1\right),$$

where use has been made of $\Delta S = -\dfrac{\mathrm{d}\sigma}{\mathrm{d}T} \cdot \Delta A$.

(3) Since

$$\Delta U = W + \Delta Q = 4\pi\sigma R^2\left(\frac{R}{r} - 1\right) - T\frac{\mathrm{d}\sigma}{\mathrm{d}T}\Delta A$$
$$= 4\pi R^2\left(\frac{R}{r} - 1\right)\left(\sigma - T\frac{\mathrm{d}\sigma}{\mathrm{d}T}\right) = 4\pi R^2 \cdot a\left(\frac{R}{r} - 1\right),$$

the increment ΔS of the entropy of the system during the process is

$$\Delta S = -\frac{\mathrm{d}\sigma}{\mathrm{d}T}\Delta A = -4\pi R^2\left(\frac{R}{r} - 1\right)(-b) = 4\pi R^2 b\left(\frac{R}{r} - 1\right).$$

Comment. For a liquid whose volume remains constant, after an isothermal process the work done by the exterior on the system can only be work done on the liquid surface. For such a system the relation $\sigma(T)$ between the surface-tension coefficient σ and the temperature T is essentially the equation of state of the system. The thermodynamic quantities describing the system are then σ, T and the surface area A. The entropy of the system is $S = -\left(\dfrac{\mathrm{d}\sigma}{\mathrm{d}T}\right)A$, and the internal energy is $U = \left[\sigma - T\left(\dfrac{\mathrm{d}\sigma}{\mathrm{d}T}\right)\right] \cdot A$.

Solution 2.23. Since, when the saturated vapour pressure of water is equal to the atmospheric pressure above the liquid surface, the temperature of the saturated vapour is just the boiling point T_c of water, and since, when water boils, the relation between its saturated vapour pressure and the temperature is given by the Clapeyron equation

$$\frac{\mathrm{d}p}{\mathrm{d}T_c} = \frac{l}{T_c\,(v_g - v_l)},$$

where l is the heat of vaporization of water at T_c, and v_g and v_l are the specific volumes of the gas and of the liquid phase respectively. Since $v_g \gg v_l$ and $Mpv_g = RT$, the above expression becomes

$$\frac{\mathrm{d}p}{\mathrm{d}T_c} = \frac{l}{T_c v_g} = \frac{p l_m}{T_c^2 R} \qquad \text{or} \qquad \frac{\mathrm{d}p}{p} = \frac{l_m \,\mathrm{d}T_c}{R T_c^2},$$

where l_m is the molar heat of vaporization of water vapour. Further, since the pressure of the air varies with height, i.e.

$$\mathrm{d}p = -\rho g \,\mathrm{d}z = -nmg \,\mathrm{d}z = -\frac{mgp}{kT_0}\,\mathrm{d}z = -\frac{M_0 g p}{RT_0}\,\mathrm{d}z,$$

where M_0 is the molar mass of air, g the acceleration of gravity and T_0 the temperature of the air. From the above two expressions we obtain

$$\frac{\mathrm{d}T_c}{T_c^2} = -\frac{M_0 g}{l_m T_0}\,\mathrm{d}z.$$

Integrating both sides gives

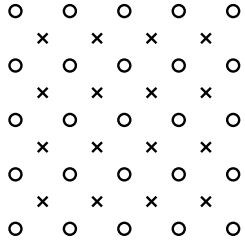
$$\frac{1}{T_c} = \frac{M_0 g}{l_m T_0} z + c.$$

When $z = 0$, $T_c = 373.15\,\mathrm{K}$; substituting this gives $c = \dfrac{1}{373.15\,\mathrm{K}}$. Hence

$$\frac{1}{T_c} = \frac{M_0 g}{l_m T_0} z + \frac{1}{373.15\,\mathrm{K}}.$$

The above expression shows that as z increases, T_c decreases. But the expression holds only in the range in which l_m does not vary with temperature.

Solution 2.24. As shown in the figure, ∘ denotes a normal lattice site and × an interstitial position.



∘ normal site, × interstitial position

Fig. 2.24

(1) Since, when N atoms are situated at normal lattice sites, their thermodynamic probability W_1^T and the entropy S_1 of the system are respectively

$$W_1^T = N!, \qquad S_1 = k \ln W_1^T = k \ln N!,$$

when n atoms are situated at interstitial positions, there are $N - n$ atoms at the N normal lattice sites, and the number of complexions is

$$(N-n)! \,\frac{N!}{n!\,(N-n)!} = \frac{N!}{n!}.$$

Similarly, the number of complexions with n atoms at the N interstitial positions is

$$\frac{N!}{(N-n)!}.$$

Therefore the total number of complexions, i.e. the thermodynamic probability W_2, for $N-n$ of the N atoms to be at normal lattice sites while n atoms are at interstitial positions, is

$$W_2^T = \frac{N!}{(N-n)!\,n!} \cdot \frac{N!}{n!\,(N-n)!}.$$

The entropy S_2 of the system is

$$S_2 = k \ln W_2^T.$$

The increment ΔS of the entropy of the system is

$$\Delta S = S_2 - S_1 = k \ln \left[\frac{N!}{(N-n)!\,n!}\right]^2 = 2k \ln \frac{N!}{(N-n)!\,n!}.$$

(2) Since the energy of an atom increases by u when it moves from a normal position to an interstitial position, when n atoms move from normal positions and become interstitial atoms the increment of the internal energy of the system is $\Delta U = nu$. Suppose that during this process the temperature of the system does not change and the work done by the exterior is zero, i.e. $\mathrm{d}T = 0$, $\mathrm{d}W = 0$; then, since

$$\mathrm{d}F = \mathrm{d}U - T\,\mathrm{d}S,$$

and the process is isothermal, $\mathrm{d}F = \mathrm{d}W = 0$, and therefore

$$\Delta U = T \Delta S.$$

Substituting $\Delta U = nu$ and the value of ΔS into the above expression gives

$$nu = 2kT \ln \frac{N!}{n!\,(N-n)!}.$$

Let $n \ll N$; applying Stirling's formula $\ln N! = N(\ln N - 1)$ to the above expression gives

$$nu = 2kT\,(n \ln N - n \ln n) = 2nkT \ln \frac{N}{n}.$$

Hence

$$n = N\mathrm{e}^{-u/2kT}.$$

In the above calculation the approximation $\ln(N-n) = \ln N$ has been used.

Solution 2.25. (1) Each segment of the one-dimensional chain is assigned a direction, either to the left or to the right. Suppose that n of the N segments point to the right; then necessarily $N-n$ segments point to the left, and the distance x between the two ends of the chain is

$$x = [n - (N-n)]\,a,$$

while n and $(N-n)$ are respectively

$$n = \frac{1}{2}\left(\frac{x}{a} + N\right), \qquad N - n = \frac{1}{2}\left(N - \frac{x}{a}\right).$$

When n of the N segments point to the right, the thermodynamic probability of the system is

$$W^T(n) = \frac{N!}{n!\,(N-n)!}.$$

Hence the entropy S of the system is

$$S = k \ln W^T = k\left[\ln N! - \ln n! - \ln(N-n)!\right].$$

Applying Stirling's formula

$$\ln N! = N\left(\ln N - 1\right),$$

we obtain

$$\begin{aligned} S &= k\left[N\ln N - n\ln n - (N-n)\ln(N-n)\right] \\ &= k\left[N\ln 2N - \frac{1}{2}\left(\frac{x}{a}+N\right)\ln\left(\frac{x}{a}+N\right) - \frac{1}{2}\left(N-\frac{x}{a}\right)\ln\left(N-\frac{x}{a}\right)\right]. \end{aligned}$$

When $x \to 0$,

$$S = k\left[N\ln 2N - N\ln N\right] = kN\ln 2.$$

At this time S is maximum, i.e. the system is in the most disordered state.

When $x \to Na$,

$$S = 0.$$

$S = 0$ indicates the greatest order.

(2) If every joint of the chain can move freely, then the internal energy of the chain system is independent of the length x. For an isothermal process the work done by the external force on the chain is equal to the increment of the free energy F of the system (the chain), i.e.

$$\mathrm{d}W = \mathrm{d}F, \qquad \text{and} \qquad \mathrm{d}W = f\,\mathrm{d}x, \quad F = U - T\,\mathrm{d}S, \quad U = U(T).$$

Therefore

$$f = \left(\frac{\partial F}{\partial x}\right)_T = -T\left(\frac{\partial S}{\partial x}\right)_T.$$

Applying the above expression $S = S(x)$, we obtain

$$f = \frac{kT}{2a}\ln\left[\frac{1+\dfrac{x}{Na}}{1-\dfrac{x}{Na}}\right].$$

When $Na \gg x$, i.e. $\dfrac{x}{Na} \ll 1$, $\ln\left(1+\dfrac{x}{Na}\right) \approx \dfrac{x}{Na}$; therefore

$$f = \frac{kT}{2a}\left(\frac{x}{Na}+\frac{x}{Na}\right) = \frac{kT}{Na^2}x.$$

This means that when the chain is folded very short, the force acting on it is proportional to the length of the chain.

Solution 2.26. This is a binomial distribution problem. Since $N = 400$, $n = 210$, $p = q = \dfrac{1}{2}$, therefore

$$W(n) = \frac{N!}{n!\,(N-n)!}\,p^n q^{N-n}.$$

Since $\bar{n} = Np = 400 \times \dfrac{1}{2} = 200$, and $\bar{n}$ is very close to n, the above expression can be expanded into a Gaussian distribution, i.e.

$$W(n) = \frac{1}{\sqrt{2\pi Npq}} \mathrm{e}^{-\frac{(n-\bar{n})^2}{2Npq}}.$$

Substituting N, n, $\bar{n}$, p and q into the above expression gives

$$\begin{aligned} W(210) &= \frac{1}{\sqrt{2\pi \times 100}} \mathrm{e}^{-\frac{10^2}{200}} = \frac{1}{10\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}} \\ &= \frac{1}{25.07} \times \frac{1}{\sqrt{\mathrm{e}}} = \frac{1}{25} \times \frac{1}{1.649} = 0.024 = 2.4\%. \end{aligned}$$

Comment. Besides being expandable into a Gaussian distribution in the neighbourhood of $n \approx \bar{n}$, the binomial distribution also has an important limiting case, the binomial–Poisson distribution. When

$$p \ll 1 \quad \text{and} \quad n \ll N,$$

the binomial distribution becomes the Poisson distribution

$$W(n) = \frac{\lambda^n}{n!} \mathrm{e}^{-\lambda},$$

where

$$\lambda = Np.$$

Solution 2.27. As shown in the figure, when the rod is in contact with two heat reservoirs at temperatures T_1 and T_2 respectively and a steady state has been reached, let the temperature inside the rod be $T(x)$; then the heat Q passing through the cross-section of area A per unit time is constant, i.e.

$$Q = \kappa \frac{\partial T}{\partial x} A = \text{constant},$$

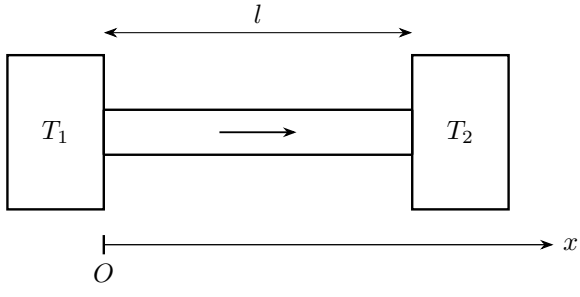


Fig. 2.27

where κ is the thermal conductivity of the rod. Hence $\dfrac{\partial T}{\partial x} = a = \text{constant}$, and

$$T = ax + b.$$

Since $T = T_1$ at $x = 0$ and $T = T_2$ at $x = l$, therefore $a = -\dfrac{T_1 - T_2}{l}$, $b = T_1$, and

$$T = T_1 - \frac{T_1 - T_2}{l} x.$$

When the heat reservoirs are removed and the rod reaches equilibrium, the temperature is the same everywhere in the rod. Let it be T_f; then

$$T_f = \frac{\int_0^l C_p \rho A T \,\mathrm{d}x}{C_p \rho A \cdot l} = \frac{1}{l}\int_0^l (ax+b)\,\mathrm{d}x = \frac{a}{2}l + b = \frac{1}{2}(T_1+T_2),$$

where ρ is the density of the rod. Here T_f is the final temperature of the rod, while T is the initial temperature of the system.

Let the increment of the entropy of the part of the rod at x, of length $\mathrm{d}x$, during the process in which the rod goes from the initial state (temperature T) to the final state (temperature T_f) be $\mathrm{d}S(x)$; then

$$\mathrm{d}S(x) = \int \frac{\mathrm{d}Q(x)}{T} = \int_T^{T_f} \frac{C_p \rho A \cdot \mathrm{d}x \cdot \mathrm{d}T}{T} = C_p \rho A \,\mathrm{d}x \cdot \ln\left[\frac{T_1+T_2}{2(T_1+ax)}\right].$$

From the initial state to the final state the entropy increment ΔS of the whole rod is

$$\Delta S = \int_0^l \mathrm{d}S_x = C_p \rho A \left[\int_0^l \ln\frac{1}{2}(T_1+T_2)\,\mathrm{d}x - \int_0^l \ln(T_1+ax)\,\mathrm{d}x\right].$$

Let $T_1 + ax = \xi$, $\ln\xi = y$; then $\xi = \mathrm{e}^y$, $\mathrm{d}\xi = \mathrm{e}^y\,\mathrm{d}y$, and therefore

$$\int \ln\xi \cdot \mathrm{d}\xi = \int y\mathrm{e}^y\,\mathrm{d}y = y\mathrm{e}^y - \int \mathrm{e}^y\,\mathrm{d}y = (y-1)\mathrm{e}^y = (\ln\xi - 1)\,\xi.$$

Hence

$$\begin{aligned}
\Delta S &= C_p \rho A \left\{ l \ln\frac{1}{2}(T_1+T_2) - \frac{1}{a}\left[\ln(T_1+ax) - 1\right]\cdot(T_1+ax)\right\} \\
&= C_p M \left\{ l \ln\frac{1}{2}(T_1+T_2) + \frac{1}{T_1-T_2}\left[(\ln T_2 - 1)\,T_2 - (\ln T_1 - 1)\,T_1\right]\right\} \\
&= C_p M \left[l \ln\frac{1}{2}(T_1+T_2) + \frac{T_2\ln T_2 - T_1 \ln T_1}{T_1 - T_2} + 1\right].
\end{aligned}$$

Solution 2.28. When n atoms in the crystal run from the N normal lattice sites to the surface and form a new layer of surface atoms, the number of lattice sites of the crystal increases from N to $N+n$, and n vacancies are left in the interior. At this time the number of ways of arranging the N atoms on the $N+n$ lattice sites, i.e. the thermodynamic probability W_2^T, is

$$W_2^T = \frac{(N+n)!}{n!},$$

while the number of ways of arranging the original N atoms on the N lattice sites, i.e. the thermodynamic probability W_1^T, is

$$W_1^T = N!.$$

Hence after n atoms have run to the surface, the increment ΔS of the entropy of the system (the crystal) is

$$\Delta S = S_2 - S_1 = k\ln W_2^T - k\ln W_1^T = k\ln\frac{(N+n)!}{n!\,N!}.$$

Since the atoms run from normal lattice sites to the surface under the condition that the temperature T remains constant, and the volume of the crystal remains constant, the work done by the exterior is zero, i.e.

$$\mathrm{d}W = 0, \qquad \mathrm{d}T = 0.$$

Since the free energy F of the system is

$$F = U - TS, \qquad \mathrm{d}F = \mathrm{d}U - T\,\mathrm{d}S - S\,\mathrm{d}T,$$

for an isothermal process $\mathrm{d}T = 0$ and $\mathrm{d}F = \mathrm{d}W = 0$, therefore

$$\mathrm{d}U = T\,\mathrm{d}S, \qquad \Delta U = T\Delta S.$$

Here U is the internal energy of the crystal, and ΔU is the increment of the internal energy when n vacancies are created, so

$$\Delta U = nu = T\Delta S = Tk\ln\frac{(N+n)!}{n!\,N!}.$$

Since n and N are both large quantities, Stirling's formula can be applied, giving

$$\ln(N+n)! = (N+n)\left[\ln(N+n) - 1\right], \quad \ln n! = n\left(\ln n - 1\right), \quad \ln N! = N\left(\ln N - 1\right).$$

Substituting into the expression for nu gives

$$nu = kT\cdot n\left[\ln(N+n) - \ln n\right], \qquad u = kT\ln\frac{N+n}{n}.$$

Therefore

$$\frac{n}{N+n} = \mathrm{e}^{-u/kT}.$$

Since $n \ll N$, therefore

$$n = N\mathrm{e}^{-u/kT}, \qquad \text{or} \qquad \frac{n}{N} = \mathrm{e}^{-u/kT}.$$

$\dfrac{n}{N}$ represents the probability of creating one Schottky defect at temperature T.

3.3 Electromagnetism

Solution 3.01. (1) Gauss's law no longer holds. It suffices to examine the field of a point charge to show this. Take the spherical surface of radius r centred on the point charge q as the Gaussian surface and calculate the flux of the electric field through this surface. From the statement of the problem,

$$\oint \vec{E}\cdot\mathrm{d}\vec{S} = E\cdot 4\pi r^2 \propto \frac{q}{r^3}\cdot r^2 = \frac{q}{r},$$

which no longer depends only on the charge inside the sphere.

(2) The concept of potential can still be introduced. Although the force between the charges is inversely proportional to the cube of the distance, it is still only a function of the distance; hence the electric force is still a conservative central force, and therefore the concept of potential can still be introduced. But the potential of a point charge is no longer inversely proportional to the distance (if the potential zero is still taken at infinity), but inversely proportional to the square of the distance.

(3) In this case an electric field will exist inside a uniformly positively charged spherical surface, and the direction of the field is along the radius towards the centre of the sphere. Consider the field at any point P inside the sphere. Take P as the vertex and construct two small opposite cones (see the figure); they cut out the surface elements $\mathrm{d}S_1$ and $\mathrm{d}S_2$ on the sphere. From the geometry, $\mathrm{d}S_1 \propto r_1^2$, $\mathrm{d}S_2 \propto r_2^2$. From the statement of the problem,

$$\mathrm{d}E_1 \propto \frac{\mathrm{d}S_1}{r_1^3} \propto \frac{r_1^2}{r_1^3} = \frac{1}{r_1}, \qquad \mathrm{d}E_2 \propto \frac{1}{r_2}.$$

It is thus seen that $\mathrm{d}E_1 > \mathrm{d}E_2$. From the superposition principle and the symmetry it follows that E_P points towards the centre of the sphere.

(4) The charge is no longer distributed on the surface. There is also a charge distribution inside the conductor. Taking a conducting sphere as an example, from the discussion in part (3) it is known that if the charge were distributed only on the spherical surface, the field inside the sphere would not be zero; hence the charge will move towards the interior of the sphere until the field inside the sphere is zero everywhere.

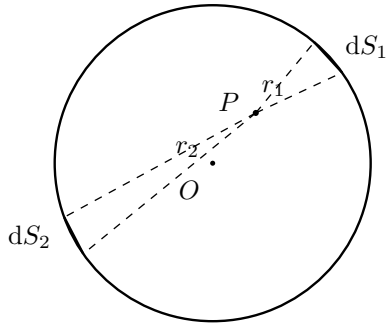


Fig. 3.01

Comment. This problem involves two basic implications of Coulomb's law: the central force and the inverse square. The former is the basis for introducing the potential, the latter the foundation for deriving Gauss's law. Hence the inverse-cube assumption of this problem does not affect the introduction of the potential, but it destroys the conditions under which Gauss's law holds, and also destroys certain results connected with Gauss's law.

Solution 3.02. (1) The electric field produced by dipole A at the position of dipole B is

$$E_r = \frac{1}{4\pi\epsilon_0}\frac{2p_1\cos\theta}{r^3},$$
$$E_\theta = \frac{1}{4\pi\epsilon_0}\frac{p_1\sin\theta}{r^3}.$$

Hence the interaction energy of the two dipoles is

$$\begin{aligned} W &= -\vec{p}_2\cdot\vec{E} = -\big[p_2\cos(\varphi-\theta)\,\hat{r} + p_2\sin(\varphi-\theta)\,\hat{\theta}\big]\cdot\big(E_r\,\hat{r} + E_\theta\,\hat{\theta}\big) \\ &= -\frac{p_1p_2}{4\pi\epsilon_0 r^3}\big[2\cos\theta\cos(\varphi-\theta) + \sin\theta\sin(\varphi-\theta)\big] \\ &= -\frac{p_1p_2}{4\pi\epsilon_0 r^3}\big\{3\cos\theta\cos(\varphi-\theta) - \big[\cos\theta\cos(\varphi-\theta) - \sin\theta\sin(\varphi-\theta)\big]\big\} \\ &= -\frac{p_1p_2}{4\pi\epsilon_0 r^3}\big[3\cos\theta\cos(\varphi-\theta) - \cos\varphi\big]. \end{aligned}$$

Therefore

$$F_r = -\frac{\partial W}{\partial r} = -\frac{3p_1p_2}{4\pi\epsilon_0 r^4}\big[3\cos\theta\cos(\varphi-\theta)-\cos\varphi\big].$$

$$F_\theta = -\frac{1}{r}\frac{\partial W}{\partial \theta} = -\frac{3p_1p_2}{4\pi\epsilon_0 r^4}\big[\sin\theta\cos(\varphi-\theta)-\cos\theta\sin(\varphi-\theta)\big] = -\frac{3p_1p_2}{4\pi\epsilon_0 r^4}\sin(2\theta-\varphi).$$

(2) The electric field produced by the electric quadrupole at the point P may be regarded as the superposition of the electric fields produced at P by two oppositely directed dipoles. The two oppositely directed dipoles have dipole moments equal in magnitude, $p_+ = p_- = ql = p$, but opposite in direction. For each dipole the potential is

$$\Phi_+ = \frac{1}{4\pi\epsilon_0}\frac{p_+\cos\theta_+}{r_+^2}, \qquad \Phi_- = \frac{1}{4\pi\epsilon_0}\frac{p_-\cos\theta_-}{r_-^2},$$

where $\theta_+ = \theta + \Delta\theta$, $\theta_- = \theta - \Delta\theta$, $r_+ = r - \dfrac{l}{2}\cos\theta$, $r_- = r + \dfrac{l}{2}\cos\theta$. Hence the potential at the point P is

$$\begin{aligned}
\Phi = \Phi_+ - \Phi_- &= \frac{p}{4\pi\epsilon_0}\left[\frac{\cos(\theta+\Delta\theta)}{\left(r-\frac{l}{2}\cos\theta\right)^2} - \frac{\cos(\theta-\Delta\theta)}{\left(r+\frac{l}{2}\cos\theta\right)^2}\right] \\
&= \frac{p}{4\pi\epsilon_0 r^2}\left[(\cos\theta-\sin\theta\,\Delta\theta)\left(1+\frac{l}{r}\cos\theta\right) - (\cos\theta+\sin\theta\cdot\Delta\theta)\left(1-\frac{l}{r}\cos\theta\right)\right] \\
&= \frac{p}{4\pi\epsilon_0 r^2}\left[-\sin\theta\cdot\Delta\theta + \frac{l}{r}\cos^2\theta - \sin\theta\cdot\Delta\theta + \frac{l}{r}\cos^2\theta\right] \\
&= \frac{p}{4\pi\epsilon_0 r^2}\left[\frac{2l}{r}\cos^2\theta - 2\sin\theta\cdot\frac{l\sin\theta}{2r}\right] \\
&= \frac{ql^2}{4\pi\epsilon_0 r^3}(3\cos^2\theta-1) = \frac{Q}{8\pi\epsilon_0 r^3}(3\cos^2\theta-1),
\end{aligned}$$

where quantities of second order and higher have been neglected and use has been made of $\Delta\theta = \dfrac{l\sin\theta}{2r}$. The electric field strength is

$$\begin{aligned}
E_r &= -\frac{\partial\Phi}{\partial r} = \frac{3ql^2}{4\pi\epsilon_0 r^4}(3\cos^2\theta-1) = \frac{3}{2}\frac{Q}{4\pi\epsilon_0 r^4}(3\cos^2\theta-1), \\
E_\theta &= -\frac{1}{r}\frac{\partial\Phi}{\partial\theta} = -\frac{1}{r}\frac{ql^2}{4\pi\epsilon_0 r^3}(-6\cos\theta\sin\theta) = \frac{6ql^2\sin\theta\cos\theta}{4\pi\epsilon_0 r^4} = \frac{3Q}{4\pi\epsilon_0 r^4}\sin\theta\cos\theta.
\end{aligned}$$

In the above expressions for the potential and the electric field, $Q = 2ql^2$ is called the quadrupole moment of the electric quadrupole.

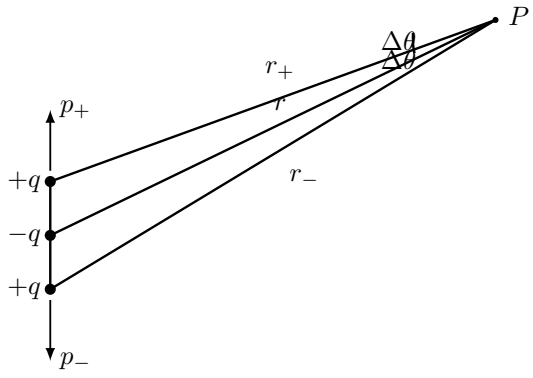


Fig. 3.02

Comment. Part (1) of this problem is solved by the energy method, which is much more convenient than solving it directly by the composition of forces. If part (2) were solved by superposition of the fields of point charges, second-order small quantities l^2/r^3 would have to be retained in the approximation of the distances, and the calculation would be more involved. The concepts of the electric (di)pole and quadrupole involved in this problem have wide applications in electricity.

Solution 3.03. (1) Since $q' < q$, the lines of force issuing from $+q$ cannot all terminate on $-q'$; a part of them will extend to infinity. Let the lines of force with $\alpha > \alpha_0$ extend to infinity; then the lines of force issuing from $+q$ and lying inside the cone of half-angle α_0 will all terminate on $-q'$. Since the number of lines of force is proportional to the electric flux, by Gauss's law, and noting that the solid angle contained in a cone of half-angle α_0 is $2\pi(1-\cos\alpha_0)$, we have

$$\frac{2\pi(1-\cos\alpha_0)}{4\pi}\,q = q'.$$

Solving this gives

$$\alpha_0 = \arccos\left(1 - 2\frac{q'}{q}\right),$$

or

$$\alpha_0 = 2\arcsin\sqrt{\frac{q'}{q}}.$$

Therefore the angle β should be discussed in the two cases $\alpha < \alpha_0$ and $\alpha > \alpha_0$.

When $\alpha < \alpha_0$, in analogy with the discussion above, when the line of force terminates on $-q'$ the angle β it makes with the line AB satisfies

$$\frac{2\pi(1-\cos\alpha)}{4\pi}\,q = \frac{2\pi(1-\cos\beta)}{4\pi}\,q'.$$

From this we obtain

$$\cos\beta = 1 - \frac{q}{q'}(1-\cos\alpha),$$

hence

$$\beta = \arccos\left[1 - \frac{q}{q'}(1-\cos\alpha)\right],$$

or

$$\beta = 2\arcsin\left(\sin\frac{\alpha}{2}\sqrt{\frac{q}{q'}}\right).$$

When $\alpha > \alpha_0$, the lines of force extend to infinity. At this time the lines of force issuing from $+q$ and lying inside the cone of half-angle $(\pi - \alpha_0)$ will ultimately be distributed uniformly within the solid angle 4π, the total number of lines of force being proportional to the charge $(q - q')$. That is to say, the lines of force issuing from $+q$ and lying within the solid angle $[4\pi - 2\pi(1 - \cos\alpha)]$ are ultimately as if emitted from a point charge of magnitude $(q - q')$, and lie within the solid angle $[4\pi - 2\pi(1 - \cos\beta)]$. Hence

$$\frac{4\pi - 2\pi(1 - \cos\alpha)}{4\pi} q = \frac{4\pi - 2\pi(1 - \cos\beta)}{4\pi}(q - q'),$$

that is,

$$q\left[1 - \frac{1}{2}(1 - \cos\alpha)\right] = (q - q')\left[1 - \frac{1}{2}(1 - \cos\beta)\right].$$

Solving this gives

$$\cos\beta = \frac{q}{q - q'}(1 + \cos\alpha) - 1,$$

or

$$\cos\frac{\beta}{2} = \cos\frac{\alpha}{2}\sqrt{\frac{q}{q - q'}},$$

hence

$$\beta = \arccos\left[\frac{q}{q - q'}(1 + \cos\alpha) - 1\right],$$

or

$$\beta = 2\arccos\left(\cos\frac{\alpha}{2}\sqrt{\frac{q}{q - q'}}\right).$$

(2) For a line of force terminating on $-q'$, the point of intersection is just at $-q'$. For a line of force going to infinity, in order to find the point of intersection of its final tangent with the x-axis, consider the electric field at a distant point P. The electric field $\vec{E}_P$ at P is the vector sum of the field $\vec{E}_q$ produced by the charge q and the field $\vec{E}_{-q'}$ produced by $-q'$.

From the figure it is not difficult to see that the extension of E_P must intersect the x-axis to the left of the point A. Let the point of intersection be C, and set $CA = x_1$, $CB = x_2$; write $AP = r_1$, $BP = r_2$, $\angle CPA = \theta$, $\angle CPB = \varphi$, $\angle PCA = \psi$. Applying the sine rule in $\triangle PCA$ and in $\triangle PCB$ respectively, we have

$$\frac{x_1}{\sin\theta} = \frac{r_1}{\sin\psi}, \qquad \frac{x_2}{\sin\varphi} = \frac{r_2}{\sin\psi}.$$

Since P is very far away, $r_1 \approx r_2$, so from the above two equations we obtain

$$\frac{x_1}{x_2} = \frac{\sin\theta}{\sin\varphi}.$$

Moreover, from the vector diagram $\vec{E}_q + \vec{E}_{-q'} = \vec{E}_P$, and from $E_q \propto q/r_1^2$, $E_{-q'} \propto q'/r_2^2$, we have

$$\frac{q'/r_2^2}{\sin\theta} = \frac{q/r_1^2}{\sin\varphi}.$$

Since also $r_1 \approx r_2$, the above equation becomes

$$\frac{\sin\theta}{\sin\varphi} = \frac{q'}{q}.$$

Comparing this with the previous equation, we obtain

$$\frac{x_1}{x_2} = \frac{q'}{q}.$$

If we put $AB = l$, it is not difficult to obtain from the above equation

$$x_1 = \frac{q'}{q - q'} l, \qquad x_2 = \frac{q}{q - q'} l.$$

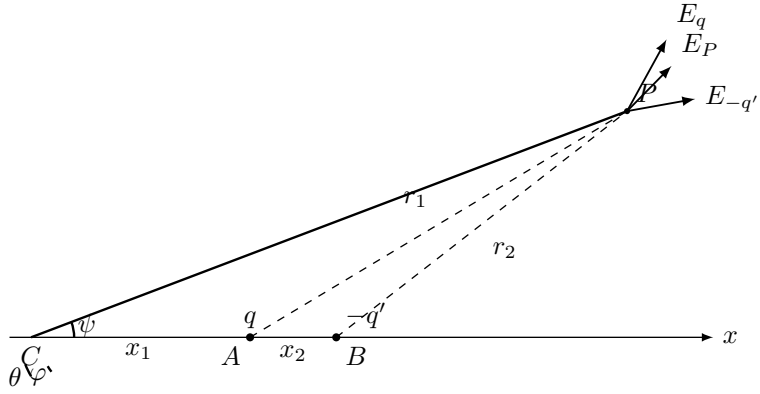


Fig. 3.03

Comment. (1) The key to solving this problem is to know that the number of lines of force is proportional to the electric flux and that the electric flux is proportional to the charge (this is Gauss's law).

(2) Comparing the expressions for x_1 and x_2 with the expression for the centre-of-mass coordinate, it is not difficult to find that, provided one takes into account that a charge has not only magnitude but also sign, these two expressions are no different from the expression for the x-coordinate of the centre of mass. Hence it may be said that, viewed from afar, the lines of force of a system of charges appear to issue from the "centre of gravity" of the system of charges. This conclusion is applicable to any system of charges. As with the centre of mass, the position of the "centre of gravity" C of a system of charges is determined by

$$\vec{r}_C = \frac{\sum q_i \vec{r}_i}{\sum q_i},$$

where $\vec{r}_i$ is the position vector of the point charge q_i. Note that q_i may be positive or negative. The above expression can also be extended to continuously distributed systems of charges, provided that the sum is replaced by an integral. This is precisely the physical basis of the assertion that any static system of charges of finite extent can be regarded as a point charge at a sufficiently great distance.

Solution 3.04. First consider the potential produced at the point P by the ring carrying $+q$. Take the vertical plane containing OP and the z-axis as the xz-plane and set up coordinates as shown in the figure. Consider the potential produced at P by the charge $\eta a \,\mathrm{d}\varphi$ of the arc element $\varphi \to \varphi + \mathrm{d}\varphi$ on the ring, where $\eta = q/2\pi a$ is the linear charge density. The distance AP from the position A of this charge element to the point P is not difficult to obtain (see the figure):

$$\begin{aligned} \overline{AP}^2 &= \overline{PP'}^2 + \overline{AP'}^2 \\ &= (r\cos\theta)^2 + a^2 + (r\sin\theta)^2 + 2a \cdot r\sin\theta\cos\varphi \\ &= r^2 + a^2 + 2ra\sin\theta\cos\varphi. \end{aligned}$$

The potential produced by this charge element at the point P (to accuracy a^2/r^2) is

$$\begin{aligned}
\mathrm{d}\Phi_+ &= \frac{1}{4\pi\epsilon_0}\frac{\eta a\,\mathrm{d}\varphi}{\sqrt{r^2+a^2+2ra\sin\theta\cos\varphi}} \\
&= \frac{\eta a}{4\pi\epsilon_0 r}\frac{\mathrm{d}\varphi}{\sqrt{1+\dfrac{a^2}{r^2}+2\dfrac{a}{r}\sin\theta\cos\varphi}} \\
&\approx \frac{\eta a\,\mathrm{d}\varphi}{4\pi\epsilon_0 r}\left(1-\frac{1}{2}\frac{a^2}{r^2}-\frac{a}{r}\sin\theta\cos\varphi+\frac{3}{2}\frac{a^2}{r^2}\sin^2\theta\cos^2\varphi\right).
\end{aligned}$$

The potential produced by the whole ring carrying $+q$ at the point P is the integral of the above expression:

$$\begin{aligned}
\Phi_+ &= \int \mathrm{d}\Phi_+ = \frac{2\pi\eta a}{4\pi\epsilon_0 r}\int_0^\pi \left(1-\frac{1}{2}\frac{a^2}{r^2}-\frac{a}{r}\sin\theta\cos\varphi+\frac{3}{2}\frac{a^2}{r^2}\sin^2\theta\cos^2\varphi\right)\mathrm{d}\varphi \\
&= \frac{2\pi\eta a}{4\pi\epsilon_0 r}\left[\left(1-\frac{1}{2}\frac{a^2}{r^2}\right)+\frac{3}{4}\frac{a^2}{r^2}\sin^2\theta\right]\pi \\
&= \frac{q}{4\pi\epsilon_0 r}\left[1-\frac{1}{2}\frac{a^2}{r^2}\left(1-\frac{3}{2}\sin^2\theta\right)\right].
\end{aligned}$$

Similarly, the potential produced at the point P by the ring carrying $-q$ is

$$\Phi_- = -\frac{q}{4\pi\epsilon_0 r}\left[1-\frac{1}{2}\frac{b^2}{r^2}\left(1-\frac{3}{2}\sin^2\theta\right)\right].$$

The total potential at the point P is

$$\begin{aligned}
\Phi = \Phi_+ + \Phi_- &= \frac{q}{4\pi\epsilon_0 r}\left[\frac{1}{2r^2}(b^2-a^2)-\frac{3}{4r^2}\sin^2\theta\,(b^2-a^2)\right] \\
&= \frac{q(b^2-a^2)}{4\pi\epsilon_0\cdot 2r^3}\left(1-\frac{3}{2}\sin^2\theta\right) \\
&= \frac{q(b^2-a^2)}{4\pi\epsilon_0\cdot 4r^3}(3\cos^2\theta-1).
\end{aligned}$$

The electric field strength is

$$\begin{aligned}
E_r &= -\frac{\partial\Phi}{\partial r} = \frac{3q(b^2-a^2)}{16\pi\epsilon_0 r^4}(3\cos^2\theta-1), \\
E_\theta &= -\frac{1}{r}\frac{\partial\Phi}{\partial\theta} = \frac{6q(b^2-a^2)}{16\pi\epsilon_0 r^4}\cos\theta\sin\theta.
\end{aligned}$$

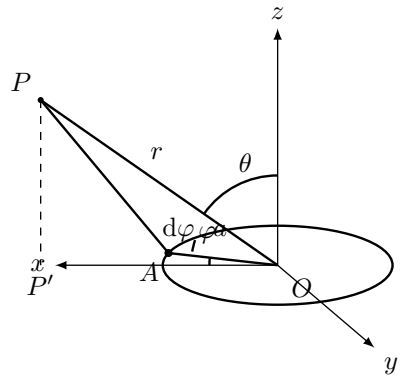


Fig. 3.04

Comment. Comparing with Problem 3.02(2), it is seen that the electric field of two concentric rings carrying equal and opposite charges is similar to that of an electric quadrupole, whose quadrupole moment is $Q = \frac{q}{2}(b^2 - a^2)$.

Solution 3.05. (1) Let the sphere have radius R and total charge Q; then the potential at a point on the spherical surface is

$$\varphi_s = \frac{1}{4\pi\epsilon_0}\frac{Q}{R}.$$

The electric field at any point inside the sphere can be found from Gauss's law. For a point at distance r from the centre,

$$E(r) = \frac{1}{4\pi\epsilon_0}\frac{r^2}{R^3}\frac{Q}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^3}r.$$

Hence the potential at the centre of the sphere is

$$\begin{aligned}\varphi_0 &= \int_0^R E\,\mathrm{d}r + \varphi_s = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^3}\int_0^R r\,\mathrm{d}r + \varphi_s \\ &= \frac{1}{4\pi\epsilon_0}\frac{Q}{2R} + \frac{1}{4\pi\epsilon_0}\frac{Q}{R} = \frac{3}{2}\cdot\frac{1}{4\pi\epsilon_0}\frac{Q}{R}.\end{aligned}$$

Therefore the required ratio of the potentials is

$$\varphi_s/\varphi_0 = \frac{2}{3}.$$

(2) Let the cube have edge length l and total charge Q, with charge density ρ. Imagine cutting the cube into eight small cubes of edge $l/2$, as shown in the figure; then the potential at the centre O is equal to the sum of the potentials at the vertices of the eight small cubes:

$$\varphi_O^l = 8\varphi^{l/2}.$$

But the potential at a vertex is obviously proportional to the total charge and inversely proportional to the edge length, hence

$$\varphi \propto \frac{Q}{l} = \frac{\rho l^3}{l} = \rho l^2.$$

With ρ unchanged, the potential is therefore proportional to the square of the edge length, so that $\varphi^{l/2} = \frac{1}{4}\varphi^l$. Substituting into the original equation,

$$\varphi_O^l = 8\varphi^{l/2} = 8 \cdot \frac{1}{4}\varphi^l = 2\varphi^l.$$

Hence the required ratio of the potentials is

$$\varphi^l/\varphi_O^l = \frac{1}{2}.$$

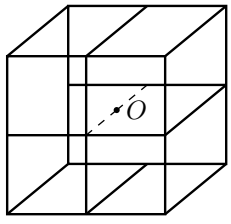


Fig. 3.05

Solution 3.06. The magnitude of the electric field at any point outside the spherical shell can be obtained from the electric field at the surface of the shell:

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{a^2}.$$

To find the charge density $\rho(r)$ at r, take two concentric spherical surfaces of radii r and $r + \Delta r$ as Gaussian surfaces. By Gauss's law,

$$E \cdot 4\pi(r + \Delta r)^2 - E \cdot 4\pi r^2 = \frac{1}{\epsilon_0}\rho \cdot 4\pi r^2 \Delta r.$$

From this we obtain

$$\rho = \rho(r) = \frac{2\epsilon_0 E}{r}.$$

Substituting the expression for E, we obtain

$$\rho(r) = \frac{q}{2\pi a^2 r}.$$

Solution 3.07. For simplicity, first assume that the positive ion layer is at rest and that the negative ion layer is displaced to the right by x, with $x \ll l$ (see the figure). To find the electrostatic force exerted on the negative ion layer by the positive ion layer, divide it into two parts: part I, which coincides with the positive ion layer, and part II, which does not coincide. Part I is acted on only by the thin positive ion layer of thickness x that does not overlap the negative ion layer, while part II is acted on by the whole positive ion layer. A positive ion layer of thickness x produces a uniform electric field on both sides, directed perpendicular to the plane and away from the thin layer on either side; its magnitude is easily found from Gauss's law:

$$E_1 = \frac{ne}{2\epsilon_0}x.$$

Hence the force per unit area on the negative ions of part I is

$$F_{\mathrm{I}} = -ne(l - x)\frac{ne}{2\epsilon_0}x \approx -\frac{n^2e^2}{2\epsilon_0}lx.$$

Here we have neglected the second-order small quantity x^2. The whole positive ion layer also produces a uniform electric field on both sides, and its strength can likewise be found from Gauss's law:

$$E_2 = \frac{ne}{2\epsilon_0} l.$$

Thus the force per unit area on the negative ions of part II is

$$F_{\mathrm{II}} = -nex \cdot \frac{ne}{2\epsilon_0} l = -\frac{n^2 e^2}{2\epsilon_0} lx.$$

The resultant force per unit area on the negative ion layer is

$$F = F_{\mathrm{I}} + F_{\mathrm{II}} = -\frac{n^2 e^2}{\epsilon_0} lx.$$

The force is proportional to the displacement x and opposite to x in direction; it is a simple harmonic force, so the negative ion layer performs simple harmonic motion with an angular frequency equal to the square root of the ratio of the force to the displacement divided by the mass:

$$\omega' = \sqrt{\frac{n^2 e^2 l}{\epsilon_0 m_- \cdot nl}} = \sqrt{\frac{ne^2}{\epsilon_0 m_-}}.$$

Taking the motion of the positive ions into account, it is only necessary to replace m_- in the above expression by the reduced mass $m_+ m_-/(m_+ + m_-)$ to obtain the actual angular frequency

$$\omega = \sqrt{\frac{ne^2(m_+ + m_-)}{\epsilon_0 m_+ m_-}}.$$

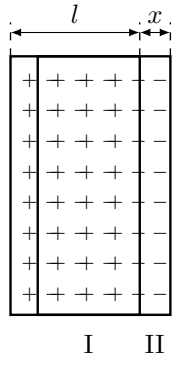


Fig. 3.07

Solution 3.08. (1) Now the potential at a distance r from the point charge q (still taking infinity as the zero of potential) is

$$\varphi = \int_r^\infty \frac{1}{4\pi\epsilon_0} \frac{q}{r^{2+\delta}} \, \mathrm{d}r = \frac{q}{4\pi\epsilon_0(1+\delta)} \frac{1}{r^{1+\delta}}.$$

Now consider the potential at a point P at distance r from the centre of the sphere. Take the line joining the centre O of the sphere to P as axis and construct cones of half-angle θ and $\theta + \mathrm{d}\theta$,

which cut out a ring band on the spherical surface. This band is at the same distance from P, denoted r', so the potential produced at P by the charge on this band is

$$\mathrm{d}\varphi = \frac{1}{4\pi\epsilon_0}\frac{\sigma\cdot 2\pi R\sin\theta\, R\,\mathrm{d}\theta}{(1+\delta)r'^{(1+\delta)}}.$$

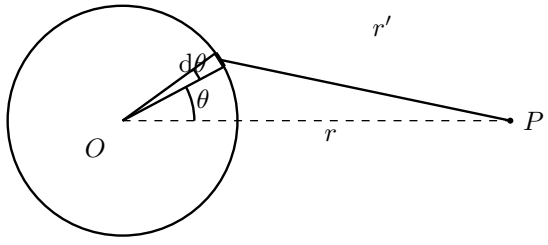


Fig. 3.08

Since $r'^2 = R^2 + r^2 - 2Rr\cos\theta$, this may be written as

$$\mathrm{d}\varphi = \frac{\sigma R^2}{2\epsilon_0(1+\delta)}\frac{\sin\theta\,\mathrm{d}\theta}{(R^2+r^2-2Rr\cos\theta)^{(1+\delta)/2}}.$$

The potential at the point P is the integral of the above expression over θ:

$$\begin{aligned}\varphi &= \frac{\sigma R^2}{2\epsilon_0(1+\delta)}\int_0^\pi \frac{\sin\theta\,\mathrm{d}\theta}{(R^2+r^2-2Rr\cos\theta)^{(1+\delta)/2}}\\ &= \frac{\sigma R}{2\epsilon_0(1-\delta^2)r}\big[(R+r)^{1-\delta} - |R-r|^{1-\delta}\big].\end{aligned}$$

When $r < R$ (the point P is inside the sphere), $|R-r| = R-r$; when $r > R$ (the point P is outside the sphere), $|R-r| = r-R$. Hence

$$\varphi = \begin{cases}\dfrac{\sigma R}{2\epsilon_0(1-\delta^2)r}\big[(R+r)^{1-\delta} - (R-r)^{1-\delta}\big], & r < R \text{ (inside the sphere)},\\ \dfrac{\sigma R}{2\epsilon_0(1-\delta^2)r}\big[(R+r)^{1-\delta} - (r-R)^{1-\delta}\big], & r > R \text{ (outside the sphere)}.\end{cases}$$

(2) Since $\delta \ll 1$,

$$\begin{aligned}(R+r)^{1-\delta} &\approx (R+r) - \delta(R+r)\ln(R+r),\\ |R-r|^{1-\delta} &\approx |R-r| - \delta|R-r|\ln|R-r|,\\ 1-\delta^2 &\approx 1.\end{aligned}$$

The potential may then be written as

$$\varphi \approx \begin{cases}\dfrac{\sigma R}{2\epsilon_0 r}\big\{2r + \delta\big[(R-r)\ln(R-r) - (R+r)\ln(R+r)\big]\big\}, & r < R,\\ \dfrac{\sigma R}{2\epsilon_0 r}\big\{2R + \delta\big[(r-R)\ln(r-R) - (R+r)\ln(R+r)\big]\big\}, & r > R,\\ \dfrac{\sigma}{2\epsilon_0}\big[2R - \delta\cdot 2R\ln 2R\big], & r = R.\end{cases}$$

Applying this result to the two spherical shells in the problem, and denoting their surface charge densities by σ_a and σ_b ($\sigma_b \ll \sigma_a$), we have, retaining first-order small quantities,

$$\begin{aligned}
\varphi_a &= \frac{\sigma_a}{2\epsilon_0}\big[2a - \delta \cdot 2a\ln 2a\big] + \frac{b\sigma_b}{2\epsilon_0 a}\big\{2b + \delta\big[(a-b)\ln(a-b) - (a+b)\ln(a+b)\big]\big\} \\
&\approx \frac{\sigma_a a}{\epsilon_0} - \delta\frac{\sigma_a a}{\epsilon_0}\ln 2a + \frac{\sigma_b b^2}{\epsilon_0 a}, \\
\varphi_b &= \frac{\sigma_b}{2\epsilon_0}\big[2b - \delta \cdot 2b\ln 2b\big] + \frac{a\sigma_a}{2\epsilon_0 b}\big\{2b + \delta\big[(a-b)\ln(a-b) - (a+b)\ln(a+b)\big]\big\} \\
&\approx \frac{\sigma_b b}{\epsilon_0} + \frac{\sigma_a a}{\epsilon_0} + \delta\frac{\sigma_a a}{2\epsilon_0 b}\big[(a-b)\ln(a-b) - (a+b)\ln(a+b)\big].
\end{aligned}$$

Putting $\varphi_a = \varphi_b = U$, we obtain

$$\frac{b\sigma_b}{\epsilon_0}\left(1 - \frac{b}{a}\right) = -\delta\left\{\frac{\sigma_a a}{\epsilon_0}\ln 2a + \frac{\sigma_a a}{2\epsilon_0 b}\big[(a-b)\ln(a-b) - (a+b)\ln(a+b)\big]\right\}.$$

But $\dfrac{\sigma_a a}{\epsilon_0} \approx U$; by the statement of the problem, $\dfrac{b\sigma_b}{\epsilon_0}\left(1 - \dfrac{b}{a}\right) \le u$. Substituting into the above equation gives

$$\begin{aligned}
u &\ge \frac{1}{2}\delta U\left\{\frac{1}{b}\big[(a+b)\ln(a+b) - (a-b)\ln(a-b)\big] - 2\ln 2a\right\} \\
&= \frac{1}{2}\delta U\left(\ln\frac{a^2-b^2}{4a^2} + \frac{a}{b}\ln\frac{a+b}{a-b}\right).
\end{aligned}$$

Therefore

$$\delta \le \frac{2u}{U\left(\ln\dfrac{a^2-b^2}{4a^2} + \dfrac{a}{b}\ln\dfrac{a+b}{a-b}\right)}.$$

Comment. The inverse-square law of the electrostatic force is of profound significance, and experimental tests of the inverse-square law have always been taken seriously by the scientific community. This problem is the theoretical basis of the experimental test method used by Maxwell in 1872. The experimental result gave a value of u equal to zero; later, from the sensitivity of the apparatus and the relevant data, it was estimated that $\delta < 5 \times 10^{-5}$. The result obtained by more precise experimental methods in modern times (the 1970s) is 2×10^{-16}.

Solution 3.09. Although the geometrical structure of this apparatus is very symmetric, the uncertainty of such factors as the distribution of ions in the air and cosmic rays makes the potentials of the aluminium cylinders A_1 and A_2 slightly different. For example, if the potential of A_1 is higher than that of A_2, then by electrostatic induction the water drops above A_1 become negatively charged and the water drops above A_2 become positively charged. After the charged water drops fall into the aluminium cups below, the cup B_1 becomes negatively charged. Since B_1 and A_2 are connected by a wire, the potential of A_2 is further lowered, and in the same way the potential of A_1 is further raised; this in turn makes the drops above A_1 carry still more negative charge and the drops above A_2 carry still more positive charge. Continuing in this way, the potential of the aluminium cup B_2 becomes higher and higher and that of B_1 lower and lower, until finally a potential difference of several thousand volts can be produced between the two aluminium cups. Of course, because of the uncertainty of the various factors, at the

beginning of the next experiment the potential of A_2 may be higher than that of A_1, and finally the potential of B_1 will be several thousand volts higher than that of B_2. But it does not change that the potential difference between A_1 and A_2 produced by accidental factors is amplified by the positive-feedback effect described above.

Solution 3.10. (1) *Capacitive type.* From the formula $u = q/C$, in order to convert the sound vibration (which causes a change of capacitance) into an output electric signal, q should be kept essentially constant; hence the circuit shown in Fig. 3.10(a) may be used, with $R \gg \dfrac{1}{C\omega}$. For this circuit we have the following equations:

$$\begin{aligned} u &= \frac{q}{C} + \frac{\mathrm{d}q}{\mathrm{d}t}R \\ &= \frac{q}{\epsilon_0 S}(d+x) + \frac{\mathrm{d}q}{\mathrm{d}t}R \\ &= u_1(t) + u_2(t). \end{aligned}$$

When $R \gg \dfrac{1}{C\omega}$, q is essentially constant and $u_2(t)$ is not easy to find, but $u_1(t)$ can be found:

$$\begin{aligned} u_1(t) &= \frac{q}{\epsilon_0 S}(d+x) = \frac{q}{\epsilon_0 S}d + \frac{q}{\epsilon_0 S}x = u_{10} + \Delta u_1(t), \\ \Delta u_1(t) &= \frac{q}{\epsilon_0 S}x = \frac{q}{\epsilon_0 S/d} \cdot \frac{x}{d} = \frac{q}{C}\frac{x}{d} = u\frac{x}{d} \\ &= u \cdot \frac{1}{d}x_0 \sin \omega t. \end{aligned}$$

This is the output signal. If R is very small $\left(R \ll \dfrac{1}{C\omega}\right)$, $u_1(t)$ is not easy to find (because q varies greatly), but $u_2(t)$ can be found:

$$\begin{aligned} u_2(t) &= \frac{\mathrm{d}q}{\mathrm{d}t}R = R \cdot \frac{\mathrm{d}(uC)}{\mathrm{d}t} = Ru\frac{\mathrm{d}C}{\mathrm{d}t} = uR\frac{\epsilon_0 S}{d^2}\frac{\mathrm{d}x}{\mathrm{d}t} \\ &\approx uRC\frac{1}{d}\omega x = uRC\omega\frac{x}{d} \ll u\frac{x}{d}. \end{aligned}$$

It is thus seen that the output signal is much smaller than that for $R \gg \dfrac{1}{C\omega}$.

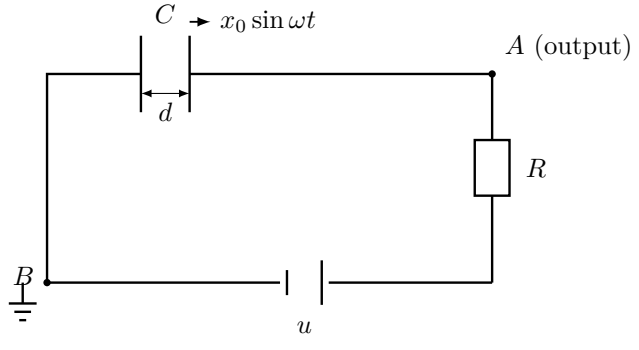


Fig. 3.10(a)

(2) *Electret type.* Since there is a constant applied charge, no external power supply is needed. The circuit shown in Fig. 3.10(b) may be used, with $R \gg \dfrac{1}{C\omega}$ (for the same reason as above). For this circuit, in the static state,

$$\frac{\sigma_1}{\epsilon_r \epsilon_0} d_1 = \frac{\sigma_2}{\epsilon_0} d \tag{1}$$

$$\sigma_1 + \sigma_2 = \sigma \tag{2}$$

From (1) and (2) we obtain

$$\sigma_2 = \frac{\sigma d_1}{\epsilon_r d + d_1}.$$

Since R is very large, this charge remains essentially unchanged when d undergoes a small change. Hence the output signal is

$$\Delta u(t) = \frac{\sigma_2}{\epsilon_0} x = \frac{d_1 \sigma}{\epsilon_0(\epsilon_r d + d_1)} x_0 \sin \omega t.$$

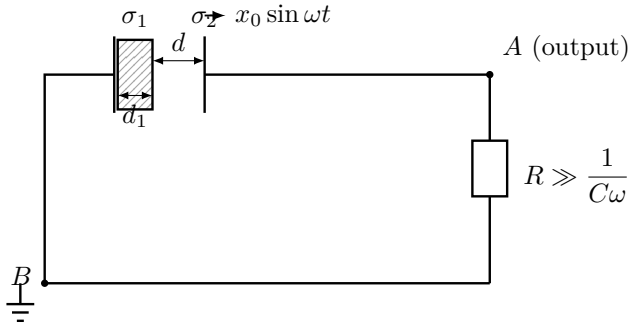


Fig. 3.10(b)

Comment. The key to both parts of this problem is to keep the charge on the moving plate essentially unchanged, so that the field strength near the moving plate is also essentially unchanged; in this way the displacement of the moving plate gives rise to a voltage change to the greatest extent.

Solution 3.11. (1) The electric field in the region where the point charge q is located (the first quadrant) can be regarded as the superposition of the fields of the point charge q and of three other image charges $-q$, $-q$, q, as shown in the figure, while in the remaining regions (the second, third and fourth quadrants) the electric field is zero. Therefore the work done by the external force in moving the point charge q from infinity to its present position is equal to one quarter of the work done by the external force in moving these four charges from infinity to their present positions (this is not difficult to understand if one imagines that during the process of moving, the positions of the four charges always remain symmetric), and the work done by the external force in moving the four charges from infinity to their present positions is equal to the interaction energy of the four charges. Hence the required work W_{in} is

$$\begin{aligned} W_{\text{in}} &= \frac{1}{4}\left[4\left(\frac{1}{4\pi\epsilon_0}\right)\frac{-q^2}{2d} + 2\left(\frac{1}{4\pi\epsilon_0}\right)\frac{q^2}{2\sqrt{2}d}\right] \\ &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{d}\left(\frac{1}{4\sqrt{2}} - \frac{1}{2}\right). \end{aligned}$$

This work is negative, which is obvious.

(2) When the point charge q is moved, the force acting on it is equivalent to the force exerted on it by the three image charges; hence the required work W_{out} is the negative of the interaction energy of this point charge with the three image charges in the present configuration:

$$\begin{aligned} W_{\text{out}} &= -\left[2 \cdot \frac{1}{4\pi\epsilon_0}\frac{-q^2}{2d} + \frac{1}{4\pi\epsilon_0}\frac{q^2}{2\sqrt{2}d}\right] \\ &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{d}\left(1 - \frac{1}{2\sqrt{2}}\right). \end{aligned}$$

Since the charge is always attracted during the displacement, this work is positive.

(3) In effect, W_{in} is equivalent to establishing the system of induced charges, that is, the system of charges on the insulating plate, and then moving the point charge q from infinity to its present position, while W_{out} is the work done by the external force in moving the point charge q from its present position to infinity in the electric field of the charge system on the insulating plate. Hence the work done by the external force in establishing the charge system on the insulating plate, that is, the electrostatic energy W of the charge system on the insulating plate, is equal to the sum of W_{in} and W_{out}:

$$\begin{aligned} W = W_{\text{in}} + W_{\text{out}} &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{d}\left(\frac{1}{4\sqrt{2}} - \frac{1}{2} + 1 - \frac{1}{2\sqrt{2}}\right) \\ &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{d}\left(\frac{1}{2} - \frac{1}{4\sqrt{2}}\right). \end{aligned}$$

This energy is obviously positive.

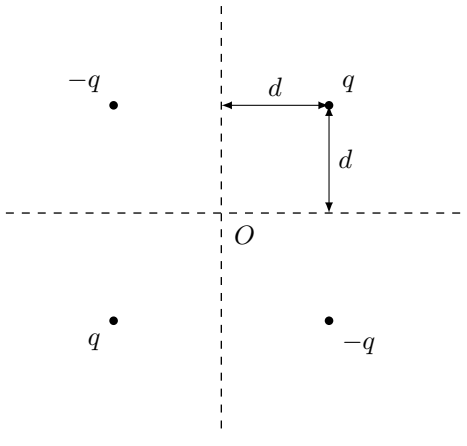


Fig. 3.11

Comment. The method of images is one of the important methods for solving electrostatic problems (especially those involving conductors). This problem is an example of the method of images.

Solution 3.12. The point charge q will induce on the inner surface of the spherical shell a charge of total amount $-q$ distributed non-uniformly, as shown in the figure, while on the outer surface of the shell a charge q will be distributed uniformly. Since the charge distributed uniformly on the outer surface produces no electric field in its interior, the point charge q inside the shell is acted on only by the force of the electric field produced by the induced charge on the inner

surface of the shell, and the induced charge on the inner surface of the shell is equivalent to that on a grounded shell. But the electric field produced inside the sphere by the induced charge on the inner surface can be replaced by the field of an image charge outside the sphere.

To find the magnitude and position of this image charge q', it suffices to examine the potentials at the two points A and B where the line joining q to the centre O of the shell intersects the inner surface of the shell. By symmetry, q' must lie on this line; let its distance from the centre O be r'. From the vanishing of the potentials at the points A and B we obtain the following equations:

$$\frac{q}{a-r}+\frac{q'}{r'-a}=0,$$
$$\frac{q}{a+r}+\frac{q'}{r'+a}=0.$$

Solving these, we obtain

$$r'=\frac{a^2}{r},$$
$$q'=-\frac{a}{r}q.$$

Hence the required force is

$$F=\frac{1}{4\pi\varepsilon_0}\frac{qq'}{(r'-r)^2}=-\frac{q^2}{4\pi\varepsilon_0}\frac{ar}{(a^2-r^2)^2}.$$

When $r\to 0$ this force is zero, and when $r\to a$ this force tends to infinity.

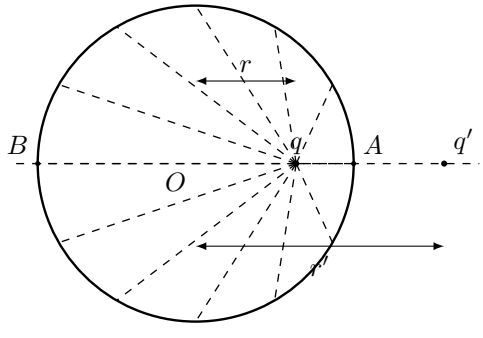


Fig. 3.12

Comment. This problem is another example of the method of images. It is not difficult to prove that the potential produced by q and q' over the whole spherical shell is zero.

Solution 3.13. (1) Let the capacitance between two adjacent conducting plates be C. Then when the positive terminal of the source is connected to the N-th plate, the source is, as it were, connected across the $(N-1)$ capacitors in series, so the N-th plate will carry the charge

$$q_N=U\cdot\frac{C}{N-1}.$$

At the same time, the left surface of the $(N-1)$-th plate carries the charge $-q_N$ and its right surface carries the charge $+q_N$; the charge distribution on each of the remaining conducting

plates is similar to that on the $(N-1)$-th plate. When the positive terminal of the battery is next connected to the $(N-1)$-th plate, the charge on the left surface of this plate, being attracted by the positive charge on the N-th plate, no longer changes, while its right surface becomes the positive surface of $(N-2)$ capacitors in series and carries the charge

$$q'_{N-1} = U \cdot \frac{C}{N-2}.$$

Hence the total charge on the $(N-1)$-th plate is

$$q_{N-1} = -U\frac{C}{N-1} + U\frac{C}{N-2} = UC\left(\frac{1}{N-2} - \frac{1}{N-1}\right).$$

By the same argument, the charge on the k-th plate is

$$q_k = UC\left(\frac{1}{k-1} - \frac{1}{k}\right),$$

and evidently

$$q_1 = -UC.$$

Therefore

$$|q_k| : |q_1| = \left(\frac{1}{k-1} - \frac{1}{k}\right) : 1.$$

(2) From the above analysis it is known that the potential difference between the N-th plate and the $(N-1)$-th plate is $U/(N-1)$, that between the $(N-1)$-th plate and the $(N-2)$-th plate is $U/(N-2)$, and so on; hence the potential of the k-th plate is

$$U_k = U\left(\frac{1}{k-1} + \frac{1}{k-2} + \cdots + \frac{1}{2} + 1\right).$$

Comment. This problem can also be solved without the idea of capacitors in series, by regarding each charged conducting plate as an infinite charged plane and applying Gauss's law; but this is far more complicated than the capacitance method.

Solution 3.14. (1) When the radius is r, the resistance of the atmosphere is

$$R = \int_r^\infty \rho \frac{\mathrm{d}r}{4\pi r^2} = \frac{\rho}{4\pi r}.$$

By Ohm's law,

$$-\frac{\rho}{4\pi r}\frac{\mathrm{d}q}{\mathrm{d}t} = V = \text{constant}, \tag{1}$$

where q is the charge carried by the balloon. The quantities q and r are connected by the following relation:

$$\frac{1}{4\pi\varepsilon_0}\frac{q}{r} = V. \tag{2}$$

From this we obtain

$$q = 4\pi\varepsilon_0 V \cdot r. \tag{3}$$

Differentiating,

$$\frac{\mathrm{d}q}{\mathrm{d}t} = 4\pi\varepsilon_0 V\frac{\mathrm{d}r}{\mathrm{d}t}. \tag{4}$$

Substituting into equation (1) gives

$$-\frac{\rho}{4\pi r}\cdot 4\pi\varepsilon_0 V\frac{\mathrm{d}r}{\mathrm{d}t}=V,$$

that is,

$$\frac{\mathrm{d}r}{r}=-\frac{1}{\rho\varepsilon_0}\mathrm{d}t.$$

Integrating both sides, we obtain

$$\ln r\Big|_a^r=-\frac{1}{\rho\varepsilon_0}t,$$

hence

$$\therefore\quad r=a\mathrm{e}^{-\frac{1}{\rho\varepsilon_0}t}.$$

(2) The Joule heat consumed in the whole leakage process is

$$Q=\int_0^\infty\left(-\frac{\mathrm{d}q}{\mathrm{d}t}\right)^2 R\,\mathrm{d}t=\int_0^\infty\left(-\frac{\mathrm{d}q}{\mathrm{d}t}\right)\cdot V\,\mathrm{d}t=Vq_0,$$

where q_0 is the charge on the balloon at $t=0$; from (3), $q_0=4\pi\varepsilon_0 aV$. The original electrostatic energy of the balloon is

$$W=\frac{1}{2}q_0V=\frac{1}{2}Q.$$

The Joule heat consumed is greater than the original electrostatic energy because, during the gradual decrease of the radius of the balloon, external forces (the atmospheric pressure and the elastic force of the balloon) do work, so that the electrostatic energy is continually replenished.

Solution 3.15. Since the outer shell is grounded and the inner shell is connected by a wire to a conducting sphere of radius c (conductor 3), the spherical capacitor formed by the inner and outer shells may be regarded as connected in parallel with the spherical conductor 3 (which acts as a capacitor). Let the charge on conductor 3 be q; then the charge on the inner shell is $Q_a=(Q-q)$, the two charges being proportional to their respective capacitances. The capacitance C_{ab} of the spherical capacitor and the capacitance C_c of conductor 3 are respectively

$$C_{ab}=\frac{4\pi\varepsilon_0}{\dfrac{1}{a}-\dfrac{1}{b}},\qquad C_c=4\pi\varepsilon_0 c.$$

Hence

$$q=\frac{c}{c+\dfrac{1}{\dfrac{1}{a}-\dfrac{1}{b}}}Q=\frac{c\left(\dfrac{1}{a}-\dfrac{1}{b}\right)}{c\left(\dfrac{1}{a}-\dfrac{1}{b}\right)+1}Q,$$

$$Q_a=Q-q=\frac{1}{c\left(\dfrac{1}{a}-\dfrac{1}{b}\right)+1}Q.$$

The inner surface of the outer shell will carry the charge $-Q_a$. Owing to the induction of the charge on conductor 3, the outer surface of the outer shell will also carry a small amount of charge. Let this charge be q'; since the potential of the outer shell is zero,

$$\frac{q'}{b}+\frac{q}{r}=0,$$

and hence

$$q' = -\frac{b}{r}q.$$

Since the inner shell and the inner surface of the outer shell carry equal and opposite charges, they produce no electric field outside themselves; therefore the electrostatic force acting on conductor 3 is in fact the force between q and q':

$$F = \frac{1}{4\pi\varepsilon_0}\frac{qq'}{r^2} = \frac{1}{4\pi\varepsilon_0 r^2}\left(-\frac{b}{r}\right)q^2 = -\frac{1}{4\pi\varepsilon_0 r^3}\frac{bc^2\left(\frac{1}{a}-\frac{1}{b}\right)^2}{\left[c\left(\frac{1}{a}-\frac{1}{b}\right)+1\right]^2}Q^2.$$

The negative sign indicates that the force is attractive.

Solution 3.16. To facilitate the solution, we first examine the behaviour of an electrically neutral conducting sphere in a uniform external field. The conducting sphere may be regarded as the superposition of two spheres carrying equal and opposite volume charges. Let the radius of the sphere be R and the volume charge density be $\pm\rho$. When an external field $\vec{E}_0$ is present, the two charged spheres undergo a small displacement in the direction of the field, so as to set up inside the sphere a reverse field that cancels the external field, ultimately making the resultant field inside the sphere zero.

Let the displacement of the two spheres be l. Consider now the electric field at an arbitrary point P inside the sphere (see the figure). Let the distance from P to the centre O_+ of the positively charged sphere be r_+, and the distance from P to the centre O_- of the negatively charged sphere be r_-. By Gauss's law, the field produced by the positively charged sphere at P is

$$\vec{E}_+ = \frac{1}{4\pi\varepsilon_0}\frac{\frac{4}{3}\pi r_+^3\rho}{r_+^3}\vec{r}_+ = \frac{\rho}{3\varepsilon_0}\vec{r}_+.$$

Similarly, the field produced by the negatively charged sphere at P is

$$\vec{E}_- = -\frac{\rho}{3\varepsilon_0}\vec{r}_-.$$

Here $\vec{r}_+$ and $\vec{r}_-$ are the vectors from O_+ and O_- respectively to the point P; hence the resultant field is

$$\vec{E} = \vec{E}_+ + \vec{E}_- = \frac{\rho}{3\varepsilon_0}(\vec{r}_+ - \vec{r}_-) = -\frac{\rho}{3\varepsilon_0}\vec{l},$$

where $\vec{l}$ is the vector from O_- to O_+. It is seen that the field inside the sphere is uniform. It is only necessary to set $\vec{E} = -\vec{E}_0$ to obtain the actual value of the displacement:

$$\frac{\rho}{3\varepsilon_0}l = E_0,$$

and hence

$$l = \frac{3\varepsilon_0 E_0}{\rho}.$$

Since ρ is very large for a conductor, l is a small quantity; therefore, as a result of the displacement of the two charged spheres, the greater part of the overlapping region inside the sphere remains electrically neutral, and a thin layer of charge appears only in the two thin non-overlapping

regions at the sides, which may be regarded as a surface charge. The surface charge density σ is a function of the angle θ (this is in fact the induced charge on the conductor):

$$\sigma = \rho d = \rho l \cos\theta.$$

Substituting the expression for l, we obtain

$$\sigma = \sigma_0 \cos\theta = 3\varepsilon_0 E_0 \cos\theta. \tag{1}$$

When $\theta < 90°$, $\sigma > 0$; when $\theta > 90°$, $\sigma < 0$; when $\theta = 90°$, $\sigma = 0$.

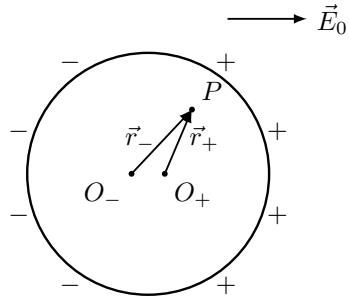


Fig. 3.16

Now we return to this problem. Imagine a conducting sphere of radius R placed in a uniform electric field $\vec{E}_0$ directed vertically upwards; then, from the above discussion, induced charges will appear on the surface of the conductor, their surface density being given by expression (1), in which the angle θ should now be understood as the angle with the vertical direction. At this time the horizontal plane through the centre of the sphere must be an equipotential surface of the conducting sphere; if a conducting flat plate is placed in this plane, it does not affect the distribution of the electric field in the upper part of the conducting sphere. Therefore, in the situation of the present problem, the charge distribution on the hemisphere is still determined by (1), and induced charges will appear on the conducting plate; their effect above the conducting plate is just like that produced by the charges on the original lower hemisphere (the charges on the lower hemisphere may be regarded as the image charges of the charges on the upper hemisphere), while below the conducting plate the electric field is zero everywhere. Thus the electric force acting on the hemisphere is the force on the surface charge of the hemisphere. This resultant force must be directed upwards, and its magnitude F can be found by integrating the vertical component of the normal force acting on each surface element of charge. We know that the force per unit area on the surface of a charged conductor is equal to $\dfrac{1}{2\varepsilon_0}\sigma^2$; hence

$$\begin{aligned} F &= \int_0^{\pi/2} \frac{1}{2\varepsilon_0}\sigma^2 \cdot 2\pi R \sin\theta \, R \, \mathrm{d}\theta \cdot \cos\theta \\ &= \frac{\pi R^2 \sigma_0^2}{\varepsilon_0} \int_0^{\pi/2} \sin\theta \cos^3\theta \, \mathrm{d}\theta \\ &= -\frac{\pi R^2 \sigma_0^2}{\varepsilon_0} \int_0^{\pi/2} \cos^3\theta \, \mathrm{d}(\cos\theta) \\ &= \frac{\pi R^2 \sigma_0^2}{4\varepsilon_0}. \end{aligned}$$

Substituting $\sigma_0 = 3\varepsilon_0 E_0$, we obtain

$$F = \frac{9}{4}\pi R^2 \varepsilon_0 E_0^2. \tag{2}$$

Setting $F = mg = \dfrac{2}{3}\pi R^3 \rho g$, we obtain

$$E_0 = \frac{2}{3}\sqrt{\frac{2R\rho g}{3\varepsilon_0}}.$$

This is the minimum value of the electric field that can lift the hemisphere off the flat plate.

Solution 3.17. (1) Imagine that the cylindrical wire carries positive charge, with a linear charge density $+\eta$. By the method of images, the electric field above the ground is excited jointly by this wire and by a cylindrical wire with linear charge density $-\eta$ located at the symmetric position below the ground.

When $r \ll d$, the charge on the wire may be regarded as uniformly distributed, and the electric field in space may be regarded as excited by two uniformly charged straight lines lying on the axes of the wires, with linear charge densities $+\eta$ and $-\eta$ respectively.

Take the ground as the yz-plane, and let the two charged straight lines run along the z direction, located at $x = \pm a$ respectively, as shown in the figure (in this part $a = d$). Take the y-axis (actually the yz-plane) as the zero-potential surface. The electric field at an arbitrary point $P(x, y)$ is

$$\begin{aligned}\vec{E}_P &= \vec{E}_+ + \vec{E}_- \\ &= \frac{\eta}{2\pi\varepsilon_0}\left(\frac{1}{r_+}\hat{r}_+ - \frac{1}{r_-}\hat{r}_-\right).\end{aligned}$$

The potential at P is the integral of the electric field from P to O (the zero point of potential). We now want the potential of the thin wire; it is most convenient to integrate along the x-axis, taking the starting point of the integration at A, the intersection of the surface of the thin wire with the x-axis, and setting $a = d$. The potential of the wire is

$$\varphi_A = \frac{\eta}{2\pi\varepsilon_0}\int_r^d \frac{\mathrm{d}r}{r} - \frac{\eta}{2\pi\varepsilon_0}\int_{2d-r}^d \frac{\mathrm{d}r}{r} \approx \frac{\eta}{2\pi\varepsilon_0}\ln\frac{2d}{r}.$$

The required capacitance is

$$C^* = \frac{\eta}{\varphi} = \frac{2\pi\varepsilon_0}{\ln\dfrac{2d}{r}}.$$

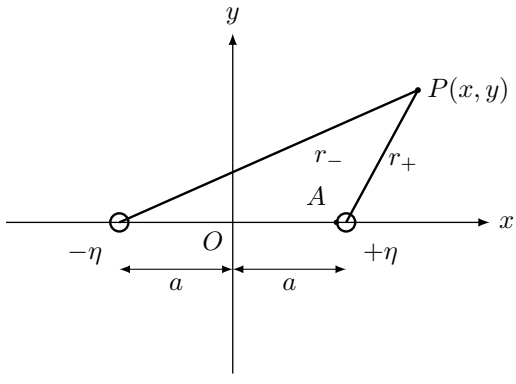


Fig. 3.17

(2) When r is of the same order of magnitude as d, the charge on the wire is no longer distributed uniformly over the cylindrical surface; being attracted by the charge of opposite sign on the image wire, it shifts towards the side of the image. The electric field in space can therefore no longer be regarded as the superposition of the fields of two lines of opposite sign placed along the axes. It can, however, still be regarded as the superposition of the fields of two lines of opposite sign parallel to the axis of the wire, the positions of both lines being shifted from the axes towards each other, while the linear charge densities remain unchanged. This is because the equipotential surfaces of two parallel straight lines carrying charges of opposite sign are a family of non-coaxial cylindrical surfaces.

For this purpose, consider the equipotential surfaces of the two charged straight lines in the figure. Let the distance of each of the two charged lines from the y-axis be a ($a \neq d$). According to the discussion above, the potential at an arbitrary point P is

$$\begin{aligned}\varphi_P &= \int_P^O \vec{E}\cdot\mathrm{d}\vec{r} = \int_P^O \vec{E}_+\cdot\mathrm{d}\vec{r} + \int_P^O \vec{E}_-\cdot\mathrm{d}\vec{r} \\ &= \frac{\eta}{2\pi\varepsilon_0}\int_{r_+}^{a}\frac{\mathrm{d}r}{r} - \frac{\eta}{2\pi\varepsilon_0}\int_{r_-}^{a}\frac{\mathrm{d}r}{r} \\ &= \frac{\eta}{2\pi\varepsilon_0}\ln\frac{a}{r_+} - \frac{\eta}{2\pi\varepsilon_0}\ln\frac{a}{r_-} = \frac{\eta}{2\pi\varepsilon_0}\ln\frac{r_-}{r_+}.\end{aligned}$$

Thus the points whose potential is φ satisfy

$$\frac{r_-}{r_+} = \mathrm{e}^{2\pi\varepsilon_0\varphi/\eta}.$$

Substituting the coordinates of the point $P(x, y)$ into the above expression and squaring, we obtain

$$\frac{(x+a)^2+y^2}{(x-a)^2+y^2} = \mathrm{e}^{4\pi\varepsilon_0\varphi/\eta}.$$

Setting $\mathrm{e}^{4\pi\varepsilon_0\varphi/\eta} = k^2$, the equation of the equipotential surface of potential φ is

$$(x+a)^2 + y^2 = k^2(x-a)^2 + k^2y^2.$$

Simplified, this gives

$$\left(x - \frac{k^2+1}{k^2-1}a\right)^2 + y^2 = \frac{4k^2}{(k^2-1)^2}a^2.$$

The above is the equation of a circle, whose centre and radius both depend on k^2. Let this circle be the cross-section of the cylindrical wire in the problem; then

$$\left(\frac{k^2+1}{k^2-1}\right)^2 a^2 = d^2, \qquad \frac{4k^2}{(k^2-1)^2}a^2 = r^2.$$

From these we solve for k^2 (noting that $k > 1$):

$$k^2 = \left[\frac{d}{r} + \sqrt{\left(\frac{d}{r}\right)^2 - 1}\right]^2.$$

From the definition $k^2 = \mathrm{e}^{4\pi\varepsilon_0\varphi/\eta}$ given above we then obtain

$$C^* = \frac{\eta}{\varphi} = \frac{4\pi\varepsilon_0}{\ln k^2} = \frac{2\pi\varepsilon_0}{\ln k} = \frac{2\pi\varepsilon_0}{\ln\left[\dfrac{d}{r} + \sqrt{\left(\dfrac{d}{r}\right)^2 - 1}\right]}.$$

When $r \ll d$, the result agrees with that of part (1).

Solution 3.18. (1) The electric field inside a uniformly charged spherical surface of radius a is zero. This can be explained by Gauss's law, or proved directly: for an arbitrary point P inside the spherical surface, draw a pair of small opposite cones with vertex at P; they cut out two surface elements on the sphere (see the figure of Solution 3.01). From the geometrical relations and the inverse-square law of the electric force it follows that the fields produced at P by the charges on these two surface elements cancel each other. The action of the other surface elements is similar. Hence the electric field at P is zero, that is, the electric field at every point inside the spherical surface is zero. Imagine that the spherical surface is made of an insulating material, and now let it be charged and then be deformed into a spheroid (keeping its volume unchanged). Because the charges repel one another, the charge will redistribute so that the spheroidal surface becomes an equipotential surface, and the electric field inside the spheroid is zero. By the condition of electrostatic equilibrium, this is just the charge distribution on a spheroidal conductor. It is only necessary to find the potential produced by this charged spheroid at its centre, and the capacitance can then be found.

Let the surface charge density on the original spherical surface of radius a be σ_0. When the charge on the ring on this spherical surface between the cone angles θ and $\theta + \mathrm{d}\theta$ is stretched onto the spheroid, its amount remains unchanged, but its distance to the centre O increases to OA, as shown in the figure. The potential it produces at O is

$$\begin{aligned}
\mathrm{d}\varphi &= \frac{1}{4\pi\varepsilon_0}\frac{\sigma_0\, 2\pi a\sin\theta\, a\,\mathrm{d}\theta}{OA} \\
&= \frac{1}{4\pi\varepsilon_0}\frac{\sigma_0\cdot 2\pi a\sin\theta\, a\,\mathrm{d}\theta}{\sqrt{a^2\sin^2\theta + \left(a\cos\theta\cdot\dfrac{b}{a}\right)^2}} \\
&= \frac{\sigma_0\, 2\pi a^2}{4\pi\varepsilon_0}\frac{\sin\theta\,\mathrm{d}\theta}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}} \\
&= -\frac{q}{8\pi\varepsilon_0 a}\frac{\mathrm{d}(\cos\theta)}{\sqrt{1 + \dfrac{b^2-a^2}{a^2}\cos^2\theta}},
\end{aligned}$$

where $q = 4\pi a^2\sigma_0$ is the total charge on the spherical surface.

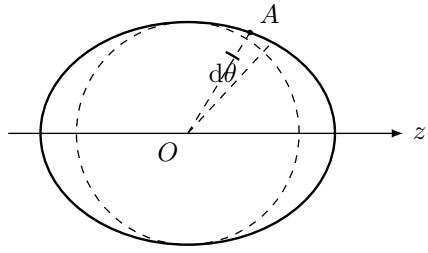


Fig. 3.18

(i) When $b > a$,

$$\mathrm{d}\varphi = -\frac{q}{8\pi\varepsilon_0 a}\cdot\frac{1}{c}\cdot\frac{\mathrm{d}(c\cos\theta)}{\sqrt{1+c^2\cos^2\theta}},$$

where $c^2 = \dfrac{b^2-a^2}{a^2}$; hence the potential at O is

$$\begin{aligned}\varphi &= -\frac{q}{8\pi\varepsilon_0 ac}\int_0^{\pi}\frac{\mathrm{d}(c\cos\theta)}{\sqrt{1+c^2\cos^2\theta}} = -\frac{q}{8\pi\varepsilon_0}\frac{1}{\sqrt{b^2-a^2}}\int_c^{-c}\frac{\mathrm{d}x}{\sqrt{1+x^2}}\\ &= \frac{q}{8\pi\varepsilon_0\sqrt{b^2-a^2}}\ln\left(x+\sqrt{1+x^2}\right)\Big|_{-c}^{c}\\ &= \frac{q}{8\pi\varepsilon_0\sqrt{b^2-a^2}}\ln\frac{b+\sqrt{b^2-a^2}}{b-\sqrt{b^2-a^2}}.\end{aligned}$$

Hence the capacitance is

$$C = \frac{q}{\varphi} = \frac{8\pi\varepsilon_0\sqrt{b^2-a^2}}{\ln\dfrac{b+\sqrt{b^2-a^2}}{b-\sqrt{b^2-a^2}}} \qquad (b>a).$$

(ii) When $b < a$,

$$\mathrm{d}\varphi = -\frac{q}{8\pi\varepsilon_0 a}\cdot\frac{1}{c}\cdot\frac{\mathrm{d}(c\cos\theta)}{\sqrt{1-c^2\cos^2\theta}},$$

where $c^2 = \dfrac{a^2-b^2}{a^2}$, and the potential at O is

$$\begin{aligned}\varphi &= -\frac{q}{8\pi\varepsilon_0 ac}\int_0^{\pi}\frac{\mathrm{d}(c\cos\theta)}{\sqrt{1-c^2\cos^2\theta}} = -\frac{q}{8\pi\varepsilon_0}\frac{1}{\sqrt{a^2-b^2}}\int_c^{-c}\frac{\mathrm{d}x}{\sqrt{1-x^2}}\\ &= -\frac{q}{8\pi\varepsilon_0\sqrt{a^2-b^2}}\arcsin x\Big|_c^{-c} = \frac{q}{4\pi\varepsilon_0\sqrt{a^2-b^2}}\arcsin\frac{\sqrt{a^2-b^2}}{a}.\end{aligned}$$

Therefore

$$\therefore\quad C = \frac{q}{\varphi} = \frac{4\pi\varepsilon_0\sqrt{a^2-b^2}}{\arcsin\dfrac{\sqrt{a^2-b^2}}{a}} \qquad (b<a).$$

(2) When $b \gg a$ (needle-like),

$$\ln \frac{b+\sqrt{b^2-a^2}}{b-\sqrt{b^2-a^2}} = \ln \frac{1+\sqrt{1-\dfrac{a^2}{b^2}}}{1-\sqrt{1-\dfrac{a^2}{b^2}}} \approx 2\ln \frac{2b}{a},$$

$$C = \frac{8\pi\varepsilon_0\sqrt{b^2-a^2}}{\ln \dfrac{b+\sqrt{b^2-a^2}}{b-\sqrt{b^2-a^2}}} \approx \frac{4\pi\varepsilon_0 b}{\ln \dfrac{2b}{a}} \qquad (b \gg a).$$

When $b \ll a$ (disk-like), $\arcsin \dfrac{\sqrt{a^2-b^2}}{a} \approx \dfrac{\pi}{2}$, and

$$C = \frac{4\pi\varepsilon_0\sqrt{a^2-b^2}}{\arcsin \dfrac{\sqrt{a^2-b^2}}{a}} \approx 8\varepsilon_0 a \qquad (b \ll a).$$

Comment. This problem skilfully uses the method of deformation to determine the charge distribution on a spheroidal conductor under electrostatic equilibrium, so that its capacitance problem is readily solved.

For a disk-like conductor, because the charge on it is not uniformly distributed, the capacitance obtained differs from the result ($2\pi\varepsilon_0 a$) obtained by assuming a uniform charge distribution. For a needle-like conductor, because the charge on it is essentially uniformly distributed, the two results are the same. The reader may compare and check them.

Solution 3.19. (1) First find the electric field inside the cylindrical surface. Let the linear charge density on the wire be λ; then

$$E = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}.$$

Also,

$$V_0 = \int_{r_0}^{R} E\,\mathrm{d}r = \frac{\lambda}{2\pi\varepsilon_0}\int_{r_0}^{R}\frac{\mathrm{d}r}{r} = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{R}{r_0},$$

$$\therefore \quad E = \frac{V_0}{\ln \dfrac{R}{r_0}}\frac{1}{r}.$$

Consider the motion of the ions. The velocity of an ion is

$$\frac{\mathrm{d}r}{\mathrm{d}t} = wE = \frac{V_0 w}{\ln \dfrac{R}{r_0}}\frac{1}{r}.$$

Then

$$r\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{V_0 w}{\ln \dfrac{R}{r_0}}.$$

Integrating,

$$\int_{r_0}^{r} r\,\mathrm{d}r = \int_0^t \frac{V_0 w}{\ln \dfrac{R}{r_0}}\,\mathrm{d}t,$$

that is,

$$\frac{1}{2}\left(r^2 - r_0^2\right) = \frac{V_0 w}{\ln\dfrac{R}{r_0}} t,$$

$$\therefore \quad r^2 = r_0^2 + 2\frac{V_0 w}{\ln\dfrac{R}{r_0}} t,$$

or

$$r = \sqrt{r_0^2 + 2\frac{V_0 w}{\ln(R/r_0)} t}.$$

(2) Between r_0 and the radial position r_i of the positive ions there is a reverse electric field. Let the counter have length L; then the linear charge density of the ions is $\lambda_i = \dfrac{Q_i}{L}$, and the potential difference of this reverse field is

$$\Delta V = \frac{Q_i}{L \cdot 2\pi\varepsilon_0} \ln\frac{r_i}{r_0}.$$

This potential difference must be cancelled by the potential difference produced by the charge Q flowing into the wire (at the same time a charge $-Q$ flows into the outer wall of the cylinder); this cancelling potential difference is

$$\Delta V' = \frac{Q}{L \cdot 2\pi\varepsilon_0} \ln\frac{R}{r_0}.$$

Setting $\Delta V = \Delta V'$, we obtain

$$Q = Q_i \frac{\ln\dfrac{r_i}{r_0}}{\ln(R/r_0)}.$$

Substituting $r_i = \sqrt{r_0^2 + 2\dfrac{V_0 w}{\ln(R/r_0)} t}$ into the above expression, we obtain

$$Q = \frac{Q_i}{2\ln(R/r_0)} \ln\left(1 + \frac{2V_0 w}{r_0^2 \ln(R/r_0)} t\right).$$

This action lasts only until $r_i = R$, that is, it applies only for

$$t \le \frac{R^2 - r_0^2}{2\dfrac{V_0 w}{\ln(R/r_0)}} = t_1.$$

After $t \ge t_1$, $Q = Q_i$.

Solution 3.20. Imagine the point charge to be distributed on a small conducting sphere, and further imagine thin conducting spherical shells of radii a and b to exist at the inner and outer surfaces of the dielectric respectively. Since the conducting shells are very thin, the charges of opposite sign on them lie very close together, and their presence may be considered not to affect the original electric field. In this way the whole system may be regarded as three capacitors connected in series: the capacitance C_1 between the innermost small conducting sphere and the conducting shell at the inner surface of the dielectric, the capacitance $\varepsilon_r C_2$ of the spherical

dielectric of inner and outer radii a and b, and the capacitance C_3 between the conducting shell of radius b and infinity. When the point charge is moved to infinity, at the place where the point charge was there still exists the capacitance between the small conducting sphere of the point charge and infinity; this capacitance can still be regarded as the series combination of the three vacuum capacitances C_1, C_2 and C_3. In this way the electrostatic energies of the two states differ only in the stored energy of the vacuum capacitance C_2 and of the dielectric capacitance $\varepsilon_r C_2$, and the work A done by the external force is just equal to this difference of electrostatic energies:

$$A = \frac{q^2}{2}\left(\frac{1}{C_2} - \frac{1}{\varepsilon_r C_2}\right) = \frac{q^2}{2C_2}\left(1 - \frac{1}{\varepsilon_r}\right).$$

Also,

$$C_2 = \frac{4\pi\varepsilon_0}{\dfrac{1}{a} - \dfrac{1}{b}}.$$

Substituting into the above expression, we obtain

$$A = \frac{q^2}{8\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)\left(1 - \frac{1}{\varepsilon_r}\right).$$

Comment. Finding the work through the change of energy is a method commonly used in electrostatic problems. But attention must be paid to whether there are other processes doing work in the course of the process (heat-transfer problems are seldom involved in electrostatics), especially work done by a source (in this problem there is no work done by any source).

This problem can also be solved by directly calculating, by integration, the difference between the energies of the electrostatic field in the two cases.

Solution 3.21. Suppose that, when the dielectric particles are absent, a uniform electric field E_0 already exists between the plates. When dielectric particles are present, since the particles are sparse, the external field polarizing a given dielectric particle may be taken to be E_0. Under the action of E_0 the spherical dielectric particle is uniformly polarized; let its polarization be P. At the same time equivalent polarization charges are formed on the surface of the dielectric sphere, with a surface density $\sigma' = P\cos\theta$ (see the figure). The field produced by these polarization charges is also uniform, opposite in direction to P, and of magnitude $P/(3\varepsilon_0)$ (see Solution 3.16); and the dielectric is polarized precisely under the action of the resultant of E_0 and the field of these polarization charges. Hence

$$P = (\varepsilon_r - 1)\varepsilon_0\left[E_0 - \frac{P}{3\varepsilon_0}\right].$$

That is,

$$\left(1 + \frac{\varepsilon_r - 1}{3}\right)P = (\varepsilon_r - 1)\varepsilon_0 E_0.$$

From this we obtain

$$P = \frac{3\varepsilon_0(\varepsilon_r - 1)}{\varepsilon_r + 2}E_0.$$

The electric moment of each particle is

$$p = \frac{4}{3}\pi a^3 P = \frac{4\pi\varepsilon_0 a^3(\varepsilon_r - 1)}{\varepsilon_r + 2}E_0.$$

The presence of the polarized particles is equivalent to a uniform polarization of the dielectric between the plates, with an equivalent polarization

$$P' = np = 4\pi\varepsilon_0 a^3 n \frac{\varepsilon_r - 1}{\varepsilon_r + 2} E_0.$$

Since the particles are sparse, P' must be very small; the reverse field produced by the polarization charges corresponding to P' may be neglected, and the field polarizing this equivalent dielectric may be taken to be the external field E_0. Thus

$$P' = (\varepsilon_r' - 1)\varepsilon_0 E_0.$$

Comparing this with the above expression, we obtain the equivalent relative permittivity

$$\varepsilon_r' = 1 + \frac{\varepsilon_r - 1}{\varepsilon_r + 2} \cdot 4\pi a^3 n.$$

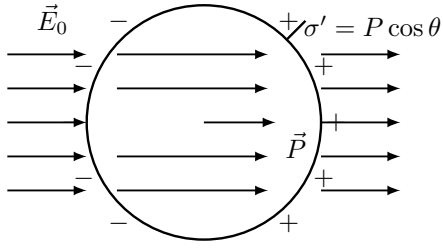


Fig. 3.21

Solution 3.22. First, around this point charge there will necessarily gather polarization charge q'' of sign opposite to q, so that $q + q'' = \dfrac{q}{\varepsilon_r}$; hence it suffices to examine the behaviour of the point charge q/ε_r.

By symmetry, it suffices to consider the surface polarization charge at a distance r from the foot O of the perpendicular from the point charge to the surface; let its density be $\sigma'(r)$. Above and below the surface the electric field is the superposition of the field $\vec{E}_1$ produced by the point charge q/ε_r and the field $\vec{E}'$ produced by σ'. At the surface, in the normal direction, it follows from Gauss's law that the normal component of the field $\vec{E}'$ produced by σ' is $\sigma'/(2\varepsilon_0)$, directed away from the surface upwards and downwards; hence

$$E_{n,\mathrm{up}} = \frac{1}{4\pi\varepsilon_0} \frac{q/\varepsilon_r\, h}{(h^2 + r^2)^{3/2}} + \frac{\sigma'}{2\varepsilon_0},$$

$$E_{n,\mathrm{down}} = \frac{1}{4\pi\varepsilon_0} \frac{q/\varepsilon_r\, h}{(h^2 + r^2)^{3/2}} - \frac{\sigma'}{2\varepsilon_0}.$$

From the continuity of the normal component of the electric displacement, D_n,

$$\varepsilon_0 E_{n,\mathrm{up}} = \varepsilon_r \varepsilon_0 E_{n,\mathrm{down}}.$$

That is,

$$\frac{1}{4\pi\varepsilon_r} \frac{qh}{(h^2 + r^2)^{3/2}} + \frac{\sigma'}{2} = \frac{1}{4\pi} \frac{qh}{(h^2 + r^2)^{3/2}} - \frac{\varepsilon_r \sigma'}{2}.$$

Solving this, we obtain

$$\sigma' = \frac{\varepsilon_r - 1}{(\varepsilon_r + 1)\varepsilon_r} \cdot \frac{qh}{2\pi(h^2 + r^2)^{3/2}}.$$

The total polarization charge is

$$q' = \int_0^\infty \sigma' \cdot 2\pi r \, \mathrm{d}r = \frac{\varepsilon_r - 1}{\varepsilon_r(\varepsilon_r + 1)} \int_0^\infty \frac{qhr \, \mathrm{d}r}{(r^2 + h^2)^{3/2}} = \frac{\varepsilon_r - 1}{\varepsilon_r(\varepsilon_r + 1)} q.$$

From the expression for $\sigma'(r)$ it is seen that the field produced by the polarization charges below the surface is the same as if there were a point charge q' at a height h above the surface; hence the electrostatic force acting on q is the repulsion between q and q':

$$F = \frac{1}{4\pi\varepsilon_0} \frac{q^2(\varepsilon_r - 1)}{\varepsilon_r(\varepsilon_r + 1) \cdot 4h^2}.$$

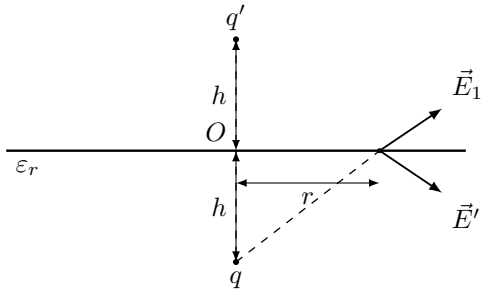


Fig. 3.22

Comment. The field above the surface is just the field of a point charge equal to $(q/\varepsilon_r + q')$. The charge q' is in essence the "image charge" in the case of a dielectric.

Solution 3.23. (1) Let the charge per unit length on the metal rod be λ. Then from Gauss's law,

$$E = \frac{\lambda}{2\pi\varepsilon_0 r},$$

so

$$U = \int_a^b E \, \mathrm{d}r = \frac{\lambda}{2\pi\varepsilon_0} \int_a^b \frac{\mathrm{d}r}{r} = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{b}{a},$$

and therefore

$$E = \frac{U}{r \ln \dfrac{b}{a}}.$$

(2) At the place where the dust lies the electric field may be regarded as uniform, and the spherical dielectric (the dust) will be uniformly polarised, the direction of the polarisation being the same as that of the field; denote the polarisation by P. Then

$$P = (\varepsilon_r - 1)\varepsilon_0 E',$$

where

$$E' = E - \frac{P}{3\varepsilon_0}.$$

(see Solution 3.21) Substituting into the above expression gives

$$P = (\varepsilon_r - 1)\varepsilon_0 \left(E - \frac{P}{3\varepsilon_0} \right) = (\varepsilon_r - 1)\varepsilon_0 E - \frac{\varepsilon_r - 1}{3} P,$$

from which we obtain

$$P = \frac{3(\varepsilon_r - 1)}{\varepsilon_r + 2} \varepsilon_0 E = \frac{3(\varepsilon_r - 1)}{\varepsilon_r + 2} \varepsilon_0 \frac{U}{r \ln \dfrac{b}{a}}.$$

Here the expression for E obtained in part (1) has already been substituted. The dust is equivalent to a dipole, and its dipole moment is

$$p = \frac{4}{3}\pi c^3 P = \frac{4}{3}\pi c^3 \frac{3(\varepsilon_r - 1)}{\varepsilon_r + 2} \varepsilon_0 \frac{U}{\ln \dfrac{b}{a}} \cdot \frac{1}{r}.$$

The force acting on the dust is

$$F = p \frac{\partial E}{\partial r} = -\frac{4\pi\varepsilon_0 c^3 (\varepsilon_r - 1) U^2}{(\varepsilon_r + 2) \left(\ln \dfrac{b}{a} \right)^2} \frac{1}{r^3}.$$

The sign "$-$" shows that the direction of the force points towards the axis.

(3) The dust must move from the wall of the cylinder towards the metal rod, and its acceleration is

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{F}{m} = -\frac{A}{r^3} \qquad \left(\text{where } A = \frac{4\pi\varepsilon_0 c^3 (\varepsilon_r - 1) U^2}{m(\varepsilon_r + 2) \left(\ln \dfrac{b}{a} \right)^2} \right).$$

Multiplying both sides of the above expression by $\mathrm{d}r$ and integrating, and noting that $\dfrac{\mathrm{d}r}{\mathrm{d}t} = v$, we have

$$\int_0^v v \, \mathrm{d}v = \int_b^r -\frac{A}{r^3} \, \mathrm{d}r,$$

that is,

$$\frac{1}{2} v^2 = \frac{A}{2} \left. \frac{1}{r^2} \right|_b^r = \frac{A}{2} \left(\frac{1}{r^2} - \frac{1}{b^2} \right),$$

and hence

$$v = -\sqrt{A \left(\frac{1}{r^2} - \frac{1}{b^2} \right)} = -\frac{\sqrt{A(b^2 - r^2)}}{br}.$$

Then

$$\mathrm{d}t = \frac{\mathrm{d}r}{v} = -\frac{br \, \mathrm{d}r}{\sqrt{A(b^2 - r^2)}}.$$

Integrating, we obtain

$$\begin{aligned} t = \int \mathrm{d}t &= -\frac{b}{\sqrt{A}} \int_b^a \frac{r \, \mathrm{d}r}{\sqrt{b^2 - r^2}} = \frac{b}{2\sqrt{A}} \int \frac{\mathrm{d}(b^2 - r^2)}{\sqrt{b^2 - r^2}} \\ &= \frac{b}{2\sqrt{A}} \left. 2\sqrt{b^2 - r^2} \right|_b^a = \frac{b}{\sqrt{A}} \sqrt{b^2 - a^2}. \end{aligned}$$

Solution 3.24. (1) Polarisation charges will be formed on the wall of the cavity, and the superposition of the field produced by these polarisation charges inside the sphere with the original field E_0 is just the field in the cavity; outside the sphere the result of superposing the field produced by the polarisation charges with the original field will make the polarisation of the medium no longer uniform. Inspired by the result that in a uniform field the field inside a dielectric sphere is still uniform, the field inside the cavity should also be uniform. Hence we may suppose the surface density of the polarisation charge to be

$$\sigma_p = -\sigma_0 \cos\theta.$$

It is only necessary to find σ_0, and then the field in the cavity can be found, because the polarisation charge with this distribution produces a uniform field inside the cavity:

$$E' = \frac{\sigma_0}{3\varepsilon_0},$$

whose direction is the same as that of E_0 (see Fig. 1). Hence the field in the cavity is

$$\vec{E} = \vec{E}_0 + \frac{\sigma_0}{3\varepsilon_0}\hat{\imath}.$$

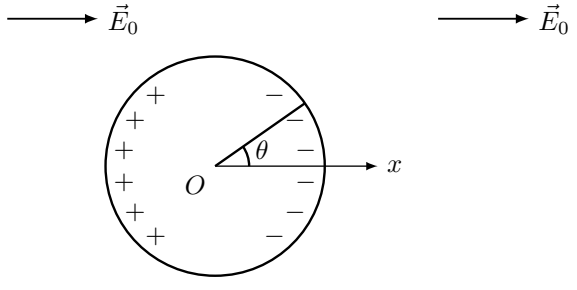


Fig. 3.24(a)

In order to find σ_0 it must be related to the polarisation, and therefore the field produced by this polarisation charge outside the cavity should first be found. This can be done by using the boundary conditions. Denote the region inside the cavity by 1 and that outside by 2; then at θ we must have

$$E_{2n} - E_{1n} = \frac{\sigma_P}{\varepsilon_0} = -\frac{\sigma_0}{\varepsilon_0}\cos\theta.$$

But

$$E_{1n} = \left(E_0 + \frac{\sigma_0}{3\varepsilon_0}\right)\cos\theta,$$

$$\therefore\ E_{2n} = E_{1n} - \frac{\sigma_0}{\varepsilon_0}\cos\theta = \left(E_0 - \frac{2}{3}\frac{\sigma_0}{\varepsilon_0}\right)\cos\theta,$$

$$\therefore\ P_{2n} = (\varepsilon_r - 1)\varepsilon_0 E_{2n} = (\varepsilon_r - 1)\varepsilon_0 E_0 \cos\theta - \frac{2}{3}(\varepsilon_r - 1)\sigma_0 \cos\theta.$$

However, from the relation between the polarisation charge and the polarisation, $P_{2n} = -\sigma' = \sigma_0 \cos\theta$; substituting this into the above expression gives

$$\sigma_0 = \frac{3(\varepsilon_r - 1)}{2\varepsilon_r + 1}\varepsilon_0 E_0.$$

Substituting into the expression for E, we obtain

$$E = E_0 + \frac{\sigma_0}{3\varepsilon_0} = E_0 + \frac{\varepsilon_r - 1}{2\varepsilon_r + 1} E_0 = \frac{3\varepsilon_r}{2\varepsilon_r + 1} E_0.$$

Written in vector form,

$$\vec{E} = \frac{3\varepsilon_r}{2\varepsilon_r + 1} \vec{E}_0. \tag{1}$$

(2) Putting a charged conducting sphere into the cavity may be imagined as first putting an uncharged conducting sphere into the cavity and then placing charge on the conducting sphere. When the uncharged conducting sphere is put into the cavity, induced charges will appear on the conducting sphere, and the field of these induced charges will in turn change the polarisation of the medium, so that finally the field produced inside the sphere by the total charge (free charge plus polarisation charge) formed on the spherical surface should exactly cancel the original external field, making the field inside the sphere zero. Hence we may suppose the surface density of the total charge to be

$$\sigma_t = \sigma_{t0} \cos\theta$$

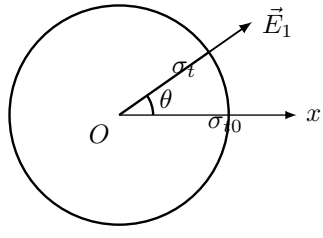


Fig. 3.24(b)

(Fig. 2) where σ_{t0} must satisfy

$$\frac{\sigma_{t0}}{3\varepsilon_0} = E_0 \qquad (\text{that is, } \sigma_{t0} = 3\varepsilon_0 E_0),$$

$$\therefore\ \sigma_t = 3\varepsilon_0 E_0 \cos\theta.$$

By the boundary conditions, the field at θ on the outer surface of the conducting sphere (inside the medium) is perpendicular to the surface, and its value is

$$E_1 = \frac{\sigma_t}{\varepsilon_0} = 3E_0 \cos\theta.$$

When a charge Q is then placed on the conducting sphere, Q must be distributed uniformly over the surface of the sphere (the field inside the sphere still being zero), and the polarisation charges caused by Q will also be distributed uniformly over the surface of the sphere; at this time the total field at θ is

$$E = E_1 + E_2 = 3E_0 \cos\theta + \frac{Q}{4\pi\varepsilon_0\varepsilon_r R^2} \tag{2}$$

and its direction is perpendicular to the spherical surface, where E_2 is the field excited after Q is added. The force density on the conductor (the force per unit area) is equal to the energy density of the field, and by symmetry the resultant force must be along the direction of E_0 (that is, the x direction); the component along the x direction of the force on the belt from θ to $\theta + \mathrm{d}\theta$ is

$$\mathrm{d}F_x = \frac{1}{2}\varepsilon_r\varepsilon_0 E^2 \cdot 2\pi R \sin\theta\, R\,\mathrm{d}\theta \cdot \cos\theta = \varepsilon_r\varepsilon_0 \pi R^2 E^2 \sin\theta\cos\theta\,\mathrm{d}\theta.$$

The resultant force is $F = \int \mathrm{d}F_x = \int_0^\pi \varepsilon_r\varepsilon_0\pi R^2 E^2 \sin\theta\cos\theta\,\mathrm{d}\theta$, and

$$\begin{aligned} E^2 &= \left(3E_0\cos\theta + \frac{Q}{4\pi\varepsilon_0\varepsilon_r R^2}\right)^2 \\ &= 9E_0^2\cos^2\theta + \left(\frac{Q}{4\pi\varepsilon_0\varepsilon_r R^2}\right)^2 + 6E_0\cos\theta\frac{Q}{4\pi\varepsilon_0\varepsilon_r R^2}. \end{aligned}$$

Only the last term contributes to the integral, so

$$\begin{aligned} \therefore\ F &= \int_0^\pi \varepsilon_r\varepsilon_0\pi R^2\cdot\frac{6E_0Q}{4\pi\varepsilon_0\varepsilon_r R^2}\cos^2\theta\sin\theta\,\mathrm{d}\theta \\ &= \frac{3}{2}E_0Q\int_0^\pi(-\cos^2\theta)\,\mathrm{d}(\cos\theta) \\ &= \frac{3}{2}E_0Q\left.\frac{\cos^3\theta}{3}\right|_\pi^0 \\ &= QE_0. \end{aligned}$$

Solution 3.25. The spherical cavity in which q' lies and the medium near it are in the electric field excited by the charge q and the surrounding medium. Since q and q' are very far apart, this field may be regarded as a uniform field, denoted by E_q. If this field did not exist, q' would experience no force in the cavity, because the polarisation charges produced by the polarisation of the medium caused by the field excited by q' are distributed uniformly on the surface of the spherical cavity and exert no force on q'. But q' is also not simply acted on by E_q, because the existence of E_q makes the medium around q' produce an additional polarisation, so that the field in the cavity is no longer equal to E_q. The problem comes down to finding the field inside a spherical cavity in a uniform electric field E_q. According to Solution 3.24, the magnitude of this field is

$$E = \frac{3\varepsilon_r}{2\varepsilon_r+1}E_q.$$

(see equation (1) of Solution 3.24) Neglecting the influence of q' and the polarisation charges around it on the polarisation of the medium near q (this is a second-order effect), E_q is evidently just the field of the point charge q partially shielded by the medium, that is,

$$E_q = \frac{1}{4\pi\varepsilon_0}\frac{q}{\varepsilon_r r^2}.$$

The required force is

$$\begin{aligned} F' = q'E &= q'\frac{3\varepsilon_r}{2\varepsilon_r+1}\cdot\frac{1}{4\pi\varepsilon_0}\frac{q}{\varepsilon_r r^2}. \\ &= \frac{1}{4\pi\varepsilon_0}\frac{3qq'}{(2\varepsilon_r+1)r^2}. \end{aligned}$$

Solution 3.26. From the hint it is known that the distribution of the polarisation charge on the surface of the dielectric sphere is the same as in the case of uniform polarisation. Let the polarisation charge be $\sigma = \sigma_0\cos\theta$, the axis of the polar coordinates being the same as the direction of the dipole p at the centre of the sphere. Then the field outside the sphere is produced jointly by the dipole moment formed by the surface polarisation charge of the sphere and the

equivalent dipole moment p/ε_r, which is the dipole at the centre of the sphere shielded by the medium; the total dipole moment is

$$p_t = \frac{p}{\varepsilon_r} + \frac{4\pi}{3}a^3\sigma_0.$$

(Note that the surface density of the polarisation charge on the surface of a uniformly polarised sphere with polarisation P is $\sigma = P\cos\theta$.) The field inside the sphere is the superposition of the field produced by the equivalent dipole p/ε_r at the centre of the sphere and the uniform field produced by the surface polarisation charge. The problem comes down to finding σ_0, which can be found from the boundary conditions.

Let the region inside the sphere be region 2 and the region outside the sphere be region 1; then at the surface (a, θ),

$$\begin{aligned}
E_{1r} &= \frac{1}{4\pi\varepsilon_0}\frac{(2p/\varepsilon_r)\cos\theta}{a^3} + \frac{1}{4\pi\varepsilon_0}\frac{2\cdot\frac{4}{3}\pi a^3\sigma_0\cos\theta}{a^3} \\
&= \frac{1}{4\pi\varepsilon_0}\frac{2p\cos\theta}{\varepsilon_r a^3} + \frac{2\sigma_0}{3\varepsilon_0}\cos\theta, \\
E_{2r} &= \frac{1}{4\pi\varepsilon_0}\frac{(2p/\varepsilon_r)\cos\theta}{a^3} - \frac{\sigma_0}{3\varepsilon_0}\cos\theta \\
&= \frac{1}{4\pi\varepsilon_0}\frac{2p\cos\theta}{\varepsilon_r a^3} - \frac{\sigma_0}{3\varepsilon_0}\cos\theta.
\end{aligned}$$

From the boundary condition $\varepsilon_r E_{2r} = E_{1r}$ we obtain

$$\frac{2p\cos\theta}{4\pi\varepsilon_0 a^3} - \frac{\varepsilon_r\sigma_0}{3\varepsilon_0}\cos\theta = \frac{2p\cos\theta}{4\pi\varepsilon_0\varepsilon_r a^3} + \frac{2\sigma_0}{3\varepsilon_0}\cos\theta.$$

From this we obtain

$$\sigma_0 = \frac{3(\varepsilon_r - 1)}{\varepsilon_r(\varepsilon_r + 2)}\frac{p}{2\pi a^3}.$$

Substituting into the expression for p_t gives

$$\begin{aligned}
p_t &= \frac{p}{\varepsilon_r} + \frac{4\pi}{3}a^3\cdot\frac{3(\varepsilon_r - 1)}{\varepsilon_r(\varepsilon_r + 2)}\frac{p}{2\pi a^3} \\
&= \left[\frac{1}{\varepsilon_r} + \frac{2(\varepsilon_r - 1)}{\varepsilon_r(\varepsilon_r + 2)}\right]p \\
&= \frac{3}{\varepsilon_r + 2}p.
\end{aligned}$$

The potential outside the sphere is just the potential of p_t. Taking the potential at infinity as zero, we have

$$\varphi_1 = \frac{1}{4\pi\varepsilon_0}\frac{p_t\cos\theta}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{3}{\varepsilon_r + 2}\frac{p\cos\theta}{r^2} \qquad (r \geq a).$$

The potential inside the sphere is the superposition of the potential of the uniform field $\dfrac{\sigma_0}{3\varepsilon_0}$ and the potential of the dipole p/ε_r:

$$\begin{aligned}
\varphi_2 &= \frac{(p/\varepsilon_r)\cos\theta}{4\pi\varepsilon_0 r^2} + \frac{\sigma_0}{3\varepsilon_0}r\cos\theta \\
&= \frac{p\cos\theta}{4\pi\varepsilon_0\varepsilon_r}\left[\frac{1}{r^2} + \frac{2(\varepsilon_r - 1)}{(\varepsilon_r + 2)a^3}r\right] \qquad (r \leq a).
\end{aligned}$$

Solution 3.27. The current in the resistive material is the superposition of the inflowing current and the outflowing current. When only the inflowing current exists, the current density in the medium is

$$j_1 = \frac{I}{2\pi r^2}.$$

From the differential form of Ohm's law, the field is

$$E_1 = \rho j_1 = \frac{\rho I}{2\pi r^2},$$

where r is the distance from the point under consideration to the point at which the current enters. Compared with the field of a point charge, $E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r^2}$, E_1 is equivalent to the field of a point charge with $q = 2\varepsilon_0\rho I$; hence the potential difference between the positions of the two ends of the voltmeter is

$$U_1 = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{d} - \frac{1}{2d}\right) = \frac{q}{4\pi\varepsilon_0}\cdot\frac{1}{2d}.$$

Substituting $q = 2\varepsilon_0\rho I$, we obtain

$$U_1 = \frac{\rho I}{2\pi}\cdot\frac{1}{2d}.$$

Similarly, when only the outflowing current exists the situation is analogous, and moreover on the line joining the points at which the current enters and leaves (that is, the straight line on which the four probe tips lie) the directions of the current are the same, and the corresponding potential differences are also the same:

$$U_2 = \frac{\rho I}{2\pi}\cdot\frac{1}{2d}.$$

When the inflowing and the outflowing currents exist simultaneously, the reading of the voltmeter is the sum of U_1 and U_2:

$$U = U_1 + U_2 = \frac{\rho I}{2\pi d}.$$

From this we obtain

$$\rho = \frac{2\pi d U}{I}.$$

Solution 3.28. Let the required resistance be xR. For convenience of calculation in what follows, we may take R as the unit of resistance; then the required resistance is set equal to x. Since resistance is proportional to length, the resistance between two opposite corners of the small network left after the two outermost squares are removed should be $x/2$, so the original network may be simplified to Fig. (1). (The values of the resistances in the lower half of the figure are the same as those in the upper half, and are not marked.)

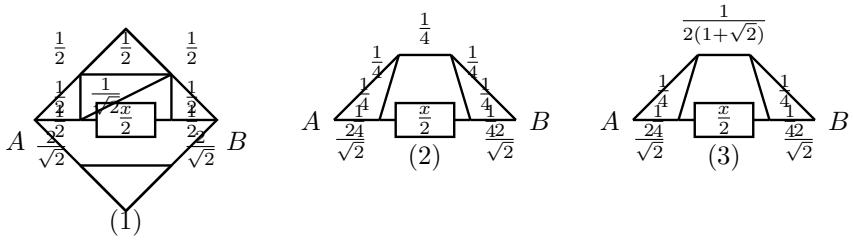


Fig. 3.28

Next fold the network along the line AB; coincident resistances are then in parallel, so their values are halved, giving Fig. (2). Then combine the two topmost resistances in parallel, giving Fig. (3).

At this point we obtain the equation

$$x = \frac{1}{2} + \left[2(1+\sqrt{2}) + \frac{2\sqrt{2}}{\sqrt{2}x+1}\right]^{-1}.$$

Simplifying,

$$2x - 1 = \frac{1}{\sqrt{2}+1+\dfrac{\sqrt{2}}{\sqrt{2}x+1}},$$

that is,

$$(2x-1)\left[(2+\sqrt{2})x + 2\sqrt{2} + 1\right] = \sqrt{2}x + 1.$$

Finally we obtain the quadratic equation for x:

$$x^2 + (\sqrt{2}-1)x - \frac{1}{\sqrt{2}} = 0.$$

Solving it, we obtain

$$x = \frac{1}{2}\left(\sqrt{3} - \sqrt{2} + 1\right) \approx 0.659.$$

$$\therefore\ R_{AB} = \frac{1}{2}\left(\sqrt{3} - \sqrt{2} + 1\right) R \approx 0.659R.$$

Solution 3.29. *Method 1. Kirchhoff's equation method.*

By symmetry we may assume the currents in the branches as shown in Fig. 1, and write down the following equations.

Node equations:

$$I_2 + I_4 - I_1 = 0,$$
$$I_3 - I_4 - I_5 = 0.$$

Loop equations:

$$I_2 - I_3 - I_4 = 0,$$
$$I_3 + I_5 - I_4 - I_1 = 0.$$

Solving the above four equations, we obtain

$$I_2 = \frac{5}{6}I_1, \qquad I_3 = \frac{2}{3}I_1, \qquad I_4 = \frac{1}{6}I_1, \qquad I_5 = \frac{1}{2}I_1.$$

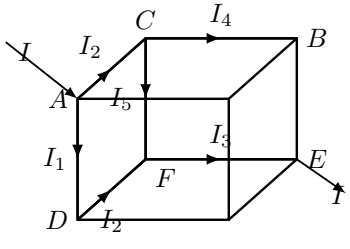


Fig. 3.29(a)

Also,

$$U_{AE} = I_1 + I_2 = \frac{11}{6} I_1, \qquad I = I_1 + I_2 + I_3 = \frac{15}{6} I_1,$$

hence

$$R_{AE} = \frac{U_{AE}}{I} = \frac{11}{15}\ \Omega.$$

Method 2. Method of superposition of currents.

Imagine a current I flowing in at A and out at D; the branch currents obtained are as shown in Fig. 2(1). Also imagine a current flowing in at D and out at E; the branch currents obtained are as shown in Fig. 2(2). Superposing the above two cases, we obtain the case in which the current flows in at A and out at E. Fig. 2(3) shows the currents in two of the branches after superposition (the remaining branches are not drawn). From this we can find

$$R_{AE} : U_{AE} = \frac{2}{5} I + \frac{5}{15} I = \frac{11}{15} I.$$

Therefore

$$R_{AE} = U_{AE}/I = \frac{11}{15}\ \Omega.$$

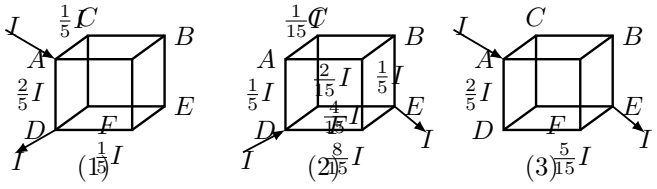


Fig. 3.29(b)

Method 3. Star-delta transformation method.

The actual transformation process is shown in Fig. 3. Finally we obtain

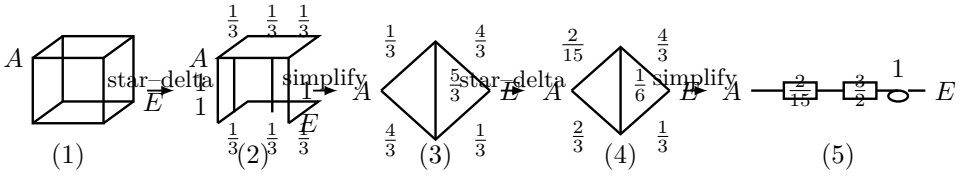


Fig. 3.29(c)

$$R_{AE} = \frac{2}{15} + \frac{\dfrac{3}{2} \cdot 1}{\dfrac{3}{2} + 1} = \frac{11}{15}\ \Omega.$$

Solution 3.30. According to the statement of the problem, the circuit shown in the figure may be drawn. The current flowing through the galvanometer may be found by Thévenin's theorem (note that quantities of order δ^2 and higher are neglected in the calculation). Imagine that what is connected across the galvanometer is an equivalent source, whose electromotive force is

$$\varepsilon_{\text{eff}} = (U_A - U_B)_{A,\,B \text{ open}}.$$

$$
\begin{aligned}
&= U\frac{R_2}{R_1+R_2} - U\frac{R_2(1-\delta)}{R_1+R_2-R_2\delta} \\
&= U\frac{R_2}{R_1+R_2} - \frac{UR_2}{R_1+R_2}\frac{(1-\delta)}{1-\delta\dfrac{R_2}{R_1+R_2}} \\
&\approx U\frac{R_2}{R_1+R_2} - U\frac{R_2}{R_1+R_2}(1-\delta)\left(1+\delta\frac{R_2}{R_1+R_2}\right) \\
&\approx \delta U\frac{R_1R_2}{(R_1+R_2)^2}.
\end{aligned}
$$

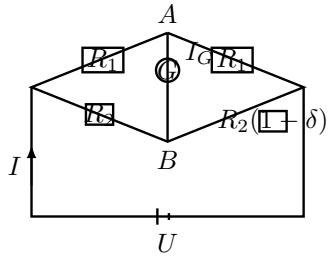


Fig. 3.30

Since ε_{eff} is already a small quantity, the internal resistance of the equivalent source may be replaced by its value when the bridge is balanced:

$$
r_{\text{eff}} = 2\frac{R_1R_2}{R_1+R_2},
$$

$$
\therefore\ I_G = \frac{\varepsilon_{\text{eff}}}{r_{\text{eff}}} = \frac{\delta U\dfrac{R_1R_2}{(R_1+R_2)^2}}{2\dfrac{R_1R_2}{R_1+R_2}} = \frac{\delta U}{2(R_1+R_2)}.
$$

Since I_G is a first-order small quantity, the current flowing through the source may be replaced by its value when the bridge is balanced:

$$
I = \frac{U}{\dfrac{1}{2}(R_1+R_2)} = \frac{2U}{R_1+R_2},
$$

$$
\therefore\ I_G/I = \frac{\delta U}{2(R_1+R_2)}\Big/\frac{2U}{R_1+R_2} = \frac{\delta}{4}.
$$

Solution 3.31. (1) According to Thévenin's theorem, looking to the left from A_n, the network may be replaced by an equivalent source with electromotive force $V_n/2$ and internal resistance R, as shown in Fig. 1. Connecting this equivalent source with the next section, it may again be replaced by an equivalent source with electromotive force $\left(\dfrac{V_{n-1}}{2}+\dfrac{V_n}{4}\right)$ and internal resistance R. This is also the equivalent source seen looking to the left from A_{n-1}. And so on: when A and B are taken as the output terminals, the network may still be replaced by an equivalent source, whose electromotive force V and internal resistance r are respectively

$$
V = \frac{V_1}{2}+\frac{V_2}{2^2}+\frac{V_3}{2^3}+\cdots+\frac{V_n}{2^n}, \qquad r = R.
$$

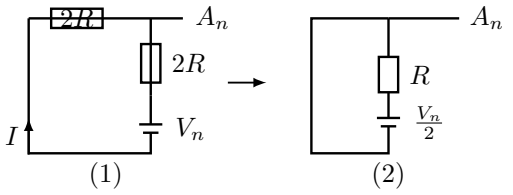


Fig. 3.31(a)

as shown in Fig. 2.

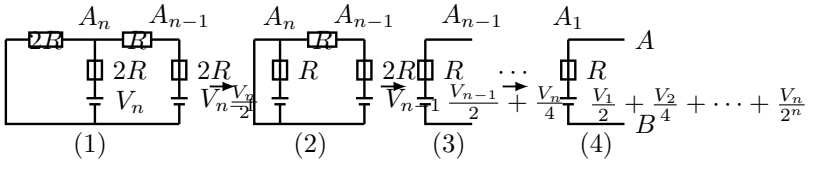


Fig. 3.31(b)

(2) At this time, looking to the left from A_i gives an equivalent source, and looking to the right from A_{i-1} also gives an equivalent source; their electromotive forces are V_{L} and V_{R} respectively, as shown in Fig. 3. Therefore which of the potentials of the two points A_i and A_{i-1} is the higher depends on which of V_{L} and V_{R} is the larger. From the results of part (1) it is known that when $i \le 6$ we must have $V_{\mathrm{L}} > V_{\mathrm{R}}$, because the electromotive force of any battery on the left is higher than that of a battery on the right. Hence the point of highest potential must be among the points $A_6, A_7, \cdots, A_{10}$. If we can find the point A_i with the smallest ordinal number i such that $V_{\mathrm{L}} < V_{\mathrm{R}}$, then A_{i-1} at this time must be the point of highest potential.

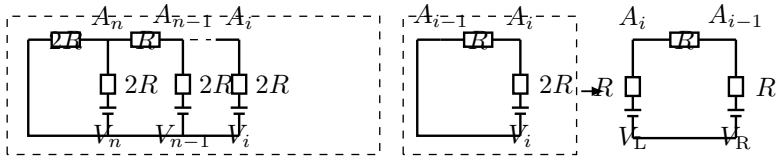


Fig. 3.31(c)

Let us first look at A_8, the middle point of $6 \sim 10$. At this time

$$V_{\mathrm{L}} = \frac{18}{2} + \frac{19}{4} + \frac{20}{8}, \qquad V_{\mathrm{R}} = \frac{17}{2} + \frac{16}{4} + \frac{15}{8} + \frac{14}{16} + \frac{13}{32} + \frac{12}{64} + \frac{11}{128},$$

then

$$V_{\mathrm{L}} - V_{\mathrm{R}} = \frac{1}{2} + \frac{3}{4} + \frac{5}{8} - \left(\frac{14}{16} + \frac{13}{32} + \frac{12}{64} + \frac{11}{128}\right) = \frac{41}{128} > 0.$$

Now look at A_9. At this time

$$V_{\mathrm{L}} = \frac{19}{2} + \frac{20}{4}, \qquad V_{\mathrm{R}} = \frac{18}{2} + \frac{17}{4} + \frac{16}{8} + \frac{15}{16} + \frac{14}{32} + \frac{13}{64} + \frac{12}{128} + \frac{11}{256},$$

$$V_{\mathrm{L}} - V_{\mathrm{R}} = \frac{1}{2} + \frac{3}{4} - \left(\frac{16}{8} + \frac{15}{16} + \frac{14}{32} + \frac{13}{64} + \frac{12}{128} + \frac{11}{256}\right) = -\frac{631}{256} < 0.$$

It can be seen that the potential of A_8 is the highest. Its potential U_8 may be found from the corresponding equivalent circuit (the right-hand part of Fig. 3):

$$\begin{aligned} U_8 &= \frac{V_\mathrm{L} - V_\mathrm{R}}{3R} \cdot R + V_\mathrm{R} = \frac{1}{3}V_\mathrm{L} + \frac{2}{3}V_\mathrm{R} \\ &= \frac{1}{3}\left(\frac{19}{2} + \frac{20}{4}\right) + \frac{2}{3}\left(\frac{18}{2} + \frac{17}{4} + \frac{16}{8} + \cdots + \frac{12}{128} + \frac{11}{256}\right) = \frac{6199}{384} = 16.14 \text{ (V)}. \end{aligned}$$

Solution 3.32. (1) It can be determined. This can be judged from whether the direction of the current in the ring is the same as or opposite to the direction of the velocity of the ring: if they are the same, the ring carries positive charge; otherwise it carries negative charge. And the direction of the current can be determined from its magnetic effect.

(2) During the deceleration of the ring, the charge carriers will be acted on by an inertial force, and this inertial force may be regarded as a force originating from a non-electrostatic source. Let the linear velocity of the ring be v, and let the field strength of the field corresponding to this force originating from a non-electrostatic source be E. Let the mass and the charge of a charge carrier be m and e respectively; then we have

$$-m\frac{\mathrm{d}v}{\mathrm{d}t} = eE,$$

$$E = -\frac{m}{e}\frac{\mathrm{d}v}{\mathrm{d}t}.$$

The equivalent electromotive force is

$$\mathcal{E} = E \cdot 2\pi b = -\frac{m}{e}\frac{\mathrm{d}v}{\mathrm{d}t} \cdot 2\pi b.$$

By Ohm's law,

$$\mathcal{E} = IR,$$

that is,

$$-\frac{m}{e}\frac{\mathrm{d}v}{\mathrm{d}t} \cdot 2\pi b = IR.$$

Integrating both sides with respect to t,

$$-\frac{m}{e} \cdot 2\pi b \int_{v_0}^{0} \mathrm{d}v = R\int_0^\infty I\,\mathrm{d}t = Rq,$$

that is,

$$\frac{m}{e} \cdot 2\pi b v_0 = Rq,$$

$$\therefore\ \frac{e}{m} = \frac{2\pi b v_0}{Rq}.$$

Solution 3.33. Since the bubble does not conduct electricity, the current will accumulate charge on the surface of the bubble, as shown in the figure. Let the surface charge density be

$$\sigma = -\sigma_0 \cos\theta$$

(only in this way can $E_r = 0$ hold on the side of the surface facing the wire). Then the electric field inside the bubble is

$$\vec{E}_\mathrm{in} = \left(E_0 + \frac{\sigma_0}{3\varepsilon_0}\right)\hat{\imath}$$

(see Solution 3.24). Here E_0 is the electric field in the wire when the bubble is absent, that is, $E_0 = \frac{1}{g} j_0$, where g is the conductivity. Outside the bubble the field produced by the surface charges is the field of a dipole. The dipole moment is

$$\vec{p} = \frac{4}{3}\pi a^3 \sigma_0 (-\hat{\imath})$$

At the outer boundary of the bubble (a, θ) (which lies inside the wire) we must have $E_r = 0$, that is,

$$E_0 \cos\theta + \frac{2p\cos\theta}{4\pi\varepsilon_0 a^3} = 0.$$

Substituting the expression for p, we obtain

$$E_0 \cos\theta - \frac{2\cos\theta}{4\pi\varepsilon_0 a^3} \cdot \frac{4\pi}{3} a^3 \sigma_0 = 0,$$

that is,

$$E_0 \cos\theta - \frac{2}{3\varepsilon_0}\sigma_0 \cos\theta = 0.$$

Hence

$$\sigma_0 = \frac{3}{2}\varepsilon_0 E_0, \qquad \vec{p} = -2\pi\varepsilon_0 a^3 E_0 = -\frac{2\pi\varepsilon_0 a^3}{g}\vec{j}_0.$$

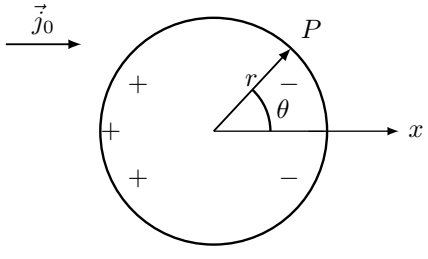


Fig. 3.33

(1) The electric field inside the bubble is

$$\vec{E}_{\text{in}} = \left(E_0 + \frac{\sigma_0}{3\varepsilon_0}\right)\hat{\imath} = \frac{3}{2}\frac{1}{g}\vec{j}_0.$$

(2) The field at the point P is the superposition of E_0 and the field of the dipole p:

$$E_r = E_0 \cos\theta + \frac{2p\cos\theta}{4\pi\varepsilon_0 r^3} = E_0 \cos\theta - \frac{a^3}{r^3}\cos\theta\, E_0 \tag{3.6}$$

$$E_\theta = -E_0 \sin\theta + \frac{p\sin\theta}{4\pi\varepsilon_0 r^3} = -E_0 \sin\theta - \frac{a^3}{2r^3}\sin\theta\, E_0 \tag{3.7}$$

Noting that $E_0 = \frac{1}{g} j_0$,

$$j_r = gE_r = j_0\cos\theta - j_0\left(\frac{a^3}{r^3}\right)\cos\theta = j_0\left(1 - \frac{a^3}{r^3}\right)\cos\theta \tag{3.8}$$

$$j_\theta = gE_\theta = -j_0\left(1 + \frac{a^3}{2r^3}\right)\sin\theta \tag{3.9}$$

(3) When $\theta = 90°$, $j_r = 0$ and $j_\theta = -j_0\left(1 + \dfrac{a^3}{2r^3}\right)$, so

$$I = \int_a^b -j_\theta \cdot 2\pi r \mathrm{d}r = \int_a^b j_0\left(1 + \frac{a^3}{2r^3}\right) 2\pi r \mathrm{d}r$$

$$= j_0\pi\left(b^2 - a^2\right) + \int_a^b \pi j_0 a^3 \frac{\mathrm{d}r}{r^2}$$

$$= j_0\pi b^2 - j_0\pi a^2 \frac{a}{b}.$$

Solution 3.34. (1) The current (strength) may be regarded as steady, and the current density j may be regarded as depending only on r; hence

$$4\pi j r^2 = I \text{ (a constant)} = j_0 \cdot 4\pi R^2 = \sigma_0 E_0 \cdot 4\pi R^2.$$

$$\therefore\ I = \sigma_0 E_0 \cdot 4\pi R^2 = 3 \times 10^{-14} \times 100 \times 4\pi \times \left(6.4 \times 10^6\right)^2 = 1.54 \times 10^3 \text{ (A)}$$

$$j = \frac{R^2}{r^2} j_0 = \frac{\sigma_0 E_0 R^2}{r^2} = \sigma E.$$

$$\therefore\ E(r) = \frac{j}{\sigma} = \frac{\sigma_0 E_0 R^2}{\sigma r^2} = \frac{\sigma_0 E_0 R^2}{r^2\left[\sigma_0 + a(r-R)^2\right]},$$

the direction of $\vec{E}$ being towards the centre of the Earth.

(2) Take the thin shell of radii $r \sim r + \Delta r$. By Gauss's law,

$$E(r+\Delta r) \cdot 4\pi(r+\Delta r)^2 - E(r) \cdot 4\pi r^2 = -4\pi r^2 \Delta r\, \rho/\varepsilon_0,$$

$$\therefore\ \rho = -\frac{\varepsilon_0}{4\pi r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left[E(r) \cdot 4\pi r^2\right] = \frac{2\varepsilon_0\sigma_0 E_0 R^2 a(r-R)}{r^2\left[\sigma_0 + a(r-R)^2\right]^2}.$$

From $E_0 = -\Sigma_0/\varepsilon_0$ we obtain

$$\Sigma_0 = -\varepsilon_0 E_0 = -\frac{100}{4\pi \times 9 \times 10^9} = -8.85 \times 10^{-10}\ \left(\mathrm{C/m^2}\right).$$

(3) The potential at the Earth's surface is

$$\varphi = \int_R^\infty -E(r)\mathrm{d}r = -\sigma_0 E_0 R^2 \int_R^\infty \frac{\mathrm{d}r}{r^2\left[\sigma_0 + a(r-R)^2\right]}.$$

By the integral formula given in the problem,

$$\int_R^\infty \frac{\mathrm{d}r}{r^2\left[\sigma_0 + a(r-R)^2\right]} = \int_R^\infty \frac{\mathrm{d}r}{ar^2\left[\frac{\sigma_0}{a} + R^2 - 2Rr + r^2\right]}$$

$$
\begin{aligned}
&= \frac{1}{a}\left[\frac{-2R}{2\left(\frac{\sigma_0}{a}+R^2\right)^2}\ln\frac{\frac{\sigma_0}{a}+R^2-2Rr+r^2}{r^2}\bigg|_R^\infty - \frac{1}{\left(\frac{\sigma_0}{a}+R^2\right)r}\bigg|_R^\infty\right.\\
&\qquad \left.+\left(\frac{4R^2}{2\left(\frac{\sigma_0}{a}+R^2\right)^2}-\frac{1}{\frac{\sigma_0}{a}+R^2}\right)\frac{2}{2\sqrt{\frac{\sigma_0}{a}}}\tan^{-1}\frac{2(r-R)}{2\sqrt{\frac{\sigma_0}{a}}}\bigg|_R^\infty\right]\\
&= \frac{1}{a}\left[\frac{R}{\left(\frac{\sigma_0}{a}+R^2\right)^2}\ln\frac{\sigma_0}{aR^2}+\frac{1}{\left(\frac{\sigma_0}{a}+R^2\right)R}+\frac{R^2-\frac{\sigma_0}{a}}{\left(\frac{\sigma_0}{a}+R^2\right)^2}\cdot\frac{\frac{\pi}{2}}{\sqrt{\frac{\sigma_0}{a}}}\right]\\
&= \frac{1}{a\left(\frac{\sigma_0}{a}+R^2\right)^2}\left[R\ln\frac{\sigma_0}{aR^2}+\frac{\frac{\sigma_0}{a}+R^2}{R}+\left(R^2-\frac{\sigma_0}{a}\right)\frac{\pi}{2\sqrt{\frac{\sigma_0}{a}}}\right] \qquad (3.10)
\end{aligned}
$$

Substituting this into the expression for φ gives

$$
\varphi = -\frac{\sigma_0 E_0 R^2}{a\left(\frac{\sigma_0}{a}+R^2\right)^2}\left[R\ln\frac{\sigma_0}{aR^2}+\frac{\frac{\sigma_0}{a}+R^2}{R}+\left(R^2-\frac{\sigma_0}{a}\right)\frac{\pi}{2\sqrt{\frac{\sigma_0}{a}}}\right].
$$

Inserting the data, and noting that $\dfrac{\sigma_0}{a} \ll R^2$, we have

$$
\varphi \approx -3.83\times 10^5 \text{ (V)}.
$$

(4) Let the total charge on the ground be q. The decrease of q is caused by the current in the ground, so

$$
-\frac{\mathrm{d}q}{\mathrm{d}t} = \sigma_0 E_0 \cdot 4\pi R^2 = \sigma_0 \frac{q}{4\pi\varepsilon_0 R^2}\cdot 4\pi R^2 = \frac{\sigma_0}{\varepsilon_0}q,
$$

that is,

$$
\frac{\mathrm{d}q}{q} = -\frac{\sigma_0}{\varepsilon_0}\mathrm{d}t.
$$

Integrating, we obtain

$$
\begin{aligned}
q &= q_0 \mathrm{e}^{-\frac{\sigma_0}{\varepsilon_0}t} = q_0 \mathrm{e}^{-t/\tau},\\
\tau &= \frac{\varepsilon_0}{\sigma_0} \approx 294 \text{ (s)} \approx 300 \text{ (s)}.
\end{aligned}
$$

This is roughly the time required for the charge on the Earth to be neutralised by the current. The mechanism which keeps the charge on the Earth from being neutralised and roughly constant is mainly lightning and thunderbolts.

Solution 3.35. (1) Let the distance between the two wires be d, let the current in the wires be I, and let the linear charge density carried by them be λ. Then, by Gauss's law, the electric field produced by one wire at the position of the other wire is $E = \dfrac{\lambda}{2\pi\varepsilon_0 d}$, and the electrostatic (attractive) force acting on unit length of the wire is

$$
F_e = \frac{\lambda^2}{2\pi\varepsilon_0 d}.
$$

Similarly, by Ampère's circuital law, the magnetic field produced by the current in one wire at the position of the other wire is $B = \dfrac{\mu_0 I}{2\pi d}$, and the magnetic (repulsive) force acting on unit length of the wire is

$$
F_m = \frac{\mu_0 I^2}{2\pi d}.
$$

For no force to act, we must have $F_e = F_m$, that is,

$$\frac{\mu_0 I^2}{2\pi d} = \frac{\lambda^2}{2\pi\varepsilon_0 d}.$$

Hence

$$I = \sqrt{\frac{1}{\varepsilon_0\mu_0}}\,\lambda = c\lambda \qquad \left[\because\ \sqrt{\frac{1}{\varepsilon_0\mu_0}} = c \text{ (the speed of light)}\right].$$

But

$$\lambda = C^* V,$$

where C^* is the capacitance per unit length between the wires and V is the voltage between the two wires. Hence the required power is

$$P = IV = cC^*V^2 = 3\times 10^8 \times 10^{-11} \times \left(33\times 10^3\right)^2$$

$$= 3.27\times 10^6 \text{ (W)}.$$

(2) The electric energy w_e and the magnetic energy w_m stored per unit length are

$$w_e = \frac{1}{2}C^*V^2 = \frac{1}{2}\times 10^{-11}\times\left(33\times 10^3\right)^2 = 5.45\times 10^{-3} \text{ (J/m)},$$

$$w_m = \frac{1}{2}L^*I^2,$$

where the self-inductance per unit length L^* can be obtained from C^*. Let the radius of the two wires be a ($a \ll d$); then the voltage between the two wires is

$$V = \int_a^d E\mathrm{d}r = 2\int_a^d \frac{\lambda}{2\pi\varepsilon_0 r}\mathrm{d}r = \frac{\lambda}{\pi\varepsilon_0}\ln\frac{d}{a},$$

$$\therefore\ C^* = \frac{\lambda}{V} = \frac{\pi\varepsilon_0}{\ln\dfrac{d}{a}}.$$

The magnetic flux per unit length is

$$\Phi = \int_a^d B\mathrm{d}r = 2\int_a^d \frac{\mu_0 I}{2\pi r}\mathrm{d}r = \frac{\mu_0 I}{\pi}\ln\frac{d}{a},$$

$$L^* = \frac{\Phi}{I} = \frac{\mu_0}{\pi}\ln\frac{d}{a}.$$

It is thus seen that

$$L^*C^* = \varepsilon_0\mu_0.$$

$$\begin{aligned}\therefore\ w_m &= \frac{1}{2}L^*I^2 = \frac{1}{2}\frac{\varepsilon_0\mu_0}{C^*}I^2 = \frac{1}{2}\frac{\varepsilon_0\mu_0}{C^*}\cdot c^2\lambda^2\\ &= \frac{1}{2}\frac{1}{C^*}C^{*2}V^2 = \frac{1}{2}C^*V^2\\ &= w_e = 5.45\times 10^{-3} \text{ (J/m)}.\end{aligned}$$

Solution 3.36. (1) The magnetic field produced at the point O by a solenoid of length $\mathrm{d}z$ can be obtained by superposing the fields of circular coils. By the Biot–Savart law (as shown in the figure),

$$\mathrm{d}B = \frac{\mu_0}{4\pi}\frac{n\mathrm{d}z\, I\cdot 2\pi R}{r^2}\cdot\frac{R}{r} = \frac{\mu_0 nIR^2}{2r^3}\mathrm{d}z.$$

But $\mathrm{d}z\sin\theta = r\mathrm{d}\theta$ and $r\sin\theta = R$, so the above expression becomes

$$\mathrm{d}B = \frac{\mu_0 nI}{2}\frac{r^2\sin^2\theta\cdot r\mathrm{d}\theta}{r^2\cdot r\sin\theta} = \frac{\mu_0 nI}{2}\sin\theta\,\mathrm{d}\theta.$$

$$\therefore\ B = \int \mathrm{d}B = \frac{\mu_0 nI}{2}\int_{\theta_0}^{\pi/2}\sin\theta\,\mathrm{d}\theta = \frac{\mu_0 nI}{2}\frac{L}{\sqrt{R^2+L^2}},$$

(where $\theta_0 = \arcsin\dfrac{R}{\sqrt{R^2+L^2}}$).

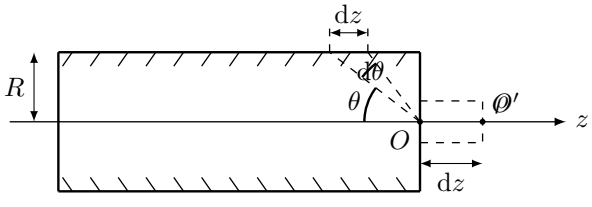


Fig. 3.36

(2) Near the point O construct a cylindrical surface of height $\mathrm{d}z$ and radius ρ ($\rho \ll R$), as shown in the figure. According to the problem, now $L\to\infty$, and the magnetic field at O is $B_O = \dfrac{\mu_0 nI}{2}$. At the point O' on the axis at a distance $\mathrm{d}z$ from O, from the above result,

$$B_{O'} = \frac{\mu_0 nI}{2}\int_0^{\theta'}\sin\theta\,\mathrm{d}\theta = \frac{\mu_0 nI}{2}(1-\cos\theta')$$

$$\approx \frac{\mu_0 nI}{2}\left(1-\frac{\mathrm{d}z}{R}\right).$$

Taking this cylindrical surface as a Gaussian surface, from the Gauss law for the magnetic field, $\oint \vec{B}\cdot\mathrm{d}\vec{s} = 0$, we obtain

$$B_\rho\cdot 2\pi\rho\mathrm{d}z + B_{O'}\pi\rho^2 = B_O\pi\rho^2,$$

that is,

$$B_\rho 2\pi\rho\mathrm{d}z = (B_O - B_{O'})\,\pi\rho^2 = \frac{\mu_0 nI}{2}\frac{\mathrm{d}z}{R}\pi\rho^2,$$

$$\therefore\ B_\rho = \frac{\mu_0 nI}{4}\frac{\rho}{R}.$$

Solution 3.37. When a dielectric moves with speed v in a magnetic field, the charges in the dielectric are subject to the Lorentz force. This action is equivalent to an electric field $\vec{E}^* = \vec{v}\times\vec{B}$. Under the action of this field the dielectric becomes polarised, and the polarisation charges produced by the surface polarisation also excite an electric field; this field in turn affects the polarisation of the dielectric, and finally the dielectric is polarised under the joint action of these

two fields. Let the polarisation intensity of the dielectric be P; then the surface density of the polarisation charge is $\sigma' = P$, and the magnitude of the electric field produced by the polarisation charges is $E' = \dfrac{\sigma'}{\varepsilon_0} = \dfrac{P}{\varepsilon_0}$, but its direction is opposite to that of $\vec{P}$. Substituting this into the relation between the polarisation intensity and the electric field gives

$$P = (\varepsilon_r - 1)\varepsilon_0 \left(vB - \frac{P}{\varepsilon_0} \right).$$

Solving this, we obtain

$$P = \left(1 - \frac{1}{\varepsilon_r} \right) \varepsilon_0 vB. \tag{1}$$

The potential difference between the two plates is caused by the electric field excited by the charges, that is, $U = E'd = \dfrac{P}{\varepsilon_0} d$; hence

$$U = \left(1 - \frac{1}{\varepsilon_r} \right) vBd. \tag{2}$$

In the situation shown in the figure, the upper plate is at the higher potential and the lower plate at the lower potential.

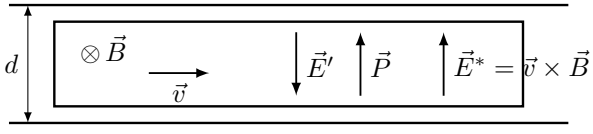


Fig. 3.37

Solution 3.38. According to the discussion of Problem 3.37, the dielectric is polarised under the joint action of the Lorentz force and the field excited by the polarisation charges. Let the electric field in the dielectric (that is, the field excited by the polarisation charges) be E, and let the equivalent electric field corresponding to the Lorentz force be $vB = \omega r B$; then the polarisation intensity (note that $\varepsilon_r = 1 + \chi$) is

$$P = (\varepsilon_r - 1)\varepsilon_0 (E + \omega r B) = \chi \varepsilon_0 (E + \omega r B). \tag{1}$$

Here both P and E are taken as positive along the radial direction outwards, so that they agree with the direction of the equivalent field of the Lorentz force (see the figure). By relating them to the polarisation charges, E and P can be connected. Take a cylindrical surface of radius r as a Gaussian surface; by Gauss's law,

$$E \cdot 2\pi r = \frac{q'}{\varepsilon_0}, \tag{2}$$

where q' is the polarisation charge inside the cylindrical surface. By the relation between the polarisation intensity and the polarisation charge, the polarisation charge inside the cylindrical surface is determined by

$$q' = -\oint \vec{P} \cdot \mathrm{d}\vec{s} = -P \cdot 2\pi r. \tag{3}$$

Substituting into (2) gives $E = -\dfrac{P}{\varepsilon_0}$; substituting this into (1) gives

$$P = \chi \varepsilon_0 \left(-\frac{P}{\varepsilon_0} + \omega r B \right),$$

from which

$$P(r) = \frac{\chi}{1+\chi}\varepsilon_0 \omega r B. \tag{4}$$

Substituting the values of P at the radii r and $r + \Delta r$ respectively into (3), we obtain the polarisation charge density

$$\rho' = \frac{q'(r+\Delta r) - q'(r)}{2\pi r \Delta r} = -2\,\frac{\chi}{1+\chi}\varepsilon_0 \omega B. \tag{5}$$

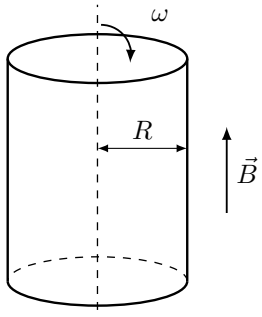


Fig. 3.38

On the surface of the cylinder, however, there is a surface distribution of polarisation charge

$$\sigma' = P(R) = \frac{\chi}{1+\chi}\varepsilon_0 \omega R B. \tag{6}$$

Solution 3.39. (1) Let the field strength at a distance r from the axis be E; then $E = E_0/r$, and

$$U = \int_a^b E \mathrm{d}r = E_0 \int_a^b \frac{\mathrm{d}r}{r} = E_0 \ln \frac{b}{a},$$

$$\therefore\ E = \frac{U}{r \ln \dfrac{b}{a}}.$$

By Ohm's law,

$$j = \frac{1}{\rho}E = \frac{U}{\rho r \ln \dfrac{b}{a}},$$

$$I = 2\pi r L j = \frac{2\pi L U}{\rho \ln \dfrac{b}{a}}.$$

(2) When a magnetic field is present, let the drift velocity of the charge carriers be u; then

$$\vec{j} = \frac{1}{\rho}\left(\vec{E} + \vec{u} \times \vec{B}\right) = \frac{1}{\rho}\vec{E} + \frac{1}{\rho n e}\vec{j} \times \vec{B}.$$

Let $\vec{j} = j_r \hat{r} + j_\theta \hat{\theta}$, $\vec{E} = E\hat{r}$ and $\vec{B} = B\hat{z}$; then

$$\begin{aligned} j_r \hat{r} + j_\theta \hat{\theta} &= \frac{1}{\rho}E\hat{r} + \frac{1}{\rho n e}\left(j_r \hat{r} + j_\theta \hat{\theta}\right) \times B\hat{z} \\ &= \left(\frac{1}{\rho}E + \frac{1}{\rho n e} j_\theta B\right)\hat{r} - \frac{1}{\rho n e} j_r B \hat{\theta}. \end{aligned}$$

Hence

$$j_r = \frac{1}{\rho}E + \frac{1}{\rho ne}j_\theta B,$$

$$j_\theta = -\frac{1}{\rho ne}j_r B.$$

Substituting the expression for j_θ into the expression for j_r, we obtain

$$j_r = \frac{1}{\rho}E - \left(\frac{B}{\rho ne}\right)^2 j_r.$$

Solving this, we obtain

$$j_r = \frac{E}{\rho\left[1 + \left(\frac{B}{\rho ne}\right)^2\right]}.$$

$$\therefore\ I' = 2\pi r L j_r = \frac{2\pi r L}{\rho\left[1 + \left(\frac{B}{\rho ne}\right)^2\right]} \cdot \frac{U}{r\ln\frac{b}{a}}$$

$$= \frac{2\pi L U}{\rho \ln\frac{b}{a}} \cdot \frac{1}{1 + \left(\frac{B}{\rho ne}\right)^2} = \frac{I}{1 + \left(\frac{B}{\rho ne}\right)^2}.$$

Solution 3.40. (1) By the Biot–Savart law, the magnetic field produced by the current-carrying coil at its centre is

$$B_0 = \frac{\mu_0}{4\pi}\frac{I \cdot 2\pi b}{b^2} = \frac{\mu_0 I}{2b}. \tag{1}$$

Since the sphere is very small, the dielectric sphere will be uniformly magnetised under the action of this field. Let the magnetisation intensity of the dielectric sphere be M; then the magnetisation surface current density is

$$i_M = M\sin\theta.$$

The meaning of θ is shown in Fig. 3.40(a). By the Biot–Savart law, the magnetic field produced by the magnetisation currents at the centre of the sphere can be obtained by integration (Fig. 3.40(a)):

$$B' = \frac{\mu_0}{4\pi}\int_0^\pi \frac{i_M \cdot a\mathrm{d}\theta \cdot 2\pi a\sin\theta}{a^2}\sin\theta = \frac{\mu_0 M}{2}\int_0^\pi \sin^3\theta\,\mathrm{d}\theta$$

$$= \frac{2}{3}\mu_0 M.$$

The resultant field is

$$B = B_0 + B' = B_0 + \frac{2}{3}\mu_0 M.$$

This is also the field throughout the interior of the sphere. From the equation of state,

$$M = (\mu_r - 1)H = (\mu_r - 1)\frac{B}{\mu_r\mu_0},$$

and substituting this into the above expression gives

$$B = B_0 + \frac{2}{3}\mu_0 \cdot \frac{\mu_r - 1}{\mu_r\mu_0}B = B_0 + \frac{2(\mu_r - 1)}{3\mu_r}B.$$

Solving, we obtain

$$B = \frac{3\mu_r}{\mu_r + 2} B_0. \tag{2}$$

Thus

$$M = \frac{\mu_r - 1}{\mu_r \mu_0} B = \frac{3(\mu_r - 1)}{\mu_0(\mu_r + 2)} B_0 = \frac{3(\mu_r - 1)}{2(\mu_r + 2)} \frac{I}{b}, \tag{3}$$

where $B_0 = \dfrac{\mu_0 I}{2b}$ from (1) has been substituted.

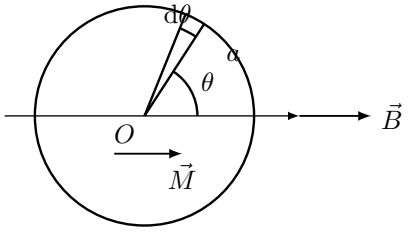


Fig. 3.40(a)

(2) The presence of the dielectric sphere increases the magnetic flux through the coil, and hence increases the self-inductance. This increase of the flux comes from the additional magnetic field produced by the magnetisation currents. Since the flux produced by this additional field has both positive and negative parts in the region $r < b$, the net increase of the flux equals the flux in the region $r > b$ (Fig. 3.40(b)). Outside the dielectric, the field of the magnetisation currents is equivalent to the field of a magnetic dipole. Hence the flux increment is

$$\Delta\Phi = \int_b^\infty \frac{\mu_0}{4\pi} \frac{m}{r^3} \cdot 2\pi r \mathrm{d}r = \frac{\mu_0}{4\pi} \cdot 2\pi m \int_b^\infty \frac{\mathrm{d}r}{r^2} = \frac{\mu_0 m}{2b},$$

where m is the magnetic moment of the magnetised sphere. Substituting (3) of part (1), we have

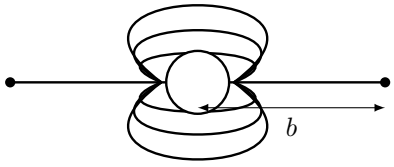


Fig. 3.40(b)

$$m = \frac{4}{3}\pi a^3 M = \frac{4}{3}\pi a^3 \cdot \frac{3(\mu_r - 1)I}{2(\mu_r + 2)b} = \frac{2\pi a^3(\mu_r - 1)I}{(\mu_r + 2)b}.$$

Substituting into the expression for $\Delta\Phi$,

$$\Delta\Phi = \frac{\mu_0}{2b} \cdot \frac{2\pi a^3(\mu_r - 1)I}{(\mu_r + 2)b} = \frac{\pi a^3 \mu_0(\mu_r - 1)}{b^2(\mu_r + 2)} I.$$

Finally we obtain the increment of the self-inductance

$$\Delta L = \frac{\Delta\Phi}{I} = \frac{\pi a^3 \mu_0(\mu_r - 1)}{b^2(\mu_r + 2)}.$$

Solution 3.41. (1) The surface current density produced on the surface of the shell by the motion of the charges is

$$i = \sigma\omega a = \sigma k t a.$$

The magnetic field inside the shell is uniform, and its magnetic flux density is

$$\vec{B} = \mu_0 i\,\hat{z} = \mu_0\sigma a k t\,\hat{z},$$

with the direction along the axis (as shown in the figure).

(2) Since the magnetic field inside the shell varies with time, by Faraday's law the electric field E inside the shell satisfies

$$E \cdot 2\pi r = -\frac{\mathrm{d}B}{\mathrm{d}t}\pi r^2.$$

Hence the electric field is

$$\vec{E} = -\frac{r}{2}\frac{\mathrm{d}B}{\mathrm{d}t}\hat{\theta} = -\frac{r}{2}\mu_0\sigma a k\,\hat{\theta}.$$

(3)

$$W_m = \frac{1}{2\mu_0}B^2 \cdot \pi a^2 l = \frac{1}{2}\mu_0\sigma^2 a^4 k^2 t^2 \pi l,$$

$$W_E = \int \frac{1}{2}\varepsilon_0 E^2 \mathrm{d}\tau = \frac{1}{2}\varepsilon_0 \int_0^a \frac{r^2}{4}\mu_0^2\sigma^2 a^2 k^2 \cdot l \cdot 2\pi r \mathrm{d}r$$

$$= \frac{\pi\varepsilon_0}{16}\mu_0^2\sigma^2 a^6 k^2 l.$$

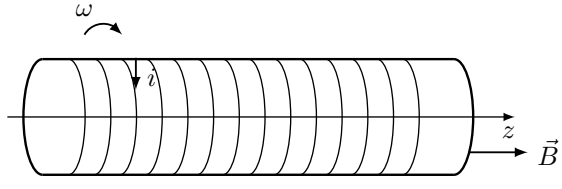


Fig. 3.41

(4) Inside the shell take a cylindrical surface of radius r ($r < a$); the Poynting vector on this surface is

$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B} = \frac{1}{\mu_0}\left(-\frac{r}{2}\mu_0\sigma a k\hat{\theta}\right) \times (\mu_0\sigma a k t\hat{z})$$

$$= \frac{\mu_0 r}{2}\sigma^2 a^2 k^2 t\,(-\hat{r}),$$

its direction being towards the inside of the cylinder. The energy flowing per second into the region inside the cylindrical surface is

$$S \cdot 2\pi r l = \mu_0\pi r^2\sigma^2 a^2 k^2 t l.$$

The electric field energy inside the cylindrical surface is constant, whereas the magnetic field energy depends on time:

$$W_m(r) = \frac{1}{2\mu_0}B^2\pi r^2 l = \frac{\mu_0}{2}\pi r^2 l\sigma^2 a^2 k^2 t^2.$$

$$\frac{\mathrm{d}W_m(r)}{\mathrm{d}t} = \mu_0\pi r^2 l\sigma^2 a^2 k^2 t = S \cdot \pi r^2 l,$$

that is, the rate of increase of the magnetic field energy is supplied by the energy flow represented by the Poynting vector.

Solution 3.42. In this case the long rod will be uniformly magnetised along the x direction, because for a rod magnetised uniformly in this way the magnetisation currents on it produce a uniform magnetic field inside the rod; this field, superposed on the uniform external field, is still a uniform magnetic field inside the rod, which is consistent with the uniform magnetisation of the rod.

Let the intensity of the uniform magnetisation of the rod along the x direction be M; then on the surface of the rod there will be a magnetisation surface current along the axis (z) direction, and the surface current density is

$$i_M = M \sin\theta,$$

as shown in Fig. 3.42(a)(1). This surface current may be imagined as formed by two opposite volume currents along the direction of the rod, displaced slightly from each other in the direction perpendicular to M, as shown in Fig. 3.42(a)(2). Let the volume current density be j and the displacement be l ($l \ll R$); then

$$jl = M.$$

According to Ampère's circuital law, the magnetic flux density at any point P inside the rod is the superposition of the magnetic fields of the two mutually displaced opposite volume currents. Let the vectors from the axes O_+ (O_-) of the volume currents along the positive (negative) z direction to the point P be $\vec{r}_+$ ($\vec{r}_-$); then the field at P can readily be found from Ampère's circuital law as

$$\vec{B}'_P = \frac{\mu_0}{2}\vec{j} \times (\vec{r}_+ - \vec{r}_-) = -\frac{\mu_0}{2}\vec{j} \times \vec{l}.$$

Here $\vec{j}$ is along the positive z direction and $\vec{l}$ points from O_- to O_+; hence $\vec{j} \times (-\vec{l}) = \vec{M}$, and therefore

$$\therefore\ \vec{B}'_{\text{in}} = \frac{\mu_0}{2}\vec{M},$$

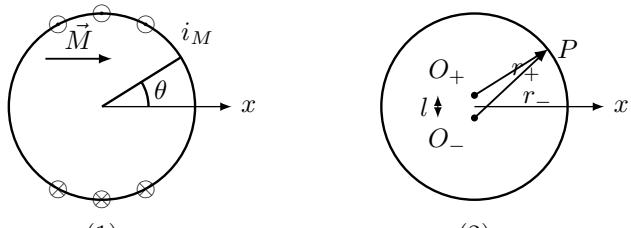
which is a uniform field. The resultant magnetic field inside the rod is $\vec{B}_1 = \vec{B}_0 + \vec{B}'_{\text{in}}$, and the equation of state is written as

$$M = (\mu_r - 1)\frac{1}{\mu_r \mu_0}\left(B_0 + B'_{\text{in}}\right) = (\mu_r - 1)\frac{1}{\mu_r \mu_0}\left(B_0 + \frac{\mu_0}{2}M\right).$$

From this we obtain

$$M = \frac{2(\mu_r - 1)}{\mu_0(\mu_r + 1)}B_0,$$

$$B_1 = B_0 + \frac{\mu_0}{2}M = \frac{2\mu_r}{\mu_r + 1}B_0.$$



(1) (2)

Fig. 3.42(a)

Outside the rod, the magnetic field produced by the magnetisation surface current can still be regarded as the superposition of the fields of the two opposite volume currents. For the positive volume current, by Ampère's circuital law,

$$\vec{B}'_{P+} = \frac{\mu_0}{2}\frac{R^2}{r_+}\vec{j} \times \hat{r}_+.$$

Similarly, for the opposite volume current,

$$\vec{B}'_{P-} = -\frac{\mu_0}{2}\frac{R^2}{r_-}\vec{j} \times \hat{r}_-.$$

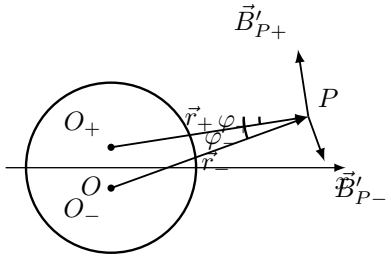


Fig. 3.42(b)

Here the meanings of $\vec{r}_+$ and $\vec{r}_-$ are as before, while $\hat{r}_+$ and $\hat{r}_-$ are the unit vectors of $\vec{r}_+$ and $\vec{r}_-$. Taking the midpoint of O_+ and O_- as the origin O, we have $r_+ \approx r - \dfrac{l}{2}\sin\theta$ and $r_- \approx r + \dfrac{l}{2}\sin\theta$ (Fig. 3.42(b)). Resolving $\vec{B}'_P = \vec{B}'_{P+} + \vec{B}'_{P-}$ into r and θ components, it is readily seen from the figure that (neglecting second-order small quantities)

$$\begin{aligned} B'_r &= \frac{\mu_0}{2}\frac{R^2}{r} j\,(\sin\varphi_+ + \sin\varphi_-) = \frac{\mu_0}{2}\frac{R^2}{r} j\left[\frac{\frac{l}{2}\cos\theta}{r} + \frac{\frac{l}{2}\cos\theta}{r}\right] \\ &= \frac{\mu_0}{2}\frac{R^2}{r^2} jl\cos\theta = \frac{\mu_0}{2}\frac{R^2}{r^2} M\cos\theta, \end{aligned}$$

$$\begin{aligned} B'_\theta &= \frac{\mu_0}{2} R^2 j\left(\frac{\cos\varphi_+}{r_+} - \frac{\cos\varphi_-}{r_-}\right) \approx \frac{\mu_0}{2} R^2 j\left(\frac{1}{r_+} - \frac{1}{r_-}\right) \\ &= \frac{\mu_0}{2} R^2 j\left[\frac{1}{r - \frac{l}{2}\sin\theta} - \frac{1}{r + \frac{l}{2}\sin\theta}\right] = \frac{\mu_0}{2}\frac{R^2}{r^2} j \cdot l\sin\theta \\ &= \frac{\mu_0}{2}\frac{R^2}{r^2} M\sin\theta. \end{aligned}$$

Noting that $M = \dfrac{2(\mu_r - 1)}{\mu_0(\mu_r + 1)} B_0$, the resultant magnetic field outside the rod is

$$B_{2r} = B_0\cos\theta + B'_r = B_0\cos\theta\left[\frac{\mu_r - 1}{\mu_r + 1}\frac{R^2}{r^2} + 1\right],$$

$$B_{2\theta} = -B_0\sin\theta + B'_\theta = B_0\sin\theta\left[\frac{\mu_r - 1}{\mu_r + 1}\frac{R^2}{r^2} - 1\right].$$

Solution 3.43. Set up plane polar coordinates, as shown in the figure; the electron must move in the plane perpendicular to the axis. Let the potential of the inner cylinder be zero; then the potential at a distance r from the axis is $U(r)$. Since the magnetic field does no work on the electron, the energy of the electron is conserved:

$$\frac{1}{2}m\left(v_r^2+v_\theta^2\right)=eU(r) \tag{1}$$

Here $v_r=\dot{r}$ and $v_\theta=r\dot{\theta}$ are respectively the components of the velocity of the electron along the radial direction ($\hat{r}$) and along the transverse direction ($\hat{\theta}$). Because the electron is acted on by the Lorentz force of the magnetic field, the angular momentum of the electron about the axis is not conserved. The angular momentum of the electron about the axis is

$$\begin{aligned}\vec{l}&=\vec{r}\times m\vec{v}=\vec{r}\times m\left(\vec{v}_r+\vec{v}_\theta\right)\\&=m\vec{r}\times\vec{v}_\theta=mr^2\dot{\theta}\,\hat{z}\end{aligned} \tag{2}$$

because $\hat{r}\times\hat{\theta}=\hat{z}$. The torque exerted on the electron by the Lorentz force (the torque of the electric force is obviously zero) is

$$\begin{aligned}\vec{\tau}&=\vec{r}\times\left(-e\vec{v}\times\vec{B}\right)=-\vec{r}\times e\left[\left(\vec{v}_r+\vec{v}_\theta\right)\times\vec{B}\right]\\&=eBrv_r\hat{z}=eBr\dot{r}\,\hat{z}\end{aligned} \tag{3}$$

because only the Lorentz force caused by the r component of the velocity exerts a torque on the electron. From the angular momentum theorem $\vec{\tau}=\mathrm{d}\vec{l}/\mathrm{d}t$ we obtain

$$eBr\dot{r}=\frac{\mathrm{d}}{\mathrm{d}t}\left(mr^2\dot{\theta}\right),$$

that is,

$$eBr\frac{\mathrm{d}r}{\mathrm{d}t}=\frac{\mathrm{d}}{\mathrm{d}t}\left(mr^2\dot{\theta}\right) \tag{4}$$

Integrating both sides,

$$\int_a^r eBr\,\mathrm{d}r=\int_0^{\dot{\theta}}\mathrm{d}\left(mr^2\dot{\theta}\right),$$

that is,

$$\frac{1}{2}eB\left(r^2-a^2\right)=mr^2\dot{\theta}.$$

In order that the electron cannot reach the anode, in the critical case we must have $\dot{r}=0$ when $r=b$. Substituting this into the above expression gives

$$\frac{1}{2}eB\left(b^2-a^2\right)=mb^2\dot{\theta}\,\big|_{r=b} \tag{5}$$

The value of $\dot{\theta}|_{r=b}$ can be obtained from conservation of energy. Substituting $v_r=\dot{r}=0$ into (1),

$$\frac{1}{2}mb^2\left(\dot{\theta}\,\big|_{r=b}\right)^2=eU,$$

we obtain

$$\dot{\theta}\,\big|_{r=b}=\frac{1}{b}\sqrt{\frac{2eU}{m}} \tag{6}$$

Substituting (6) into (5), we obtain the required magnetic flux density

$$B = \frac{2b}{b^2 - a^2}\sqrt{\frac{2mU}{e}}.$$

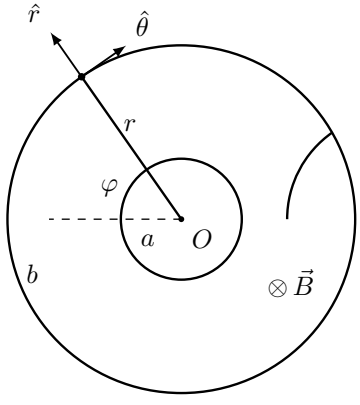


Fig. 3.43

Solution 3.44. (1) Let the speed of the particle be v. From

$$\frac{mv^2}{R} = qvB_z$$

we obtain

$$\omega_c = \frac{v}{R} = \frac{qB_z}{m}.$$

(2) At the position of the particle, since the magnetic field weakens gradually along the radial direction, the lines of magnetic induction must be curved in the shape of a barrel, as shown in the figure. Hence at $z \neq 0$ the magnetic field will have a radial component. At $z > 0$ the radial component is negative, while at $z < 0$ it is positive. If the particle carries a positive charge, then at the position shown in the figure the particle must move out of the plane of the page. Hence when it deviates upwards, under the action of the negative radial component of the magnetic field it experiences a downward restoring force; when it deviates downwards, under the action of the positive radial component of the magnetic field it experiences an upward restoring force. Therefore the particle will vibrate in the vertical direction.

In order to find the radial component of the magnetic field, take the rectangular loop shown in the figure, whose sides are Δr and z. From $\oint \vec{B} \cdot \mathrm{d}\vec{l} = 0$ we obtain

$$[B_z(r + \Delta r) - B_z(r)]\, z - B_r \cdot \Delta r = 0,$$

that is,

$$\frac{\partial B_z}{\partial r}\Delta r \cdot z - B_r \Delta r = 0,$$

$$\therefore\ B_r = \frac{\partial B_z}{\partial r} \cdot z \qquad \left(\text{note that } \frac{\partial B_z}{\partial r} < 0\right).$$

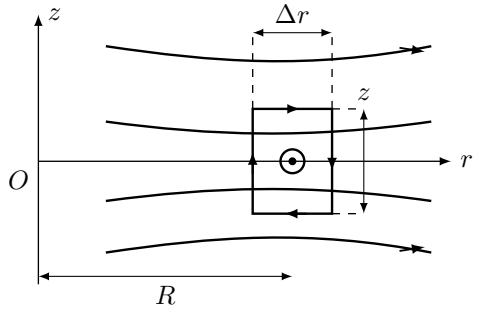


Fig. 3.44

Hence the equation of motion of the particle in the z direction is

$$m\ddot{z} = qvB_r = qv\frac{\partial B_z}{\partial r}z.$$

Since $\dfrac{\partial B_z}{\partial r} < 0$, the particle performs simple harmonic motion.

(3) From the equation of motion, the angular frequency of the simple harmonic motion of the particle is

$$\omega = \sqrt{-\frac{q}{m}v\frac{\partial B_z}{\partial r}}.$$

But $\omega_c = \dfrac{qB_z}{m}$ and $mv = qB_zR$, so that

$$\omega = \sqrt{-\frac{q}{m}\frac{qB_zR}{m}\frac{\partial B_z}{\partial r}} = \sqrt{-\frac{q^2B_z^2}{m^2}\frac{R}{B_z}\frac{\partial B_z}{\partial r}}$$
$$= \omega_c\sqrt{-\frac{R}{B_z}\frac{\partial B_z}{\partial r}},$$
$$\therefore\ \alpha = \sqrt{-\frac{R}{B_z}\frac{\partial B_z}{\partial r}}.$$

Solution 3.45. With the coordinates shown in the figure of the problem, the force on the particle (we may take $v_r \approx 0$) is

$$\vec{F} = q\vec{v}\times\vec{B} = q\begin{vmatrix}\hat{r} & \hat{\theta} & \hat{z}\\ 0 & v_\theta & v_z\\ B_r & 0 & B_z\end{vmatrix}$$

Hence

$$F_z = -qv_\theta B_r. \tag{1}$$

The quantity B_r can be related to $\dfrac{\partial B_z}{\partial z}$. By Gauss's law for the magnetic field, take as the Gaussian surface the cylindrical surface of radius r and height $\mathrm{d}z$ shown in the figure; then

$$B_z(z)\Delta s = B_z(z+\Delta z)\Delta s + B_r\cdot 2\pi r\,\mathrm{d}z.$$

Then

$$B_r\cdot 2\pi r\,\mathrm{d}z = -\frac{\partial B_z}{\partial z}\mathrm{d}z\,\Delta s,$$

$$B_r = -\frac{r}{2}\frac{\partial B_z}{\partial z}.$$

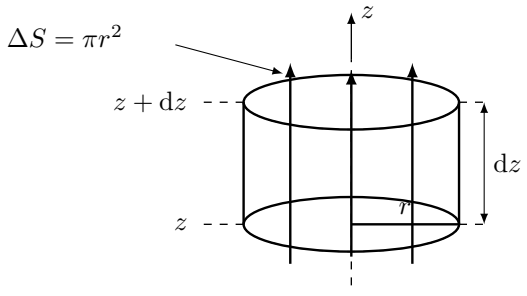


Fig. 3.45

Substituting into (1), and noting that $v_\theta = -v_\perp$, we have

$$F_z = -qv_\perp \frac{r}{2}\frac{\partial B_z}{\partial z}.$$

But $r = \dfrac{mv_\perp}{qB_z}$; substituting this into the above expression gives

$$F_z = -qv_\perp \cdot \frac{1}{2}\frac{mv_\perp}{qB_z}\frac{\partial B_z}{\partial z} = -\frac{1}{2}\frac{mv_\perp^2}{B_z}\frac{\mathrm{d}B_z}{\mathrm{d}z}. \tag{2}$$

And from Newton's law,

$$F_z = m\frac{\mathrm{d}v_z}{\mathrm{d}t} = m\frac{\mathrm{d}v_z}{\mathrm{d}z}\frac{\mathrm{d}z}{\mathrm{d}t} = mv_z\frac{\mathrm{d}v_z}{\mathrm{d}z}. \tag{3}$$

And from conservation of energy,

$$v_z^2 + v_\perp^2 = v_0^2. \tag{4}$$

Substituting (4) and (3) into (2) gives

$$mv_z\frac{\mathrm{d}v_z}{\mathrm{d}z} = -\frac{1}{2}m\left(v_0^2 - v_z^2\right)\frac{1}{B_z}\frac{\mathrm{d}B_z}{\mathrm{d}z}.$$

But $v_z\dfrac{\mathrm{d}v_z}{\mathrm{d}z} = \dfrac{1}{2}\dfrac{\mathrm{d}v_z^2}{\mathrm{d}z}$, so the above expression becomes

$$\frac{\mathrm{d}v_z^2}{v_0^2 - v_z^2} = -\frac{\mathrm{d}B_z}{B_z},$$

that is,

$$\frac{\mathrm{d}\left(v_0^2 - v_z^2\right)}{v_0^2 - v_z^2} = \frac{\mathrm{d}B_z}{B_z}.$$

Integrating both sides, and supposing that the particle is reflected at the place where the magnetic field is B, at which $v_z = 0$, we have

$$\ln\left(v_0^2 - v_z^2\right)\Big|_0^B = \ln B_z\Big|_{B_1}^B,$$

that is,

$$\ln \frac{v_0^2}{v_0^2 - v_z^2} = \ln \frac{B}{B_1}.$$

Noting that $v_0^2 - v_z^2 = v_\perp^2$, the above expression is

$$\frac{v_0^2}{v_\perp^2} = \frac{B}{B_1}.$$

But $v_\perp / v_0 = \sin \varphi_0$, so the above expression is

$$\sin^2 \varphi_0 = \frac{B_1}{B} \geqslant \frac{B_1}{B_2},$$

or

$$\sin \varphi_0 \geqslant \sqrt{\frac{B_1}{B_2}} = \sin \varphi_c.$$

The required φ_c is

$$\varphi_c = \arcsin \sqrt{\frac{B_1}{B_2}}.$$

Solution 3.46. Take the positive direction of circulation of the coil to be the same as the direction of the magnetic field.

(1) The equation of motion of the coil is

$$m \frac{\mathrm{d}v}{\mathrm{d}t} = IBb \tag{1}$$

The circuit equation of the coil is

$$IR = -vBb \tag{2}$$

Substituting (2) into (1) gives

$$m \frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{B^2 b^2}{R} v,$$

that is,

$$\frac{\mathrm{d}v}{v} = -\frac{B^2 b^2}{mR} \mathrm{d}t.$$

Integrating both sides, we have

$$\int_{v_0}^{v} \frac{\mathrm{d}v}{v} = \int_0^t -\frac{B^2 b^2}{mR} \, \mathrm{d}t \quad \Rightarrow \quad \ln \frac{v}{v_0} = -\frac{B^2 b^2}{mR} t,$$

that is,

$$v = v_0 \, \mathrm{e}^{-\frac{B^2 b^2}{mR} t}. \tag{3}$$

Integrating once more with respect to time, we have

$$\begin{aligned} x &= \int_0^t v \, \mathrm{d}t = \int_0^t v_0 \, \mathrm{e}^{-\frac{B^2 b^2}{mR} t} \, \mathrm{d}t \\ &= \frac{mR}{B^2 b^2} v_0 \left(1 - \mathrm{e}^{-\frac{B^2 b^2}{mR} t} \right). \end{aligned} \tag{4}$$

After the coil enters the magnetic field its speed gradually decreases, and finally it stops.

(2) The equation of motion of the coil is the same as (1), while the circuit equation becomes

$$-L\frac{\mathrm{d}I}{\mathrm{d}t} - vBb = 0. \tag{5}$$

Differentiating (1), we obtain

$$m\frac{\mathrm{d}^2v}{\mathrm{d}t^2} = Bb\frac{\mathrm{d}I}{\mathrm{d}t}.$$

Substituting (5) into this, the above expression becomes

$$m\frac{\mathrm{d}^2v}{\mathrm{d}t^2} = Bb\left(-\frac{Bbv}{L}\right) = -\frac{B^2b^2}{L}v.$$

It is seen that v is a simple harmonic function; let

$$v = V\cos(\omega t + \varphi), \tag{6}$$

where

$$\omega = \sqrt{\frac{B^2b^2}{mL}} = \frac{Bb}{\sqrt{mL}}.$$

Integrating (6),

$$x = \int_0^t v\,\mathrm{d}t = \frac{V}{\omega}\left[\sin(\omega t + \varphi) - \sin\varphi\right].$$

Here V and φ can be determined from the initial conditions:

$$x_0 = \frac{V}{\omega}(\sin\varphi - \sin\varphi) = 0, \qquad v_0 = V\cos\varphi.$$

We may take $\varphi = 0$; then $V = v_0$, and

$$\therefore\ x = \frac{v_0}{\omega}\sin\omega t \qquad \text{when } t < \frac{\pi}{\omega} = \pi\frac{\sqrt{mL}}{Bb} = \frac{T}{2},$$

$$x = -v_0\left(t - \frac{T}{2}\right) \qquad \text{when } t > \frac{T}{2}.$$

The coil behaves as if it were repelled back by the magnetic field.

Solution 3.47. (1) Let the time taken by the ring to fall down be T, and let the moment of inertia of the ring about the pivot be I. Then the kinetic energy of the ring when it has fallen down is estimated as

$$W_k = \frac{1}{2}I\omega^2 = \frac{1}{2}I\left(\frac{\pi/2}{T}\right)^2 = \frac{1}{2}\left(\frac{1}{2}mr^2 + mr^2\right)\cdot\frac{\pi^2}{4T^2}$$
$$= \frac{3\pi^2}{16}\frac{mr^2}{T^2}.$$

The Joule heat is estimated as

$$W_h = i^2RT = \left(\frac{\varepsilon}{R}\right)^2 RT = \frac{\varepsilon^2}{R}T = \frac{(\Phi/T)^2}{R}T$$
$$= \frac{\Phi^2}{RT} = \frac{(B\cdot\pi r^2)^2}{RT} = \frac{B^2\pi^2r^4}{RT}.$$

The ratio of the two is

$$\frac{W_k}{W_h} = \frac{3\pi^2 mr^2}{16T^2} \frac{RT}{B^2\pi^2 r^4} = \frac{3mR}{16B^2 T r^2}.$$

With

$$m = \rho \cdot 2\pi r \cdot \pi r_1^2, \qquad R = \frac{1}{\sigma}\frac{2\pi r}{\pi r_1^2} = \frac{1}{\sigma}\frac{2r}{r_1^2}$$

substituted into the above expression,

$$\frac{W_k}{W_h} = \frac{3}{16}\frac{\rho \cdot 2\pi^2 r r_1^2}{B^2 T r^2} \cdot \frac{1}{\sigma}\frac{2r}{r_1^2} = \frac{3\pi^2}{4}\frac{\rho}{\sigma B^2 T}.$$

But $T \geqslant T_0$, where T_0 is the time taken by the ring to fall down in the absence of a magnetic field. It is not difficult to see that

$$T_0 \approx \frac{\pi}{2}\sqrt{\frac{I}{mgr}} = \frac{\pi}{2}\sqrt{\frac{\frac{3}{2}mr^2}{mgr}} = \frac{\pi}{2}\sqrt{\frac{3r}{2g}}.$$

$$\therefore \ \frac{W_k}{W_h} = \frac{3\pi^2}{4}\frac{\rho}{\sigma B^2 T} < \frac{3\pi^2}{4}\frac{\rho}{\sigma B^2 T_0} = \frac{3\pi^2}{4}\frac{\rho}{\sigma B^2}\frac{2}{\pi}\sqrt{\frac{2g}{3r}} = \frac{3\pi}{2}\frac{\rho}{\sigma B^2}\sqrt{\frac{2g}{3r}}.$$

Substituting the data, we obtain

$$\frac{W_k}{W_h} < \frac{3\pi}{2}\frac{19.3 \times 10^3}{4.25 \times 10^7 \times 1^2}\sqrt{\frac{2 \times 9.8}{3 \times 0.01}} = 5.5 \times 10^{-2}.$$

It is seen that the released potential energy is mainly converted into Joule heat.

(2) Neglecting the kinetic energy, the Ampere torque is at every instant balanced with the gravitational torque during the falling of the ring. During the falling of the ring, when the plane of the ring makes an angle θ with the vertical plane, the magnetic flux through the ring is

$$\Phi = BS\sin\theta,$$

where $S = \pi r^2$. Then the electromotive force is

$$\varepsilon = \left|\frac{\mathrm{d}\Phi}{\mathrm{d}t}\right| = BS\cos\theta \cdot \dot{\theta}.$$

The current is

$$i = \frac{\varepsilon}{R} = \frac{BS\cos\theta \cdot \dot{\theta}}{R}.$$

The Ampere torque is

$$\tau_B = iSB\cos\theta = \frac{B^2 S^2 \cos^2\theta}{R}\dot{\theta}.$$

The gravitational torque is

$$\tau_G = mgr\sin\theta.$$

Setting $\tau_B = \tau_G$,

$$\frac{B^2 S^2 \cos^2\theta}{R}\frac{\mathrm{d}\theta}{\mathrm{d}t} = mgr\sin\theta,$$

so

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{mgr\sin\theta\, R}{B^2 S^2 \cos^2\theta},$$

$$\mathrm{d}t = \frac{B^2S^2}{mgrR}\frac{\cos^2\theta}{\sin\theta}\mathrm{d}\theta.$$

The time taken to fall down is

$$t = \frac{B^2S^2}{mgrR}\int_{0.1}^{\frac{\pi}{2}}\frac{\cos^2\theta}{\sin\theta}\mathrm{d}\theta,$$

where

$$\begin{aligned}\int_{0.1}^{\frac{\pi}{2}}\frac{\cos^2\theta}{\sin\theta}\mathrm{d}\theta &= \int_{0.1}^{\frac{\pi}{2}}\frac{\mathrm{d}\theta}{\sin\theta} - \int_{0.1}^{\frac{\pi}{2}}\sin\theta\,\mathrm{d}\theta = \left[\ln\tan\frac{\theta}{2} + \cos\theta\right]_{0.1}^{\frac{\pi}{2}} \\ &\approx 2.99 - 0.99 = 2.\end{aligned}$$

And

$$\begin{aligned}\frac{B^2S^2}{mgrR} &= \frac{B^2\left(\pi r^2\right)^2}{\rho\cdot 2\pi r\cdot \pi r_1^2 gr\cdot\frac{1}{\sigma}\cdot\frac{2\pi r}{\pi r_1^2}} = \frac{\sigma B^2 r}{4\rho g} \\ &= \frac{4.25\times 10^7\times 1^2\times 0.01}{4\times 19.3\times 10^3\times 9.8} = 0.56\ (\mathrm{s}).\end{aligned}$$

Substituting into the expression for t, we obtain

$$t = 0.56\times 2 = 1.12\ (\mathrm{s}).$$

Solution 3.48. (1) The magnetic field of the small bar magnet may be regarded as the field of a magnetic dipole. Take the centre of the bar as the origin and take the vertical axis of the bar magnet as the z-axis, as shown in the figure of the problem. On the axis the magnetic flux density may be written as

$$B_z = \frac{B_0}{z^3}. \tag{1}$$

For, by the statement of the problem, when $z = 1$, $B = B_0$. Since the coil is superconducting and its resistance is $R = 0$, from the law of electromagnetic induction and Ohm's law, for the circuit of the coil we have

$$-\frac{\mathrm{d}\Phi}{\mathrm{d}t} = IR = 0.$$

Integrating gives

$$\Phi = \text{constant} = 0, \tag{2}$$

because when the coil was originally far away, $\Phi = 0$. When the coil is in equilibrium at $z = z_0$, let the current in the coil be I_0; then

$$\frac{B_0}{z_0^3}\pi a^2 - I_0 L = 0. \tag{3}$$

From this we obtain

$$I_0 = \frac{B_0\pi a^2}{Lz_0^3}. \tag{4}$$

The direction of I_0 is as shown in the figure.

In equilibrium the weight of the coil is balanced by the Ampere force, and the vertical component of the Ampere force acting on the coil is determined by the component B_r of the

magnetic field at the position of the coil along the radial direction of the coil. This radial component can be found from the rate of change of the z component by means of Gauss's law for the magnetic field. At z construct a cylindrical surface of height Δz and radius a as the Gaussian surface, as shown in the figure; then by Gauss's law,

$$B_z(z)\pi a^2 = B_z(z+\Delta z)\pi a^2 + B_r \cdot 2\pi a \cdot \Delta z,$$

that is,

$$-\frac{\partial B_z}{\partial z}\Delta z \cdot \pi a^2 = B_r \cdot 2\pi a \cdot \Delta z.$$

Substituting (1) into the above expression, we find

$$B_r = \frac{3}{2}\frac{B_0 a}{z^4}. \tag{5}$$

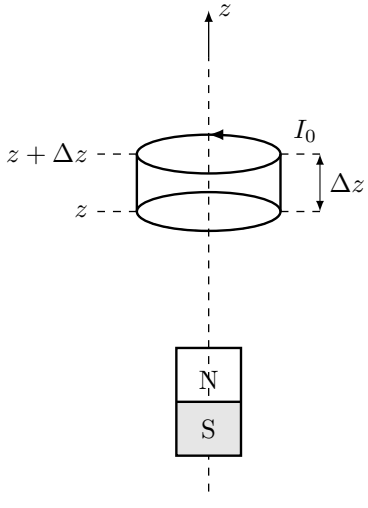


Fig. 3.48

Substituting (4) and (5) into the force equilibrium equation of the coil,

$$I_0 B_r(z_0) 2\pi a = mg, \tag{6}$$

we obtain

$$\frac{B_0\pi a^2}{Lz_0^3} \cdot \frac{3}{2}\frac{B_0 a}{z_0^4} \cdot 2\pi a = mg.$$

From this we find

$$z_0 = \sqrt[7]{\frac{3B_0^2\pi^2 a^4}{Lmg}}. \tag{7}$$

(2) When the coil is moved upwards from the equilibrium position by a small amount z', let the current in the coil become $I_0 + i$. From $\Phi = 0$, equation (2), we obtain

$$\frac{B_0}{(z_0+z')^3}\pi a^2 - L(I_0+i) = 0.$$

Retaining first-order small quantities in z'/z_0, the above expression becomes

$$\frac{B_0\pi a^2}{z_0^3}\left(1-3\frac{z'}{z_0}\right)-LI_0-Li=0.$$

Using (3) and (4), we obtain

$$i=-3\frac{z'}{z_0}\frac{B_0\pi a^2}{Lz_0^3}=-3I_0\frac{z'}{z_0}. \tag{8}$$

The force on the coil at this time is

$$F=(I_0+i)B_r(z_0+z')2\pi a-mg.$$

From (5),

$$B_r(z_0+z')=\frac{3}{2}\frac{B_0a}{(z_0+z')^4}=\frac{3}{2}B_r(z_0)\left(1-4\frac{z'}{z_0}\right).$$

Substituting into the above expression,

$$F=I_0\left(1+\frac{i}{I_0}\right)B_r(z_0)\left(1-4\frac{z'}{z_0}\right)2\pi a-mg.$$

Substituting (8) into the above expression, neglecting small quantities of second and higher order, and using (6), we obtain

$$F=-7mg\frac{z'}{z_0}.$$

Then the equation of motion of the coil is

$$m\ddot{z}'=F=-7mg\frac{z'}{z_0}.$$

It is seen that the coil performs simple harmonic motion, with angular frequency

$$\omega=\sqrt{\frac{7g}{z_0}},$$

and period

$$T=2\pi\sqrt{\frac{z_0}{7g}}.$$

Solution 3.49. When the magnet falls, induced currents (eddy currents) are produced in the metal tube by electromagnetic induction. According to Lenz's law, the polarity of the magnetic field produced by the induced current is opposite to that of the magnet, which hinders the fall of the magnet, reduces the acceleration of the falling magnet, and finally makes the magnet fall uniformly with a certain speed. The constant speed of fall of the magnet can be found from energy relations.

To simplify the calculation, the magnet is regarded as a magnetic dipole of magnetic moment μ, and the metal tube is imagined to consist of a series of identical circular coils arranged coaxially (with μ on the axis). Let the radius of each coil be a, its resistance be R, and the density of arrangement of the coils along the axial direction (that is, the number of turns per unit length) be n. Suppose that the direction of the magnetic moment μ of the magnetic dipole is vertically downwards. At a certain instant when μ falls with constant speed v, consider the induced current

in a certain coil. Take the axis of the metal tube as the z-axis, with the downward direction positive and the origin at the position of μ; the centre of the coil is at a distance z below μ, as shown in the figure. Let the distance of the wire of the coil from the centre of μ (that is, from the origin) be r, and let the angle between the vector r directed from the origin to the wire and the z-axis be θ. Then the induced electromotive force ε in the coil is equal to the ratio of the magnetic flux $\mathrm{d}\Phi$ of the field of the magnetic moment μ through the annular strip of circumference $2\pi a$ and width $\mathrm{d}z = v\mathrm{d}t$ to $\mathrm{d}t$, that is,

$$|\varepsilon| = \frac{\mathrm{d}\Phi}{\mathrm{d}t} = (B_r \sin\theta + B_\theta \cos\theta)\, 2\pi a\, \mathrm{d}z/\mathrm{d}t.$$

For the field of a magnetic dipole,

$$B_r = \frac{\mu_0}{4\pi}\frac{2\mu\cos\theta}{r^3},$$

$$B_\theta = \frac{\mu_0}{4\pi}\frac{\mu\sin\theta}{r^3}.$$

And

$$\mathrm{d}z/\mathrm{d}t = v.$$

Substituting into the above expression, we obtain

$$\begin{aligned}|\varepsilon| &= \frac{\mu_0}{4\pi}\left(\frac{2\mu\cos\theta}{r^3}\sin\theta + \frac{\mu\sin\theta}{r^3}\cos\theta\right)\cdot 2\pi a v \\ &= \frac{3}{2}\mu_0\mu a v\frac{\sin\theta\cos\theta}{r^3}.\end{aligned}$$

Then the Joule heating power dissipated in the coil is

$$p = I^2R = \varepsilon^2/R = \frac{9\mu_0^2\mu^2a^2v^2}{4R}\frac{\sin^2\theta\cos^2\theta}{r^6}.$$

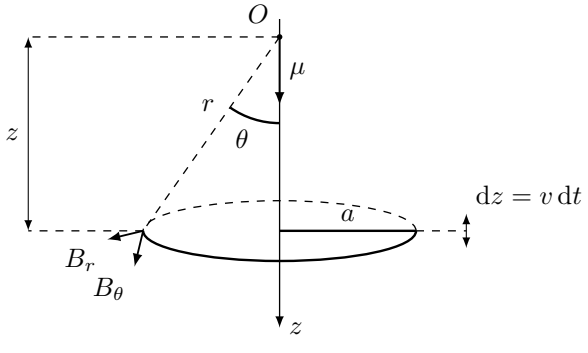


Fig. 3.49

In the interval of height $\mathrm{d}z$ there are altogether $\mathrm{d}N = n\mathrm{d}z = n\dfrac{r\,\mathrm{d}\theta}{\sin\theta}$ turns of coils, and the Joule heating power dissipated in these coils is

$$\begin{aligned}\mathrm{d}P = p\,\mathrm{d}N = pn\,\mathrm{d}z &= \frac{9\mu_0^2\mu^2a^2v^2}{4R}\frac{\sin^2\theta\cos^2\theta}{r^6}\cdot n\frac{r\,\mathrm{d}\theta}{\sin\theta} \\ &= \frac{9\mu_0^2\mu^2a^2v^2}{4R}\frac{\sin\theta\cos^2\theta}{r^5}n\,\mathrm{d}\theta \\ &= \frac{9\mu_0^2\mu^2v^2n}{4Ra^3}\sin^6\theta\cos^2\theta\,\mathrm{d}\theta.\end{aligned}$$

Here the relation $r = a/\sin\theta$ has been used. The Joule heating power dissipated in all the coils is the integral of the above expression. For simplicity we may suppose that the coils are arranged to an infinite length, so that the limits of integration are taken as 0 and π, that is,

$$P = \int \mathrm{d}P = \int_0^\pi \frac{9\mu_0^2\mu^2 v^2 n}{4Ra^3} \sin^6\theta\cos^2\theta\,\mathrm{d}\theta.$$

But

$$\int_0^\pi \sin^6\theta\cos^2\theta\,\mathrm{d}\theta = \int_0^\pi \sin^6\theta\left(1-\sin^2\theta\right)\mathrm{d}\theta = \int_0^\pi \left(\sin^6\theta - \sin^8\theta\right)\mathrm{d}\theta,$$

and from the integration formula given in the problem, we have

$$\begin{aligned}\int_0^\pi \sin^6\theta\,\mathrm{d}\theta &= -\frac{\sin^5\theta\cos\theta}{6}\bigg|_0^\pi + \frac{5}{6}\int_0^\pi \sin^4\theta\,\mathrm{d}\theta \\ &= \frac{5}{6}\left[-\frac{\sin^3\theta\cos\theta}{4}\bigg|_0^\pi + \frac{3}{4}\int_0^\pi \sin^2\theta\,\mathrm{d}\theta\right] \\ &= \frac{3}{5}\left[\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta\right]_0^\pi = \frac{3\pi}{10}.\end{aligned}$$

$$\begin{aligned}\int_0^\pi \sin^8\theta\,\mathrm{d}\theta &= -\frac{\sin^7\theta\cos\theta}{8}\bigg|_0^\pi + \frac{7}{8}\int_0^\pi \sin^6\theta\,\mathrm{d}\theta \\ &= \frac{7}{8}\cdot\frac{3\pi}{10}.\end{aligned}$$

$$\therefore\ \int_0^\pi \sin^6\theta\cos^2\theta\,\mathrm{d}\theta = \int_0^\pi \left(\sin^6\theta - \sin^8\theta\right)\mathrm{d}\theta = \left(1-\frac{7}{8}\right)\frac{3\pi}{10} = \frac{3\pi}{80}.$$

Substituting into the expression for P, we obtain the total thermal power dissipated in all the coils:

$$P = \frac{9\mu_0^2\mu^2 v^2 n}{4Ra^3}\cdot\frac{3\pi}{80}.$$

From conservation of energy, when the magnet (magnetic dipole) falls steadily, the power of the work done on it by gravity must be equal to the total thermal power dissipated in all the coils, that is,

$$P = \frac{9\mu_0^2\mu^2 v^2 n}{4Ra^3}\cdot\frac{3\pi}{80} = mgv.$$

From this we obtain the speed of fall of the magnetic dipole:

$$v = \frac{320Ra^3 mg}{27\pi\mu_0^2\mu^2 n}.$$

Solution 3.50. Let the current through the appliance be I_L; then I_L lags behind U in phase by φ. Let the capacitance connected in parallel be C; then the current through the capacitor is

$$I_C = \frac{U}{1/(C\omega)} = UC\omega,$$

and its phase leads U by $\pi/2$. The vector relation among I_L, I_C and U is as shown in the figure. In order to raise the power factor to 1, the current $I = I_L + I_C$ must be in phase with U. Hence

$$UC\omega = I_L\sin\varphi.$$

From the conditions given,

$$P = I_L U \cos\varphi.$$

Dividing the two expressions gives

$$C = \frac{P \sin\varphi}{\omega U^2 \cos\varphi} = \frac{P\sqrt{1-\cos^2\varphi}}{\omega U^2 \cos\varphi}.$$

Substituting the data,

$$C = \frac{100\sqrt{1-0.6^2}}{100\pi \times 220^2 \times 0.6} = 8.77 \times 10^{-6}\,\mathrm{F} = 8.77\,\mu\mathrm{F}.$$

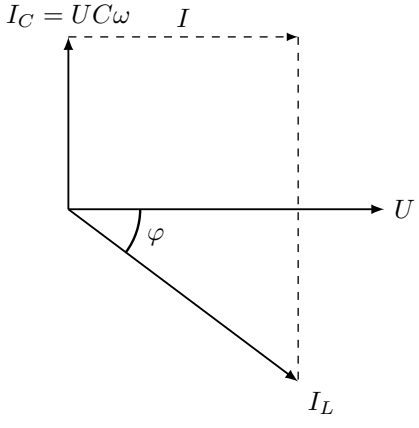


Fig. 3.50

Solution 3.51. Let the currents in the four loops be as shown in the figure. Since $\omega = 1/\sqrt{LC}$ and $R = \sqrt{L/C}$, the complex impedances of the resistor, the inductor and the capacitor are respectively

$$\tilde{Z}_R = R,$$

$$\tilde{Z}_L = j\omega L = j\sqrt{L/C} = jR,$$

$$\tilde{Z}_C = \frac{1}{j\omega C} = -\sqrt{L/C} = -jR.$$

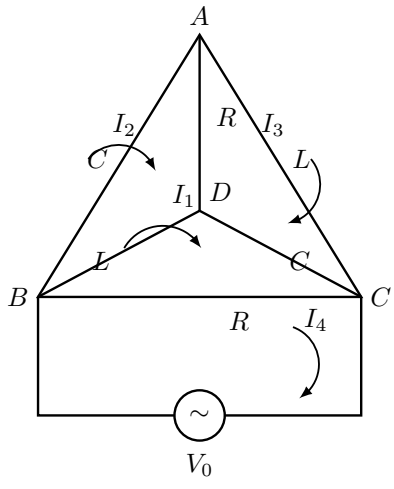


Fig. 3.51

According to Kirchhoff's laws, the loop equations are as follows:

$$(I_1 - I_4)R + (I_1 - I_2)(jR) + (I_1 - I_3)(-jR) = 0,$$

$$I_2(-jR) + (I_2 - I_3)R + (I_2 - I_1)(jR) = 0,$$

$$I_3(jR) + (I_3 - I_1)(-jR) + (I_3 - I_2)R = 0,$$

$$(I_4 - I_1)R = V_0.$$

Simplifying gives

$$I_1 - I_4 + j(I_3 - I_2) = 0,$$

$$I_2 - I_3 - jI_1 = 0,$$

$$I_3 - I_2 + jI_1 = 0,$$

$$I_4 - I_1 = V_0/R.$$

The second and third equations are in fact the same. Multiplying the second equation by j and adding it to the first gives $I_1 = I_4/2$; substituting this into the fourth equation gives the required total current

$$I_4 = 2V_0/R.$$

Thus the equivalent impedance of the circuit is $R/2$: the five branches above are equivalent to a single resistance R, connected in parallel with the resistance R at the bottom.

Solution 3.52. (1) The equivalent circuit is as shown in the figure, where $\tilde{Z}_1 = \dfrac{1}{j\omega C}$ and $\tilde{Z}_2 = jL\omega$. Let the complex impedance between the terminals A and B be $\tilde{Z}$. By the conditions of the problem,

$$\tilde{Z} = \tilde{Z}_1 + \frac{\tilde{Z}_2\tilde{Z}}{\tilde{Z} + \tilde{Z}_2},$$

that is,

$$\tilde{Z}^2 - \tilde{Z}_1\tilde{Z} - \tilde{Z}_1\tilde{Z}_2 = 0.$$

Solving gives

$$\tilde{Z} = \frac{\tilde{Z}_1}{2}\left(1 + \sqrt{1 + 4\frac{\tilde{Z}_2}{\tilde{Z}_1}}\right) = \frac{1}{2j\omega C}\left(1 + \sqrt{1 - 4LC\omega^2}\right).$$

When $4LC\omega^2 \leq 1$, i.e. $\omega \leq \dfrac{1}{2\sqrt{LC}}$, $\tilde{Z}$ is purely imaginary. When $4LC\omega^2 > 1$, i.e. $\omega > \dfrac{1}{2\sqrt{LC}}$, $\tilde{Z}$ has both imaginary and real parts. Since the real part cannot be negative, only the "+" sign can be taken before the square root in the expression for $\tilde{Z}$.

(2) The average power is

$$\bar{P} = \frac{1}{2}\mathrm{Re}\left[V_0 I_0^*\right] = \frac{1}{2}|V_0|^2 \mathrm{Re}\left[\frac{1}{\tilde{Z}}\right].$$

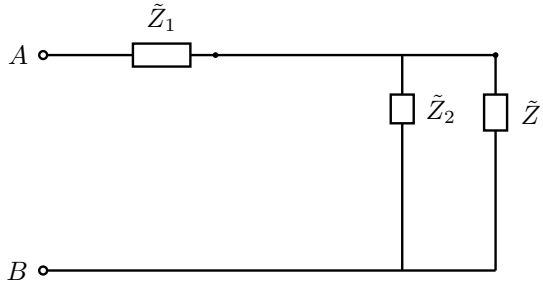


Fig. 3.52

Hence when

$$\omega \leq \frac{1}{2\sqrt{LC}},$$

$$\bar{P} = 0,$$

and when

$$\omega > \frac{1}{2\sqrt{LC}},$$

$$\bar{P} = \frac{1}{2}V_0^2 \,\mathrm{Re}\left[\frac{1}{\tilde{Z}}\right].$$

From part (1),

$$\frac{1}{\tilde{Z}} = \frac{2j\omega C}{1 + j\sqrt{4LC\omega^2 - 1}} = \frac{2j\omega C\left(1 - j\sqrt{4LC\omega^2 - 1}\right)}{4LC\omega^2},$$

$$\therefore\ \mathrm{Re}\left[\frac{1}{\tilde{Z}}\right] = \frac{C\omega}{2LC\omega^2}\sqrt{4LC\omega^2 - 1} = \frac{1}{2L\omega}\sqrt{4LC\omega^2 - 1},$$

$$\therefore\ \bar{P} = \frac{V_0^2}{4L\omega}\sqrt{4LC\omega^2 - 1}.$$

Solution 3.53. (1) Since the quasi-steady condition is satisfied, at any instant the charge on the two plates is distributed uniformly. The electric field is

$$E = \frac{V}{d} = \frac{V_0}{d}\sin\omega t.$$

The magnetic field is excited by the total current. By the Ampère circuital law,

$$B \cdot 2\pi r = \mu_0 \varepsilon_0 \frac{\partial E}{\partial t} \pi r^2.$$

From the expression for E,

$$\frac{\partial E}{\partial t} = \frac{V_0 \omega}{d} \cos \omega t.$$

Substituting into the above expression gives

$$B = \frac{\mu_0 \varepsilon_0 V_0 \omega}{2d} r \cos \omega t.$$

(2) At the edge of the capacitor, $r = a$, we have

$$E = \frac{V_0}{d} \sin \omega t,$$

$$B = \frac{\mu_0 \varepsilon_0 V_0 \omega}{2d} a \cos \omega t.$$

The energy flux density is

$$S = EH = \frac{1}{\mu_0} EB = \frac{\varepsilon_0 V_0^2 \omega a}{2d^2} \sin \omega t \cos \omega t.$$

The energy flux entering the capacitor through its edge is

$$I = S \cdot 2\pi a d = \frac{\pi a^2}{d} \varepsilon_0 V_0^2 \omega \sin \omega t \cos \omega t.$$

(3) The electric field energy between the plates is

$$W_e = \int \frac{1}{2} \varepsilon_0 E^2 \, \mathrm{d}\tau = \frac{1}{2} \varepsilon_0 E^2 \cdot \pi a^2 d,$$

$$\begin{aligned} \frac{\mathrm{d}W_e}{\mathrm{d}t} &= \varepsilon_0 E \frac{\partial E}{\partial t} \pi a^2 d = \varepsilon_0 \frac{V_0}{d} \sin \omega t \cdot \frac{V_0 \omega}{d} \cos \omega t \cdot \pi a^2 d \\ &= \frac{\pi a^2}{d} \varepsilon_0 V_0^2 \omega \sin \omega t \cos \omega t = I. \end{aligned}$$

The magnetic field energy is

$$\begin{aligned} W_m &= \int \frac{1}{2\mu_0} B^2 \, \mathrm{d}\tau = \frac{\mu_0 \varepsilon_0^2 V_0^2 \omega^2}{8 \mu_0 d^2} \cos^2 \omega t \, r^2 \cdot 2\pi r \, \mathrm{d}r \cdot d \\ &= \frac{\mu_0 \varepsilon_0^2 V_0^2 \omega^2 \pi}{4d} \cos^2 \omega t \int_0^a r^3 \, \mathrm{d}r \\ &= \frac{\mu_0 \varepsilon_0^2 V_0^2 \omega^2 \pi a^4}{16 d} \cos^2 \omega t, \\ \frac{\mathrm{d}W_m}{\mathrm{d}t} &= -\frac{\mu_0 \varepsilon_0^2 V_0^2 \omega^3 \pi a^4}{8d} \sin \omega t \cos \omega t. \end{aligned}$$

According to the conditions of the problem, $a \ll c/\omega$, and noting that $\varepsilon_0\mu_0 = 1/c^2$, we have

$$\left|\frac{\mathrm{d}W_m/\mathrm{d}t}{\mathrm{d}W_e/\mathrm{d}t}\right| = \frac{1}{8}\frac{\omega^2 a^2}{c^2} \ll 1,$$

$$\therefore\ \frac{\mathrm{d}W_e}{\mathrm{d}t} + \frac{\mathrm{d}W_m}{\mathrm{d}t} \approx \frac{\mathrm{d}W_e}{\mathrm{d}t} = I.$$

It is thus seen that under the quasi-steady condition the energy flowing in through the edge of the capacitor is equal to the rate of increase of the electromagnetic field energy between the two plates.

Solution 3.54. (1) Solve by the complex method. Let the current flowing through R_1 be $\tilde{I}_1$. The input impedance of the circuit is

$$\tilde{Z} = R_1 + \frac{\left(-j\dfrac{1}{\omega C}\right)R_2}{R_2 - j\dfrac{1}{\omega C}},$$

$$\tilde{I}_1 = \frac{\tilde{V}}{\tilde{Z}}, \qquad \tilde{V}_1 = \tilde{I}_1 R_1 = \frac{R_1}{\tilde{Z}}\tilde{V}.$$

Substituting the expression for $\tilde{Z}$,

$$\tilde{V}_1 = \frac{R_1}{R_1 + \dfrac{\left(-j\dfrac{1}{\omega C}\right)R_2}{R_2 - j\dfrac{1}{\omega C}}}\tilde{V} = \frac{R_1\left(R_2 - j\dfrac{1}{\omega C}\right)}{R_1R_2 - j\dfrac{R_1+R_2}{\omega C}}\tilde{V}$$

$$= \frac{R_1\sqrt{R_2^2 + \dfrac{1}{\omega^2C^2}}}{\sqrt{R_1^2R_2^2 + \left(\dfrac{R_1+R_2}{\omega C}\right)^2}} \cdot \frac{\mathrm{e}^{-j\varphi_1}}{\mathrm{e}^{-j\varphi_2}}\tilde{V} = \frac{R_1\sqrt{R_2^2 + \dfrac{1}{\omega^2C^2}}}{\sqrt{R_1^2R_2^2 + \left(\dfrac{R_1+R_2}{\omega C}\right)^2}}\mathrm{e}^{j(\varphi_2-\varphi_1)}\tilde{V}.$$

$$\therefore\ V_1 = V_0\frac{R_1\sqrt{R_2^2 + \dfrac{1}{\omega^2C^2}}}{\sqrt{R_1^2R_2^2 + \left(\dfrac{R_1+R_2}{\omega C}\right)^2}}\cos(\omega t + \varphi),$$

where

$$\tan\varphi = \tan(\varphi_2 - \varphi_1) = \frac{\tan\varphi_2 - \tan\varphi_1}{1 + \tan\varphi_2\tan\varphi_1}$$

$$= \frac{\dfrac{R_1+R_2}{\omega C}\cdot\dfrac{1}{R_1R_2} - \dfrac{1}{\omega CR_2}}{1 + \dfrac{R_1+R_2}{\omega C}\cdot\dfrac{1}{R_1R_2}\cdot\dfrac{1}{\omega CR_2}}$$

$$= \frac{\omega CR_2^2}{R_1 + R_2 + \omega^2C^2R_1R_2^2}.$$

(2) When the pulse is just applied, C may be regarded as a short circuit (the proof is given below); hence C is charged with a current U_0/R_1, and after a duration Δt the charge accumulated at the two ends of the capacitor is

$$q_0 = \frac{U_0}{R_1}\Delta t.$$

Afterwards the capacitor discharges through the parallel resistors R_1 and R_2, and its charge varies with time as

$$q = q_0 \mathrm{e}^{-t'/\frac{R_1R_2}{R_1+R_2}C},$$

$$\therefore\ V_1 = \frac{q}{C} = \frac{q_0}{C}\mathrm{e}^{-t'\frac{R_1+R_2}{R_1R_2C}} = \frac{U_0\Delta t}{R_1C}\mathrm{e}^{-t'\frac{R_1+R_2}{R_1R_2C}}.$$

Here t' is the time of discharge of the capacitor through the parallel resistors R_1 and R_2. In summary, the result is

$$V_1 = \begin{cases} 0, & t < 0, \\ U_0, & 0 < t < \Delta t, \\ -\dfrac{U_0\Delta t}{R_1C}\mathrm{e}^{-(t-\Delta t)\frac{R_1+R_2}{R_1R_2C}}, & t > \Delta t. \end{cases}$$

The negative sign indicates that at this time the potential at the right end of R_1 is higher than that at the left end.

The proof that C may be regarded as a short circuit when $V(t)$ is a short pulse is as follows. When $V = U_0$ (constant), we have (see the figure)

$$U_0 = i_1R_1 + i_2R_2 \tag{1}$$

$$i_1 - i_2 = \frac{\mathrm{d}q}{\mathrm{d}t} \tag{2}$$

$$\frac{q}{C} = i_2R_2 \tag{3}$$

Differentiating (3) and substituting into (2) gives

$$i_1 - i_2 = R_2C\frac{\mathrm{d}i_2}{\mathrm{d}t}.$$

Multiplying both sides by R_1 and using (1) gives

$$U_0 - (R_1 + R_2)i_2 = R_1R_2C\frac{\mathrm{d}i_2}{\mathrm{d}t}, \tag{4}$$

that is,

$$\frac{\mathrm{d}i_2}{U_0 - (R_1 + R_2)i_2} = \frac{\mathrm{d}t}{R_1R_2C}.$$

Integrating the above expression gives

$$i_2 = \frac{U_0}{R_1 + R_2}\left(1 - \mathrm{e}^{-t/RC}\right), \tag{5}$$

where $R = \dfrac{R_1R_2}{R_1 + R_2}$. It is seen that as $t \to 0$, $i_2 \to 0$. Substituting (5) into (1) gives

$$i_1 = \frac{U_0}{R_1} - \frac{U_0R_2}{R_1(R_1 + R_2)}\left(1 - \mathrm{e}^{-t/RC}\right).$$

It is seen that as $t \to 0$, $i_1 \approx \dfrac{U_0}{R_1}$, and the charging current is

$$i_c = i_1 - i_2 \approx i_1 = \frac{U_0}{R_1}.$$

This proves the statement.

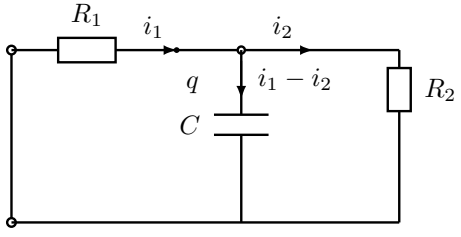


Fig. 3.54

Solution 3.55. (1) After K is closed, let the charges on the capacitors be Q_1 and Q_2 respectively, and let the current flowing through the resistor and the inductor be I (see the figure). Then the circuit equation is

$$IR + L\frac{\mathrm{d}I}{\mathrm{d}t} + \frac{Q_2}{C_2} = \frac{Q_1}{C_1}.$$

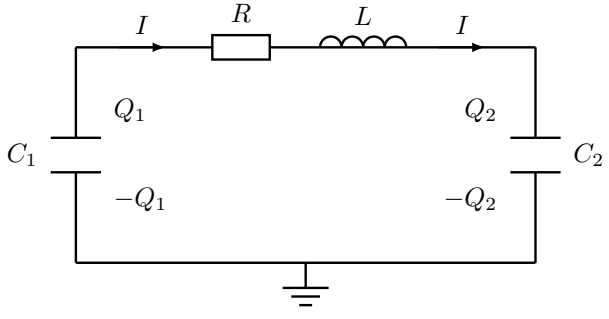


Fig. 3.55

Differentiating both sides with respect to t, and noting that $\dfrac{\mathrm{d}Q_2}{\mathrm{d}t} = I$ and $-\dfrac{\mathrm{d}Q_1}{\mathrm{d}t} = I$, we obtain

$$L\frac{\mathrm{d}^2 I}{\mathrm{d}t^2} + R\frac{\mathrm{d}I}{\mathrm{d}t} + \left(\frac{1}{C_1} + \frac{1}{C_2}\right) I = 0.$$

The roots of the characteristic equation corresponding to this equation are

$$D = -\frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} - \frac{1}{L}\left(\frac{1}{C_1} + \frac{1}{C_2}\right)}.$$

According to the conditions of the problem, the process should be in the critically damped state; hence

$$\frac{R^2}{4L^2} = \frac{1}{L}\left(\frac{1}{C_1} + \frac{1}{C_2}\right).$$

Solving gives

$$R = R_0 = 2\sqrt{L\left(\frac{1}{C_1} + \frac{1}{C_2}\right)}.$$

(2) At this time the solution of the original equation is

$$I = (a + bt)\mathrm{e}^{-\frac{R}{2L}t}.$$

Here a and b are determined by the boundary conditions. When $t = 0$, $I = 0$, so $a = 0$, and

$$I = bt\,\mathrm{e}^{-\frac{R}{2L}t}.$$

To find the maximum of I, set $\dfrac{\mathrm{d}I}{\mathrm{d}t} = 0$:

$$\frac{\mathrm{d}I}{\mathrm{d}t} = b\mathrm{e}^{-\frac{R_0}{2L}t} + bt\left(-\frac{R_0}{2L}\right)\mathrm{e}^{-\frac{R_0}{2L}t} = b\left(1 - \frac{R_0 t}{2L}\right)\mathrm{e}^{-\frac{R_0}{2L}t} = 0,$$

which gives

$$t = \frac{2L}{R_0} = \sqrt{\frac{LC_1C_2}{C_1 + C_2}}.$$

When $t = \dfrac{2L}{R_0}$,

$$\begin{aligned}\frac{\mathrm{d}^2I}{\mathrm{d}t^2} &= -\frac{R_0 b}{2L}\mathrm{e}^{-\frac{R_0}{2L}t} - \frac{R_0 b}{2L}\mathrm{e}^{-\frac{R_0}{2L}t} + \frac{R_0^2}{4L^2}bt\,\mathrm{e}^{-\frac{R_0}{2L}t} \\ &= -\frac{R_0 b}{L}\mathrm{e}^{-\frac{R_0}{2L}t} + \frac{R_0^2}{4L^2}bt\,\mathrm{e}^{-\frac{R_0}{2L}t}.\end{aligned}$$

Substituting $t = \dfrac{2L}{R_0}$ into the above expression gives

$$\left.\frac{\mathrm{d}^2I}{\mathrm{d}t^2}\right|_{t=\frac{2L}{R_0}} = \left(-\frac{R_0 b}{L} + \frac{R_0^2}{4L^2}b\frac{2L}{R_0}\right)\mathrm{e}^{-1} = -\frac{R_0 b}{2L}\frac{1}{\mathrm{e}} < 0.$$

It is thus seen that I has a maximum when $t = \dfrac{2L}{R_0}$.

To find the maximum value I_m of I, the value of b must be found. This can be obtained from the integral and the final-state condition:

$$\begin{aligned}\int_0^\infty I\,\mathrm{d}t &= \int_0^\infty bt\,\mathrm{e}^{-\frac{R_0}{2L}t}\,\mathrm{d}t = b\left(-\frac{2L}{R_0}\right)^2\int_0^\infty\left(-\frac{R_0 t}{2L}\right)\mathrm{e}^{-\frac{R_0 t}{2L}}\,\mathrm{d}\left(-\frac{R_0 t}{2L}\right) \\ &= b\frac{4L^2}{R_0^2}\left[-\frac{R_0 t}{2L}\mathrm{e}^{-\frac{R_0 t}{2L}} - \mathrm{e}^{-\frac{R_0 t}{2L}}\right]_0^\infty = b\frac{4L^2}{R_0^2}.\end{aligned}$$

On the other hand, $\displaystyle\int_0^\infty I\,\mathrm{d}t = Q_2 = Q_0\frac{C_2}{C_1 + C_2}$; hence

$$Q_0\frac{C_2}{C_1 + C_2} = b\frac{4L^2}{R_0^2}.$$

Substituting the expression for R_0 gives

$$b = Q_0 \frac{C_2}{C_1 + C_2} \frac{R_0^2}{4L^2} = \frac{Q_0}{LC_1}.$$

Substituting into the expression for I gives

$$I = \frac{Q_0}{LC_1} t \, \mathrm{e}^{-\frac{R_0}{2L} t}.$$

Substituting $t = \dfrac{2L}{R_0}$ into the above expression, and inserting the expression for R_0, the maximum value of the current is obtained:

$$I_m = \frac{Q_0}{LC_1} \frac{2L}{R_0} \frac{1}{\mathrm{e}} = \frac{2Q_0}{\mathrm{e}R_0C_1} = \frac{Q_0}{\mathrm{e}C_1 \sqrt{L\left(\dfrac{1}{C_1} + \dfrac{1}{C_2}\right)}}.$$

Solution 3.56. (1) Obviously, the propagation factor is

$$\alpha = \frac{\dfrac{1}{j\omega C}}{j\omega L + \dfrac{1}{j\omega C}} = \frac{1}{1 - LC\omega^2}.$$

For $|\alpha| < 1$ we must have $LC\omega^2 > 2$, that is,

$$\omega > \sqrt{\frac{2}{LC}}.$$

(2) According to the conditions of the problem, we must have

$$jL\omega + \frac{1}{j\omega C + \dfrac{1}{\tilde{Z}_0}} = \tilde{Z}_0,$$

that is,

$$\tilde{Z}_0^2 - jL\omega \tilde{Z}_0 - \frac{L}{C} = 0.$$

Solving gives

$$\tilde{Z}_0 = \frac{1}{2} jL\omega + \sqrt{\frac{L}{C} - \frac{\omega^2 L^2}{4}}.$$

When $\dfrac{L}{C} > \dfrac{\omega^2 L^2}{4}$, the quantity under the square root is real. Since the real part cannot be negative, only the "+" sign can be taken before the square root.

(3) Consider the propagation factor $\alpha_n = U_{n+1}/U_n$ between two adjacent sections (n being the serial number from left to right). Obviously,

$$\alpha_n = \frac{U_{n+1}}{U_n} = \frac{\tilde{Z}_0 - jL\omega}{\tilde{Z}_0}.$$

Substituting the expression for $\tilde{Z}_0$,

$$\alpha_n = \frac{\sqrt{\dfrac{L}{C} - \dfrac{\omega^2 L^2}{4}} - \dfrac{1}{2} jL\omega}{\sqrt{\dfrac{L}{C} - \dfrac{\omega^2 L^2}{4}} + \dfrac{1}{2} jL\omega}.$$

When $\dfrac{L}{C} > \dfrac{\omega^2 L^2}{4}$, i.e. $\omega < \dfrac{2}{\sqrt{LC}}$, $|\alpha_n| = 1$, and the electromagnetic disturbance passes through the ladder network without attenuation.

$$\therefore\ \omega_c = \frac{2}{\sqrt{LC}}.$$

(4) When $\omega < \omega_c$, we have

$$\alpha_n = \frac{\sqrt{\dfrac{L}{C} - \dfrac{\omega^2 L^2}{4}} - \dfrac{1}{2} jL\omega}{\sqrt{\dfrac{L}{C} - \dfrac{\omega^2 L^2}{4}} + \dfrac{1}{2} jL\omega} = \mathrm{e}^{-j2\delta_1} = \mathrm{e}^{-j\delta}.$$

Here

$$\tan\delta_1 = \frac{\dfrac{1}{2} L\omega}{\sqrt{\dfrac{L}{C} - \dfrac{\omega^2 L^2}{4}}}, \qquad \delta = 2\delta_1,$$

so

$$\tan\delta = \frac{2\tan\delta_1}{1 - \tan^2\delta_1} = \frac{\dfrac{L\omega}{\sqrt{\dfrac{L}{C} - \dfrac{\omega^2 L^2}{4}}}}{1 - \dfrac{\dfrac{1}{4}\omega^2 L^2}{\dfrac{L}{C} - \dfrac{\omega^2 L^2}{4}}} = \frac{L\omega\sqrt{\dfrac{L}{C} - \dfrac{\omega^2 L^2}{4}}}{\dfrac{L}{C} - \dfrac{1}{2}\omega^2 L^2}.$$

When $\omega \ll \omega_c$, $\dfrac{\omega^2 L^2}{4} \ll \dfrac{L}{C}$, and

$$\tan\delta \approx \frac{L\omega\sqrt{\dfrac{L}{C}}}{\dfrac{L}{C}} = \omega\sqrt{LC} \ll 1,$$

$$\therefore\ \delta \approx \omega\sqrt{LC}.$$

The time required for the disturbance to propagate from one section to the next is

$$\Delta t = \frac{\delta}{\omega} = \sqrt{LC}.$$

Hence the propagation speed of the signal along the network is

$$v = \frac{\Delta r}{\Delta t} = \frac{\Delta r}{\sqrt{LC}}.$$

Solution 3.57. Using the Y-Δ transformation, the original circuit can be converted into the circuit shown in Fig. 3.57(a), in which the unprimed quantities are the transformed complex impedances of the lower T-shaped circuit (two C's and one $R/2$) in the original circuit, and the primed quantities are the transformed complex impedances of the upper T-shaped circuit (two R's and one C') in the original circuit. By symmetry, $\tilde{Z}_2 = \tilde{Z}_3$ and $\tilde{Z}'_2 = \tilde{Z}'_3$. Moreover, the parallel combination of $\tilde{Z}_2$ and $\tilde{Z}'_2$ cannot be zero (otherwise the input would be short-circuited); hence, for the output voltage U_{out} to be zero, the parallel impedance of $\tilde{Z}_1$ and $\tilde{Z}'_1$ can only be made infinite. The parallel impedance of $\tilde{Z}_1$ and $\tilde{Z}'_1$ is

$$\tilde{Z} = \frac{\tilde{Z}_1 \tilde{Z}'_1}{\tilde{Z}_1 + \tilde{Z}'_1}.$$

For $\tilde{Z} \to \infty$, we can only have $\tilde{Z}_1 + \tilde{Z}'_1 = 0$.

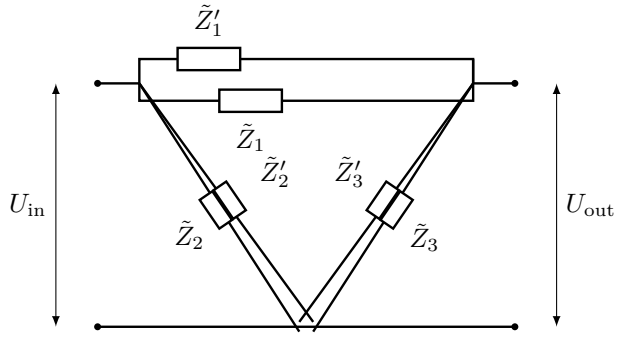


Fig. 3.57(a)

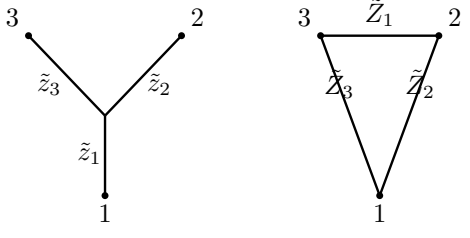


Fig. 3.57(b)

Corresponding to the d.c. Y-Δ transformation (Fig. 3.57(b)), we have

$$\begin{aligned}
\tilde{Z}_1 &= \frac{\tilde{z}_1\tilde{z}_2 + \tilde{z}_2\tilde{z}_3 + \tilde{z}_3\tilde{z}_1}{\tilde{z}_1} = \tilde{z}_2 + \tilde{z}_3 + \frac{\tilde{z}_2\tilde{z}_3}{\tilde{z}_1} \\
&= \frac{1}{j\omega C} + \frac{1}{j\omega C} + \frac{\dfrac{1}{j\omega C} \cdot \dfrac{1}{j\omega C}}{R/2} \\
&= -j\frac{2}{\omega C} - \frac{2}{R\omega^2 C^2}.
\end{aligned}$$

Similarly,

$$\tilde{Z}'_1 = R + R + \frac{R^2}{\dfrac{1}{j\omega C'}} = 2R + jR^2\omega C'.$$

For $\tilde{Z}_1 + \tilde{Z}_1' = 0$, we have

$$2R = \frac{2}{R\omega^2 C^2},$$

$$R^2 \omega C' = \frac{2}{\omega C}.$$

From these we obtain

$$\omega = \frac{1}{RC},$$

$$C' = 2C.$$

At this time,

$$\tilde{Z}_2 = \frac{\tilde{z}_1\tilde{z}_2 + \tilde{z}_2\tilde{z}_3 + \tilde{z}_3\tilde{z}_1}{\tilde{z}_2} = \tilde{z}_1 + \tilde{z}_3 + \frac{\tilde{z}_1\tilde{z}_3}{\tilde{z}_2}$$

$$= \frac{R}{2} + \frac{1}{j\omega C} + \frac{\dfrac{R}{2} \cdot \dfrac{1}{j\omega C}}{\dfrac{1}{j\omega C}} = R - j\frac{1}{\omega C} = \tilde{Z}_3.$$

Similarly,

$$\tilde{Z}_2' = \frac{1}{j\omega C'} + R + \frac{\dfrac{1}{j\omega C'} \cdot R}{R} = R - j\frac{2}{\omega C'} = \tilde{Z}_3'.$$

The parallel impedance of $\tilde{Z}_2$ and $\tilde{Z}_2'$, and that of $\tilde{Z}_3$ and $\tilde{Z}_3'$, are both

$$\frac{1}{2}\left(R - j\frac{2}{\omega C'}\right) \neq 0.$$

Solution 3.58. (1) The magnetic field of the incident wave must be along the y direction, and the magnetic field strength can be written as

$$\vec{H}_i(z,t) = \sqrt{\frac{\varepsilon_0}{\mu_0}} E_{i0} \cos(\omega t - kz)\, \hat{j}.$$

Here use has been made of

$$H = \frac{1}{\mu_0} B = \frac{1}{\mu_0 c} E = \sqrt{\frac{\varepsilon_0}{\mu_0}} E.$$

The reflected wave must propagate along the negative z direction and can be written as

$$\vec{E}_r(z,t) = E_{r0} \cos(\omega t + kz)\, \hat{i},$$

$$\vec{H}_r(z,t) = -H_{r0} \cos(\omega t + kz)\, \hat{j}.$$

The minus sign in front of the expression for H_r is due to the fact that the Poynting vector $\vec{S}_r = \vec{E}_r \times \vec{H}_r$ of the reflected wave must be along the negative z direction. After the incident and reflected waves are superposed at the conductor surface, since $E = 0$ inside an ideal conductor while the tangential component of the electric field must be continuous, we obtain

$$E_{i0} + E_{r0} = 0,$$

that is,

$$E_{r0} = -E_{i0}.$$

Hence the total electric field outside the conductor is

$$\begin{aligned} E_x(z,t) &= E_i(z,t) + E_r(z,t) \\ &= E_{i0}\cos(\omega t - kz) + E_{r0}\cos(\omega t + kz) \\ &= E_{i0}\cos(\omega t - kz) - E_{i0}\cos(\omega t + kz) \\ &= E_{i0}\big[(\cos\omega t\cos kz + \sin\omega t\sin kz) - (\cos\omega t\cos kz - \sin\omega t\sin kz)\big] \\ &= 2E_{i0}\sin kz\sin\omega t. \end{aligned}$$

The total magnetic field outside the conductor is then

$$\begin{aligned} H_y(z,t) &= H_i(z,t) + H_r(z,t) \\ &= H_{i0}\cos(\omega t - kz) - H_{r0}\cos(\omega t + kz) \\ &= \sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\cos(\omega t - kz) - \sqrt{\frac{\varepsilon_0}{\mu_0}}E_{r0}\cos(\omega t + kz) \\ &= \sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\cos(\omega t - kz) + \sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\cos(\omega t + kz) \\ &= \sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\big[(\cos\omega t\cos kz + \sin\omega t\sin kz) + (\cos\omega t\cos kz - \sin\omega t\sin kz)\big] \\ &= 2\sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\cos kz\cos\omega t. \end{aligned}$$

(2) On the two sides of the conductor surface H is discontinuous, which shows that there is a surface current on the conductor surface. Applying the Ampère circuital law to the loop *abcda* in the figure, we have

$$\oint \vec{H}\cdot \mathrm{d}\vec{l} = H\cdot\overline{ab} = 2\sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\cos\omega t\cdot\overline{ab} = i_c\cdot\overline{ab},$$

$$\therefore\ i_c = 2\sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\cos\omega t\,\hat{i},$$

with its direction along the x axis. The force acting on unit area of the conductor surface is

$$\begin{aligned} \vec{f} &= \vec{i}_c\times\bar{\vec{B}} = \vec{i}_c\times\mu_0\bar{\vec{H}} = \frac{1}{2}\vec{i}_c\times\mu_0\vec{H}_{\text{surface}} \\ &= \frac{1}{2}\cdot 2\sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\cos\omega t\,\hat{i}\times\mu_0\cdot 2\sqrt{\frac{\varepsilon_0}{\mu_0}}E_{i0}\cos\omega t\,\hat{j} \\ &= 2\varepsilon_0 E_{i0}^2\cos^2\omega t\,\hat{k}. \end{aligned}$$

Because the average magnetic field acting on the current should be half the magnetic field at the conductor surface, the expression for $\vec{f}$ has an extra factor "$\frac{1}{2}$" in front. Since the time average of $\cos^2\omega t$ equals $1/2$, the average value of $\vec{f}$ is

$$\begin{aligned} \bar{f} &= 2\varepsilon_0 E_{i0}^2\overline{\cos^2\omega t}\,\hat{k} \\ &= \varepsilon_0 E_{i0}^2\hat{k}. \end{aligned}$$

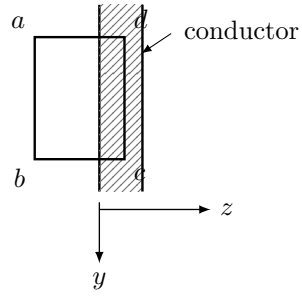


Fig. 3.58

Comment. In this problem, since the electromagnetic wave is completely reflected by the metal surface, the force per unit area on the conductor surface equals twice the product of the average momentum density G of the electromagnetic field and the speed of light c, i.e. $\bar{f} = 2Gc$; it also equals twice the ratio of the Poynting vector to the speed of light, i.e. $\bar{f} = 2S/c$, and is numerically exactly twice the average energy density of the electromagnetic field.

3.4 Optics

Note. The copy of the Chinese original used for this edition is defective in this chapter: book pages 249–255 are missing, so Solutions 4.03–4.09 could not be reproduced, and Solution 4.02 is preserved only as far as the surviving page allows.

Solution 4.01. (1) δ should be the sum of the angles of deviation produced in the two refractions. From Fig. 4.01-1,

$$\delta = i - r + r' - i'.$$

From the geometry,

$$r + i' = A. \tag{1}$$

Hence

$$\delta = i + r' - A. \tag{2}$$

By the law of refraction,

$$\sin i = n \sin r, \tag{3}$$
$$\sin r' = n \sin i'. \tag{4}$$

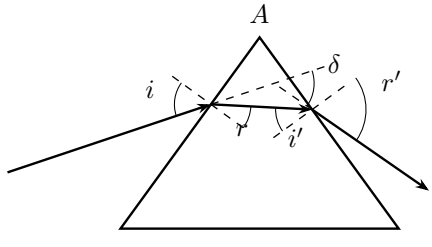


Fig. 4.01-1

From (1), $i' = A - r$; substituting this into (4) and using (3), we have

$$\begin{aligned}\sin r' &= n\sin(A-r) = n\left(\sin A\cos r - \cos A\sin r\right)\\ &= \sin A\sqrt{n^2-\sin^2 i} - \cos A\sin i.\end{aligned}$$

Substituting this into (2), we obtain

$$\delta = i + \arcsin\left(\sin A\sqrt{n^2-\sin^2 i} - \cos A\sin i\right) - A.$$

(2) For δ to be minimum we must have $\mathrm{d}\delta/\mathrm{d}i = 0$. Differentiating (2) and setting the derivative equal to zero,

$$\frac{\mathrm{d}\delta}{\mathrm{d}i} = 1 + \frac{\mathrm{d}r'}{\mathrm{d}i} = 0,$$

we obtain

$$\frac{\mathrm{d}r'}{\mathrm{d}i} = -1. \tag{5}$$

Differentiating (3) and (4),

$$\begin{aligned}\cos i\,\mathrm{d}i &= n\cos r\,\mathrm{d}r,\\ \cos r'\,\mathrm{d}r' &= n\cos i'\,\mathrm{d}i'.\end{aligned}$$

Dividing the first of these by the second, and using the result $\mathrm{d}r = -\mathrm{d}i'$ obtained by differentiating (1) together with (5), we obtain

$$\frac{\cos i}{\cos r} = \frac{\cos r'}{\cos i'}.$$

Substituting (3) and (4) into this, we obtain

$$\frac{1-\sin^2 i}{1-\sin^2 r'} = \frac{n^2-\sin^2 i}{n^2-\sin^2 r'}.$$

Since $n \neq 1$, for the above equation to hold we must have

$$i = r',$$

and hence

$$r = i'.$$

It is thus seen that when the incident and emergent rays are symmetric, the deviation δ takes an extremal value δ_m. Further verification shows that δ then takes its minimum value, i.e. $\delta_m = \delta_{\min}$.

In the case $\delta = \delta_m$, from (1) and (2),

$$\begin{aligned}r &= i' = A/2,\\ i &= r' = (\delta_m + A)/2.\end{aligned}$$

Substituting into (3), we obtain

$$n = \frac{\sin\left[(\delta_m + A)/2\right]}{\sin(A/2)}, \tag{6}$$

or

$$\delta_m = 2\arcsin\left(n\sin\frac{A}{2}\right) - A.$$

Comment. In experiments the minimum angle of deviation can be measured precisely with ease, so equation (6) is often used to determine the refractive index of the material of a prism.

If it is only required to prove that the deviation takes an extremal value when the light path is symmetric, a simpler method can also be used; it is described briefly below.

Let the symmetric path be 1–1′, as shown in Fig. 4.01-2. For this path the angle of incidence and the angle of emergence are equal ($i = r'$); let its deviation be δ. Now let the light be incident at a slightly larger angle of incidence; then its angle of emergence is slightly smaller. Denote this path by 2–2′ and its deviation by $\delta' \neq \delta$. By the principle of reversibility of the light path, the light can also travel in the reverse direction along the same path 2′–2. This is equivalent to reducing the angle of incidence of the light relative to its value for the symmetric path (and increasing the angle of emergence), while the deviation of 2′–2 remains δ'. Since the deviation is the same whether the angle of incidence is increased or decreased relative to its value for the symmetric path, it is evident that the deviation of the symmetric path must be extremal.

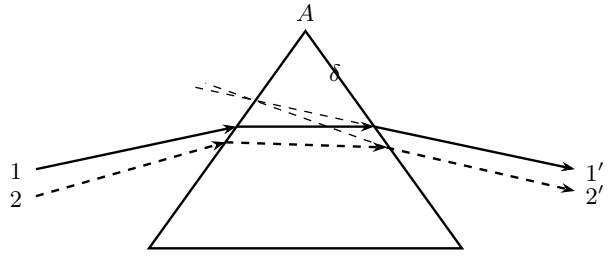


Fig. 4.01-2

Solution 4.02. (1) From symmetry it is known that a ray of light travelling in an isosceles prism must be parallel to the base, as shown in Fig. 4.02-1. Applying the law of refraction at the first and the second interface, we have

$$\sin i_1 = n_1 \sin r_1, \tag{1}$$

$$n_1 \sin i_2 = n_2 \sin r_2. \tag{2}$$

From the geometry,

$$i_1 = \frac{\pi}{2} - \frac{\alpha}{2},$$

$$i_2 = \frac{\pi}{2} - r_1,$$

$$r_2 = \frac{\alpha}{2}.$$

Substituting these into (1) and (2), we obtain

$$\cos\frac{\alpha}{2} = n_1 \sin r_1,$$

$$n_1 \cos r_1 = n_2 \sin\frac{\alpha}{2}.$$

From these two equations we can solve for

$$\sin\frac{\alpha}{2} = \sqrt{\frac{n_1^2 - 1}{n_2^2 - 1}},$$

that is,

$$\alpha = 2\arcsin\sqrt{\frac{n_1^2 - 1}{n_2^2 - 1}}. \tag{3}$$

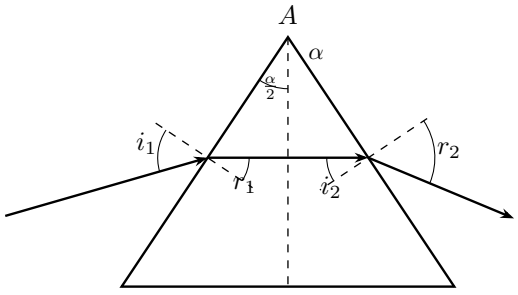


Fig. 4.02-1

(2) From the table, for the F line,

$$n_1^F = 1.515\,65, \qquad n_2^F = 1.774\,75.$$

Substituting into (3), we obtain

$$\alpha = 2\arcsin\sqrt{\frac{1.515\,65^2 - 1}{1.774\,75^2 - 1}} = 101.94^\circ.$$

For the C line and the G line,

$$n_1^C = 1.507\,60, \qquad n_2^C = 1.747\,32,$$
$$n_1^G = 1.520\,16, \qquad n_2^G = 1.792\,05.$$

The angles of refraction are calculated one after another and listed in the table below. The last column gives the deviation δ of the emergent light with respect to the incident light, taken positive in the anticlockwise sense.

Solution 4.11. (1) Consider an arbitrary ray PA issuing from P, and let the angle it makes with the principal axis be θ, as shown in Fig. 4.11-1. Let the angle of incidence of the ray at the spherical surface be i and the angle of refraction be r. By the law of refraction,

$$\frac{\sin i}{\sin r} = \frac{1}{n}. \tag{1}$$

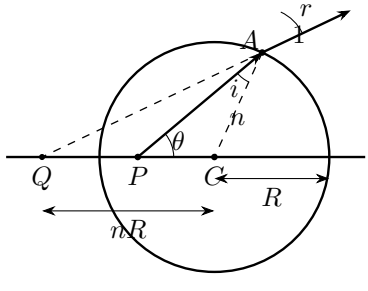


Fig. 4.11-1

In $\triangle PCA$, applying the sine rule, we have

$$\frac{\sin i}{R/n} = \frac{\sin\theta}{R},$$

or

$$\frac{\sin i}{\sin \theta} = \frac{1}{n}. \tag{2}$$

Comparing (1) and (2), in the case where θ is an acute angle (r must be acute), we have

$$\theta = r.$$

But $r = \angle QAC$, where Q is the point of intersection of the backward extension of the refracted ray with the principal axis; hence $\triangle QAC \sim \triangle APC$, and $\overline{QC}/R = R/(R/n) = n$, i.e.

$$\overline{QC} = nR,$$

independently of the angle θ, which was to be proved.

(2) The geometrical configuration of the two lenses is shown in Fig. 4.11-2.

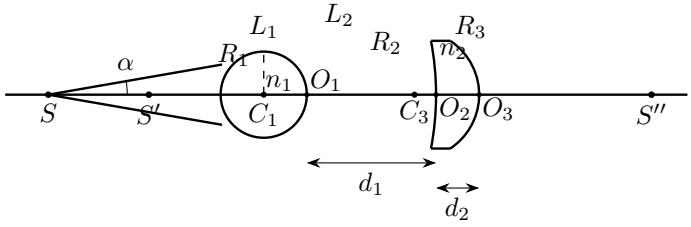


Fig. 4.11-2

(i) Let C_1 be the centre of the sphere of the objective lens (L_1). The light emitted from S (the object) is refracted by L_1 and forms an image at S'. From part (1) of this problem, $\overline{S'C_1} = n_1R_1$, so

$$\overline{S'O_1} = n_1R_1 + R_1 = (n_1 + 1)R_1.$$

The light emitted from S' should enter the front spherical surface of the second lens (L_2) without refraction. Hence S' is the centre of the front spherical surface, and the radius of the front spherical surface is

$$R_2 = \overline{S'O_2} = \overline{S'O_1} + d_1 = (n_1 + 1)R_1 + d_1.$$

In order that S' be located at the aplanatic point of L_2, and that the central thickness of L_2 be d_2, we must have

$$\overline{S'O_3} = R_3 + \frac{R_3}{n_2}$$

and

$$\overline{S'O_3} = R_2 + d_2 = (n_1 + 1)R_1 + d_1 + d_2.$$

From these two equations we can solve for the radius of the back spherical surface:

$$R_3 = \frac{(n_1 + 1)R_1 + d_1 + d_2}{1 + \dfrac{1}{n_2}}.$$

In Fig. 4.11-2, S'' is the image of S' formed by L_2 [$\overline{S''O_3} = (n_2 + 1)R_3$], and C_3 is the centre of the back spherical surface.

(ii) The aperture angle α of the light emitted from S satisfies

$$\tan \alpha = \frac{R_1}{R_1/n_1} = n_1.$$

Let the aperture angle of the image S' formed by S through L_1 be β_1; then

$$\tan\beta_1 = \frac{R_1}{n_1 R_1} = \frac{1}{n_1}.$$

β_1 is also the aperture angle of the object point of L_2, and β is the aperture angle of the final image S''. From Fig. 4.11-2 it is easily seen that

$$\tan\beta = \frac{\overline{S'A}\sin\beta_1}{\overline{S'A}\cos\beta_1 + \overline{S'S''}} \approx \frac{R_2\sin\beta_1}{R_2\cos\beta_1 + \overline{S'S''}} \qquad \text{(the edge thickness of the lens being neglected).}$$

Moreover,

$$\overline{S'S''} = n_2 R_3 - \frac{1}{n_2} R_3,$$

therefore

$$\tan\beta = \frac{R_2\sin\beta_1}{R_2\cos\beta_1 + \left(n_2 - \dfrac{1}{n_2}\right) R_3}.$$

(iii) $k = k_1 \cdot k_2 = \dfrac{n_1 R_1}{R_1/n_1} \cdot \dfrac{n_2 R_2}{R_2/n_2} = n_1^2 \cdot n_2^2$.

(iv) Substituting the data, we obtain $R_2 = 9.5\,\text{mm}$, $R_3 = 6.77\,\text{mm}$, $\beta = 20.0°$, $k = 5.76$.

Comment. This problem is an application of the concept of aplanatic points to the microscope objective. Such an objective is called an oil-immersion objective and is one of the main types of microscope objective.

Solution 4.12. (1) From Fig. 4.12-1, the deviation of the light emitted from S after two refractions by the prism is

$$\delta = i - r + r' - i' = i + r' - (r + i'). \tag{1}$$

But

$$r + i' = \alpha. \tag{2}$$

From the statement of the problem, i is a small angle, hence r, i' and r' are also small angles; therefore, by the law of refraction,

$$i = nr, \qquad r' = ni'.$$

Substituting these into (1) and using (2), we obtain

$$\begin{aligned} \delta &= nr + ni' - (r + i') \\ &= (n-1)\alpha. \end{aligned} \tag{3}$$

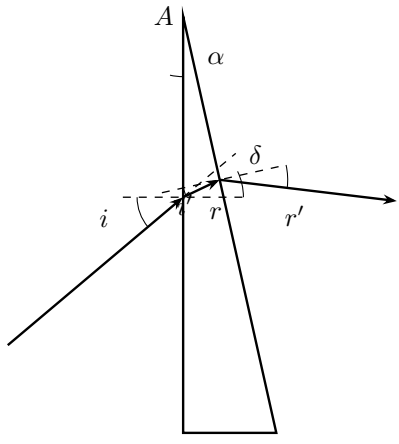


Fig. 4.12-1

Since δ is independent of the angle of incidence, the backward extensions S_1A and S_1B of any two rays SA and SB emitted from S with an angle $\Delta\theta$ between them, after refraction through the upper half of the prism, still make the angle $\Delta\theta$ (see Fig. 4.12-2), and the arc which they subtend is the same arc AB; hence the distance of S_1 (i.e. the image of S) from the prism is also a. In the same way, the image S_2 formed by S through refraction at the lower half of the prism is also at a distance a from the prism. Thus (using (3)) the distance between the two coherent sources is

$$\overline{S_1S_2} = 2a\delta = 2a(n-1)\alpha.$$

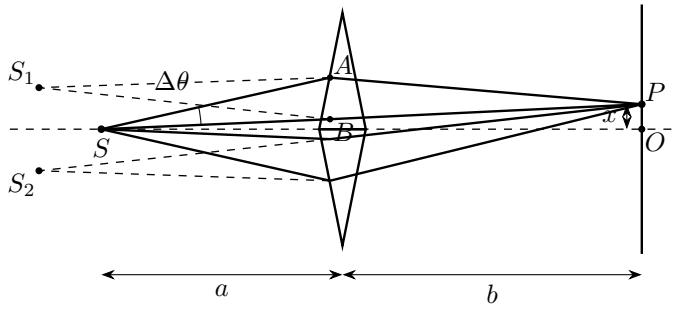


Fig. 4.12-2

It is not difficult to see that the optical path difference from S_1 and S_2 to the point P (the distance of P from the central point O being x) is

$$\Delta = \overline{S_1S_2} \cdot \frac{x}{a+b} = \frac{2a(n-1)\alpha x}{a+b}. \tag{4}$$

Setting $\Delta = m\lambda$, we obtain the distance x of the m-th bright fringe:

$$x = \frac{m\lambda(a+b)}{2a(n-1)\alpha}.$$

(2) When S is moved up to S', imagine that a thin plate of thickness d is cut away from the biprism along the dashed line; then the zero-order fringe would move up to O' (Fig. 4.12-3). The

presence of the thin plate of thickness d produces for S' an additional optical path difference $(n-1)d$ in the lower half of the prism, so that the zero-order fringe moves down from O' by a distance y', and [cf. (4)]

$$\frac{2a(n-1)\alpha y'}{a+b} = (n-1)d.$$

From the geometrical relation $d = (h/\cos\alpha)\cdot\sin 2\alpha \approx 2h\alpha$; substituting this into the above equation gives $y' = (a+b)h/a$. Hence the upward displacement of S to S' finally makes the zero-order fringe move down from the point O by a distance

$$y = y' - h = \frac{b}{a}h.$$

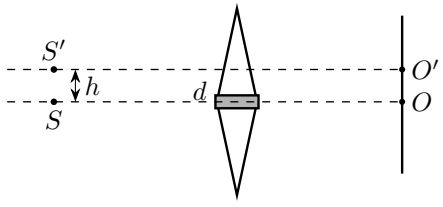


Fig. 4.12-3

Comment. There are three keys to the solution of this problem. First, to prove that the light emitted from the source S is deviated through a small angle $(n-1)\alpha$ after refraction by the prism; second, to show that the image of S formed by refraction through the prism is still at a distance a from the prism, and thereby to obtain the distance between the two coherent sources; finally, to find the distance through which the zero-order fringe moves down when the source moves up, by imagining a thin plate cut away from the prism.

Solution 4.13. (1) From the conditions of the problem the geometrical light path is as shown in Fig. 4.13(a). On the screen M one obtains the interference fringes of the parallel beam 1, formed from the light of S after refraction by L_1, and of the divergent beam 2, emitted from the image point S' after refraction and imaging by L_2; interference occurs only in the region above the optical axis, so the fringes are concentric semicircles centred on the optical axis, as shown on the right of Fig. 4.13(a). At the centre O the optical path difference of the two beams is originally zero, but the converging beam 2 undergoes a phase jump of π when it passes through its focal point S', so that the optical path difference of the two beams at the centre O is $\lambda/2$; the centre is therefore dark.

The occurrence of a π phase jump when a converging beam passes through its focal point can be explained as follows. For the divergent wave emitted from the point S (Fig. 4.13(b)), in order to obtain by Fresnel's half-period-zone method the correct phase of the wave disturbance arriving at P, one must take the phase of every secondary wavelet emitted from the wavefront Σ to be ahead of the phase of the wave arriving at Σ from S by $\pi/2$. For the converging wave Σ' focused at S', when the observation point P lies between Σ' and S' (Fig. 4.13(c)(1)), it is enough to assume that the phase of the secondary wavelets on Σ' is ahead of the phase of the wave on Σ' by $\pi/2$; the correct phase of the wave disturbance at P is still obtained, because the distances from the half-period zones on Σ' to P still increase gradually. But when P lies outside Σ' and S' (Fig. 4.13(c)(2)), since, counting from the first half-period zone, the distances from the successive half-period zones to P decrease one after another, their phases advance one after another; adding the $\pi/2$ by which the wavelets on Σ' were originally ahead, the phase of the wave disturbance at P will differ by π from the phase obtained from Σ' together with the optical path. This is the origin of the π phase jump.

(2) The two beams have equal optical paths at S'. At a point P of the screen M at a distance x from O, beam 1 has in addition traversed the optical path $S'O$, while beam 2 has traversed $S'P$. Taking further into account the π phase jump of beam 2 at S', the optical path difference of the two beams at P is

$$\Delta = S'P + \frac{\lambda}{2} - S'O = \sqrt{f^2 + x^2} + \frac{\lambda}{2} - f.$$

Putting $\Delta = m\lambda$, we obtain the position of the m-th bright fringe:

$$x = \sqrt{\left(m - \frac{1}{2}\right)^2 \lambda^2 + 2\left(m - \frac{1}{2}\right)\lambda f}.$$

It is seen that the interference fringes become denser towards the outside.

Comment. The π phase jump undergone by light on passing through a focus is easily overlooked, and special attention should be paid to it.

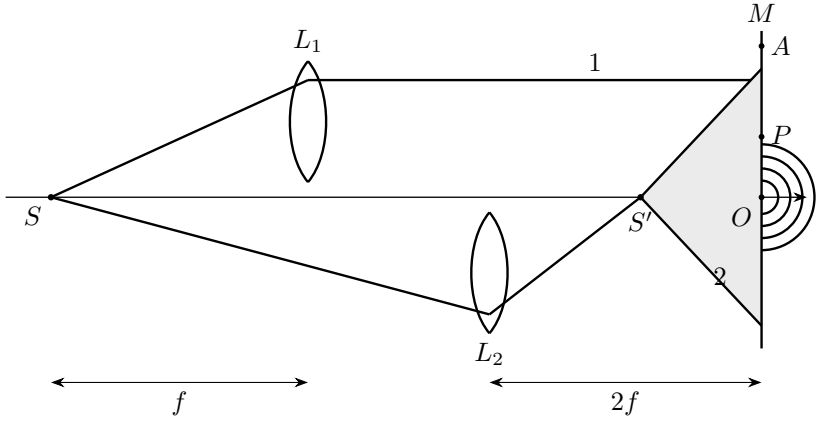


Fig. 4.13(a)

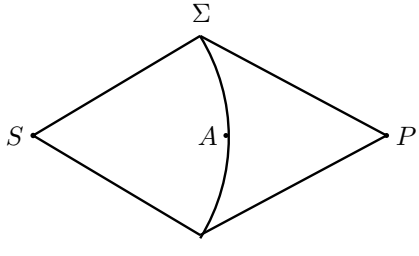


Fig. 4.13(b)

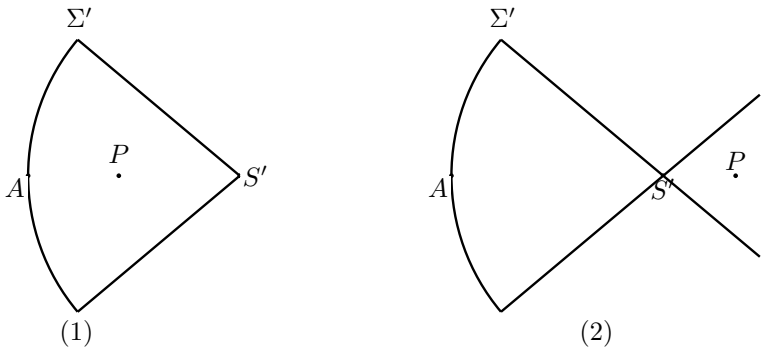


Fig. 4.13(c)

Solution 4.14. (1) If the amplitudes of the incident, reflected and transmitted waves are as shown in Fig. 4.14(a)(1) (the amplitude of the incident wave has already been set equal to 1), then time-reversal invariance requires that the situation of Fig. 4.14(a)(2) is also possible. The superposition principle, on the other hand, allows the situation of Fig. 4.14(a)(2) to be regarded as the sum of Fig. 4.14(a)(3) and Fig. 4.14(a)(4). Making the incident and emergent waves of this sum equal to the incident and emergent waves of Fig. 4.14(a)(2) gives the Stokes relations

$$\begin{aligned} r^2 + tt' &= 1, \\ r' + r &= 0. \end{aligned}$$

(2) Let A denote the amplitude of the incident wave I, and let r_1, r_1', t, t' denote respectively the (amplitude) reflection and transmission coefficients at the first interface, while r_2 denotes the reflection coefficient at the second interface. Then the amplitudes of the reflected waves $R_0, R_1, R_2, R_3, \ldots$ (Fig. 4.14(b); for clarity the incidence is drawn obliquely) are as given in the following table:

R_0	R_1	R_2	R_3	R_4	$\cdots$
Ar_1	$Att'r_2$	$Att'r_2r_1'r_2$	$Att'r_2(r_1'r_2)^2$	$Att'r_2(r_1'r_2)^3$	$\cdots$

Taking into account the phase changes of the various reflected waves, the total amplitude of the wave reflected into vacuum is

$$\begin{aligned} A_R &= Ar_1 + Att'r_2\mathrm{e}^{-2\mathrm{i}kd}\left[1 + r_1'r_2\mathrm{e}^{-2\mathrm{i}kd} + (r_1'r_2)^2\mathrm{e}^{-4\mathrm{i}kd} + \cdots\right] \\ &= Ar_1 + \frac{Att'r_2\mathrm{e}^{-2\mathrm{i}kd}}{1 - r_1'r_2\mathrm{e}^{-2\mathrm{i}kd}} \\ &= \frac{A\left(r_1 - r_1r_1'r_2\mathrm{e}^{-2\mathrm{i}kd} + tt'r_2\mathrm{e}^{-2\mathrm{i}kd}\right)}{1 - r_1'r_2\mathrm{e}^{-2\mathrm{i}kd}} \\ &= \frac{A\left[r_1 + r_2\mathrm{e}^{-2\mathrm{i}kd}(tt' - r_1r_1')\right]}{1 - r_1'r_2\mathrm{e}^{-2\mathrm{i}kd}}. \end{aligned}$$

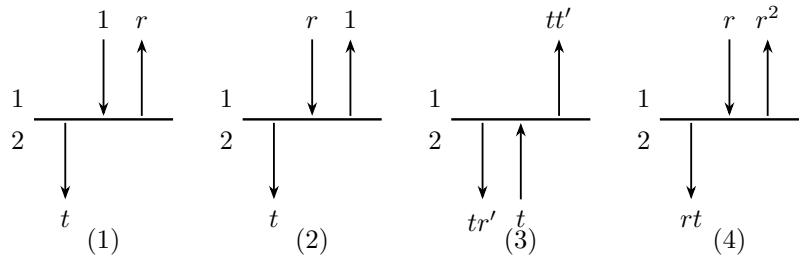


Fig. 4.14(a)

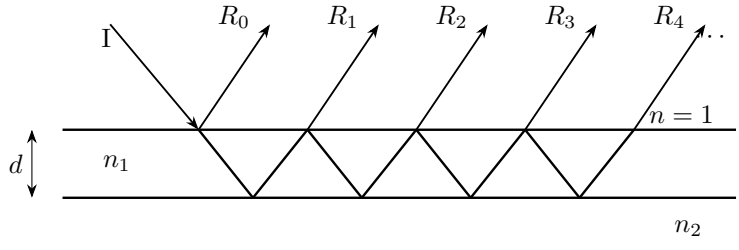


Fig. 4.14(b)

Here $k = 2\pi n_1/\lambda$, λ being the wavelength in vacuum. The factor $\mathrm{e}^{-2\mathrm{i}kd}$ is the phase change of the wave after it has traversed a distance $2d$ in the medium of refractive index n_1. Using the Stokes relations derived in (1), we obtain

$$\frac{A_R}{A} = \frac{r_1 + r_2\mathrm{e}^{-2\mathrm{i}kd}}{1 + r_1 r_2 \mathrm{e}^{-2\mathrm{i}kd}}.$$

From the above expression, the condition for the reflected wave to vanish is

$$1) \quad \begin{cases} r_1 = -r_2, \\ \mathrm{e}^{-2\mathrm{i}kd} = 1, \end{cases} \qquad \text{or} \qquad 2) \quad \begin{cases} r_1 = r_2, \\ \mathrm{e}^{-2\mathrm{i}kd} = -1. \end{cases}$$

Let us connect the coefficients r_1, r_2 with the refractive indices. By the Fresnel formulae,

$$r_1 = \frac{1 - n_1}{1 + n_1}, \qquad r_2 = \frac{n_1 - n_2}{n_1 + n_2}.$$

Expressed in terms of the refractive indices, the condition for the reflected wave to vanish becomes

$$1) \quad \begin{cases} n_2 = 1, \\ \dfrac{n_1 d}{\lambda} = \dfrac{p}{2}, \end{cases} \qquad \text{or} \qquad 2) \quad \begin{cases} n_2 = n_1^2, \\ \dfrac{n_1 d}{\lambda} = \dfrac{2p-1}{4}, \end{cases}$$

where p is a positive integer. The physical meaning of the above conditions is evident.

Comment. This problem is precisely the physical principle of the antireflection coating. The result agrees with that obtained by considering only the interference of the two reflected waves from the upper and lower surfaces. This is because at normal incidence the reflection coefficients are very small, so that apart from these first two reflected waves, which are relatively strong, all the remaining reflected waves are very weak.

Solution 4.15. The speed of light in the lower tube is c/n, while its speed in the upper tube (relative to the laboratory frame) is given by the relativistic velocity-addition formula as

$$v' = \frac{\dfrac{c}{n} + u}{1 + \dfrac{1}{c^2}\dfrac{c}{n}u} = \frac{\dfrac{c}{n} + u}{1 + \dfrac{u}{cn}}.$$

Hence the equivalent refractive index of the upper tube is

$$\begin{aligned} n' = \frac{c}{v'} &= \frac{c}{\dfrac{c}{n} + u}\left(1 + \frac{u}{cn}\right) = \frac{n}{1 + \dfrac{nu}{c}}\left(1 + \frac{u}{cn}\right) \\ &\approx n\left(1 + \frac{u}{cn}\right)\left(1 - \frac{u}{cn}\right) \approx n - \frac{u}{c}(n^2 - 1). \end{aligned}$$

Therefore the optical path difference of the light in the upper and lower tubes is

$$\Delta = L(n - n') = \frac{u}{c}(n^2 - 1)L.$$

When $\Delta = (2p-1)\frac{\lambda}{2}$ (p a positive integer), destructive interference occurs at S'. Setting $p = 1$, we obtain the minimum value of u:

$$u_{\min} = \frac{1}{2}\frac{\lambda}{L}\frac{c}{n^2 - 1}.$$

Solution 4.16. Let $\delta = \frac{2\pi}{\lambda} d \sin\theta$ be the phase difference between the light rays emitted from two adjacent slits in the direction θ, and let a be the amplitude coming from a single slit in that direction. Then the total amplitude is

$$A = a\left[(1+\alpha) + (1-\alpha)\mathrm{e}^{\mathrm{i}\delta} + (1+\alpha)\mathrm{e}^{\mathrm{i}2\delta} + (1-\alpha)\mathrm{e}^{\mathrm{i}3\delta} + \cdots + (1-\alpha)\mathrm{e}^{\mathrm{i}(N-1)\delta}\right].$$

Here we take N to be even. After rearranging, we have

$$\begin{aligned} A &= a\left[\sum_{n=0}^{N-1}\mathrm{e}^{\mathrm{i}n\delta} + \alpha\sum_{n=0}^{N-1}(-1)^n\mathrm{e}^{\mathrm{i}n\delta}\right] \\ &= a\left[\frac{1-\mathrm{e}^{\mathrm{i}N\delta}}{1-\mathrm{e}^{\mathrm{i}\delta}} + \alpha\frac{1-\mathrm{e}^{\mathrm{i}N\delta}}{1+\mathrm{e}^{\mathrm{i}\delta}}\right] \\ &= a\,\frac{\mathrm{e}^{\mathrm{i}N\delta/2}}{\mathrm{e}^{\mathrm{i}\delta/2}}\left[\frac{\mathrm{e}^{-\mathrm{i}N\delta/2} - \mathrm{e}^{\mathrm{i}N\delta/2}}{\mathrm{e}^{-\mathrm{i}\delta/2} - \mathrm{e}^{\mathrm{i}\delta/2}} + \alpha\frac{\mathrm{e}^{-\mathrm{i}N\delta/2} - \mathrm{e}^{\mathrm{i}N\delta/2}}{\mathrm{e}^{-\mathrm{i}\delta/2} + \mathrm{e}^{\mathrm{i}\delta/2}}\right] \\ &= a\mathrm{e}^{\mathrm{i}(N-1)\delta/2}\left[\frac{\sin(N\delta/2)}{\sin(\delta/2)} - \mathrm{i}\alpha\frac{\sin(N\delta/2)}{\cos(\delta/2)}\right]. \end{aligned}$$

The intensity is proportional to

$$AA^* = a^2\left[\frac{\sin^2(N\delta/2)}{\sin^2(\delta/2)} + \alpha^2\frac{\sin^2(N\delta/2)}{\cos^2(\delta/2)}\right].$$

Noting that $I_0 \propto N^2a^2$ is the central maximum, the above expression can be written as

$$I = I_0\,\frac{\sin^2(N\delta/2)}{N^2\sin^2(\delta/2)}\left[1 + \alpha^2\tan^2\frac{\delta}{2}\right].$$

When $\delta \approx (2p+1)\pi$ (p an integer), the effect of α is most easily detected.

Solution 4.17. The displacement of the i-th step of the drunkard can be represented by the vector $\vec{L}_i$. After a certain time, the vector $\vec{R}$ from the hotel to the drunkard is the sum of many vectors $\vec{L}_i$ of various directions:

$$\vec{R} = \sum_{i=1}^{N} \vec{L}_i .$$

N is the number of steps taken by the drunkard during this time and is a large number. What we require is actually the modulus of the vector $\vec{R}$. Taking the hotel as the origin and choosing x, y coordinates arbitrarily, we have

$$\vec{L}_i = L\cos\alpha_i\,\hat{x} + L\sin\alpha_i\,\hat{y},$$

where α_i is the angle between $\vec{L}_i$ and the x-axis (as shown in the figure). Then

$$\begin{aligned} R^2 = R_x^2 + R_y^2 &= \left(\sum_{i=1}^{N} L\cos\alpha_i\right)^2 + \left(\sum_{i=1}^{N} L\sin\alpha_i\right)^2 \\ &= \sum_{i=1}^{N} L^2\cos^2\alpha_i + \sum_{i\neq j} L^2\cos\alpha_i\cos\alpha_j + \sum_{i=1}^{N} L^2\sin^2\alpha_i + \sum_{i\neq j} L^2\sin\alpha_i\sin\alpha_j \\ &= L^2\sum_{i=1}^{N}\left(\cos^2\alpha_i + \sin^2\alpha_i\right) + L^2\sum_{i\neq j}\left(\cos\alpha_i\cos\alpha_j + \sin\alpha_i\sin\alpha_j\right). \end{aligned}$$

Since the α_i are arbitrary and N is very large, the second term above evidently vanishes through the cancellation of positive and negative contributions, while the first term is evidently NL^2.

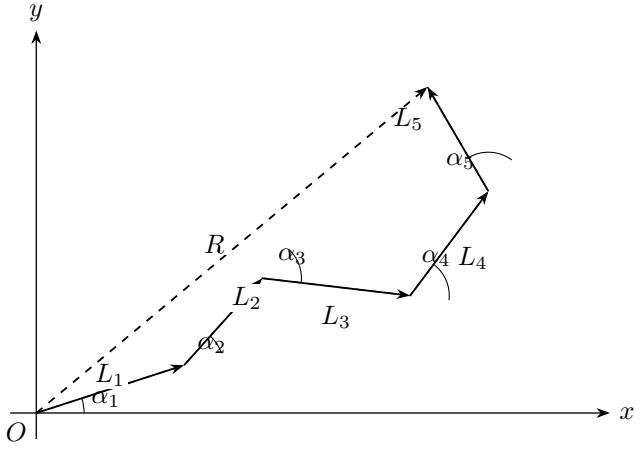


Fig. 4.17

Hence

$$R^2 = NL^2,$$

so that

$$R = \sqrt{N}L.$$

If the drunkard takes the same time for every step, then $N \propto t$, and hence

$$R \propto \sqrt{t}\, L,$$

that is, the distance between the drunkard and the hotel is proportional to the square root of the time.

Comment. The physical ideas and methods of this problem are manifested in many fields of physics. In optics it corresponds to the superposition of N light waves whose phases vary at random, the result being the absence of any interference phenomenon; in molecular physics it is the physical basis of Brownian motion.

Solution 4.18. Let O be the geometrical focus of the rays (as shown in the figure). Let M be the plane through O perpendicular to the principal axis of the lens, and let r_0 be the shortest distance from the lens to the point O. Draw a spherical surface S with centre O and radius r_0, its edge being defined by the aperture stop. When calculating the light field on the surface S, geometrical optics may be applied approximately. In this way the light field on S can be written in complex form as

$$E_S = \frac{1}{r_0} \mathrm{e}^{\mathrm{i}(\omega t + k r_0)}.$$

Let P be an observation point on the plane M, $\mathrm{d}S$ an area element on S, and r the distance between $\mathrm{d}S$ and P. According to Huygens' principle, the light field at the point P can be obtained from the following formula:

$$E_P = \int_S \frac{1}{r_0 r} \mathrm{e}^{\mathrm{i}[\omega t - k(r - r_0)]}\, \mathrm{d}S. \tag{1}$$

Putting $OP = \rho$, we have $\vec{r} = \vec{r}_0 + \vec{\rho}$, so that $r^2 = r_0^2 + \rho^2 + 2\vec{\rho} \cdot \vec{r}_0$ and $r = r_0 \sqrt{1 + \frac{\rho^2}{r_0^2} + \frac{2\vec{\rho} \cdot \vec{r}_0}{r_0^2}}$. Since ρ is a small quantity, Newton's binomial formula is applicable, and to accuracy of order ρ^2,

$$r - r_0 = \vec{\rho} \cdot \hat{r}_0 + \frac{\rho^2}{2r_0} - \frac{1}{2} \frac{(\vec{\rho} \cdot \vec{r}_0)^2}{r_0^3}. \tag{2}$$

When $\rho \ll r$, r_0, to first approximation,

$$r - r_0 = \vec{\rho} \cdot \hat{r}_0,$$

where $\hat{r}_0$ is the unit vector in the direction of $\vec{r}_0$. The error permitted in calculating $(r - r_0)$ must be much smaller than the wavelength; from (2), $r - r_0 - \vec{\rho} \cdot \vec{r}_0 \approx \rho^2 / r_0$, so when calculating $(r - r_0)$ the following condition must be satisfied:

$$\rho^2 / r_0 \ll \lambda. \tag{3}$$

We now calculate the light field in the neighbourhood of O under condition (3). Introduce rectangular coordinates in the observation plane, with the origin at O, the coordinate axes x and y parallel to adjacent sides of the square aperture, and the optical axis as the z-axis. Let x and y denote the coordinates of P. The direction of the unit vector $\hat{r}_0$ can be expressed by the angles φ and ψ, where φ and ψ are respectively the angles made by r_0 with the yz-plane and the xz-plane. Thus

$$r - r_0 = \vec{\rho} \cdot \hat{r}_0 = x \sin\varphi + y \sin\psi \approx x\varphi + y\psi,$$

$$\mathrm{d}S = r_0^2\, \mathrm{d}\varphi\, \mathrm{d}\psi.$$

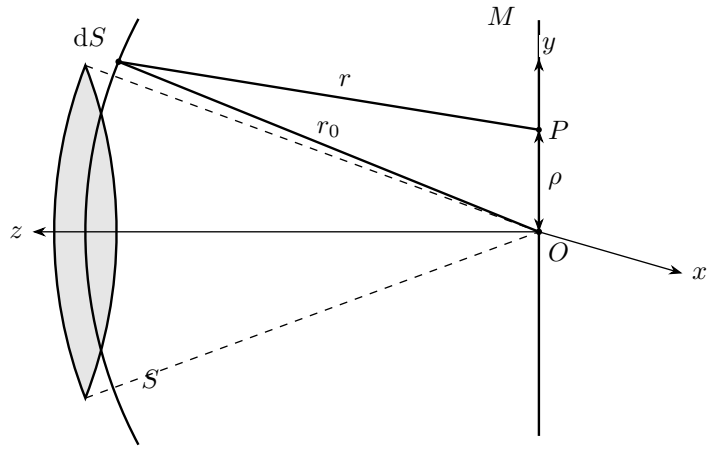


Fig. 4.18

We assume that the half-angle subtended at O by the square aperture, $\alpha = \arctan \frac{D}{2r_0}$, is very small, so that $\sin\varphi$ and $\sin\psi$ may be replaced by φ and ψ. Apart from this, since $\rho \ll r$, r_0, the quantity r in the denominator of (1) may be replaced approximately by r_0. Finally we obtain

$$E_P = \mathrm{e}^{\mathrm{i}\omega t} \int_{-\alpha}^{+\alpha}\int_{-\alpha}^{+\alpha} \mathrm{e}^{-\mathrm{i}k(x\varphi+y\psi)}\,\mathrm{d}\varphi\,\mathrm{d}\psi.$$

Integrating the above expression, taking the amplitude at O as unity and noting that $k = \frac{2\pi}{\lambda}$, we obtain the amplitude at P:

$$A_P = \frac{\sin\dfrac{2\pi x\alpha}{\lambda}}{\dfrac{2\pi x\alpha}{\lambda}} \cdot \frac{\sin\dfrac{2\pi y\alpha}{\lambda}}{\dfrac{2\pi y\alpha}{\lambda}},$$

and the intensity is

$$I_P = \left[\frac{\sin\dfrac{2\pi x\alpha}{\lambda}}{\dfrac{2\pi x\alpha}{\lambda}}\right]^2 \left[\frac{\sin\dfrac{2\pi y\alpha}{\lambda}}{\dfrac{2\pi y\alpha}{\lambda}}\right]^2.$$

The diffraction pattern obtained is the same as the Fraunhofer diffraction pattern of a square aperture. The distance from the centre of the central maximum to the first minimum, and the distance between two adjacent minima, are both equal to

$$\Delta x = \Delta y = \frac{\lambda/2}{\alpha}.$$

It remains to check whether condition (3) is satisfied. Since the range of perceptible intensity is Δx, we may take $\rho \approx \lambda/\alpha$; substituting into (3) changes the condition into

$$\alpha \gg \sqrt{\lambda/r_0}.$$

This condition is satisfied in all optical instruments constructed from lenses or plane mirrors.

Comment. This problem concerns the basic method for calculating a diffracted light field. The result shows that it is not necessary to satisfy strictly the Fraunhofer diffraction conditions (the source and

the image field both at infinity, or both in the focal plane of a lens) in order to observe a Fraunhofer diffraction pattern; it can also be observed in the geometrical image plane of the source, provided that the geometrical conditions for imaging are satisfied and the observation region lies near the geometrical image point.

Solution 4.19. (1) Let the intensities of the elliptically polarised component along the x- and y-axes be I_{ex} and I_{ey} respectively; the unpolarised component has the same intensity along the x- and y-axes, which we denote by I_u. Then from the conditions of the problem,

$$I_u + I_{ex} = 1.5I_0, \tag{1}$$
$$I_u + I_{ey} = 1.0I_0. \tag{2}$$

When the transmission axis of the polariser makes an angle θ with the x-axis, the unpolarised component passing through the polariser is still I_u, while the amplitude of the elliptically polarised component passing through it is

$$E_\theta = E_x \cos\theta + E_y \sin\theta.$$

Since x and y are the principal axes of the ellipse, the phase difference between E_x and E_y is $90°$, and hence $E_\theta^2 = E_x^2 \cos^2\theta + E_y^2 \sin^2\theta$; therefore

$$\begin{aligned} I_\theta &= I_{ex}\cos^2\theta + I_{ey}\sin^2\theta + I_u \\ &= (I_u + I_{ex})\cos^2\theta + (I_u + I_{ey})\sin^2\theta \\ &= 1.5I_0\cos^2\theta + 1.0I_0\sin^2\theta. \end{aligned}$$

The result is independent of the unpolarised component.

(2) After passing through the quarter-wave plate, the phase difference between E_x and E_y is changed by $90°$, so that they become in phase. Since the transmitted intensity is maximum when the polariser makes an angle of $30°$ with the x-axis, it is seen that

$$\frac{E_x}{E_y} = \cot 30° = \sqrt{3}, \qquad \text{i.e.} \qquad \frac{I_{ex}}{I_{ey}} = 3.$$

Substituting into (1), we obtain

$$\begin{aligned} I_{ex} &= 0.75I_0, \\ I_{ey} &= 0.25I_0, \\ I_u &= 0.75I_0. \end{aligned}$$

Hence the maximum intensity at $\theta = 30°$ is

$$I = I_{ex} + I_{ey} + I_u = 1.75I_0.$$

The percentage of the unpolarised component in the total incident intensity is

$$\frac{2I_u}{2I_u + I_{ex} + I_{ey}} = \frac{2 \times 0.75I_0}{2 \times 0.75I_0 + 0.75I_0 + 0.25I_0} = 60\%.$$

Solution 4.20. In a birefringent crystal, light waves polarised parallel and perpendicular to the optic axis travel with different phase velocities, so that the linearly polarised wave $E_0(\hat{x} + \hat{y})/\sqrt{2}$ emerging from the polarising film becomes, after entering the crystal and travelling a distance z,

$$\vec{E} = \frac{E_0}{\sqrt{2}}\left(\hat{x}\,\mathrm{e}^{\mathrm{i}kn_+z} + \hat{y}\,\mathrm{e}^{\mathrm{i}kn_-z}\right)\mathrm{e}^{-\mathrm{i}\omega t},$$

where $\hat{y}$ is chosen parallel to the optic axis, $\hat{x}$ perpendicular to $\hat{y}$ and to the direction of propagation $\hat{z}$, the phase velocity of the wave polarised along x is c/n_+, and that of the wave polarised along y is c/n_-; the plane of the polariser is parallel to $\vec{l} = (\hat{x} + \hat{y})/\sqrt{2}$. After travelling the whole length $2^N d$ of the crystal, the amplitude of the wave transmitted through the polariser is $\vec{E} \cdot \vec{l}$, namely

$$\vec{E} \cdot \vec{l} = \frac{E_0}{2} \left(\mathrm{e}^{\mathrm{i}kn_+ 2^N d} + \mathrm{e}^{\mathrm{i}kn_- 2^N d} \right).$$

With

$$n_+ = \frac{1}{2}\left[(n_+ + n_-) + (n_+ - n_-)\right], \qquad n_- = \frac{1}{2}\left[(n_+ + n_-) - (n_+ - n_-)\right],$$

substitution into the above gives

$$\begin{aligned}
\vec{E} \cdot \vec{l} &= \frac{E_0}{2} \left\{ \mathrm{e}^{\mathrm{i}k2^{N-1}d[(n_+ + n_-) + (n_+ - n_-)]} + \mathrm{e}^{\mathrm{i}k2^{N-1}d[(n_+ + n_-) - (n_+ - n_-)]} \right\} \\
&= \frac{E_0}{2} \mathrm{e}^{\mathrm{i}k2^{N-1}d(n_+ + n_-)} \left[\mathrm{e}^{\mathrm{i}k2^{N-1}d(n_+ - n_-)} + \mathrm{e}^{-\mathrm{i}k2^{N-1}d(n_+ - n_-)} \right] \\
&= \frac{E_0}{2} \mathrm{e}^{\mathrm{i}k2^{N-1}d(n_+ + n_-)} \left[\mathrm{e}^{\mathrm{i}k2^{N}d(n_+ - n_-)/2} + \mathrm{e}^{-\mathrm{i}k2^{N}d(n_+ - n_-)/2} \right] \\
&= E_0 \, \mathrm{e}^{\mathrm{i}k2^{N-1}d(n_+ + n_-)} \cos\left(2^N \varphi\right).
\end{aligned}$$

3.5 Atomic Physics

Solution 5.01. According to Einstein's photoelectric equation, the maximum kinetic energy $E_{k\,\max}$ of the photoelectrons is

$$E_{k\,\max} = h\nu - \phi_0,$$

where $h\nu$ is the energy of the incident photon and ϕ_0 is the work function of the metal. Let the stopping voltage be V_s; then $eV_s = E_{k\,\max}$. Substituting this into the above, the work function of the metal is found to be

$$\phi_0 = h\nu - eV_s = (1240/200) - 2.6 = 3.6\,\mathrm{eV}.$$

When ultraviolet light of wavelength $\lambda = 300\,\mathrm{nm}$ is used instead, the stopping voltage V_s is

$$V_s = (hc/\lambda - \phi_0)/e = 1240/300 - 3.6 = 0.53\,\mathrm{V}.$$

If visible light is used to irradiate the surface of this metal, then, since the wavelength of visible light is greater than 400 nm and its energy is less than 3.1 eV, while the minimum work necessary for an electron to escape from the metal is $\phi_0 = 3.6\,\mathrm{eV}$, no photoelectric effect can occur.

Solution 5.02. Method 1.

For part (h), if the production of a large number of protons is regarded as the result of γ photons striking the hydrogen nuclei in paraffin (this phenomenon is usually called the Compton effect of the proton), then, analogously to the Compton effect, we may regard the process as an elastic collision between a γ photon of energy $h\nu$ and a free proton initially at rest. Since a photon is involved and the recoil velocity of the proton may approach the speed of light, relativistic expressions must be used.

From the conservation of energy and mass,

$$\frac{hc}{\lambda} + M_p c^2 = \frac{hc}{\lambda'} + \sqrt{(pc)^2 + (M_p c^2)^2}.$$

From the conservation of momentum,

$$p^2 = \left(\frac{h}{\lambda}\right)^2 + \left(\frac{h}{\lambda'}\right)^2 - 2\,\frac{h^2}{\lambda\lambda'}\cos\theta,$$

where λ and λ' are respectively the wavelengths of the γ photon before and after the collision; M_pc^2 is the rest energy of the proton, $M_pc^2 = 0.511 \times 1836 = 938\,\text{MeV}$; p is the recoil momentum of the proton; and θ is the scattering angle of the photon. Combining the above two equations and eliminating p, we obtain

$$\Delta\lambda = \lambda' - \lambda = \frac{hc}{M_pc^2}(1 - \cos\theta).$$

When $\theta = \pi$, the recoil energy of the proton is maximum, namely

$$E_k = \left(\frac{hc}{\lambda}\right) - \left(\frac{hc}{\lambda'}\right) = \left(\frac{hc}{\lambda}\right) - \frac{hc}{\lambda + \dfrac{2hc}{M_pc^2}} = 5.7\,\text{MeV}.$$

From the above equation the energy $h\nu$ of the incident photon can be solved:

$$\begin{aligned} h\nu = \frac{hc}{\lambda} &= \frac{1}{2}\left(E_k + \sqrt{E_k^2 + 2E_kM_pc^2}\right) \\ &= \frac{1}{2} \times \left(5.7 + \sqrt{5.7^2 + 2 \times 5.7 \times 938}\right) = 55\,\text{MeV}. \end{aligned}$$

That is, to produce, after scattering, a proton with the maximum recoil energy of 5.7 MeV, the incident γ photon must have an energy of 55 MeV. This result is inconsistent with the result of (a); therefore this particle is not a γ photon, unless the conservation of energy and the conservation of momentum are abandoned.

Method 2.

Analogously to Method 1, we obtain

$$\Delta\lambda = \lambda' - \lambda = \frac{hc}{M_pc^2}(1 - \cos\theta).$$

When $\theta = \pi$, the recoil energy of the proton is maximum, namely

$$E_k = \left(\frac{hc}{\lambda}\right) - \left(\frac{hc}{\lambda'}\right) = \left(\frac{hc}{\lambda}\right) - \frac{hc}{\lambda + \dfrac{2hc}{M_pc^2}} = h\nu - \frac{h\nu}{1 + \dfrac{2h\nu}{M_pc^2}}.$$

From part (a) we know that $h\nu = 15\,\text{MeV}$. Substituting this into the above equation gives

$$E_k = 15 - 15/(1 + 2 \times 15/938) = 0.46\,\text{MeV}.$$

This result is inconsistent with the result of (b); therefore this particle is not a γ photon, unless the conservation of energy and the conservation of momentum are abandoned.

Comment. The problem is based on some experiments connected with the discovery of the neutron. In 1920, after much reflection, Rutherford put forward his famous neutron hypothesis:[1] he considered it

[1] E. Rutherford, *Proc. Roy. Soc.* A97 (1920) p. 374.

possible that there exists a new kind of neutral atom, perhaps a double particle formed by an electron and a hydrogen nucleus bound together more tightly. He imagined that this atom would have many novel properties, properties that are necessary for explaining the composition of the nuclei of heavy elements.

In order to test Rutherford's hypothesis, scientists carried out a large amount of experimental work. In 1930 the German W. Bothe and his student H. Becker were the first to publish the results of experiments in which particles produced by radioactive polonium bombarded beryllium: when α particles strike beryllium, the beryllium emits what was then regarded as a more penetrating kind of γ radiation. In 1932 the Joliot-Curies repeated these experiments with a very strong polonium α source and measured the penetrating power of this unknown radiation through lead. They found that the intensity of the radiation emitted by beryllium decreased by only one half in passing through 4.7 cm of lead. If this radiation were γ rays, its energy would be about 15 MeV. This value is much higher than the energy of the γ rays emitted by natural radioactive substances. They also found that this unknown radiation, on falling on hydrogen-containing paraffin, produces many protons of very high speed. Unfortunately, they did not carefully check whether, according to the conservation of momentum and the conservation of energy, γ particles of energy 15 MeV striking hydrogen nuclei could produce such a high-speed proton stream. Like Bothe, they regarded this radiation emitted by beryllium as γ rays, and considered the production of the proton stream to be the result of γ particles striking hydrogen nuclei, a kind of special phenomenon similar to the Compton effect. In this way both Bothe and the Joliot-Curies missed the opportunity of discovering the neutron.

Rutherford's early student and capable assistant J. Chadwick had been engaged in experiments searching for the neutron ever since 1921. In 1932, when he read the report of the Joliot-Curies on the astonishing properties of the beryllium radiation, he told Rutherford of the views of the Joliot-Curies. Rutherford did not believe that this was the Compton effect, and thought it very likely that the neutron, sought for many years, had appeared here.

Chadwick quickly rechecked the experimental results of Bothe and of the Joliot-Curies. By measuring the range of these protons in air, he found the energy of the protons to be 5.7 MeV. He also extended the experiment, using the beryllium radiation to irradiate many other substances, including the gases hydrogen, helium, nitrogen and argon, and determined the energy of the recoil atoms produced thereby, from which he calculated the energy of the incident γ rays. Chadwick found that this result depended on the gas used. For example, for recoil nitrogen atoms of energy 1.4 MeV, the energy of the incident γ rays should be 90 MeV, while to produce recoil protons of energy 5.7 MeV the energy of the γ rays should be 55 MeV. If the beryllium radiation consisted of neutral particles of mass approximately equal to the proton mass, then when these particles (called neutrons by Chadwick, following Rutherford) make a head-on collision with the atoms of a gas, the maximum recoil energy will appear. As for the recoil protons, since the proton and the neutron have equal masses, in a head-on collision the proton will acquire the whole kinetic energy of the neutron; hence the energy of the neutrons emitted from beryllium equals the maximum recoil energy of the protons, 5.7 MeV. Neutrons of this energy, incident on nitrogen gas, can produce recoil nitrogen atoms of energy 1.4 MeV.

In the same year Chadwick published a paper entitled "The Existence of the Neutron",[2] in which he described his experimental results and theoretical analysis in detail. He pointed out in the paper that only by abandoning the conservation of energy and momentum could the hypothesis that the beryllium radiation consists of γ rays be supported. The neutron hypothesis, on the other hand, gives a direct and simple explanation of the experimental facts. It is self-consistent and offers new insight into the structure of atomic nuclei. Using the nuclear reaction ${}^4\mathrm{He} + {}^{11}\mathrm{B} \to {}^{14}\mathrm{N} + \mathrm{n}$, Chadwick concluded that the mass of the neutron lies roughly between 1.005 and 1.008.

Solution 5.03. (1) From the Compton shift formula

$$\Delta\lambda = \lambda_c(1 - \cos\theta),$$

where $\lambda_c = h/(m_e c) = 0.0243\,\text{Å}$ is the Compton wavelength of the electron (for the derivation of

[2] J. Chadwick, *Proc. Roy. Soc.* (London) A136 (1932) p. 692.

the formula see Solution 5.02), we obtain:

(a) If after scattering the electron moves along the direction of incidence of the photon, then $\theta = \pi$, and

$$\Delta\lambda = 2\lambda_c = 0.00486\,\text{nm}.$$

(b) If the photon still moves along the original direction of incidence, then $\theta = 0$ and $\Delta\lambda = 0$.

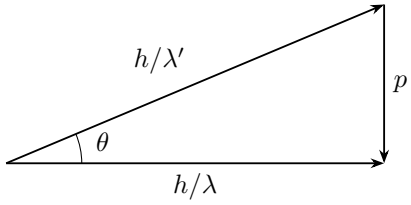


Fig. 5.03

(2)(a) From the conservation of energy and mass,

$$\frac{hc}{\lambda} + \sqrt{p_e^2c^2 + E_0^2} = \frac{hc}{\lambda'} + E_0.$$

From the conservation of momentum,

$$p_e^2 = \left(\frac{h}{\lambda}\right)^2 + \left(\frac{h}{\lambda'}\right)^2 - 2\,\frac{h^2}{\lambda\lambda'}\cos\theta,$$

where λ and λ' are respectively the wavelengths of the photon before and after the collision; E_0 is the rest energy of the electron; p_e is the initial momentum of the electron; and θ is the scattering angle of the photon. Combining the above equations and eliminating p_e, we obtain

$$\begin{aligned}\Delta\lambda = \lambda - \lambda' &= \lambda_c(1 - \cos\theta)\\ &= \lambda_c(1 - \cos 60^\circ) = 0.0012\,\text{nm}.\end{aligned}$$

(b) It is known that the energy of the incident photon equals the rest energy of the electron; hence the wavelength of the incident photon is $\lambda = \lambda_c$. From (a) we have $\Delta\lambda = \lambda_c/2$. Substituting λ and $\Delta\lambda$ into the following expression, we obtain the energy $h\nu'$ of the scattered photon:

$$h\nu' = \frac{hc}{\lambda'} = \frac{hc}{\lambda - \Delta\lambda} = 2E_0 = 1.02\,\text{MeV}.$$

From the conservation of energy, the kinetic energy E_k of the electron before the collision is found to be

$$E_k = \left(\frac{hc}{\lambda'}\right) - \left(\frac{hc}{\lambda}\right) = E_0 = 0.511\,\text{MeV}.$$

Comment. Part (1) of this problem discusses the Compton effect: a high-frequency γ photon makes an elastic collision with a free electron that is approximately at rest; the photon transfers part of its energy to the electron, this becoming the recoil kinetic energy of the electron, while the wavelength of the scattered photon becomes longer. Part (2) discusses the reverse process of the Compton effect: the collision of a high-energy relativistic electron with a low-frequency photon, after which the energy (frequency) of the scattered photon becomes larger and its wavelength shorter; it is therefore called the inverse Compton effect. The inverse Compton effect is an important non-thermal radiation mechanism for the production of γ rays.

The Compton effect was discovered by A. H. Compton in 1923, who also gave its theoretical explanation. Following the photoelectric effect, it provided further confirmation of the particle nature of radiation; hence it is of extremely profound significance both theoretically and experimentally. For his series of experiments on the Compton effect and his theoretical explanation of it, Compton shared the 1927 Nobel Prize in Physics with Wilson of Britain.[3] During the ten years from 1930 to 1940 Compton devoted himself to the study of cosmic rays and discovered the inverse Compton effect. The Compton effect and the inverse Compton effect are of great importance in astrophysics, because the interaction of γ photons and electrons in cosmic rays is an important aspect of the Compton effect, while the inverse Compton effect of relativistic electrons is an important topic of contemporary astrophysics. For example, in April 1991, following the Hubble telescope, NASA launched the second giant space observatory, the Compton Gamma Ray Observatory, which can image cosmic γ-ray sources and their background and measure their spectra and their time variation. The Compton imaging telescope with which it is equipped uses the Compton effect to detect γ rays of energy 1–30 MeV in the universe. As another example, the large number of high-temperature electrons present in the ionised gas of clusters of galaxies produce inverse Compton scattering of the photons of the cosmic microwave background, giving rise to the so-called Sunyaev–Zel'dovich (SZ) effect. The SZ effect is a very effective means of studying clusters of galaxies; in particular, the SZ effect depends only on the intrinsic properties of the cluster and not on its position, which will be favourable for discovering clusters of high redshift. It is therefore of extremely great significance for studying the evolution of clusters of galaxies. Astronomical telescopes built on the basis of the SZ effect will help cosmologists to find clusters of galaxies that are too distant or too faint to be discovered by other methods. These observations in turn help cosmologists to discover how clusters evolve, and may make it possible to locate the dark matter in the universe. The SZ cluster survey telescope now under construction will use the SZ effect and provide an entirely new and effective way of determining cosmological parameters.

Solution 5.04. (1) As shown in Fig. 5.04(a), before the collision the energy of the electron is E, its momentum is $p_x = p$, $p_y = 0$; the energy of the photon is $h\nu$, its momentum is $p_x = -h\nu/c$, $p_y = 0$. As shown in Fig. 5.04(b), after the collision the momentum of the electron is p', directed at an angle to the x-direction; the energy of the photon is $h\nu'$, its momentum is $h\nu'/c$, making an angle θ with the x-direction.

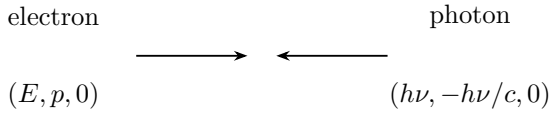


Fig. 5.04(a)

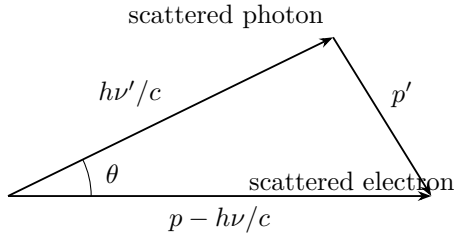


Fig. 5.04(b)

[3]The Chinese physicist Wu Youxun made outstanding contributions to Compton scattering experiments; see Guan Weiyan, "A brief account of Professor Wu Youxun", *Physics*, 11 (1982) 457.

From the conservation of energy,

$$E + h\nu = \sqrt{p'^2c^2 + E_0^2} + h\nu', \tag{1}$$

where E_0 is the rest energy of the electron. From the conservation of momentum,

$$p'^2c^2 = (h\nu')^2 + (pc - h\nu)^2 - 2(h\nu')(pc - h\nu)\cos\theta. \tag{2}$$

Substituting (2) into (1), we obtain

$$h\nu' = \frac{(E + pc)h\nu}{E + h\nu + h\nu\cos\theta - pc\cos\theta}.$$

Since

$$E = \sqrt{p^2c^2 + E_0^2} = 20\,\text{GeV} \gg E_0,$$

we have $E \simeq pc$, and hence

$$h\nu' \simeq \frac{2h\nu E}{h\nu(1 + \cos\theta) + E - pc\cos\theta}.$$

(2) When $\theta = 0$, the energy of the scattered photon is maximum, its value being

$$(h\nu')_{\max} = \frac{2h\nu E}{2h\nu + E - pc}.$$

In the extreme relativistic case,

$$E = \sqrt{p^2c^2 + E_0^2} \simeq pc + \frac{E_0^2}{2pc},$$

$$E - pc \simeq \frac{E_0^2}{2E} \simeq \frac{(0.5 \times 10^6)^2}{2 \times 2 \times 10^{10}} \approx 6\,\text{eV}.$$

For a typical visible photon,

$$h\nu = \frac{hc}{\lambda} \approx \frac{12\,400}{5\,000} \approx 2.5\,\text{eV}.$$

Thus the maximum photon energy obtainable by this technique is

$$(h\nu')_{\max} = \frac{2 \times 2.5 \times 20}{2 \times 2.5 + 6} = 9\,\text{GeV}.$$

Solution 5.05. Suppose that the electron moves along the y-direction with momentum p_y. From the relativistic momentum–energy relation,

$$cp_y = \sqrt{E^2 - E_0^2},$$

where c is the speed of light, E is the energy of the electron, $E = 1\,\text{MeV}$, and E_0 is the rest energy of the electron, $E_0 = 0.511\,\text{MeV}$.

When a fast-moving electron travels through a cloud chamber, the ions produced along its path form condensation centres, causing the water vapour to condense into small droplets. The linear size of a droplet is about $10^{-5}\,\text{m}$; therefore in this case $\Delta x = 10^{-5}\,\text{m}$. From the uncertainty

relation we may assume that the uncertainty in the momentum of the electron in the x-direction is about

$$\Delta p_x = \hbar/\Delta x,$$

where $\hbar = h/2\pi = 1973\,\text{eV.Å}$.

Consequently the motion of the electron will deviate from the y-direction by an angle θ, and the uncertainty $\Delta\theta$ of this angle is about

$$\begin{aligned}\Delta\theta = \frac{\Delta p_x}{p_y} &= \frac{\hbar c}{\Delta x\sqrt{E^2 - E_0^2}} \\ &= \frac{1973}{10^{-5} \times 10^{10} \times \sqrt{10^{12} - (0.511 \times 10^6)^2}} \\ &= 2.3 \times 10^{-8},\end{aligned}$$

$$\theta = 2.3 \times 10^{-8}\,\text{rad}.$$

It is thus seen that the track of the electron will not differ noticeably from a straight line.

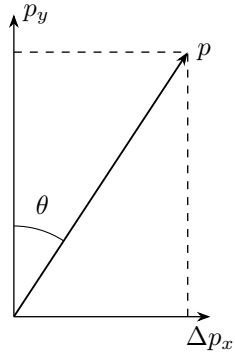


Fig. 5.05

Comment. (1) The estimate of Δp_x in the solution can evidently be applied to the momentum components in every direction; here, in order to estimate the maximum deviation of the electron track from a straight line, only the transverse momentum uncertainty is considered.

(2) The particles existing in nature have wave properties. When a particle moves with a definite momentum p, its wavelength is $\lambda = h/p$. This relation between wavelength and momentum holds for all real particles. Under certain conditions, however, we may still regard a particle as a classical particle; the uncertainty relation helps us to estimate the range of validity of the classical description of particles. For example, under the conditions of this problem the energy of the electron is very high and the region to which it is confined is relatively large, so that $\Delta p/p \ll 1$; hence the electron may be regarded as a classical particle moving along a straight track. At this time the de Broglie wavelength of the electron is very short, $\lambda = h/p = 1.4 \times 10^{-12}\,\text{m} \ll 10^{-5}\,\text{m}$, which is similar to the condition that makes geometrical optics valid in optics. If an electron with a kinetic energy of a few electron-volts were confined in a space of atomic size, then in this case $\Delta p \sim p$ for the electron, and the electron would have no definite track. The classical description of the particle would no longer be applicable.

(3) The p and E in the above formulae are both relativistic, which also shows that there is a close relation between quantum theory and relativity. Only when the kinetic energy E_k of the particle is much smaller than its rest energy E_0 may the non-relativistic momentum–energy relation $pc = \sqrt{2E_0E_k}$ be used.

Solution 5.06. Let the distance of the electron from the nucleus be r, and take the order of magnitude of the position uncertainty to be $\Delta x = r$. From the uncertainty relation $\Delta x\,\Delta p \geq \hbar$ we have $(\Delta p)^2 = p^2 \geq \hbar^2/r^2$; hence

(1)

$$E = \frac{\hbar^2}{2mr^2} - \frac{Ze^2}{4\pi\varepsilon_0 r}.$$

When E takes its minimum value the hydrogen-like ion is in its ground state. Let its radius be r_0 and its energy be E_1; from $\left(\dfrac{\mathrm{d}E}{\mathrm{d}r}\right)_{r=r_0} = 0$ we obtain r_0 and E_1 as

$$r_0 = \frac{4\pi\varepsilon_0\hbar^2}{Ze^2m},$$

$$E_1 = -\frac{mZ^2e^4}{2(4\pi\varepsilon_0\hbar)^2}.$$

(2)

$$E = \sqrt{p^2c^2 + m^2c^4} - \frac{Ze^2p}{4\pi\varepsilon_0\hbar}.$$

Let the momentum of the electron when the hydrogen-like ion is in its ground state be p_0, and its energy be E_1'. From $(\mathrm{d}E/\mathrm{d}p)_{p=p_0} = 0$ we obtain p_0 and E_1' as

$$p_0 = \frac{mcZe^2}{\sqrt{(4\pi\varepsilon_0\hbar c)^2 - (Ze^2)^2}},$$

$$E_1' = mc^2\sqrt{1 - \left(\frac{Ze^2}{4\pi\varepsilon_0\hbar c}\right)^2}.$$

(3) When $\dfrac{Ze^2}{4\pi\varepsilon_0\hbar c} \ll 1$, that is, $Z \ll \alpha^{-1} = 137$, expanding the right-hand side of the above expression in series and neglecting the higher-order infinitesimals, the above expression can be written as

$$E_1' = mc^2\left[1 - \frac{1}{2}\left(\frac{Ze^2}{4\pi\varepsilon_0\hbar c}\right)^2\right] = E_1 + mc^2.$$

E_1 and E_1' differ by a constant mc^2; this is caused by the different choices of the zero of energy in the two cases. Therefore E_1 is the classical limit of E_1'. When Z is small the non-relativistic approximation may be used.

Comment. (1) The r_0 and E_1 obtained in part (1) agree exactly with the results of Bohr's theory. This is because $\Delta x = r$ was wisely chosen rather than $2r$ or $r/2$; such a choice is precisely reasonable. Of course, any reasonable choice of Δx will give the correct order of magnitude of the atomic size and of the atomic energy.

(2) In part (3), if one more term is retained in the series expansion of E_1', one obtains

$$E_1' = mc^2\left[1 - \frac{1}{2}\left(\frac{Ze^2}{4\pi\varepsilon_0\hbar c}\right)^2 - \frac{1}{8}\left(\frac{Ze^2}{4\pi\varepsilon_0\hbar c}\right)^4\right].$$

To make the zero of energy agree with the non-relativistic case, we subtract mc^2 from the above expression and thus obtain

$$E_1' = -mc^2\left[\frac{1}{2}\left(\frac{Ze^2}{4\pi\varepsilon_0\hbar c}\right)^2 + \frac{1}{8}\left(\frac{Ze^2}{4\pi\varepsilon_0\hbar c}\right)^4\right] = E_1 - \frac{mc^2}{8}\left(\frac{Ze^2}{4\pi\varepsilon_0\hbar c}\right)^4.$$

The term $-\frac{mc^2}{8}\left(\frac{Ze^2}{4\pi\varepsilon_0\hbar c}\right)^4 = -|E_1|\frac{Z^2\alpha^2}{4} = \Delta E_r$ is the relativistic correction to the kinetic energy of the ground state of the hydrogen-like ion. The result for this term derived rigorously in quantum mechanics is

$$\Delta E_r = -|E_1|\,\frac{5Z^2\alpha^2}{4}.$$

For the hydrogen atom, $Z = 1$ and $\Delta E_r = -6.66 \times 10^{-5}|E_1| \ll |E_1|$; it is therefore called the fine-structure correction of the energy level. But for hydrogen-like ions with $Z = 50$, this correction already exceeds 16%.

Solution 5.07. (1) The circumference S_n of the n-th orbit of the hydrogen atom and the de Broglie wavelength λ_n of the electron moving in that orbit are respectively

$$S_n = 2\pi r_n, \qquad \lambda_n = \frac{h}{mv_n},$$

where r_n and v_n are respectively the radius of the n-th orbit and the orbital speed of the electron, and m is the mass of the electron. From the quantisation condition for the angular momentum of the electron, $mv_n r_n = n\hbar$, we obtain

$$\frac{h}{\lambda_n} \cdot 2\pi r_n = n\hbar.$$

Finally,

$$\frac{S_n}{\lambda_n} = n.$$

(2) Substituting $n = 1$ into the above expression gives

$$\lambda_1 = S_1/1 = 2\pi a_0,$$

where $a_0 = 0.0529\,\text{nm}$ is the Bohr radius. Substituting this value into the above expression gives

$$\lambda_1 = 0.332\,\text{nm}.$$

Comment. The solution of part (1) shows that in the hydrogen atom only those orbits whose circumference is an integral multiple of the de Broglie wavelength of the electron are allowed. One may imagine that, just as a standing wave is formed on a string fixed at both ends, the de Broglie wave of the electron is here a standing wave on a circle. Bohr's quantisation condition for angular momentum is precisely the necessary condition for the formation of such a standing wave. In 1924 de Broglie gave a physical explanation of Bohr's quantisation condition for angular momentum; he pointed out that Bohr's quantisation condition for angular momentum is equivalent to the standing-wave condition.

In fact, according to the uncertainty principle the orbit of an electron cannot be determined precisely; moreover, the orbital angular momentum of the lowest energy state of the hydrogen atom is zero, and not the value $\hbar$ predicted by Bohr. Later, Bohr's model of the hydrogen atom was replaced by the quantum-mechanical theory of the hydrogen atom. Nevertheless, in quantum mechanics the orbital angular momentum is still quantised.

Solution 5.08. The wave numbers of the spectral lines of the hydrogen atom and of the helium ion are respectively given by the following formulae:

$$\tilde{\nu}_{\text{H}} = R_{\text{H}}\left(\frac{1}{m^2} - \frac{1}{n^2}\right), \tag{1}$$

$$\tilde{\nu}_{\mathrm{He}} = R_{\mathrm{He}} Z^2 \left(\frac{1}{m^2} - \frac{1}{n^2} \right) = R_{\mathrm{He}} \left[\frac{1}{(m/2)^2} - \frac{1}{(n/2)^2} \right], \tag{2}$$

where $m = 1, 2, 3, \ldots$; $n = m+1, m+2, m+3, \ldots$; and for the helium ion $Z = 2$. Since the mass of the helium nucleus is greater than that of the hydrogen nucleus, the Rydberg constant $R_{\mathrm{H}} \simeq R_{\mathrm{He}}$, but R_{He} is slightly greater than R_{H}.

Comparing (1) and (2), it is seen that when the quantum numbers m and n of the helium ion and the quantum numbers of the hydrogen atom satisfy the conditions $(m/2)_{\mathrm{He}} = (m)_{\mathrm{H}}$ and $(n/2)_{\mathrm{He}} = (n)_{\mathrm{H}}$, we have $\tilde{\nu}_{\mathrm{H}} \simeq \tilde{\nu}_{\mathrm{He}}$, but $\tilde{\nu}_{\mathrm{H}}$ is slightly smaller than $\tilde{\nu}_{\mathrm{He}}$. Therefore some almost coincident doublets will appear in the spectrogram: the line of slightly smaller wave number in each doublet is the spectral line emitted by the hydrogen atom; apart from these, all the remaining lines belong to the spectrum of the helium ion. Since the intensity and the spacing of the spectral lines both decrease in the direction of increasing wave number, the magnitude of the wave number can be judged from their trend of variation.

Comment. In 1897 the astronomer W. H. Pickering discovered in the spectrum of the star ξ in the constellation Puppis a series of spectral lines very similar to the Balmer series of the hydrogen atom; it is called the Pickering series. The figure below shows the first five lines H_α, H_β, H_γ, H_δ and H_ε of the Pickering series and of the Balmer series of the hydrogen atom. From the figure it can be seen that every other line of the Pickering series almost coincides with a line of the Balmer series, but the wavelengths of the two still differ slightly. There are other lines of the Pickering series lying between adjacent lines of the Balmer series. Rydberg pointed out that the Pickering series can be represented by

$$\tilde{\nu}_P = R \left(\frac{1}{2^2} - \frac{1}{k^2} \right), \qquad k = 2.5,\ 3,\ 3.5,\ 4,\ 4.5,\ \ldots$$

where $\tilde{\nu}_P$ is the wave number of the spectral line. The Pickering series cannot be observed in hydrogen on the earth. At first people thought that it was emitted by a special kind of hydrogen in the celestial bodies; later it was found that the Pickering series appears when some helium is mixed with hydrogen, and only then was it made clear that the Pickering series is emitted by the helium ion He^+. In the solution of this problem it is only necessary to set $m = 4$, $n = 5, 6, 7, \ldots$ in equation (2) to obtain the expression for the Pickering series.

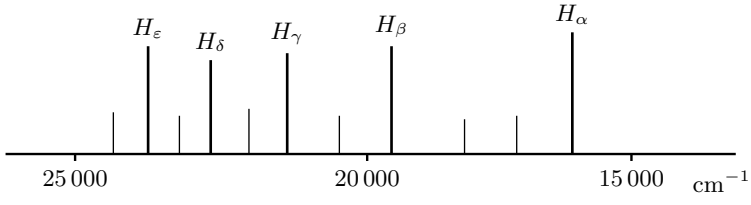


Fig. 5.08

Solution 5.09. (1) Since the masses of the μ^- meson and the π^+ meson are of the same order of magnitude, both mesons move about their common centre of mass under the Coulomb force exerted by the other. Introducing the reduced mass

$$\mu = \frac{m_\mu m_\pi}{m_\mu + m_\pi},$$

the lowest energy E_1 of the atom is

$$E_1 = -\frac{1}{2}\mu(\alpha c)^2,$$

where α is the fine-structure constant and c is the speed of light. Substituting m_μ and m_π into the above expression gives

$$E_1 = -\frac{1}{2}(\alpha c)^2 m_e \times \frac{106 \times 140}{0.511 \times (106 + 140)} = -13.6 \times 118\,\text{eV} = -1.60\,\text{keV}.$$

(2) In the ground state the distance r_1 between the two mesons is

$$r_1 = \frac{a_0 m_e}{\mu} = \frac{0.529\,\text{Å}}{118} = 4.48 \times 10^{-4}\,\text{nm}.$$

(3) In the ground state the relative speed of the two mesons is $v_1 = \alpha c$. Neglecting the overall motion of the atom, we have

$$m_\pi v_\pi = m_\mu v_\mu,$$
$$v_1 = v_\pi + v_\mu.$$

From these the speed of the μ^- meson is found to be

$$v_\mu = \frac{\alpha c m_\pi}{m_\mu + m_\pi} = \frac{3 \times 10^8 \times 140}{137 \times (106 + 140)} = 1.25 \times 10^6\,\text{m/s}.$$

Comment. An ordinary atom is composed of electrons, protons and neutrons. A so-called exotic atom is an atom-like system formed, through electromagnetic interaction, when particles such as $\mu^\pm$ mesons, $\tau^\pm$ mesons, $\pi^\pm$ mesons, $K^\pm$ mesons, positrons, antiprotons, $\Sigma^\pm$ hyperons and Ω^- hyperons replace respectively the electrons or the nucleus of an ordinary atom, or replace both. For example, a μ^- meson replacing an electron of an atom forms a muonic atom, and a positron replacing the nucleus forms positronium. The average lifetime of an exotic atom is not long, but there is enough time for the exotic atom to be formed and detected before the replacing particle itself decays. The best known of these is the antihydrogen atom, composed of an antiproton and a positron.

Solution 5.10. According to Gauss's law in electrostatics it can be shown that the potential $V(r)$ of a uniformly charged sphere of radius R and total charge Ze is

$$V(r) = \begin{cases} \dfrac{Ze}{4\pi\epsilon_0 r}, & r \geq R, \\ \dfrac{Ze}{4\pi\epsilon_0 R}\left(\dfrac{3}{2} - \dfrac{r^2}{2R^2}\right), & r \leq R. \end{cases}$$

Since the mass of the lead nucleus is $M_{\text{Pb}} \simeq \dfrac{208 \times 1836\, m_e}{207\, m_e} = 1.8 \times 10^3 m_\mu$, the motion of the lead nucleus may be neglected. When $r > R$, the expression for the Bohr orbit radius of the μ^- meson is

$$r_n = \frac{m_e a_0}{m_\mu}\frac{n^2}{Z} = \frac{0.529}{207 \times 82} n^2 = 3.12\, n^2\ (\text{fm}), \tag{1}$$

and the expression for the energy levels is

$$E_n = -\frac{1}{2} m_\mu (\alpha c)^2 \frac{Z^2}{n^2} = -\frac{13.6 \times 207 \times 82^2}{n^2} = -18.93\,\frac{1}{n^2}\ (\text{MeV}). \tag{2}$$

When $r < R$, the μ^- meson moves in a circle inside the nucleus; from the centripetal force formula

$$m_\mu \frac{v^2}{r} = -e\frac{\mathrm{d}V(r)}{\mathrm{d}r} = \frac{Ze^2}{4\pi\epsilon_0 R^3} r \tag{3}$$

the kinetic energy of the μ^- meson is found to be

$$\frac{1}{2}m_\mu v^2 = \frac{Ze^2r^2}{4\pi\epsilon_0\, 2R^3}.$$

The potential energy of the μ^- meson inside the nucleus is

$$-eV(r) = -\frac{Ze^2}{4\pi\epsilon_0 R}\left(\frac{3}{2} - \frac{r^2}{2R^2}\right).$$

Hence the total energy of the lead μ^- meson atom is

$$\begin{aligned} E &= \frac{1}{2}m_\mu v^2 - eV(r) \\ &= \frac{Ze^2}{4\pi\epsilon_0 R}\left(\frac{r^2}{R^2} - \frac{3}{2}\right). \end{aligned} \tag{4}$$

From the angular momentum quantization condition $m_\mu vr = n\hbar$, $n = 1, 2, 3, \ldots$, combined with (3), the expression for the Bohr orbit radius of the μ^- meson inside the nucleus is solved to be

$$r_n = \left(\frac{4\pi\epsilon_0 R^3\hbar^2}{Ze^2 m_\mu}n^2\right)^{1/4} = \left(\frac{137 \times 7.7^3 \times 197.3}{82 \times 207 \times 0.511}n^2\right)^{1/4} = 6.14\sqrt{n}\ \text{(fm)}. \tag{5}$$

Substituting into (4) gives the expression for the energy levels:

$$\begin{aligned} E_n &= \frac{Ze^2}{4\pi\epsilon_0 R}\left(n\hbar\sqrt{\frac{4\pi\epsilon_0}{Ze^2 m_\mu R}} - \frac{3}{2}\right) \\ &= \frac{82 \times 197.3}{137 \times 7.7}\left(\frac{6.14^2}{7.7^2}n - \frac{3}{2}\right) \\ &= (9.75n - 23.00)\ \text{(MeV)}. \end{aligned} \tag{6}$$

(1) From (5) and (1) the first two Bohr orbit radii of the μ^- meson are found to be

$$r_1 = 6.14\ \text{fm} \quad \text{(inside the nucleus)}$$

and

$$r_2 = 12.48\ \text{fm} \quad \text{(outside the nucleus)}.$$

(2) From (6) and (2) the energies of the first two levels of the lead μ^- meson atom are found to be

$$E_1 = -13.25\ \text{MeV}$$

and

$$E_2 = -\frac{18.93}{4} = -4.73\ \text{(MeV)}.$$

(3) The wavelength of the first line of the Lyman series of the lead μ^- meson atom is

$$\lambda = \frac{hc}{E_2 - E_1} = \frac{1240 \times 10^{-6}}{13.25 - 4.73} = 1.46 \times 10^{-4}\ \text{(nm)}.$$

(4) If the lead nucleus is regarded as a point charge, then from (2) the energies of the first two levels of the lead μ^- meson atom are found to be

$$E_1' = -18.93 \text{ MeV}$$

and

$$E_2' = -\frac{18.93}{4} = -4.73 \text{ (MeV)}.$$

Hence, in the point-nucleus model, the wavelength of the first line of the Lyman series of the lead μ^- meson atom is

$$\lambda' = \frac{hc}{E_2' - E_1'} = \frac{12400 \times 10^{-6}}{18.93 - 4.73} = 8.73 \times 10^{-5} \text{ (nm)}.$$

Comment. (1) In using Bohr theory to calculate the energy levels of an atom (or ion), we usually regard the nucleus as a point charge. In reality every nucleus has a finite size, so only at the surface of the nucleus or outside it is the potential of the nucleus that of a point charge. This problem uses Bohr's circular-orbit model to estimate the influence of the finite size of the nucleus on the energy levels and spectra of exotic atoms. From the results of parts (3) and (4) it can be seen that, since in the ground state the μ^- meson has entered the nucleus so that the effective nuclear charge it experiences is $Z^*e < Ze$, the ground-state level is raised, and hence $\lambda > \lambda'$. The correction to the ground-state energy of the atom is about 30%; the correction to the wavelength of the first line of the Lyman series is about 67%. In fact, although $R > r_1$, the μ^- meson does not stay inside the nucleus all the time, so the correction obtained by the method of this problem is on the large side.

(2) When the nuclear radius $R < r_1$, according to the method of this problem the above correction is zero. According to quantum mechanics, even if $R < r_1$, the electron still has a certain probability of entering the nucleus, and the correction of the finite nuclear size to the ground-state energy is not zero. Quantum mechanics uses perturbation theory to obtain the correction of this effect to the ground-state energy of a one-electron atom (or ion) as

$$\delta E = \frac{4}{5}\left(\frac{ZR}{a_0}\right)^2 \frac{Z^2e^2}{4\pi\epsilon_0 \cdot 2a_0} = \frac{4}{5}\left(\frac{ZR}{a_0}\right)^2 \left|E_1'\right|,$$

where E_1' is the ground-state energy in the point-nucleus model, the nuclear radius is $R = 1.3 \times A^{1/3}$ fm, and the Bohr radius is $a_0 = 0.0529$ nm.

For the neon ion Ne^{9+} ($Z = 10$, $A = 20$), $\delta E = 3.6 \times 10^{-7}|E_1'|$; for the lead ion Pb^{82+} ($Z = 82$, $A = 208$), $\delta E = 1.1 \times 10^{-4}|E_1'|$. It can be seen that for light elements this effect may be neglected altogether, while for heavy elements, although the effect is very small, it is nevertheless detectable. For the neon μ^- meson atom, since the mass of the μ^- meson is 207 times the mass of the electron, $\delta E = 0.015|E_1'|$, which is $(207)^2$ times larger than that for the neon ion. For the above three atoms (ions) the nuclear radius R is smaller than r_1.

(3) From the data in comment 2 it can be seen that, since the mass of the μ^- meson is 207 times the mass of the electron, the μ^- meson will come very close to the nucleus. In this way we may regard the μ^- meson as a detector with which to probe the nucleus. The so-called finite-size effect produced by the size of the nucleus is all the more clearly displayed in the level structure of such exotic atoms. Hence, if we measure the energy levels of μ^- meson atoms, we can infer the charge distribution in the nucleus and the size of the nucleus. For example, for the μ^- meson hydrogen atom, provided the level energies are known to a certain accuracy, the size of the proton can be deduced accurately. In the past the method of measuring the size of the proton was electron scattering, in which the size was obtained by analysing the scattering angle.

As an important constituent of matter, the proton's size is of great importance in physics. Only independent measurements of the proton's size can genuinely make use of the hydrogen spectrum for a further test of quantum electrodynamics; they can also serve in the future to test calculations of lattice quantum chromodynamics.

Solution 5.11. (1) The first term on the right-hand side of the formula corresponds to the ground state of the sodium atom, and the second term to an excited state of the sodium atom; the larger m is, the higher the excited state. The core of the sodium atom may be regarded as an effective nucleus. When the sodium atom is in a highly excited state m, the valence electron is far from the nucleus and the other 10 inner electrons, and what it experiences is the Coulomb interaction of a nucleus completely screened by the inner electrons, so the effective nuclear charge may be taken as 1. Only in this way does the second term on the right-hand side of the formula have exactly the same form as the hydrogen spectral term; therefore the formula holds only when m is large.

(2) The ionization energy E_∞ of sodium is

$$E_\infty = 0.378Rhc = 0.378 \times 13.6 = 5.14 \text{ (eV)}.$$

Comment. Now let us use the ionization energy E_∞ of sodium to calculate the effective nuclear charge Z^* felt by the valence electron when the sodium atom is in the ground state. Similarly to the hydrogen atom, for the sodium atom

$$E_\infty = \frac{Z^{*2}Rhc}{n^2},$$

where $n = 3$ is the principal quantum number of the valence electron when the sodium atom is in the ground state. From this we obtain

$$Z^* = \sqrt{\frac{n^2}{Rhc}E_\infty} = \sqrt{\frac{3^2}{13.6} \times 5.14} = 1.84.$$

The degree to which the Coulomb potential of the nucleus is screened (that is, the effective nuclear charge exerted by the core on the valence electron) depends on the state in which the valence electron finds itself. In different states the probability of the valence electron appearing at various places in space is different. As n increases, the distance of the valence electron from the nucleus also increases, while the probability of the valence electron entering the core becomes smaller, so the degree of screening of the Coulomb potential of the nucleus by the inner electrons becomes more and more pronounced, and the effective nuclear charge felt by the valence electron becomes smaller and smaller. From the above calculation it can be seen that the valence electron of the sodium atom in the ground state (in the $3s$ state) has a comparatively large probability of appearing near the nucleus, and therefore feels a comparatively large effective nuclear charge ($Z^* = 1.84$).

Solution 5.12. (1) The ground-state energy of the helium ion (He^+) and the first Bohr orbit radius of the electron are

$$E_1(e) = -\frac{1}{2}m(\alpha c)^2 Z^2 = -13.6 \times 4 = -54.4 \text{ (eV)},$$

$$a_1(e) = \frac{4\pi\epsilon_0\hbar^2}{me^2Z} = \frac{0.0529}{2} = 0.026 \text{ (nm)}.$$

(2) Similarly to the helium ion, the ground-state energy of the μ^- meson helium ion and the first Bohr orbit radius of the μ^- meson are

$$E_1(\mu) = -\frac{1}{2}m_\mu(\alpha c)^2 Z^2 = -13.6 \times 4 \times 200 = -1.1 \times 10^4 \text{ (eV)},$$

$$a_1(\mu) = \frac{4\pi\epsilon_0\hbar^2}{m_\mu e^2 Z} = \frac{0.0529}{200 \times 2} = 1.3 \times 10^{-4} \text{ (nm)}.$$

(3) In the ground state of the μ^- meson helium atom both the μ^- meson and the electron are in the state $n = 1$. Since the probability of the μ^- meson appearing near the helium nucleus is far greater than that of the electron, it may be said approximately that the electron is acted on by a nucleus completely screened by the μ^- meson, i.e. $Z^* \simeq 1$, whereas the μ^- meson is acted on by a completely unscreened nucleus, i.e. $Z^* \simeq 2$. Hence

(a) for $Z^* \simeq 1$, the first Bohr orbit radius of the electron is

$$a_1 = \frac{4\pi\epsilon_0\hbar^2}{me^2Z^*} = a_0 = 0.053 \text{ (nm)};$$

(b) the ground-state energy of the μ^- meson helium atom is

$$E = E_1(\mu) + E_1(e) = -1.1 \times 10^4 - 13.6 = -1.1 \times 10^4 \text{ (eV)};$$

(c) the chemical properties of this atom are most similar to those of hydrogen. Since the μ^- meson is very close to the nucleus, is tightly bound to the helium nucleus and is not easily excited, it acts like an effective nucleus of effective nuclear charge $Z^* \simeq 1$. Outside this effective nucleus there is one electron, and the chemical properties of an atom are determined by the number of electrons in its outer shell; therefore the μ^- meson helium atom resembles hydrogen rather than helium.

Comment. The μ^- meson atom is a kind of exotic atom. Compared with other exotic atoms, the lifetime of the μ^- meson atom is comparatively long. It was discovered in the late 1940s by the famous Chinese high-energy physicist Zhang Wenyu, academician of the Chinese Academy of Sciences, and has therefore been called the "Zhang atom" and "Zhang radiation". The μ^- meson atom broke through the Rutherford–Bohr atomic model and opened a new field of research on exotic atoms.

Solution 5.13. Let the mass of the particle be m and the mass of the hydrogen atom be M, and let the relative speeds of the particle and the hydrogen atom before and after the collision be v_{rel} and v'_{rel} respectively. Since the momentum of the system is conserved, the velocity of the centre of mass remains unchanged throughout; then energy conservation gives

$$\frac{1}{2}\mu v_{\text{rel}}^2 = \frac{1}{2}\mu v_{\text{rel}}'^2 + \Delta E,$$

where μ is the reduced mass, $\mu = \dfrac{mM}{m+M}$, and ΔE is the excitation energy acquired by the hydrogen atom. When $v'_{\text{rel}} = 0$, the excitation energy acquired by the hydrogen atom after the collision is maximum, and from the above equation

$$\Delta E = (\Delta E)_{\max} = \frac{1}{2}\mu v_{\text{rel}}^2. \tag{1}$$

(1) The first excitation energy of the hydrogen atom is about 10 eV, while at room temperature the average kinetic energy of the thermal motion of a hydrogen atom is about 1/25 eV; hence before the collision the hydrogen atom at room temperature is in the ground state. The Hα line is emitted when the hydrogen atom makes a transition from the $n = 3$ state to the $n = 2$ state. According to the above analysis, in order that the hydrogen atom emit the Hα line, the hydrogen atom must be excited from the ground state ($n = 1$) to the $n = 3$ state. Therefore the excitation energy acquired by the hydrogen atom after the collision must satisfy $(\Delta E)_{\max} \geq E_3 - E_1$. From (1) it is known that this condition is

$$\frac{1}{2}\mu v_{\text{rel}}^2 \geq E_3 - E_1. \tag{2}$$

Let the speed of the electron before the collision be v. Compared with v, the hydrogen atom at room temperature before the collision may be regarded as at rest, so $v_{\text{rel}} = v$; and since $m \ll M$, $\mu \simeq m$. Therefore the kinetic energy of the electron before the collision must satisfy

$$\frac{1}{2}mv^2 \geq E_3 - E_1 = -\frac{13.6}{9} + 13.6 = 12.09 \text{ (eV)},$$

that is, the minimum kinetic energy of the electron should be 12.09 eV.

(2) Since the mass m of the proton and the mass M of the hydrogen atom may be regarded as equal, $\mu = \dfrac{m}{2}$. From (2) we obtain

$$\frac{1}{2}\mu v_{\text{rel}}^2 = \frac{1}{4}mv^2 \geq E_3 - E_1.$$

Therefore the minimum kinetic energy E_k of the proton should be

$$E_k = \frac{1}{2}mv^2 = 2(E_3 - E_1) = 24.18 \text{ (eV)}.$$

(3) As stated in the problem, $\mu = \dfrac{m}{2}$ and $v_{\text{rel}} = 2v$. Substituting into (2) gives

$$\frac{1}{2}\mu v_{\text{rel}}^2 = mv^2 \geq E_3 - E_1.$$

Therefore the minimum kinetic energy E_k of the proton should be

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(E_3 - E_1) = 6.05 \text{ (eV)}.$$

Comment. An important physical concept involved in this problem is the following: in the above inelastic collision process, how much of the kinetic energy of the particle system can be converted through the collision into the excitation energy of the atom? According to König's theorem, the kinetic energy of a system of particles is equal to the sum of the kinetic energy of the centre of mass and the kinetic energy of the system relative to the centre of mass. Since this system is not acted on by external forces, the momentum of the system is conserved, and throughout the process neither the velocity of the centre of mass nor the kinetic energy of the centre of mass changes. Therefore only the kinetic energy of the system relative to the centre of mass can possibly be converted through the collision into the excitation energy of the atom.

Solution 5.14. (1) If there were no interaction between the electrons, the ground-state energy of the helium atom would be

$$E_1 = -2R_{\text{He}}hc\frac{Z^2}{n^2} = -2 \times 13.6 \times \frac{2^2}{1} = -108.8 \text{ (eV)}.$$

(2) Neglecting the interaction between the electrons and the motion of the helium nucleus, the orbit radius of the electron in the ground state is

$$r_1 = a_0\frac{n^2}{Z} = \frac{a_0}{2},$$

where a_0 is the Bohr radius. Suppose that the two electrons, owing to their electrostatic repulsion, tend to move apart, so that their separation is $2r_1$. Then the Coulomb interaction between the electrons is

$$E_c = \frac{e^2}{4\pi\epsilon_0 \cdot 2r_1} = \frac{14.4}{0.53} = 27.2 \text{ (eV)}.$$

Taking the Coulomb interaction between the electrons into account, the ground-state energy of the helium atom is

$$E_1' = E_1 + E_c = -108.8 + 27.2 = -81.6 \text{ (eV)}.$$

(3) Since the $4f$ electron is far from the nucleus while the $1s$ electron is close to it, the $4f$ electron is in the Coulomb field of a completely screened nucleus, with effective nuclear charge $Z^* \simeq 1$, whereas the $1s$ electron is in the Coulomb field of an unscreened nucleus, with effective nuclear charge $Z^* \simeq 2$. Using the hydrogen energy-level formula, the energy of the helium atom is

$$E_{1s4f} = -13.6 \times \frac{2^2}{1^2} - 13.6 \times \frac{1^2}{4^2} = -55.25 \text{ (eV)}.$$

Comment. Experimentally the ground-state energy of the helium atom is about 79 eV. Comparing this with the answers to parts (1) and (2), it is seen that in discussing the energy-level structure of a multi-electron atom one must take into account, besides the Coulomb interaction between the electrons and the nucleus, the Coulomb interaction between the electrons themselves. It is evident that the energy levels of helium are not the same as those of hydrogen, but when one electron of the helium atom is excited into a state far from the nucleus (a state with large n and l, for example the $4f$ state), its energy level coincides with the corresponding level of hydrogen (for example the $4f$ level), differing only by a constant (-54.4 eV).

Solution 5.15. Since the half-life $T_{1/2}(^{238}\text{U})$ of the parent nucleus is much longer than that of all the daughter nuclei, and among the daughter nuclei the one with the longest half-life is ^{234}U, with $T_{1/2}(^{234}\text{U}) = 2.45 \times 10^5$ years, after about $5 \times T_{1/2}(^{234}\text{U}) \approx 1.2 \times 10^6$ years the whole decay series reaches secular equilibrium, that is, the activities of the parent and of all the daughters are equal: $\lambda_1 N_1 = \lambda_2 N_2 = \cdots$. After this, the amount of the final stable nuclide continuously increases, while the sum of the numbers of daughter nuclei remains essentially unchanged.

Suppose that the present time is t_1 and the initial time is t_0. Then at time t_1 the sum of the residual parent nuclei $N_P(t_1)$, the stable nuclei $N_D(t_1)$ and the total number $\sum_i N_i(t_1)$ of nuclei of all generations of daughters equals the number $N_P(t_0)$ of parent nuclei at the initial time, i.e.

$$N_D(t_1) + N_P(t_1) + \sum_i N_i(t_1) = N_P(t_0).$$

Since $N_P(t_1) = N_P(t_0)\mathrm{e}^{-\lambda(t_1 - t_0)}$, that is, in the above equation $N_P(t_0) = N_P(t_1)\mathrm{e}^{\lambda(t_1 - t_0)}$. In addition, considering that the geological age is very long and the time required to reach secular equilibrium is much shorter than it, after secular equilibrium the sum of the daughter nuclei remains unchanged, so approximately $N_D(t_1) \gg \sum_i N_i(t_1)$. We therefore obtain

$$N_D(t_1) = N_P(t_1)\left[\mathrm{e}^{\lambda(t_1 - t_0)} - 1\right].$$

That is,

$$\Delta t = t_1 - t_0 = \frac{1}{\lambda} \ln\left[1 + \frac{N_D(t_1)}{N_P(t_1)}\right].$$

Substituting the known quantities, the age of the Earth is estimated to be

$$\Delta t = \frac{4.468 \times 10^9 \text{ yr}}{0.693} \ln\left[1 + \frac{1}{2.785}\right] = 1.98 \times 10^9 \text{ (yr)}.$$

Comment. Through this problem one may gain a good understanding of the laws of radioactive decay, of the characteristics of the secular equilibrium of radioactive series, and of their practical applications.

Solution 5.16. (1) The disintegrations per minute of 1 gram of carbon come from the ^{14}C contained in it. Let the activity be A'; then

$$\begin{aligned} A' &= \lambda N(^{14}\mathrm{C}) = \lambda \cdot \eta \cdot \frac{1}{12} N_A \\ &= \frac{0.693}{5730 \times 365 \times 24 \times 60} \times 1.2 \times 10^{-12} \times \frac{1}{12} \times 6.02 \times 10^{23} = 14 \text{ decays/(gram} \cdot \text{min)}, \end{aligned}$$

where N_A is Avogadro's constant.

(2) $A(t) = A(0)\mathrm{e}^{-\lambda t}$. Now $A(0) = A' = 14$ decays/(gram·min), so the age of the ancient biological sample is

$$t = \frac{1}{\lambda} \ln \frac{A(0)}{A(t)} = \frac{5730 \text{ yr}}{0.693} \ln 1.831 \approx 5000 \text{ (yr)}.$$

(3) $\delta = 1\%$ requires a count $N = 10000$. Let m be the required number of grams of the sample; the count of the ancient biological sample in 1 hour is now required to be 10000, hence

$$14 \times 60 \times m \cdot \frac{1}{1.831} = 10000.$$

From this we obtain

$$m = \frac{10000}{459} = 22 \text{ (grams)}.$$

Let the error in the age determined under this error be Δt; then from the formula $t = \dfrac{1}{\lambda} \ln \dfrac{A(0)}{A(t)}$ we obtain

$$\Delta t = \frac{1}{\lambda} \frac{\Delta A(t)}{A(t)} = \frac{5730 \text{ yr}}{0.693} \times \frac{1}{100} = 83 \text{ (yr)}.$$

Comment. This is a problem on archaeology with ^{14}C, and it is of practical significance.

Solution 5.17. (1) The β^- decay formula is

$$^{A}_{Z}\mathrm{X} \to {}^{A}_{Z+1}\mathrm{Y} + \mathrm{e}^- + \bar{\nu}_e.$$

The decay energy is the energy that can be released in the decay process (that is, the maximum kinetic energy of the emitted electron). Denoting the nuclear mass by m, the decay energy is

$$E_d(\beta^-) = [m_X - (m_Y + m_e)]\, c^2.$$

Considering that what is actually known are atomic masses, then, in the case where the difference between the electron binding energies of the parent (X) and daughter (Y) atoms is neglected, we have

$$E_d(\beta^-) = [M(\mathrm{X}) - M(\mathrm{Y})]\, c^2,$$

where M denotes the atomic mass.

(2) In this problem

$$E_d(\beta^-) = \left[M(^{17}\mathrm{N}) - M(^{17}\mathrm{O}^*)\right] c^2 = 3.72 \text{ (MeV)},$$

that is,

$$M(^{17}\mathrm{O}^*)c^2 = M(^{17}\mathrm{N})c^2 - 3.72 \text{ (MeV)}.$$

If $^{17}\mathrm{O}^*$ can emit a neutron and reach the excited state $^{16}\mathrm{O}^*$, then energetically the following inequality must be satisfied:

$$M(^{17}\mathrm{O}^*)c^2 - \left[M(^{16}\mathrm{O})c^2 + E^*(^{16}\mathrm{O})\right] - m_n c^2 > 0,$$

where $E^*(^{16}\mathrm{O})$ is the excitation energy of $^{16}\mathrm{O}$. Substituting the expression for $M(^{17}\mathrm{O}^*)c^2$ into the above gives

$$\begin{aligned}
&\left[M(^{17}\mathrm{N}) - M(^{16}\mathrm{O}) - m_n\right] c^2 - 3.72 \text{ MeV} - E^*(^{16}\mathrm{O}) \\
&= (17.008449 - 15.994915 - 1.008665)uc^2 - 3.72 \text{ MeV} - E^*(^{16}\mathrm{O}) \\
&= 0.82 \text{ MeV} - E^*(^{16}\mathrm{O}) > 0.
\end{aligned}$$

Since all the excitation energies $E^*(^{16}\mathrm{O})$ are greater than 0.82 MeV, in this problem $^{17}\mathrm{O}^*$ can emit a neutron in only one way, namely by decaying to the ground state of $^{16}\mathrm{O}$.

Let the energy of the emitted neutron, with the recoil of $^{16}\mathrm{O}$ taken into account, be E_n; then

$$E_n + E_n \cdot \frac{m_n}{M(^{16}\mathrm{O})} = 0.82 \text{ (MeV)}.$$

Hence

$$E_n \approx 0.82 \times \frac{16}{17} \text{ MeV} = 0.77 \text{ (MeV)}.$$

Comment. The decay energy is an important physical quantity in nuclear decay. This problem is concerned with how to use energy relations to decide whether a particular nuclear reaction can occur.

Solution 5.18. (1) The semi-empirical formula for the binding energy of a nucleus with Z protons and N neutrons ($A = Z + N$) is (the famous Weizsäcker formula)

$$B(Z, A) = a_V A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_a \frac{(N - Z)^2}{A} + B_P,$$

where a_V, a_s, a_c, a_a are adjustable parameters (constants), $a_V A$ is the volume energy, $a_s A^{2/3}$ the surface energy, $a_c \dfrac{Z^2}{A^{1/3}}$ the Coulomb energy, $a_a \dfrac{(N - Z)^2}{A}$ the symmetry energy, and B_P the pairing energy. The empirical formula for B_P is $B_P = \delta a_P A^{-1/2}$, where a_P is a constant,

$$\delta = \begin{cases} 1 & \text{even–even nucleus,} \\ 0 & \text{odd–even nucleus,} \\ -1 & \text{odd–odd nucleus.} \end{cases}$$

The last two terms in the formula arise from quantum effects and have no classical counterpart.

(2) In order to use the semi-empirical formula to calculate the nuclear radius constant, the Coulomb energy term in it should be replaced by the more accurate formula for the Coulomb energy that contains the nuclear radius R:

$$B_c = \frac{3}{5} \frac{Z(Z - 1)e^2}{4\pi\epsilon_0 R},$$

in which $R = \gamma_0 A^{1/3}$, where γ_0 is the nuclear radius constant; the corresponding value of γ_0 can be determined from the mass difference of mirror nuclei.

Knowing the binding energy of a nucleus, one obtains the corresponding semi-empirical formula for the atomic mass:

$$M(Z,A) = ZM(^1\mathrm{H}) + Nm_n - \frac{B(Z,A)}{c^2}.$$

Then for the mirror nuclei ${}^{11}_{5}\mathrm{B}$ and ${}^{11}_{6}\mathrm{C}$ we have

$$M({}^{11}_{5}\mathrm{B}) = 5M(^1\mathrm{H}) + 6m_n - B({}^{11}_{5}\mathrm{B})/c^2,$$

$$M({}^{11}_{6}\mathrm{C}) = 6M(^1\mathrm{H}) + 5m_n - B({}^{11}_{6}\mathrm{C})/c^2.$$

That is,

$$M({}^{11}_{6}\mathrm{C}) - M({}^{11}_{5}\mathrm{B}) = M(^1\mathrm{H}) - m_n + B({}^{11}_{5}\mathrm{B})/c^2 - B({}^{11}_{6}\mathrm{C})/c^2.$$

Since the two mirror nuclei have the same A and the same $(N-Z)^2$, we have

$$\left[M({}^{11}_{6}\mathrm{C}) - M({}^{11}_{5}\mathrm{B})\right]c^2 = \left[M(^1\mathrm{H}) - m_n\right]c^2 - \frac{3}{5}(5\times 4)\frac{e^2}{4\pi\epsilon_0\gamma_0 A^{1/3}} + \frac{3}{5}(6\times 5)\frac{e^2}{4\pi\epsilon_0\gamma_0 A^{1/3}}.$$

That is,

$$\frac{6e^2}{4\pi\epsilon_0\gamma_0 A^{1/3}} = \left[M({}^{11}_{6}\mathrm{C}) - M({}^{11}_{5}\mathrm{B}) - M(^1\mathrm{H}) + m_n\right]c^2 = 2.76\ (\mathrm{MeV}),$$

$$\gamma_0 = \frac{6\times 1.44\ \mathrm{MeV}\cdot\mathrm{fm}}{2.76\ \mathrm{MeV}\times 11^{1/3}} = 1.41\ (\mathrm{fm}).$$

Comment. The liquid-drop model description of the nucleus is very important in nuclear physics. Through this problem the reader is required to become familiar with the expression of the semi-empirical binding-energy formula and with its simple applications.

Solution 5.19. In this problem one may introduce the representation of the physical quantities in the centre-of-mass frame; combining the centre-of-mass frame with the laboratory frame makes the discussion convenient and clear. As shown in the figure, the incident particle M_1 is $M({}^4_2\mathrm{He})$ and the target nucleus M_2 is $M({}^1_1\mathrm{H})$. The speed of the incident particle is v_1, and the target nucleus is assumed to be at rest. The speed of the outgoing particle in the L frame is v_1', and in the centre-of-mass frame it is V_1; the recoil speed of the target nucleus in the L frame is v_2', and in the centre-of-mass frame it is V_2.

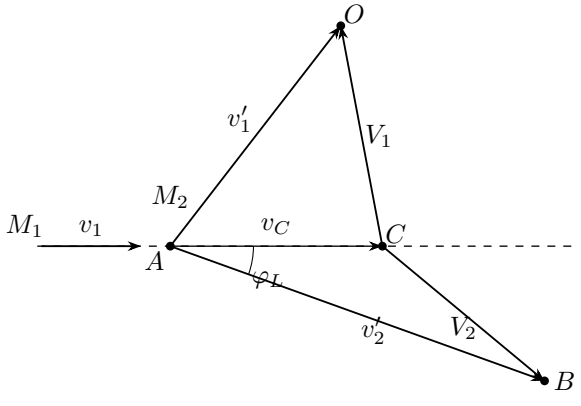


Fig. 5.19

(1) Before the collision the speed of the centre of mass is $v_C = v_1 \dfrac{M_1}{M_1 + M_2}$, in the same direction as v_1. The speed of the target nucleus is also v_C, but its direction is opposite to that of v_1.

Since in the elastic scattering process the magnitudes of the centre-of-mass velocities of the two colliding particles remain unchanged, the speed of the recoiling target nucleus is $V_2 = v_C$; hence from the figure it is seen that triangle ABC is isosceles. Since the velocity of the recoiling target nucleus in the laboratory frame is $v_2' = v_C + V_2$, the magnitude of v_2' is

$$v_2' = 2v_C \cos\varphi_L.$$

Accordingly the kinetic energy of the recoiling hydrogen nucleus is

$$\begin{aligned} E_{\text{recoil}} &= \frac{1}{2} M_2 v_2'^2 = \frac{1}{2} M_2 (2v_C \cos\varphi_L)^2 \\ &= \frac{1}{2} M(^1\text{H}) \left[\frac{2M(^4_2\text{He}) v_1}{M(^1\text{H}) + M(^4_2\text{He})} \right]^2 \cos^2 30^\circ \\ &= \frac{4M(^1\text{H}) M(^4_2\text{He})}{[M(^1\text{H}) + M(^4_2\text{He})]^2} E(^4_2\text{He}) \cos^2 30^\circ \\ &\approx \frac{4 \times 1 \times 4}{(1+4)^2} \cdot 2 \text{ MeV} \cdot \left(\frac{\sqrt{3}}{2} \right)^2 = 0.96 \text{ (MeV)}. \end{aligned}$$

(2) The first step is to find the expression for the differential cross section of the recoil particle in the centre-of-mass frame, $\sigma_C(\varphi_C)$. Using $\sigma_C(\varphi_C)\mathrm{d}\Omega(\varphi_C) = -\sigma_C(\theta_C)\mathrm{d}\Omega(\theta_C)$ (the minus sign arises because when $\mathrm{d}\Omega(\varphi_C)$ increases, the corresponding $\mathrm{d}\Omega(\theta_C)$ decreases), we obtain

$$\sigma_C(\varphi_C) = -\frac{\sin\theta_C \mathrm{d}\theta_C}{\sin\varphi_C \mathrm{d}\varphi_C} \sigma_C(\theta_C).$$

Since $\theta_C = \pi - \varphi_C$, we have $\mathrm{d}\theta_C = -\mathrm{d}\varphi_C$ and $\sin\theta_C = \sin\varphi_C$, and hence

$$\sigma_C(\varphi_C) = \left(\frac{a}{4} \right)^2 \frac{1}{\sin^4 \left(\dfrac{\pi - \varphi_C}{2} \right)} = \left(\frac{a}{4} \right)^2 \frac{1}{\cos^4 \left(\dfrac{\varphi_C}{2} \right)}.$$

The second step is to find the expression for the differential cross section of the recoil particle in the laboratory frame, $\sigma_L(\varphi_L)$. Using $\sigma_L(\varphi_L)\mathrm{d}\Omega(\varphi_L) = \sigma_C(\varphi_C)\mathrm{d}\Omega(\varphi_C)$, we obtain

$$\sigma_L(\varphi_L) = \frac{\sin\varphi_C \mathrm{d}\varphi_C}{\sin\varphi_L \mathrm{d}\varphi_L} \sigma_C(\varphi_C).$$

Since $\varphi_C = 2\varphi_L$, we have $\sin\varphi_C \mathrm{d}\varphi_C = 4\sin\varphi_L \cos\varphi_L \mathrm{d}\varphi_L$, and hence

$$\sigma_L(\varphi_L) = 4\cos\varphi_L \, \sigma_C(\varphi_C) = 4\cos\varphi_L \left(\frac{a}{4} \right)^2 \frac{1}{\cos^4\varphi_L} = \left(\frac{a}{4} \right)^2 \frac{4}{\cos^3\varphi_L}.$$

Now

$$a = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 E_C} = \frac{2 \times 1 \times 1.44 \text{ eV} \cdot \text{nm}}{2 \times 10^6 \text{ eV} \cdot \dfrac{1}{1+4}} = 7.2 \times 10^{-15} \text{ (m)}$$

(the above has made use of the centre-of-mass energy of the incident particle $E_C = \dfrac{M_2}{M_1 + M_2} E_L$).

Therefore

$$\sigma_L(\varphi_L) = \left(\frac{7.2 \times 10^{-15}}{4}\right)^2 \cdot \frac{4}{\left(\frac{\sqrt{3}}{2}\right)^3} \text{ (m}^2\text{/sr)}$$
$$= 0.20 \text{ b/sr} \qquad (1 \text{ b (barn)} = 10^{-28} \text{ m}^2).$$

Comment. This problem involves a comprehensive calculation of the kinematics and dynamics of two-particle elastic scattering, and also involves the transformation of the differential cross section between the centre-of-mass frame and the laboratory frame.

Experiments

Chapter 4

Mechanics and Thermal Experiments

Experiment 6.01 Measuring the density of paraffin wax by Archimedes' principle

[Available apparatus]

A physical balance, a thermometer, a glass beaker, fine thread, a copper block, a paraffin block, water, etc.

[Requirements] Write briefly the principle of the experiment, the experimental procedure, the recorded data and their processing. (The density of water is known to be $1.000\,\mathrm{g/cm^3}$.)

Solution 6.01. For paraffin wax, whose density is less than that of water, a part would float above the surface if it were immersed in water. In order to measure the volume of such a body, a heavy copper block may be suspended below the body to be measured, as shown in the figure. The principle of the measurement and the procedure are as follows. First immerse the copper block in water, so that the paraffin block is entirely above the water surface, and place the whole beaker, copper block, paraffin, etc. on the "special support" of the physical balance (note that they must not be placed on the pan of the balance); connect the thread carrying the paraffin and the copper block to the hook at the end of the balance arm. When the balance is in equilibrium, read the mass M_1 of the weights, as shown in Fig. 6.01(a). Then let the copper block and the paraffin block be immersed in water together; when the balance is in equilibrium, read the corresponding mass M_2 of the weights. The buoyant force acting on the paraffin block, the body to be measured, is

$$F = (M_1 - M_2)g,$$

where g is the acceleration due to gravity. If the volume of the paraffin block is V and the density of water is ρ_0, then by Archimedes' principle

$$(M_1 - M_2)g = \rho_0 V g.$$

If the mass M of the paraffin block in air has been determined beforehand, then the density ρ of the paraffin is

$$\rho = \frac{M}{M_1 - M_2}\,\rho_0.$$

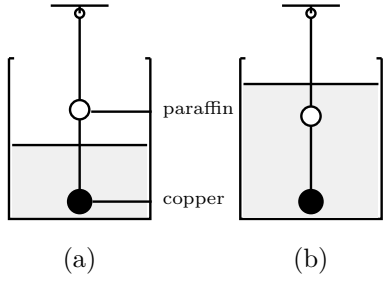


Fig. 6.01

Experiment 6.02 Measuring the density of solder wire with a pycnometer

[Available apparatus]

A physical balance, a pycnometer, a small coil of solder wire, scissors, water, a beaker, a cloth, etc.

[Requirements]

(1) Observe the structure of the pycnometer and describe briefly the function of the capillary in it;

(2) Explain the principle of measuring the density of a solid with a pycnometer, and write down the calculation formula; the meaning of every symbol in the formula must be explained;

(3) Measure the density ρ of a certain solder wire. The density of the water used in the experiment is $\rho_0 = 1.000\,\mathrm{g/cm^3}$;

(4) Measure the scale interval (or division value) of the balance when it is unloaded, and write down the calculation formula and the result.

Solution 6.02. **1.** Observe and analyse the structural features of the pycnometer; its structure is shown in the figure. The stopper of the pycnometer is a ground-glass stopper with a capillary in its middle. In use, liquid is poured into the pycnometer up to the mouth and the stopper is pressed in firmly; the excess liquid then flows out through the capillary. In this way the volume of the liquid contained in the vessel is kept fixed.

2. Let the mass of the solder wire to be measured be m_x, its density ρ_x and its volume V_x. Then the relation $m_x = \rho_x V_x$ holds, where m_x can be weighed on the balance. Let the mass of the pycnometer be M, its mass when filled with pure water be m_0, and its volume be V_0. The volume of the water is just the volume of the pycnometer (during the experiment one must constantly pay attention to the height of the water column in the capillary). There is also the relation $m_0 = V_0\rho_0 + M$, in which $\rho_0 = 1.000 \times 10^3\,\mathrm{kg/m^3}$; the value of m_0 can be weighed on the balance.

Immerse as much of the solder wire of mass m_x as possible in the pycnometer; part of the pure water overflows. Keeping the water column at its original height in the capillary, the total mass is

$$m_1 = \rho_x V_x + \rho_0(V_0 - V_x) + M.$$

m_1 can be weighed on the balance. From the three equations for m_x, m_0 and m_1, after simplification,

$$\rho_x = \rho_0 \frac{m_x}{m_x + m_0 - m_1}.$$

3. Data measurement and calculation

$$m_x = 93.50 \times 10^{-3}\,\mathrm{kg}, \qquad m_0 = 38.55 \times 10^{-3}\,\mathrm{kg}, \qquad m_1 = 118.80 \times 10^{-3}\,\mathrm{kg},$$

then

$$\rho_x = \frac{1.000 \times 10^3 \times 93.50}{93.50 + 38.55 - 118.80} = 7.060 \times 10^3\,\mathrm{kg/m^3}.$$

4. The scale interval of a balance is its division value; it represents the mass difference corresponding to two adjacent graduations indicated by the pointer. The scale interval is $\Delta m/\Delta L$, where ΔL is the displacement increment of the observed variable and Δm is the increment of the mass being measured, while $\Delta L/\Delta m$ is called the sensitivity of the balance.

If the balance is in equilibrium when unloaded, and an added mass $\Delta m = 0.020\,\mathrm{g}$ makes the pointer swing so that the difference between the deviations on the two sides is $\Delta L = 5.0 - 3.0 = 2.0$ divisions, then the scale interval of the balance when unloaded is

$$\frac{\Delta m}{\Delta L} = \frac{0.020\ \mathrm{g}}{2.0\ \text{divisions}} = 0.010\,\mathrm{g/division}.$$

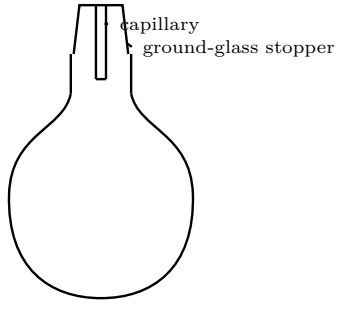


Fig. 6.02

Experiment 6.03 Measuring the volume ratio of a body in a mixture to a single impurity contained in it

Two cylinders of equal volume, each provided with a suspension thread, are available; one is made of wax, and the other is made of wax but contains a heavy-metal impurity. The density of this heavy-metal impurity is known to be $\rho_{\mathrm{metal}} = 7.67\,\mathrm{g/cm^3}$. Find the volume ratio n of the heavy-metal impurity to the wax contained in the mixture cylinder.

[Available apparatus]

A wax cylinder, a wax cylinder containing a heavy-metal impurity, a physical balance and weights, a beaker filled with water (the density of water is $1.000\,\mathrm{g/cm^3}$).

[Requirements]

(1) Write down the principle and the measurement formula.

(2) Measure and record the data, and write down the experimental result.

Solution 6.03. Let the volume of the paraffin part in the mixture cylinder be V_1 and the density of paraffin be ρ_1; let the volume of the heavy-metal part be V_2 and the density of the heavy metal be $\rho_2 = \rho_{\mathrm{metal}} = 7.67\,\mathrm{g/cm^3}$. Then the following two equations hold:

$$\begin{cases} V_1\rho_1 + V_2\rho_2 = M_2, \\ V\rho_1 = V_1\rho_1 + V_2\rho_1 = M_1. \end{cases} \tag{1}$$

In equations (1), V is the volume of the cylinder to be measured, M_2 is the mass of the mixture cylinder to be measured, and M_1 is the mass of the pure paraffin cylinder. Solving these equations

gives

$$n = \frac{V_2}{V_1} = \frac{(M_2 - M_1)\rho_1}{M_1\rho_2 - M_2\rho_1}. \tag{2}$$

In equation (2), M_1 and M_2 can both be determined with the physical balance. The density of the paraffin can be measured by the method for measuring the density of paraffin described in Solution 6.01; in the experiment the mixture cylinder may be used as the heavy body.

Experiment 6.04 Measuring the density of salt water by two methods

Using the apparatus and materials provided, measure the density of salt water by two methods.
[Available apparatus and materials]

A physical balance and weights, two 500 mL beakers, a plastic cylinder with a hole in its bottom, a number of steel balls, nylon thread, pure water, a pycnometer, a thermometer, a cloth, the salt water to be measured, and a table of the density of water at different temperatures.
[Requirements]

(1) Write down the two methods and formulas for measuring the density of salt water, and explain the meaning of each quantity in the formulas.

(2) The measurement results are required to have four significant figures.

Solution 6.04. The two different methods are: (1) measurement of the density of a liquid by the method of hydrostatic weighing, using Archimedes' principle; (2) measurement of the density of a liquid by the pycnometer method.

1. Measurement of the density of an unknown liquid by hydrostatic weighing

The principle and the method of the experiment are as follows.

Put a certain amount of small steel balls into a small plastic cylinder, suspend the plastic cylinder with nylon thread, and hang it on the hook above the pan of the balance. Weigh the mass of the "cylinder" body as M. Then place a beaker containing pure water on the special support of the physical balance. Let the "cylinder" body be completely immersed in the pure water, and weigh its apparent mass with the physical balance as M_1. Let the volume of the "cylinder" body be V and the density of pure water be ρ_0; then

$$M - M_1 = V\rho_0. \tag{1}$$

Similarly, replace the pure water in the beaker with salt water. When the "cylinder" body is completely immersed in the salt water, its apparent mass weighed with the physical balance is M_2. Let the density of the salt water to be measured be ρ_x; then

$$M - M_2 = V\rho_x. \tag{2}$$

Dividing equation (2) by equation (1) term by term gives

$$\rho_x = \frac{M - M_2}{M - M_1}\,\rho_0. \tag{3}$$

If the four quantities M, M_2, M_1 and ρ_0 are known, the value of ρ_x can be found. In order to ensure that the experimental result has four significant figures, the following must be done. (1) The values of $M - M_2$, $M - M_1$ and ρ_0 must all have more than four significant figures. Therefore the number of steel balls in the small plastic cylinder should be rather large, i.e. the volume V should be rather large, so as to ensure that both $M - M_1$ and $M - M_2$ have four significant figures. At the same time the water temperature must be measured, and the value of

ρ_0 obtained from the table must have four significant figures. (2) When changing the water or when measuring the salt water, the heavy body (the plastic cylinder and the steel balls) and the beaker must be wiped dry with a cloth, and the concentration of the salt water to be measured must not be changed during the experiment.

2. Measurement of the density of salt water by the pycnometer method

Let the mass of the pycnometer be M, and its mass when filled with pure water, weighed with the physical balance, be m_1. The density of pure water is ρ_0, obtained from a table after measuring the water temperature with the thermometer. Empty the water from the pycnometer, fill it with the salt water to be measured, and weigh its mass with the physical balance as m_2. Then

$$m_2 - M = V \cdot \rho_x, \tag{1}$$

$$m_1 - M = V \cdot \rho_0. \tag{2}$$

In the above equations, V is the volume of the pycnometer. Solving (1) and (2) gives

$$\rho_x = \frac{m_2 - M}{m_1 - M} \cdot \rho_0. \tag{3}$$

In the experiment of measuring the density of salt water by the pycnometer method, in order that the result for ρ_x have four significant figures, the following must be done. 1) The quantities $m_2 - M$, $m_1 - M$ and ρ_0 must all reach four significant figures; hence it is necessary to use a pycnometer of appropriate size. 2) When the pure water in the pycnometer is replaced by salt water for the measurement, it is also important that the pure water be poured out completely before the salt water is introduced.

Experiment 6.05 Measurement of the density of air

Air under ambient conditions is sealed with a rubber stopper in a glass vessel (a small ampoule bottle). The air in the vessel can be withdrawn with a medical syringe; as the number of suction strokes increases, the pressure in the bottle gradually decreases. Neglecting leakage past the rubber stopper during and after the suction strokes, devise a measurement scheme to determine the density of the air sealed in the vessel under ambient conditions, and require the systematic error (uncertainty) of the measurement to be kept below 3%.

[Available apparatus]

A glass vessel sealed with a rubber stopper (its volume is $V_0 = 312\,\mathrm{cm}^3$), a medical syringe ($30\,\mathrm{cm}^3$), a physical balance and weights, adhesive tape for sealing, and a sheet of graph paper measuring 20 cm × 25 cm.

[Requirements]

(1) Write down the principle of the measurement and the measurement formula on which it is based.

(2) Record the measured data and obtain the measurement result.

(3) Explain why the systematic error (uncertainty) is less than 3%.

Solution 6.05. 1. Principle and formulas of the measurement

Let the volume of the vessel be V_0, the pressure of the air in the vessel be p_0, the room temperature be T_0 (T_0 being the thermodynamic temperature), and the mass of the air in the vessel be M_0. Then

$$p_0 V_0 = \frac{M_0}{\mu} R T_0, \tag{1}$$

where R is the universal constant of ideal gases.

Insert the syringe and make the first suction stroke; the volume and pressure become $V_0 + V_c$ and p_1, where V_c is the suction volume of the syringe, $V_c = 30.0\,\mathrm{cm}^3$. Then, on reaching equilibrium,

$$\frac{p_1}{p_0} = \frac{V_0}{V_0 + V_c}.$$

Similarly, after n suction strokes the gas pressure is p_n:

$$\frac{p_n}{p_0} = \left(\frac{V_0}{V_0 + V_c}\right)^n. \tag{2}$$

Let the mass of the air remaining in the bottle after n suction strokes be M_n. Then

$$p_n V_0 = \frac{M_n}{\mu} R T_0,$$

that is,

$$\frac{p_n}{p_0} = \frac{M_n}{M_0} = \frac{M_0 - \Delta M_n}{M_0}, \tag{3}$$

where ΔM_n is the mass of the air removed by n suction strokes.

From (2) and (3) we obtain

$$\Delta M_n = M_0 \left[1 - \left(\frac{V_0}{V_0 + V_c}\right)^n\right]. \tag{4}$$

Equation (4) shows that ΔM_n varies linearly with $\left[1 - \left(\frac{V_0}{V_0 + V_c}\right)^n\right]$, the slope being M_0. Measure respectively the values ΔM_5, ΔM_{10}, ΔM_{15} and ΔM_{20} corresponding to 5, 10, 15 and 20 suction strokes, and find M_0 by linear regression or graphically. Then

$$\rho_0 = \frac{M_0}{V_0}. \tag{5}$$

2. Measurement data and results

The measured data are given in the table below.

$$V_0 = 312\,\mathrm{cm}^3, \qquad V_c = 30.0\,\mathrm{cm}^3.$$

Table 6.05

n	$1 - \left(\frac{V_0}{V_0 + V_c}\right)^n$	ΔM_n/g
5	0.368 3	115×10^{-3}
10	0.462	152×10^{-3}
15	0.606	195×10^{-3}
20	0.711	243×10^{-3}

Fitting a straight line to ΔM_n against $1 - \left(\frac{V_0}{V_0 + V_c}\right)^n$ gives:

slope $M_0 = 363 \times 10^{-3}\,\mathrm{g}$; correlation coefficient $r = 0.997$.

The density of the ambient air is

$$\rho = \frac{363 \times 10^{-3}}{312}\ \mathrm{g/cm^3} = 1.16 \times 10^{-3}\,\mathrm{g/cm^3}.$$

Comment. Because the slope M_0 is obtained by fitting a straight line, the influence of the systematic error due to the small amount of gas remaining in the bottle after the 20 suction strokes is eliminated, and the measurement result is made more accurate.

Experiment 6.06 Measurement of the volume of a rectangular thin plate

Measure the volume of a rectangular thin plate, making the accuracy of the measurement result as high as possible.

[Available apparatus and measuring instruments]

A metre rule, a micrometer, a vernier calliper, the thin plate to be measured.

[Requirements]

(1) Choose the measuring instruments correctly to measure the length, width and height of the body, so that the experimental result has four significant figures; the length, width and height need be measured only once each.

(2) Calculate the volume V of the thin plate and estimate the uncertainty $U(V)$ of the experimental measurement result.

(3) Express the measurement result in terms of the uncertainty.

Solution 6.06. The length x of the thin plate is measured with a metre rule whose division value is 1 mm, giving $x = 14.01\,\mathrm{cm}$; the width y is measured with a vernier calliper whose division value is 0.02 mm, giving $y = 4.998\,\mathrm{cm}$; the thickness z is measured with a micrometer whose division value is 0.01 mm, giving $z = 0.2491\,\mathrm{cm}$.

The volume of the thin plate is

$$V = x \cdot y \cdot z = 14.01 \times 4.998 \times 0.2491 = 17.44\,\mathrm{cm^3}.$$

Let the uncertainty in the measurement of the length x be $U(x)$, that in the measurement of the width y be $U(y)$, and that in the measurement of the height z be $U(z)$. Then the uncertainty $U(V)$ of the volume V of the thin plate is

$$U(V) = \sqrt{\left(\frac{\partial V}{\partial x}\right)^2 U^2(x) + \left(\frac{\partial V}{\partial y}\right)^2 U^2(y) + \left(\frac{\partial V}{\partial z}\right)^2 U^2(z)}.$$

Taking $U(x) = 0.05\,\mathrm{cm}$, $U(y) = 0.001\,\mathrm{cm}$ and $U(z) = 0.0005\,\mathrm{cm}$, we have

$$U(V) = \sqrt{(yz)^2U^2(x) + (xz)^2U^2(y) + (xy)^2U^2(z)} \approx 0.07\,\mathrm{cm^3}.$$

Hence the measurement result is

$$V = (17.44 \pm 0.07)\ \mathrm{cm^3},$$

the relative uncertainty being about 0.4%.

Experiment 6.07 Measurement of the internal diameter of a glass capillary

[Available apparatus]

A reading microscope, a glass capillary, etc.

[Requirements]

(1) Adjust the reading microscope correctly, and explain how the parallax is eliminated.

(2) Explain how the screw-pitch error is eliminated.

(3) Measure the internal diameter of the glass capillary.

Solution 6.07. The principal errors in measuring a length or a displacement with a reading microscope are the parallax error and the screw-pitch error.

The parallax is eliminated by making the image of the object and the scale of the reading microscope (engraved on the reticle) lie in the same plane. In actual operation, first adjust the eyepiece so that the reticle is sharp, that is, so that the reticle lies in the focal plane of the eyepiece. Then adjust the object distance so that the image of the object to be measured is also formed on the reticle; the reticle is then also sharp. In this way the parallax, which arises because the scale and the image are not in the same plane, is eliminated.

The screw-pitch error is eliminated as follows. When the micrometer screw of the reading microscope moves, both position measurements (left and right) should be made with the cross-wire moving in one direction, and the readings should be taken when the wire is tangent to the circle (as shown in the figure); in this way the screw-pitch error is avoided.

Table 6.07 Capillary diameter measurement data

Capillary end face	ϕ_1/mm	ϕ_2/mm	Diameter d/mm
A end	6.280	5.496	0.784
	6.196	5.390	0.806
	6.496	5.711	0.785
	6.457	5.652	0.805
	6.182	5.354	0.808
B end	3.098	2.303	0.795
	4.168	3.354	0.814
	4.440	3.644	0.796
	4.486	3.701	0.785
	3.344	2.560	0.784

The diameter is $\bar{d} = 7.96 \times 10^{-2}\,\text{cm}$.

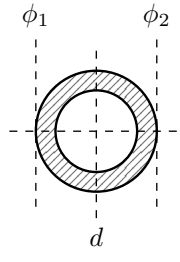


Fig. 6.07

Experiment 6.08 Measurement of the thickness of a thin plate with an optical lever

[Available apparatus and materials]

An optical lever, a support, a plumb line, the thin plate to be measured.

[Requirements]

(1) Adjust the optical lever correctly, so that the scale is parallel to the reflecting mirror and the principal axis of the telescope is perpendicular to the scale.

(2) Focus the telescope correctly.

(3) Measure the thickness h of the thin plate.

Solution 6.08. The optical lever consists of a reflecting mirror M whose base has three support points, a base plate, a scale, a telescope T, a plumb line and an iron stand, as shown in the figure.

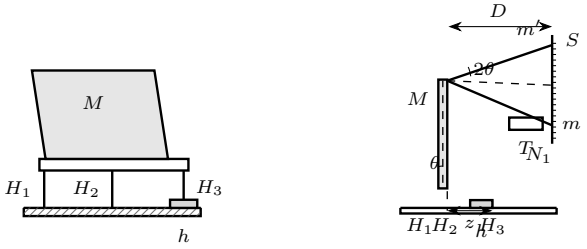


Fig. 6.08

The adjustment is made with the aid of the plumb line, aligning the parts by eye so that the scale is vertical and the telescope horizontal. Then adjust the eyepiece of the telescope so that the cross-wire is sharp, and then adjust the object distance so that the image of the scale formed through the reflecting mirror is seen sharply through the eyepiece of the telescope. At this time both the cross-wire and the image of the scale lie in the focal plane of the eyepiece. If the telescope is perpendicular to the scale, the angle of the reflecting mirror may be adjusted so that the value of the scale graduation corresponding to the principal axis of the telescope is the same as the value seen through the reflecting mirror; this shows that the mirror surface is parallel to the scale and also vertical. Then, keeping the feet H_1 and H_2 fixed, lift the foot H_3 and insert the thin plate to be measured under it. The support point of foot H_3 is raised by a height h. Then

$$\frac{h}{z} = \tan\theta, \tag{1}$$

where z is the distance from the support point of foot H_3 to the line joining H_1 and H_2, and θ is the angle through which the base of the reflecting mirror turns after the support point of foot H_3 is raised by h.

Moreover,

$$\tan(2\theta) = \frac{\Delta m}{D}, \tag{2}$$

where D is the distance between the scale and the reflecting mirror, and Δm is the distance between the position m' of the scale seen in the telescope after the mirror has turned through θ and the original reading m of the telescope. Since θ is rather small,

$$\arctan\theta = \theta\left(1 - \frac{1}{3}\theta^2 + \frac{1}{5}\theta^4 - \cdots\right) \approx \theta\left(1 - \frac{1}{3}\theta^2\right),$$

and therefore

$$h = \frac{z \cdot \Delta m}{2D}\left[1 - \frac{1}{3}\left(\frac{\Delta m}{D}\right)^2\right]. \tag{3}$$

In equation (3), Δm is the difference between the readings of the image of the scale seen in the telescope, produced by the change in position of H_3.

Table 6.08

Number	Original reading m/mm	Later reading m'/mm	D/cm	z/cm
1	70.2	71.9	173.5	5.03
2	70.3	72.0	173.0	5.03
3	70.4	71.8	173.3	5.02

$$\Delta\bar{m} = 71.9 - 70.3 = 1.6 \text{ (mm)},$$

$$h = \frac{\bar{z} \cdot \Delta\bar{m}}{2\bar{D}} = \frac{5.03 \times 1.6}{2 \times 173.3} = 0.023\,\text{mm}.$$

Experiment 6.09 Measurement of Young's modulus with a Hall position sensor

Measure the displacement of the centre of a thin plate when it is under tension and the dimensions of the plate, and hence calculate the Young's modulus of the plate material.
[Available apparatus]

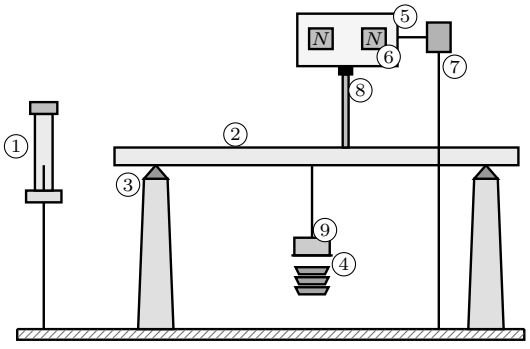
An FD-HY-1 type Young's modulus measuring instrument using the Hall position sensor (the FD in the model designation indicates that it is produced by Fudan Tianxin Scientific Instrument Company), weights of 10.00 g, a vernier calliper, a micrometer, and one brass sample and one cast-iron sample.
[Requirements]

(1) Use the reading microscope to measure the displacement of the centre of the brass sample under a tensile force, measure the dimensions of the sample, and calculate the Young's modulus of brass.

(2) Calibrate the Hall position sensor and determine its sensitivity.

(3) Measure the Young's modulus of cast iron with the Hall position sensor.

Solution 6.09. The experimental arrangement for measuring Young's modulus with a Hall position sensor is shown in Fig. 6.09(a).



1. Reading microscope.
2. Crossbeam.
3. Knife edge.
4. Weights.
5. Magnet box.
6. Magnets (two N poles placed facing each other).
7. Three-dimensional adjusting frame.
8. Brass lever (a type 95A integrated Hall sensor is attached to its top and extends into the magnet box).
9. Copper hanger (with graduation lines on it).
10. Leads of the integrated Hall sensor.

Fig. 6.09(a)

When the crossbeam is bent, the Young's modulus E is expressed by

$$E = \frac{d^3 mg}{4a^3 b\,\Delta z}, \tag{4}$$

where d is the distance between the two knife edges, a is the thickness of the beam, b is the width of the beam, m is the mass of the added weights, Δz is the distance through which the centre of the beam descends under the action of the external force, and g is the acceleration due to gravity.

The structure of the Hall position sensor is shown in Fig. 6.09(b). It uses two identical magnets, placed with their N poles facing each other; the two magnets are strictly parallel, and the Hall element is placed parallel to the magnets on the central axis of the gap between them.

From the Hall effect, the Hall potential difference U_H is

$$U_H = KIB, \tag{5}$$

where K is the sensitivity of the Hall element, I is the current through the Hall element, and B is the magnetic flux density. If the current I is constant and the magnetic field produced by the magnets is a uniform gradient field in the region at the centre of the gap, that is, $\mathrm{d}B/\mathrm{d}z$ is constant, then

$$\Delta U_H = KI\,\frac{\mathrm{d}B}{\mathrm{d}z}\cdot\Delta z, \tag{6}$$

where Δz is the displacement; in a uniform gradient field ΔU_H is proportional to Δz.

The experimental method is: (1) adjust the magnet box to be horizontal; (2) adjust the upright post below the magnet box and move the magnets relative to the Hall element so that the digital voltmeter indicates the output voltage of the Hall element $U_H = 0$ V; (3) adjust the eyepiece and the object distance of the reading microscope so that the graduation line on the groove of the copper knife edge is imaged sharply on the reticle; (4) after each addition of a 10.00 g weight,

read the position of the graduation line and the corresponding reading of the digital voltmeter; (5) measure the distance d between the two knife edges of the beam with a metre rule, the width b of the beam with a vernier calliper, and the thickness a of the beam with a micrometer; (6) calculate the experimental result. The experimental results are shown in Tables 1 and 2.

Table 6.09-1 Displacement after loading and sensor output

Serial number i	1	2	3	4	5	6	7	8
M_i/g	20.00	30.00	40.00	50.00	60.00	70.00	80.00	90.00
U_i/mV	0	25	52	78	103	131	157	183
z_i/mm	2.297	2.396	2.523	2.653	2.778	2.895	3.004	3.129

Table 6.09-2 Measured dimensions of the brass sample

											Mean
a'/mm	0.997	1.010	1.007	1.011	1.011	1.011	1.008	1.003	1.000	0.995	
a/mm	0.993	1.006	1.003	1.007	1.007	1.007	1.004	0.999	0.996	0.991	1.001
b/cm	2.302	2.298	2.296	2.300	2.296	2.298	2.300	2.300	2.298	2.300	2.299

The zero point of the micrometer is $a_0 = 0.004\,\text{mm}$, so $a = a' - a_0$; $d = 23.10\,\text{cm}$. Moreover,

$$\Delta z_1 = z_8 - z_4 = 0.476\,\text{mm}, \quad \Delta z_2 = z_7 - z_3 = 0.481\,\text{mm},$$

$$\Delta z_3 = z_6 - z_2 = 0.499\,\text{mm}, \quad \Delta z_4 = z_5 - z_1 = 0.481\,\text{mm},$$

hence $\Delta\bar{z} = 0.484\,\text{mm}$. Taking the acceleration due to gravity in Shanghai as $g = 9.794\,\text{m/s}^2$,

$$E = \frac{d^3\bar{M}g}{4a^3b\,\Delta\bar{z}} = \frac{(23.10\times10^{-2})^3\times40.00\times10^{-3}\times9.794}{4\times(1.001\times10^{-3})^3\times2.299\times10^{-2}\times0.484\times10^{-3}} = 10.8\times10^{10}\,\text{N/m}^2.$$

The accepted value of Young's modulus of this brass material is 10.55×10^{10} N/m^2, and the percentage difference between the two is 2.4%.

For the sensitivity $\Delta U/\Delta z$ of the Hall position sensor: fitting a straight line U versus z by the method of least squares gives the slope $\Delta U/\Delta z = 218\,\text{mV/mm}$ and the correlation coefficient $r = 0.9996$. The sensitivity of this position sensor is therefore 218 mV/mm.

If the brass sample is replaced by a cast-iron sample, then, with the Hall position sensor moving in a straight line in its original direction, it is only necessary to add a 90.00 g weight to read the corresponding voltage change ΔU, and the displacement Δz can be found from the sensitivity of the sensor. Measuring the dimensions of the cast-iron sample in the same way as for the brass sample, the Young's modulus of cast iron can also be calculated.

Experiment 6.10 Measurement of the volume of a solid with a Jolly balance

[Available apparatus and materials]

An FD-GLB-2 new-type Jolly balance apparatus, a beaker, a thermometer, water and the organic-glass block to be measured, and a table of the density of water as a function of temperature.

[Requirements]

(1) Write down the experimental procedure for measuring the stiffness coefficient of a spring with the new-type Jolly balance.

(2) Measure the stiffness coefficient of the spring.

(3) Measure the volume of the organic-glass block.

Solution 6.10. The new-type Jolly balance and the counter-timer are shown in the figure.

In Fig. 6.10, part (a) shows the Jolly balance, part (b) the small magnet and the Hall-switch sensor, and part (c) the counter-timer. In this experiment only the Jolly balance is needed; the timer is not used, and the stiffness coefficient of the spring can be measured.

The experimental procedure for measuring the stiffness coefficient k of the spring with the Jolly balance is as follows.

(1) Adjust the levelling screws of the base plate so that the upright column of the Jolly balance is vertical.

(2) Hang the hook at the top of the main scale, then mount the spring and the balancing cylinders (two cylinders), so that the small pointer is clamped between the two balancing cylinders; the lower end of the balancing cylinders is hooked to the weight pan through the hook. The spring is now stretched by a certain distance.

(3) Adjust the height of the small vernier so that the base-line graduation on the left side of the vernier is approximately aligned with the pointer. Tighten the locking screw that fixes the vernier, and adjust the fine-adjustment screw so that the pointer coincides with the graduation on the frame of the mirror. Only when the graduation on the mirror frame, the pointer and the image all coincide can the observer read the position of the pointer on the main scale and the vernier scale.

(4) First place a 1 g weight in the weight pan and, following step (3), read the position of the pointer. Then add another 1 g weight and read the new position of the pointer, and so on until all 10 g of weights have been placed on the pan; read the corresponding pointer positions.

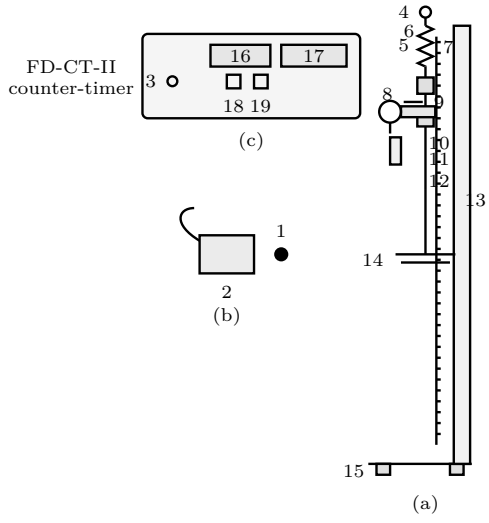
(5) From the mass M_i of the weights placed in or removed from the pan each time and the corresponding pointer position Y_i, find the stiffness coefficient of the spring by the method of successive differences or graphically.

The experimental results are given in the table below.

The acceleration due to gravity in the Shanghai region is $g = 9.794\,\mathrm{m/s^2}$.

Table 6.10 Data for the relation between M and y

M/g	y/mm (weights added)	y'/mm (weights removed)	$\bar{y}$/mm
1.000	211.98	212.36	212.17
2.000	214.22	214.62	214.42
3.000	216.42	216.84	216.63
4.000	218.46	218.96	218.71
5.000	220.52	221.12	220.77
6.000	222.92	223.14	223.03
7.000	224.98	225.24	225.11
8.000	227.16	227.56	227.36
9.000	229.36	229.58	229.46
10.000	231.64	231.64	231.64



1. Small magnet; 2. Hall switch; 3. Trigger indicator; 4–15. Jolly balance; 16–19. Counter-timer.

Fig. 6.10

Processing the data by the method of successive differences, the average extension of the spring for each additional 5.000 g of weights in the pan is $\Delta\bar{y} = 10.780\,\text{mm}$. By Hooke's law,

$$\Delta M \cdot g = k \cdot \Delta y. \tag{1}$$

Substituting $\Delta M = 5.000\,\text{g}$, $\Delta\bar{y}$ and g into (1) gives the stiffness coefficient of the spring $k = 4.543\,\text{N/m}$. Once the stiffness coefficient of the spring has been measured, suspend one body to be measured from the spring, measure the extension Δy, and the weight of the body, or its mass, can be calculated. Then, using Archimedes' principle and the method of weighing by buoyancy, the volume of the organic-glass block to be measured is obtained; the principle and the experimental method may be found in the relevant part of Solution 6.01.

Comment. For experiments on the measurement of length and volume, we note the following.

(1) The length of an ordinary body is usually measured with a metre rule; for the length of a shorter body, a vernier calliper is also often used in order to read more accurately. For lengths of about 1 mm or less, however, a micrometer, a reading microscope, an optical lever, etc. are commonly used. For vernier callipers, micrometers, reading microscopes, optical levers and the like, one must master their working principles and methods of operation and know their ranges and division values, so that the instruments are chosen and read correctly.

(2) For regular bodies such as metal wires and metal plates, although their diameter, or their length, width and height, are essentially uniform, these quantities form part of a measurement of a property of the material (for example, the measurement of Young's modulus), and their measurement accuracy greatly affects the experimental result. Therefore the diameter of a steel wire, the thickness of a thin plate, etc. must be measured several times at different positions and the average taken.

The following problem is similar to Experiment 6.08 in its method of measuring a small displacement. Can you solve it?

Experiment 6.11 Measurement of Young's modulus of a fine steel wire with an optical lever

[Available apparatus]

A Young's modulus measuring instrument (telescope, straight scale, optical lever, weights, etc.), a micrometer, a tape measure, a set square, a fine steel wire.
[Requirements]

(1) Draw up a table for the quantities to be measured.

(2) Adjust the state of the measuring instrument.

(3) Choose the measuring instruments correctly, and enter the measured data in the table.

(4) Calculate the mean value of Young's modulus and the relative error (ΔN being treated by the method of successive differences).

Hint 6.11. 1. Let the original length of the metal wire be L; under the action of a force its extension is ΔL, and its relative extension $\Delta L/L$ is called the strain. According to Hooke's law, within the elastic limit the strain is proportional to the stress (the force acting on unit cross-sectional area of the wire).

Let the cross-sectional area of the wire be S. Under the action of a force F the strain produced is $\Delta L/L$; then $F/S \propto \Delta L/L$, which may be written as an equation $F/S = E\,\Delta L/L$, where E is Young's modulus. Hence

$$E = \frac{FL}{S \cdot \Delta L}.$$

Also

$$S = \pi \left(\frac{d}{2}\right)^2,$$

where d is the diameter of the wire.

2. The quantities of the optical lever are marked as shown in the figure, and their relation is given by

$$\Delta L = \frac{h\,\Delta N}{2D},$$

where ΔN is the difference between the scale readings before and after the weights are suspended, obtained by the method of successive differences; h is the perpendicular distance from the rear foot of the optical lever to the line joining the two front feet; D is the distance from the mirror to the scale.

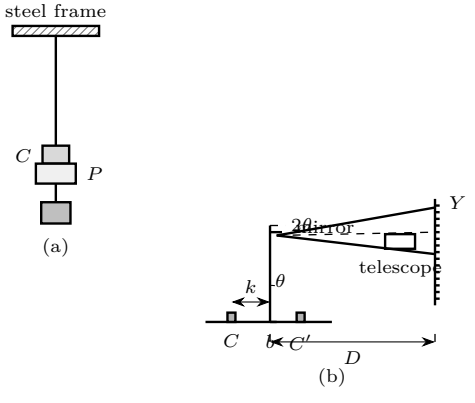


Fig. 6.11

3. The formula for the measurement of Young's modulus is

$$E = \frac{8FLD}{\pi d^2 h\,\Delta N},$$

where F is the force applied to the steel wire after the successive-difference treatment.
4. When measuring ΔN, the method of adding weights and the method of removing weights must both be used.

Experiment 6.12 Measurement of the speed of sound in air

[Available apparatus]

An oscilloscope, a low-frequency signal generator, an ultrasonic sound-velocity measuring instrument, a three-way connector (electrical divider), and several coaxial cables.
[Requirements]

(1) Write down the relevant formulas.

(2) Write down brief experimental steps.

(3) Measure several times and take the average.

Explanation: An alternating voltage is applied to a piezoelectric transducer, which produces mechanical vibrations and radiates sound waves of the same frequency. After the sound wave has travelled a certain distance in air, it is received by the receiver and converted into an output signal of the same frequency. Compared with the input signal, the output signal has a corresponding phase delay $\Delta\varphi$. This phase difference depends on the speed of the sound wave, its frequency and the distance of propagation. By measuring the relevant quantities, the speed of the sound wave can be obtained.

While receiving the sound wave, the receiver also reflects part of it back. When the distance between the two transducers is a certain particular value, the signal received by the receiver reaches a maximum (minimum) value. When the distance changes by half a sound wavelength, the receiver again receives a signal of another maximum (minimum) value. As the distance between the receiver and the transmitter changes, the amplitude of the signal received by the receiver varies periodically.

Solution 6.12. The experimental arrangement is shown in the figure. Two transducers of the same natural frequency are mounted facing each other on the jaws of a vernier calliper. By moving the vernier and the fine-adjustment device, the distance between the two transducers can be adjusted precisely.

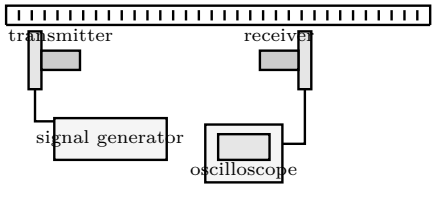


Fig. 6.12

(1) *Resonance interference method.* If the frequency f is kept constant and the distance between the receiving surfaces at two successive maxima of the received signal is measured, this distance being $\lambda/2$, where λ is the wavelength of the sound wave, then the speed of sound can be calculated from $v = f\lambda$.

(2) *Phase comparison method.* At the same instant the phase of the wave at the transmitter differs from that at the receiver. Observing the Lissajous figures, the phase φ and the propagation time t corresponding to the angular frequency are related by

$$\varphi = \omega t,$$

or

$$\varphi = \frac{2\pi l}{\lambda},$$

where l is the distance between the receiver and the transmitter. When $l \approx n(\lambda/2)$ $(n = 1, 2, 3, \dots)$, one obtains $\varphi = n\pi$. When the phase difference changes by π, the corresponding change in the distance l is half a wavelength. Once f and λ are known, the speed of sound can be found from $v = f\lambda$.

Experimental steps:

(1) Adjust the resonant frequency of the measuring system.

(2) Measure the half-wavelength $\lambda/2$ by the resonance method and by the phase method; a fairly large number of data should be taken.

(3) Process the data by the method of successive differences to obtain the speed of sound at the time, and record the air temperature at that time.

The experimental data are shown in Tables 1 and 2.

Table 6.12-1 Data measured by the resonance interference method

Serial number	f/Hz	s_n/cm	$\Delta s = s_{n+5} - s_n$/cm
1	35 676	5.688	2.400
2	35 669	6.172	2.406
3	35 670	6.654	2.391
4	35 662	7.136	2.401
5	35 672	7.615	2.393
6	35 668	8.088	
7	35 671	8.578	
8	35 671	9.045	
9	35 663	9.537	
10	35 672	10.008	

Mean value $\bar{f} = 35\,669.4\,\text{Hz}$; $\Delta\bar{s} = 2.3977\,\text{cm}$;

$$\frac{\bar{\lambda}}{2} = \frac{\Delta\bar{s}}{5}; \qquad \text{wavelength } \bar{\lambda} = 0.9591\,\text{cm};$$

$$v_1 = \bar{f} \cdot \bar{\lambda} = 35\,669.4 \times 0.9591 = 342.1\,\text{m/s}.$$

Table 6.12-2 Data measured by the phase method

Serial number	s_n/cm	$\Delta s = s_{n+5} - s_n$/cm	Serial number	s_n/cm	$\Delta s = s_{n+5} - s_n$/cm
1	9.500	2.411	6	11.911	
2	9.984	2.404	7	12.388	
3	10.464	2.405	8	12.869	
4	10.950	2.387	9	13.337	
5	11.425	2.401	10	13.826	

$f = 35\,669\,\text{Hz}$; $\dfrac{\bar{\lambda}}{2} = \dfrac{\Delta\bar{s}}{5}$; the mean wavelength $\bar{\lambda} = 0.9608\,\text{cm}$;
the speed of sound $v = \bar{f} \cdot \bar{\lambda} = 35\,669 \times 0.9608 = 342.7\,\text{m/s}$.

The temperature is $\theta = 14.8\,^{\circ}\mathrm{C}$; the sky is clear and the influence of water vapour is neglected. The accepted value of the speed of sound is

$$v = 331.45\sqrt{1+\frac{\theta}{\theta_0}} = 331.45 \times \sqrt{1+\frac{14.8}{273.15}} = 340.31\,\mathrm{m/s}.$$

The uncertainty of the speeds of sound measured by the two methods relative to the accepted value is less than 1%.

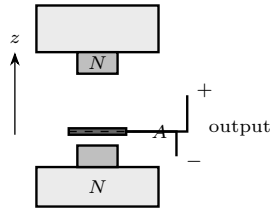


Fig. 6.09(b)

Experiment 6.13 Measuring the speed and the acceleration of a slider sliding on an inclined air track

On an air track adjusted to a certain inclination, use a flat light-blocking plate to measure the speed of the slider after it has slid 50.0 cm from rest and the acceleration of the slider during its descent.

[Available apparatus]

An air track already adjusted to an inclination and connected to compressed air (a metre rule is pasted along one side of the track), a slider, a flat light-blocking plate, two light gates, a photoelectric timer, a vernier calliper.

[Requirements]

(1) Write down the formulas of the experimental principle.

(2) Write down the plan of the experimental design.

(3) Draw up a table of the original data.

(4) Obtain the experimental result graphically.

Hint 6.13. 1. The flat light-blocking plate has already been fixed on the slider and need not be adjusted in the experiment.

2. One of the two light gates need not be moved during the experiment.

Instructions for the photoelectric timer:

(1) *Light gate.* It consists of one light-emitting tube and one photodiode, and is fixed on the air track by a bracket (held by two screws; to move the bracket, loosen the screws and move it by grasping the base of the bracket with both hands).

(2) *Timer.* Plug in the power supply; the light-emitting tube of the light gate lights up and the display of the timer shows "0". When the flat light-blocking plate passes through the light gate and blocks the light once, the display shows "1" and at the same time the timer starts timing. If the plate passes through the light gate a second time, the display shows "2" and at the same time the timer stops timing; a moment later it displays a number, and this number is the time interval between the two passages of the plate through the light gate (that is, between the first and the second blocking of the light). The unit of the digits before the decimal point on the display of the timer is the second.

If the count is to be restarted, press the "zero clearing" key; the data recorded by the timer are then erased and the display again shows "0".

Solution 6.13. **1.** If a body slides down a straight smooth inclined plane and the air resistance and the friction of the plane may be neglected, the body moves with uniform acceleration under the action of a constant force (the component of gravity along the plane), and its law of motion satisfies

$$v^2 = v_0^2 + 2as, \tag{1}$$

where v_0 and v are the instantaneous speeds of the body at $t = 0$ and at $t = t$ respectively, s is the distance travelled by the body during the time interval t, and a is the acceleration of the body. The experimental arrangement is shown in Fig. 6.13(a).

One end of the air track, which is horizontal when unloaded, is raised by a metal block of height h;

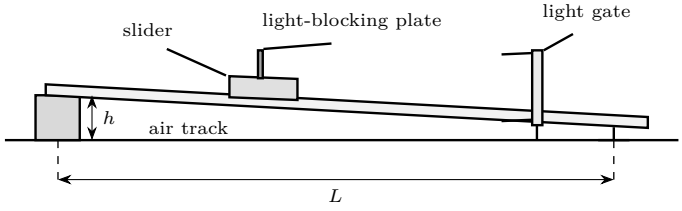


Fig. 6.13(a)

let the distance between the foot screws at the two ends of the air track be L. Then the acceleration of the body sliding down the inclined air track is

$$a = g \sin\theta = g \frac{h}{L}, \tag{2}$$

where g is the acceleration due to gravity and θ is the angle of inclination of the track surface to the horizontal.

If the body slides down from rest from the top of the air track, and the speeds $v_1, v_2, \ldots, v_7$ of the body at different positions $A_1, A_2, \ldots, A_7$ are measured in turn by the photoelectric timing method, as shown in Fig. 6.13(b), and the corresponding distances $s_1, s_2, \ldots, s_7$ travelled are recorded, then a v^2–s graph is plotted on graph paper. If a straight line is obtained, this shows that the body moves with uniform acceleration under the action of a constant force, and the slope of the graph is $2a$. Once the acceleration a has been found, the acceleration due to gravity g can also be found, provided the values of h and L have been measured.

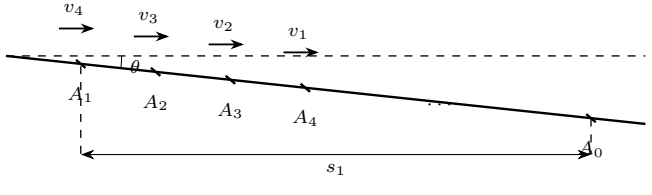


Fig. 6.13(b)

Usually the length of the light-blocking plate on the slider is Δl and the blocking time displayed by the digital timer is Δt; then the average speed of the slider as it passes through the light gate is

$$v = \frac{\Delta l}{\Delta t}.$$

If the length Δl of the light-blocking plate is made sufficiently small, so that the time during which the plate passes through the light gate is very short, the above average speed is approximately the instantaneous speed.

2. Experimental data

Determination of the v^2–s relation

The length of the light-blocking plate is $\Delta l = 1.000\,\text{cm}$, and the speed of the slider is $v = \Delta l/\Delta t$, where Δt is the blocking time of the slider. The point A_0 is the position of the fixed light gate, whose reading on the scale of the air track is 130.00 cm; A_i are the positions of the light gate placed in turn at the various observation points. The distance between A_i and A_0 is $s_i = A_0 - A_i$. The height of the block is $h = 4.26\,\text{mm}$. The measured data are given in the table below.

Table 6.13

No.	A_i/cm	s/cm	$\Delta t/10^{-4}\,\text{s}$	$v/\text{cm}\cdot\text{s}^{-1}$	$v^2/\text{cm}^2\cdot\text{s}^{-2}$
1	40.00	90.00	123.2	81.17	6 589
2	50.00	80.00	129.0	77.52	6 009
3	60.00	70.00	136.4	73.31	5 375
4	70.00	60.00	143.2	69.83	4 877
5	80.00	50.00	153.6	65.10	4 239

Plotting the v^2–s graph (Fig. 6.13(c)) gives a straight line. The slope of the line may also be found by the method of least squares.

By the method of least squares the slope is $k = 58.32\,\text{cm}^2/\text{s}^2$ and the intercept $b = 1335.4\,\text{cm}^2/\text{s}^2$; hence the acceleration is $a = 29.16\,\text{cm/s}^2$, and the correlation coefficient is $r = 0.99943$. The linearity is good, that is, the body moves with uniform acceleration when acted on by a constant force. From the equation of the line

$$v^2 = k \cdot s + b$$

it follows that when $v = 0$, $s_i = -\dfrac{b}{k} = -22.89\,\text{cm}$; that is, the value of $s_i = A_0 - A_i$ at the point where the speed of the slider is zero is $-22.89\,\text{cm}$ ($A_i = 152.89\,\text{cm}$). The speed of the body at 50.00 cm from its starting point at rest is

$$v = \sqrt{k \cdot s' + b} = \sqrt{58.32 \times [130.00 - (152.89 - 50.00)] + 1335.4} = 54.00\,\text{cm/s}.$$

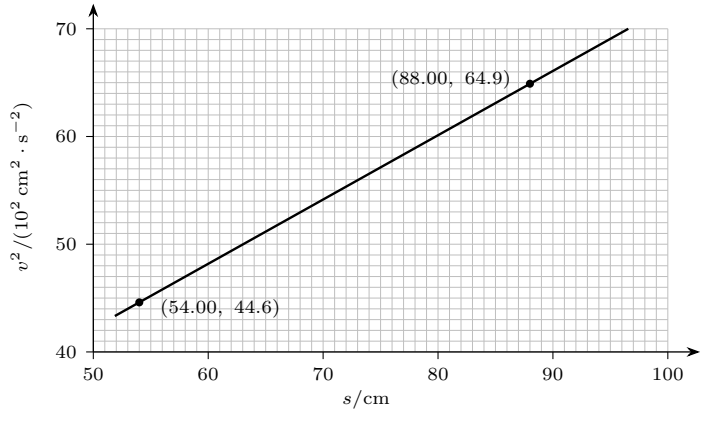


Fig. 6.13(c)

Experiment 6.14 Measuring the acceleration due to gravity with a simple pendulum

The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{l/g}$, where the pendulum length l is the distance from the point of suspension to the centre of gravity of the pendulum bob. Use the following apparatus to measure the acceleration due to gravity g.

[Available apparatus]

A support stand, an irregular ball, a metre rule, a stopwatch, fine thread.

[Requirements]

(1) Derive the formula for the determination of the acceleration due to gravity g.

(2) Write down the experimental procedure briefly.

(3) Record the measured data and obtain the experimental result.

Solution 6.14. From the formula for the period of oscillation of a simple pendulum,

$$T^2 = \frac{4\pi^2}{g}(l_1 + r). \tag{1}$$

In equation (1), l_1 is the length of the thread from the upper point of suspension of the pendulum to the point where the thread is attached to the ball, while r is the distance from the point of attachment to the centre of gravity of the irregular ball, and $l = l_1 + r$. If the length l_1 is changed to six different values, six corresponding values of the period can be measured. Taking l_1 as the independent variable and the period T as the dependent variable, draw a straight line on graph paper; its slope gives $4\pi^2/g$, and the reciprocal of this value multiplied by $4\pi^2$ gives the acceleration due to gravity. Besides processing the data graphically, one may also process them by the method of successive differences: measure the six corresponding thread lengths $l_1, l_2, \ldots, l_6$, and measure the corresponding periods $T_1, T_2, \ldots, T_6$ with the stopwatch. The method of successive differences gives

$$T_6^2 - T_3^2 = \frac{4\pi^2}{g}(l_6 - l_3),$$

$$T_5^2 - T_2^2 = \frac{4\pi^2}{g}(l_5 - l_2),$$

$$T_4^2 - T_1^2 = \frac{4\pi^2}{g}(l_4 - l_1),$$

$$g = 4\pi^2\left[\frac{l_6 + l_5 + l_4 - l_3 - l_2 - l_1}{T_6^2 + T_5^2 + T_4^2 - T_3^2 - T_2^2 - T_1^2}\right].$$

Experimental procedure:

(1) Suspend the irregular ball from the clamping fixture of the support stand with fine thread, and make the distance l_1 between the point of suspension and the ball as large as possible, for example 100.0 cm.

(2) Take the passage of the irregular ball through the vertical position of the thread as the starting point and the end point of the timing, and record with the stopwatch the time of 50 oscillations; divide by 50 to obtain the period of oscillation T_6 of the simple pendulum. The angle of swing is less than 5°.

(3) Change the length of the thread in turn to 90.0 cm, 80.0 cm, 70.0 cm, 60.0 cm and 50.0 cm, and measure respectively the periods of oscillation T_5, T_4, T_3, T_2 and T_1 of the pendulum.

(4) Process the data by the method of successive differences and find the acceleration due to gravity g.

Experiment 6.15 Accurate measurement of the acceleration due to gravity with a simple pendulum

[Available apparatus]

An experimental simple-pendulum apparatus with adjustable pendulum length, in which the pendulum is provided with a metre rule, a small magnet, a Hall-switch sensor and a counter-timer (produced by Fudan Tianxin Company), one sheet of graph paper, one ruler, and one vernier calliper.

[Requirements]

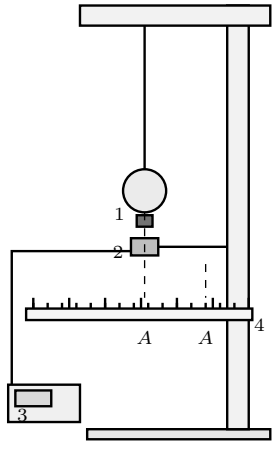
(1) Measure the period of oscillation of the simple pendulum with the Hall switch and the timer.

(2) It is known that the period of oscillation T of a simple pendulum depends on the angle of swing θ:

$$T = 2\pi\sqrt{\frac{L}{g}}\left[1 + \frac{1}{4}\sin^2\left(\frac{\theta}{2}\right)\right].$$

Design an experimental method by which the acceleration due to gravity g of this region can be measured accurately.

Solution 6.15. When the period can be measured accurately with the Hall-switch sensor together with the timer, the influence of the angle of swing θ on the period T must be taken into account, that is, the second-order approximation $\frac{1}{4}\sin^2(\theta/2)$ cannot be neglected; moreover, the influence of air damping on the decay of the oscillation period must also be considered, so the number of oscillations should not be too large. In this experiment the double period $2T$ corresponding to different amplitudes θ_m is measured, and a graph of $2T$ against $\sin^2(\theta_m/2)$ is plotted. Extrapolating this straight line to find the intercept (the value of $2T_0$) at $\theta_m = 0$ gives the period T_0, and hence the correct value of the acceleration due to gravity g. The experimental arrangement is shown in the figure.



1. Small magnet; 2. Hall switch; 3. Counter-timer; 4. Horizontal ruler.

Fig. 6.15

Experimental method:

(1) Adjust the pendulum length so that, at rest, the magnetic field produced by the NdFeB magnet at the lower end of the small iron ball is aimed exactly at the position of the Hall switch and can make the Hall switch conduct (the distance between the two is about 0.8 cm to 1 cm).

(2) Record the distance L from the point of suspension of the thread to the centre of gravity of the small ball. Adjust the timer; the preset count of the switch should not be large. In this experiment a count of 4 may be used, that is, two periods are measured.

(3) Pull the small ball from the middle position to the position A, measure the position of point A with the adjusted horizontal ruler, and calculate the angle of swing by means of trigonometry.

The experimental results are given in the table below.

Table 6.15 Measured values of $2T$

x/cm	$\sin^2\left(\frac{\theta}{2}\right)$	$2T$/s					
		1	2	3	4	5	Mean
15.00	0.018 14	3.052	3.052	3.051	3.052	3.053	3.052 4
20.00	0.032 73	3.064	3.064	3.064	3.063	3.063	3.063 6
25.00	0.052 19	3.077	3.077	3.076	3.077	3.077	3.076 8
30.00	0.077 20	3.097	3.096	3.096	3.096	3.096	3.096 2
35.00	0.108 80	3.122	3.122	3.123	3.123	3.123	3.122 6
40.00	0.148 78	3.156	3.156	3.154	3.156	3.156	3.155 6

Length of the suspension thread $L_1 = (56.20 \pm 0.05)$ cm;
radius of the pendulum ball $r = (1.000 \pm 0.002)$ cm;
pendulum length $L = L_1 + r = (57.20 \pm 0.05)$ cm; x is the horizontal distance of the small ball from the central position.

Fitting a straight line to the data of $2T$ against $\sin^2\left(\frac{\theta}{2}\right)$ in the table gives the correlation coefficient $r = 0.9995$, the slope (0.790 ± 0.013) s and the intercept (3.0369 ± 0.0011) s; that is, $T_0 = 1.5185$ s. Substituting this value into $T_0 = 2\pi\sqrt{L/g}$ gives the acceleration due to gravity $g = 9.794\,\mathrm{m/s^2}$. Compared with the acceleration due to gravity in the Shanghai region, $g = 9.794\,07\,\mathrm{m/s^2}$, the error is extremely small.

Points to be noted in the experiment:

(1) The small ball must move in one plane. This can be checked by observing whether the indicator lamp of the timer suddenly lights up each time the small ball passes through the equilibrium position.

(2) If the Hall switch is too close to the magnet, a timing error will also arise; if it is too far, counts may be missed. The distance between the Hall switch and the magnet should be set at the critical distance for conduction (the light-emitting diode lights up).

Comment. This experiment adopts the experimental idea of Galileo's extrapolation method for studying physical laws: by measuring the relation between the period and the angle of swing, the period of oscillation at an extremely small angle of swing is obtained by extrapolation, and the acceleration due to gravity is thereby determined accurately. This important idea of experimental design is frequently used in the teaching of physics experiments.

Besides the air track and the simple pendulum described above, a spring oscillator can also be used to measure the acceleration due to gravity; the following problem may serve as a reference.

Experiment 6.16 Measuring the acceleration due to gravity with a spring oscillator

[Available apparatus]

A Jolly balance (spring, weight pan, adjustable support, vernier scale, weights, small magnet); a switch-type Hall sensor and a counter-timer, etc.

[Requirements]

(1) Determine the stiffness coefficient k of the spring by measuring the period of oscillation of the spring oscillator, and write down the corresponding formula.

(2) When the period T of the spring oscillator is measured, the relative uncertainty of the measurement is required to be $\leq 0.5\%$. Suppose that, with the Hall switch working together with the timer, the total uncertainty of the starting and stopping operations in each measurement of the period is $\leq \pm 0.02\,\mathrm{s}$.

(3) The mass of the spring itself is $m_0 = 14.79\,\mathrm{g}$; one third of it should be counted as the effective mass ρm_0 of the oscillator ($\rho \approx \frac{1}{3}$). The total mass of the weights plus the magnet is $20.80\,\mathrm{g}$.

(4) Write down the formula for measuring the acceleration due to gravity with this spring, and carry out the measurement. Compare the measured result with the acceleration due to gravity in the Shanghai region.

Solution 6.16. **1.** Fix one end of the spring at the upper support of the stand, hang a $20.00\,\mathrm{g}$ weight (made of iron) on the lower end of the spring, and attach a small NdFeB magnet to the lower end of the weight. A Hall switch is fixed below the small magnet, the distance between them being about $1\,\mathrm{cm}$. Pull the weight downwards by hand so that it leaves the equilibrium position slightly, and then release it; the body then performs simple harmonic motion about the equilibrium point, and its period is

$$T = 2\pi\sqrt{\frac{m + \rho m_0}{k}}, \tag{1}$$

where ρ is a coefficient to be determined, whose value is approximately $1/3$, m_0 is the mass of the spring itself, ρm_0 is the effective mass of the spring, and k is the stiffness coefficient of the spring. Pull the spring gently to set it oscillating, and the time of 50 oscillations can be measured. Repeating the measurement three times, the mean value of 50 periods is $43.54\,\mathrm{s}$, so the period of oscillation of the spring is $T = 0.8708\,\mathrm{s}$. Obviously the uncertainty of this period is less than 0.5%. Substituting $m = 20.80\,\mathrm{g}$, $m_0 = 14.79\,\mathrm{g}$, $\rho = \frac{1}{3}$ and $T = 0.8708\,\mathrm{s}$ gives the stiffness coefficient $k = 1.340\,\mathrm{N/m}$.

2. A spring is deformed (extended or compressed) under the action of an external force. Within the elastic limit, Hooke's law states that the external force F is proportional to its deformation Δy, that is,

$$F = k \cdot \Delta y, \tag{2}$$

where k is the stiffness coefficient of the spring. A small plane mirror is pasted on the movable jaw of the vernier calliper, so that the graduation line on the small mirror is aligned with the base plate of the weight pan, and the mass m_i of the weights and the reading r_i on the calliper are recorded. For each addition of $1\,\mathrm{g}$ (or removal of $1\,\mathrm{g}$) of weights, the data of the m_i–y_i relation are recorded. The experimental data are given in the table below.

Table 6.16

m_i/g	y_i/mm (weights added)	y_i'/mm (weights removed)	$\bar{y}_i$/mm
11.00	19.80	19.88	19.84
12.00	26.96	26.80	26.88
13.00	34.36	34.28	34.32
14.00	41.56	41.70	41.63
15.00	48.80	49.12	48.96
16.00	56.34	56.30	56.32
17.00	63.22	63.80	63.51
18.00	70.88	71.00	70.94
19.00	77.96	78.04	78.00
20.00	85.50	85.50	85.50

After processing by the method of successive differences, it is found that for each addition of 5.00 g of weights to the pan the spring extends on average by $\Delta\bar{y} = 36.528\,\text{mm}$. From $m_i g = k \cdot \Delta y_i$, where g is the acceleration due to gravity, $\Delta y_i = \dfrac{g}{k} \cdot m_i$. Substituting $\Delta\bar{y}$ and 5.00 g into this expression gives

$$\frac{g}{k} = \frac{36.528}{5.000} = 7.306\,\text{m/kg}.$$

Substituting $k = 1.340\,\text{N/m}$ gives

$$g = k \cdot 7.306 = 9.7895\ (\text{m/s}^2) = 9.790\,\text{m/s}^2.$$

Compared with the acceleration due to gravity in the Shanghai region, $g = 9.794\,07\,\text{m/s}^2$, the percentage uncertainty is less than 0.1%.

Experiment 6.17 Determination of the density of oil (mechanics experiment, 1st Asian Physics Olympiad)

[Available apparatus]

A test tube (with a uniform cross-section over most of its middle, except near the two ends), a large vessel containing water, a ruler, a dropper, graph paper, absorbent cotton wool, absorbent paper, a rubber band used as a marker of a fixed position, distilled water (density $1.00\,\text{g/cm}^3$), and oil (in a plastic cup).

[Requirements]

In the experiment the density of the oil is to be determined without measuring the dimensions of the test tube, and oil and water must not be put into the test tube at the same time. The report on the experiment should include the following:

(1) the theoretical basis of the analysis of the experiment (write down the relevant formulas);

(2) a brief account of the method and the steps of the experiment;

(3) the final result for the density of the oil;

(4) the errors and their sources.

Solution 6.17. 1. Method of the experiment

This experiment is an application of Archimedes' law. The experimental arrangement is shown in Fig. 6.17(a). In the figure the test tube is assumed to be upright (that is, perpendicular to the water surface), and the same reference point is used in measuring the levels of the water

and of the oil. In Fig. 6.17(a), l_z is the distance from the water surface to the reference point, and l_c is the distance from the surface of the liquid to be measured to the reference point. In order to apply Archimedes' law correctly in the experiment, the volume occupied by the liquid inside the tube and the volume of water displaced by the test tube must be expressed accurately; Fig. 6.17(b) therefore marks in more detail the parameters relating to the dimensions of the test tube.

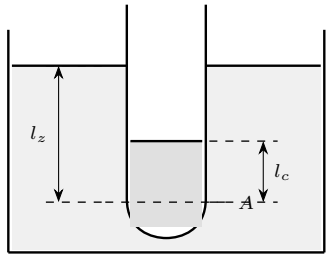
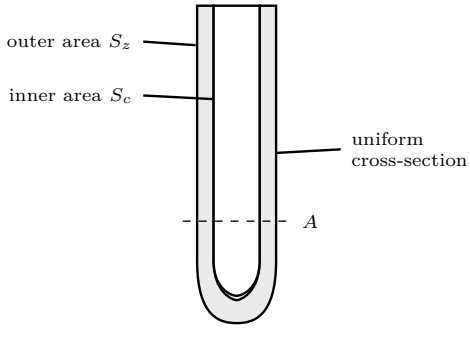


Fig. 6.17(a)



A is the common fixed reference point.

Fig. 6.17(b)

2. Theoretical derivation

All the physical quantities needed in the theoretical derivation and their definitions are as follows.

S_c is the inner cross-sectional area of the test tube at point A; S_z is the outer cross-sectional area of the test tube at point A; V_o is the inner volume of the test tube below point A; V_e is the outer volume of the test tube below point A (V_o plus the volume of the glass below point A).

l_z is the distance from point A to the outside of the test tube, and l_c is the distance from point A to the surface of the liquid inside the test tube; ρ_c is the density of the liquid inside the test tube; ρ_w is the density of water; ρ_o is the density of the oil; and M is the mass of the empty test tube.

When equilibrium is reached, the Archimedean buoyant force F_A is equal to the total weight W of the test tube and the liquid inside it. From Figs. 6.17(a) and 6.17(b) and from the physical quantities given above, the following expressions are obtained:

$$F_A = (V_e + S_z l_z)\rho_w g, \tag{1}$$

$$W = (M + V_o\rho_c + S_c l_c \rho_c)g. \tag{2}$$

From the condition of equilibrium,

$$(V_e + S_z l_z)\rho_w = M + (V_o + S_c l_c)\rho_c.$$

This relation may be written in the form

$$l_z = C + Dl_c, \tag{3}$$

where

$$C = \frac{M + V_o\rho_c - V_e\rho_w}{\rho_w S_z},$$

$$D = \frac{\rho_c S_c}{\rho_w S_z}.$$

From the above equations it can be seen that the coefficient D in (3) relies on l_c and l_z having a common zero point; therefore the reference point A may be chosen at any convenient point in the part of the test tube where the cross-section is uniform.

In the measurement, a certain amount of water is first put into the test tube, and l_c and l_z are measured with the ruler; this gives one set of data. Adding some more water gives another set of experimental data l'_c and l'_z. Plotting five such points gives the value of D_1, that is,

$$D_1 = \frac{S_c}{S_z}. \tag{4}$$

Then pour the water out of the test tube, wipe the tube dry with absorbent cotton wool and absorbent paper, drip oil into the test tube and repeat the above steps; the value of D_2 is obtained, that is,

$$D_2 = \frac{\rho_o S_c}{\rho_w S_z}. \tag{5}$$

Solving equations (4) and (5) then gives

$$\rho_o = \left(\frac{D_2}{D_1}\right)\rho_w. \tag{6}$$

3. Experimental results

The data measured in the experiment are given in the table below. Figures 6.17(c) and 6.17(d) are plotted from the data of the table.

Table 6.17 Measured experimental data

water in tube		oil in tube		water in tube		oil in tube	
l_c/cm	l_z/cm	l_c/cm	l_z/cm	l_c/cm	l_z/cm	l_c/cm	l_z/cm
3.70	11.70	5.70	12.50	5.20	12.90	6.40	13.00
4.50	12.30	6.00	12.60	5.30	13.00	6.80	13.30
4.90	12.60	6.00	12.80	5.70	13.30	7.20	13.50

The slope of the straight line in Fig. 6.17(c) gives $D_1 = 0.809$, and that of the straight line in Fig. 6.17(d) gives $D_2 = 0.686$; finally the density of the oil is

$$\rho_o = 0.848\,\mathrm{g/cm^3}.$$

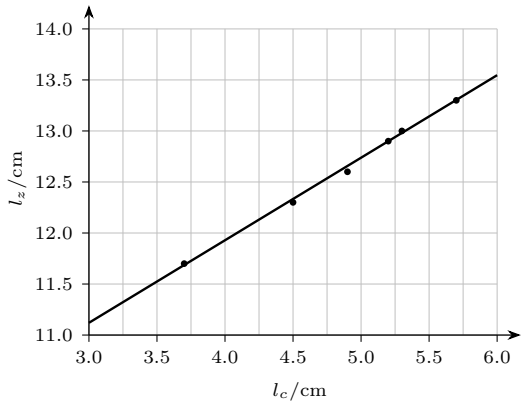


Fig. 6.17(c)

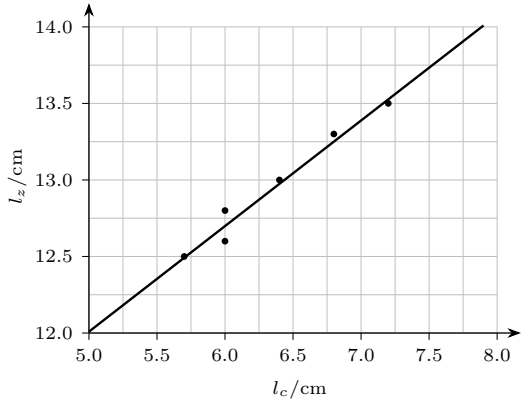


Fig. 6.17(d)

The sources of error in this experiment are mainly two:

(1) During the experiment the test tube is not always kept vertical.

(2) Owing to the surface tension of the liquid and similar effects, the liquid surface is slightly curved, and this affects the readings.

Points to note:

(1) For a typical test tube the ratio S_c/S_z is 0.8 and not 1.

(2) The surfaces of the water and of the oil whose positions are measured must lie in the part of the test tube where the cross-section is uniform.

(3) In order to determine the values of D_1 and D_2, at least five points must be measured so that a well-fitting straight line can be drawn. In the experiment the data must be taken over a range of 2 cm to 2.5 cm.

Comment. This problem requires the candidates to apply Archimedes' law to solve a practical problem, to set up a correct mathematical model, and to possess a certain ability in theoretical derivation: how to establish a linear equation and, by finding its slope, to obtain the relation between the quantity to be measured and the known quantities, and finally to determine the density of the oil. The design idea of the problem and its requirements on the experimental operation are both outstanding. Because the five

Chinese contestants had paid attention to training in experimental ability, they all scored relatively high marks on this problem.

Experiment 6.18 Investigation of the relation between the vibration frequency of a tuning fork and the mass on its symmetric arms (mechanics experiment, 3rd Asian Physics Olympiad)

[Aim]

To investigate how the vibration frequency of a tuning fork varies with the mass of identical blocks attached at a fixed position on the two arms of the fork, and hence to determine the unknown mass of a pair of blocks X attached to the fork.

[Basic concepts and knowledge]

(1) *Stroboscope*

This experiment uses a stroboscope, a simple electronic device with a discharge lamp which emits intense light for a short time at very uniform intervals. With a stroboscope the frequency of rotation or vibration of a body can be measured without touching the moving body directly.

[Points to note]

A stroboscope has a limited life, that is, a maximum number of flashes; it should be switched off when not in use.

Consider a particle moving in a circle, illuminated by a stroboscope. If the flash frequency is a multiple, or a submultiple, of the frequency of the particle's motion, the particle will appear to be stationary. Hence the period of the circular motion of the particle can be determined by adjusting the frequency of the flashing light.

Suppose that a particle rotates with frequency x Hz and that the flash frequency of the stroboscope is y Hz. Then during the time interval $1/y$ s between two successive flashes the body turns through $2\pi x/y$ rad.

If x/y is irrational, it cannot be expressed as the ratio of two integers; the particle then does not appear stationary but seems to rotate slowly forwards or backwards, according to whether the ratio x/y is somewhat smaller or somewhat larger than a nearby rational number.

If $y/x = q/p$, where p and q are integers, the stroboscope flashes q times in every p complete periods. Furthermore, if p and q have no common divisor (as is assumed throughout the experiment), each flash illuminates the body in a different position, so that under the flashing light the body presents q different stationary positions.

If q becomes very large, it is difficult to count the number of stationary positions presented by the rotating body.

The theory given above for a rotating body may be applied equally to a tuning fork executing simple harmonic motion, because the vibration may be regarded as the projection of circular motion on a given diameter. In this case, however, the vibrating body retraces the same path after every half period, so that a coincidence (although unlikely) may occur in which the pattern in one half period is the same as that in the next half period; this leads to only one figure being recorded instead of two (although the intensity is still doubled). Such coincidences should be avoided in the observations.

(2) *Determination of fundamental synchronism*

When the stroboscope flashes exactly once for each revolution or vibration of the body, the body appears to stand at one fixed position; this is synchronism of the two. It must be pointed out, however, that similar indistinguishable results also occur when the flash frequency is a submultiple of the frequency of the body's motion. Thus, if the frequency of the moving body is completely unknown, a safe procedure for adjusting to fundamental synchronism is to start from a high frequency and, after many images have been seen, to lower the flash frequency slowly

until a single image appears for the first time. In all the measurements this procedure is used to check whether fundamental synchronism has been reached.

(3) *Multiple frequency*

When the flash frequency of the stroboscope is higher than the vibration frequency of the observed body, this flash frequency is called a multiple frequency of the vibration frequency of the body. The opposite case, in which the flash frequency is lower than the frequency of the moving body, is called a submultiple frequency.

If the flash frequency of the stroboscope is q times the frequency of the rotating body, several images are observed. In this case a single rotating particle appears as several stationary images distributed at equal intervals around the circle. For twice the frequency, $q/p = 2$, two images separated by an angle π are produced. For three times the frequency, $q/p = 3$, three images are produced at angular distances $2\pi/3$, and so on. The frequency of rotation of the particle is then equal to the flash frequency divided by the number of images observed. Generally, if $q \gg p > 1$, the stroboscope flashes q times while the particle makes p revolutions, so that q stationary positions still appear.

(4) *Submultiple frequency*

Here q/p is less than 1. If the flash frequency of the stroboscope is exactly $1/p$ of the frequency of the body's motion, where $p > 1$, then the body moves through p revolutions during each flash interval, yet only one image of the body is presented. If $p \gg q > 1$, the stroboscope flashes q times while the body makes p complete revolutions, and the number of stationary images seen is q.

(5) *Adjustable-frequency tuning fork*

Once an adjustable-frequency tuning fork (equivalent to an ordinary tuning fork) is set vibrating, it vibrates at a certain fundamental frequency without harmonics. The two arms of the fork are symmetric when viewed from either side, so that the motions of the two arms are exactly opposite, and at any instant they exert equal and opposite forces on the central stem. The net force on the central stem is zero and it does not vibrate; hence the stem may be gripped firmly without causing any unwanted damping of the vibration. For the same reason the two arms of the fork cannot vibrate in phase, because in-phase vibration would produce a finite oscillating force on the central stem, and this force would make such a vibration decay very quickly.

The fundamental frequency of a tuning fork can be reduced by loading identical blocks symmetrically on its two arms (the blocks loaded on the two arms of the fork must be symmetric). For such a loaded fork the period of vibration T is given by

$$T^2 = A(m + B),$$

where A is a constant which depends on the mechanical properties of the material of the fork, on its size and on its shape; B is a constant related to the effective mass of each vibrating arm; and m is the mass of the loading (including the mass of the clamp).

[Available apparatus]

(1) A stroboscope with a digital frequency display.

(2) A small flashlight.

(3) A tuning fork, on each of whose arms a block of 31.6 g marked with numbers and lines is symmetrically loaded; the centre of mass of the block coincides with point P (P is clearly marked on each arm but is covered by the block).

(4) Two clamps with detachable handles; the handles are used only when the clamps have to be opened, and are removed during the experiment.

(5) One pair of identical blocks of unknown mass.

(6) Blocks of the following known masses (in pairs): blocks of 5 g, 10 g, 15 g, 20 g and 25 g.

(7) Five sheets of standard graph paper.

Solution 6.18. *Experimental steps and experimental results*

(1) *Measurement of fundamental synchronism and of multiple frequencies*

Make the flashes of the stroboscope reach fundamental synchronism with the vibration of the tuning fork whose two arms carry the 31.6 g blocks. Temporarily take this pair of blocks off the arms, check and confirm that the centre of mass of the blocks falls on point P, and then mount them again and carry out the measurement. Record the flash frequency of the stroboscope at synchronism. Keeping the flash frequency higher than the frequency of the fork, find as many flash frequencies as possible at which stationary images can be observed (the arrows drawn on the arms), and determine the values of q/p corresponding to them. The f–q/p relation for $q \geq p$ is shown in Fig. 6.18(a).

Table 6.18-1

Stroboscope frequency f/Hz	Number of stationary images	q/p
64.5	1	1
86.3	4	4/3
96.9	3	3/2
129.4	2	2
193.9	3	3
258.3	4	4

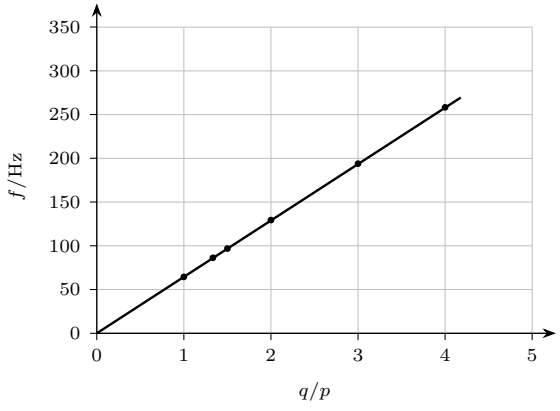


Fig. 6.18(a)

(2) *Measurement of submultiple frequencies*

Keep the flash frequency lower than the fundamental frequency of the fork (whose two arms carry the 31.6 g blocks) and find all the observable flash frequencies at which stable stationary images are produced. Tabulate the readings (in order of decreasing q/p) and plot a straight-line graph of all the observed flashes against q/p (for the fork whose arms carry the 31.6 g blocks).

The f–q/p graph for $q \leq p$ is given in Table 6.18-2 and Fig. 6.18(b).

Table 6.18-2

Stroboscope frequency f/Hz	Number of stationary images	q/p
64.5	1	1
48.5	3	3/4
43.1	2	2/3
38.8	3	3/5
32.2	1	1/2
21.5	1	1/3

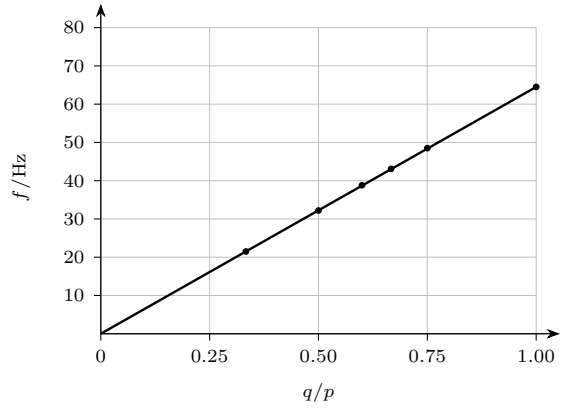


Fig. 6.18(b)

(3) *Determination of the mass of the pair of unknown blocks x*

Take the 31.6 g blocks off the two arms of the tuning fork and measure the vibration frequency of the fork with no blocks on its arms; then measure the vibration frequencies of the fork when blocks of known masses 5.0 g, 10.0 g, 15.0 g, 20.0 g and 25.0 g are loaded on both arms. Note that the mass marked on a block is the total mass of the clamp holding the block together with the block. Finally replace the blocks of known mass by the unknown blocks, measure the vibration frequency of the fork at that time, and determine the mass x. Note that the mass x includes the mass of the clamp (the handles of the clamps have now been removed). The experimental results are given in Table 6.18-3 and Fig. 6.18(c). Fitting a straight line to the data of Table 6.18-3 according to the formula $T^2 = A(m + B)$ gives $A = 5.85 \times 10^{-6}\,\mathrm{s^2/g}$, $B = 9.85\,\mathrm{g}$ and a correlation coefficient $r = 0.9997$. From the T^2–m graph the slope is $A = 5.85 \times 10^{-6}\,\mathrm{s^2/g}$ and the intercept on the x axis is $-B = -9.85\,\mathrm{g}$. For the body of unknown mass x, $f = 85.6\,\mathrm{Hz}$ and $T^2 = f^{-2} = 13.6 \times 10^{-4}\,\mathrm{s^2}$; substituting into the formula $T^2 = 5.85 \times 10^{-6}(m + 9.85)$ gives the mass of the body to be measured as $m_x = 13.4\,\mathrm{g}$.

Table 6.18-3

m/g	5.0	10.0	15.0	20.0	25.0
f/Hz	106.9	92.0	82.8	75.6	70.6
$T^2/\mathrm{s^2}$	8.75×10^{-5}	1.18×10^{-4}	1.46×10^{-4}	1.75×10^{-4}	2.01×10^{-4}

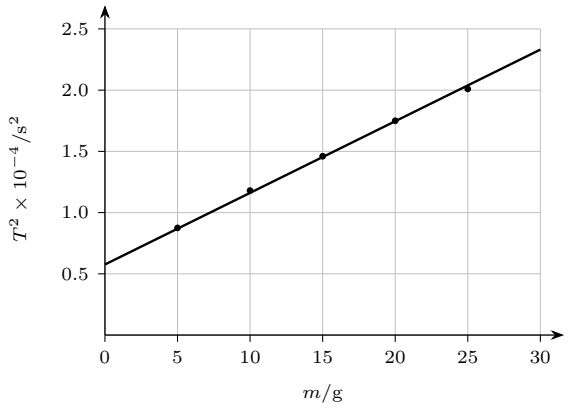


Fig. 6.18(c)

Comment. In this experiment the fundamental frequency must be measured accurately and confirmed repeatedly before the experiment can be done well. A few students treated twice the multiple frequency as the fundamental frequency when processing the data, which easily leads to errors in the measured result. Since the concepts of multiple and submultiple frequencies are not studied in the basic physics courses of secondary school and university, the students had to understand the material on the spot, carry out the investigation and complete a design-type experiment; this reflects their ability of analysis and experiment.

Experiment 6.19 Measurement of the surface tension coefficient of water by the pull-off method

[Available apparatus and materials]

A liquid surface tension coefficient measuring instrument (produced by Fudan Tianxin Company), weights, water, a glass dish, a pull ring, a thermometer, cleaning paper.

[Requirements] Use the liquid surface tension measuring instrument to measure the surface tension coefficient of a liquid at room temperature by the pull-off method.

[Explanation]

(1) The pull-off method used in this experiment is based on the wetting phenomenon between a liquid and a solid. A metal ring of outer diameter D_1 and inner diameter D_2 is slowly pulled up out of water; the water film spreads over the whole circumference of the ring and is lifted with it. Taking into account that the lifted water film is a double film in contact with the ring, the pull-off force equals the surface tension coefficient multiplied by the perimeter of the surface of the departing liquid, that is,

$$F = \alpha \cdot \pi(D_1 + D_2),$$

where F is the surface tension and α the surface tension coefficient.

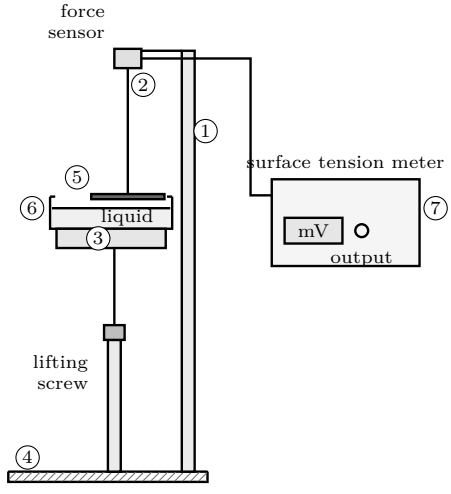
(2) The ring and the water must be kept clean during the experiment.

(3) The surface tension of water depends on temperature; the acceleration due to gravity in the Shanghai region is $g = 9.794\,\mathrm{m/s^2}$.

Solution 6.19. The experimental apparatus is shown in the figure. A surface tension measuring instrument with a silicon piezoresistive force-sensitive sensor is used to measure the surface tension at the contact between the liquid and the metal. A thin metal ring serves as the contact body, and the sensor is connected to a digital voltmeter. If the sensitivity of the sensor is known, the surface tension can be calculated from the reading of the digital voltmeter.

Water is poured into a clean culture dish, and then the metal ring, whose surface has been cleaned, is immersed in the liquid, care being taken to keep the ring perpendicular to the liquid surface. The liquid surface is made to descend slowly by means of the lifting platform. An annular liquid film then forms between the metal ring and the liquid surface; just before detachment the middle part of the film becomes thinner and thinner, and finally the film is broken.

Since the lifting platform descends sufficiently slowly during the whole process, the motion may be regarded as uniform. The metal ring is in equilibrium; at the moment of detachment from the liquid surface, the weight mg of the metal ring, the pulling force F and the surface tension f of the liquid satisfy the following relation (m is the mass of the ring and g the acceleration due to gravity):



1. Upright post; 2. Force-sensitive sensor; 3. Lifting platform; 4. Base; 5. Metal ring; 6. Large culture dish; 7. Digital voltmeter.

Fig. 6.19

$$F = mg + f\cos\theta. \tag{1}$$

Just before the metal ring leaves the liquid surface, $\theta \approx 0$ (here θ is the angle between the direction of the surface tension and the vertical), that is, $\cos\theta \approx 1$ and $F = F_1$. Hence, to a good approximation,

$$f = F_1 - mg = \alpha\pi(D_1 + D_2),$$

where D_1 is the outer diameter and D_2 the inner diameter of the metal ring. After the metal ring has been pulled clear of the water surface, $F = F_2$ and $f = 0$, so that $F_2 = mg$.

Since the pulling force acts on the elastic beam in contact with the silicon piezoresistive force-sensitive sensor, the reading of the digital voltmeter just before the liquid film breaks is $U_1 = kF_1$, and the reading immediately after the film breaks (when f vanishes) is $U_2 = kF_2$. Their difference is

$$\Delta U = U_1 - U_2 = kf,$$

or

$$\alpha = \frac{\Delta U}{k\pi(D_1 + D_2)}. \tag{2}$$

Here k is the sensitivity of the dynamometer, in mV/N.

The experimental results are shown in Tables 1 and 2. The calibration data of the force-sensitive sensor are given in Table 1; the acceleration due to gravity in the Shanghai region is $g = 9.794\,\mathrm{m/s^2}$.

Table 6.19-1 Sensor calibration data

m/g	U/mV
0.500	14.7
1.000	29.5
1.500	44.3
2.000	59.0
2.500	74.7
3.000	88.4
3.500	100.3

Table 6.19-2 Data for pure water at 5.0 °C

ΔU/mV	$\alpha/(10^{-3}\ \mathrm{N\cdot m^{-1}})$
48.6	75.5
48.6	75.5
48.1	74.8
48.1	74.8
48.7	75.7
47.9	74.5
48.2	74.9

A straight-line fit gives the sensitivity of the dynamometer $k = 3007\,\mathrm{mV/N}$, with correlation coefficient $r = 0.999\,998$. The measured diameters are $\bar{D}_1 = 3.316\,\mathrm{cm}$ and $\bar{D}_2 = 3.491\,\mathrm{cm}$, and the temperature is $\theta = 5.0\,^\circ\mathrm{C}$.

From Table 2 the surface tension coefficient of pure water at 5.0 °C is $\bar{\alpha} \approx 75.10 \times 10^{-3}$ N/m. According to the handbook of fundamental physical constants, the surface tension coefficient of water at 5.0 °C is 74.9×10^{-3} N/m, and the percentage difference between the two values is 0.27%.

Experiment 6.20 Measurement of the viscosity coefficient of a liquid by the falling-ball method

When a small ball falls in a liquid, forces act on it, and the viscous force is one of them. If the motion of the ball satisfies the following conditions: (1) the velocity of the ball falling in the liquid is very small; (2) the volume of the ball is very small, then the viscous force is

$$f = 6\pi\eta v \cdot r,$$

where η is the viscosity coefficient of the liquid, v the falling velocity of the ball and r the radius of the ball. It should be noted that the temperature has a great influence on the viscosity coefficient of a liquid.

[Available apparatus and materials]

A metre rule, a micrometer, a stopwatch, a measuring cylinder containing castor oil, a thermometer, a rubber band, a small steel ball.

[Requirements]

(1) Analyse the forces acting on the ball as it falls, and derive the formula for measuring the viscosity coefficient of the liquid. (Hint: during the fall of the ball, when the resultant force on it is zero, the ball begins to move uniformly.)

(2) Choose suitable measuring instruments and measure the physical quantities in the formula. The density of the ball is known to be $\rho = 7.90\,\mathrm{g/cm^3}$; the density of the liquid to be measured is $\rho_0 = 0.961\,\mathrm{g/cm^3}$.

(3) Calculate the viscosity coefficient of the liquid.

Solution 6.20. When a small sphere of radius r falls in an infinitely extensive viscous liquid without producing vortices, three vertical forces act on it: its weight mg, the buoyant force

exerted by the liquid $\frac{4}{3}\pi r^3\rho_0 g$, and the viscous resistance $6\pi\eta rv$. At the beginning of the fall the velocity is small and the viscous force is small, so the ball accelerates until, at a certain velocity, the three forces balance and the ball falls uniformly:

$$mg = \frac{4}{3}\pi r^3\rho_0 g + 6\pi\eta rv.$$

Therefore

$$\eta = \frac{m - \frac{4}{3}\pi r^3\rho_0}{6\pi rv} \cdot g, \tag{1}$$

where η is the viscosity coefficient of the liquid to be measured, m the mass of the ball, r its radius, ρ_0 the density of the liquid to be measured, v the falling velocity of the ball and g the acceleration due to gravity.

The diameter of the ball is measured with the micrometer, the diameter of the measuring cylinder and the distance fallen by the ball with the metre rule, and the temperature with the thermometer.

The oil temperature is $\theta = 15.90\,^\circ\mathrm{C}$; the density of the oil is $\rho_0 = 0.961\,\mathrm{g/cm^3}$; the diameter of the measuring cylinder is $D = 6.72\,\mathrm{cm}$; the half-course distance is $s_1 = 10.70\,\mathrm{cm}$ and the full-course distance $s_2 = 20.12\,\mathrm{cm}$. The other measured data are given in Table 1.

Table 6.20-1

Diameter of the ball $2r$/mm	Half-course time t_1/s	Full-course time t_2/s
0.993	39.76	74.14

The mean velocity over the half course is $v_1 = 0.269\,\mathrm{cm/s}$, and the mean velocity over the full course is $v_2 = 0.271\,\mathrm{cm/s}$. The difference between the two mean velocities is very small (less than 1.6%). Since the experiment is not carried out under infinitely extensive conditions, the measured velocity data must be corrected. The inner diameter of the measuring cylinder is $D = 6.72\,\mathrm{cm}$, its height $h = 53.40\,\mathrm{cm}$, and $d = 0.993\,\mathrm{mm}$. The velocity correction formula is

$$v = v_0\left(1 + \frac{2.4d}{D}\right)\left(1 + \frac{3.3d}{2h}\right),$$

where v_0 is the measured velocity. Substituting into formula (1) gives $\eta = 1.331\,\mathrm{Pa\cdot s}$. Taking into account that at $\theta = 15.90\,^\circ\mathrm{C}$ the accepted value of the viscosity coefficient of castor oil is $\eta_{\text{accepted}} = 1.370\,\mathrm{Pa\cdot s}$, the percentage difference between the two is 2.8%.

Experiment 6.21 Measurement of the coefficient of thermal expansion of a solid

[Available apparatus and materials]

A linear expansion apparatus, a steam generator, a metal test rod, a micrometer, a metre rule, an optical lever, a thermometer, etc.

[Requirements]

(1) Write down the principle and the formula for measuring the coefficient of thermal expansion of a metal.

(2) Write down the experimental method.

(3) Measure the coefficient of linear expansion of a brass material between room temperature and 100 °C.

Solution 6.21. **1.** When the temperature of a solid substance changes by 1 °C, the ratio of the change in its length to its length at 0 °C is called the coefficient of linear expansion of that substance. If a metal rod has length L_0 at 0 °C and length L_1 at temperature t_1 °C, then

$$L_1 = L_0(1 + \alpha t_1). \tag{1}$$

In reality the coefficient of linear expansion α of a metal is not a constant; it may be regarded as approximately constant only within a certain temperature range. If the lengths of the metal rod at two different temperatures t_1 and t_2 ($t_2 > t_1$) are L_1 and L_2, then from (1)

$$\begin{cases} L_1 = L_0(1 + \alpha t_1), \\ L_2 = L_0(1 + \alpha t_2). \end{cases} \tag{2}$$

Hence

$$\frac{L_2}{L_1} = \frac{1 + \alpha t_2}{1 + \alpha t_1} \approx (1 + \alpha t_2)(1 - \alpha t_1) \approx 1 + \alpha(t_2 - t_1) - \alpha^2 t_1 t_2.$$

In this expression the second term $\alpha^2 t_1 t_2$ may be neglected in comparison with the first two terms, so

$$\alpha = \frac{L_2 - L_1}{L_1(t_2 - t_1)} = \frac{\Delta L}{L_1(t_2 - t_1)}. \tag{3}$$

In formula (3), $\Delta L = L_2 - L_1$ is the change in length of the metal rod when it is heated.

2. The instrument for measuring the coefficient of linear expansion of a metal is shown in the figure.

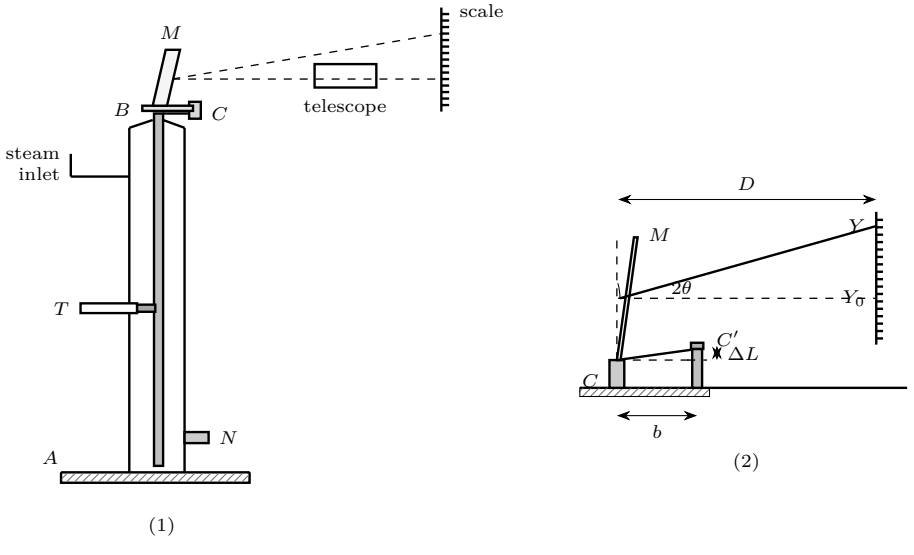


Fig. 6.21

In the figure, (1) is the vertical linear expansion instrument used with the optical lever, and (2) is the reflecting mirror of the optical lever. The reflecting mirror of the optical lever consists of a plane mirror and a base with three feet. At one end of the base there are two feet H_1 and H_2, and the plane mirror M is mounted above them; under the other end of the base there is one foot H_3. The three feet form an isosceles triangle. A scale S and a telescope are placed at a

distance D from the mirror surface. In the experiment, first adjust the three feet H_1, H_2, H_3 of the optical lever so that they are level, and turn the plane mirror M and the scale S so that they are vertical, with the telescope aimed horizontally at the plane mirror. Then adjust the eyepiece and the objective of the telescope so that the cross-wire is sharp and the image of the target scale S is sharp; in this way the image of the scale S formed by the objective lies in the same plane as the cross-wire, and the parallax is eliminated. Adjust the optical lever and the mirror–scale assembly so that the image of the scale graduation falling on the cross-wire is at the same height as the principal axis of the telescope.

If the length of the metal rod is L, then when the temperature of the metal rod rises from t_1 to t_2, its length increases by ΔL. If the perpendicular distance between the rear foot H_3 of the optical lever base and the line joining the front feet H_1, H_2 is b, the mirror tilts through an angle $\Delta L/b$. The displacement of the image of the scale graduation on the cross-wire, observed through the eyepiece of the telescope, is then d; hence the beam originally perpendicular to the mirror is deflected through an angle d/D, and this angle is twice the tilt angle of the mirror. That is,

$$\frac{d}{D} = 2\,\frac{\Delta L}{b},$$

and substituting $\Delta L = bd/2D$ into (3) gives the coefficient of linear expansion of the metal:

$$\alpha = \frac{bd}{2DL_1(t_2 - t_1)}. \tag{4}$$

In the experiment, the length L of the metal test rod and the perpendicular distance D between the scale and the mirror are measured with a metre rule, and the temperature is measured with a mercury thermometer. The bulb of the thermometer should be inserted into the middle part of the metal test rod; one end of the metal rod is fixed and the other end can expand.

3. Experimental data and calculation

The elongation ΔL of the metal rod on heating is measured with the optical lever. The material tested is brass, in the form of a rod. The mercury thermometer has a range of 0–100 °C and a division value of 1 °C.

The length of the rod is $L = 49.80$ cm. The initial temperature of the rod is $t_1 = 26.2\,^\circ$C. When steam is passed through and thermal equilibrium is reached, the mean final temperature of the rod is $\bar{t}_2 = 98.6\,^\circ$C; when the final temperature is measured, t_2 fluctuates slightly, so t_2 is taken as the mean of these values.

Table 6.21

Number	Position of the cross-wire/cm	
	x_1	x_2
1	+0.05	+1.88
2	+0.05	+1.89
3	+0.05	+1.88
4	+0.05	+1.90
5	+0.05	+1.90
Mean	+0.05	+1.89

$\Delta t = 98.6 - 26.2 = 72.4\,^\circ$C; $d = 1.89 - 0.05 = 1.84$ cm; $D = 103.3$ cm; $b = 7.40$ cm.

$$\alpha = \frac{bd}{2DL(t_2 - t_1)} = \frac{7.40 \times 1.84}{2 \times 103.3 \times 49.80 \times 72.4} = 1.83 \times 10^{-5}\ (^\circ\text{C})^{-1},$$

that is, between 26.2 °C and 98.6 °C the mean coefficient of thermal expansion of brass is $1.83 \times 10^{-5}\ (°C)^{-1}$.

Experiment 6.22 Measurement of the volume expansion coefficient of a liquid

[Available apparatus and materials]

A pycnometer, a beaker containing water at room temperature, an iron stand, an asbestos gauze, one gas burner (or stove), an analytical balance, a glass mercury thermometer (0–100 °C), a stirrer, fine thread, etc.

[Requirements]

(1) Write down the principle and the formula for measuring the coefficient of volume expansion of a liquid with a pycnometer.

(2) Write down the steps of the experiment.

(3) Measure and record the experimental data, and write down the experimental results.

Solution 6.22. **1.** The pycnometer is filled with the liquid to be measured, and is suspended by a metal wire and immersed in a beaker containing the same liquid; when the liquid is poured in, care must be taken that no air bubbles enter the liquid. The temperature of the liquid in the beaker at room temperature is recorded, and the pycnometer is then carefully taken out and wiped dry on the outside; the total mass M_1 of the pycnometer together with the liquid is then weighed with the analytical balance. After that the pycnometer is put back into the beaker.

2. The temperature of the liquid in the beaker is raised to a certain temperature, the gas burner is turned off, and the liquid to be measured in the beaker is stirred continuously. After about 10–15 minutes the temperature θ_2 of the liquid in the beaker is recorded. The pycnometer is then taken out, the liquid on its outside is wiped off, and the total mass M_2 of the pycnometer and the liquid is weighed with the analytical balance. Substituting the measured results into the formula then gives the coefficient of volume expansion of the liquid.

3. Principle of the experiment

If the mass of the glass pycnometer is M, and at temperatures θ_1 and θ_2 ($\theta_2 > \theta_1$) the volumes of the pycnometer are V_1 and V_2 respectively, while the total masses of the pycnometer are M_1 and M_2 respectively, then

$$M_1 - M = \rho_1 V_1 = \frac{\rho_0}{1+\beta\theta_1} \cdot V_0(1+3\alpha\theta_1), \tag{1}$$

$$M_2 - M = \rho_2 V_2 = \frac{\rho_0}{1+\beta\theta_2} \cdot V_0(1+3\alpha\theta_2). \tag{2}$$

In these two equations, ρ_0, ρ_1 and ρ_2 are the densities of the liquid at 0 °C, θ_1 °C and θ_2 °C respectively, and α is the coefficient of linear expansion of glass; the coefficient of volume expansion of glass is then 3α (the coefficient of volume expansion of a solid material is about three times its coefficient of linear expansion).

Hence

$$\frac{M_1 - M}{M_2 - M} = \frac{1+\beta\theta_2}{1+\beta\theta_1} \cdot \frac{1+3\alpha\theta_1}{1+3\alpha\theta_2} \approx \frac{1+\beta(\theta_2-\theta_1)}{1+3\alpha(\theta_2-\theta_1)} \approx 1 + (\beta - 3\alpha)(\theta_2 - \theta_1),$$

and therefore

$$\beta - 3\alpha = \frac{M_1 - M_2}{(M_2 - M)(\theta_2 - \theta_1)}. \tag{3}$$

In (3), $\beta' = \beta - 3\alpha$ is the apparent coefficient of volume expansion of the liquid in the glass vessel.

4. Experimental data and calculation

The mass of the empty pycnometer is $M = 7.6797\,\mathrm{g}$, and the coefficient of linear expansion of glass is $\alpha = 3.9 \times 10^{-6}\ (^\circ\mathrm{C})^{-1}$.

At temperature $\theta_1 = 21.70\,^\circ\mathrm{C}$ the total mass of the pycnometer and water is $M_1 = 17.8298\,\mathrm{g}$; at temperature $\theta_2 = 40.79\,^\circ\mathrm{C}$ the total mass of the pycnometer and water is $M_2 = 17.7716\,\mathrm{g}$.

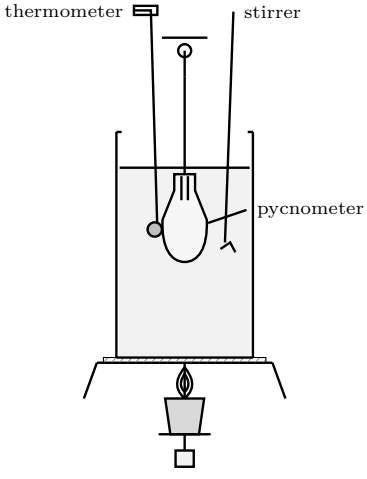


Fig. 6.22

Hence the mean coefficient of volume expansion of water between 21.70 °C and 40.79 °C is

$$\beta = 3\alpha + \frac{M_1 - M_2}{(M_2 - M)(\theta_2 - \theta_1)} = 3 \times 3.9 \times 10^{-6} + \frac{17.8298 - 17.7716}{(17.7716 - 7.6797) \times (40.79 - 21.70)}$$

$$= 0.314 \times 10^{-3}\ (^\circ\mathrm{C})^{-1}.$$

Comment. The coefficient of thermal expansion of a liquid depends not only on the material but also on the temperature interval; different intervals give different values. In the experiment it must therefore be stated within which temperature interval the result is the mean coefficient of thermal expansion.

Experiment 6.23 Measurement of the specific heat of fusion of ice by the method of mixtures

[Available apparatus and materials]

A calorimeter, a mercury thermometer, a physical balance, pure ice, water, an iron stand with clamps, etc.

[Requirements]

(1) Write down the formula for measuring the heat of fusion of ice by the method of mixtures.

(2) Choose the best experimental parameters so as to reduce the systematic error of the experiment as far as possible.

(3) Measure the specific heat of fusion of ice.

Solution 6.23. **1.** A piece of ice of mass m at 0 °C is put into water of mass M and temperature θ_1 °C contained in the inner cylinder of a calorimeter. Suppose that after the ice has completely melted the water temperature has fallen to θ_2 °C. The total heat absorbed by this piece of ice in

melting into water and then warming to θ_2 °C is $m\lambda + mc\theta_2$, where c is the specific heat capacity of water and λ the specific heat of fusion of ice. If the inner cylinder of the calorimeter and the stirrer are made of the same material, their total mass is M_1 and their specific heat capacity is c_1, then in the experiment the heat given out by the calorimeter, the stirrer and the water is $(Mc + M_1c_1)(\theta_1 - \theta_2)$. Assuming that no heat is exchanged between the calorimeter and the surrounding space during the mixing process, we have

$$(Mc + M_1c_1)(\theta_1 - \theta_2) = m\lambda + mc\theta_2,$$

hence

$$\lambda = \frac{Mc + M_1c_1}{m}(\theta_1 - \theta_2) - c\theta_2. \tag{1}$$

2. Points of procedure. Ice is made in a refrigerator. After the ice box has been taken out of the refrigerator it should be left in the room for some time; the ice is then taken out and the water on its surface is wiped off with a dry gauze. The masses of the inner cylinder and of the stirrer of the calorimeter are weighed with the balance. The mass of the ice is obtained at the end of the experiment, after all the ice has melted, by subtracting the masses of the other parts from the total mass of the inner cylinder of the calorimeter. As regards the choice of the mass of the pure ice and of the initial temperature and mass of the water, the initial temperature of the water should be higher than room temperature. After the ice has been put into the water and has completely melted, the final temperature of the water should be lower than room temperature, and the difference between the initial temperature and room temperature should be roughly equal to the difference between room temperature and the final temperature; in this way the influence of the heat exchange between the system and the surroundings on the experiment can be effectively reduced. It should also be noted that the final temperature of the water must not be lower than the dew point at that time, otherwise the water vapour in the surroundings will condense on the outer wall of the inner cylinder of the calorimeter and release heat to the system. The dew point of the environment must also be measured in the experiment.

3. Experimental results

The room temperature is 23.8 °C and the dew point 19.5 °C.

The total mass of the inner cylinder of the calorimeter and of the stirrer (both made of copper) is

$$M_1 = 144.160\,\text{g}.$$

The mass of the water is

$$M = \text{mass of vessel with water} - \text{mass of empty vessel} = 345.400 - 144.160 = 201.240\,\text{g}.$$

The measured variation of the temperature of the water in the calorimeter with time is shown in the table.

Table 6.23 Temperature of the water in the calorimeter: before the ice is added

t/min	0	0.50	1.00	1.50	2.00	2.50	3.00	3.50	4.00	4.50
θ/°C	31.75	31.70	31.66	31.62	31.60	31.57	31.52	31.50	31.48	31.46
After the ice is added										
t/min	5.00	5.50	6.00	6.50	7.00	7.50	8.00	8.50	9.00	9.50
θ/°C	25.70	22.20	20.80	20.00	19.80	19.80	19.80	19.85	19.85	19.88

Plotting the temperature of the calorimeter against time on 20 cm×25 cm graph paper gives $\theta_1 = 31.39\,^\circ\mathrm{C}$ and $\theta_2 = 19.77\,^\circ\mathrm{C}$, as shown in the figure.

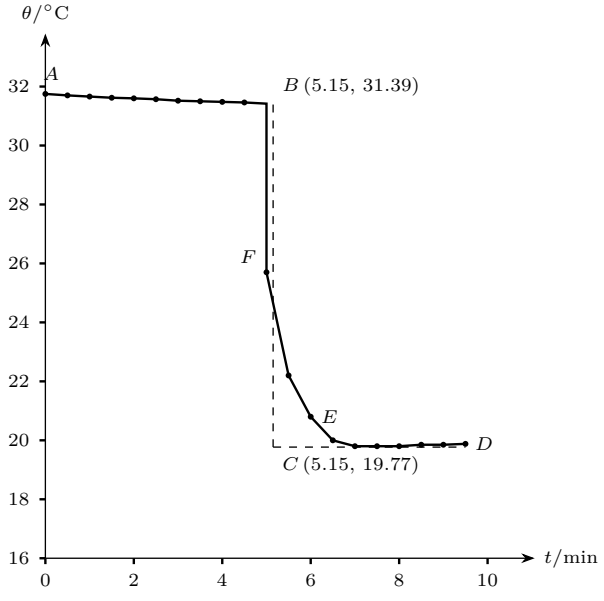


Fig. 6.23

The mass of the ice is $m = 370.800 - 345.400 = 25.400\,\mathrm{g}$. Given that at 20 °C the specific heat capacity of copper is $c_1 = 385\,\mathrm{J/(kg \cdot K)}$ and that of water is $c = 4.180 \times 10^3\,\mathrm{J/(kg \cdot K)}$, the specific heat of fusion of ice is

$$\begin{aligned}\lambda &= \frac{Mc + M_1 c_1}{m}(\theta_1 - \theta_2) - c\theta_2 \\ &= \left[\frac{(201.240 \times 4.180 + 144.160 \times 0.385) \times (31.39 - 19.77)}{25.400} - 4.180 \times 19.77\right] \times 10^3 \\ &= 327.9 \times 10^3\ (\mathrm{J/kg}).\end{aligned}$$

The standard value of the specific heat of fusion of ice is $\lambda = 332.9 \times 10^3$ J/kg. It is seen that the error of the result of this experiment compared with the standard value is rather small. Since the ice used in this experiment contained small cavities, a small amount of water was unavoidably carried in with it, so it is normal that the experimental result is on the low side.

Comment. The method of correcting the temperature for the systematic error introduced by the heat exchange between the experimental system and the surroundings by means of a graph is frequently used in thermal experiments.

Experiment 6.24 Determination of the empirical law of heat transfer in free and forced convection of air

Under conditions of forced or free convection, the heat transfer satisfies the following functional relation:

$$\frac{\Delta Q}{\Delta t} = E(\theta - \theta_0)^m, \tag{1}$$

where $\Delta Q/\Delta t$ is the heat lost by the body per unit time, θ is the surface temperature of the heated body, θ_0 is the temperature of the surrounding fluid (usually room temperature), and E is a constant independent of $(\theta - \theta_0)$. When $m = 1$, relation (1) is called Newton's law of cooling.
[Available apparatus and materials]

A mercury thermometer, a small tubular electric furnace (a 25 W wire-wound resistor may be used instead), a variable-ratio transformer, a stopwatch, an iron stand with clamps, etc.
[Requirements]

(1) Write down the formula used for the experimental measurement.

(2) Describe the experimental method briefly.

(3) Measure and calculate the data, and find the empirical formulas for Newtonian natural cooling and for forced convection.

Solution 6.24. **1.** It is known from calorimetry that for a given body the heat lost per unit time is proportional to the fall in temperature per unit time, that is,

$$\frac{\Delta Q}{\Delta t} = Mc\,\frac{\Delta \theta}{\Delta t}, \tag{2}$$

where M is the mass of the body and c its specific heat capacity. Substituting (2) into (1) gives

$$\frac{\Delta \theta}{\Delta t} = \frac{E}{Mc}(\theta - \theta_0)^m.$$

Putting $k = E/Mc$, the above expression can be written as

$$\frac{\Delta \theta}{\Delta t} = k(\theta - \theta_0)^m. \tag{3}$$

Taking the natural logarithm of both sides of (3) gives

$$\ln\left(\frac{\Delta \theta}{\Delta t}\right) = m \ln(\theta - \theta_0) + \ln k.$$

By measuring the relation between θ and t and the value of θ_0, a straight line can be fitted by means of this expression; the slope gives m and the intercept gives $\ln k$.

2. Free convection experiment

The experiment is performed with a mercury thermometer of range 0–100 °C and division value 1 °C. The variable-ratio transformer is connected to the tubular electric furnace and the supply is switched on; a suitable voltage is applied to the ends of the furnace so that the input electric power of the furnace is ≤ 25 W. First the room temperature is measured with the mercury thermometer: $\theta_0 = 10.0\,^\circ\mathrm{C}$. The thermometer is then heated uniformly in the tubular electric furnace up to 85.0 °C, and finally taken out of the furnace and allowed to cool freely in the air. A reading is taken every 10 s; the readings θ of the thermometer (in °C) are given in Table 1.

Table 6.24-1

t/s	0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0
θ/°C	85.0	79.1	73.8	69.0	64.6	60.6	57.0	53.7	50.7

Substituting the data of Table 1 into formula (3) and performing a curve fit with a calculator gives

$$m = 1.25, \qquad \text{correlation coefficient } r = 0.9998, \qquad k = 0.0027.$$

Under the condition of free cooling, the cooling formula obtained experimentally is

$$\frac{\Delta\theta}{\Delta t} = 0.0027(\theta - \theta_0)^{1.25}.$$

3. Forced convection experiment

A small fan is switched on. The room temperature is $\theta_0 = 14.8\,^\circ\mathrm{C}$. The mercury thermometer is heated uniformly in the tubular electric furnace up to 100.0 °C, then taken out of the furnace and cooled under the forced convection of the small fan; the readings of the thermometer are recorded every 10 s. The experimental data are given in Table 2.

Table 6.24-2

t/s	0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0
θ/°C	100.0	79.0	63.9	52.0	43.5	36.9	31.8	28.0	24.9

Substituting the data of Table 2 into formula (3) and fitting in the same way gives

$$m = 1.05, \qquad r = 0.9998, \qquad k = 0.023.$$

Under the condition of forced convection, the cooling formula obtained experimentally is

$$\frac{\Delta\theta}{\Delta t} = 0.023(\theta - \theta_0)^{1.05}.$$

Experiment 6.25 Measurement of the specific heat capacity of a metal by the cooling method

[Available apparatus and materials]

Samples of copper, iron and aluminium (small cylindrical metal samples 5 mm in diameter and 30 mm long, each with a deep hole drilled in its base), a copper–constantan thermocouple, a heating furnace with a lifting support, a metal shield against draughts, an ice–water mixture and a vacuum flask, a physical balance, a digital voltmeter, a stopwatch, etc.

[Requirements]

(1) Write down the principle and the experimental formula for measuring the specific heat capacity of a metal by the cooling method.

(2) Given that the specific heat capacity of copper at 100 °C is $c_1 = 0.3929 \times 10^3\,\mathrm{J/(kg \cdot {}^\circ C)}$, measure the specific heat capacities of iron and aluminium at 100 °C.

Solution 6.25. 1. If the metal sample is a small cylinder whose diameter and length are very small and whose thermal conductivity is very good, the temperature may be regarded as approximately the same everywhere in the sample. Then the following relation holds:

$$\frac{\Delta Q}{\Delta t} = c_1 M_1 \frac{\Delta\theta_1}{\Delta t}, \tag{1}$$

where c_1 is the specific heat capacity of the metal sample at temperature θ_1, $\Delta\theta_1/\Delta t$ is the rate of fall of the temperature of the metal sample at θ_1, and $\Delta Q/\Delta t$ is the heat lost per unit time. From the law of cooling,

$$\frac{\Delta Q}{\Delta t} = \alpha_1 S_1(\theta_1 - \theta_0)^m, \tag{2}$$

where α_1 is the heat transfer coefficient, S_1 the area of the outer surface of the sample, m a constant, θ_1 the temperature of the metal sample and θ_0 the temperature of the surrounding medium. From (1) and (2),

$$c_1 M_1 \frac{\Delta\theta_1}{\Delta t} = \alpha_1 S_1 (\theta_1 - \theta_0)^m. \tag{3}$$

Similarly, for another metal sample of mass M_2 and specific heat capacity c_2 the same expression may be written:

$$c_2 M_2 \frac{\Delta\theta_2}{\Delta t} = \alpha_2 S_2 (\theta_2 - \theta_0)^m. \tag{4}$$

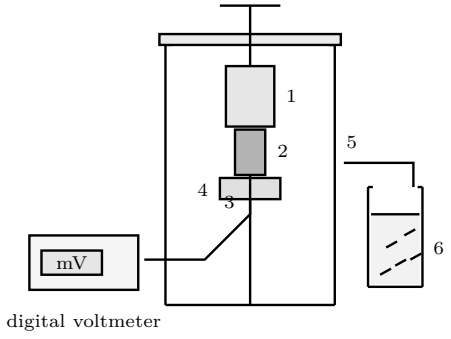
If the two samples have the same shape and dimensions, that is $S_1 = S_2$, and the same surface condition (coating, colour, etc.), and the properties of the surrounding medium are unchanged, then $\alpha_1 = \alpha_2$. If the temperature θ_0 of the surrounding medium is constant and the two samples are at the same temperature $\theta_1 = \theta_2$, then from (3) and (4)

$$c_2 = c_1 \frac{M_1 \left(\dfrac{\Delta\theta}{\Delta t}\right)_1}{M_2 \left(\dfrac{\Delta\theta}{\Delta t}\right)_2}. \tag{5}$$

If the specific heat capacity c_1 and the mass M_1 of the standard sample are known, and the mass M_2 of the sample to be measured and the ratio of the cooling rates of the two samples at temperature θ have been measured, the specific heat capacity c_2 of the metal to be measured can be found.

2. The experimental arrangement is shown in the figure. The masses of the three different samples of copper, iron and aluminium are measured separately with the physical balance. Since the samples are coated with a layer of metal film, they must be distinguished according to $M_{\mathrm{Cu}} > M_{\mathrm{Fe}} > M_{\mathrm{Al}}$. The measuring junction of the thermocouple is fixed at the centre of the metal draught shield, and the hole in the sample is slipped over the measuring junction of the thermocouple. The heating furnace A is lowered by means of the lifting support so that it covers the sample to be heated. When the sample has been heated to the required temperature, the furnace is raised and the metal lid of the draught shield is put in place. The sample then cools freely, and the time Δt required for the sample to cool from 102 °C to 98 °C is measured. The experimental data are given in Table 1.

The masses of the samples are $M_1 = 4.830\,\mathrm{g}$, $M_2 = 4.280\,\mathrm{g}$ and $M_3 = 1.500\,\mathrm{g}$. Since the samples have the same shape and dimensions, it follows from the magnitudes of the masses that M_1 is the copper sample, M_2 the iron sample and M_3 the aluminium sample. The times required for the samples to cool from 102 °C to 98 °C are given in Table 1.



1. Heating furnace; 2. Sample to be heated; 3. Thermocouple; 4. Support; 5. Draught shield; 6. Ice–water mixture.

Fig. 6.25

Table 6.25-1

Sample	Number					$\overline{\Delta t}$
	1	2	3	4	5	
Copper Δt_1/s	8.11	8.36	8.50	8.33	8.60	8.38
Iron Δt_2/s	8.86	8.99	8.94	8.58	9.09	8.89
Aluminium Δt_3/s	6.26	6.22	6.35	6.29	6.44	6.31

Taking the specific heat capacity of copper as the standard: $c_1 = 0.3929 \times 10^3\,\mathrm{J/(kg \cdot °C)}$ (100 °C).

The specific heat capacity of iron at 100 °C is

$$c_2 = c_1 \frac{M_1 \cdot \Delta t_2}{M_2 \cdot \Delta t_1} = 0.3929 \times 10^3 \times \frac{4.830 \times 8.89}{4.280 \times 8.38} = 0.470 \times 10^3\,\mathrm{J/(kg \cdot °C)}.$$

The specific heat capacity of aluminium at 100 °C is

$$c_3 = c_1 \frac{M_1 \cdot \Delta t_3}{M_3 \cdot \Delta t_1} = 0.3929 \times 10^3 \times \frac{4.830 \times 6.31}{1.500 \times 8.38} = 0.953 \times 10^3\,\mathrm{J/(kg \cdot °C)}.$$

Experiment 6.26 Measurement of the specific heat capacity of a 10% salt solution at room temperature

[Available apparatus and materials]

A physical balance, a calorimeter, a thermometer, a vacuum flask, two small glass beakers, a sufficient amount of water, ice, salt and gauze.

[Requirements]

(1) Write down the principle and the steps of the experiment.

(2) Measure and obtain the specific heat capacity of the salt solution.

Notes:

(1) The inner cylinder and the stirrer of the calorimeter are both made of copper, with specific heat capacity $c_{\mathrm{Cu}} = 0.3929 \times 10^3\,\mathrm{J/(kg \cdot °C)}$.

(2) The specific heat capacity of water at 20 °C is 4180 J/(kg · °C). The specific heat capacity of a material depends on temperature; when the temperature change does not exceed 20 °C, the specific heat capacity may be regarded as approximately constant.

Solution 6.26. Take the inner cup of the calorimeter and the insulating support out, weigh the mass of the inner cup and of the stirrer with the physical balance, and use the balance to prepare a 10% salt solution in a beaker. The prepared 10% salt solution is put into the inner cup of the calorimeter, and the cup is placed, on its insulating support, on the ice in the vacuum flask. The stirrer and the thermometer are then placed in the inner cup of the calorimeter, and the lid of the vacuum flask is put on. The temperature of the calorimeter and of the pure water must be about 20 °C above the freezing point, so the calorimeter system cools naturally inside the vacuum flask. The contents are stirred uniformly, continuously and slowly so that they cool uniformly, and the cooling curve θ_2–t of the calorimeter containing the salt solution is measured. Here θ_2 is the temperature of the salt solution and of the calorimeter system, and t is the time, measured with a stopwatch. Pure water of the same initial temperature and the same volume is then put into the inner cup of the calorimeter, and, with the same continuous, uniform and slow stirring, another cooling curve θ_1–t is obtained.

From the formula of the law of cooling, the heat lost by the calorimeter during the first cooling is

$$\frac{\Delta Q_2}{\Delta t} = -aS(\theta_2 - \theta_0)^m, \tag{1}$$

where θ_2 is the surface temperature of the calorimeter, S its surface area, a a coefficient related to the surface condition of the calorimeter, and m a coefficient related to the condition of the surrounding medium. From the definition of specific heat capacity,

$$\frac{\Delta Q_2}{\Delta t} = (m_2 c_2 + q)\,\frac{\Delta \theta_2}{\Delta t}, \tag{2}$$

where q is the water equivalent of the inner cup of the calorimeter, the stirrer and the thermometer. Combining (1) and (2) gives

$$(m_2 c_2 + q)\,\frac{\Delta \theta_2}{\Delta t} = -aS(\theta_2 - \theta_0)^m. \tag{3}$$

Similarly, after the same volume of pure water has been put into the calorimeter, the following expression is obtained in the same way:

$$(m_1 c_1 + q)\,\frac{\Delta \theta_1}{\Delta t} = -aS(\theta_1 - \theta_0)^m. \tag{4}$$

From (3) and (4),

$$(m_1 c_1 + q)\left.\frac{\Delta \theta_1}{\Delta t}\right|_{\theta=\theta_3} = (m_2 c_2 + q)\left.\frac{\Delta \theta_2}{\Delta t}\right|_{\theta=\theta_3}.$$

Here $\theta = \theta_3$ indicates that the specific heat capacity of the liquid to be measured is required at the temperature θ_3 °C. The masses m_1 and m_2 can be weighed on the balance, c_1 is known, the values of $\Delta\theta_1/\Delta t|_{\theta=\theta_3}$ and $\Delta\theta_2/\Delta t|_{\theta=\theta_3}$ can be found from the cooling curves, and q can also be found experimentally. In this way the specific heat capacity of the liquid to be measured can be determined. Rearranging and combining the above expression gives

$$c_2 = \frac{c_1 m_1 \left(\dfrac{\Delta \theta_1}{\Delta t}\right)_{\theta=\theta_3}}{m_2 \left(\dfrac{\Delta \theta_2}{\Delta t}\right)_{\theta=\theta_3}} - \frac{q}{m_2}\left(1 - \frac{\Delta \theta_1}{\Delta t} \Big/ \frac{\Delta \theta_2}{\Delta t}\right). \tag{5}$$

Using (5), the specific heat capacity of the liquid to be measured at temperature θ_3 can be obtained.

Experiment 6.27 Determination of the specific heat of vaporisation of a liquid by the method of mixtures

[Available apparatus and materials]

A steam boiler, a gas burner (or an electric furnace), a calorimeter (with a stirrer), a physical balance, a digital thermometer, etc.

[Requirements]

(1) Write down the principle and the calculation formula for measuring the specific heat of vaporisation of water.

(2) Write down the experimental method.

(3) Measure and calculate the specific heat of vaporisation of water at 100 °C.

Solution 6.27. **1.** The usual method of determining the specific heat of vaporisation of water is to pass steam at nearly 100 °C from a steam boiler through a rubber tube into a calorimeter. If the initial temperature of the calorimeter and of the water in it is θ_1 °C, and steam of mass M enters the water in the calorimeter and condenses there, then, when the water and the inner cup of the calorimeter have reached a common temperature θ_2 °C, the specific heat of vaporisation of water is given by

$$ML + Mc_{\mathrm{w}}(\theta_3 - \theta_2) = (mc_{\mathrm{w}} + m_1 c_{\mathrm{c}})(\theta_2 - \theta_1), \tag{1}$$

where c_{w} is the specific heat capacity of water; m is the mass of the water originally in the calorimeter; c_{c} is the specific heat capacity of copper; m_1 is the mass of the copper inner cup of the calorimeter and of the stirrer; θ_3 is the temperature of the steam; and L is the specific heat of vaporisation of water. According to (1), it is only necessary to measure m_1, m, θ_2, θ_1, etc., and M, in order to obtain the specific heat of vaporisation L of water.

2. The apparatus for measuring the specific heat of vaporisation of water is shown in the figure.

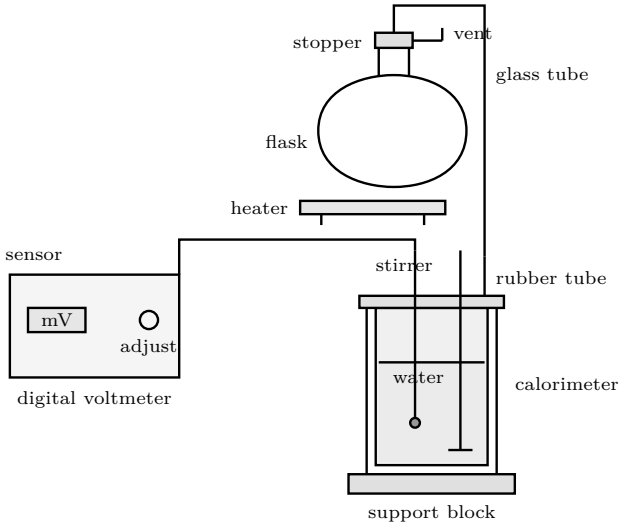


Fig. 6.27

The concrete measuring method and steps are as follows.

(1) The flask is placed on a gas burner (or an electric furnace) and a certain amount of water is put into it. The mass m_1 of the copper inner cup of the calorimeter together with the copper

stirrer is weighed with the physical balance. A certain amount of water is then poured into the inner cup of the calorimeter; the mass m of the water is obtained by weighing the total mass of the inner cup of the calorimeter containing water together with the stirrer and subtracting m_1.

(2) Before the measurement, the inner cup of the calorimeter and the water in it are placed on ice and pre-cooled to a temperature somewhat below room temperature. This temperature must, however, be higher than the dew point of the air at that time, because otherwise a thin layer of water might condense on the outer wall of the inner cup of the calorimeter and affect the measurement. The pre-cooled calorimeter is placed under the steam boiler, and the delivery tube of the steam boiler is inserted 1 cm deep into the water.

(3) The steam boiler is heated and the vent on its lid is opened, so that steam at a temperature below 100 °C escapes from the neck. When the water in the steam boiler is boiling, the water in the calorimeter is stirred gently and continuously, and the reading θ_1 of the thermometer is taken. The vent hole in the lid of the steam boiler is then closed, so that the steam passes into the inner cup of the calorimeter; stirring must be continued throughout the passage of the steam. The duration of the steam passage should be chosen so that the difference between the final temperature θ_2 of the water in the calorimeter and room temperature is as close as possible to the difference between room temperature and the initial temperature θ_1; in this way the heat exchange between the inner cup of the calorimeter and the surroundings during the experiment is cancelled out.

(4) The vent hole in the lid of the steam boiler is opened to stop the admission of steam into the calorimeter, and the water in the calorimeter is stirred continuously. The final temperature θ_2 is read when the temperatures of the water and of the inner cup have just become uniform and equal. By weighing the inner cup of the calorimeter and the water in it, the mass M of the water that has entered the calorimeter can be calculated.

(5) Substitute the values into formula (1) and calculate the specific heat of vaporisation of water at 100 °C.

3. Experimental data and calculation

The mass of the calorimeter and stirrer (made of copper) is

$$m_1 = 143.965\,\mathrm{g}.$$

The mass of the calorimeter, stirrer and water is

$$m_1 + m = 406.060\,\mathrm{g},$$

so the mass of the water is $m = 262.095\,\mathrm{g}$.

The total mass of the calorimeter, stirrer, original water and steam is $m_1+m+M = 415.480\,\mathrm{g}$, so the mass of the steam is $M = 9.420\,\mathrm{g}$. The specific heat capacity of copper is $c_\mathrm{c} = 0.387\,\mathrm{J/(g\cdot °C)}$, the specific heat capacity of water is $c_\mathrm{w} = 4.188\,\mathrm{J/(g\cdot °C)}$ (its value at about 13 °C), the initial temperature is $\theta_1 = 2.91\,°\mathrm{C}$ and the final temperature $\theta_2 = 23.88\,°\mathrm{C}$. Substituting these data into (1) gives

$$\begin{aligned} L &= \frac{mc_\mathrm{w} + m_1 c_\mathrm{c}}{M}(\theta_2 - \theta_1) - c_\mathrm{w}(\theta_3 - \theta_2) \\ &= \frac{4.188 \times 262.095 + 0.387 \times 143.965}{9.420} \times (23.88 - 2.91) - 4.188 \times (100.0 - 23.88) \\ &= 2248\,\mathrm{J/g} = 2.248 \times 10^6\,\mathrm{J/kg}. \end{aligned}$$

The standard value of the specific heat of vaporisation of water at 100 °C is $L_\mathrm{st} = 2.257 \times 10^6\,\mathrm{J/kg}$; the experimental result lies within the limits of error.

Experiment 6.28 Measurement of the specific heat of vaporisation of liquid nitrogen

[Available apparatus and materials]

A physical balance, a calorimeter, a stopwatch, a mercury thermometer (0–50 °C, division value 0.1 °C), a copper sample (in the form of a small rod), a small Dewar flask (with an outer casing of polystyrene foam for thermal insulation), liquid nitrogen and a vacuum flask, etc. The small Dewar flask may be a vacuum-cup inner vessel.

[Requirements]

(1) Write down the principle and the formula of the experiment.

(2) Write down the experimental method.

(3) Measure and calculate the specific heat of vaporisation of liquid nitrogen.

Solution 6.28. 1. A small hole is made in the lid of a small vacuum flask containing a certain mass of liquid nitrogen. The liquid nitrogen gradually vaporises by absorbing heat from the surrounding atmosphere and escapes through the small hole. The mass of liquid nitrogen vaporised per unit time can be weighed with the physical balance. If a copper sample of known mass is put through the hole into the liquid nitrogen, then, since the temperature of the liquid nitrogen is very low, the copper sample gives out heat and the vaporisation process is accelerated, until the temperature of the copper sample becomes equal to that of the liquid nitrogen and the heat exchange between them stops. The mass of the Dewar flask containing liquid nitrogen is weighed with the physical balance, and its variation with time is shown in Fig. 1. In the figure, t is the time and M is the total mass of the copper sample, the liquid nitrogen and the Dewar flask. Join bc and extend the straight lines ab and cd. Through the midpoint e of the line bc draw a line parallel to the M axis; it meets the extensions of the lines ab and dc at f and g. The difference between the values of f and g on the M axis is then the mass m_N of liquid nitrogen vaporised by the heat released at the instant when the copper sample at room temperature is put into the liquid nitrogen.

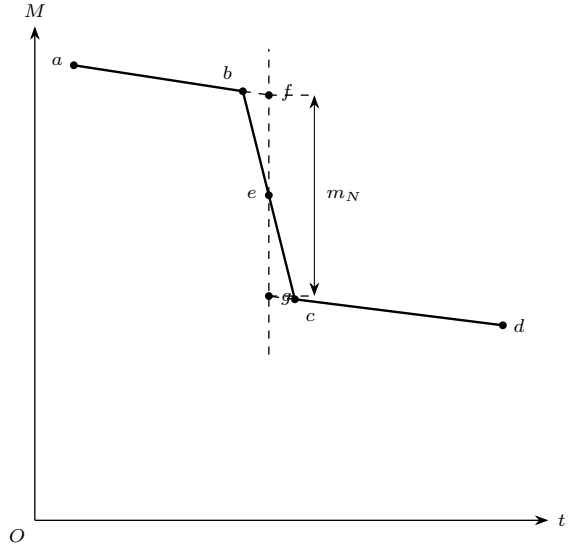


Fig. 6.28(a)

Let the specific heat of vaporisation of liquid nitrogen be L; the heat absorbed by the vaporisation of a mass m_N of liquid nitrogen is $Q = m_N L$. This heat can be measured by putting a copper

sample at the temperature of liquid nitrogen into a calorimeter. Let the room temperature be θ_1 and the initial temperature of the water and of the calorimeter be θ_2. The copper sample, at the boiling point of liquid nitrogen, is put into the water, and when it has reached equilibrium with the inner cup of the calorimeter, the stirrer and the water, the temperature is θ_3. The copper sample at the boiling point of liquid nitrogen absorbs heat Q_1, and the copper sample absorbs heat Q_2 in rising from θ_3 to room temperature. From the law of conservation of energy,

$$Q = Q_1 + Q_2.$$

From the principle of heat exchange in the method of mixtures,

$$Q_1 = (m_{\mathrm{w}} c_{\mathrm{w}} + m_{\mathrm{c}} c_{\mathrm{c}} + h_t)(\theta_2 - \theta_3),$$

where m_{w} and c_{w} are the mass and specific heat capacity of the water; m_{c} and c_{c} are the mass of the calorimeter together with the stirrer and their specific heat capacity; and h_t is the heat capacity of the immersed part of the thermometer. Moreover,

$$Q_2 = m_b c_b (\theta_1 - \theta_3),$$

where m_b is the mass of the copper sample and c_b the specific heat capacity of the sample material. Hence

$$m_N \cdot L = (m_{\mathrm{w}} c_{\mathrm{w}} + m_{\mathrm{c}} c_{\mathrm{c}} + h_t)(\theta_2 - \theta_3) + m_b c_b (\theta_1 - \theta_3),$$

so the specific heat of vaporisation of liquid nitrogen at its boiling point is

$$L = \frac{1}{m_N} \left[(m_{\mathrm{w}} c_{\mathrm{w}} + m_{\mathrm{c}} c_{\mathrm{c}} + h_t)(\theta_2 - \theta_3) + m_b c_b (\theta_1 - \theta_3) \right]. \tag{1}$$

2. Experimental method

(1) *Measurement of the mass m_N vaporised when the copper sample at room temperature is put into liquid nitrogen.*

The curve of the total mass M of the liquid nitrogen, the vessel and the copper sample against time t is measured with the physical balance. While the straight segment ab is being measured the copper sample is on the pan at room temperature. At the time t_b the copper sample is quickly put into the flask of liquid nitrogen, whereupon the liquid nitrogen rapidly vaporises and escapes. It is only necessary to record the time t_b at which the copper is put into the liquid nitrogen and the time t_c at which thermal equilibrium with the liquid nitrogen is reached, and then to measure the straight segment cd. Finally the M–t graph is drawn, a line fg is drawn through the midpoint of the straight segment bc, and it meets the extensions of the straight lines ab and cd at the points f and g. The value of the mass m_N is read from the line fg.

(2) *Measurement of the heat released by the copper sample in cooling from room temperature to the temperature of liquid nitrogen with the calorimeter.*

The mass of the metal inner cup of the calorimeter together with the stirrer is weighed with the physical balance, and then the mass of the water together with the inner cup of the calorimeter, etc., is weighed, so as to obtain the mass of the water in the cup. The ambient temperature θ_1 and the temperature θ_2 of the water and of the inner cup are measured with the thermometer. The copper sample is then taken out of the liquid nitrogen and quickly put into the calorimeter; when thermal equilibrium has been reached the temperature is θ_3. Substituting the measured results into formula (1) gives the specific heat of vaporisation of liquid nitrogen.

3. Experimental data and calculation

The mass of the copper sample is $m_b = 12.530\,\text{g}$, the mass of the inner cup of the calorimeter (aluminium) is $m_a = 27.870\,\text{g}$, and the mass of the brass stirrer is $m_c = 6.510\,\text{g}$. The specific heat capacity of brass is $c_c = 0.384\,\text{J/(g}\cdot{}^\circ\text{C)}$ and that of aluminium $c_a = 0.900\,\text{J/(g}\cdot{}^\circ\text{C)}$.

The data for the relation between M and t are given in the table below.

Table 6.28

t/s	0	49	109	166	224	327	391	518	580
M/g	227.00	226.50	226.00	225.50	225.00	220.00	219.50	218.50	218.00

In the table above, the copper sample is put in at 224 s and reaches the boiling temperature of liquid nitrogen at 327 s.

The room temperature is $\theta_1 = 16.05\,^\circ\text{C}$; the initial temperature of the water is $\theta_2 = 15.10\,^\circ\text{C}$; when the copper sample, at the boiling temperature of liquid nitrogen, has been put into the calorimeter and thermal equilibrium has been reached, the temperature is $\theta_3 = 12.35\,^\circ\text{C}$. The heat capacity of the immersed part of the thermometer is $h_t = 1.7\,\text{J/}^\circ\text{C}$ (it must be measured beforehand). The relation between M and t is plotted on graph paper (Fig. 2), from which

$$m_N = 224.60 - 220.40 = 4.20\,\text{g}.$$

The specific heat capacity of water is known to be $c_\text{w} = 4.188\,\text{J/(g}\cdot{}^\circ\text{C)}$.

Substituting the above measured data into formula (1) gives

$$\begin{aligned} L &= \frac{1}{m_N}\left[(m_\text{w}c_\text{w} + m_a c_a + m_\text{c}c_\text{c} + h_t)(\theta_2 - \theta_3) + m_b c_b(\theta_1 - \theta_3)\right] \\ &= \frac{1}{4.20}\Big[(63.64 \times 4.188 + 27.87 \times 0.900 + 6.51 \times 0.384 + 1.7) \\ &\qquad \times (15.10 - 12.35) + 12.53 \times 0.384 \times (16.05 - 12.35)\Big] = 197.9 \times 10^3\,\text{J/kg}. \end{aligned}$$

The standard value of the specific heat of vaporisation of liquid nitrogen at one atmosphere is 199.6×10^3 J/kg; the percentage difference between the two is 0.9%.

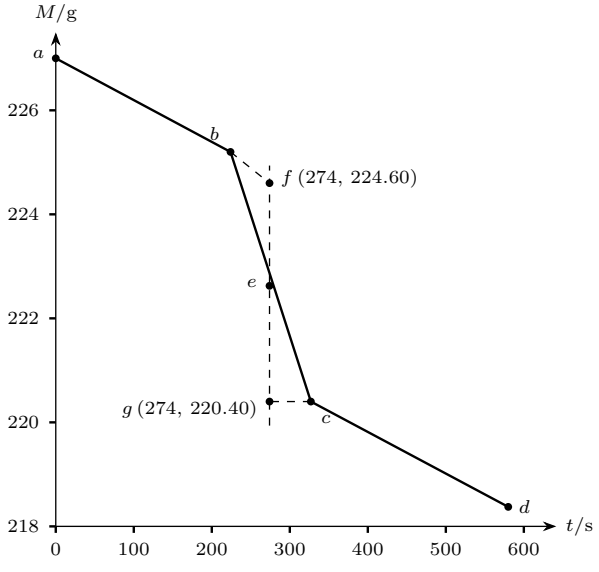


Fig. 6.28(b)

Comment. A flask of liquid nitrogen with a small hole vaporises naturally in the atmosphere and loses mass; in the experiment this part of the vaporisation unavoidably consumes liquid nitrogen. When the amount of liquid nitrogen vaporised by the heat given out by the copper sample is measured by the heat-transfer method, the "amount of liquid nitrogen vaporised by natural absorption of heat" must be eliminated. In this experiment the systematic error is eliminated by a graphical method, which is worth imitating in other thermal experiments.

Experiment 6.29 Thermal radiation and measurement of the Stefan–Boltzmann constant

(Experimental problem of the First Asian Physics Olympiad.) Measure the Stefan–Boltzmann constant with the apparatus provided.

[Available apparatus and materials]

A stabilised DC power supply, a heater mounted on a ceramic base plate, a digital voltmeter, a digital ammeter, a digital millivoltmeter (for measuring the potential difference of an iron–constantan thermocouple), a table of the potential difference of an iron–constantan thermocouple as a function of temperature, several connecting wires, a candle and matches (for blackening the outer surface of an aluminium cylindrical sleeve).

[Requirements]

(1) Describe the principle of the experiment and the relevant formulas.

(2) Describe briefly the experimental method and steps.

(3) Measure the value of the Stefan–Boltzmann constant σ.

(4) Analyse the errors and their sources.

Solution 6.29. **1.** According to the Stefan–Boltzmann principle, for a body of surface area S which has reached a steady thermal equilibrium with its surroundings, if its thermodynamic temperature is T, the effective power radiated is

$$P = e\sigma S(T^4 - T_0^4), \tag{1}$$

where σ is the Stefan–Boltzmann constant, T_0 the temperature of the surroundings and e a coefficient. For an ideal black body $e = 1$; for an ideal reflector $e = 0$.

According to the theory of electromagnetic radiation of solids, a polished aluminium cylinder may be regarded as an ideal reflector: it absorbs no heat and radiates no heat, that is, $e = 0$. The same cylinder, after being uniformly coated with a thin layer of lampblack, may be regarded as an ideal black body, which absorbs all the heat incident on it, and at the same time it is an ideal radiator, that is, $e = 1$. Hence, under ideal conditions, a polished cylinder, when heated, may be regarded as losing heat only through non-radiative processes, such as heat conduction or air convection. A blackened hot cylinder, on the other hand, loses heat at the same time through the process of thermal radiation, and this process obeys the Stefan–Boltzmann principle.

There are three different methods of measuring the Stefan–Boltzmann constant.

(1) *Method of measurement at the same temperature.*

Suppose that the cylinder is heated to the same temperature both when it is blackened and when it is not blackened. The electric energy consumed in maintaining thermal equilibrium must then be equal to the energy lost through radiation, that is,

$$P_r = P_t(T) - P_h(T),$$

where P_r is the energy lost by the blackened cylinder through thermal radiation, $P_t(T)$ is the total energy loss of the blackened cylinder at temperature T, and $P_h(T)$ is the energy loss of the

smooth cylinder at temperature T caused by non-radiative processes. Assuming that the value of $P_h(T)$ is the same in the two processes, we have

$$\sigma = \frac{P_t(T) - P_h(T)}{S(T^4 - T_0^4)},$$

where T_0 is room temperature. Suppose that $P_h(T)$ is approximately proportional to $(T - T_0)$, that is,

$$P_h = k(T - T_0),$$

where k is a constant independent of T. On the basis of this relation the following two methods of measuring σ may be set up (these two methods are less accurate than the first).

(2) *Method of measurement at constant power.*

In this method the power P supplied to heat the ideal black body and the ideal reflector is kept constant, and their temperatures at equilibrium are denoted by T_b and T_p respectively. Then

$$P = k(T_p - T_0),$$

$$P = k(T_b - T_0) + P_r(T_b).$$

Eliminating k gives

$$P_r(T_b) = \frac{(T_p - T_b)P}{T_p - T_0}.$$

Substituting this expression into (1) gives

$$\sigma = \frac{(T_p - T_b)P}{S(T_b^4 - T_0^4)(T_p - T_0)}.$$

(3) *Method of measurement at two different temperatures.*

In this method only the blackened cylinder is used, and the measurements are made at two equilibrium temperatures T_1 and T_2. If the powers required to heat it to T_1 and T_2 are denoted by P_1 and P_2 respectively, then

$$P_1 = k(T_1 - T_0) + \sigma S(T_1^4 - T_0^4),$$

$$P_2 = k(T_2 - T_0) + \sigma S(T_2^4 - T_0^4).$$

Eliminating k gives

$$\sigma = \frac{(T_2 - T_0)P_1 - (T_1 - T_0)P_2}{S\left[(T_1^4 - T_0^4)(T_2 - T_0) - (T_2^4 - T_0^4)(T_1 - T_0)\right]}.$$

2. Experimental method

The experimental arrangement is shown in the figure. The heater is mounted on a ceramic base and connected to the supply and to the meters. During the measurement the heater is completely enclosed in a hollow cylinder (the cylinder is also mounted on the ceramic base). An iron–constantan thermocouple is connected to the cylinder and at the same time to the digital millivoltmeter, and the temperature is obtained from the table. The length of the cylinder is measured with a vernier calliper as 60.00 mm; its outer diameter is 12.50 mm, and the surface area of the cylinder is $S = 24.80\,\mathrm{cm}^2$. The wall thickness of the cylinder is 1.0 mm and the thickness of its base is 3.0 mm. The digital millivoltmeter is replaced by a digital multimeter. When the voltage U and the current I of the heater are read, the cylinder and the surroundings are required to have reached thermal equilibrium; reaching thermal equilibrium generally takes 25 min to 30 min.

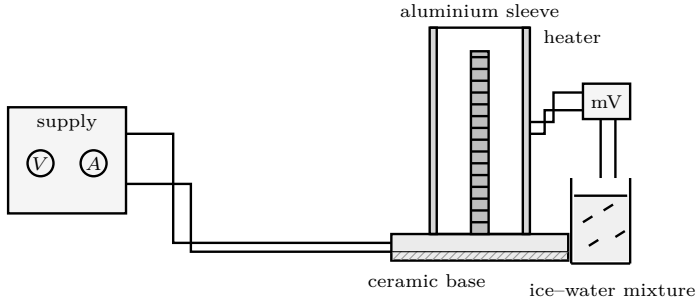


Fig. 6.29

3. The experiment is carried out at a room temperature of 298.3 K. The experimental data are given in Table 1 and the experimental results in Table 2. The table of the potential difference of the iron–constantan thermocouple as a function of temperature is omitted.

Table 6.29-1

Data number	Nature of the cylinder surface	U/V	I/A	T/K
A	Polished	9.80	1.50	485.5
B	Blackened	9.80	1.50	433.5
C	Blackened	11.90	1.82	485.5

Table 6.29-2 Values of $\sigma_{\exp}$ obtained by the three different methods

Method used	Data used	$\sigma_{\exp}/\mathrm{W}\cdot\mathrm{m}^{-2}\cdot\mathrm{K}^{-4}$	$\sigma_{\exp}/\sigma$
T unchanged	A + C	5.945×10^{-8}	1.05
P unchanged	A + B	6.087×10^{-8}	1.07
Different temperatures	B + C	5.386×10^{-8}	0.95

In Table 2, $\sigma_{\exp}$ is the measured experimental value, and $\sigma = 5.669\times10^{-8}\ \mathrm{W}\cdot\mathrm{m}^{-2}\cdot\mathrm{K}^{-4}$ is the accepted value.

The main factors affecting the accuracy of the experimental results are as follows.

(1) The cylinder with a smooth surface is not an ideal reflector, and when blackened it is not an ideal black body either. The absorption coefficient of the former is greater than 0, and that of the latter is less than 1; both make the value of σ smaller.

(2) The heat lost through the ceramic base cannot be calculated; neglecting this loss will cause the measured value of σ to deviate from the true value.

(3) Neglecting the resistance of the connecting wires makes the measured value of σ smaller.

(4) The assumption that the smooth cylinder and the blackened cylinder have the same non-radiative energy loss is only an approximation. For example, when the difference in heat conduction between the lampblack and the aluminium is neglected, the value of σ is made smaller. The energy lost through the ceramic base cannot be controlled, and this also makes the non-radiative energy loss uncorrected, thus affecting the experimental results.

Chapter 5

Electromagnetic Experiments

Electromagnetic Experiments

Experiment 7.01 Measurement of the resistance of a medium-value resistor by two different methods, with a comparison

[**Apparatus and materials available**]

Three 1.5 V dry cells (type-A dry cells), an ammeter of range 0–500 μA (scale division 10 μA per division), a voltmeter of range 0–1 V (0.02 V per division), a resistance box, three single-pole switches, one resistor R_x to be measured, one galvanometer, one slide-wire rheostat, and a number of connecting wires.

Solution 7.01 1. Measurement of a resistance by the voltmeter–ammeter method

Measuring a medium-value resistance with pointer-type voltmeters and ammeters is a common method of resistance measurement. However, the internal resistance R_V of a practical pointer-type voltmeter is not infinite, and the internal resistance R_A of the ammeter is not zero; moreover, when a resistance is measured, whether the ammeter is connected inside or outside has a comparatively large influence on the result for an unknown resistance. The measuring circuit is shown in Fig. 7.01(a). Throwing the blade of the single-pole double-throw switch to 1 or to 2 represents respectively the ammeter-inside or the ammeter-outside connection. When the ammeter is connected inside, the voltage measured is the sum of the voltages across the resistor R_x under test and across the ammeter; when the ammeter is connected outside, the current measured by the ammeter is the total current through the resistor R_x under test and the voltmeter. If the internal resistance R_V of the voltmeter is known, or the internal resistance R_A of the ammeter is known, then the magnitude of the error of the measured result can be estimated and a correction can be made.

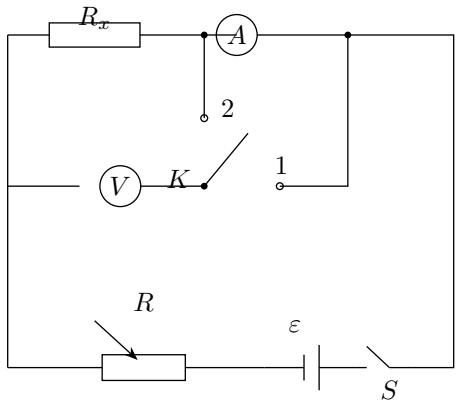


Fig. 7.01(a)

Following the wiring of Fig. 7.01(a), the experimental results are as shown in Table 7.01-1.

Table 7.01-1

	U/V			U/V	
I/μA	K at position 1	K at position 2	I/μA	K at position 1	K at position 2
50	0.180	0.102	250	0.920	0.516
100	0.362	0.204	300	> 1	0.628
150	0.542	0.305	400	> 1	0.836
200	0.740	0.414			

From the slope of the straight line in Table 7.01-1, the apparent resistance with the key K at position 1 is $(R_x)_1 = 3.70\,\mathrm{k\Omega}$; with the key K at position 2 the apparent resistance is $(R_x)_2 = 2.09\,\mathrm{k\Omega}$. It can be seen that for a resistor of a few kilo-ohms, because the internal resistance of the ammeter cannot be neglected (a 0–500 μA DC ammeter has an internal resistance of about 200 Ω) or the internal resistance of the voltmeter is not large enough (the 0–1 V range of an MF30 multimeter has an internal resistance of 20 000 Ω), the voltmeter–ammeter method of measuring resistance introduces a comparatively large error.

2. Measurement of the voltage by the compensation method and of the current by the ammeter, to find the value of the unknown resistance R_x

The basic principle of measuring a voltage by the compensation method is shown in Fig. 7.01(b). When the terminals of like polarity of two DC sources are connected together and their electromotive forces are exactly equal in magnitude, the indication of the sensitive galvanometer G is zero, no current flows in the loop, and the circuit is then in balance. Hence a circuit possessing this property (called an outside-compensation circuit) can be used to measure an unknown voltage or electromotive force by means of a known voltage or electromotive force. Its advantage is that, when a voltage is measured, no current has to be drawn from the circuit under test, so that the systematic error introduced by the current flowing through the voltmeter is avoided.

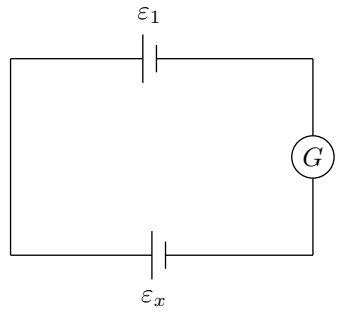


Fig. 7.01(b)

The circuit for measuring a voltage by the compensation method is shown in Fig. 7.01(c). The circuit inside the dashed frame replaces the voltmeter of Fig. 7.01(a); in it ε_1 is the compensating source and R_0 is a voltage divider (a slide-wire rheostat). By adjusting the position of the sliding contact C, different voltages U_{BC} can be obtained, and these are measured with the voltmeter V. In the experiment the position of C should be adjusted until the indication of the galvanometer G is zero; then U_{BC} must compensate the voltage U_x across R_x, and the reading of the voltmeter V is just U_x. A reliable sign that balance has been reached is that, on closing and opening the switch K, the needle of the galvanometer shows no tiny tremor whatever. The voltage drop across the resistor R_x was measured by the compensation method, the current through R_x was measured with the ammeter, and the data obtained are shown in Table 7.01-2.

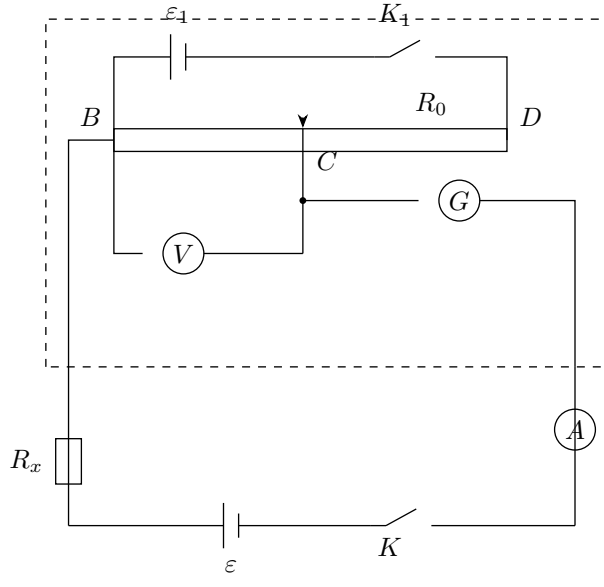


Fig. 7.01(c)

Table 7.01-2

I/μA	50	100	200	300	390
U/V	0.117	0.234	0.466	0.700	0.912

From the slope of the I–U graph (see Fig. 7.01(d)) one obtains $R_x = 2.33\,\mathrm{k}\Omega$.

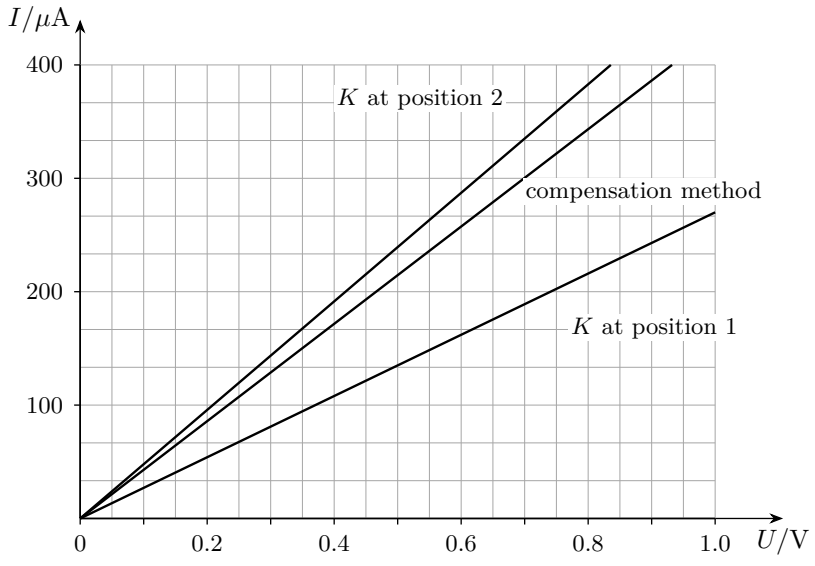


Fig. 7.01(d)

From the figure it can be seen that with the key K at position 1 the slope of the I–U straight line is small and the measured resistance is too large, whereas with K at position 2 the measured resistance is too small. When the voltage is measured by compensation, the resistance value obtained can truly reflect the resistance of this resistor.

Experiment 7.02 Measurement of the resistance of a medium-value resistor with a Wheatstone bridge

[Apparatus and materials available]

Two type-A dry cells, three resistance boxes, one galvanometer, one resistor to be measured, a number of connecting wires, etc.

[Requirements]

(1) Draw the DC bridge circuit and write down the working principle of the DC bridge.

(2) Use the DC bridge to measure the resistance of the unknown resistor.

(3) Analyse the experimental error and estimate the standard uncertainty of a resistance measurement made with the DC bridge.

Solution 7.02 1. The Wheatstone bridge is also called the DC bridge; its working principle is shown in Fig. 7.02(a).

R_A, R_B, R_x and R_s are four resistors, and each side is called one arm of the bridge; the opposite corners 1 and 3 are connected to a DC source, while between the opposite corners 2 and 4 a galvanometer is connected, in order to detect whether there is a current between them. When the potentials at the two points 2 and 4 are equal, no current flows through the galvanometer, and the bridge is said to be balanced. That is, $i_{AB}R_A = i_{xs}R_x$; in the same way $i_{AB}R_B = i_{xs}R_s$, and hence

$$\frac{R_A}{R_B} = \frac{R_x}{R_s}. \tag{1}$$

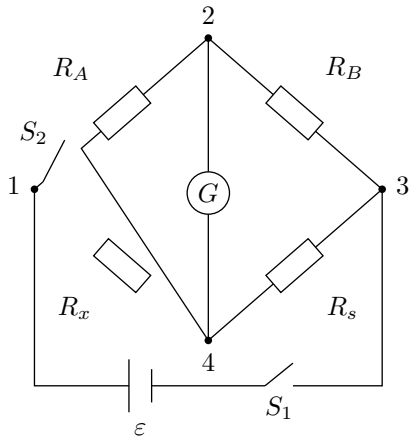


Fig. 7.02(a)

This is the balance equation of the bridge. From the values of R_A, R_B and R_s, the resistance R_x to be measured can be calculated by means of (1).

2. Measurement error of the bridge

(1) *Error introduced by the sensitivity of the bridge.* When the state of unbalance of the bridge cannot be detected by the galvanometer, the measured result R_x has a corresponding uncertainty. In (1), R_A and R_B are fixed values. If balance is reached at $R_s = 1000\,\Omega$, and when R_s is adjusted to $1010\,\Omega$ the galvanometer deviates by one division from the zero point, then the error of R_s caused by the adjustment is $1\,\Omega$; hence the uncertainty of the adjustment of R_s is U_s, and from (1) one obtains

$$\frac{U_{R_x}}{R_x} = \frac{U_{R_s}}{R_s}.$$

Hence the uncertainty introduced by the sensitivity of the bridge can be found as

$$U_{R_x} = \frac{R_x}{R_s} \cdot U_{R_s}.$$

As regards the bridge, the ways of raising the sensitivity and reducing the error are: (a) generally speaking, when the four arms R_A, R_B, R_s, R_x are approximately equal, the measurement error is smallest; (b) the larger the source voltage, the smaller the measurement error, but the permissible power of the elements of each arm must be taken into account; (c) a galvanometer of somewhat higher sensitivity should be chosen, though of course the sensitivity must not be too high, which would make the operation troublesome.

(2) *Error introduced by unequal arm resistances.* If the nominal values of the arms R_A and R_B are equal, the two can be interchanged to eliminate part of the error. That is, the readings R_{s1} and R_{s2} of the arm R_s at balance before and after the interchange are measured separately; then from (1) one obtains

$$R_x = \sqrt{R_{s1} \cdot R_{s2}} \approx \frac{R_{s1} + R_{s2}}{2}.$$

This eliminates the systematic error introduced by the deviation of the nominal values of R_A and R_B from their actual values.

The resistor to be measured is a metal-film resistor. The arms R_A and R_B both use resistance boxes of class 0.1, 0–9999 Ω, and R_s is a resistance box of class 0.1, 0–99 999.9 Ω; the source is a dry cell. The sensitivity of the galvanometer is $5\,\mu$A per division. The measured results are given in Table 7.02-1.

Table 7.02-1 Measured results for different bridge-arm ratios

R_A/Ω	R_B/Ω	R_s/Ω	$\frac{\Delta R_s}{\Delta\theta}$/(Ω per division)	R_x/Ω	U_{R_x}/Ω
1000	1000	355.0	12	355.0	1.2
100.0	100.0	354.9	11	354.9	1.1
100.0	1000	3530	340	353.0	3.4
10.00	1000	3500×10	2000×10^{-1}	3.5×10^2	0.2×10^2

Table 7.02-2 Results obtained after interchanging the bridge arms

	R_A/Ω	R_B/Ω	R_s/Ω	R_A/Ω	R_B/Ω	R_s/Ω
Before interchange	1000	1000	354.8	100.0	100.0	356.0
After interchange	1000	1000	354.6	100.0	100.0	353.4
$R_s = \frac{R_{s1} + R_{s2}}{2}$			354.7			354.7

Calculation of the uncertainty: in the above example the error is introduced by the sensitivity of the bridge, hence

$$\frac{U_{B1}(R_x)}{R_x} = \frac{U_{B1}(R_s)}{R_s},$$

and the maximum possible uncertainty introduced by the class of the bridge-arm elements is

$$\frac{U_{B2}(R_x)}{R_x} = \sqrt{\left[\frac{U_{B2}(R_A)}{R_A}\right]^2 + \left[\frac{U_{B2}(R_B)}{R_B}\right]^2 + \left[\frac{U_{B2}(R_s)}{R_s}\right]^2} = \sqrt{\left(\frac{0.1}{100}\right)^2 \times 3} = 0.17\%.$$

The total uncertainty is

$$U(R_x) = \sqrt{[U_{B1}(R_x)]^2 + [U_{B2}(R_x)]^2} = \sqrt{(1.2)^2 + (0.6)^2} = 1.4\,(\Omega).$$

Hence the resistance measured with the bridge is

$$R_x = (354.7 \pm 1.4)\,\Omega.$$

Comment. The balanced bridge is a method of measuring resistance accurately by means of comparison, and it is very widely used. Besides the source voltage and the sensitivity of the galvanometer, the ratio of the bridge-arm resistances also has a great influence on the sensitivity of the bridge. Therefore one usually first makes a rough measurement of the resistance to be measured, and then makes the other three arms all of a resistance close to that of the resistor under test, so as to reduce the measurement error. In recent years the unbalanced bridge has become very common in electronic circuits and in the design of sensors. The points discussed in this experiment on raising the sensitivity of a bridge are also very useful for the design of an unbalanced bridge.

Experiment 7.03 Measurement of a low resistance by the compensation method

[Apparatus and materials available]

One DC constant-current power supply, one type-A cell (its electromotive force is accurately given; when the external resistance is above a few kilo-ohms its internal resistance may be neglected), two resistance boxes, one galvanometer, one switch, connecting wires, and a low resistance to be measured (of the order of $10^{-2}\,\Omega$).

[Points for attention]

(1) When the external resistance of the power supply is very small, the supply does not show an output voltage; the DC constant-current supply may then be set to current output, and by adjusting the current knob the current output value is displayed. The current should be chosen to be less than 2 A.

(2) The low resistance to be measured has four terminals: the J terminals denote the current terminals and the P terminals denote the voltage terminals; the resistance between P and P is required to be measured.

Solution 7.03 The principle and the method of measuring a voltage by compensation are given in Solution 7.01, and the measuring circuit is similar. The difference is that the low resistance to be measured must have four terminals, and is called a four-terminal resistance. When a standard resistance or a low-value resistance is used, two further terminals should be added; the structure of a four-terminal resistance is shown in Fig. 7.03(a). When a resistance R_{AB} is connected into a circuit, lead resistances and contact resistances are inevitably introduced; these are marked by dashed lines in Fig. 7.03(a), and the resistance of these additional parts is about $0.001\,\Omega$ or more. In most cases such a small additional resistance has little influence on the measurement; but if this resistance is a standard resistance, or if the resistance value is very small, then the effect of this factor should be considered. Suppose a low-value resistance has a resistance of $0.01\,\Omega$; then, because of the existence of the additional resistances, an error of at least 10% is introduced, so four terminals must be used for standard resistances and low-value resistances. In Fig. 7.03(b), J and J' are the two current lead-in terminals, R_{AB} denotes the standard resistance (made of an alloy wire) or the low-value resistance itself, and P and P' are two added potential lead-out terminals, which can be used for measuring a voltage.

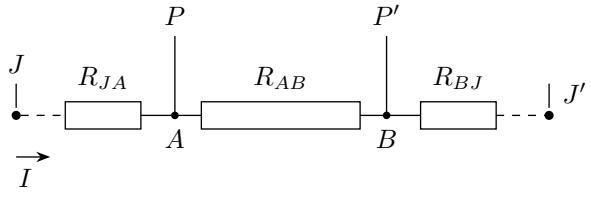


Fig. 7.03(a)

When a current i is passed, the voltage between P and P' is iR_{AB} and is independent of the voltages across the additional resistances R_{JA} and R_{JB}. In this experiment, when the two voltages are compensated, no current flows through P_A or P_B, that is, there is no voltage across the resistance of the potential leads, so the voltage measured must be the voltage of the resistor under test or of the standard resistor itself. If the current i through this low-value resistor under test is known, the value of this low resistance can be calculated from the measured voltage.

The wiring diagram of this experiment is shown in Fig. 7.03(b). When the indication of the galvanometer is zero, the voltage across the two ends of the voltage divider that has been set is equal to the voltage across the two ends A and B (compensation); hence, by the principle of voltage division, the voltage U_{AB} across A and B is calculated, and the current i through the low resistance can be read directly from the meter on the DC constant-current source; then

$$R_{AB} = \frac{U_{AB}}{i}.$$

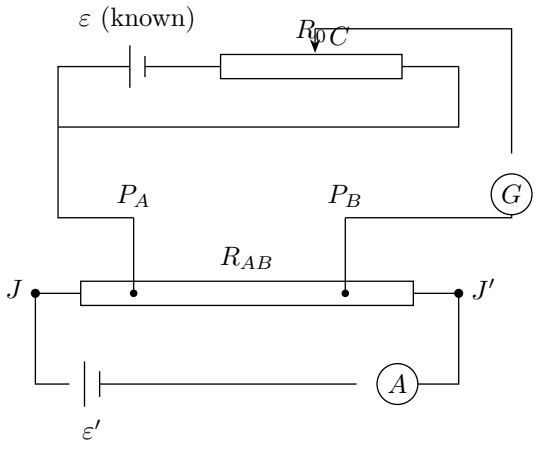


Fig. 7.03(b)

Experiment 7.04 Measurement of the resistance of a high-value resistor by the discharge method

[**Apparatus and materials available**]

A DC power supply, a capacitor ($235\,\mu$F/10 V non-polar capacitor, formed by connecting two $470\,\mu$F/10 V electrolytic capacitors of opposite polarity in series), a pointer-type multimeter, a stopwatch, a switch, connecting wires, etc.

[**Requirements**]

(1) Write down the measuring formula for the resistance of a high-value resistor, the meaning of each symbol in the formula, and the experimental circuit diagram.

(2) Measure the internal resistance of the 5 V DC voltage range of a pointer-type multimeter.

(3) Analyse the experimental error.

Solution 7.04 The experimental circuit is as shown in Fig. 7.04(a). Throw the switch to 1 and connect the 5 V DC supply; when the pointer multimeter on the 5 V range indicates a voltage of 5 V, disconnect the DC supply, connect the switch to 2 and at the same time press the stopwatch to start timing; when the voltage indication U of the pointer multimeter has fallen to 2.5 V, press the stopwatch button to stop timing. Record the half-life $T_{1/2}$ corresponding to the decrease of the capacitor voltage from 5 V to 2.5 V. After the capacitor has been charged, the law of discharge of the capacitor is

$$U = U_0 \mathrm{e}^{-t/RC},$$

where U_0 is the capacitor voltage at the beginning, U the capacitor voltage at the time t from the beginning of the discharge, R the discharge resistance between the two plates of the capacitor (that is, the internal resistance of the 5 V range), and C the capacitance of the capacitor. When

$$\frac{U}{U_0} = \frac{1}{2},$$

we have $T_{1/2}/RC = \ln 2$, and therefore $RC = T_{1/2}/\ln 2 = T_{1/2}/0.693$. If the value of the capacitance C is known, this formula can be used to find the internal resistance R of the 5 V range of the pointer multimeter.

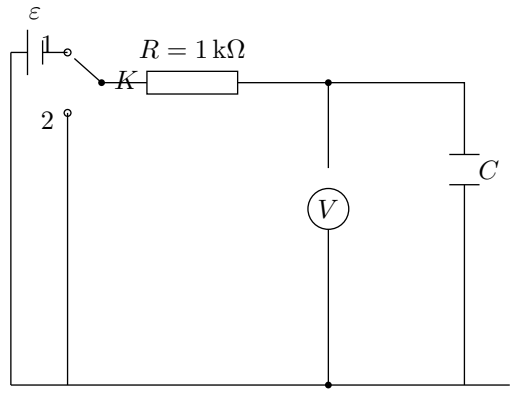


Fig. 7.04(a)

Experimental data example:

The half-life was measured five times in all, and the mean value $\overline{T}_{1/2} = 19.5\,\mathrm{s}$ was taken; it is known that $C = 235 \times 10^{-6}\,\mathrm{F}$.

$$\therefore\ R = \frac{T_{1/2}}{0.693C} = \frac{19.5}{0.693 \times 235 \times 10^{-6}} = 120 \times 10^{3}\,(\Omega).$$

In this experiment the main errors come from the accuracy of the capacitance of the two electrolytic capacitors in series and from the error that may be caused by timing with the stopwatch. The error of the nominal value of an electrolytic capacitor is about 5%, while the error of the stopwatch may reach 0.2 s. These two items will affect the accuracy of the measured result.

In this experiment two $470\,\mu\mathrm{F}/10\,\mathrm{V}$ electrolytic capacitors are used, with their negative plates connected together and the two positive plates led out as the electrodes, forming one non-polar capacitor whose capacitance is $235\,\mu\mathrm{F}/10\,\mathrm{V}$.

Experiment 7.05 Measurement of the volt–ampere characteristic curve of a non-linear resistor

Measure the relation between the current I through a 2CW-type silicon Zener diode and the voltage U across its terminals, and give this curve.

[**Apparatus and materials available**]

A multirange voltmeter, a multirange ammeter, a galvanometer, a DC stabilised power supply, three resistance boxes, a slide-wire rheostat, the diode to be measured, a switch, graph paper, and connecting wires.

[**Requirements**]

(1) Draw the best measuring circuit.

(2) Measure the forward and reverse current values through the diode and the corresponding voltage values (the current measuring range is not to exceed ±10 mA).

(3) Plot the I–U curve on graph paper.

(4) Explain the features of this characteristic curve section by section.

Solution 7.05 The forward and reverse voltages of the 2CW diode are measured by the compensation method, so that the measurement uncertainty is smaller than that of the volt–ampere method. The particular measuring circuit is shown in Fig. 7.05. Note that the voltmeter is not connected directly to the diode: when K_1 and K_2 are closed, R_3 is adjusted until the

current of the galvanometer G is zero, and the voltage indicated by the voltmeter is then just the voltage across the diode. The advantage of this method is that the current of the diode is not used to deflect the meter, so the accuracy of the measurement is high.

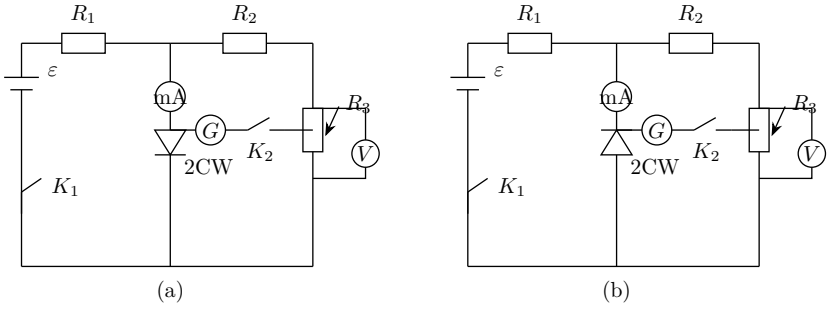


Fig. 7.05

The experimental data are shown in Tables 7.05-1 and 7.05-2.

Table 7.05-1 Forward I–U data of the diode under test

I/mA	0.050	0.100	0.200	0.500	1.00	2.00	3.00	4.00	7.00	10.00
U/V	0.545	0.575	0.600	0.625	0.650	0.675	0.680	0.690	0.700	0.710

Table 7.05-2 Reverse I–U data of the diode under test

I/mA	0.100	0.300	0.500	1.00	2.00	3.00	4.00	6.00	8.00	10.00
U/V	1.65	2.08	2.30	2.55	2.90	3.05	3.20	3.45	3.55	3.70

The above I–U curves are omitted. From the data tables it can be seen that when a forward current is passed through the diode, the current increases slowly with the voltage as long as the terminal voltage is less than 0.7 V, but once the voltage is greater than about 0.7 V the current increases abruptly. When a reverse current is passed through the diode, the current increases slowly with the voltage as long as the voltage across its terminals is less than 3.2 V, but once the voltage is greater than 3.2 V the current increases abruptly. This is the reverse characteristic of the Zener diode, that is, its voltage-stabilising characteristic.

Experiment 7.06 Measurement of the basic characteristics of a solar cell

A solar cell can absorb light energy (electromagnetic waves) and convert the energy of the absorbed photons into electrical energy; it is used in spacecraft, artificial satellites and various environment-friendly instruments, and is an important energy converter in machinery. This experiment requires the accurate measurement of some characteristic parameters of a solar cell, so that solar energy may be used more efficiently.

[Introduction of the basic terms]

(1) Short-circuit current I_{sc}: the output current of the solar cell when the external circuit is short-circuited, that is, when the load resistance is zero.

(2) Open-circuit voltage U_{oc}: the output voltage of the solar cell when the external circuit is open, that is, when the load resistance is infinite.

(3) The maximum output power P_m of the solar cell.

(4) The fill factor FF is defined as $P_m/(I_{sc} \cdot U_{oc})$; it is an important parameter representing the quality of a solar cell.

[**Apparatus and materials available**]

One 60 W tungsten filament lamp, two multimeters, one slide-wire rheostat, one optical bench, one solar cell with its holder (including leads), a number of connecting wires, two sheets of graph paper, a stabilised DC power supply, etc.

[**Requirements**]

(1) Measure the I–U characteristic of the solar cell under forward bias in complete darkness, and draw the measuring circuit.

(2) Measure the I–U characteristic of the solar cell under constant illumination, and draw the measuring circuit.

(3) Find the short-circuit current I_{sc}, the open-circuit voltage U_{oc} and the maximum output power P_m of the solar cell.

(4) Calculate the fill factor of the solar cell.

Solution 7.06 **1.** Place the solar cell in a dark box, and connect the DC supply, the slide-wire rheostat, the voltmeter, the ammeter and the solar cell as in Fig. 7.06(a); measure the I–U curve of the solar cell under forward bias.

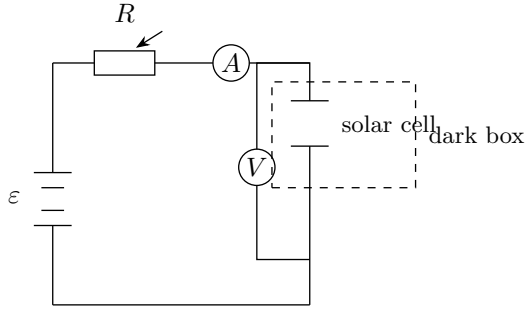


Fig. 7.06(a)

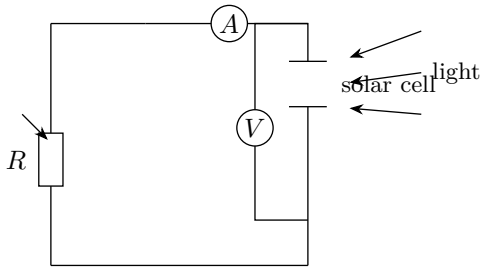


Fig. 7.06(b)

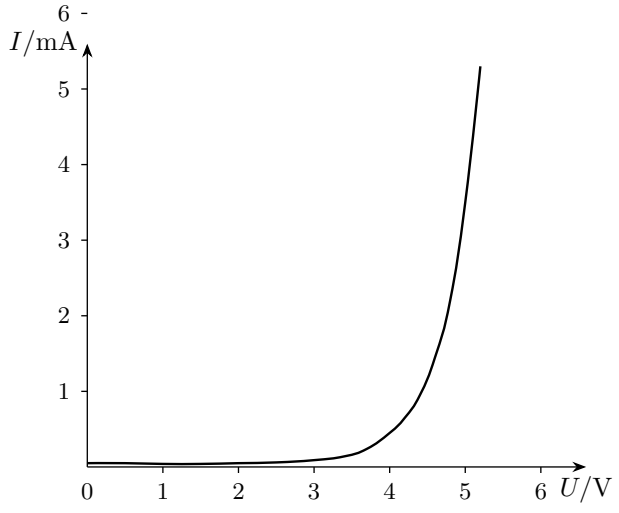
2. Connect the solar cell as a source into the circuit, as shown in Fig. 7.06(b). The solar cell is placed under constant illumination (a 60 W incandescent lamp is used for the illumination in this experiment); measure the current I through the load resistance and the output voltage U for different load resistances, calculate its output power P under different loads, and hence determine the load resistance R_m at the maximum output power P_m; obtain the short-circuit current I_{sc} at $U = 0$ and the open-circuit voltage U_{oc} at $I = 0$ from the I–U graph, and calculate the fill factor FF.

Experimental results:

Under completely dark conditions, the current I flowing through the solar cell and the voltage U across the solar cell were measured when a forward bias was applied, and the I–U graph shown in Fig. 7.06(c) was obtained. From the graph it can be seen that in the region below 4 V the

characteristic is non-linear, and that the I–U relation fitted by a curve satisfies an exponential relation. When the voltage is larger, owing to the influence of other factors, I and U are basically in a linear relation.

With no bias applied, the solar cell is placed under constant illumination. The solar cell is placed at a distance of 20 cm from a 60 W incandescent lamp.



I–U characteristic curve of the solar cell in complete darkness

Fig. 7.06(c)

For different load resistances R, the output power P of the solar cell is related to the load resistance R as shown in Fig. 7.06(d), and the relation between the output current I and the output voltage U of the solar cell is shown in Fig. 7.06(e). From Fig. 7.06(e) it is found that the short-circuit current is $I_{sc} = 1.85\,\text{mA}$ and the open-circuit voltage is $U_{oc} = 3.60\,\text{V}$; from Fig. 7.06(d) it is found that when the load resistance is $R = 1.8\,\text{k}\Omega$ the maximum output power of the solar cell is $P_m = 4.37\,\text{mW}$. Finally the fill factor is calculated to be $FF = 65.6\%$.

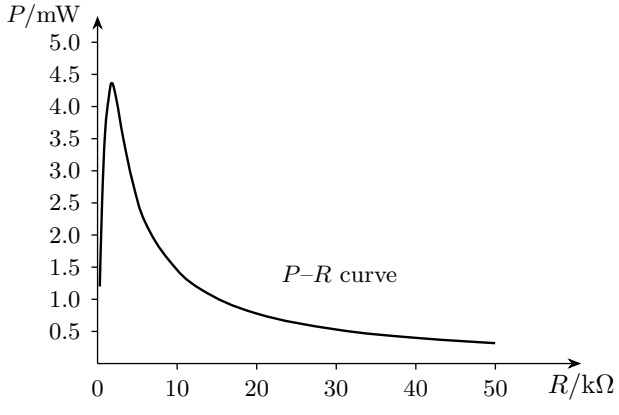


Fig. 7.06(d)

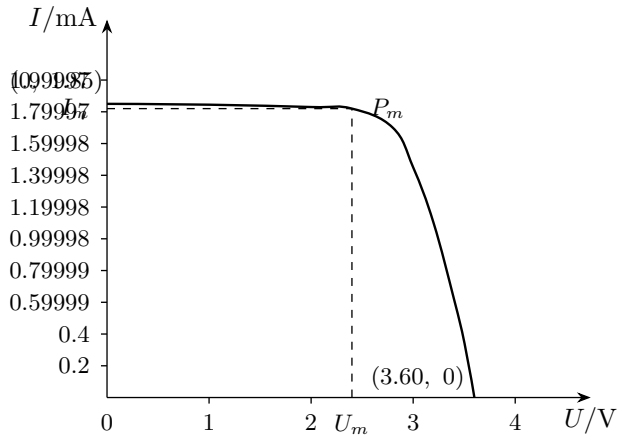


Fig. 7.06(e)

Comment. In measuring the volt–ampere characteristic curves of transistor devices such as PN junctions and solar cells, if pointer-type voltmeters and ammeters are used, then in order to eliminate the error introduced by the insufficiently large internal resistance of the pointer voltmeter, the best circuit measures the voltage by the compensation method. When a digital voltmeter is used, however, to measure the volt–ampere characteristic curve of a non-linear resistor, since the internal resistance of a digital voltmeter is mostly between $10^6\,\Omega$ and $10^8\,\Omega$, the compensation method need not be used; instead, a sampling resistor of suitable resistance may be connected in series in the circuit. The digital voltmeter can measure directly the voltage across the non-linear resistor and at the same time the voltage across the sampling resistor; if the resistance of the sampling resistor is known, the volt–ampere characteristic of the non-linear resistor can then be obtained.

Experiment 7.07 Measurement of the internal resistance of a microammeter with a DC bridge, and its conversion into a voltmeter

[Apparatus and materials available]

Three resistance boxes, one DC ammeter of range $100\,\mu$A, one variable resistor (0–22 kΩ), one class-0.5 standard voltmeter (range 0–3 V), two switches, two type-A cells (1.5 V), and a number of connecting wires.

[Requirements]

(1) Draw the circuit diagram for measuring the internal resistance of a meter with a DC bridge.

(2) Write down the points for attention, the measurement steps and the calculation formula.

(3) Measure the internal resistance of the meter.

(4) Convert the $100\,\mu$A microammeter into a voltmeter of range 3 V.

(5) Find the percentage error when the reading of the converted meter is 2.40 V.

Solution 7.07 The circuit for measuring the internal resistance of the meter is shown in Fig. 7.07(a). A variable resistor is connected with one type-A cell to form a current-limiting circuit whose output current can be varied continuously; three resistance boxes, the ammeter to be measured and a switch are connected by wires into a bridge circuit. At the beginning, adjust the variable resistor so that the output current is rather small, and take the resistances of the bridge arms as 2–3 kΩ. Close and then break S_1, and observe whether the pointer of the meter deflects; if it does, adjust R_4 so that the pointer of the meter does not move when S_1 is closed and when it is opened. This shows that the potentials of the ends a and b of the bridge are equal. Then adjust the output current of the source to be larger, and continue the above

process, observing whether the meter in the bridge stays still when S_1 is closed and opened. If the pointer does not move, the bridge is balanced. At this time R_1, R_2, R_x and R_4 satisfy

$$\frac{R_1}{R_2} = \frac{R_x}{R_4}.$$

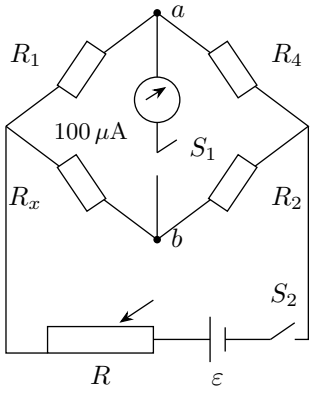


Fig. 7.07(a)

Hence

$$R_x = \frac{R_1 \cdot R_4}{R_2}.$$

This bridge is usually called a Thomson DC bridge.

Experimental data: when the bridge is balanced, $R_1 = 2000\,\Omega$, $R_2 = 2000\,\Omega$, $R_4 = 1950\,\Omega$; then the internal resistance of the meter is $R_x = 1950\,\Omega$.

Connecting an additional resistance R_M in series with the $100\,\mu$A ammeter of known internal resistance R_g forms a voltmeter whose range is U, and then

$$\frac{U}{R_M + R_g} = \frac{U_g}{R_g}.$$

The design requires $U = 3.000\,\text{V}$, $R_g = R_x = 1950\,\Omega$ and $U_g/R_g = 100\,\mu\text{A}$, so that

$$R_M = \frac{U}{100 \times 10^{-6}} - R_g = \frac{3.000}{100 \times 10^{-6}} - 1950 = 28.1 \times 10^3\,(\Omega).$$

Connecting the resistance R_M in series as in Fig. 7.07(b) forms a voltmeter of range 3 V. The variable resistor, a type-A cell, the standard voltmeter and the self-made voltmeter are connected into the circuit of Fig. 7.07(c). The variable resistor in the voltage-divider circuit is adjusted; when the reading of the standard voltmeter is 2.450 V, the reading of the self-made voltmeter is 2.40 V. Then the percentage difference between the readings of the self-made meter and of the standard meter is 2%.

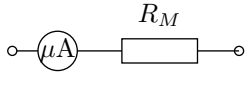


Fig. 7.07(b)

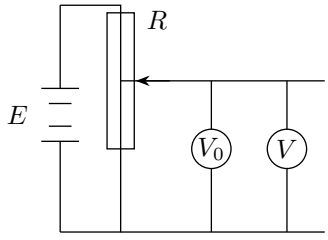


Fig. 7.07(c)

Experiment 7.08 Conversion of a meter head into a multirange ammeter

Try to convert a meter head of internal resistance 360 Ω and range 500 μA into a multirange ammeter, with current ranges $I_1 = 100\,\text{mA}$, $I_2 = 10\,\text{mA}$ and $I_3 = 1\,\text{mA}$.

[**Apparatus and materials available**]

One DC ammeter (range 0–500 μA, internal resistance $R_g = 360\,\Omega$), three resistance boxes, a number of connecting wires, and one multimeter (used as a standard meter).

[**Requirements**]

(1) Draw the circuit diagram of the converted multirange ammeter.

(2) Calculate the values of the shunt resistances R_1, R_2, R_3.

(3) Connect the 500 μA meter head and the shunt resistances with wires to form a multirange ammeter.

(4) Use the 1 mA range of the multimeter as the standard value to calibrate the 1 mA range of the converted ammeter.

Solution 7.08 1. Design of the current ranges and calculation of the shunt resistances

Connecting a shunt resistance R_s in parallel with the ammeter can expand the range I_g to I, that is,

$$(I - I_g) \cdot R_s = I_g \cdot R_g,$$

$$I = I_g \frac{R_g + R_s}{R_s} = N I_g.$$

Hence, after a shunt resistance R_s is connected in parallel with the ammeter, the current range is enlarged by a factor $N = (R_g + R_s)/R_s$. Conversely, if the enlarged range I is known, the value of the required shunt resistance R_s can be calculated, and from the above formula one obtains

$$R_s = \frac{I_g R_g}{I - I_g} = \frac{R_g}{N - 1}.$$

The current ranges of a multimeter usually comprise several ranges, and the so-called closed shunt circuit shown in Fig. 7.08(a) is commonly adopted. In the figure the resistances R_1, R_2, R_3 are connected in series with one another and then connected in parallel with the meter head; I_1, I_2, I_3 are three different extended ranges, among which the range I_3 has the smallest current.

The steps for calculating the shunt resistances are as follows.

(1) According to the range I_3 of smallest current, first calculate the total resistance R_s of the shunts. Because

$$R_s(I_3 - I_g) = R_g I_g,$$

therefore

$$R_s = \frac{R_g I_g}{I_3 - I_g}.$$

(2) Calculate the resistance values of the individual shunts. For the range I_1, the shunt resistance is R_{s1}, and R_{s1} is in parallel with $(R_g + R_s - R_{s1})$, so

$$R_{s1}(I_1 - I_g) = (R_g + R_s - R_{s1})I_g,$$

and hence

$$R_{s1} = \frac{I_g(R_g + R_s)}{I_1}.$$

From the expression for R_s there is also

$$I_g(R_g + R_s) = R_s I_3,$$

so R_{s1} can be expressed by the simpler formula

$$R_{s1} = \frac{R_s I_3}{I_1}.$$

In the same way, for the shunt resistance R_{s2} of the range I_2 one obtains

$$R_{s2} = \frac{R_s I_3}{I_2}.$$

(3) Calculation of the values of the elements. From Fig. 7.08(a),

$$R_1 = R_{s1}, \qquad R_2 = R_{s2} - R_{s1}, \qquad R_3 = R_s - R_{s2}.$$

Given $R_g = 360\,\Omega$, $I_g = 500\,\mu\text{A}$, $I_3 = 1\,\text{mA}$, $I_2 = 10\,\text{mA}$ and $I_1 = 100\,\text{mA}$, then

$$R_s = \frac{I_g R_g}{I_3 - I_g} = \frac{360 \times 0.5}{1 - 0.5} = 360\,(\Omega),$$

$$R_1 = \frac{R_s I_3}{I_1} = \frac{360 \times 1}{100} = 3.6\,(\Omega),$$

$$R_2 = \frac{R_s I_3}{I_2} - R_1 = \frac{360 \times 1}{10} - 3.6 = 32.4\,(\Omega),$$

$$R_3 = \frac{R_s I_3}{I_3} - (R_1 + R_2) = \frac{360 \times 1}{1} - (3.6 + 32.4) = 324\,(\Omega).$$

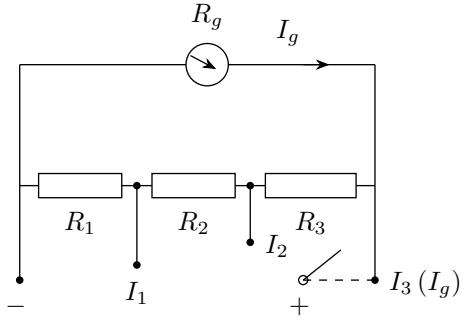


Fig. 7.08(a)

(4) The converted 1 mA ammeter is connected in series with the standard meter (the 1 mA range of the multimeter) to form the circuit of Fig. 7.08(b). On the 1 mA range, 6 to 7 points must be checked, and these must include the two points 0 and full scale. During the calibration the current in the circuit is adjusted so that the reading of the meter under calibration is an integer, and the reading of the standard meter is recorded at the same time; finally the calibration value of the meter under calibration is obtained: reading of the standard meter minus reading of the meter to be calibrated equals the calibration value.

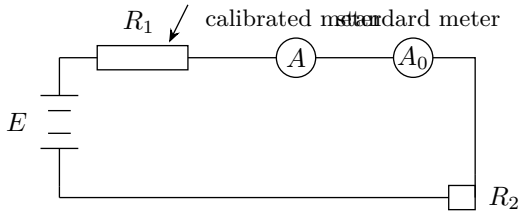


Fig. 7.08(b)

Experiment 7.09 Conversion of a microammeter into an ohmmeter

Try to convert a 200 μA microammeter into an ohmmeter with a mid-scale resistance of 1500 Ω.

[**Apparatus and materials available**]

One 200 μA DC ammeter (internal resistance unknown), a 1.5 V (DC) supply, three resistance boxes, two single-pole switches, a number of connecting wires, and one sheet of rectangular coordinate paper.

[**Requirements**]

(1) Measure the internal resistance R_g of the microammeter.

(2) Draw the circuit diagram of the ohmmeter; it is required that the error caused by a 20% decrease of the electromotive force of the supply be no greater than 5%.

(3) Build the ohmmeter and make a calibration curve.

Solution 7.09 1. Measurement of the internal resistance R_g of the microammeter

From the given experimental conditions it is known that the internal resistance of the 200 μA ammeter is measured with the circuit shown in Fig. 7.09(a), the bridge method being used to measure the internal resistance of the ammeter; the measurement procedure is omitted. Experimental result: the internal resistance of the 200 μA ammeter is $R'_g = 975\,\Omega$.

2. Drawing of the ohmmeter circuit

The mid-scale resistance is the value of the resistance measured when the pointer of the ohmmeter points to the centre of the scale of the meter head, that is, it is equal to the total internal resistance of the ohmmeter; at this time the measurement error of the ohmmeter is the smallest.

In this problem the mid-scale resistance of the ohmmeter is required to be 1500 Ω, that is, when the resistance to be measured is $R_x = 0$ (the two test rods short-circuited) the pointer of the meter head has a full-scale deflection, and the current in the circuit at this time is

$$I_g = \frac{E}{1500} = \frac{1.50}{1500} = 1.00\,(\text{mA}).$$

Therefore the 200 μA meter head must first be converted into an ammeter of range 1.00 mA.

The circuit for the conversion into an ammeter of range 1 mA is shown inside the dashed frame in Fig. 7.09(a). By Ohm's law,

$$(I_g - I'_g) \cdot R_1 = I'_g \cdot R'_g,$$

so the shunt resistance R_1 for the conversion of the meter I'_g is

$$R_1 = \frac{I'_g \cdot R'_g}{I_g - I'_g} = \frac{200 \times 10^{-6} \times 975}{800 \times 10^{-6}} = 243.8\,(\Omega).$$

Let the resistance of R'_g and R_1 connected in parallel be R_T; then R_T is

$$R_T = \frac{I'_g \cdot R'_g}{I_g} = \frac{200 \times 10^{-6} \times 975}{1000 \times 10^{-6}} = 195\,(\Omega).$$

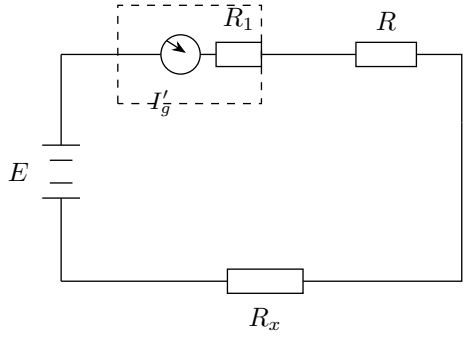


Fig. 7.09(a)

So the series resistance is

$$R = 1500 - R_T = 1500 - 195 = 1305\,(\Omega).$$

If the voltage of the DC supply falls to 1.2 V and the test rods of the ohmmeter are short-circuited, the indication of the ammeter also falls by 20%; if at this time the resistance of the resistance box R_1 is increased to R'_1, the current indication I'_g of the ammeter can still be brought to full scale. The value of R'_1 is calculated below, and the mid-scale resistance of the ohmmeter after zero adjustment is also calculated.

When the ammeter I'_g is at full scale, the voltage is

$$U_{ab} = I'_g \cdot R'_g = 200 \times 10^{-6} \times 975 = 0.195\,(\mathrm{V}),$$

and at this time the current through the resistance R is

$$I' = \frac{E - U_{ab}}{R} = \frac{1.2 - 0.195}{1305} = 0.770\,(\mathrm{mA}).$$

The value of the parallel resistance R'_1 can then be calculated as

$$R'_1 = \frac{U_{ab}}{I' - I'_g} = \frac{0.195}{(0.770 - 0.200) \times 10^{-3}} = 342\,(\Omega).$$

The parallel resistance of R'_1 and R'_g is

$$\frac{U_{ab}}{I'} = \frac{0.195}{0.770 \times 10^{-3}} = 253.2\,(\Omega).$$

Hence, when the supply voltage has fallen to 1.2 V and zero adjustment has been made by short-circuiting (full-scale indication of the ammeter), the mid-scale resistance of the ohmmeter

is $1305 + 253.2 = 1558.2\,\Omega$. Compared with the mid-scale resistance $1500\,\Omega$ for the original 1.5 V supply, the percentage difference is 3.9%, which is less than 5%.

3. Calibration of the ohmmeter

The circuit of Fig. 7.09(b) is used to calibrate the self-made ohmmeter. On the scale plate of the self-made ohmmeter, the relation between the panel current values and the corresponding resistance values can be marked according to the supply voltage 1.5 V and the mid-scale resistance $1500\,\Omega$ (calculated by Ohm's law); in this way it is assembled into an ohmmeter. A resistance box is used to vary the resistance to be measured, and the relation between the indication of the ohmmeter and the standard value (the reading of the resistance box) is found in order to calibrate the self-made ohmmeter; 10 to 12 data points are required, and these must include the two points of zero resistance and of the mid-scale resistance. A calibration curve is drawn on rectangular coordinate paper, that is, a curve with the indication of the ohmmeter as the independent variable and the correction value ΔR as the dependent variable.

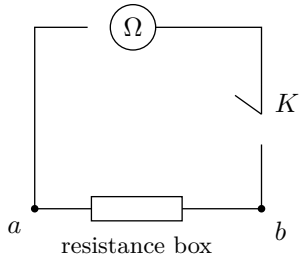


Fig. 7.09(b)

Experiment 7.10 Accurate measurement of the internal resistance of a DC microammeter by two bridge-balance methods

[**Apparatus and materials available**]

Three resistance boxes R_1, R_3, R_4 of certain error (none of which can be used as an accurate resistance), one resistance box R_2 of accurate value, a 1.5 V stabilised supply, two single-pole switches, one DC ammeter of internal resistance about $2000\,\Omega$ and range $100\,\mu$A, and one slide-wire rheostat.

[**Requirements**]

(1) Write down the experimental method and the calculation formula.

(2) Draw the experimental circuit diagram.

(3) Measure the experimental result (the internal resistance R_g of the meter).

Solution 7.10

Method 1

1. The method of Solution 7.07 (this method is often called the Thomson bridge method) is used to measure the internal resistance R_g of the microammeter; the circuit is shown in Fig. 7.10(a), that is, one arm of the bridge is connected to the microammeter and the other three arms to R_1, R_3 and R_4, a switch is connected across the diagonal, and the bridge is adjusted to balance. Since the accurate resistance values of R_1, R_3 and R_4 are not known, R_g cannot be calculated.

2. The substitution method is then used to measure R_g accurately. Since the resistance of the resistance box R_2 is accurate, it is used to replace the microammeter, the microammeter serving as the null detector of the bridge, as shown in Fig. 7.10(b). R_1, R_3 and R_4 are all kept unchanged, and R_2 is adjusted so that the microammeter indicates zero; the value of R_2 at this time is then the internal resistance R_g of the microammeter.

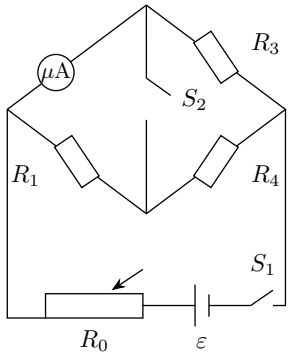


Fig. 7.10(a)

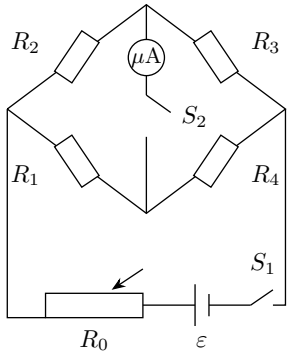


Fig. 7.10(b)

Method 2

The Thomson bridge method is used to measure R_g, as shown in Fig. 7.10(a). After the bridge has been balanced, R_2 is used in turn to replace R_1, R_3 and R_4, and the bridge is balanced again so as to obtain their accurate values R_1', R_3' and R_4'; then, using the formula

$$\frac{R_g}{R_1} = \frac{R_3'}{R_4'},$$

one obtains

$$R_g = \frac{R_3'}{R_4'} \cdot R_1'.$$

The experimental data are omitted.

Experiment 7.11 Study of the characteristics of half-wave rectification

[Apparatus and materials available]

A signal generator, a dual-trace oscilloscope, a rectifying diode, a multimeter, and a number of connecting wires.

[Requirements]

(1) Draw the circuit diagram of a half-wave rectifier.

(2) Observe with the oscilloscope the waveform of the input voltage and the waveform of the voltage across the load; draw the waveforms and calculate the frequency and the voltage amplitude.

(3) Measure the voltage across the load with the multimeter, compare it with the voltage value measured with the oscilloscope, and explain the relation between them.

Solution 7.11 **1.** The circuit diagram of the half-wave rectifier is shown in Fig. 7.11(a). In the figure CH_1 is the Y_1-axis input of the oscilloscope (channel 1), and CH_2 is the Y_2-axis input of the oscilloscope (channel 2).

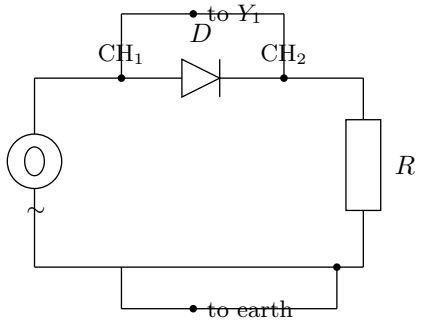


Fig. 7.11(a)

2. With CH_1 the input-voltage wave is observed, and the sinusoidal waveform shown in Fig. 7.11(b)(1) is seen; with CH_2 the waveform of the voltage across the load is observed, and the half-wave waveform shown in Fig. 7.11(b)(2) is seen.

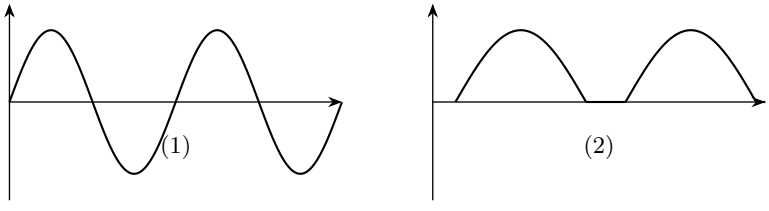


Fig. 7.11(b)

Measurement of the frequency f of an unknown sine wave with the oscilloscope.

If on the oscilloscope screen a length l (in cm) contains n complete periods of the sine wave, and the sweep setting of the oscilloscope is m (in ms/cm), then the period T of the sine wave to be measured is

$$T = \frac{l \cdot m}{n}.$$

Hence the frequency of the sine wave to be measured is

$$f = \frac{1}{T} = \frac{n}{l \cdot m}.$$

For example, $n = 4$ complete periods of the sine wave, $l = 8.00\,\text{cm}$, $m = 10.0\,\text{ms/cm}$. Then

$$f = \frac{n}{l \cdot m} = \frac{4}{8.00 \times 10.0}\,(\text{kHz}) = 50.0\,(\text{Hz}).$$

The method of measuring the amplitude of a sine wave with the oscilloscope is as follows: read from the oscilloscope screen the peak-to-peak value $l_{\text{p–p}}$ of the sine wave and the scale value of the y-axis (in V/cm); then the amplitude of the sine wave ($U_{\text{peak}} = U_{\text{p–p}}/2$) is

$$U_{\text{peak}} = (\text{scale value}) \times \frac{l_{\text{p–p}}}{2}.$$

[**Note**] When reading the sweep setting of the oscilloscope and the scale value of the Y-axis, the fine-adjustment knob must in both cases be turned to its maximum position.

3. The amplitude U_{peak} of the half-wave rectified voltage across the load is measured with the oscilloscope, while the effective value U_{rms} of the half-wave rectified voltage is measured with the AC range of the multimeter; these two satisfy

$$U_{\text{rms}} = \frac{U_{\text{peak}}}{2\sqrt{2}}.$$

Comment. An oscilloscope can display the waveform of an alternating current correctly. In the experiment one should master the principle of sweep synchronisation of the oscilloscope, its use and the method of reading it: for instance, how to display one stable alternating-current waveform, how to measure with the oscilloscope the period (or frequency) of a waveform to be measured, and how to measure the amplitude of an alternating-current waveform. These are all basic uses and experimental methods of the oscilloscope, and these essential points must be mastered in study.

Experiment 7.12 Measurement of the capacitance of a given capacitor

[**Apparatus and materials available**]

An oscilloscope, a signal generator (using a square wave), a resistance box, the capacitor to be measured, graph paper, a ruler, a switch and connecting wires.

[**Requirements**]

(1) State the measuring principle briefly, and draw the measuring circuit.

(2) Adjust the oscilloscope to its working state, and observe the waveform so that it is displayed stably on the fluorescent screen.

(3) Give the experimental data.

(4) Find the capacitance to be measured by a graphical method.

Solution 7.12 **1.** It is known that when a capacitor C discharges through a resistor R connected in series with it, the voltage on the capacitor and the discharge time t satisfy the relation

$$U_C = U_0 \mathrm{e}^{-t/RC},$$

where U_0 is the voltage on the capacitor at time $t = 0$. Taking logarithms of both sides of this functional equation gives

$$\ln U_C(t) = \ln U_0 - \frac{t}{RC}.$$

The relation $U_C(t)$–t is measured, the straight line $\ln U_C(t)$–t is plotted, and its slope k is obtained; then

$$C = -\frac{1}{R \cdot k}.$$

2. The measuring circuit is shown in Fig. 7.12(a).

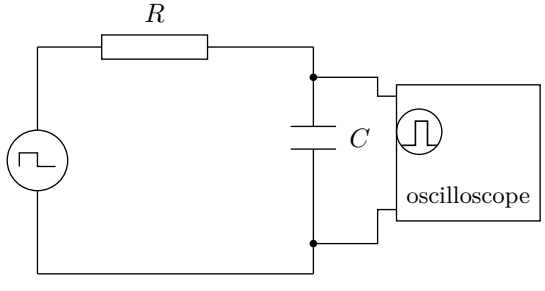


Fig. 7.12(a)

3. The waveform of $U_C(t)$ is shown in Fig. 7.12(b).

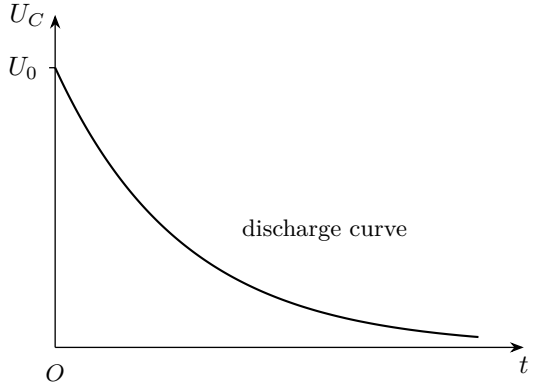


Fig. 7.12(b)

4. The data of the relation $U_C(t)$–t are measured, at least 6 points being taken; the method of recording the readings of $U_C(t)$ and t is the method described in Solution 7.11 for reading the Y-axis voltage value and the X-axis time value.

5. The straight-line graph $\ln U_C(t)$–t is plotted, its slope k is found, and then the value of the capacitance C is obtained from the known value of the resistance R.

Comment. The relation $U_C(t)$–t is measured by the discharge method, and from the straight line $\ln U_C$–t the value of RC (or of C) is found by calculating the slope of the line; this is a method often used in experiment. In this experiment, however, the period T of the input square wave must satisfy $T \geqslant 10RC$, that is,

$$\frac{T}{2} \geqslant 5RC,$$

for the discharge to be complete. That is, at $t = 0$, $U_C(t) = U_0$, and at $t = T/2$, $U_C(t) = 0$. In an examination, if the problem states that only one measurement is to be made, one may also measure directly the half-life $T_{1/2}$ in the discharge curve: from

$$RC = \frac{T_{1/2}}{\ln 2} = \frac{T_{1/2}}{0.693}$$

the value of RC (or of C) is obtained.

Experiment 7.13 Measurement of the inductance of a given inductor by two different methods

[Apparatus and materials available]

An oscilloscope, a signal generator (using a sine wave), a resistance box, a capacitance box, the inductor to be measured, and connecting wires.

[Requirements]

(1) Draw the circuit diagrams of two different methods of measuring an inductance.

(2) Write down the experimental method and the calculation formulas.

(3) Measure the inductance of the inductor to be measured and its loss resistance.

Solution 7.13 **1.** The circuit for measuring the inductance of the unknown inductor by the RLC series resonance method is shown in Fig. 7.13(a); the circuit for measuring the inductance of the unknown inductor by the vector-synthesis method (that is, the AC volt–ampere method) is shown in Fig. 7.13(b).

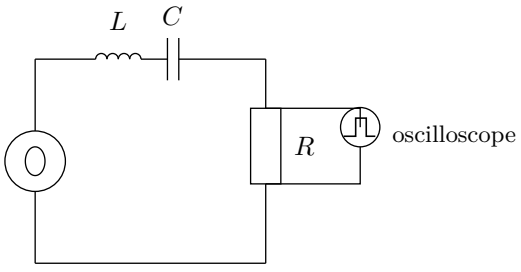


Fig. 7.13(a)

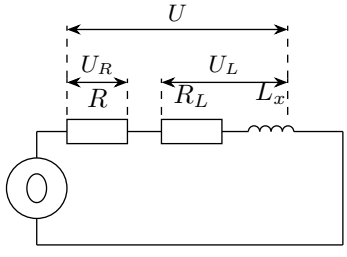


Fig. 7.13(b)

2. Measurement of the inductance and of the loss resistance by the series resonance method.

In Fig. 7.13(a) the earth lead of the oscilloscope must be connected to the earth lead of the signal generator, so as to prevent the short-circuiting of a component or of the source. In the experiment the capacitance of the capacitance box may be taken in the range $0.1\,\mu$F–$0.5\,\mu$F, and the resistance R is generally taken in the range $10\,\Omega$–$30\,\Omega$, so as to ensure a comparatively large Q of the series circuit and a comparatively small experimental error. The output frequency f of the signal generator is changed so that the voltage U_R across the resistor R reaches a maximum; the circuit is then in resonance. We have

$$f_0 = \frac{1}{2\pi\sqrt{LC}}.$$

Knowing the capacitance C and the resonant frequency f_0, the unknown inductance can be found as

$$L = \frac{1}{4\pi^2 f_0^2\, C}.$$

The loss resistance R_L of the inductor can be found by the following method. When the RLC series circuit is in resonance, the output voltage U of the signal generator and the voltage U_R across the resistor are measured. Since the circuit is then purely resistive, it satisfies

$$\frac{U_R}{R} = \frac{U}{R + R_L},$$

so

$$R_L = \frac{U}{U_R} \cdot R - R.$$

The value of R_L so obtained is the loss resistance R_L of the inductor at the frequency f_0.

3. Measurement of the inductance and of the loss resistance by the AC volt–ampere method.

The inductance of the coil is measured by the AC volt–ampere method using Fig. 7.13(b). The resistor R and the inductor L to be measured are connected in series in an alternating-current

circuit, in which the resistance value of R is known. A practical inductor may be regarded as an ideal inductance L_x and a loss resistance R_L in series. With the oscilloscope the voltages U_R and U_z across the resistor R and across the inductor L to be measured, and the total voltage U_T, are measured separately; using the vector diagram of Fig. 7.13(c), one obtains

$$(U_R + IR_L)^2 + (IX_L)^2 = U_T^2 \tag{1}$$

$$(IR_L)^2 + (IX_L)^2 = U_z^2 \tag{2}$$

where I is the current, X_L the reactance, and R_L the loss resistance of the inductor. Note that when U_R is measured the resistor and the inductor must be interchanged, to prevent a short circuit.

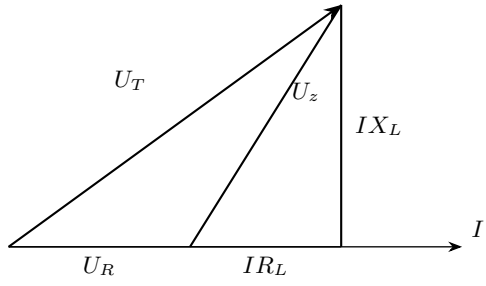


Fig. 7.13(c)

Solving the two equations above and eliminating IX_L, one obtains

$$R_L = \frac{U_T^2 - U_R^2 - U_z^2}{2U_R I}.$$

Substituting R_L into (1) and (2) gives

$$X_L = \frac{\sqrt{[U_T^2 - (U_R - U_z)^2]\,[(U_R + U_z)^2 - U_T^2]}}{2U_R I} \tag{3}$$

in which I can be calculated from U_R/R. Then, from X_L and the frequency f of the alternating voltage, the inductance can be calculated, namely

$$L = \frac{X_L}{2\pi f}. \tag{4}$$

Comment. The series resonance method and the AC volt–ampere method for measuring the inductance of a coil are two of the basic methods of measuring inductance. In measuring inductance by the resonance method attention must be paid to the Q of the circuit: if Q is too small, a comparatively large measurement error will be introduced, and therefore the series resistance R should not be chosen too large. In measuring inductance by the AC volt–ampere method, a suitable sine-wave frequency must be chosen so that the values of U_z and U_R are approximately equal (the same is true if the resistance R is changed while the frequency is kept constant), and in the experiment U_T should be as close as possible to the full scale of the AC voltmeter. Only in this way can the inductance and the loss resistance be measured with comparatively high accuracy. The two methods described above for measuring the inductance and the loss resistance of a coil are often used in competitions.

Experiment 7.14 Measurement of the DC resistance, the AC impedance and the inductance of an inductor

Design a circuit to measure the DC resistance of a certain inductor and its AC impedance and inductance when a sinusoidal current is passed through it.

[**Apparatus and materials available**]

A low-frequency signal generator, a dry cell, a resistance box, two MF30 multimeters (with no battery inside), the inductor to be measured (with a ceramic dielectric), connecting wires, a switch, etc.

[**Requirements**]

(1) Draw the experimental circuit, and measure the DC resistance R_L of the inductor.

(2) Measure the AC impedance Z of the inductor when a sinusoidal current of frequency $f = 50.0\,\text{Hz}$ is passed through it.

(3) Calculate the inductance L of the inductor.

Solution 7.14 The circuit for measuring the DC resistance R_L (the loss resistance of the inductor) is shown in Fig. 7.14(a); during the experiment the multimeter should be set to the DC voltage range. The circuit shown in Fig. 7.14(b) is used to measure the AC impedance Z of the inductor, but before measuring, the multimeters must be set to the AC voltage range.

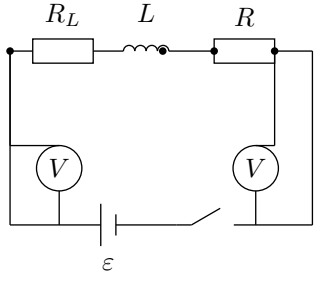


Fig. 7.14(a)

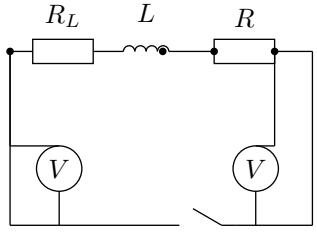


Fig. 7.14(b)

1. Measurement of the DC resistance R_L

Take $R = 35.00\,\Omega$; the measured values are $U_1 = 0.705\,\text{V}$ and $U_2 = 0.680\,\text{V}$.

$$\therefore \quad R_L = \frac{U_1}{I} = \frac{U_1}{U_2} \cdot R = \frac{0.705}{0.680} \times 35.00 = 36.3\,(\Omega).$$

2. Measurement of the AC impedance Z of the inductor

The frequency of the sinusoidal alternating current is $f = 50.00\,\text{Hz}$, $R = 50.00\,\Omega$; the measured values are $U_1 = 9.00\,\text{V}$ and $U_2 = 9.35\,\text{V}$.

$$Z = \frac{U_1}{I} = \frac{U_1}{U_2} \cdot R = 50.00 \times \frac{9.00}{9.35} = 48.1\,(\Omega).$$

3. Since $U^2 = U_{R_L}^2 + U_L^2$, we have $Z^2 = R_L^2 + (\omega L)^2$. At the low frequency $f = 50.0\,\text{Hz}$ the loss resistance R_L of the inductor is approximately equal to the loss resistance in the DC case.

$$\text{Hence,}\quad L = \frac{\sqrt{Z^2 - R_L^2}}{\omega} = \frac{\sqrt{48.1^2 - 36.3^2}}{2\pi \cdot 50.00} = 0.100\,(\text{H}).$$

4. From Fig. 7.14(c) it can be seen that, when a sinusoidal alternating current passes through the inductor, the phase angle α between the voltage across the inductor and the current through it is

$$\cos\alpha = \frac{R_L}{Z} = \frac{36.3}{48.1} = 0.7547,$$

so that $\alpha = 41.0°$.

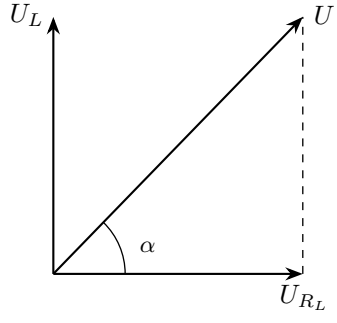


Fig. 7.14(c)

Experiment 7.15 Determination of the relative permittivity of the insulating medium between the two conductors of a coaxial cable

A coaxial cable is a transmission line commonly used in industry and in research laboratories. The relative permittivity of the medium between the conductors of a coaxial cable (the conductors consisting of a single solid wire and a copper braid) determines the capacitance formed by the two conductors. It is now required to determine experimentally the relative permittivity ε_r of this cable.

[**Apparatus and materials available**]

A low-frequency signal generator, an oscilloscope, a resistance box, a standard inductor, a section of the cable to be tested (about 1 m long, which may be cut as required), scissors, a ruler, a micrometer, connecting wires, and 2 sheets of graph paper.

[**Requirements**]

(1) Write down the test scheme or the experimental steps.

(2) Draw the experimental circuit.

(3) Measure the necessary data (repeated measurements are required).

(4) Process the data.

(5) Calculate the value of the relative permittivity ε_r of the insulating medium.

Hint 7.15

1. The relative permittivity of a material is $\varepsilon_r = C_x/C_0$, where C_x is the capacitance of the capacitor when it is filled with a certain medium, and C_0 is the capacitance of the capacitor in vacuum.

2. For the coaxial cable to be tested, neglecting the edge effects at its two ends, the capacitance C_0 when the medium is vacuum can be calculated by the following formula:

$$C_0 = \frac{2\pi\varepsilon_0 l}{\ln \dfrac{b}{a}},$$

where a is the inner radius and b the outer radius of the cable. The structure of the cable is shown in the figure.

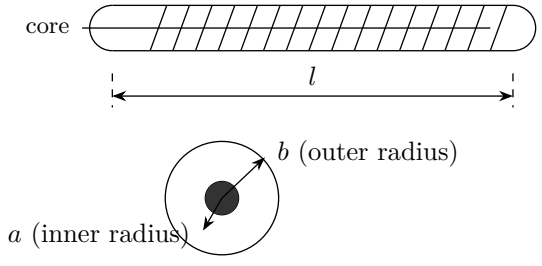


Fig. 7.15(a)

The permittivity of vacuum is $\varepsilon_0 = 8.90 \times 10^{-12}\,\mathrm{C}^2/(\mathrm{N}\cdot\mathrm{m}^2)$.

3. In the experiment, the distributed capacitance between the leads and the distributed capacitance in the standard inductor, etc., form a background capacitance C'. When the relative positions and shapes of the leads are unchanged, this value is a constant. In the experiment the influence of this value on the measurement result must be eliminated.

Solution 7.15 1. Test scheme and experimental steps.

(1) The key of this experiment is to obtain the result for C_x; the measuring method may be the RLC series resonance method. With scissors strip the centre conductor and the outer shielding braid of the cable, then connect them in series with R and L and connect the combination to the output terminals of the low-frequency signal generator; by changing the frequency, measure the resonant frequency f_0 of this series circuit.

(2) At the other end of the cable cut off 5–10 cm at a time, and measure the resonant frequency corresponding to each different length of cable.

(3) From the formula $f_0 = 1/(2\pi\sqrt{LC})$, find the corresponding capacitance C values.

(4) Plot the straight-line graph of C against l, and from it obtain the value C' at $l = 0$ (the intercept); C' is the background capacitance.

(5) $C_x = C - C'$ is then the capacitance of the cable for each different length.

(6) Use the micrometer to measure the values of a and b of the cable, and the ruler to measure the value of l; from the formula for the capacitance of a coaxial cable obtain the value of C_0.

(7) From the values of C_x and C_0 measured for cables of different lengths, the result for ε_r can be calculated.

2. The experimental circuit is shown in the figure.

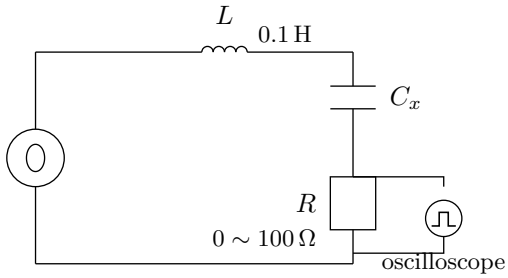


Fig. 7.15(b)

3. The measured experimental data are as follows.
$l = 1.000\,\text{m}$, $a = 0.740\,\text{mm}$, $b = 2.200\,\text{mm}$.

Table 7.15-1

l/m	1.000	0.900	0.800	0.700	0.600	0.500
f_{01}/Hz	30 430	30 970	31 660	32 370	33 080	33 880
f_{02}/Hz	30 440	30 940	31 640	32 380	33 050	33 840
f_{03}/Hz	30 450	30 950	31 640	32 300	33 100	33 900
$\overline{f}_0$/Hz	30 440	30 953	31 647	32 350	33 080	33 873

4. Data processing
(1) The intercept C' is obtained graphically: $C' = 166.0\,\text{pF}$. The C–l graph is omitted.
(2) The values of C_0 for the different lengths l are calculated.

Table 7.15-2

l/m	1.000	0.900	0.800	0.700	0.600	0.500
$C_0/10^{-12}\,\text{F}$	51.3	46.2	41.0	35.9	30.8	25.6

(3) For $l = 1.000$, 0.900, 0.800, 0.700, 0.600 and $0.500\,\text{m}$, the value C' is subtracted from the measured capacitance C, and the value C_x is then obtained.

Table 7.15-3

l/m	1.000	0.900	0.800	0.700	0.600	0.500
$C/10^{-12}\,\text{F}$	276.7	264.7	253.2	242.3	231.8	221.0
$C_x/10^{-12}\,\text{F}$	110.7	98.7	87.2	76.3	65.8	55.0

5. Calculation of the result.

Table 7.15-4

l/m	1.000	0.900	0.800	0.900	0.700	0.600	0.500
ε_r	2.16	2.14	2.13	2.13	2.13	2.14	2.15

$$\varepsilon_r = \frac{C_x}{C_0}; \qquad \overline{\varepsilon}_r = 2.14.$$

Comment. The content learnt in this experiment is quite rich; it contains several important experimental items. 1. Measurement of capacitance by the resonance method. If the object to be measured is a small

capacitance (a capacitance of 1×10^2 pF to 1×10^3 pF), the resonant frequency may then be greater than 1×10^4 Hz; if the experimenter makes a correct analysis and estimate, the resonance point can be found comparatively quickly. 2. The distributed capacitance of an inductor means that in a practical coil the potential difference existing between one turn and another also forms a capacitance, so that a practical coil may be regarded as a pure inductance in parallel with a distributed capacitance. When the resonant frequency is greater than 1×10^4 Hz, the effect of the distributed capacitance of the inductor cannot be neglected. In this experiment the extrapolation method is used to measure the distributed capacitance of the inductor (including the distributed capacitance of the leads), which is an important experimental method. 3. This experiment is also a method of measuring the permittivity of a solid.

Experiment 7.16 Measurement of the characteristic of the resistance R of a thermistor versus the temperature t

[**Apparatus and materials available**]

Three resistance boxes, one type-A cell (1.5 V), a galvanometer, a thermistor (its resistance at room temperature is about 10 kΩ; during the experiment the current through it must be less than 500 μA), a vacuum flask, hot water, cold water, a switch, connecting wires, a thermometer, a stirrer, drawing paper, etc.

[**Requirements**]

(1) Draw the experimental circuit.

(2) Measure the data of the resistance R of the thermistor versus the temperature t.

(3) Find the empirical formula for the relation between the resistance of the thermistor and the thermodynamic temperature T.

Solution 7.16 1. The resistance of a thermistor is

$$R_T = A\mathrm{e}^{B/T},$$

where T is the thermodynamic temperature and A and B are constants to be determined, with $T = 273.15 + t$.

The experimental circuit is shown in the figure.

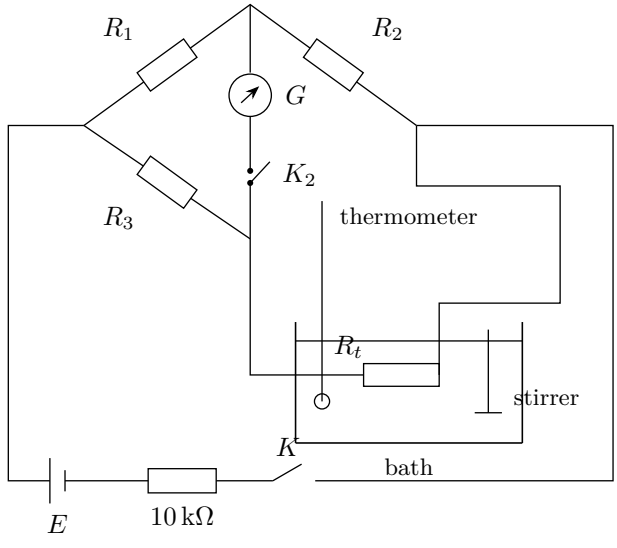


Fig. 7.16(a)

2. The experimental data are given in Table 7.16-1.

$$R_1 = 1000\,\Omega, \qquad R_2 = 1000\,\Omega, \qquad R_T = \frac{R_1}{R_2} \cdot R_4.$$

Table 7.16-1

R_T/Ω	$t/^\circ\mathrm{C}$	$\ln R_T$	T/K	$\frac{1}{T}/10^{-3}\,\mathrm{K}^{-1}$
1 630	94.71	7.3963	367.86	2.7184
2 136	82.57	7.6667	355.72	2.8112
2 565	75.11	7.8497	348.26	2.8714
3 747	62.29	8.2287	335.44	2.9812
5 184	50.25	8.5533	323.40	3.0921
7 324	38.70	8.8989	311.85	3.2067
9 486	30.43	9.1576	303.58	3.2940
6.163×10^4	24.50	9.3613	297.65	3.3597

Plotting the straight line $\ln R_T$–$1/T$ gives $B = 3.05 \times 10^3\,\mathrm{K}$, $\ln A = -0.888$, $A = 0.411\,\Omega$, and hence

$$R_T = 0.411\,\mathrm{e}^{3.05\times10^3/T}\ \Omega.$$

If this thermistor is used to measure a certain temperature, it is only necessary to measure the resistance at that temperature, and the temperature T can then be calculated from the above formula.

Comment. In this experiment the required temperature is obtained in the thermostat by mixing hot and cold water; the sensing parts of the thermometer and of the thermistor should be placed as close to each other as possible, and the mixture must be stirred continuously, so that an accurate temperature is obtained. In data processing, the curve equation is transformed into a linear equation, and the undetermined constants are then found by fitting a straight line; this is also often used in physics experiments.

Experiment 7.17 Measurement of the characteristics of a new two-terminal integrated-circuit temperature sensor

When the temperature is constant, the output current of the integrated temperature sensor supplied in this problem is a constant value. Its working voltage ranges from U_r to 20 V, where U_r is the minimum working voltage. Measure the volt–ampere characteristic of this sensor at constant temperature, and its temperature characteristic at constant working voltage.

[**Apparatus and materials available**]

A DC stabilised power supply, a digital DC voltmeter (with automatic range switching), three resistance boxes, one integrated temperature sensor, one 0–50 °C thermometer, one beaker (containing water at room temperature), two thermos cups (containing an ice–water mixture and hot water), two sheets of drawing paper, a switch, a stirring rod, connecting wires, etc.

[**Requirements**]

(1) At 0 °C measure the output current I of the integrated temperature sensor and the voltage U applied across its terminals; plot the I–U graph and find the minimum voltage U_r at which the current is constant at 0 °C.

(2) In the temperature range from room temperature to 50 °C, measure the relation between the current I and the temperature θ, find the empirical formula, and state the meaning of each constant in the empirical formula.

(3) Using a 12 V DC source, a digital DC voltmeter, etc., assemble a Celsius thermometer such that the reading of the digital voltmeter is 0 mV when the sensor temperature is 0 °C, and 50 mV when the sensor temperature is 50 °C. Use the Celsius thermometer so assembled to measure the temperature of water at room temperature, and compare the result with the value measured by a mercury thermometer. Draw the circuit of your design.

Solution 7.17 **1.** The circuit for measuring the integrated temperature sensor AD590 is shown in Fig. 7.17(a).

The resistance is $R = 1000\,\Omega$, and $I = U/R$. The experimental data are shown in Table 7.17-1.

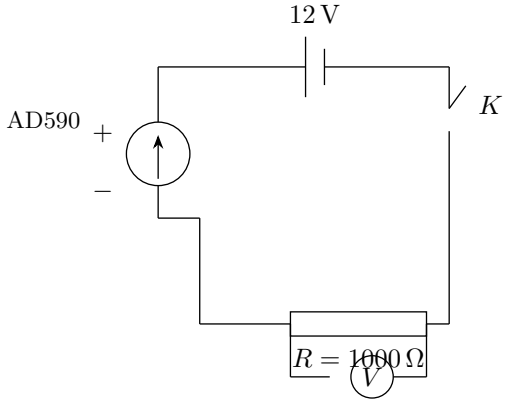


Fig. 7.17(a)

Table 7.17-1 (temperature $\theta = 3.0\,^\circ\mathrm{C}$)

U_Σ/V	1.326	1.589	1.736	2.065	2.342	2.456	2.533	2.680	2.713	2.880	3.010	3.252	3.643
U_r/V	0.200	0.777	1.130	1.795	2.306	2.495	2.616	2.770	2.775	2.760	2.759	2.759	2.760
I/μA	20.0	77.7	113.0	179.5	230.6	249.5	261.6	277.0	276.0	275.9	275.9	275.9	276.0

The relation between I and U_r is shown in Fig. 7.17(b).

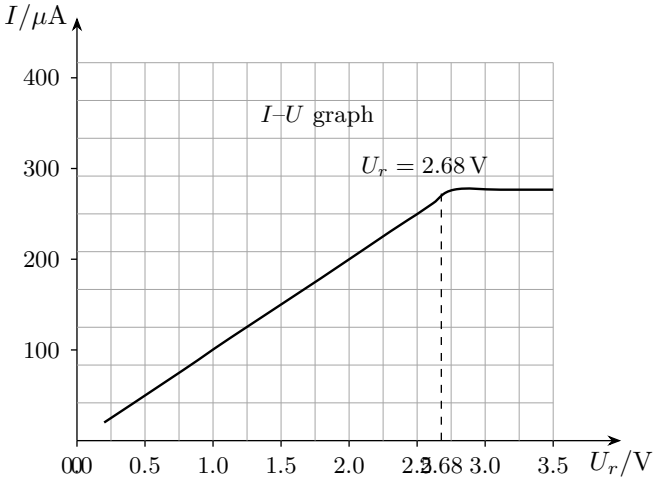


Fig. 7.17(b)

The minimum working voltage is $U_r = 2.68\,\mathrm{V}$.

2. Determination of the empirical formula and its constants

Table 7.17-2

$\theta/^\circ\mathrm{C}$	29.0	34.2	37.0	40.0	42.8	45.8	49.2
U_r/V	3.036	3.086	3.112	3.143	3.168	3.196	3.239
$I/\mu\mathrm{A}$	303.6	308.6	311.2	314.3	316.8	319.6	323.9

A straight line is fitted to the I–θ data in order to find the empirical formula. By the method of least squares, the correlation coefficient is $r = 0.999$, the slope is $B = 0.9873\,\mu\mathrm{A}/^\circ\mathrm{C}$, and the intercept is $A = 274.8\,\mu\mathrm{A}$. Hence

$$I = 0.9873\,\theta + 274.8.$$

The current–temperature sensitivity of this sensor is approximately $1\,\mu\mathrm{A}/^\circ\mathrm{C}$, and the numerical value of the intercept is approximately $273\,\mu\mathrm{A}$.

3. Construction of a Celsius thermometer

The compensation circuit of Fig. 7.17(c) is used. In it R_4 and R_3 form a voltage divider; $R_2 = 1000.9\,\Omega$, $R_3 = 1000\,\Omega$, and the supply voltage is 12 V. When the self-made Celsius temperature sensor AD590 is placed in an ice–water mixture, $\theta = 1.2\,^\circ\mathrm{C}$ and the voltmeter reading is 1.2 mV.

When the AD590 sensor is placed in water at room temperature, the reading of the thermometer is 28.8 mV (taken as 28.8 °C), while a mercury thermometer measures 28.7 °C.

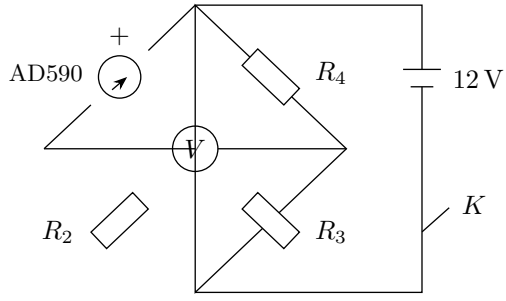


Fig. 7.17(c)

Comment. The reading of such a thermometer at 0 °C is not zero. In this case the compensation method may be used to make it zero, and this is also a commonly used experimental method.

Experiment 7.18 Investigation of an electromagnetic black box with DC meters

Inside a black box with three terminals, 1, 2 and 3, three electronic components of different kinds are connected. Each of them may be a battery, a resistor greater than $100\,\Omega$, a capacitor greater than $1\,\mu\mathrm{F}$, or a diode, as shown in the figure.

[**Apparatus and materials available**]

The DC voltage range 0–6 V (internal resistance $120\,\mathrm{k}\Omega$) and the DC current range 0–30 mA (internal resistance $10\,\Omega$) of a multimeter, one DC stabilised power supply, one black box, one resistance box, one stopwatch, and a number of connecting wires.

[**Requirements**]

(1) If there is a battery in the box, determine its electromotive force and draw the measuring circuit.

(2) If there is a resistor in the box, determine its resistance and draw the measuring circuit.

(3) If there is a capacitor in the box, determine its capacitance and draw the measuring circuit.

(4) If there is a diode in the box, mark its polarity and measure its forward threshold voltage; the forward resistance of the diode is comparatively small and may be neglected.

(5) Draw the distribution of the relevant electronic components inside the box and indicate their polarities.

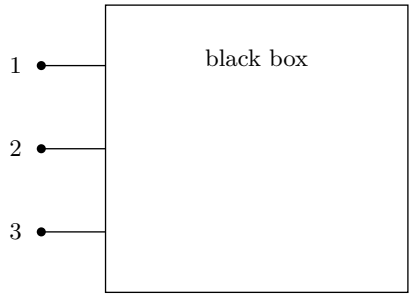


Fig. 7.18(a)

Solution 7.18 **1.** The voltmeter was used to measure each pair of terminals in order to check whether a battery exists; the result of the check was that none exists.

2. The internal structure is determined and drawn; the circuit to be checked is shown in Fig. 7.18(b).

(1) A unidirectional continuous conduction appears between terminals 1 and 3, which shows that inside there is a diode in series with a resistor.

(2) A pulsed unidirectional conduction appears between terminals 1 and 2, which shows that the diode is at terminal 1.

(3) A pulsed bidirectional conduction appears between terminals 2 and 3, which shows that the capacitor is at terminal 2. (Note: before the test the capacitor must first be short-circuited with a wire so that it is completely discharged, and only then may the test be made.)

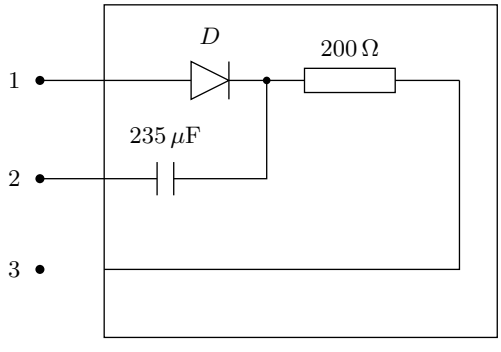


Fig. 7.18(b)

From the above tests and judgements it is known that the three different components are a diode, a capacitor and a resistor, and their connection diagram is shown in Fig. 7.18(b).

3. The resistance and the diode threshold voltage are measured by the volt–ampere method. The experimental circuit is shown in Fig. 7.18(c). The volt–ampere characteristic curve (I–U curve) of the series combination of the diode and the resistor to be measured is measured; by extrapolating the current I to $I = 0$, the threshold voltage of the diode, U_0, is obtained (Fig. 7.18(d)). At the same time the slope of the straight portion of the curve gives the resistance of the resistor in the box, $R = \Delta U/\Delta I = 200\,\Omega$.

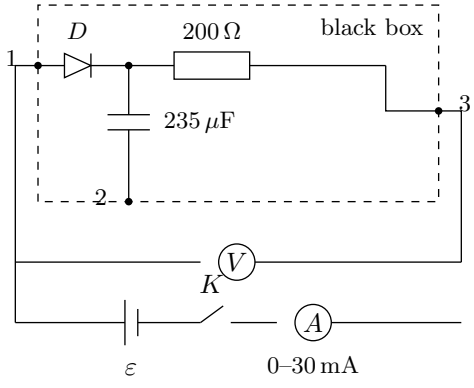


Fig. 7.18(c)

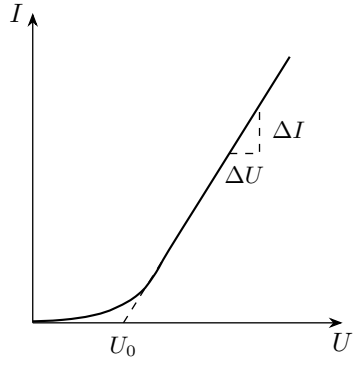


Fig. 7.18(d)

4. Measurement of the capacitance by the discharge method

The wiring is shown in Fig. 7.18(e)(1) or (2). By measuring the discharge time constant and knowing the resistance R', the value of the unknown capacitance C is obtained.

In Fig. 7.18(e)(1), $R' = 120.2\,\mathrm{k\Omega}$; in Fig. 7.18(e)(2), $R' = 99.2\,\mathrm{k\Omega}$. These are the total resistances of the discharge loop: in (1) the 120 kΩ voltmeter in series with the 200 Ω resistor, and in (2) the 99 kΩ resistance box in series with the 200 Ω resistor.

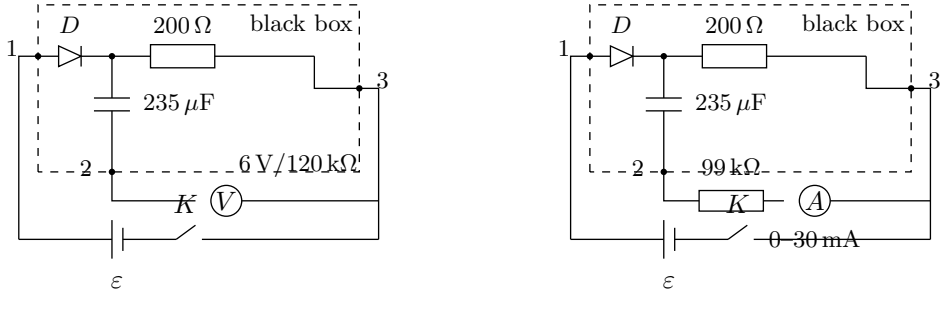


Fig. 7.18(e)

The capacitor $C = 235\,\mu\mathrm{F}$ in Fig. 7.18(b) is a non-polar capacitor; it is formed by connecting two 470 μF electrolytic capacitors in series with opposite polarities.

Comment. The use of a voltmeter (or an ammeter), a source, a resistance box and similar instruments to detect the components of a black box and their parameters is often required in scientific research. To judge the components and their parameters correctly one must have a correct understanding of the properties of the basic components and of electric circuits, and be familiar with the properties of basic electrical circuits; the black-box exploration experiment is therefore a comprehensive electrical experiment.

Experiment 7.19 Investigation of an electromagnetic black box with an oscilloscope, experiment I

The upper panel of a black box carries 8 terminals, to which four components are connected: an audio-frequency transformer, an inductor, a diode and a capacitor. Determine experimentally which terminals each of these four components is connected to, and measure the relevant parameters.

[**Apparatus and materials available**]

A black box (containing an audio-frequency transformer, an inductor, a diode and a capacitor; the input and output terminals of every component are all brought out to the panel of the black box, and the connections between the components do not form a closed loop), a dual-trace oscilloscope, a signal generator, a resistance box, a number of connecting wires, and one sheet of graph paper.

[**Requirements**]

(1) Determine to which terminals the input and output ends of the transformer are connected, and measure the ratio of the number of turns of the secondary winding to that of the primary winding (assume that the winding with fewer turns is the primary).

(2) Determine to which numbered terminals the anode and the cathode of the diode are connected.

(3) Determine which two terminals the capacitor is connected to, and measure its capacitance.

(4) Determine which two terminals the inductor is connected to, whether it is an air-core inductor or an inductor with a magnetic medium, and measure its inductance and loss resistance.

(5) Draw the overall wiring diagram of the components inside the black box.

(6) If the parameter characterising the value of some component varies appreciably with another physical quantity, measure the relation between them and plot the curve of the variation on graph paper.

Solution 7.19 The terminals of the black box are arranged in two rows: the upper row is numbered 1, 2, 3, 4 from left to right, and the lower row 5, 6, 7, 8 from left to right, eight terminals in all.

1. Determination of the components between the terminals

The experimental circuit is shown in Fig. 7.19(a). In the figure, i and j inside the dashed frame denote terminals on the panel of the black box. A sine-wave signal generator is used as the voltage input, and the resistance box is given a suitable value according to the circumstances. Channel CH1 of the dual-trace oscilloscope measures the voltage across the resistance box, and CH2 monitors the output voltage of the signal generator; when the output frequency of the signal generator is adjusted, the output voltage is kept constant. In Fig. 7.19(a) the terminal j on the panel of the black box corresponds to the red output terminal of the signal generator.

Each time, i and j are connected to different terminals, and the voltage across the resistance box, that is the waveform displayed on the CH1 channel of the oscilloscope, is observed. The results are collected in Table 7.19-1 (the pairs of terminals i, j for which the voltage across the resistance box is zero are not listed; this means that the terminals i and j are open-circuited).

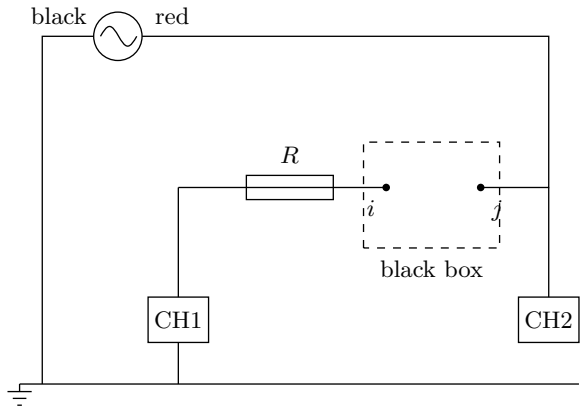


Fig. 7.19(a)

Table 7.19-1

j	i	Change of U_R as f increases	Component between i, j	Remarks
2	3	unchanged	diode	see Fig. 7.19(b)
4	8	decreases	inductor	
7	8	increases	capacitor	
1	5	decreases	inductor	
2	6	decreases	inductor	

Using the oscilloscope waveform shown in Fig. 7.19(b): when terminal 2 is connected to the red terminal of the signal generator, the lower half of the sine wave is clipped flat. Hence between 2 and 3 there is a diode, with terminal 2 connected to the anode and terminal 3 to the cathode.

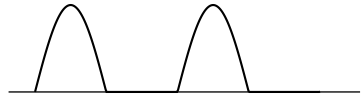


Fig. 7.19(b)

When the signal generator is connected to terminals 1, 5 and the oscilloscope to 2, 6, it is found that the voltage across 2, 6 varies markedly with the voltage across 1, 5; conversely, when a signal is fed in at 2, 6, a signal is likewise observed at 1, 5. Hence 1, 5 and 2, 6 form a transformer.

At this point the positions of the components inside the black box can be drawn as in Fig. 7.19(c).

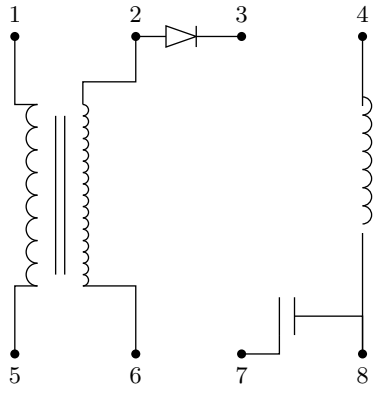


Fig. 7.19(c)

2. Measurement of the transformer parameters

The two ends of the signal generator are connected to 1, 5, the input CH1 to 1, 5 and the input CH2 to 2, 6; the data obtained are shown in Table 7.19-2.

Table 7.19-2

Input	Peak-to-peak value/cm	Voltage range/(V cm^{-1})	Voltage U/V
CH1	6.00	1	6.00
CH2	6.00	2	12.00

In Table 7.19-2, U is the peak-to-peak value. The measurement results show that 1, 5 are the primary of the transformer and 2, 6 are the secondary. The fine-adjustment knobs of the voltage ranges of CH1 and CH2 were both turned to maximum.

$$\text{turns ratio} \quad n = \frac{U_{\mathrm{CH2}}}{U_{\mathrm{CH1}}} = \frac{12.00}{6.00} = 2.00.$$

3. Measurement of the capacitance parameters

Assume that the loss resistance of the capacitor is negligible. Use the circuit of Fig. 7.19(a), with i, j connected to 7, 8; CH1 is set to the 1 V/cm range and CH2 to the 2 V/cm range.

Take a certain value of the resistance box, and adjust the output frequency and the output voltage of the signal generator so that both the CH1 and the CH2 waveforms reach full scale on the screen; then

$$\frac{U_{\mathrm{CH1}}}{U_{\mathrm{CH2}}} = \cos 60.0^\circ = 0.500,$$

that is,

$$\frac{R}{1/(\omega C)} = \tan 30.0^\circ,$$

and hence $C = \frac{1}{2\sqrt{3}\,\pi f R}$. The measured data are shown in Table 7.19-3. The vector sum of U_R and U_C is shown in Fig. 7.19(d).

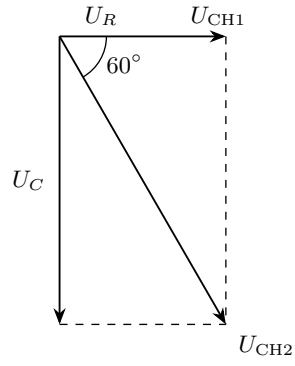


Fig. 7.19(d)

Table 7.19-3

R/Ω	f/kHz	C/μF
500.0	0.855	0.215

Hence $C = 0.215\,\mu$F.

4. Measurement of the inductance parameters

The experimental circuit is shown in Fig. 7.19(a), with i, j connected to terminals 4, 7. The inductance is measured by RLC series resonance: $L = \frac{1}{4\pi^2 f_0^2 C}$, where f_0 is the resonant frequency. Let U_R and U_0 denote the voltage across the resistance R and the output voltage of the signal generator; then at resonance the loss resistance of the inductor is $R_L = \left(\frac{U_0}{U_R} - 1\right) R$. The experimental data are shown in Table 7.19-4.

Table 7.19-4

R/Ω	U_R/V	U_0/V	I/mA	f_0/kHz	L/mH	R_L/Ω
100.0	12.0	16.0	42.4	1.995	29.6	33.3

Changing the output voltage of the signal generator gives Table 7.19-5:

$$I_{\text{eff}} = \frac{1}{2\sqrt{2}} I_{\text{p–p}} = \frac{1}{2\sqrt{2}} \cdot \frac{U_R}{R}.$$

Table 7.19-5

R/Ω	U_R/V (peak-to-peak)	I_{eff}/mA	f_0/kHz	L/mH
100.0	4.00	14.1	2.038	28.3

From Tables 7.19-4 and 7.19-5 it can be seen that the inductance L varies with the current I; therefore this inductor is not an air-core inductor, and there is ferromagnetic material inside the inductor. The data for the variation of the inductance L with the current I are as follows:

Table 7.19-6

U_R/V	12.0	10.0	8.00	6.00	4.00	2.00
I_{eff}/mA	42.4	35.3	28.3	21.2	14.1	7.07
f_0/kHz	1.995	1.980	1.982	2.000	2.038	2.110
L/mH	29.6	30.1	30.0	29.5	28.3	26.5

The graph of Table 7.19-6 is omitted.

Comment. This experimental problem has several important features. (1) Analysing the nature and the quality of components by observing with an oscilloscope the waveforms of the components or of parts of a circuit is very useful in scientific experiments, and this problem is closely related to practice. (2) The experiment is comprehensive: although there are many methods of measuring capacitance and inductance, within a limited time one must choose the method that is accurate, simple and quick, and this also reflects the depth of the student's knowledge and the strength of his practical ability.

Experiment 7.20 Detection of an electromagnetic black box with an oscilloscope (II)

A black box is given which contains three electromagnetic elements (a resistance R, a capacitance C and an inductance L) connected in a delta configuration. The three elements are connected respectively to terminals 1, 2 and 3 on the panel of the black box, the element between terminals 1 and 2 being Z_1, that between terminals 2 and 3 being Z_2, and that between terminals 1 and 3 being Z_3, as shown in the figure. The available instruments are to be used to identify and to measure them.

[**Apparatus and materials available**]

One black box, one dual-trace oscilloscope, one low-frequency signal generator, one resistance box, one single-pole double-throw switch, one capacitance box, graph paper, etc.

[**Requirements**]

(1) Determine what kind of elements the three elements in the black box are, and between which two terminals each of them is connected; draw the circuit diagram of their connection and briefly describe the method of identification.

(2) Measure the values of these three elements.

(3) If the value of an electrical element depends on some physical quantity, the relation between them must be measured.

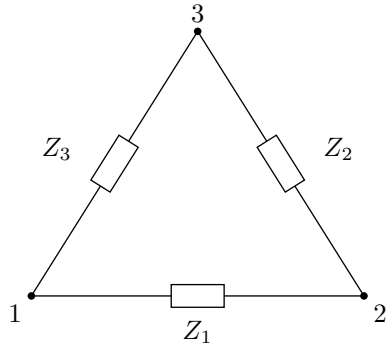


Fig. 7.20

Solution 7.20 1. Determination of the types of the elements in the black box

Test 1. Terminals 1 and 2 are connected with a wire, and the resistance R is connected in series to terminal 2. The circuit is then simplified to that of Fig. 7.20(a)(1). The output voltage of the source is kept unchanged, the output frequency f of the signal source is varied, and the variation of the voltage U_R across R is observed.

Test 2. Terminals 2 and 3 are short-circuited, and the resistance R is connected in series to terminal 3; the circuit is simplified to that of Fig. 7.20(a)(2). The output voltage of the source is

kept unchanged, the output frequency f of the signal source is varied, and the variation of the voltage U_R across R is observed.

Test 3. Terminals 1 and 3 are short-circuited, and the resistance R is connected in series to terminal 3; the circuit is simplified to that of Fig. 7.20(a)(3). The output voltage of the source is kept unchanged, the output frequency f of the signal source is varied, and the variation of the voltage U_R across R is observed.

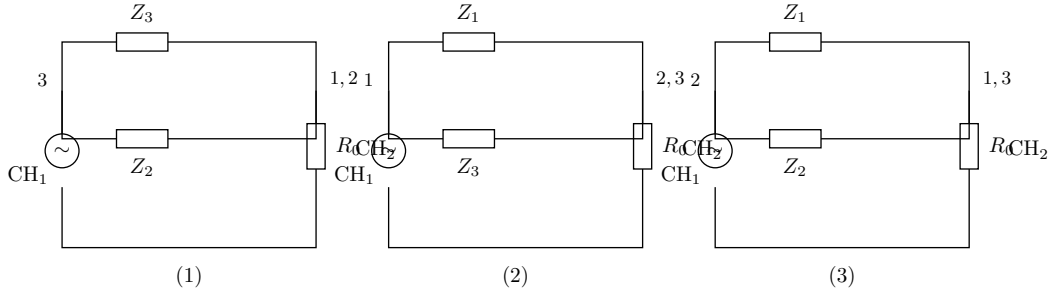


Fig. 7.20(a)

Table 7.20-1

	Test 1	Test 2	Test 3
Resonance observed?	yes	no	no
Variation of U_R as f increases	at the resonant frequency, U_R is a minimum	U_R decreases	U_R increases
Possible composition	Z_2 and Z_3 contain an inductance and a capacitance	Z_1 and Z_3 contain an inductance and a resistance	Z_1 and Z_2 contain a capacitance and a resistance

From the above measurements the circuit and the relevant elements can be summarised as in Fig. 7.20(b), namely a resistance R between terminals 1 and 2, a capacitance C between terminals 2 and 3, and an inductance L between terminals 1 and 3.

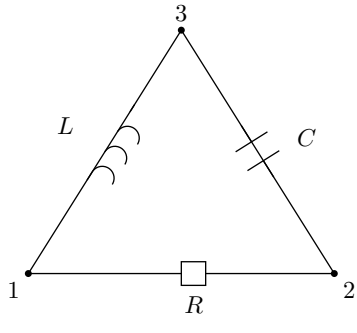


Fig. 7.20(b)

2. Method of measuring the resistance R and the capacitance C

Terminals 1 and 3 are short-circuited with a wire, and the black box, the resistance box, the oscilloscope and the signal generator are connected as in Fig. 7.20(c). When the frequency f of the signal source is rather low, for example $f = 20\,\mathrm{Hz}$, we may take approximately $1/(\omega C) \gg R$.

The resistance box is adjusted to R_0 so that $U_2 = \frac{1}{2}U_1$ (U_2 being the voltage measured by CH_2 and U_1 that measured by CH_1); then $U_{R_0} = U_R$, and hence

$$R = R_0 = 204\,\Omega.$$

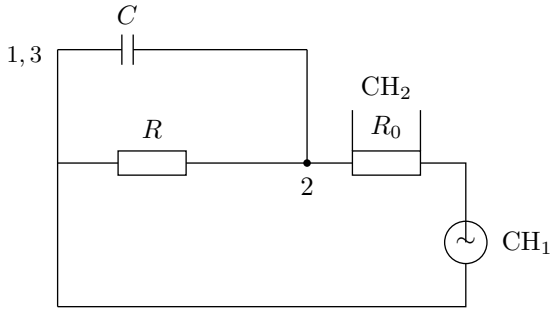


Fig. 7.20(c)

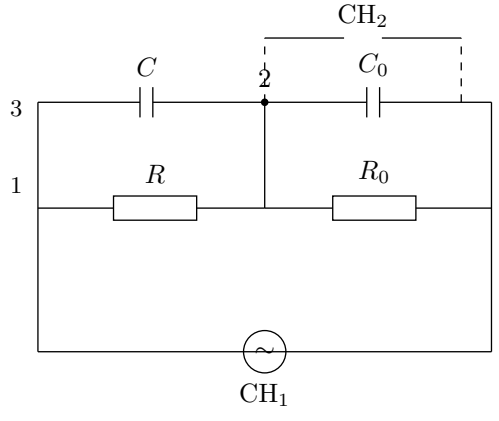


Fig. 7.20(d)

After the value of the resistance R has been measured, the black box, the low-frequency signal generator, the oscilloscope and the capacitance box are connected as in Fig. 7.20(d). With the resistance box set to $R_0 = R = 204\,\Omega$ and the frequency taken as $f = 1000\,\text{Hz}$, the value C_0 of the external capacitance box is adjusted until $U_2 = \frac{1}{2}U_1$. The bridge is then balanced, the voltage drops across the capacitance C and across the capacitance box C_0 are equal, and therefore

$$C = C_0 = 0.100\,\mu\text{F}.$$

3. Measurement of the inductance L and of its characteristics

The black box, the low-frequency signal generator and the oscilloscope are connected as in Fig. 7.20(e), and the frequency f of the signal generator is adjusted.

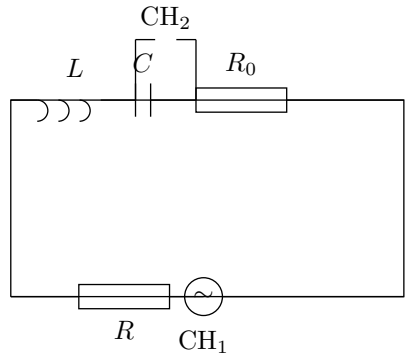


Fig. 7.20(e)

When the LC circuit reaches resonance, the voltage U_C measured across the capacitance C is a maximum. From $\omega^2 = 1/(LC)$, with $C = 0.100\,\mu\text{F}$ and resonant frequency $f_0 = 3.30\,\text{kHz}$, the inductance is obtained as $L = 23.3\,\text{mH}$ (the current through the inductance being $I = 8.29\,\text{mA}$).

Since this inductance is a ferrite-core coil, the permeability of the magnetic medium is related to the current through the coil, and hence the inductance varies as the current increases. The data of the inductance against the current, and the corresponding graph, are omitted.

Comment. A practical circuit is often a series or parallel combination of several elements and is comparatively complex. In order to examine the nature, the value and the quality of a given electronic element, a simple method is to short-circuit part of the elements, so that a complex circuit is reduced to a simple one; the circuit is then tested with an oscilloscope and with resistors, and the required results are obtained. This experimental method is very useful in physics experiments and in the testing of electronic circuits.

Experiment 7.21 Identification of the elements in an electromagnetic black box with a multimeter

A black box has four terminals on its lid, numbered on the lid. Inside the box there are three elements connected in a certain way; between any two terminals there is at most one element, and possibly none or a short circuit. The three elements may be a battery, a resistance, a capacitance (of comparatively large capacitance), an inductance or a semiconductor diode. A multimeter is to be used to examine the box and to determine the names of the three elements and their connection diagram.

[Apparatus and materials available]

The black box to be examined, a multimeter, a low-frequency signal generator, connecting wires, etc. The terminals on the panel of the black box are numbered as shown in the figure.

[Requirements]

(1) Determine the names of the three elements in the black box and draw their connection diagram.

(2) Explain the method of identification.

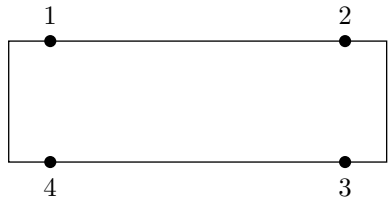


Fig. 7.21

Solution 7.21 **1.** *Determination of whether there is a battery in the box.* The voltage range of the multimeter shows that there is no battery in the box.

2. *Check whether there is a diode.* The ohmmeter range of the multimeter is used to measure the resistance between two terminals, and the positive and negative test leads are then interchanged. If the forward and reverse resistances differ greatly, a diode is present. For this black box no such phenomenon was found at any pair of terminals, so there is no diode in the box.

3. *Determination of how many capacitances and resistances are present.* The $\times 100\,\Omega$ and $1\,\mathrm{k}\Omega$ ranges of the multimeter were used for the tests, and the measured results are shown in the table.

Table 7.21-1

Terminal numbers	1, 2	1, 3	1, 4	2, 3	2, 4	3, 4
Charge–discharge phenomenon	none	yes	none	yes	none	yes
Resistance between terminals	R_{12}	R_{13}	R_{14}	R_{23}	R_{24}	R_{34}
Measured resistance/Ω	180	∞	200	∞	20	∞

From the table it is seen that resistances exist between terminals 1 and 2, between 1 and 4 and between 2 and 4, and that $R_{14} = R_{12} + R_{24}$. According to these measured results there are three possible cases, as shown in Fig. 7.21(a)(1), (2) and (3).

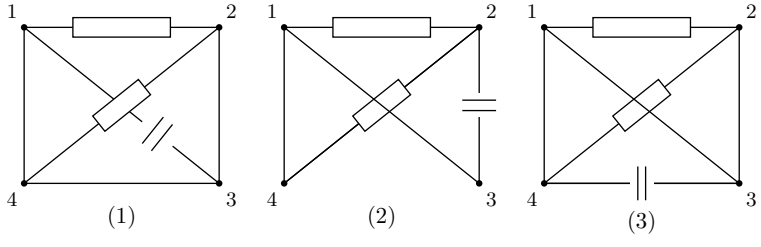


Fig. 7.21(a)

The output of the signal generator is applied between terminals 3 and 4, and the voltage between terminals 1 and 2 is measured. It is found that there is a voltage between 1 and 2, and one can therefore be certain that the capacitance is connected between terminals 1 and 3.

4. *Determination of whether the elements between 1 and 2 and between 2 and 4 are resistances or inductances.*

The low-frequency signal generator (producing a sine wave), the AC current range of the multimeter and the element between terminals 1 and 2 on the panel are connected in series with wires. Keeping the sine-wave output voltage constant, observe whether the current in this series circuit varies; the experimental circuit is shown in Fig. 7.21(b). If, when the frequency of the sine wave output of the signal generator decreases, the AC current reading increases markedly, this shows that the element between 1 and 2 is an inductance. If the current does not vary with

the frequency, it is a resistance. In the same way one can also distinguish whether the element between 2 and 4 is a resistance or an inductance.

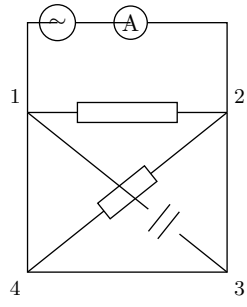


Fig. 7.21(b)

The final result of this experiment is: the three elements in the black box are a resistance, an inductance and a capacitance, connected as shown in Fig. 7.21(c).

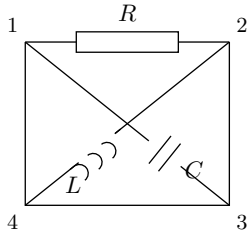


Fig. 7.21(c)

Experiment 7.22 Measurement of capacitance and inductance with an AC bridge

[Apparatus and materials available]

A low-frequency signal generator, three resistance boxes, a standard capacitance $0.200\,\mu$F, a transistor voltmeter (for AC use), the capacitance to be measured, the inductance to be measured, connecting wires, etc.

[Requirements]

(1) Write down the experimental principle and the formulae for measuring the unknown capacitance and the unknown inductance.

(2) Write down the experimental method.

(3) Measure the data and calculate the experimental results.

Solution 7.22 1. The AC bridge circuit is shown in Fig. 7.22(a), in which the bridge arms Z_1, Z_2, Z_3 and Z_4 are complex impedances. An AC sine-wave source is used, and a transistor AC voltmeter is used to detect whether the bridge is balanced. When the AC bridge is balanced, the balance equation is

$$\frac{Z_1}{Z_4} = \frac{Z_2}{Z_3}. \tag{1}$$

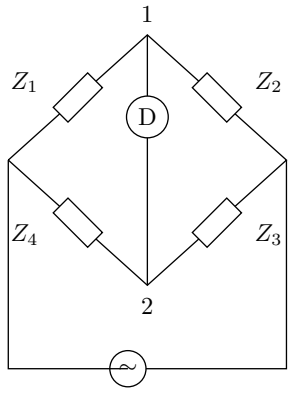


Fig. 7.22(a)

When the unknown capacitance is measured, the circuit is connected as the capacitance bridge shown in Fig. 7.22(b), in which the unknown capacitance is represented by the capacitance C_3 and the loss resistance r_3. When the AC bridge is balanced,

$$R_1\left(r_3 - \frac{\mathrm{i}}{\omega C_3}\right) = R_2\left(R_4 - \frac{\mathrm{i}}{\omega C_4}\right),$$

where ω is the angular frequency of the alternating current and $\mathrm{i} = \sqrt{-1}$ is the imaginary unit. Equating the real and the imaginary parts separately gives

$$C_3 = \frac{R_1}{R_2} \cdot C_4, \tag{2}$$

$$r_3 = \frac{R_2}{R_1} \cdot R_4. \tag{3}$$

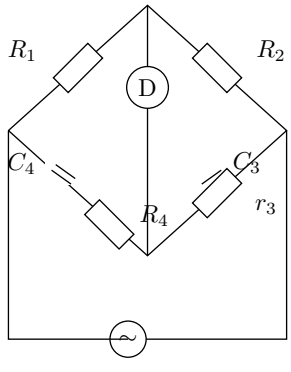


Fig. 7.22(b)

When the unknown inductance (an air-cored coil) is measured, the circuit is as shown in Fig. 7.22(c), in which L_3 and r_3 are respectively the inductance to be measured and its loss resistance, and C_1 is the standard capacitance. The balance equation of this AC bridge is

$$r_3 + \mathrm{i}\omega L_3 = \left(\frac{1}{R_1} + \mathrm{i}\omega C_1\right) R_2 R_4.$$

Equating the real and the imaginary parts separately gives

$$L_3 = C_1 R_2 R_4, \tag{4}$$

$$r_3 = \frac{R_2}{R_1} \cdot R_4. \tag{5}$$

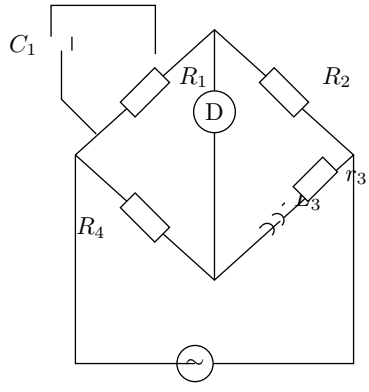


Fig. 7.22(c)

2. Measurement of the unknown capacitance and of the loss resistance

(1) The resistance boxes, the standard capacitance, the capacitance to be measured, the low-frequency signal generator and the transistor AC voltmeter are connected as in the circuit of Fig. 7.22(b).

(2) The output frequency of the low-frequency signal generator is set to 1000 Hz, suitable resistance values are chosen for the bridge arms, and the arm resistances are adjusted repeatedly until the indication of the transistor AC voltmeter is a minimum.

(3) When the bridge has reached balance, the readings of the arms are recorded, and the capacitance and the loss resistance of the capacitance to be measured are found.

(4) After the AC bridge has been balanced, the resistance of the adjusting element is changed slightly, so that the AC detector deviates from the balance position by a small amount, in order to measure the sensitivity of adjustment of the AC bridge and to find the uncertainty in the measurement of the capacitance to be measured.

The method of measuring the inductance and the loss resistance of an inductor whose support is a non-magnetic dielectric is similar to the method of adjustment for measuring a capacitance. Only in Solution 7.22 the value of R_4 must be taken very small. But in Fig. 7.22(c) the value of the resistance R_1 connected in parallel with the standard capacitance must be taken large, so that the bridge can be balanced.

3. Experimental data and calculations

(1) *Measurement of a capacitance with the capacitance bridge*

The frequency of the source is $f = 1000\,\text{Hz}$, the effective value of the output voltage is $U = 3\,\text{V}$, the standard capacitance is $C_4 = 0.200\,\mu\text{F}$, and $R_1 = 1000\,\Omega$. R_2 and R_4 are adjusted; at bridge balance $R_2 = 370.2\,\Omega$ and $R_4 = 3.7\,\Omega$. At this setting a change of $0.4\,\Omega$ in R_2, or of $0.6\,\Omega$ in R_4, changes the indication of the detector by one division.

$$C_3 = \frac{R_1}{R_2} \cdot C_4 = \frac{1000}{370.2} \times 0.200 = 0.540\,(\mu\text{F}),$$

$$r_3 = \frac{R_2}{R_1} \cdot R_4 = \frac{370.2}{1000} \times 3.7 = 1.37\,(\Omega).$$

The error of the capacitance measurement caused by the sensitivity of adjustment is

$$\frac{\Delta C_3}{C_3} = \frac{0.04}{370.2} = 0.011\%,$$

and the error of the measurement of the loss resistance is

$$\frac{\Delta r_3}{r_3} = \sqrt{\left(\frac{\Delta R_1}{R_1}\right)^2 + \left(\frac{\Delta R_4}{R_4}\right)^2} = \sqrt{(0.011\%)^2 + \left(\frac{0.06}{3.7}\right)^2} = 1.6\%,$$

$$\Delta C_3 = C_3 \times 0.011\% = 0.540 \times 0.011\% = 0.6 \times 10^{-4}\,(\mu\text{F}).$$

The class of the resistance boxes is 0.1 (0.1%), so the measurement error caused by the sensitivity of adjustment is far smaller than the systematic error introduced by the resistance boxes; the balance condition of this bridge therefore meets the requirements of the experiment.

(2) *Measurement of an inductance with the inductance bridge*

The frequency is $f = 1000\,\text{Hz}$, the output voltage of the signal source is $U = 3\,\text{V}$, the standard capacitance is $C_1 = 0.200\,\mu\text{F}$, and $R_2 = 1000\,\Omega$. At bridge balance $R_1 = 14\,500\,\Omega$ (adjustment sensitivity 0.9 division per $100\,\Omega$) and $R_4 = 1683.0\,\Omega$ (adjustment sensitivity 1 division per Ω). From (3) and (4),

$$L_3 = C_1 R_2 R_4 = 0.200 \times 10^{-6} \times 1000 \times 1683 = 336.6\,(\text{mH}).$$

C_1 is of class 0.05, and R_2 and R_4 are of class 0.1; the relative error $\Delta L_3/L_3$ of L_3 caused by these is about 0.7%, while ΔL_3 itself is very small, only about 0.02 mH. The balance of this bridge meets the requirement that the measurement error (the error of adjustment sensitivity) be only 1/10 of the total error, which shows that the bridge is well balanced. Moreover,

$$r_3 = \frac{R_2 R_4}{R_1} = 116.1\,\Omega.$$

Experiment 7.23 Characteristics and empirical formula of an InSb magnetoresistive element

A new type of semiconductor magnetoresistive sensor has a resistance which increases as the external magnetic field increases. When the external magnetic field is zero (apart from the Earth's field), its resistance is R_0. If the sensor is placed in the gap of an electromagnet with its sensitive surface perpendicular to the direction of the magnetic field in the gap, and the current through the electromagnet is gradually increased from a small value, the resistance R of the sensor also increases. The change of resistance is $\Delta R = R - R_0$, and the relative change $\Delta R/R_0$ of the resistance of the sensor satisfies

$$\frac{\Delta R}{R_0} = aB^n, \tag{1}$$

where B is the magnetic flux density in the gap, perpendicular to the pole faces, measured with a digital teslameter; a is a constant and n is an exponent.

[**Apparatus and materials available**]

An electromagnet with a gap, a constant-current source (DC), a digital teslameter, one slide-wire rheostat, one voltmeter, one ammeter, a magnetoresistive sensor (with leads), one dry cell, one $400\,\Omega$ current-limiting resistor, and several connecting wires.

[Requirements]

(1) Since the core of the electromagnet is made of die steel, it may retain residual magnetism. Before the experiment the magnet must first be demagnetised; the method of demagnetisation is shown in the figure. After demagnetisation, record the value of B_0.

(2) Measure the resistance R_0 of the magnetoresistive sensor when the magnetic flux density in the gap of the electromagnet is $B = 0$.

(3) Measure the relation R–B of the resistance of the magnetoresistive sensor and enter the data in the table. Note that the current through the magnetoresistive sensor must always be less than 3 mA.

(4) Enter the data of the relation between $\Delta R/R$ and B in the table, plot the graph of $\Delta R/R$ against B, find the magnetic flux density B_r at the turning point of the curve, and, from the trend of the curve, find the best empirical formula.

(5) If the output of the signal generator is connected to the electromagnet, with an output frequency in the range $f = 10.0$–100 Hz, and the voltage across the energised electromagnet is applied to the x-axis of the oscilloscope while the voltage across the energised magnetoresistive sensor is applied to the y-axis, observe and record the Lissajous figure formed by the two sine voltages when the magnetic flux density in the middle of the gap is small, $B < B_r$, and explain the reason for this phenomenon.

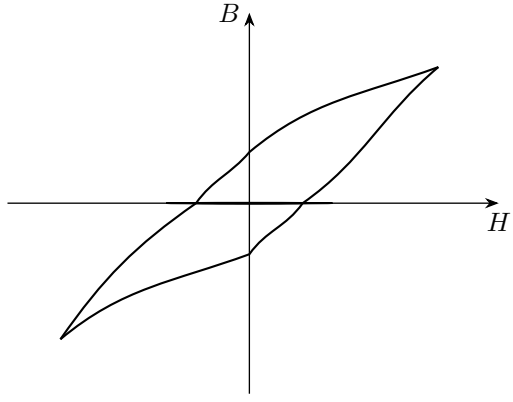


Fig. 7.23(a)

Solution 7.23 **1.** Before the measurement the material of the electromagnet is first demagnetised. For example, the current through the coil of the electromagnet is increased from 0 to 600 mA and then reduced to 0; the direction of the current is then reversed with a double-pole double-throw switch, the current being increased from 0 to 500 mA and then reduced from 500 mA to 0. In this way the magnetising current is repeatedly reversed, the maximum current being reduced by 100 mA each time. When the residual magnetism has become small, the current may be reduced in smaller steps each time; finally the residual magnetism is eliminated and $B_0 = 0$.

2. The arrangement for measuring the magnetoresistance effect is shown in Fig. 7.23(b). By adjusting the exciting current, different magnetic fields are produced; the current I through the magnetoresistive sensor and the voltage U across it are read, together with the corresponding magnetic flux density B in the gap of the electromagnet. The experimental data are shown in the table below. The sensor is supplied from a constant-voltage source, $U = 0.400$ V.

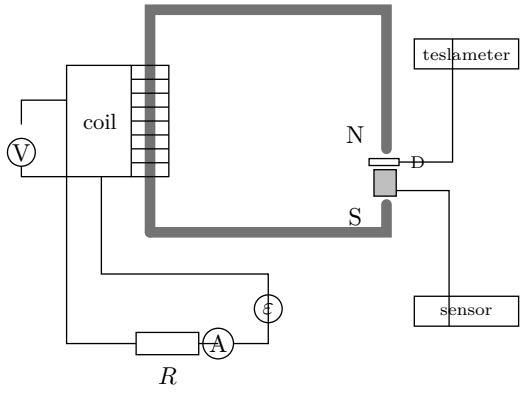


Fig. 7.23(b)

Table 7.23-1 Relation between $\Delta R/R_0$ and B

I/mA	1.09	1.07	1.05	1.02	0.97	0.95	0.93	0.89	0.86	0.83	0.80	0.76	0.71	0.66	0.62
B/mT	0	16	25	34	46	51	58	67	79	98	125	185	252	327	406
R/Ω	367	374	381	392	412	421	430	449	465	482	500	526	563	606	645
$\Delta R/R_0$	0.00	0.02	0.04	0.07	0.12	0.15	0.17	0.22	0.27	0.31	0.36	0.43	0.54	0.65	0.76

Taking logarithms of both sides of (1) gives

$$\ln \frac{\Delta R}{R} = n \ln B + \ln a. \tag{2}$$

Below 0.06 T a straight-line fit of $\ln(\Delta R/R)$ against $\ln B$ gives a correlation coefficient $r = 0.9990$, $n = 1.724$ and $\ln a = -8.725$. From a large number of measurements on different elements, the value of n lies between 1.7 and 2.2. Above 0.12 T a straight-line fit of $\Delta R/R$ against B gives a correlation coefficient $r = 0.9992$, that is $n = 1$ and $a = 1.45 \times 10^{-3}$.

From these data it follows that for the InSb magnetoresistive sensor the relation between $\Delta R/R_0$ and B is approximately quadratic when $B < 0.06$ T, whereas the relation between $\Delta R/R_0$ and B is linear when $B > 0.12$ T.

3. The electromagnet is connected in the circuit of Fig. 7.23(c), in which the magnet is excited by an alternating sinusoidal magnetic field of frequency 10–100 Hz. When $|B| < 0.05$ T, a Lissajous figure formed by the voltage divider across the magnetoresistor and the input signal of the electromagnet can be displayed on the oscilloscope, as shown in Fig. 7.23(d). This is the AC frequency-doubling effect in a weak magnetic field.

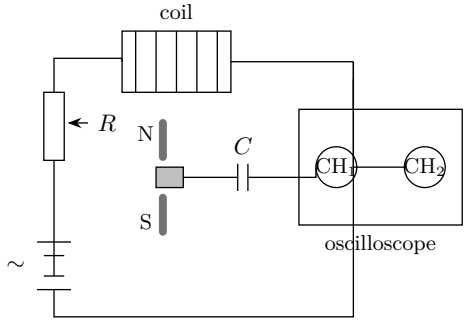


Fig. 7.23(c)

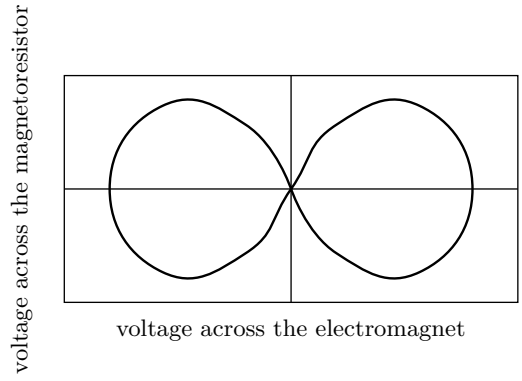


Fig. 7.23(d)

As regards the AC frequency-doubling effect: when the magnetoresistive element is in a weak AC magnetic field, $\Delta R/R(0)$ is approximately proportional to B^2, so the resistance R in the field also varies periodically with time, which in turn causes the voltage division across R to vary. Suppose that the current I through the magnetoresistive element is constant and let $B = B_0 \cos \omega t$; from $\Delta R/R(0) = aB^2$ (where a is a constant) we have

$$\begin{aligned} R(B) &= R(0) + \Delta R = R(0) + R(0) \times \frac{\Delta R}{R(0)} = R(0) + R(0) a B_0^2 \cos^2 \omega t \\ &= R(0) + \frac{1}{2} R(0) a B_0^2 + \frac{1}{2} R(0) a B_0^2 \cos 2\omega t. \end{aligned}$$

The voltage across R is

$$\begin{aligned} U(B) &= I_0 R(B) = I_0 \left[R(0) + \frac{1}{2} R(0) a B_0^2 \right] + \frac{1}{2} I_0 R(0) a B_0^2 \cos 2\omega t \\ &= U(0) + \tilde{U} \cos 2\omega t. \end{aligned}$$

Hence, although the angular frequency of the working current of the electromagnet is ω, the angular frequency of the AC signal output by the magnetoresistive sensor is 2ω.

Experiment 7.24 Measurement of the Earth's magnetic field

The Earth itself is magnetic, so a magnetic field exists in the Earth and in the near-Earth space; it is called the geomagnetic field. The strength and the direction of the geomagnetic field vary with location (and even with time). The projection of the geomagnetic field vector $\vec{B}$ on the horizontal plane is called the horizontal component $B_{\parallel}$ of the geomagnetic field. Given Helmholtz coils, a compass, a DC supply, a DC ammeter, etc., the horizontal component $B_{\parallel}$ of the geomagnetic field can be measured.

[Apparatus and materials available]

Helmholtz coils (number of turns of each coil $N = 310$; diameter of the coils $R = 16.2\,\text{cm}$), a stabilised DC supply, a DC ammeter, a compass with an angular scale, a spirit level, connecting wires, etc., as shown in the figure.

[Requirements]

(1) State the principle of the experiment and the experimental formulae.

(2) Briefly describe the experimental method.

(3) Measure and calculate the results.

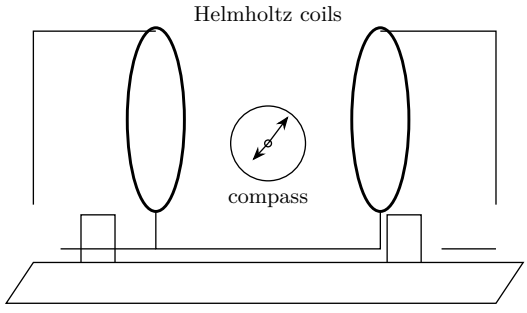


Fig. 7.24

Solution 7.24 The magnetic field at the midpoint of the common axis of the Helmholtz coils is

$$B' = \frac{\mu_0 NI}{\bar{R}} \cdot \frac{8}{5^{3/2}},$$

where N is the number of turns of the coil, I the current through the coil, $\bar{R}$ the mean radius of the Helmholtz coils, and μ_0 the permeability of vacuum, $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T \cdot m/A}$.

1. The base plate of the Helmholtz coils is adjusted to be horizontal, and a compass is placed at the centre of the common axis of the coils, as shown in the figure. The plane of the coils, before any current is passed, is first made parallel to the direction of the magnetic needle of the compass. The DC supply, the ammeter and the Helmholtz coils are then connected in series. When a current is passed through the Helmholtz coils, the magnetic flux density B' on their axis is necessarily perpendicular to the horizontal component $B_{\parallel}$ of the geomagnetic field, as shown in Fig. 7.24(a). The magnetic needle of the compass is acted upon simultaneously by the magnetic torques produced by the two fields $B_{\parallel}$ and B', so that it deviates from the horizontal component of the geomagnetic field and makes an angle θ with it. From Fig. 7.24(a),

$$\frac{B'}{B_{\parallel}} = \tan\theta. \tag{1}$$

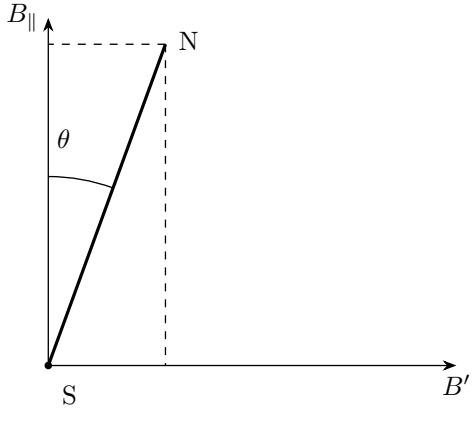


Fig. 7.24(a)

Combining (1) with the formula for the magnetic field on the axis of the Helmholtz coils gives

$$B_{\parallel} = \frac{8\mu_0}{5^{3/2}} \cdot \frac{NI}{\bar{R}\tan\theta},$$

that is,

$$I = \frac{5^{3/2}\bar{R}B_{\parallel}}{8\mu_0 N}\tan\theta = C\tan\theta. \tag{2}$$

In (2), $C = \dfrac{5^{3/2}\bar{R}}{8\mu_0 N}\cdot B_{\parallel}$. At a given place $B_{\parallel}$ is a fixed value, and $\bar{R}$ and N are also fixed, so C is a constant. Equation (2) shows that the current I through the galvanometer is proportional to the tangent of the deflection angle θ of the magnetic needle; for this reason the arrangement is also called a tangent galvanometer. It is only necessary to measure the relation between I and $\tan\theta$ and to obtain C from the slope; the horizontal component $B_{\parallel}$ of the geomagnetic field can then be found from the formula for C.

2. Experimental method

The experimental circuit is shown in Fig. 7.24(b). In the figure G is the tangent galvanometer (actually the Helmholtz coils of the tangent galvanometer), K is a reversing switch, and ε is a stabilised DC supply. The experimental steps are as follows.

(1) Connect the circuit as in Fig. 7.24(b).

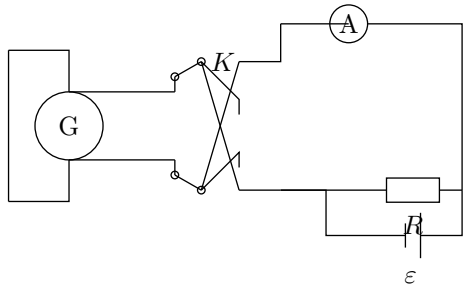


Fig. 7.24(b)

(2) Adjust the levelling screws of the base of the tangent galvanometer so that the bubble in the middle of the spirit level is at the centre. Place the compass at the midpoint of the common axis of the coils, so that the plane of the coils is parallel to the direction of the magnetic needle and at the same time the magnetic needle coincides with the "0" line of the graduated circle.

(3) Switch on the supply and increase the input current I of the tangent galvanometer step by step, obtaining a series of corresponding deflection angles θ of the magnetic needle. For each deflection angle θ two readings θ' and θ'' can be obtained simultaneously from the compass, as shown in Fig. 7.24(c). The direction of the current I is also reversed with the reversing switch, and the deflection angles before and after the reversal, θ_+ and θ_-, are obtained, so that

$$\theta = \frac{1}{4}\left(\theta'_+ + \theta'_- + \theta''_+ + \theta''_-\right).$$

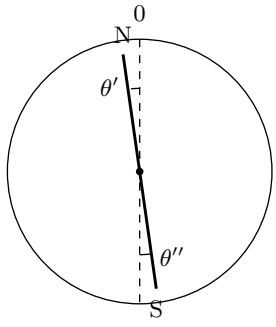


Fig. 7.24(c)

(4) Plot the graph of I against $\tan\theta$, and calculate the value of $B_{\parallel}$ from the slope of the straight line.

3. The radius of the Helmholtz coils is $\bar{R} = 16.2\,\text{cm}$ and the number of turns is $N = 310$. The data of the relation between I and $\tan\theta$ are omitted. The measured horizontal component of the geomagnetic field is $B_{\parallel} = 2.91 \times 10^{-5}\,\text{T}$; compared with the value measured in the open air outside the building in Shanghai, this value is about 20% smaller, which is caused by the screening effect of the steel material of the building.

Experiment 7.25 Measurement of the magnetisation curve and the hysteresis loop of a ferromagnetic material with a Hall sensor

[Apparatus and materials available]

A stabilised DC current source, a multirange ammeter, a digital teslameter (with a probe mounted on a graduated travelling device), a ring-shaped magnetic material to be measured (with a 0.2 cm gap; 2000 turns of wire are wound on the sample, whose cross-section is $2.00\,\text{cm} \times 2.00\,\text{cm}$), a double-pole double-throw switch, a ruler, connecting wires, etc.

[Requirements]

(1) Measure the relation between the residual magnetic flux density B in the gap of the sample and the position x, and find the extent of the region of uniform field.

(2) Measure the initial magnetisation curve of the sample; demagnetise the sample before the measurement.

(3) Measure the hysteresis loop of the die-steel sample; magnetically train the sample before the measurement.

(4) Measure the mean magnetic path length $\bar{l}$ of the sample and the width l_g of the gap, correct the coercive force H_c and the field strength H_m at saturation, and find the saturation magnetic flux density B_m.

Solution 7.25 **1.** The wiring of the instruments is shown in Fig. 7.25(a).

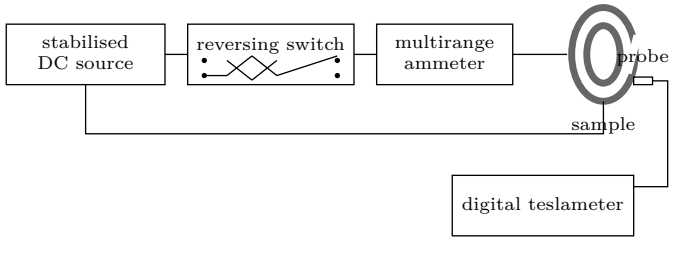


Fig. 7.25(a)

The relation B–x measured with the teslameter is given in the following table.

Table 7.25-1 Relation between B and x

x/mm	B/mT	x/mm	B/mT	x/mm	B/mT	x/mm	B/mT
−10.0	130.7	−4.0	168.1	1.0	168.4	6.0	167.4
−9.0	165.3	−3.0	168.3	2.0	168.4	7.0	166.8
−8.0	166.1	−2.0	168.4	3.0	168.4	8.0	165.3
−7.0	167.0	−1.0	168.4	4.0	168.4	9.0	160.6
−6.0	167.4	0.0	168.4	5.0	168.0	10.0	123.1

From the data in the table it is seen that the magnetic flux density is uniform in the range from $x_1 = -4.0\,\text{mm}$ to $x_2 = 5.0\,\text{mm}$, that is, over $\Delta x = 0.9\,\text{cm}$. In the following measurements the sensitive element of the teslameter probe should be placed within this uniform region, with its sensing surface perpendicular to the direction of B.

2. The initial magnetisation curve of the sample is measured after the sample has first been demagnetised. The magnetising current is repeatedly reversed while its amplitude is gradually reduced from the maximum value to zero, so that finally the residual magnetism B of the sample is zero. The B–H relation is then measured.

3. Magnetic training is carried out before the hysteresis loop of the die steel is measured. From the initial magnetisation curve one can find the current I_m through the magnetising coil when B increases very slowly. This current I_m is then kept constant, and the double-pole reversing switch is thrown back and forth 50 times for magnetic training; when the switch is operated, the time from contact to separation of the contacts should be made rather long, so that sparking is avoided. Magnetic training makes the hysteresis loop symmetrical.

4. The mean magnetic path length $\bar{l}$ and the gap width l_g of the die-steel sample are measured, and H is corrected by means of formula (1). The sample is shown in Fig. 7.25(b).

If there is a small gap l_g in the magnetic circuit of the iron core, the mean magnetic path length in the iron core is $\bar{l}$, the number of turns of the coil on the core is N, and the current through it is I, then by the circuital law

$$H\bar{l} + H_g l_g = NI. \tag{1}$$

In (1), H_g is the magnetic field strength in the gap. Generally speaking, this is the magnetic flux density in the iron core. When the gap is very narrow, that is, when the length and the width of the rectangular cross-section of the iron core are $\gg l_g$ and the mean magnetic path length $\bar{l} \gg l_g$, we have

$$B_g \cdot S_g = B \cdot S. \tag{2}$$

In (2), S_g is the cross-sectional area of the magnetic path in the gap and S that in the iron core. Under the above conditions $S_g \approx S$, so $B = B_g$, where B_g is the magnetic flux density B at the middle of the iron core to be measured. Moreover, in the gap,

$$B_g = \mu_0 \mu_r H_g. \tag{3}$$

In (3), μ_0 is the permeability of vacuum and μ_r the relative permeability; in the gap $\mu_r = 1$. Hence $H_g = B/\mu_0$, and (1) can be written as

$$H\bar{l} + \frac{1}{\mu_0} B l_g = NI, \tag{4}$$

$$\therefore \quad H = \left(NI - \frac{1}{\mu_0} \cdot B l_g \right) \Big/ \bar{l}. \tag{5}$$

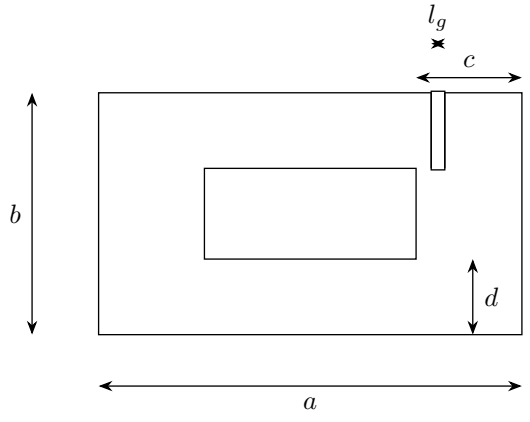
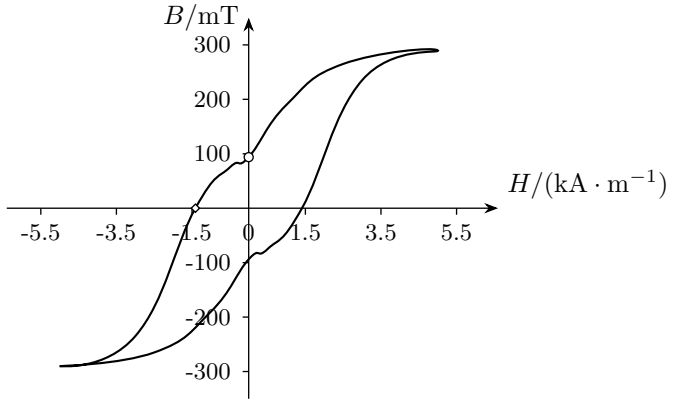


Fig. 7.25(b)

The sample was measured with a metre rule: $a = 10.00\,\text{cm}$, $b = 6.00\,\text{cm}$, $c = d = 2.00\,\text{cm}$;

$$\bar{l} = 10.00 \times 2 + 6.00 \times 2 - 2.00 \times 4 - 0.20 = 23.8\,(\text{cm});$$

and $l_g = 0.20\,\text{cm}$. In (5), B is measured directly, N is known, I is read from the ammeter, and μ_0 is a constant. The uncorrected B–H curve is shown in Fig. 7.25(c).



$\circ$ marks the remanence: 93.8 mT; $\diamond$ marks the coercive force: $-1410\,\text{A/m}$

Fig. 7.25(c)

Substituting $N = 2000$ turns, I, $\bar{l}$, l_g and B into formula (5) gives the corresponding H; the B–H curve obtained in this way is the true static magnetisation curve of the die-steel material. The corrected B–H curve is omitted.

Chapter 6

Optical Experiments

Experiment 8.01 Measurement of the focal length of a convex lens by three different methods

[**Apparatus and materials available**] Incandescent lamp, optical bench, object screen (a plate with an arrow-shaped aperture), thin convex lens, image screen, plane mirror, small electric bulb, etc.

[**Requirements**]

(1) Write down the principles and formulas of three methods for measuring the focal length of a thin convex lens, and draw the ray diagrams.

(2) Analyse the characteristics of these measurement methods and estimate the magnitude of the measurement error.

Solution 8.01 The imaging formula of a lens under paraxial conditions is

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \tag{1}$$

where u is the object distance, v the image distance and f the focal length of the lens. This is the Gaussian formula.

Since a convex lens can form a real image, it is sufficient to measure the object distance and the image distance directly and then substitute them into formula (1) to calculate the focal length f of the convex lens.

1. Rough measurement

Find an object at "infinity", such as a distant window or a building outside the window, and use a wall as the image screen. As the object distance u tends to infinity, formula (1) gives $f \approx v$; that is, after passing through the lens the image of an infinitely distant object is formed with its image plane coinciding with the focal plane of the lens. The measurement error of this method is about 10%, so it is used mainly for rough estimates, for example when choosing a lens.

2. Autocollimation method

When the arrow-shaped light source S_0 is placed in the focal plane of the lens L, the light rays become, after passing through L, a beam parallel to the principal axis, as shown in Fig. 8.01(a). If a plane mirror M perpendicular to the principal axis of the lens is placed behind the lens, this beam is reflected back. By the principle of reversibility of light paths, S_0 must form an image in the focal plane of the lens. The image of S_0 is equal in size to S_0, inverted, and reversed left to right. The distance between the optical centre O of the lens and the light source S_0 is then the focal length f of the lens under test.

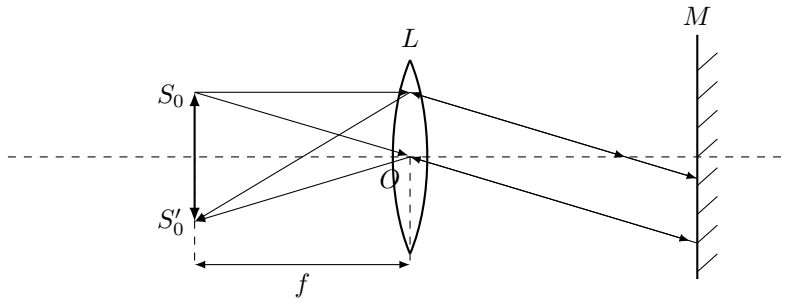


Fig. 8.01(a)

3. Measurement of the focal length of a convex lens by the two-image method

The two-image method is also called the conjugate method. According to the laws of image formation by a convex lens, if the distance D between the object and the image screen is kept unchanged and $D > 4f$, then when the convex lens is moved between the object and the screen two images can be formed, as shown in Fig. 8.01(b). S_0 is the object screen and P the image screen; 1 denotes the object, 2 the small image and 3 the large image. When the lens is at position x_1 it forms an inverted, magnified real image; when the lens is at position x_2 it forms an inverted, reduced real image. Let the absolute value of the distance between these two positions be d, and let the distance between position x_2 and the image screen be v_2. From Fig. 8.01(b), for position x_1,

$$u_1 = D - d - v_2, \qquad v_1 = d + v_2,$$

and for position x_2,

$$u_2 = D - v_2.$$

Using the lens imaging formula (1), one can show that the focal length of the lens is

$$f = \frac{D^2 - d^2}{4D}. \tag{2}$$

In formula (2) the experimental condition $D > 4f$ must be satisfied.

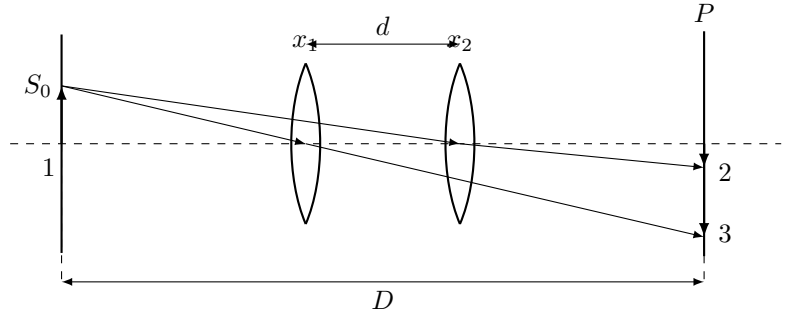


Fig. 8.01(b)

Usually, in measuring the focal length of a convex lens, the object, the lens holder and the screen are placed on an optical bench. However, the optical centre of the lens and the reading line on the lens holder base may not lie in the same plane, and hence a certain uncertainty exists. When the two-image method is used, $d = |x_2 - x_1|$, which eliminates the systematic error

introduced by the possible non-coplanarity of the optical centre of the lens and the reading line on the lens holder base. Of course, when the light source and the screen are not at the positions of the base reading lines, the value of the distance D must still be corrected.

Experiment 8.02 Measurement of the focal length of a concave lens by two different methods

[**Apparatus and materials available**] Optical bench, convex lens, concave lens, object screen, image screen, plane mirror, etc.

[**Requirements**]

(1) Write down the principles and formulas of two different methods for measuring the focal length of a concave lens.

(2) Draw the experimental ray diagrams.

(3) Measure and calculate the focal length of the concave lens.

Solution 8.02 1. Measurement of the focal length of a concave lens by the object-distance and image-distance method

First use the convex lens L_1 to form a reduced, inverted real image $A'B'$ of the object AB, and then place the concave lens L_2 under test between the convex lens L_1 and the image $A'B'$, as shown in Fig. 8.02(a). If $O'A' < |f_c|$ (where f_c is the focal length of the concave lens), then the beam that has passed through L_1, after refraction by L_2, still forms a real image $A''B''$. Note that for L_2 the image $A'B'$ is a virtual object and $A''B''$ is its real image, so that $u_2 = -O'A'$ and $v_2 = O'A''$. Substituting u_2 and v_2 into the lens focal-length formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v},$$

one can calculate the focal length f_c of the concave lens. The value of f_c is negative.

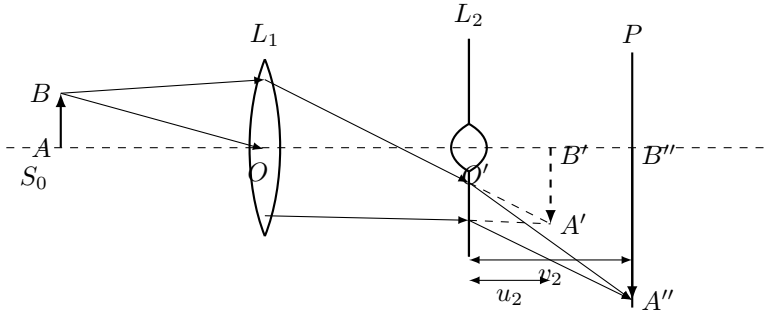


Fig. 8.02(a)

In measuring the focal length of a concave lens, note the following. (1) Before the experiment, adjust the lenses so that the convex lens and the concave lens are coaxial and their principal axes are parallel to the scale of the optical bench. (2) Take care that the difference between the virtual object distance and the distance of the real image formed is not too small, so as to avoid too few significant figures. (3) Take care whether the optical centre of the lens and the screen lie on the reading line of the base; this can be checked by turning the lens through 180°, and the error can be eliminated or reduced by readjusting for a sharp image.

2. Measurement of the focal length of a concave lens by the autocollimation method

A concave lens is a diverging lens. If a parallel beam is to be obtained with a concave lens, a convex lens must be used. First the convex lens L_1 forms an image of the point light source located at S_0 at S_0'. Record the reading of the position of S_0' on the scale of the optical bench, as shown in Fig. 8.02(b). If the distance OS_0' from the optical centre O of L_1 to S_0' satisfies $OS_0' > |f_c|$, then place the concave lens L_2 under test and the plane mirror M between L_1 and S_0'. When L_2 is moved so that the distance from the optical centre O' of L_2 to S_0' is $O'S_0' = |f|$, the beam emitted by the point source at S_0 becomes parallel after passing through L_1 and L_2, and after reflection by the plane mirror M it forms a clear real image again at S_0. Once the positions of S_0' and of L_2 (that is, O') are determined, the focal length f_c of the concave lens can be measured.

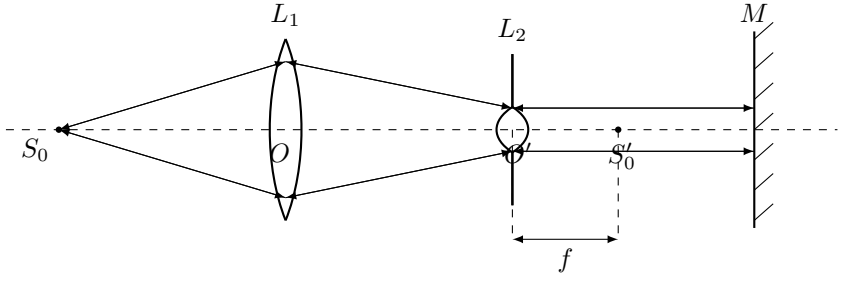


Fig. 8.02(b)

Experiment 8.03 Measurement of the focal length of a concave lens by the telescope method

[**Apparatus and materials available**] Optical bench, light source, ground glass, convex lens, concave lens, Kepler telescope (to be adjusted by the experimenter), object and image screen, etc.

[**Requirements**]

(1) Draw the ray diagram.

(2) Write down the experimental method and the experimental steps.

(3) Obtain the experimental results.

Solution 8.03 1. The experimental ray diagram is shown in the figure. Since a telescope is usually used to observe distant objects, which produce approximately parallel light, the real image formed by the convex lens is a virtual object for the concave lens. If this virtual object of the concave lens is located at the focal point of the concave lens, the emergent light will be parallel. In this way the focal length f_c of the concave lens can be determined.

Method and steps:

(1) Adjust the light path so that the light source, the object, the convex lens, the concave lens, the telescope and so on are at the same height and coaxial, as shown in the figure.

(2) First use the convex lens to form an image of the object, and record the reading of the image position on the surface of the optical bench. Turn the lens round, form a clear image again, read the two readings and take their average.

(3) Direct the telescope at an object outside the window and make the image clear, that is, focus it at infinity; the parallax must be eliminated.

(4) Insert the concave lens under test between the convex lens and the telescope. Move the concave lens and change the distance until a clear image of the object is seen through the telescope; record the position of the concave lens. It must also be turned round; repeat this operation and read the scale. Take the average of the two readings.

(5) The difference between the position of the image formed by the convex lens and the reading of the position of the optical centre of the concave lens is then the focal length of the concave lens. Since the telescope forms an image only of parallel light, this shows that the object image lies in the focal plane of the concave lens.

The experimental data are omitted.

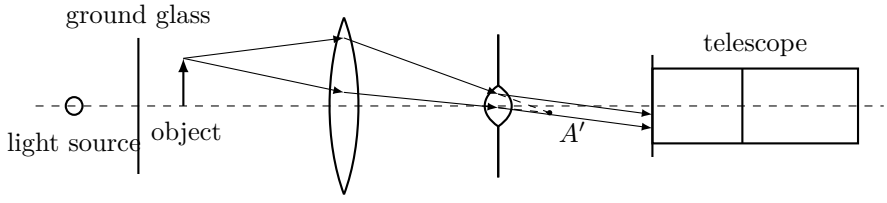


Fig. 8.03

Experiment 8.04 Measurement of the focal length of the objective of the telescope and of the lens of the collimator on a spectrometer

[**Apparatus and materials available**] Spectrometer, light source, plane reflecting mirror, ruler.

[**Requirements**]

(1) The optical axis of the telescope and the optical axis of the collimator on the spectrometer have already been adjusted perpendicular to the rotation axis of the instrument, and the two are coaxial; this adjustment is not to be made again.

(2) Draw the ray diagram of the measurement. For simplicity the object may be represented by a point object.

(3) Briefly describe the experimental procedure.

(4) Give the measurement results.

Solution 8.04 1. Measurement of the focal length of the telescope objective

(1) The experimental ray diagram is shown in Fig. 8.04(a).

(2) Procedure:

(i) Adjust the eyepiece so that the crosshairs are sharp, that is, the crosshairs lie in the focal plane of the eyepiece.

(ii) Adjust the objective and the platform so that the bright cross line forms a clear image, by reflection in the plane mirror, on the reticle on which the crosshairs lie. (The bright cross line is provided on the spectrometer telescope itself.)

(iii) Use a ruler to measure the distance from the telescope objective to the reticle several times and take the average value. This distance is the focal length f_1 of the telescope objective.

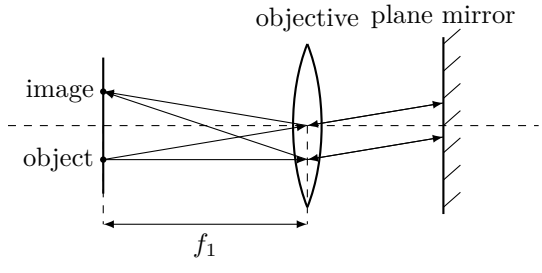


Fig. 8.04(a)

2. Measurement of the focal length of the collimator lens

(1) The ray diagram of the formation of the slit image on the reticle of the telescope is shown in Fig. 8.04(b). The telescope has previously been focused at infinity.

(2) Procedure:

Adjust the distance between the slit of the collimator and the lens so that the image of the slit observed through the telescope is sharpest. At this time the slit lies in the focal plane of the collimator. Use a ruler to measure the distance from the slit to the collimator lens several times and take the average value; this value is the focal length f_2 of the collimator lens.

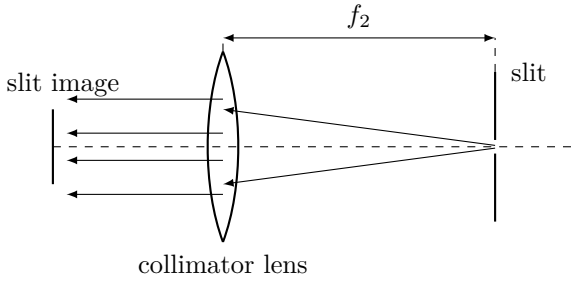


Fig. 8.04(b)

Experiment 8.05 Measurement of the focal length of a concave mirror, and of the radii of curvature of a convex mirror and a concave mirror

[**Apparatus and materials available**] Optical bench, screen with a "1"-shaped aperture, arrow (pin), convex lens, convex mirror, concave mirror, lamp, etc.

[**Requirements**]

(1) Measure the focal length of the concave mirror, draw the ray diagram and write down the measurement formula.

(2) Measure the radius of curvature of the concave mirror, draw the ray diagram and write down the measurement formula.

(3) Measure the radius of curvature of the convex mirror.

Solution 8.05 1. Measurement of the focal length of a concave mirror by the object-distance and image-distance method

The experimental light path is shown in Fig. 8.05(a). The light source is an incandescent lamp. The object screen is a white screen with a hollow "1"-shaped arrow cut in it; the arrow is above the optical axis. Another white half-screen is placed below the optical axis. With the object distance S fixed, the white half-screen is moved until an inverted real image can be observed; the distance from the white half-screen to the spherical mirror is then the image distance S'. Using the spherical mirror imaging formula

$$\frac{1}{S}+\frac{1}{S'}=\frac{1}{f},$$

the focal length f of the concave mirror can be obtained.

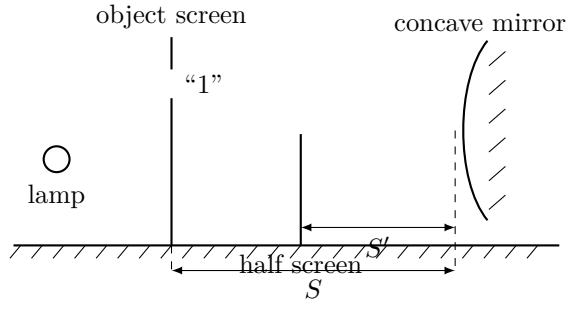


Fig. 8.05(a)

2. Measurement of the radius of curvature of a concave mirror by the parallax method

A pin and the spherical mirror under test are placed perpendicular to the optical bench, as shown in Fig. 8.05(b). A small electric bulb illuminates the pin from the side. The observer looks along the surface of the optical bench (the inclined scale surface of the bench) at the object and at the real image formed by reflection in the concave mirror. When the concave mirror is moved and the observer sees the inverted real image formed by reflection in the mirror coincide with the original object, the distance between the pin and the concave mirror can be read from the scale on the bench; this is the radius of curvature of the concave mirror.

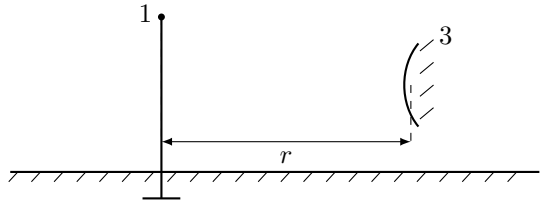


Fig. 8.05(b)

3. Determination of the radius of curvature of a convex mirror

A convex mirror diverges a light beam, so the parallax method described above is not applicable to a convex mirror. A converging beam must be used to illuminate the convex mirror. The light path formed by the light source (object screen), the convex lens and the convex mirror is shown in Fig. 8.05(c). Move the convex mirror so that the centre of curvature of the convex mirror lies exactly at the convergence point of the original converging beam; then the converging beam, after reflection by the convex mirror, returns along its original path and forms a real image at the position of the object screen. Then move the convex mirror so that the centre of the mirror surface is exactly at the convergence point of the converging beam; then the light, after reflection by the convex mirror, still forms an image at the position of the original object screen. In this way the distance r' through which the convex mirror is moved, taken with a negative sign, is the radius of curvature of the convex mirror:

$$r = -r'.$$

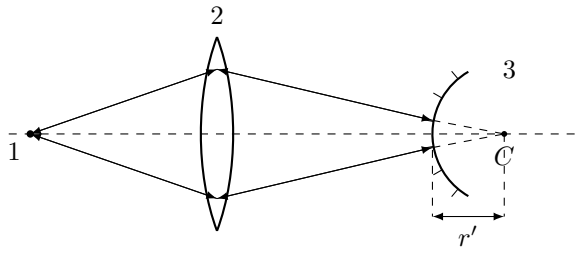


Fig. 8.05(c)

Experiment 8.06 Measurement of the focal length of a convex lens group

[**Apparatus and materials available**] One convex lens, one convex lens group, millimetre ruler, object screen, image screen, light source.

[**Requirement**] Use the magnification formula $V = f/x$ to measure the focal length f of the convex lens group, where V is the transverse magnification of the convex lens group, x is the distance from the object to the object-side focal point of the convex lens group, and f is the focal length of the convex lens group.

(1) Write down the measurement method and the steps.

(2) Draw a schematic diagram of the positions of the experimental measuring instruments.

(3) Record the original data; measure three times.

Solution 8.06 1. Principle and steps of the measurement

First the focal length of the single lens L is measured by the displacement method. Then the distance from the single lens to the object screen is made equal to the focal length of the lens, the lens group under test is placed behind it, and the imaging position, that is, the position of the image-side focal point, is found with the image screen. The single lens is removed, the imaging position is found again, the sizes of the image and of the object are measured, and the transverse magnification V is obtained. The difference between the two imaging positions is x'; substituting this into the measurement formula, the focal length f can be calculated.

The measurement formula is $V = f/x$, where V is the transverse magnification of the convex lens group, x is the distance from the object to the object-side focal point of the lens group, and

$$f = Vx = \frac{y'}{y}x',$$

where y' is the size of the image, y the size of the object, $V = y'/y$, and x' is the difference between the two imaging positions; it is more convenient to measure x'.

2. Diagram of the measuring light path and arrangement

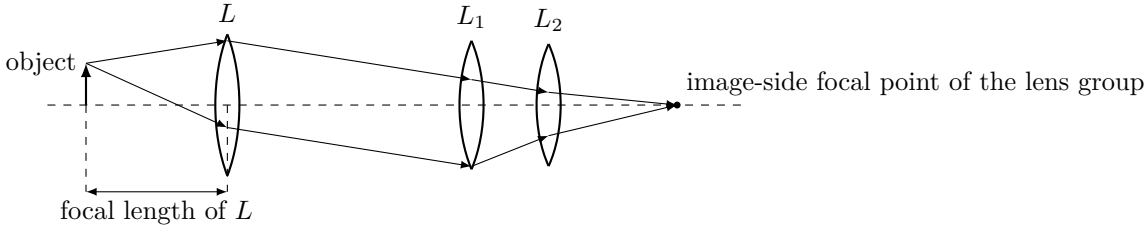


Fig. 8.06(a)

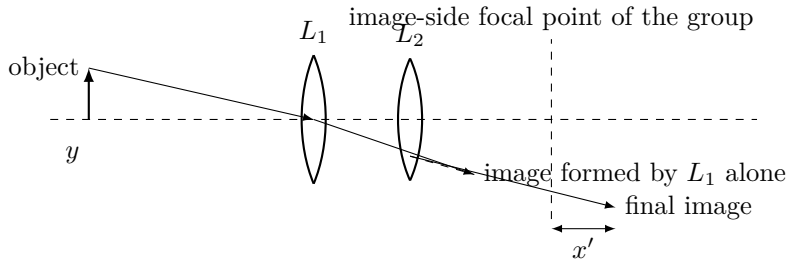


Fig. 8.06(b)

3. Example of experimental data

Table 8.06-1 Two-image method for measuring the focal length of the convex lens

Number	Object–screen distance D/cm	Displacement d/cm	Focal length/cm
1	52.36	23.29	10.51
2	52.35	23.56	10.48
3	52.36	22.88	10.59
Average	52.36	23.24	10.52

Table 8.06-2 Measurement of the focal length of the lens group

Number	Object height y/cm	Image height y'/cm	Magnification V	Displacement x'/cm	Focal leng
1	1.62	0.83	1.952	3.70	7.
2	1.63	0.80	2.038	3.68	7.
3	1.62	0.78	2.077	3.57	7.
Average	1.623	0.803	2.022	3.65	7.

The result is a focal length $f = 7.38\,\text{cm}$ for the convex lens group.

Comment. This problem uses the magnification formula to measure the focal length of a lens group, which is a fairly typical method in the measurement of the focal length of lens groups.

Experiment 8.07 Measurement of the focal length of a convex lens

[**Apparatus and materials available**] Optical bench (with a short scale only near the light source, and no scale on the rest), light source, object screen, image screen (with a graduated scale on it), convex lens under test.

[**Requirements**]

(1) Draw the ray diagram and briefly write down the principle of the measurement and the formula.

(2) Measure the focal length of the convex lens.

Solution 8.07 1. Principle of the experiment

Since the optical bench has graduations only on the section near the object screen, and the image screen has graduations, one can use only the readings of the large image and the small image on the image screen, together with the displacement of the lens read on a section of the guide rail of the bench, to calculate the focal length of the convex lens.

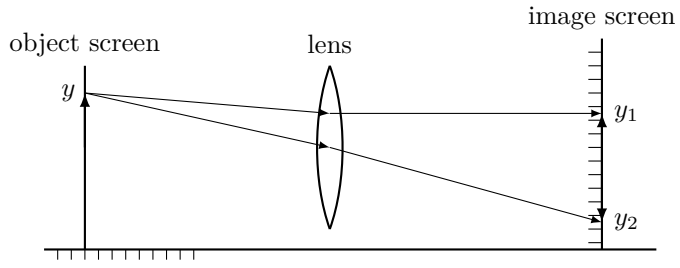


Fig. 8.07(a)

2. Experimental method

(1) Find the object height y by the conjugate method:

$$y = \sqrt{y_1 y_2}, \tag{1}$$

where y_1 and y_2 are the image heights in the two conjugate imagings.

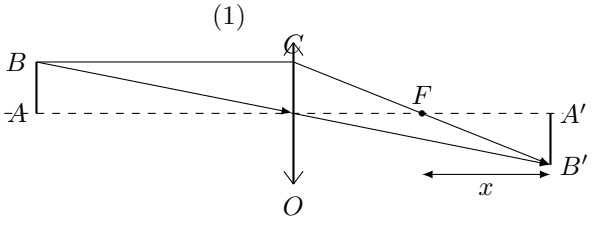
(2) The first imaging is shown in Fig. 8.07(b)(1); we have

$$\triangle OCF \sim \triangle A'B'F,$$

so

$$\frac{x}{f} = \frac{A'B'}{AB} = \frac{A'B'}{y}, \tag{1}$$

where x is the length FA'.



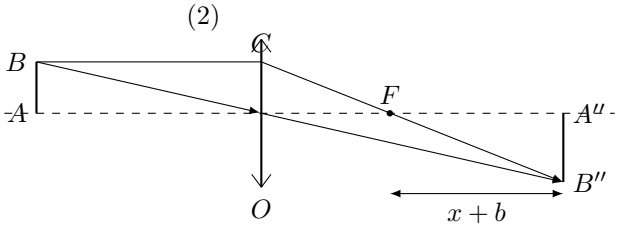


Fig. 8.07(b)

(3) The second imaging is shown in Fig. 8.07(b)(2); we have $\triangle OCF \sim \triangle A''B''F$, so

$$\frac{x+b}{f} = \frac{A''B''}{AB} = \frac{A''B''}{y}, \tag{2}$$

where b is the displacement of the convex lens in the two imagings.

From (1) and (2),

$$\frac{A'B'}{y} - \frac{A''B''}{y} = \frac{x}{f} - \frac{x+b}{f},$$

so

$$f = \frac{by}{A''B'' - A'B'}.$$

Comment. A lens is a basic element of all kinds of optical instruments. Mastering the laws of lens imaging and learning to analyse and adjust light paths is useful for understanding the construction and the correct use of optical instruments. The focal length is an important characteristic parameter of a lens; lenses of suitable focal length must be chosen on different occasions, so the measurement of the focal length of a lens is very important.

Experiment 8.08 Measurement of the focal length f and the radius of curvature R of a thin symmetric biconcave lens, and calculation of the refractive index n from them

[**Apparatus and materials available**] Optical bench, light source, pinhole object screen, image screen, thin convex lens, thin symmetric biconcave lens under test whose two surfaces have the same radius of curvature, one fine thread, white paper, adhesive tape, scissors, etc.

[**Requirements**]

(1) Briefly describe the measuring method and write down the measuring formula.

(2) Measure the focal length f of the thin biconcave lens.

(3) Measure the radius of curvature R of the thin biconcave lens.

(4) From the focal length f and the radius of curvature R of the thin biconcave lens, find the value of the refractive index n.

Solution 8.08 1. In this experiment the two-image method is first used to measure the focal length f_1 of the convex lens; the ray diagram is as shown in Fig. 8.01(b). Then the concave lens is placed between the convex lens and the image. The image of the convex lens is a virtual object for the concave lens; the focal length f_2 of the concave lens is measured by the object-distance and image-distance method. The radii of curvature R_1 and R_2 of the biconcave lens can be measured by the method of Fig. 8.05(b). Applying the thin-lens focal-length formula

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right),$$

we obtain

$$n = 1 + \frac{1}{f\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)}.$$

2. Experimental results

Table 8.08-1 Measurement of the focal length of the convex lens

Number	Object–screen distance D/cm	Displacement d/cm	Focal length f_1/cm
1	40.00	10.39	
2	40.00	10.42	
3	40.00	10.36	
Average	40.00	10.39	9.33

Table 8.08-2 Focal length of the concave lens

Object distance/cm			Image distance/cm			Focal length/cm
S_4	S_3	$S_4 - S_3$	S_5	S_3	$S_5 - S_3$	f
42.50	35.00	7.50	47.40	35.00	12.40	−18.98

Table 8.08-3 Measurement of the radii of curvature of the concave lens (parallax method)

Position of the concave lens/cm	Position of the image of the left concave surface/cm	Position of the imag
5.00	27.76	

$$R_1 = -(27.76 - 5.00) = -22.76 \text{ (cm)}; \qquad R_2 = 28.89 - 5.00 = 23.89 \text{ (cm)}.$$

Hence the refractive index of the glass is

$$n = 1 + \frac{1}{f\left(\frac{1}{R_1} - \frac{1}{R_2}\right)} = 1 + \frac{1}{(-18.98)\left(-\frac{1}{22.76} - \frac{1}{23.89}\right)} = 1.614.$$

The radius of curvature of a concave mirror is twice its focal length, $R = 2f$. Therefore one may also use the collimator method or another method to measure the focal length of the concave mirror, and multiply it by 2 to obtain the radius of curvature.

Comment. The method described in this problem can measure the refractive index of the material of a concave lens. With a different line of thought, if the refractive index of the glass is known, the focal length of a biconcave lens can also be measured by the above method.

Experiment 8.09 Measurement of the refractive index of a liquid

[**Apparatus and materials available**] Opaque square box, ruler, set square, liquid under test (in a beaker).

[**Requirements**]

(1) Draw the ray diagram and derive the measuring formula.

(2) Calculate the result.

Solution 8.09 **1.** The ray diagram is shown in Fig. 8.09(a).

Make a mark on the bottom edge of the square box (for example, put a coin there) and observe it with the eye along the upper edge of the box. When there is no water (liquid under test), the angle α is

$$\sin\alpha = \frac{d}{\sqrt{h^2 + d^2}},$$

and when the liquid under test is present, the angle β is

$$\sin\beta = \frac{s}{\sqrt{s^2 + l^2}}.$$

By the refraction formula

$$n_{\text{water}} \sin\alpha = n_{\text{air}} \sin\beta. \tag{1}$$

In formula (1), $\sin\alpha = \dfrac{d}{\sqrt{h^2+d^2}}$, $\sin\beta = \dfrac{s}{\sqrt{s^2+l^2}}$, and $n_{\text{air}} = 1$, so the refractive index is

$$n_{\text{water}} = \frac{\sin\beta}{\sin\alpha} = \frac{s}{\sqrt{s^2+l^2}} \cdot \frac{\sqrt{h^2+d^2}}{d}. \tag{2}$$

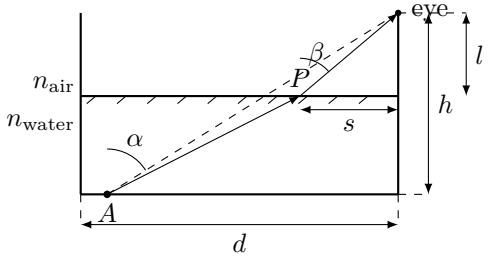


Fig. 8.09(a)

2. The second method is as follows. Press a ruler tightly against the inner wall of the square box on the opposite side, and place the box at a certain position beside a support or the edge of a set square. Bring the eye close to the support or the set square and observe. When the box contains water, the eye observes the point A at the bottom edge of the box, and the light path is the straight line CBA. Then pour all the water out of the box and observe with the eye from the original position C; the light path is the straight line CBA'.

From Fig. 8.09(b),

$$n_{\text{water}} = \frac{\sin i}{\sin\alpha} = \frac{D/A'B}{D/AB} = \frac{AB}{A'B} = \frac{\sqrt{D^2+h^2}}{\sqrt{D^2+(h-h')^2}}. \tag{3}$$

In formula (3), h' is the height AA'.

The experimental data are omitted.

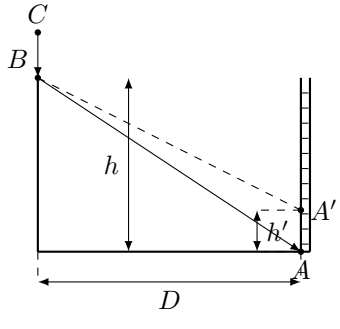


Fig. 8.09(b)

Experiment 8.10 Determination of the refractive index of a craft glass ball

[**Apparatus and materials available**] Optical bench, one small bulb (12 V), one power supply, object screen, image screen, stage, steel tape, vernier caliper, glass ball under test, ground glass, etc.

[**Requirements**]

(1) Write down the experimental method and derive the measuring formula.

(2) Briefly describe the experimental steps.

(3) Record the experimental data and calculate the result.

(4) Explain the main sources of experimental error.

Solution 8.10 1. Experimental method and derivation of the measuring formula

With the apparatus provided, the object can be made to form an image on the screen; the object distance, the image distance and the diameter of the ball are measured separately and substituted into the measuring formula. The experimental light path is as shown in the figure.

Let $SO = u$ and $OS' = v$.

By the law of refraction,

$$n = \frac{\sin i}{\sin \varphi} = \frac{\sin i'}{\sin \varphi'}. \tag{1}$$

By the sine rule,

$$\frac{u}{\sin i} = \frac{r}{\sin \alpha}, \qquad \frac{v}{\sin i'} = \frac{r}{\sin \beta}.$$

By the geometrical relations,

$$\varphi + \varphi' = (i - \alpha) + (i' - \beta). \tag{2}$$

Under paraxial conditions the angles α, β, i, i', φ, φ' are all small, so

$$n = \frac{i}{\varphi}, \qquad n = \frac{i'}{\varphi'}, \qquad \frac{i}{\alpha} = \frac{u}{r}, \qquad \frac{i'}{\beta} = \frac{v}{r}. \tag{3}$$

Substituting (3) into (2), we obtain

$$n = \frac{\alpha u v}{\alpha u v - r(u + v)}. \tag{4}$$

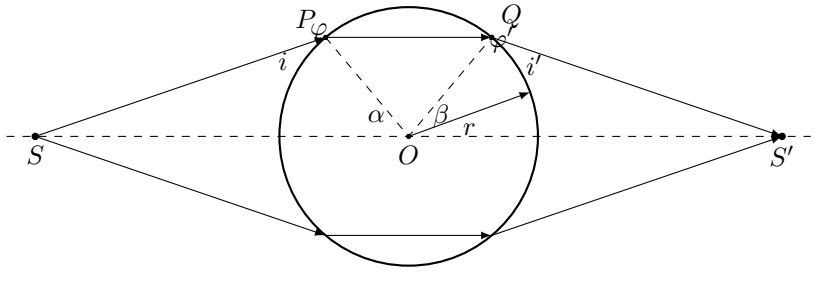


Fig. 8.10

2. Experimental steps

Adjust the apparatus on the optical bench so that the light source, the object screen, the craft glass ball and the image screen are at the same height and coaxial, and the light emitted by the object screen is imaged clearly on the image screen by the glass ball (lens). Measure the object distance and the image distance, and use a vernier caliper to measure the diameter $2r$ of the glass ball.

3. The experimental data are omitted. The main error of the experiment comes from the inaccurate determination of the image plane.

Experiment 8.11 Measurement, with a spectrometer, of the angle of incidence on a prism and the angle of refraction of the emergent ray when the condition of minimum deviation is satisfied, and measurement of the minimum deviation angle

[**Apparatus and materials available**] Spectrometer, prism, sodium lamp.

[**Requirements**]

(1) Measure the minimum deviation angle δ_0 of the prism and the corresponding angles of incidence and emergence.

(2) Briefly explain the measuring method and draw a simple schematic diagram.

(3) Give the measurement results.

Note: the spectrometer has already been adjusted, so measurements can be made directly. On no account adjust the levelling screws of the telescope, of the collimator or of the stage, lest the measurement conditions be destroyed.

Solution 8.11 **1.** The angle of incidence is $55°30'0''$, the angle of refraction is $55°30'0''$, and the minimum deviation angle is $51°0'0''$.

2. The experimental light path is as shown in the figure.

3. Minimum deviation angle: place the prism on the stage, find the spectral line, and rotate the stage to change the angle of incidence while the telescope follows the emergent spectral line. If the line moves in the direction in which the deviation angle decreases, the stage is being rotated in the correct direction; when a certain position is reached, the spectral line no longer moves forward but begins to move in the opposite direction. This turning position is the position of minimum deviation. Measure its azimuth, then measure the azimuth of the incident light; subtracting the two gives the minimum deviation angle.

4. Angle of incidence: measure the azimuths of the incident light and of the normal to the corresponding face of the prism, and subtract them to obtain the angle between them; the supplement of this angle is the angle of incidence.

Angle of refraction: measure the azimuths of the emergent light from the prism and of the corresponding normal, and subtract them to obtain the angle between them, which is the angle of refraction.

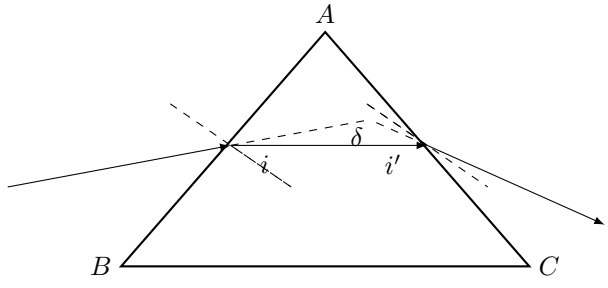


Fig. 8.11

Experiment 8.12 Measurement of the refractive index of prism material

[**Apparatus and materials available**] Prism, set square, white paper, etc.

[**Requirements**]

(1) The optical surfaces must not be measured in direct contact with paper, and the optical surfaces must not be touched with the hands.

(2) Draw the measuring ray diagram and write down the measuring formula.

(3) Measure and calculate the result.

Solution 8.12 **1.** The ray diagram is as shown in the figure.

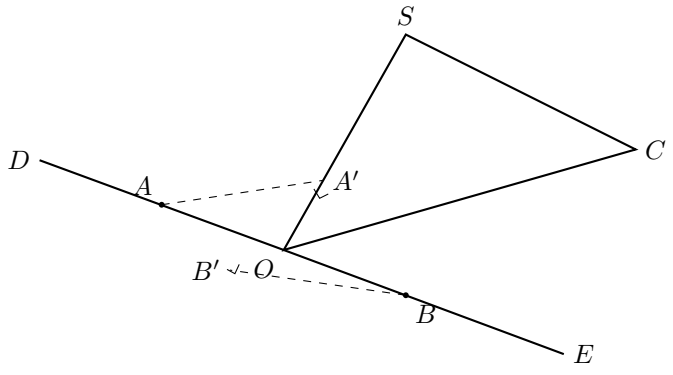


Fig. 8.12

2. Experimental method

The vertex O of the prism lies on the straight line DE. The prism is rotated about O and observed from the direction of D until the other vertex S coincides with the straight line DO (that is, lies on the same straight line). Take $OA = OB$; then the refractive index of the prism material is

$$n = \frac{\sin r}{\sin i} = \frac{BB'}{AA'},$$

where BB' is the length of the perpendicular from B to the straight line OD, and AA' is the length of the perpendicular from A to the straight line OS.

By the above graphical construction the refractive index n of the prism material can be obtained. The experimental data are omitted.

Experiment 8.13 Measurement of the refractive index of a prism (minimum deviation method)

[**Apparatus and materials available**] Spectrometer, mercury lamp, prism.

[**Requirements**]

(1) Measure the minimum deviation angle $\delta_{\min}$ of the prism.

(2) Measure the apex angle of the prism.

(3) Calculate the refractive index of the prism for sodium light.

Note: the spectrometer has already been adjusted, so measurements can be made directly.

Solution 8.13 **1.** The experimental light path is as shown in Fig. 8.11.

2. The scale division of the spectrometer is $1'$, and the prism under test is No. 7.

The apex angle of the prism measured with the spectrometer is $A = 60°1.8'$.

3. $\delta_{\min}$ is the minimum deviation angle. For an equilateral prism, $\delta_{\min}$ is the angle between the incident ray and the emergent ray when the angle of incidence is equal to the angle of emergence. The refractive index of the prism for sodium light is

$$n = \frac{\sin\dfrac{\delta_{\min} + A}{2}}{\sin\dfrac{A}{2}},$$

where A is the apex angle of the prism.

The minimum deviation angle of the prism was measured for $\lambda = 577.0\,\text{nm}$ (yellow light); the data are given in the table below.

Table 8.13 Measurement of the minimum deviation angle

Number	Position of the incident light		Position of the refracted light	
	φ_0'	φ_0''	φ_1'	φ_1''
1	163°58′	344°0′	225°9′	45°7′
2	163°57′	343°59′	225°8′	45°8′
3	163°58′	344°0′	225°9′	45°8′
4	163°57′	344°0′	225°9′	45°7′
5	163°58′	343°59′	225°8′	45°8′
Average	163°57.6′	343°59.6′	225°8.6′	45°7.6′

From the data in the table, calculate the minimum deviation angle $\delta_{\min}$:

$$\delta_{\min} = \frac{(\varphi_1' - \varphi_0') + (\varphi_1'' - \varphi_0'')}{2} = \frac{61°11' + 61°8'}{2} = 61°9.5'.$$

The refractive index of the prism material is

$$n = \frac{\sin\dfrac{A+\delta_{\min}}{2}}{\sin\dfrac{A}{2}} = \sin\frac{60°1.8' + 61°9.5'}{2} \Big/ \sin\frac{60°1.8'}{2} = 1.7415 \qquad (\lambda = 577.0\,\text{nm}).$$

Experiment 8.14 Measurement of the refractive index of prism material by the grazing incidence method

The principle formula for the refractive index is

$$n = \sqrt{1 + \left(\frac{\cos A + \sin\varphi}{\sin A}\right)^2},$$

where A is the apex angle and φ is the minimum emergence angle.

[**Apparatus and materials available**] Spectrometer, prism under test, ground glass.

[**Requirements**]

(1) Explain the principle of the measurement.

(2) Draw the ray diagram.

(3) Measure the refractive index of the prism.

Solution 8.14 1. The principle of measuring the refractive index of a medium by the grazing incidence method is as shown in the figure. The medium under test is made into a prism; an extended source (a large ground glass illuminated by a sodium lamp) illuminates the refracting face AB of the prism, and the telescope is directed at the other refracting face AC for observation. At the interface AB the angles of incidence of the rays a, b, c in the figure increase successively, and ray c is the grazing ray (angle of incidence 90°); hence the corresponding angle of refraction is the critical angle i_c. No ray inside the prism can have an angle of refraction greater than i_c. At the interface AC the angles of emergence of the rays a, b, c decrease successively, and the angle of emergence i' of ray c is the smallest. Therefore the field of view observed through the telescope is half bright and half dark, with a clear boundary between light and dark. This boundary

corresponds to ray c, whose angle of emergence is i'. It can be shown that the refractive index n of the prism is related to the prism angle A and the minimum emergence angle i' as follows:

$$n = \sqrt{1 + \left(\frac{\sin i' + \cos A}{\sin A}\right)^2}.$$

Using the spectrometer, measure the apex angle A of the prism and the angle of emergence i' corresponding to the boundary between light and dark; the refractive index of the prism material then follows from the above formula.

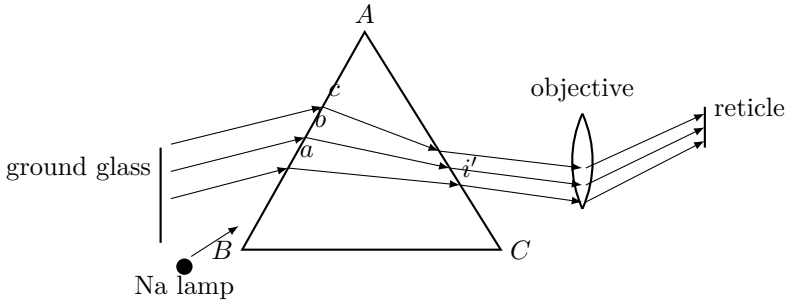


Fig. 8.14

2. The wavelength of sodium light is $\lambda = 589.3\,\text{nm}$. The scale division of the spectrometer is $30''$. With the spectrometer the apex angle of the prism was measured as $A = 60°11'8''$, and the angle of emergence (minimum) as $i' = 47°34'14''$. Substituting into the formula gives $n = 1.7399$ (for a wavelength of 589.3 nm). The apex angle A and the minimum emergence angle i' are both obtained by averaging several measurements.

Experiment 8.15 Measurement of the wavelength of laser light

[**Apparatus and materials available**] One semiconductor laser, tape measure, steel ruler (with 0.5 mm graduations), retort stand, optical bench, white paper screen.

[**Requirements**]

(1) Briefly describe the experimental steps and derive the formula.

(2) Record the experimental data and calculate the wavelength of the laser light.

Note: when a laser beam is incident obliquely on the graduations of a steel ruler, diffraction occurs. Let the angle between the incident laser beam and the surface of the steel ruler be θ_0, and the angle between the diffracted ray of order K and the steel ruler be θ_K; then the diffraction spots satisfy the relation

$$d(\cos\theta_0 - \cos\theta_K) = K\lambda,$$

where λ is the laser wavelength and d the grating constant, that is, the distance between adjacent graduations of the steel ruler.

Solution 8.15 **1.** A laser beam is directed at the end of the steel ruler. Part of the light passes directly above the ruler and falls on the observation screen at the point $-S_0$; the rest is reflected from the surface of the ruler onto the screen. Besides the reflected bright spot at S_0, which is symmetric to $-S_0$, a whole series of bright spots S_1, S_2, S_3, S_4, ... can be seen on the screen. This is because the ruler bears scratches (the spacing of the scratches is $d = 0.5\,\text{mm}$). Light is reflected from the many smooth surfaces between two scratches. If these reflected beams have the same phase (that is, crest meets crest and trough meets trough), they superpose and

reinforce each other, forming a bright spot; otherwise they cancel and the light is weakened. As shown in Fig. 1, the laser beam is incident on the end face of the steel ruler, and the reflected light forms the diffraction pattern.

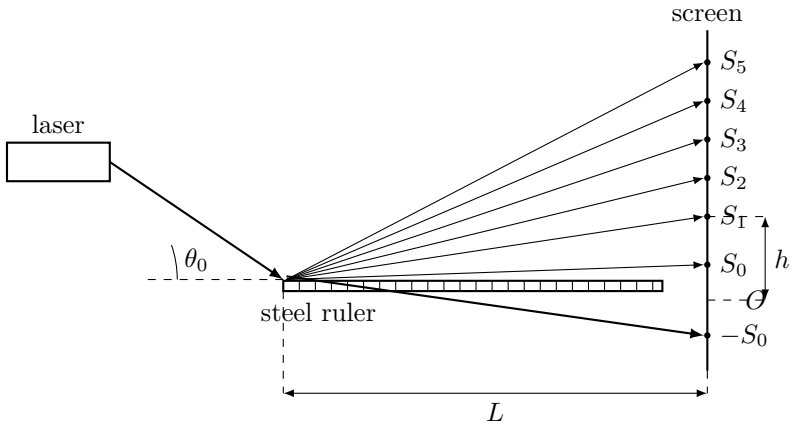


Fig. 8.15(a)

From Fig. 2 it can be seen that the path difference (called the optical path difference) accumulated by light emitted from a point A of the source and reflected at two adjacent smooth surfaces B and B' in reaching the point C on the screen is

$$\Delta = AB'C - ABC = DB' - BD' = d(\cos\alpha - \cos\beta) \tag{1}$$

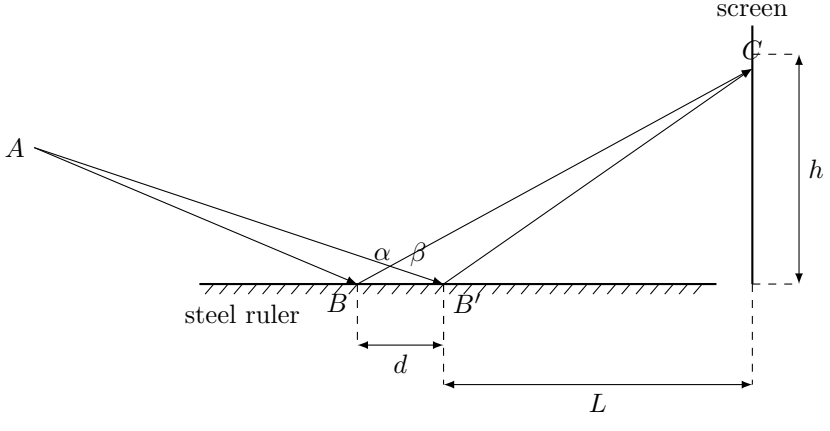


Fig. 8.15(b)

If Δ happens to be zero or an integral multiple of the wavelength λ, the phases of these reflected beams are the same, and a bright spot appears at the point C on the screen. Obviously, at $\beta = \alpha$ one has $\Delta = 0$, and this is the bright spot at S_0. At S_1, S_2, S_3, S_4, ... one must have $\Delta_1 = \lambda$, $\Delta_2 = 2\lambda$, $\Delta_3 = 3\lambda$, $\Delta_4 = 4\lambda$, ...; hence from (1) it follows that

$$d(\cos\alpha - \cos\beta_1) = \lambda, \tag{2}$$

$$d(\cos\alpha - \cos\beta_2) = 2\lambda, \tag{3}$$

$$d(\cos\alpha - \cos\beta_3) = 3\lambda, \tag{4}$$

$$d(\cos\alpha - \cos\beta_4) = 4\lambda. \tag{5}$$

Here $d = 0.500\,\text{mm}$ is known. Therefore, provided that α and β_1, β_2, β_3, β_4, ... are measured, the wavelength λ can be calculated from each of these equations. In the experiment the ruler is set perpendicular to the screen, so that

$$\tan\beta = h/L. \tag{6}$$

Here L is the distance from the end of the ruler to the screen, and h is the distance from each bright spot to the point O, where O lies midway between S_0 and $-S_0$. By measuring the distances of the bright spots one obtains the values of β_k; for the bright spot S_0 the angle β is just α.

2. Experimental steps

(1) Place the steel ruler on the optical bench in a horizontal position. Fix the laser on a retort stand and let the semiconductor laser beam fall on the end face of the steel ruler at an angle of 2° to the surface of the bench. Place the white screen at a distance of about 2 m from the steel ruler, with the ruler perpendicular to the screen. (A wall may be used instead of the screen.)

(2) Paste a strip of white paper on the screen (or wall) and mark on it the positions of the bright spots $-S_0$, S_0, S_1, S_2, S_3, S_4, etc. (Take care to identify the position of S_0 correctly and never mistake it; the position of S_0 can be identified by allowing the laser to fall on a part of the steel ruler that bears no scratches.)

3. Use the tape measure to find the distance L from the front end of the steel ruler to the wall.

4. Take the paper strip down from the wall and take the midpoint of $-S_0$ and S_0 as the point O. Measure the distances of the points S_0, S_1, S_2, S_3, S_4, ... from O (these are the values of h), calculate α and β from (6) (the β corresponding to S_0 is α), and then calculate λ from each of equations (2)–(5); take the average as the wavelength of the semiconductor laser.

Comment. In this experiment the graduations of a steel ruler serve as a "grating", so the experiment belongs to the class of "reflection grating" diffraction experiments.

Experiment 8.16 Making a double slit, measuring its interference fringes and finding the slit separation d

[**Apparatus and materials available**] One helium–neon laser ($\lambda = 633\,\text{nm}$), light screen, support, tape measure, coordinate paper, card paper, two double-edged razor blades.

[**Requirements**]

(1) Make a suitable double slit and write down the procedure.

(2) Draw the arrangement of the apparatus and the ray diagram, and write down the relevant formulas.

(3) Perform the experiment and find the separation d of the two slits.

Solution 8.16 **1.** Place the two razor blades together with one or two sheets of paper between them, and draw them lightly once across the card paper supplied, so as to make a double slit of width d of about 0.3–0.5 mm.

2. The experimental arrangement is shown in Fig. 1.

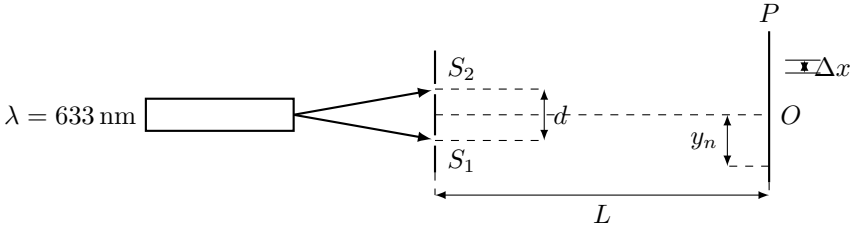


Fig. 8.16(a)

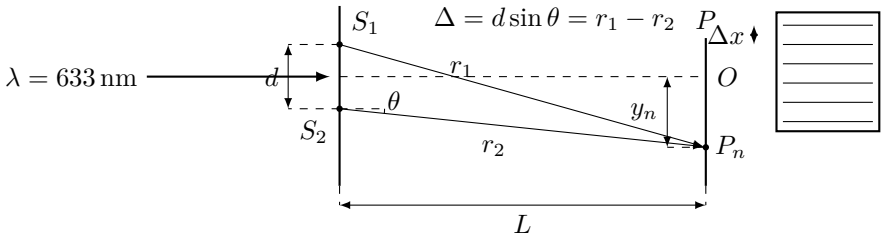


Fig. 8.16(b)

As shown in Fig. 1, place the laser, the double slit and the light screen on the table, and let the narrow laser beam fall exactly on the double slit; suitable interference fringes then appear on the screen.

In the double-slit interference experiment the slit separation is

$$d = \frac{\lambda L}{\Delta x}, \tag{1}$$

where Δx is the width of an interference fringe, L the distance from the slits to the screen, and λ the laser wavelength. Generally speaking, the distance y_n between bright (or dark) fringes should be measured over as large a length as possible, so that the measurement error is small. Hence for the positions of the bright fringes

$$d = \frac{n\lambda L}{y_n}, \qquad n = 0, 1, 2, 3, \ldots, \tag{2}$$

and for the positions of the dark fringes

$$d = \frac{\left(n + \dfrac{1}{2}\right)\lambda L}{y_n}. \tag{3}$$

In the experiment one may measure the distance y_n from the zeroth-order to the n-th order bright fringe, substitute it into formula (2) and obtain the slit separation d; in general n should be not less than 10, and the measurement must be repeated several times and averaged.

Experiment 8.17 Measurement of the diameter of a fine thread by the interference method

[**Apparatus and materials available**] Reading microscope, two flat glass plates, sodium lamp (sodium wavelength 589.3 nm), the fine thread under test, adhesive tape.

[**Requirements**]

(1) Briefly describe the principle of the experiment and write down the calculation formula.

(2) Write down the main experimental steps.

(3) Measure the data and obtain the diameter of the fine thread.

Solution 8.17 1. As shown in the figure, two glass plates are laid one on top of the other, and the fine thread whose diameter is to be measured is clamped between them at one end, so that an air wedge is formed between the two plates. When monochromatic light falls on it perpendicularly, interference occurs. Since the places of equal path difference are straight lines parallel to the line of intersection of the two plates, the fringes of equal thickness are a set of straight bright and dark lines parallel to the line of intersection. The interference arises because,

when monochromatic light falls perpendicularly, the beams 1 and 2 reflected from the upper and lower surfaces of the air layer meet at the upper surface and interfere. Then the thickness of the air wedge at the m-th dark fringe is

$$d = m\frac{\lambda}{2}. \tag{1}$$

From (1), when $m = 0$ one has $d = 0$, that is, the zeroth-order dark fringe lies at the line of intersection of the two glass plates. If $m = N$ fringes appear at the position of the fine thread, then the diameter of the thread is $d = N\frac{\lambda}{2}$.

2. Experimental steps

(1) Lay the two glass plates one on top of the other and clamp the fine thread between them at one end, so that an air wedge is formed between the plates. Place the glass wedge on the glass stage of the reading microscope.

(2) Adjust the inclination of the glass reflector below the reading microscope so that the field of view is brightest when observed through the eyepiece.

(3) Adjust the eyepiece until the crosshairs in its field of view are seen clearly; then lower the microscope tube until it is close to the glass specimen, and afterwards raise it slowly until the interference fringes are observed. Then finely adjust the inclination of the glass reflector below the microscope so that the fringes become clearer.

(4) Read the number N of fringes appearing at the fine thread, substitute it into formula (1) with $\lambda = 589.3\,\text{nm}$, and obtain the diameter d of the thread.

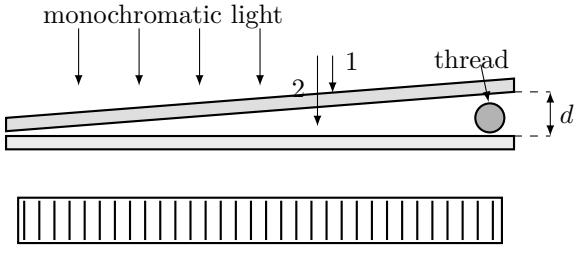


Fig. 8.17

Experiment 8.18 Measurement of the radius of curvature of a spherical surface with Newton's rings

[**Apparatus and materials available**] Reading microscope, sodium lamp ($\lambda = 589.3\,\text{nm}$), glass plate, plano-convex lens and flat glass (the specimen under test).

[**Requirements**]

(1) Write down the formula for measuring the radius of curvature of a concave lens with Newton's rings.

(2) Briefly describe the experimental steps.

(3) Measure and calculate the radius of curvature of the concave lens.

Solution 8.18 1. A plano-convex (or plano-concave) lens of very long focal length is placed on a flat glass plate, with its convex surface in contact with the plane, so that a thin air layer is formed near the point of contact. Monochromatic light is incident from the front on the thin air layer; the light reflected from its upper and lower surfaces gives two coherent beams with a fixed path difference, and equal-thickness interference produces a set of bright and dark rings centred on the point of contact, namely Newton's rings, as shown in Figs. (a) and (b).

The diameter d_k of the k-th dark ring of Newton's rings is related to the radius of curvature R of the convex surface of the plano-convex lens and to the wavelength λ by

$$d_k = \sqrt{4kR\lambda}. \tag{1}$$

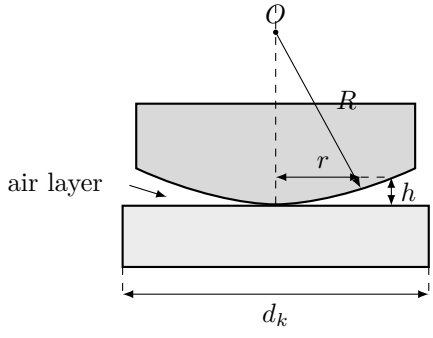


Fig. 8.18(a)

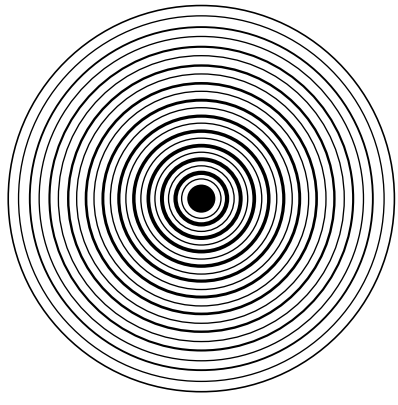


Fig. 8.18(b)

If λ is known, then by measuring d_k and reading off the order k one can find the radius of curvature R.

Because the contact between the convex lens and the flat glass deforms the glass and so on, the centre of the rings is not necessarily a dark point, which makes the order k difficult to determine correctly. In practice one usually measures the diameters d_k and d_{k+1} of two adjacent dark rings; from (1),

$$\Delta(d^2) = d_{k+1}^2 - d_k^2 = 4R\lambda. \tag{2}$$

To reduce the measurement error it is very necessary to make $\Delta(d^2)$ as large as possible. Usually the difference of the squares of the diameters of two dark rings separated by m rings is used in the calculation:

$$\Delta(d^2)_m = d_{k+m}^2 - d_k^2 = 4mR\lambda. \tag{3}$$

The minus sign added here indicates that the order number of an outer ring is smaller than that of an inner ring (air gap between a plano-concave lens and a plane mirror).

2. Experimental data and calculation

Table 8.18 Measurement of the diameters of Newton's rings

Ring order	x_{left}/mm	x_{right}/mm	d/mm	$d^2/(\text{mm})^2$	$\Delta(d^2)_{20}/(\text{mm})^2$
30	8.840	27.360	18.520	342.99	
29	9.004	27.192	18.188	330.80	
28	9.178	27.038	17.860	318.98	
27	9.330	26.868	17.538	307.58	
26	9.500	26.702	17.202	295.91	
10	12.848	23.352	10.504	110.33	232.66
9	13.128	23.062	9.934	98.68	232.12
8	13.430	22.750	9.320	86.86	232.12
7	13.756	22.442	8.686	75.44	232.14
6	14.110	22.084	7.974	63.58	232.33

From the table above,

$$\Delta(d^2) = 2.323 \times 10^{-4}\,\text{m}^2.$$

Therefore

$$R = 4.927\,\text{m}.$$

The following points must be noted in the measurement: (1) the ring number (order) of the interference fringes must not be miscounted; (2) during the measurement, vibrations of the table must be prevented from changing the interference rings; (3) in the measurement the crosshair must be aimed at the centre of the dark fringe of Newton's rings; (4) the screw-pitch error of the reading microscope must be eliminated; (5) in the experiment the measurement of a chord length may be used instead of the measurement of the diameter.

Comment. This experiment enables students to understand, from experimental observation, the principle of equal-thickness interference and the concept of localised interference, to learn the use of the reading microscope, and to process the data by the method of successive differences so as to reduce the measurement error.

Experiment 8.19 Measurement of the wavelength of light with a biprism

[**Apparatus and materials available**] Optical bench, sodium lamp, adjustable slit (its holder can be translated and the slit can be rotated), biprism and its holder, convex lens, micrometer eyepiece, etc.

[**Requirements**]

(1) Write down the principle and the calculation formula for measuring the wavelength of light in the biprism interference experiment.

(2) Assemble the optical elements on the optical bench and adjust the interference fringes until clear fringes can be seen through the micrometer eyepiece; measure the fringe spacing x.

(3) Measure the separation d of the two virtual sources by the two-image method.

(4) Measure the distance D from the plane of the two virtual sources to the screen (the reticle of the micrometer eyepiece) and calculate the wavelength of the sodium light.

Solution 8.19 1. The Fresnel biprism is a thin glass plate whose upper surface is worked into two wedge-shaped faces with a very small wedge angle, the ridge being perpendicular to the end face, as shown in Fig. 1. The principle of the interference experiment with a biprism is shown in Fig. 2. A sodium lamp illuminates the slit s (parallel to the ridge of the biprism). After the light wave emerging from the slit passes through the biprism P, its wavefront is divided into two parts, which propagate in different directions. These may be regarded as equivalent to the

cylindrical waves emitted by two virtual sources s_1 and s_2 that satisfy the coherence conditions. If an observation screen is placed anywhere in the region where the two waves overlap, bright and dark interference fringes are seen, with their direction parallel to the slit. If the distance d between s_1 and s_2, the distance D from the plane containing s_1 and s_2 to the observation screen, and the fringe width x are known, the wavelength λ of the light source is given by

$$\lambda = x\frac{d}{D}.$$

Usually D is several hundred millimetres, d is about 1–2 mm and x is about a few tenths of a millimetre. D may be measured with a metre rule, d by the two-image method with a convex lens and the micrometer eyepiece, and x with the micrometer eyepiece. The wavelength λ of the monochromatic light can then be found.

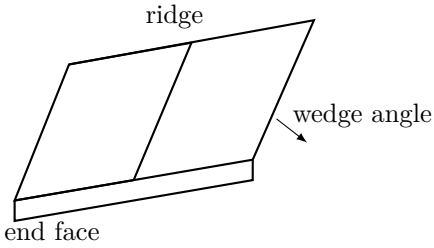


Fig. 8.19(a)

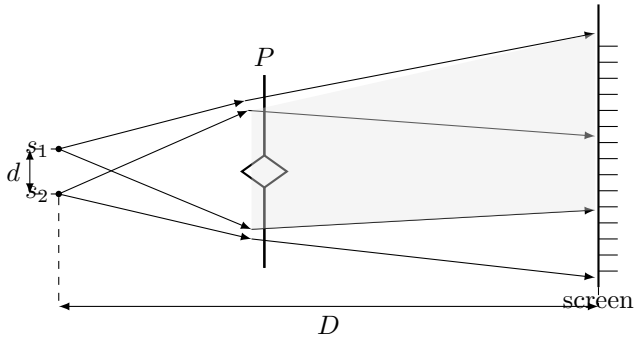


Fig. 8.19(b)

2. Experimental method

(1) Adjusting the light path. Place the sodium lamp N, the slit s, the biprism P and the micrometer eyepiece E on the optical bench and make them coaxial, as shown in Fig. 3. First open the slit s somewhat wide and adjust the left–right positions of the slit and of the eyepiece so that the region where the waves from the two virtual sources overlap lies within the aperture of the micrometer eyepiece. Then adjust the position of the sodium lamp so that the illumination on the two sides of the overlap region is equal. If interference fringes are to appear in the field of view of the eyepiece, the slit must first be made as narrow as possible, but not closed; at the same time the orientation of the slit must be rotated slowly, and when the slit is parallel to the ridge of the biprism clear interference fringes are seen.

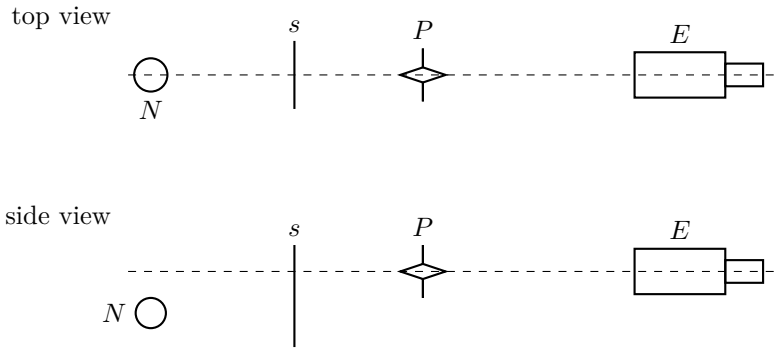


Fig. 8.19(c)

(2) Measure the width x of the interference fringes with the micrometer eyepiece. Measure the total width of 10 (or 20) adjacent interference fringes and take the average as x.

(3) Measure the distance D between the slit and the reticle of the micrometer eyepiece. Note that neither the slit nor the reticle lies in the same plane as the reading mark of its sliding carriage, so the value of D must be corrected.

(4) Measure the distance d between the two virtual sources by the two-image method. Make the distance between the reticle of the micrometer eyepiece and the slit slightly larger than $4f$, where f is the focal length of the convex lens. Two images are formed, one magnified with size d_1 and the other reduced with size d_2; repeat five times and take the average. The separation of the two virtual sources s_1 and s_2 is then $d = \sqrt{d_1 d_2}$.

3. Experimental data and calculation

Table 8.19-1 Measurement of the fringe width x

Order	Reading 1/mm	Order	Reading 2/mm	Width of 20 fringes $20x$/mm
k_1	1.658	$k_1 + 20$	6.182	4.524
k_2	1.908	$k_2 + 20$	6.400	4.492
k_3	2.118	$k_3 + 20$	6.642	4.524
k_4	2.340	$k_4 + 20$	6.878	4.538
k_5	2.570	$k_5 + 20$	7.084	4.514

$$20\bar{x} = 4.518\,\text{mm}; \qquad \bar{x} = 0.2259\,\text{mm}.$$

The reading indicated on the carriage of the adjustable slit is 1240.0 mm (in fact the position of the slit is 1240.0 mm plus a correction of 42.5 mm); the reading on the carriage of the micrometer eyepiece is 634.0 mm (in fact the position of the crosshair in the micrometer eyepiece is 634.0 mm plus a correction of −27.0 mm). Hence

$$D = 1240.0 - 634.0 + 42.5 + 27.0 \approx 676.0 \text{ (mm)}.$$

Table 8.19-2 Measurement of the separation of the virtual sources by the two-image method

Order	Magnified image/mm			Reduced image/mm		
	Reading 1	Reading 2	d_1/mm	Reading 1	Reading 2	d_2/mm
1	7.358	5.190		6.370	4.950	
2	7.376	5.210		6.398	4.954	
3	7.348	5.174		6.400	4.960	
4	7.362	5.188		6.430	4.970	
5	7.326	5.180		6.412	4.972	
Average	7.354	5.188	2.166	6.402	4.961	1.437

$$d_1 = 2.166\,\text{mm}; \qquad d_2 = 1.437\,\text{mm}.$$

The wavelength of the sodium light is

$$\lambda = \frac{x\sqrt{d_1 d_2}}{D} = 5896\ (\text{Å}) = 589.6\ (\text{nm}).$$

Experiment 8.20 Measurement of the grating constant by the principle of multiple-slit diffraction

[**Apparatus and materials available**] Spectrometer, grating, mercury lamp, etc.

[**Requirements**]

(1) Adjust the spectrometer and write down the requirements for its adjustment.

(2) Adjust the platform screws so that the scratches of the grating are parallel to the axis of the spectrometer.

(3) Measure the data and calculate the grating constant of the grating.

Solution 8.20 1. Place the plane grating vertically on the object stage and make the plane of the grating coincide as nearly as possible with the perpendicular bisector of the line joining the platform adjusting screws b and c. Using the reflection from the glass backing of the grating, adjust the spectrometer so that the telescope is focused at infinity, its optical axis is perpendicular to the axis of rotation of the spectrometer, and the plane of the grating is parallel to that axis. With the telescope already adjusted, adjust the collimator so that it emits parallel light and its optical axis is parallel to the optical axis of the telescope.

2. Adjust the platform screw a so that the scratches of the grating are parallel to the axis of rotation of the spectrometer. The screws a, b and c are distributed on the platform as shown in the figure. The spectral lines seen through the telescope should then all lie at the same height. Finally, rotate the entrance slit of the collimator slightly until the spectral lines are parallel to the vertical crosshair of the telescope.

3. After the spectrometer has been adjusted, measure the diffraction angle φ corresponding to the bright lines of the three spectral lines with $m = \pm 1$, repeat the measurement five times and take the average. Substitute into the formula

$$d\sin\varphi = m\lambda \qquad (m = 0, \pm 1, \pm 2, \ldots), \tag{1}$$

where d is the grating constant, λ the wavelength of the light, and m the order of the spectral line.

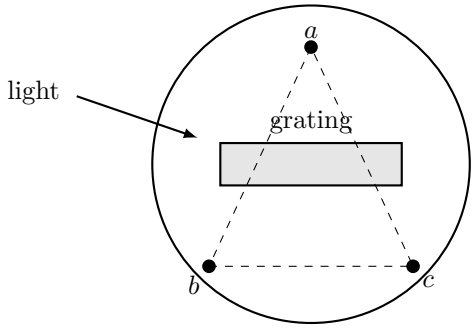


Fig. 8.20

Table 8.20-1 Positions of the zeroth-order and ±1-order diffracted lines

λ/nm	Zeroth order		+1 order		−1 order	
	window "1"	window "2"	window "2"	window "1"	window "1"	window "2"
546.07 (green)	223°13′	43°13.5′	215°22′	35°23′	231°4.5′	51°5.5′
	223°13.5′	43°14.5′	215°22′	35°23′	231°4.5′	51°5.5′
	223°13′	43°13.5′	215°22′	35°22.5′	231°5′	51°6′
	223°12.5′	43°13.5′	215°22.5′	35°22.5′	231°4.5′	51°5.5′
	223°12.5′	43°13.5′	215°22.5′	35°23′	231°5′	51°5.5′
576.96 (yellow)			214°55′	34°55.5′	231°31′	51°32′
			214°54.5′	34°55.5′	231°31′	51°31.5′
			214°54.5′	34°55′	231°31.5′	51°32′
			214°54.5′	34°55.5′	231°31′	51°32′
			214°55′	34°55.5′	231°31.5′	51°32.5′

Table 8.20-2 Data and calculated results

λ/nm	φ	d/nm
546.07	7°51.3′	3996
576.96	8°18.3′	3994

$\lambda = 576.96\,\text{nm}$ is the shorter-wavelength line of the yellow doublet.

The grating constant is $d = 3995\,\text{nm}$.

Experiment 8.21 Observation of single-slit and single-thread diffraction, and measurement of the slit width and the thread diameter

[**Apparatus and materials available**] Optical bench, sliders, single slit and its holder, single fine thread and its holder, white screen, ruler and semiconductor laser.

[**Requirements**]

(1) Briefly describe the principle of single-slit diffraction and write down the single-slit diffraction formula.

(2) Measure the width of the single slit and the diameter of the fine thread.

Solution 8.21 1. By Fraunhofer diffraction, parallel light from the source falls perpendicularly on the single slit (or single thread). According to the Huygens–Fresnel principle, every point of the slit may be regarded as a new source emitting spherical waves in all directions, and the waves superpose on the receiving screen to form a set of bright and dark fringes parallel to the slit. In experiments the parallel light is obtained either by placing the source at infinity or by means of two convex lenses; when a laser is used as the source, the two lenses may be omitted

because of the collimation of the laser beam. The experimental light path is as shown in the figure.

When the distance Z from the slit to the screen is much greater than a, where a is the slit width, the angle θ is very small, and then $\sin\theta \approx \tan\theta = x_k/Z$. The diffraction angles of the dark fringes of various orders are

$$\sin\theta = \frac{K\lambda}{a} \approx \frac{x_k}{Z}, \qquad K = 0, 1, 2, 3, \ldots \tag{1}$$

Hence the width of the single slit is

$$a = \frac{K\lambda Z}{x_k}, \tag{2}$$

where K is the order of the dark fringe, Z the distance from the slit to the screen, and x_k the distance of the K-th order dark fringe from the centre of the central maximum.

If a single thread replaces the single slit, formulas (1) and (2) still hold.

2. Experimental data

The data for the single slit are given in Table 1; the wavelength of the semiconductor laser is $\lambda = 650.0\,\text{nm}$.

Table 8.21-1

K	Z/cm	$\bar{x}_k$/cm	a/mm
4	102.0	0.82	0.323

The slit width measured with a JCD3 reading microscope is $\bar{a} = 0.324\,\text{mm}$, which is quite close to the above value.

The measurements of the thread diameter are given in Table 2.

Table 8.21-2

K	Z/cm	$\bar{x}_k$/cm	a/mm
5	102.0	1.95	0.170
10	102.0	3.92	0.169

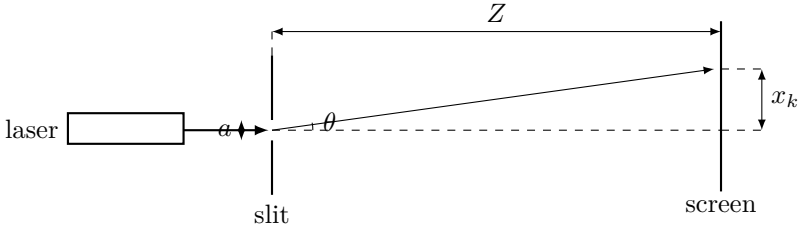


Fig. 8.21

The average diameter of the single thread measured with the reading microscope is $\bar{x} = 0.167\,\text{mm}$; the percentage difference between the two values is 1.8%.

Experiment 8.22 Assembly of a telescope and measurement of its magnifying power

[**Apparatus and materials available**] Optical bench, two convex lenses (one of focal length 200–250 mm, the other of 20–25 mm), one metre rule.

[**Requirements**]
(1) Draw the measuring light path and assemble a Kepler telescope.
(2) Write down the measuring method and the experimental steps.
(3) Measure the magnifying power of the self-assembled telescope.

Solution 8.22 **1.** Use the converging lens of focal length 200–250 mm as the objective and the converging lens of focal length 20–25 mm as the eyepiece. Mount them on an optical bench as shown in the figure, with the distance between the objective and the eyepiece equal to the sum of the two focal lengths, and adjust them until they are coaxial. A Kepler telescope is thus assembled.

2. Place a metre rule in a plane perpendicular to the optical axis at a large distance (greater than 2 m). The experimenter observes the metre rule through the self-assembled telescope with one eye and looks at the distant metre rule directly with the other eye; the visual system then superimposes the two images. Adjust the distance between the objective and the eyepiece slightly until there is no parallax between the two images. The reciprocal of the ratio of a fixed graduation value on the image of the metre rule magnified by the telescope (for example the image of a 1 cm interval) to the graduation value seen directly by the eye on the metre rule is the magnifying power M of the self-assembled telescope.

The main source of measurement error is that the metre rule is not placed at infinity but at a finite distance S from the objective of the telescope. If the focal length of the objective is f_0 and that of the eyepiece is f_e, the magnifying power of the telescope focused at infinity is $M = \dfrac{f_0}{f_e}$. If it is focused at the finite distance S, its magnifying power M' must be corrected as follows:

$$M' = M\frac{S}{S + f_0}.$$

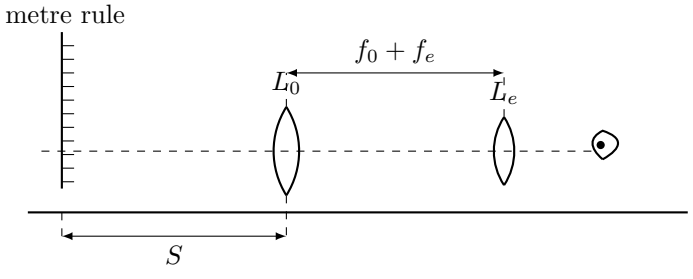


Fig. 8.22

Experiment 8.23 Assembly of a microscope and measurement of its magnifying power

[**Apparatus and materials available**] Optical bench, small electric bulb, ground glass, a millimetre scale ruled on a transparent plate, convex lens (focal length 40–50 mm), convex lens (focal length 25 mm), semi-transparent semi-reflecting mirror, metre rule, lens holders, supports, illuminating lamp, etc.

[**Requirements**]
(1) Assemble a microscope with the apparatus supplied.
(2) Measure the magnifying power M of the assembled microscope.

Solution 8.23 **1.** Use the convex lens of focal length 40–50 mm as the objective and the convex lens of focal length 25 mm as the eyepiece, with an optical interval of 170 mm (the optical

interval is the distance from the second focal point of the objective to the first focal point of the eyepiece). Place them on the optical bench to form a microscope, as shown in the figure.

2. From left to right, place the small electric bulb, the ground glass and the transparent plate (bearing the millimetre scale) in front of the objective. Behind the eyepiece place a semi-transparent semi-reflecting mirror (a plane glass plate coated with a chromium film) whose reflectance is 50%, inclined at 45° to the principal axis. At a distance of about 250 mm from the semi-transparent semi-reflecting mirror, in the direction perpendicular to the optical axis, place a metre rule illuminated by the light source (the illuminating lamp), as shown in the figure. When the light source illuminates the millimetre scale through the ground glass, the experimenter can observe simultaneously with the eye the virtual image of the millimetre scale formed by the assembled microscope and the virtual image of the metre rule formed by the semi-transparent semi-reflecting mirror. When the two images are adjusted until no parallax exists between them, the magnifying power M of the assembled microscope can be found from the ratio of their sizes.

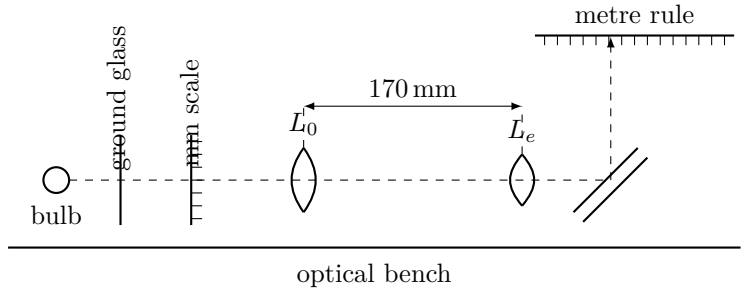


Fig. 8.23

Experiment 8.24 Measurement of the relation between the emergent intensity I of a "polarised-light intensity regulator" and the angle of rotation θ

Light is an electromagnetic wave; its electric and magnetic vectors are mutually perpendicular and both are perpendicular to the direction of propagation. Usually the electric vector is used to represent the light vector. In a plane perpendicular to the direction of propagation the light vector may have various states of oscillation; these are called the polarisation states of light. If the direction of the light vector is arbitrary and the average magnitude of the light vector is the same in all directions, the light is called natural light. If the direction of the light vector never changes and only its amplitude changes with a phase shift, so that the end point of the light vector traces a straight line, the light is called linearly polarised light or plane-polarised light. A device that converts natural light into polarised light is called a polariser; a device used to examine whether light is polarised is called an analyser. Polariser and analyser are collectively called polaroids.

Consider a beam of natural light which, after passing through two polaroids making a certain angle θ with each other, has an emergent intensity related to the angle θ by the following relation:

$$I = DI_0 \cos^n \theta,$$

where I_0 is the intensity of the polarised light incident on the analyser and is a definite value once the instrument is fixed, D is the transmittance of the analyser, and n is a coefficient to be determined. Using the apparatus supplied, measure the relation between I and θ, and fit the above relation by a curve.

[**Apparatus and materials available**] Two polaroids with graduated discs, optical bench, one laser and its power supply, optical power meter with a silicon photocell, graph paper, etc.

[**Requirements**]

(1) The direction of the optical axis of a polaroid is not necessarily at the $\theta = 0°$ position of the graduated disc; the directions of the optical axes of the two polaroids must be measured by the experimenter. (The laser beam emitted by the semiconductor laser is partially polarised.)

(2) Draw the experimental arrangement and measure the data of the I–θ relation.

(3) Find the coefficient n to be determined by a graphical method.

(4) Write down the functional relation between I and θ.

Solution 8.24 **1.** Let the beam emitted by the laser be perpendicular to the centres of the polariser and the analyser and be incident perpendicularly on the silicon photocell of the optical power meter. Measure the light intensity I with the optical power meter, read the angle θ on the graduated disc of the analyser, and measure the relation between I and θ.

2. Turn the disc of the analyser slowly, that is, change the direction of the optical axis of the analyser. When the reading of the optical power meter reaches its minimum or zero, the optical axis of the analyser is perpendicular to that of the polariser. Taking logarithms of both sides of the formula $I = DI_0 \cos^n \theta$ gives

$$\ln I = \ln DI_0 + n \ln \cos \theta.$$

Plot $\ln I$ against $\ln \cos \theta$; the slope of the straight line is the coefficient n to be determined.

3. Experimental data

Table 8.24-1

$\theta/°$	4.5	15.5	24.5	35.5	44.5	55.5
I/mW	1.710	1.587	1.431	1.129	0.870	0.550

Table 8.24-2

$\ln \cos \theta$	-3.09×10^{-3}	-0.0370	-0.0944	-0.206	-0.338	-0.568
$\ln I$	0.536	0.462	0.358	0.121	-0.139	-0.598

Linear regression of the data in Table 2 gives the slope $n = 2.007$ and the correlation coefficient $r = 0.99994$. The graphical method gives the slope of the straight line as $n = 2.01$, so $n \approx 2$ (the graph is omitted).

The fitted empirical formula is

$$I = 1.710 \cos^2 \theta \quad \text{(unit: mW)}.$$

The diagram of the experimental arrangement is omitted.

Comment. The results of this experiment verify Malus's law. This law shows that an analyser may be used as a device for adjusting the intensity of the incident polarised light; moreover, Malus's law may also be used to test whether the output working state of a silicon photocell is linear in the light intensity.

Experiment 8.25 Measurement of the refractive index of semiconductor silicon by the extrapolation method

[**Brief introduction to the principle**]

When light falls on the surface of a semiconductor, part of it is reflected, part absorbed and part transmitted. The ratio R of the reflected light power I_r to the incident light power I_i is called the reflectance, that is,

$$R = \frac{I_r}{I_i}. \tag{1}$$

The incident light may be resolved into two polarisation components: the component P with its polarisation parallel to the plane of incidence, and the component S with its polarisation perpendicular to the plane of incidence. In this experiment a semiconductor laser of wavelength 650 nm is used, and its absorption in the silicon surface may be neglected. Under this condition the reflectances R_P and R_S for light incident from air onto the surface of the material are given by the Fresnel formulas

$$\pm\sqrt{R_P} = \frac{n\cos\theta_i - \cos\theta_t}{\cos\theta_i + n\cos\theta_t}, \tag{2}$$

$$-\sqrt{R_S} = \frac{\cos\theta_i - n\cos\theta_t}{\cos\theta_i + n\cos\theta_t}, \tag{3}$$

where n is the refractive index of the material, θ_i the angle of incidence, θ_r the angle of reflection and θ_t the angle of refraction, as shown in Fig. 1.

At $\theta_i = 0$ it is in practice impossible to measure R_P and R_S directly. The Fresnel formulas, however, allow n to be calculated from the values of R_P and R_S obtained at an arbitrary angle of incidence.

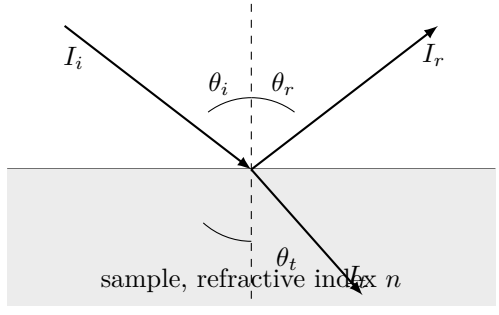


Fig. 8.25(a)

[**Apparatus and materials available**] The experimental arrangement is shown in Fig. 2.

(1) A semiconductor laser and its power supply, fixed on a rotating stage. The laser wavelength is 650 nm.

(2) A linear polaroid of diameter 20 mm, fixed on a rotating stage. Note that the zero reading of the stage is not necessarily the direction of the polarisation axis of the polaroid.

(3) A glass plate (refractive index 1.57 at the wavelength 650 nm) and a thin semiconductor silicon wafer (the sample), fixed on opposite sides of a rectangular metal block (the sample block).

(4) An optical rotating stage on which the sample block can be fixed.

(5) A digital laser power meter connected to a photodetector fixed on a support that can rotate about the sample.

(6) An optical bench, a light shield, graph paper, a flashlight and a soft tape, etc.

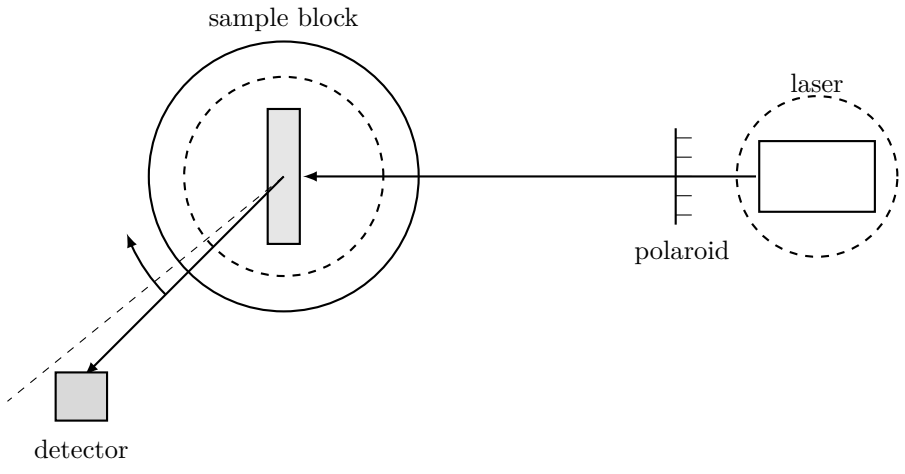


Fig. 8.25(b)

[**Requirements**]

(1) Determine the plane of polarisation of the incident laser light.

In the experiment the polarisation axis of the polaroid should coincide with the direction of the strongest linear polarisation of the laser light.

(2) Measure the reflectances R_P and R_S of the thin semiconductor silicon wafer.

(3) Calculate the refractive index n of the semiconductor silicon material.

Solution 8.25 1. In this experiment the plane of polarisation of the incident laser light must first be determined. It should be noted that the light emitted by a semiconductor laser is partially polarised. To obtain the best results in the experiment, the polarisation axis of the polaroid is made to coincide with the strongest component of the laser light. The laser, the polaroid and the detector are placed on one straight line. Fix either the laser or the polaroid and rotate the other; observe the light intensity measured by the photodetector until the intensity reaches its maximum. In the subsequent measurements the polaroid and the laser are treated as one system and are rotated together when necessary.

From the refractive index of the glass, the Brewster angle of the glass is calculated by the formula

$$\theta_B = \arctan n. \tag{4}$$

The glass sample is fixed on the rotating stage at this position. Rotate the polarising system and measure the power of the reflected laser light at different polarisation angles. The experimental results are shown in Fig. 3. From them the angle of the polaroid relative to the incident glass sample at which the reflected laser power is minimum can be found. According to Brewster's principle, the plane of polarisation of the laser light is then parallel to the plane of incidence on the sample.

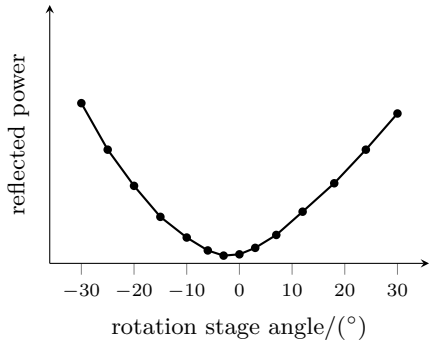


Fig. 8.25(c)

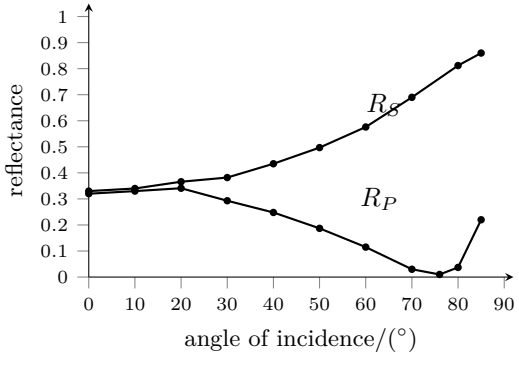


Fig. 8.25(d)

2. With the direction of the plane of polarisation of the laser light determined as above, make it perpendicular and parallel respectively to the plane of incidence on the semiconductor silicon, and measure the reflected laser power at different angles, as shown in Fig. 4. In this figure the relations R_P–θ_i and R_S–θ_i are extrapolated to the reflectance at zero angle of incidence; the two should converge to one point.

When plotting, take care to measure more points near the minimum so as to ensure that the graph is smooth and correct.

3. The average of several measured values of the refractive index n is given in the table below.

Table 8.25

$\theta_i/(^\circ)$	R_P	R_S	n	$\bar{n}$
20.0	0.341	0.366	3.93	
30.0	0.293	0.382	3.82	
40.0	0.248	0.435	3.81	
50.0	0.187	0.497	3.82	
60.0	0.115	0.576	3.85	
80.0	0.037	0.812	3.61	
Average				3.81

Comment. This problem is taken from the optics experimental paper of the 3rd Asian Physics Olympiad. The experiment requires students to master thoroughly the theory of polarised light and the operating

skills of experimental adjustment. Moreover, during the measurement more points should be taken near the extremum so that this examination problem can be done well.

Experiment 8.26 Measurement of the refractive index of glass by measuring the Brewster angle

[**Apparatus and materials available**] Spectrometer, helium–neon laser with polarisation, polaroid with an angular graduated disc, prism under test, white screen, etc.

[**Requirements**]

(1) Write down the principle of the experiment and the experimental formula.

(2) Draw a simplified diagram of the experiment.

(3) Measure the experimental data and calculate the result.

(4) Estimate the experimental error.

Solution 8.26 1. A beam of plane-polarised light of wavelength λ whose electric vector is parallel to the plane of incidence (called a TM wave for short) is incident from air on the surface of a medium of refractive index n at the Brewster angle i_B. At this interface the reflection coefficient of the TM wave is zero, and the incident light enters the medium completely. According to Brewster's law, the tangent of the angle of incidence i_B is then equal to the refractive index n of the medium at the wavelength λ, that is,

$$\tan i_B = n. \tag{1}$$

It follows that, if the angle of incidence of the TM wave is varied, the Brewster angle can be determined when the intensity of the reflected light becomes zero, and the refractive index of the optical medium under test can then be found from (1).

2. As shown in the figure, the prism is placed vertically on the small platform of the spectrometer. The laser light, whose polarisation has been adjusted beforehand, that is, polarised light whose electric vector makes a certain angle with the plane of incidence, is reflected from the glass surface; the reflected light is followed with the analyser, and its intensity can be observed on the white screen. During the experiment the analyser must be rotated continuously in order to observe the state of the reflected polarisation. If the turntable of the spectrometer is rotated to change the angle of incidence of the laser light, and the analyser and the turntable are carefully adjusted, then when the intensity of the reflected light passing through the analyser and falling on the white screen is zero, the reading can be taken on the spectrometer and the Brewster angle i_B determined.

3. Experimental results and error calculation

The measured Brewster angle is $i_B = 57°0'$, so the refractive index is $n = \tan 57°0' = 1.54$.

From $n = \tan i_B$, the uncertainty Δi_B is estimated to be $0.5°$, that is, $\Delta i_B = \dfrac{\pi}{360} = 0.009$. Then

$$\Delta n = \Delta(\tan i_B) = \frac{\Delta i_B}{(\cos i_B)^2} = \frac{0.009}{(\cos 57°0')^2} = \frac{0.009}{0.297} = 0.03.$$

Hence the refractive index of the glass is $n = 1.54 \pm 0.03$.

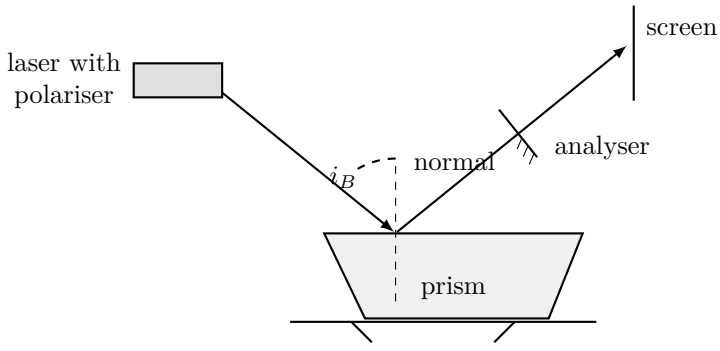


Fig. 8.26

Experiment 8.24 Measurement of the relation between the emergent intensity I and the rotation angle θ of a "polarised-light intensity regulator"

Light is an electromagnetic wave; its electric and magnetic field vectors are mutually perpendicular and both are perpendicular to the direction of propagation. The electric vector is usually taken to represent the light vector. In a plane perpendicular to the direction of propagation the light vector may assume various states of vibration, which are called the polarisation states of light. If the direction of the light vector is arbitrary and the average magnitude of the light vector is the same in every direction, the light is called natural light. If the direction of the light vector remains unchanged and only its amplitude varies with phase, so that the end point of the light vector traces out a straight line, the light is called linearly polarised light or plane-polarised light. A device that can convert natural light into polarised light is called a polariser; a device used to test whether light is polarised is called an analyser. Polariser and analyser are collectively called polarising plates. Consider a beam of natural light passing through two polarising plates set at a certain angle θ to each other; the emergent intensity and the angle θ satisfy the following relation:

$$I = DI_0 \cos^n \theta,$$

where I_0 is the intensity of the polarised light incident on the analyser, which has a definite value once the apparatus is fixed, D is the light transmission coefficient of the analyser, and n is a coefficient to be determined. Using the apparatus provided, measure the relation between I and θ, and obtain the above relation by curve fitting.

[**Apparatus and materials available**] Two polarising plates with graduated discs, optical bench, one laser and its power supply, optical power meter with a silicon photocell, graph paper, etc.

[**Requirements**]

(1) The direction of the optic axis of a polarising plate is not necessarily at the 0° position of its graduated disc; the directions of the optic axes of the two plates must be determined by the experimenter. (The light emitted by a semiconductor laser is partially polarised.)

(2) Draw the experimental arrangement and measure the data of the I–θ relation.

(3) Determine the coefficient n to be found by a graphical method.

(4) Write down the functional relation between I and θ.

Solution 8.24 1. Make the laser beam pass perpendicularly through the centres of the polariser and the analyser and fall normally on the silicon photocell of the optical power meter detector. Measure the intensity I with the optical power meter and read the corresponding angle θ on the graduated disc of the analyser, thus measuring the relation between I and θ.

2. Rotate the disc of the analyser slowly, that is, change the direction of the optic axis of the analyser. When the reading of the optical power meter reaches its minimum or zero, the optic axis of the analyser is perpendicular to that of the polariser. Taking logarithms of both sides of the formula $I = DI_0 \cos^n \theta$ gives

$$\ln I = \ln DI_0 + n \ln \cos \theta.$$

Plot $\ln I$ against $\ln \cos \theta$; the slope of the resulting straight line is the coefficient n to be determined.

3. Experimental data

Table 8.24-1

$\theta/^\circ$	4.5	15.5	24.5	35.5	44.5	55.5
I/mW	1.710	1.587	1.431	1.129	0.870	0.550

Table 8.24-2

$\ln \cos \theta$	-3.09×10^{-3}	-0.0370	-0.0944	-0.206	-0.338	-0.568
$\ln I$	0.536	0.462	0.358	0.121	-0.139	-0.598

A linear regression of the data in Table 8.24-2 gives the slope $n = 2.007$ and the correlation coefficient $r = 0.99994$. The graphical method gives the slope of the straight line as $n = 2.01$, hence $n \approx 2$ (the graph is omitted).

The fitted empirical formula is

$$I = 1.710 \cos^2 \theta \qquad \text{(unit: mW)}.$$

The diagram of the experimental arrangement is omitted.

Comment. The results of this experiment verify Malus's law. This law shows that an analyser can be used as a device for adjusting the intensity of the incident polarised light; in addition, Malus's law can be used to test whether the output of a silicon photocell varies linearly with the light intensity.

Experiment 8.25 Measurement of the refractive index of semiconducting silicon by the extrapolation method

[Brief introduction to the principle]

When light falls on the surface of a semiconductor, part of it is reflected, part absorbed and part transmitted. The ratio R of the reflected light power I_r to the incident light power I_i is called the reflection coefficient, that is,

$$R = \frac{I_r}{I_i}. \tag{1}$$

The incident light can be resolved into two polarisation components: the component P, whose electric vector is parallel to the plane of incidence, and the component S, whose electric vector is perpendicular to the plane of incidence. In this experiment a semiconductor laser of wavelength 650 nm is used, and its absorption at the silicon surface can be neglected. Under this condition the reflection coefficients R_p and R_s for light incident from air on the surface of the material are given by the Fresnel formulae

$$\pm\sqrt{R_p} = \frac{n \cos \theta_i - \cos \theta_t}{n \cos \theta_i + \cos \theta_t}, \tag{2}$$

$$-\sqrt{R_s} = \frac{\cos\theta_i - n\cos\theta_t}{\cos\theta_i + n\cos\theta_t}, \tag{3}$$

where n is the refractive index of the material, θ_i the angle of incidence, θ_r the angle of reflection and θ_t the angle of refraction, as shown in Fig. 8.25(a).

At $\theta_i = 0$ it is in practice impossible to measure R_p and R_s directly. The Fresnel formulae, however, allow n to be calculated from the values of R_p and R_s obtained at an arbitrary angle of incidence.

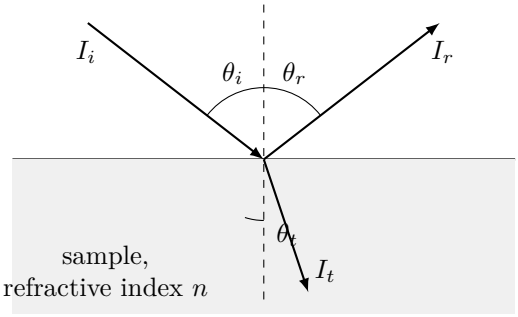


Fig. 8.25(a)

[**Apparatus and materials available**] The experimental arrangement is shown in Fig. 8.25(b).

(1) A semiconductor laser and its power supply, fixed on a rotating stage. The laser wavelength is 650 nm.

(2) A linear polarising plate of diameter 20 mm, fixed on a rotating stage. Note that the zero reading of the stage is not necessarily the direction of the polarisation axis of the plate.

(3) A glass plate (refractive index 1.57 at the wavelength 650 nm) and a thin semiconductor silicon wafer (the sample), fixed on opposite side faces of a rectangular metal block (the sample block).

(4) An optical rotating stage on which the sample block can be fixed.

(5) A digital laser power meter connected to a photodetector fixed on a support that can rotate about the sample.

(6) An optical bench, a light shield, graph paper, a flashlight, a soft tape, etc.

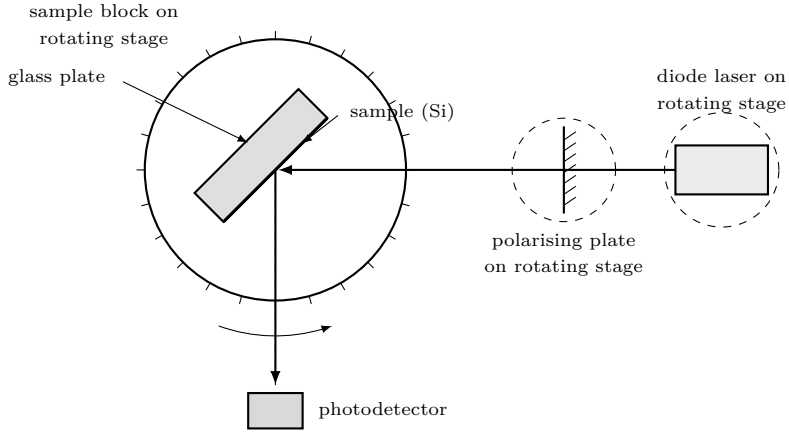


Fig. 8.25(b)

[Requirements]

(1) Determine the plane of polarisation of the incident laser light. During the experiment the polarisation axis of the polarising plate should coincide with the direction of the strongest linear polarisation of the laser light.

(2) Measure the reflection coefficients R_p and R_s of the thin silicon wafer.

(3) Calculate the refractive index n of the silicon semiconductor material.

Solution 8.25 1. In this experiment the plane of polarisation of the incident laser light must first be determined. Note that the light emitted by a semiconductor laser is partially polarised. To obtain the best results, the polarisation axis of the polarising plate should coincide with the strongest component of the laser light. The laser, the polarising plate and the detector are placed in a straight line. Fix either the laser or the polarising plate and rotate the other. Observe the intensity measured by the laser detector until the intensity reaches a maximum. In the subsequent measurements the polarising plate and the laser are treated as one system and are rotated together when necessary.

From the refractive index of the glass, calculate the Brewster angle of the glass from the formula

$$\theta_B = \arctan n \tag{4}$$

and fix the glass sample on the rotating stage at this position. Rotate the polarisation system and measure the power of the reflected laser light at different polarisation angles. The experimental results are shown in Fig. 8.25(c). From this the orientation of the polarising plate, relative to the glass sample, at which the reflected laser power is minimum can be found. According to Brewster's principle, it can be determined that at this position the plane of polarisation of the laser light is parallel to the plane of incidence on the sample.

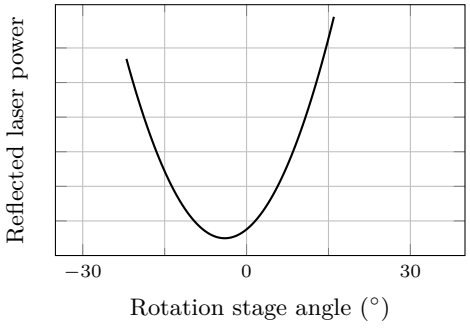


Fig. 8.25(c)

2. With the direction of the laser polarisation plane determined above, make it respectively perpendicular and parallel to the plane of incidence on the silicon semiconductor, and measure the reflected laser power at different angles, as shown in Fig. 8.25(d). In this figure the relations R_p–θ_i and R_s–θ_i, extrapolated to zero angle of incidence θ_i, should meet at one point.

When plotting, note that more points should be measured near the minimum so as to ensure the smoothness and correctness of the graph.

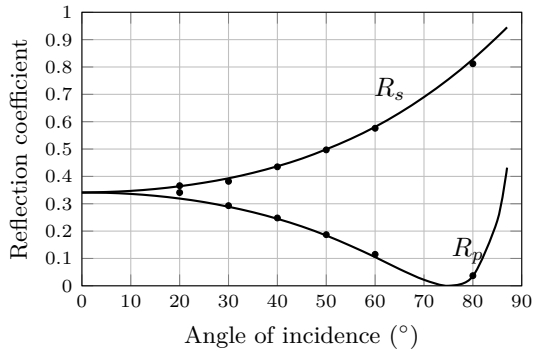


Fig. 8.25(d)

3. The average values of several measurements of the refractive index are shown in the following table.

Table 8.25

$\theta_i/^\circ$	R_p	R_s	n	$\bar{n}$
20.0	0.341	0.366	3.93	
30.0	0.293	0.382	3.82	
40.0	0.248	0.435	3.81	
50.0	0.187	0.497	3.82	
60.0	0.115	0.576	3.85	
80.0	0.037	0.812	3.61	
			Average	3.81

Comment. This problem is an optics experimental problem from the 3rd Asian Physics Olympiad. The experiment requires students to master thoroughly the theory of polarised light and the operational skills of experimental adjustment; in addition, more points should be measured near the extremum so that this problem can be done well.

Experiment 8.26 Measurement of the refractive index of glass by the Brewster-angle method

[**Apparatus and materials available**] Spectrometer, helium–neon laser with polarised output, polarising plate with an angular scale, prism under test, white screen, etc.

[**Requirements**]

(1) Write down the principle of the experiment and the experimental formula.

(2) Draw a simple diagram of the experiment.

(3) Measure the experimental data and calculate the result.

(4) Estimate the experimental error.

Solution 8.26 **1.** A beam of plane-polarised light of wavelength λ whose electric vector is parallel to the plane of incidence (called a TM wave) is incident from air at the Brewster angle i_B on the surface of a medium of refractive index n. At this interface the reflection coefficient of the TM wave is zero, and the incident light enters the medium completely. According to Brewster's law, the tangent of the angle of incidence i_B is equal to the refractive index n of the medium at the wavelength λ, that is,

$$\tan i_B = n. \tag{1}$$

It follows that if the angle of incidence of the TM wave is varied, the Brewster angle can be determined when the intensity of the reflected light is zero, and the refractive index of the optical medium under test follows from formula (1).

2. As shown in the figure, the prism is placed vertically on the small platform of the spectrometer. The laser light, whose polarisation has been adjusted beforehand—that is, plane-polarised light whose electric vector is incident at a certain angle—is reflected from the glass surface, and the reflected light is followed with an analyser, so that the intensity of the reflected light can be observed on the white screen. During the experiment the analyser must be rotated continuously in order to observe the state of the reflected polarisation. If the turntable of the spectrometer is rotated to change the angle of incidence of the laser light, and the analyser and the turntable are adjusted carefully, then when the intensity of the reflected light passing through the analyser on the white screen is zero, the reading can be taken from the spectrometer and the Brewster angle i_B determined.

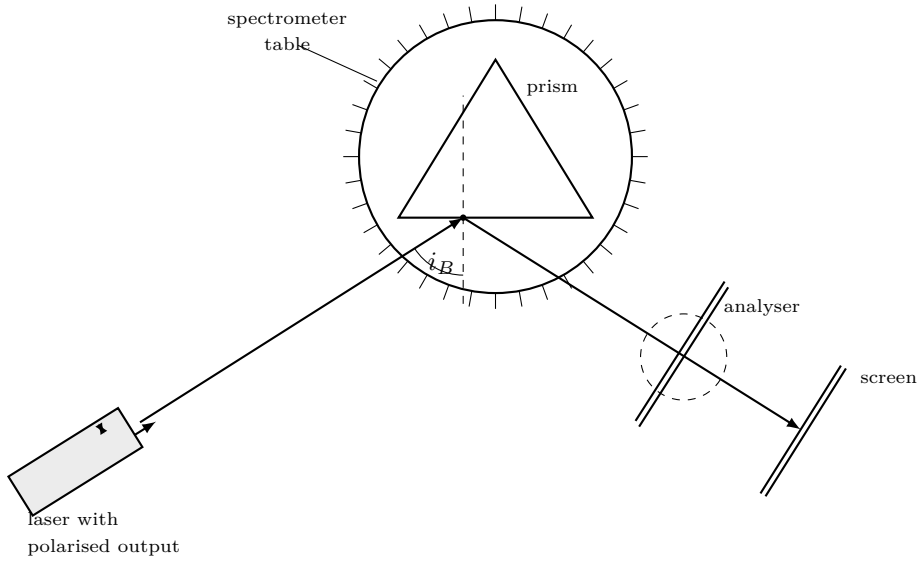


Fig. 8.26

3. Experimental results and error calculation

The measured Brewster angle is $i_B = 57°0'$, so the refractive index is

$$n = \tan 57°0' = 1.54.$$

From $n = \tan i_B$, Δi_B is estimated to be $0.5°$, that is, $\Delta i_B = \frac{\pi}{360} = 0.009$.

$$\Delta n = \Delta(\tan i_B) = \frac{\Delta i_B}{(\cos i_B)^2} = \frac{0.009}{(\cos 57°0')^2} = \frac{0.009}{0.297} = 0.03.$$

Therefore the refractive index of the glass is $n = 1.54 \pm 0.03$.

Appendices

Appendix A

The International Physics Olympiad

A.1 The Syllabus of the International Physics Olympiad

1. Mechanics

1. **Kinematics.** Motion of a point mass; position, displacement, velocity and acceleration; motion with uniform velocity and uniformly accelerated motion; motion under gravity, projectile motion; motion in a circle with constant speed; angular velocity and angular acceleration.
2. **Dynamics.** Newton's laws of motion; inertial frames of reference; momentum and its conservation; centre of mass; variable mass (rocket motion); friction; elastic and inelastic collisions.
3. **Work and energy.** Work done by a force; kinetic energy; conservative forces and potential energy; conservation of mechanical energy; power; efficiency.
4. **Rotational motion.** Rotation of a rigid body about a fixed axis; torque; moment of inertia; angular momentum and its conservation; conditions of equilibrium of a rigid body.
5. **Gravitation.** Newton's law of gravitation; gravitational field and potential; Kepler's laws; motion of planets and satellites; escape velocity.
6. **Hydrostatics and hydrodynamics.** Pressure in a fluid; Pascal's law; Archimedes' principle and floating; steady flow of an ideal fluid; Bernoulli's equation; viscosity; Stokes' law; Reynolds number.
7. **Elasticity.** Hooke's law; Young's modulus; shear modulus; elastic energy.
8. **Oscillations.** Simple harmonic motion; pendulum and spring oscillator; composition of oscillations; damped and forced oscillations; resonance.
9. **Waves.** Travelling waves; longitudinal and transverse waves; wave equation; superposition; reflection, refraction and interference of waves; standing waves; sound waves and the Doppler effect.

2. Thermodynamics and Molecular Physics

1. Ideal gas; equation of state; kinetic theory of gases; pressure and temperature; equipartition of energy; mean free path; diffusion.
2. First law of thermodynamics; internal energy; work and heat; specific heats of gases; isothermal, isobaric, isochoric and adiabatic processes.

3. Second law of thermodynamics; reversible and irreversible processes; entropy; Carnot cycle; efficiency of heat engines; refrigerators and heat pumps.
4. Real gases; van der Waals equation; phase transitions; saturated vapour; humidity.
5. Heat transfer: conduction, convection and radiation; thermal expansion; calorimetry.

3. Electromagnetism

1. **Electrostatics.** Electric charge; Coulomb's law; electric field; Gauss's law; electric potential; conductors and dielectrics; capacitance and capacitors; energy of the electric field.
2. **Direct-current circuits.** Electric current; Ohm's law; resistance and resistivity; series and parallel connections; Kirchhoff's rules; work and power of a current; electrolysis.
3. **Magnetic field.** Magnetic field of currents; Biot–Savart law; Ampère's circuital law; force on a current and on a moving charge; magnetic properties of matter.
4. **Electromagnetic induction.** Faraday's law; Lenz's law; self-inductance and mutual inductance; energy of the magnetic field.
5. **Alternating currents.** RLC circuits; impedance; resonance; power; transformers.
6. **Electromagnetic waves.** Production and properties; the electromagnetic spectrum; energy and momentum of electromagnetic radiation.

4. Optics

1. **Geometrical optics.** Laws of reflection and refraction; total internal reflection; plane mirrors and spherical mirrors; lenses; image formation; optical instruments; the eye.
2. **Wave optics.** Coherence; interference of light; Young's experiment; thin films; diffraction; diffraction gratings; resolving power; polarisation; dispersion of light.

5. Modern Physics

1. Atomic structure; spectra of atoms; the Bohr model of the hydrogen atom; energy levels; X-rays; the Moseley law.
2. Photoelectric effect; photons; the Compton effect; de Broglie waves; the uncertainty principle.
3. Nuclear physics: the atomic nucleus; radioactivity; the law of radioactive decay; half-life; nuclear reactions; binding energy; fission and fusion; interaction of radiation with matter.
4. The special theory of relativity: the principle of relativity; Lorentz transformations; time dilation and length contraction; relativistic momentum and energy; the equivalence of mass and energy.

6. Experimental Skills

1. Use of measuring instruments: vernier calipers, micrometer screw gauge, balances, thermometers, ammeters and voltmeters, galvanometers, oscilloscopes, optical instruments.
2. Measurement of physical quantities and estimation of errors; systematic and random errors; error propagation; significant figures.
3. Presentation of results: tables, graphs, linearisation of relations, least-squares fitting.
4. Design of simple experiments and critical discussion of the results.

The 31st International Physics Olympiad

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Theoretical Problems

Problem 1

Part A: The bungee jumper

A bungee jumper is attached to one end of a long elastic rope, the other end of which is fixed to a high bridge. He falls from rest from the bridge towards the river below without touching the water. His mass is m, the natural length of the rope is L, the force constant of the rope (the force required to stretch it by 1 m) is k, and the strength of the gravitational field is g. Find expressions for the following quantities:

(a) the distance y that the jumper falls before coming to instantaneous rest for the first time;

(b) the maximum speed v that he attains during the fall;

(c) the time t taken for the fall up to the first instantaneous rest.

The jumper may be regarded as a particle attached to one end of the rope, and the mass of the rope is negligible compared with m. When the rope is stretched it obeys Hooke's law, and the resistance of the air may be neglected throughout the fall.

Part B: The heat engine

A heat engine operates between two bodies made of the same material. The temperatures of the two bodies are T_A and T_B ($T_A > T_B$), the mass of each body is m, and their specific heat capacities are constant and equal to s. The pressure of the two bodies is kept constant and no phase change occurs.

(a) Assuming that the engine extracts the maximum mechanical energy theoretically permitted from the system, find an expression for the final temperature T reached by both bodies A and B. Give the complete reasoning of the solution.

(b) Hence obtain an expression for the maximum work that can be obtained.

(c) Suppose that the heat engine operates between two tanks of water, each of volume 2.50 m^3, one at a temperature of 350 K and the other at 300 K. Calculate the maximum mechanical energy that can be obtained.

Given: specific heat capacity of water $= 4.19 \times 10^3\ \mathrm{J \cdot kg^{-1} \cdot K^{-1}}$, density of water $= 1.00 \times 10^3\ \mathrm{kg \cdot m^{-3}}$.

Part C: Radioactivity and the age of the Earth

Assume that when the Earth was formed the isotopes ^{238}U and ^{235}U were already present, but that their decay products were not. The decays of ^{238}U and ^{235}U are used to determine the age T of the Earth.

(a) The isotope ^{238}U decays with a half-life of 4.50×10^9 years. The half-lives of the remaining radioactive decay products in the decay chain are all much shorter, so that to a first approximation the presence of these products may be ignored; the chain terminates with the lead isotope ^{206}Pb. Using the half-life of ^{238}U and the present number ^{238}N of ^{238}U nuclei, express the number ^{206}n of ^{206}Pb atoms produced by the radioactive decay (it is convenient to work in units of 10^9 years).

(b) Similarly, ^{235}U decays, after passing through a series of shorter-lived products, with a half-life of 0.710×10^9 years, terminating in the stable isotope ^{207}Pb. Write down the relation between ^{207}n, ^{235}N and the half-life of ^{235}U.

(c) A mixed ore of lead and uranium is analysed with a mass spectrometer, and the relative concentrations of the lead isotopes ^{204}Pb, ^{206}Pb and ^{207}Pb in this ore are found to be in the ratio $1.00 : 29.6 : 22.6$. Since the isotope ^{204}Pb is not radioactive, it can be used as a reference in the analysis. Analysis of a pure lead ore gives the relative concentrations of these three isotopes as $1.00 : 17.9 : 15.5$. Given that the ratio $^{238}N :^{235} N$ is $137 : 1$, derive a relation involving T.

(d) Assuming that the age T of the Earth is much greater than the half-lives of both uranium isotopes, obtain an approximate value of T.

(e) Clearly the above approximation is not markedly greater than the longer of the half-lives of the uranium isotopes, but a more accurate value of T can be obtained from this approximation. Hence estimate the age T of the Earth to within 2%.

Part D: A spherical charge distribution

In vacuum a charge Q is distributed uniformly throughout a sphere of radius R.

(a) Derive the electric field strength at a distance r from the centre of the sphere for the two cases $r \leq R$ and $r > R$.

(b) Derive an expression for the total electrical energy associated with this charge distribution.

Part E: A rotating copper ring

A circular ring made of fine copper wire rotates about its vertical diameter in the Earth's magnetic field. The magnetic flux density of the Earth's field at the position of the ring is $44.5\,\mu$T, and its direction makes an angle of 64° downwards with the horizontal. Given that the density of copper is $8.90 \times 10^3\,\mathrm{kg \cdot m^{-3}}$ and its resistivity is $1.70 \times 10^{-8}\,\Omega \cdot \mathrm{m}$, calculate the time required for its angular velocity to fall from its initial value to half of that value. Show your working. This time is much longer than the time of one revolution. Assume that friction due to the air and at the bearings is negligible, and neglect self-induction (although these effects ought not really to be neglected).

Solutions

Problem 1

Part A: The bungee jumper

(a) By conservation of energy,

$$mgy = \frac{1}{2}k(y - L)^2.$$

Solving,

$$y = \frac{kL + mg + \sqrt{2mgkL + m^2g^2}}{k}$$

(the positive sign must be taken before the square root).

(b) The maximum speed occurs when the acceleration is zero, that is, when the forces balance. Let the extension of the rope at that instant be x; then

$$mg = kx.$$

By the work–energy theorem,

$$\frac{1}{2}mv_m^2 = mg(L+x) - \frac{1}{2}kx^2.$$

Solving these two equations,

$$v_m = \sqrt{2gL + mg^2/k}.$$

(c) The time of the fall is the sum of the time t_f of free fall and the time t_s of simple harmonic motion of the elastic rope up to its maximum extension:

$$t = t_f + t_s,$$

$$t_f = \sqrt{\frac{2L}{g}}.$$

The jumper enters the simple harmonic motion with the speed $v_t = gt_f = \sqrt{2gL}$ acquired in free fall, and his speed may be written as

$$v = v_m \sin \omega t,$$

where

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}},$$

and

$$v_t = v_m \sin \omega\tau,$$

$$\tau = \frac{1}{\omega} \arcsin \frac{v_t}{v_m} = \frac{1}{\omega} \arcsin \frac{\sqrt{2gL}}{v_m}.$$

$$t_s = \frac{T}{2} - \tau = \pi\sqrt{\frac{m}{k}} - \tau.$$

Hence

$$t = t_f + t_s = \sqrt{\frac{2L}{g}} + \pi\sqrt{\frac{m}{k}} - \frac{1}{\omega} \arcsin \frac{\sqrt{2gL}}{v_m},$$

$$t = \sqrt{\frac{2L}{g}} + \sqrt{\frac{m}{k}} \left[\pi - \arcsin \frac{\sqrt{2gL}}{\sqrt{2gL + mg^2/k}} \right].$$

It may also be written in other equivalent forms.

Part B: The heat engine

In calculating the work obtainable, the friction losses in the operation of the engine are neglected.

Let ΔQ_1 be the heat absorbed from body A, and ΔQ_2 the heat released to body B; then

$$\Delta Q_1 = -ms\,\Delta T_1 \qquad (\Delta T_1 < 0),$$

$$\Delta Q_2 = ms\,\Delta T_2 \qquad (\Delta T_2 > 0).$$

(a) To obtain the maximum mechanical energy, the engine is taken to be a Carnot engine. By the second law of thermodynamics, throughout the working process,

$$\frac{\Delta Q_1}{T_1} = \frac{\Delta Q_2}{T_2},$$

that is,

$$-ms\int_{T_A}^{T_0}\frac{\mathrm{d}T_1}{T_1}=ms\int_{T_B}^{T_0}\frac{\mathrm{d}T_2}{T_2},$$

$$\ln\frac{T_A}{T_0}=\ln\frac{T_0}{T_B}.$$

Hence

$$T_0=\sqrt{T_AT_B}.$$

(b)

$$\begin{aligned}W&=Q_1-Q_2=ms(T_A-T_0)-ms(T_0-T_B)\\&=ms\left(T_A+T_B-2\sqrt{T_AT_B}\right),\end{aligned}$$

or

$$W=ms\left(\sqrt{T_A}-\sqrt{T_B}\right)^2.$$

(c) Numerical result:

$$\begin{aligned}W&=2.50\times1.00\times10^3\times4.19\times10^3\left[350+300-2\sqrt{350\times300}\right]\ \mathrm{J}\\&=2.0\times10^7\ \mathrm{J}=20\,\mathrm{MJ}.\end{aligned}$$

Part C: Radioactivity and the age of the Earth

(a)

$$N=N_0\,2^{-t/\tau},$$

where N_0 is the original number of atoms and τ is the half-life.

$$n=N_0-N_0\,2^{-t/\tau}=N_0\left(1-2^{-t/\tau}\right).$$

Expressed in terms of the present number N of atoms, this becomes

$$n=N\left(2^{t/\tau}-1\right).$$

Therefore

$$^{206}n=\,^{238}N\left(2^{t/4.50}-1\right),$$

where t is measured in units of 10^9 years.

(b) In the same way,

$$^{207}n=\,^{235}N\left(2^{t/0.710}-1\right).$$

(c)

$$\frac{^{206}n}{^{207}n}=\frac{^{238}N}{^{235}N}\cdot\frac{2^{t/4.50}-1}{2^{t/0.710}-1},$$

that is,

$$\frac{29.6-17.9}{22.6-15.5}=137\,\frac{2^{T/4.50}-1}{2^{T/0.710}-1},$$

or

$$0.0120\left(2^{T/0.710}-1\right)=\left(2^{T/4.50}-1\right).$$

(d) Since $T \gg 4.50$ is assumed, the 1 may be neglected in the above equation, giving

$$0.0120 \cdot 2^{T/0.710} = 2^{T/4.50},$$

that is,

$$0.0120 = 2^{T(1/4.50 - 1/0.710)}.$$

$$T = \frac{\log 0.0120}{\log 2 \cdot \left(\dfrac{1}{4.50} - \dfrac{1}{0.71}\right)} = 5.38,$$

$$T = 5.38 \times 10^9 \text{ years}.$$

(e) T is not in fact $\gg 4.5 \times 10^9$ years, but it is $> 0.710 \times 10^9$ years. Substituting the approximate value of T (call it $T^* = 5.38 \times 10^9$ years) into the term $2^{T/4.50}$ of the equation of part (c), in which the 1 has not been neglected, one may calculate the T occurring in the term $2^{T/0.71}$, and so obtain a better approximation to T; repeating the above calculation then gives an even better approximation.

$$0.0120 \left(2^{T/0.71} - 1\right) = 2^{T^*/4.50} - 1,$$

$$2^{T/0.71} - 1 = \frac{2^{5.38/4.50} - 1}{0.0120} = 107.5.$$

Hence

$$T = 0.71 \, \frac{\log 92.2}{\log 2} = 4.80.$$

Taking next $T^* = 4.80$,

$$2^{T/0.71} - 1 = \frac{2^{4.80/4.50} - 1}{0.0120} = 91.2,$$

$$T = 0.71 \, \frac{\log 92.2}{\log 2} = 4.63.$$

One more iteration gives $T = 4.58$. Hence the more accurate answer for T lies between 4.5×10^9 and 4.6×10^9 years (both values are counted as correct).

Part D: A spherical charge distribution

(a) The charge density is

$$\rho = \frac{Q}{(4/3)\pi R^3} = \frac{3Q}{4\pi R^3}.$$

By Gauss's law,

$$E = \frac{Qr}{4\pi\varepsilon_0 R^3} \qquad (r \le R),$$

$$E = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad (r > R).$$

(b) The energy density is $(1/2)\varepsilon_0 E^2$. For $r \le R$, the energy within a spherical shell of radius r and thickness $\mathrm{d}r$ is

$$\mathrm{d}E = \frac{1}{2}\varepsilon_0 E^2 \, 4\pi r^2 \, \mathrm{d}r = \frac{1}{2} \frac{Q^2}{4\pi\varepsilon_0 R^6} r^4 \, \mathrm{d}r.$$

The energy stored inside the sphere is

$$E_{\text{in}} = \frac{1}{2} \frac{Q^2}{4\pi\varepsilon_0 R^6} \int_0^R r^4 \, \mathrm{d}r = \frac{1}{40} \frac{Q^2}{\pi\varepsilon_0 R}.$$

For $r > R$, the energy within a shell is

$$\mathrm{d}E = \frac{1}{2}\frac{Q^2}{4\pi\varepsilon_0 r^4} r^2 \,\mathrm{d}r.$$

The energy in the region $r > R$ is

$$E_{\text{out}} = \frac{1}{2}\frac{Q^2}{4\pi\varepsilon_0}\int_R^\infty \frac{1}{r^2}\,\mathrm{d}r = \frac{1}{8}\frac{Q^2}{4\pi\varepsilon_0 R}.$$

Hence the total energy associated with the charge is

$$E = E_{\text{in}} + E_{\text{out}} = \frac{3}{20}\frac{Q^2}{4\pi\varepsilon_0 R}.$$

The result may also be obtained by summing the work done in assembling the charge; this is omitted here.

Part E: The rotating copper ring

The kinetic energy of rotation of the ring gradually decreases because of the heat losses due to the induced current. The horizontal component of the Earth's magnetic field that produces the induced current in the ring is

$$B = 44.5 \times 10^{-6} \cos 64^\circ \ \text{T}.$$

When the plane of the ring makes an angle $\theta = \omega t$ with the horizontal component of the Earth's field, the magnetic flux through the ring is

$$\Phi = B\pi a^2 \sin \omega t,$$

where a is the radius of the ring.

The instantaneous induced emf is

$$\frac{\mathrm{d}\Phi}{\mathrm{d}t} = B\pi a^2 \omega \cos \omega t.$$

The average thermal power dissipated in the resistance R of the ring during one revolution is

$$P = \frac{B^2\pi^2 a^4 \omega^2}{2R}.$$

The moment of inertia of the ring is $I = (1/2)ma^2$, where m is the mass of the ring. The kinetic energy of rotation is therefore

$$W = \frac{1}{2}I\omega^2 = \frac{1}{4}ma^2\omega^2.$$

Hence the energy equation is

$$\frac{\mathrm{d}W}{\mathrm{d}t} = -P,$$

that is,

$$\frac{1}{2}ma^2\omega\frac{\mathrm{d}\omega}{\mathrm{d}t} = -\frac{B^2\pi^2 a^4\omega^2}{2R},$$

or

$$\frac{\mathrm{d}\omega}{\omega} = -\frac{B^2\pi^2 a^2}{mR}\,\mathrm{d}t.$$

If T is the time required for the angular velocity to fall to half its value, then

$$\int_{\omega}^{\omega/2} \frac{\mathrm{d}\omega}{\omega} = -\int_0^T \frac{B^2\pi^2a^2}{mR}\,\mathrm{d}t,$$

that is,

$$\ln 2 = \frac{B^2\pi^2a^2}{mR}T.$$

But $R = 2\pi a\rho/A$ and $m = 2\pi aAd$, where A is the cross-sectional area of the copper wire, d is the density of copper and ρ is its resistivity. Substituting these into the above gives

$$T = \frac{4\rho d \ln 2}{B^2}.$$

Inserting the data,

$$T = 1.10 \times 10^6\,\mathrm{s} = 306\,\mathrm{hr} = 12\,\mathrm{days}\ 18\,\mathrm{hr}.$$

This part may also be solved by a dynamical method; this is omitted here.

Problem 2

(a) A cathode-ray tube consists of an electron gun and a fluorescent screen and is placed in a uniform, constant magnetic field of flux density B, as shown in Fig. 31.1. The direction of the magnetic field is parallel to the axis of the electron beam of the gun. The electron beam emerges from the anode of the gun along the axis, but with a divergence angle of at most 5° with the axis, as shown in Fig. 31.2. Normally the electron beam produces a diffuse spot on the fluorescent screen, but for a certain magnitude of the magnetic field a well-focused spot is obtained. Consider an electron which leaves the gun with its direction of motion at an angle β $(0 < \beta < 5°)$ to the axis, and resolve its motion into components perpendicular and parallel to the axis. Using the following quantities, derive an expression for the specific charge e/m of the electron: the minimum magnetic flux density that focuses the beam to a point; the accelerating potential difference V of the gun (note that $V < 2\,\mathrm{kV}$); the distance D from the anode to the screen.

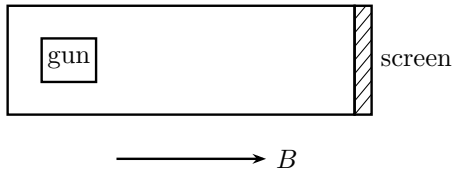


Fig. 31.1

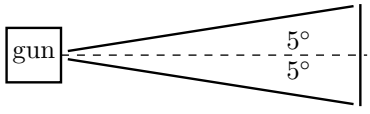


Fig. 31.2

(b) Consider another method of measuring the specific charge of the electron. Fig. 31.3 shows the side view and the plan view of the experimental arrangement, and also the direction of the magnetic field B. Two copper discs separated by a very small distance t are placed in this uniform magnetic field; the radius of the discs is ρ. A potential difference V is maintained between the

two discs. The discs are parallel and coaxial, and their axis is perpendicular to the magnetic field. A photographic plate is placed coaxially on a cylinder of radius $\rho + s$, on its inner surface; in other words, the plate is at a radial distance s from the edge of the discs. The whole apparatus is placed in vacuum. Note that t is much smaller than s and ρ.

A point source of β particles is placed between the centres of the two discs and emits β particles uniformly in all directions, the speeds of the particles being distributed over a certain range. The same plate is exposed under the following different experimental conditions:

Case 1 $B = 0$, $V = 0$;
Case 2 $B = B_0$, $V = V_0$;
Case 3 $B = -B_0$, $V = -V_0$.

Here V_0 and B_0 are positive constants. Note that when $V > 0$ the upper disc is positively charged (when $V < 0$ the upper disc is negatively charged). When $B > 0$ the direction of the magnetic field is defined as shown in Fig. 31.3 (when $B < 0$ the field is in the opposite direction). In solving this part the separation of the two discs may be neglected.

In Fig. 31.3 the two regions of the plate are marked A and B respectively. After the plate has been exposed and developed, the β-particle tracks displayed by the developed plate in one of the two regions are as shown in Fig. 31.4. From which region (A or B) was this part of the plate taken? Justify your answer by indicating the direction of the force acting on the electron.

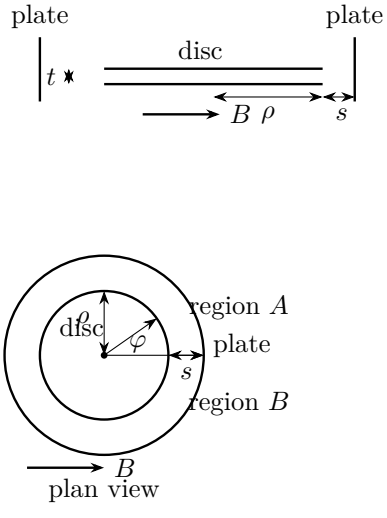


Fig. 31.3

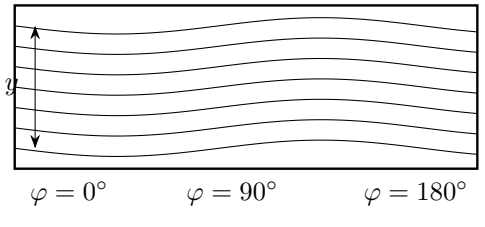


Fig. 31.4

(c) After exposure and development, the plate, unrolled, is as shown in Fig. 31.4. The separation of the two outermost tracks is measured with a microscope [the separation y at a

particular angle is marked in Fig. 31.4]. The results of these measurements are given in the table below. The angle φ is defined as the angle between the direction of the magnetic field and the line joining a point on the plate to the centre of the discs, as shown in Fig. 31.3.

Table 31.1

Angle with magnetic field/°	φ	90	60	50	40	30	23
Separation/mm	y	17.4	12.7	9.7	6.4	3.3	end of track

The numerical values of the parameters of the measuring system are as follows:

$$B_0 = 6.91\,\mathrm{mT} \qquad V_0 = 580\,\mathrm{V} \qquad t = 0.80\,\mathrm{mm} \qquad s = 41.0\,\mathrm{mm}.$$

In addition, the speed of light in vacuum may be taken as $3.00 \times 10^8\,\mathrm{m \cdot s^{-1}}$ and the rest mass of the electron as $9.11 \times 10^{-31}\,\mathrm{kg}$. Calculate the maximum kinetic energy of the observed β particles (in eV).

(d) Using the information given in part (c) above, find the ratio of the charge of the electron to its rest mass. This should be done by plotting a suitable graph on the graph paper provided. Write down the algebraic expressions for the quantities represented by the horizontal and vertical axes of your graph, and give the value of the specific charge e/m.

Note that, because of systematic errors of observation, the value you obtain may differ from the accepted standard value.

Problem 2

(a) According to the problem, when an electron completes exactly one revolution just as it reaches the screen, it will be sharply focused on the screen.

The angular velocity of the circular motion is $\omega = eB/m$, and the period of one revolution is $T = 2\pi m/eB$.

The speed of the electron is $u = \sqrt{2eV/m}$ (since $V < 2\,\mathrm{kV}$, it is not difficult to calculate that $u < 2.65 \times 10^7\,\mathrm{m \cdot s^{-1}}$, so relativistic effects need not be considered).

The distance travelled by the electron is

$$D = Tu\cos\beta \approx Tu = (2\pi m/eB)\sqrt{2eV/m}.$$

Hence the specific charge of the electron is

$$e/m = 8\pi^2 V/B^2 D^2.$$

(b) Consider Case 2. The electric force on the electron is directed upwards.

In region A the magnetic force on the electron is also directed upwards, so the electron strikes the disc and cannot reach the plate.

In region B the magnetic force is directed downwards; if the magnetic force equals the electric force, the electron experiences no net force and emerges from between the discs, exposing the plate.

Case 3 is similar. Hence the plate was taken from region B.

(c) An electron that emerges from between the discs is in force balance. The electric force on it is eV/t and the magnetic force is $euB\sin\varphi$; hence

$$eV/t = euB\sin\varphi,$$

giving the electron speed

$$u = V/Bt\sin\varphi.$$

The maximum speed u_m corresponds to the smallest angle φ, namely 23°; hence

$$\begin{aligned} u_m &= \frac{V_0}{B_0 t\sin 23^\circ} = \frac{580}{6.91\times10^{-3}\times0.80\times10^{-3}\times\sin 23^\circ}\,\mathrm{m\cdot s^{-1}} \\ &= 2.685\times10^8\,\mathrm{m\cdot s^{-1}} = 0.895c. \end{aligned}$$

This is relativistic, so the maximum kinetic energy of the electron is

$$\begin{aligned} E_m = (\gamma-1)mc^2 &= \left(\frac{1}{\sqrt{1-u^2/c^2}}-1\right)mc^2 \\ &= 1.02\times10^{-13}\,\mathrm{J} = 637\,\mathrm{keV}. \end{aligned}$$

(d) An electron emerging from between the discs is acted on only by the magnetic force, and this force is approximately vertical, because the angle between the velocity of the electron and the horizontal is always very small.

The acceleration due to the magnetic force is $a = Beu\sin\varphi/\gamma m$. The initial horizontal speed of the electron is u, and the time taken by the electron to travel from between the discs to the plate is s/u. During this time the vertical displacement is $y/2 = (1/2)a(s/u)^2$, that is,

$$y = \frac{Bes^2\sin\varphi}{\gamma mu}.$$

From (c), an electron emerging from between the discs must have $u = \dfrac{V}{Bt\sin\varphi}$. Substituting this to eliminate u [note that $\gamma = 1/\sqrt{1-u^2/c^2}$] gives

$$y^2 = \left(\frac{eBs\sin\varphi}{m}\right)^2\left[\left(\frac{Bt\sin\varphi}{V}\right)^2-\left(\frac{s}{c}\right)^2\right].$$

Plotting $(y/Bs\sin\varphi)^2$ as ordinate against $(Bt\sin\varphi/V)^2$ as abscissa, the resulting straight line has slope $(e/m)^2$ and intercept on the ordinate axis $-(es/mc)^2$.

From the graph drawn (figure omitted) the intercept is read as $-537.7\,(\mathrm{C\cdot s/kg})^2$, giving $e/m = 1.70\times10^{11}\,\mathrm{C/kg}$.

The slope is read as $2.826\times10^{22}\,(\mathrm{C/kg})^2$, giving $e/m = 1.68\times10^{11}\,\mathrm{C/kg}$.

Problem 3

Gravitational waves and the effects of gravity on light

Part A

This part is concerned with the difficulty of detecting the gravitational waves produced by astronomical events. It is known that a distant supernova explosion may produce a disturbance of about $10^{-19}\,\mathrm{N\cdot kg^{-1}}$ in the gravitational field strength at the Earth's surface. A model of a gravitational-wave detector (see Fig. 31.5) consists of two metal bars each 1 m long, mounted at right angles. One end of each bar is polished to an optical flat, and the other end is rigidly fixed. The position of one of the bars is adjusted so that the signal received by the phototube is a minimum (see Fig. 31.5).

A very short longitudinal pulse is produced in the bar by a piezoelectric device, with the result that the free end of the bar vibrates with a longitudinal displacement Δx_1,

$$\Delta x_1 = ae^{-\mu t}\cos(\omega t + \varphi),$$

where a, μ, ω and φ are constants.

(a) If the amplitude of the displacement decreases by 20% in a time interval of 50 seconds, find the value of μ.

(b) Suppose that both bars are made of aluminium, with density $\rho = 2700\,\mathrm{kg}\cdot\mathrm{m}^{-3}$ and Young's modulus $E = 7.1\times 10^{10}\,\mathrm{Pa}$. Given that the speed of longitudinal waves is $v = \sqrt{E/\rho}$, find the minimum value of ω.

(c) It is in general impossible to make the two bars exactly the same length, so that the phototube signal shows a beat frequency of 0.005 Hz. What is the difference in length of the two bars?

(d) For a bar of length l, derive an algebraic expression for the change in length Δl caused by a change Δg in the gravitational field strength g, in terms of l and other constants of the bar material. Assume that the response of the detector to a change in gravitational field strength occurs only along the axis of one bar.

(e) A laser produces monochromatic light of wavelength 656 nm. If the smallest fringe shift that can be measured is 10^{-4} of the laser wavelength, what is the minimum value of l for the bar if the system is to be able to detect a change in g of $10^{-19}\,\mathrm{N}\cdot\mathrm{kg}^{-1}$?

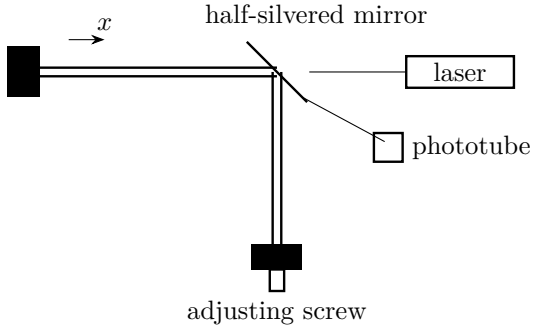


Fig. 31.5

Part B

This part examines the effects produced by the gravitational field on the propagation of light in space.

(a) A photon emitted from the surface of the Sun (mass M, radius R) will be red-shifted. Assuming that the mass of a photon is equivalent to its energy, use Newton's theory of gravitation to show that the effective (or measured) frequency of the photon at infinity is reduced by a factor $(1 - GM/Rc^2)$ (that is, it is red-shifted).

(b) A decrease in the frequency of light is equivalent to an increase in its period. When light is used as a standard clock, this is equivalent to a dilation of time. Moreover, a dilation of time is always accompanied by a contraction of length by the same factor.

We now attempt to study the effect of this on light propagating at the edge of the Sun. First define the equivalent refractive index n_r at a distance r from the centre of the Sun:

$$n_r = c/c'_r,$$

where c is the speed of light measured in a coordinate system far from the gravitational influence of the Sun ($r \to \infty$), and c'_r is the speed of light measured in the coordinate system at a distance r from the centre of the Sun.

When GM/rc^2 is very small, show that n_r may be expressed approximately as

$$n_r = 1 + aGM/rc^2,$$

where a is a constant; determine this constant.

(c) Using the above expression for n_r, calculate the angle (in radians) by which light is deflected from a straight-line path when passing the edge of the Sun.

Data:

$$\begin{aligned} \text{gravitational constant } G &= 6.67 \times 10^{-11}\,\mathrm{N \cdot m^2 \cdot kg^{-2}}, \\ \text{mass of the Sun } M &= 1.99 \times 10^{30}\,\mathrm{kg}, \\ \text{radius of the Sun } R &= 6.95 \times 10^{8}\,\mathrm{m}, \\ \text{speed of light } c &= 3.00 \times 10^{8}\,\mathrm{m \cdot s^{-1}}. \end{aligned}$$

You may need the following integral:

$$\int_{-\infty}^{+\infty} \frac{\mathrm{d}x}{(x^2 + a^2)^{3/2}} = \frac{2}{a}.$$

Problem 3

Part A

(a) Since

$$\Delta x_1 = ae^{-\mu t}\cos(\omega t + \varphi),$$

$$0.8 = \mathrm{e}^{-50\mu} \quad \Longrightarrow \quad \mu = -\frac{\ln 0.8}{50}\,\mathrm{s^{-1}} = 4.5 \times 10^{-3}\,\mathrm{s^{-1}}.$$

(b)

$$v = (E/\rho)^{1/2} = (7.1 \times 10^{10}/2700)^{1/2}\,\mathrm{m \cdot s^{-1}} = 5.1 \times 10^{3}\,\mathrm{m \cdot s^{-1}}.$$

For the fundamental mode of vibration,

$$\lambda = 4l = 4\,\mathrm{m},$$

$$v = \lambda f, \qquad f = v/\lambda,$$

$$\omega = 2\pi f = 2\pi v/\lambda = 8.06 \times 10^{3}\,\mathrm{rad \cdot s^{-1}}.$$

(c)

$$f = v/\lambda = v/4l, \qquad |\Delta f| = v\Delta l/4l^2,$$

$$\Delta l = 4l^2\Delta f/v = 4 \times 1^2 \times 0.05/5.1 \times 10^{3}\,\mathrm{m} = 3.9 \times 10^{-6}\,\mathrm{m}.$$

(d) At a distance x from the free end, the additional gravitational force on the bar due to the change in gravitational field strength is

$$m\Delta g = \rho x A \Delta g,$$

where A is the cross-sectional area of the bar. The corresponding additional stress is $m\Delta g/A = \rho x \Delta g$, and the additional strain produced by it is $\delta(\mathrm{d}x)/\mathrm{d}x = \rho x \Delta g/E$. Hence the additional deformation of the bar is

$$\Delta l = \sum \delta(\mathrm{d}x) = \int_0^l \frac{\rho \Delta g}{E} x \, \mathrm{d}x = \frac{\rho \Delta g}{2E} l^2.$$

(e) When one bar is deformed by Δl because of the change in gravitational field strength, the optical path difference of the laser light reflected from the free ends of the two bars changes by $2\Delta l$. According to the problem, $2\Delta l = 10^{-4}\lambda$; substituting the expression for Δl from (d) gives

$$10^{-4}\lambda = (\rho \Delta g/E) l^2.$$

Hence

$$\begin{aligned} l &= \sqrt{\frac{10^{-4}\lambda E}{\rho \Delta g}} = \sqrt{\frac{10^{-4} \times 656 \times 10^{-9} \times 7.1 \times 10^{10}}{2700 \times 10^{-19}}}\ \mathrm{m} \\ &= 1.31 \times 10^{8}\ \mathrm{m}. \end{aligned}$$

Part B

(a) Since $mc^2 = hf$, we have $m = hf/c^2$. Let the frequency of the photon at the surface of the Sun be f_R and its frequency at infinity be f. By conservation of energy,

$$hf = hf_R - GMm/R = hf_R - \frac{GMhf_R}{Rc^2} = hf_R \left(1 - \frac{GM}{Rc^2}\right).$$

Hence

$$f = f_R \left(1 - GM/Rc^2\right).$$

(b) Let the frequency of the photon at a distance r from the centre of the Sun be f_r; in the same way,

$$f = f_r \left(1 - GM/rc^2\right).$$

The above relation shows that the frequency of a photon at infinity is lower than its frequency at a distance r from the centre of the Sun by a factor, and its period is longer by the same factor. In other words, the frequency of a photon at a distance r from the centre of the Sun is higher than the frequency of light at infinity by a factor, and its period is shorter by the same factor. When a photon is used as a standard clock, the time reading of the same process at a distance r from the centre of the Sun will be greater than the time reading at infinity by a factor, and the length reading will be smaller by the same factor. Therefore the speed of light measured at a distance r from the centre of the Sun is smaller than the speed of light measured at infinity by the square of this factor, that is,

$$c_r' = c\left(1 - GM/rc^2\right)^2.$$

Hence, when $GM/rc^2 \ll 1$,

$$n_r = c/c_r' = \left(1 - GM/rc^2\right)^{-2} \approx 1 + 2GM/rc^2.$$

Therefore

$$a = 2.$$

(c) Since the refractive index decreases with increasing distance r from the centre of the Sun, light passing the edge of the Sun is bent towards the centre of the Sun by refraction (Fig. 31.6). By Snell's law,

$$n(r + \delta r)\sin(\theta + \delta\theta) = n(r)\sin\theta,$$

that is,

$$\left[n(r) + \frac{\mathrm{d}n}{\mathrm{d}r}\delta r\right](\sin\theta + \cos\theta \cdot \delta\theta) = n(r)\sin\theta,$$

$$\sin\theta\,\frac{\mathrm{d}n}{\mathrm{d}r}\delta r + n(r)\cos\theta \cdot \delta\theta = 0,$$

$$\delta\theta = -\tan\theta \cdot \frac{1}{n} \cdot \frac{\mathrm{d}n}{\mathrm{d}r}\delta r.$$

But

$$n = 1 + 2GM/rc^2 \approx 1, \qquad \frac{\mathrm{d}n}{\mathrm{d}r} = -2GM/r^2c^2.$$

Substituting into the above,

$$\delta\theta = \tan\theta\left(2GM/r^2c^2\right)\delta r.$$

From the figure, $r^2 = x^2 + R^2$, $r\,\mathrm{d}r = x\,\mathrm{d}x$ and $\tan\theta = R/x$; hence the deflection angle is

$$\Delta\theta = \int \delta\theta = \frac{2GM}{c^2}\int \frac{\tan\theta\,\mathrm{d}r}{r^2} = \frac{2GM}{c^2}\int \frac{R}{x}\frac{x\,\mathrm{d}x}{r^3}$$
$$= \frac{2GMR}{c^2}\int_{-\infty}^{+\infty} \frac{\mathrm{d}x}{(x^2+R^2)^{3/2}} = \frac{2GMR}{c^2}\cdot\frac{2}{R^2} = \frac{4GM}{Rc^2}.$$

Inserting the data,

$$\Delta\theta = \frac{4 \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30}}{6.95 \times 10^{8} \times (3 \times 10^{8})^2}\,\mathrm{rad} = 8.5 \times 10^{-6}\,\mathrm{rad}.$$

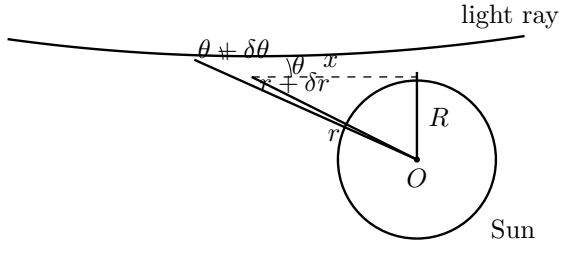


Fig. 31.6

Experimental Problems

Problem 4

The CDROM spectrometer

The aim of this experiment is to represent graphically how the conductance of a photoresistor varies with wavelength over the visible spectrum. You are not required to state the uncertainty of your measurements.

* Conductance $G = 1/\text{resistance}$ (unit: siemens, $1\,\text{S} = 1\,\Omega^{-1}$).

[**Contents of the experiment**]

The experiment consists of the following five parts:

(1) Use a concave reflection grating (made from a piece of CDROM) to produce a focused first-order spectrum from lamp A (a 12 V, 50 W tungsten filament lamp).

(2) Measure and plot the conductance of the photoresistor as a function of wavelength within this first-order spectrum.

(3) Show that the light emission of the filament of lamp A behaves approximately as an ideal black body.

(4) Find the temperature of the lamp filament when it is connected to a 12 V supply.

(5) After calculating the energy distribution of the spectrum emitted by lamp A, correct the curve of conductance against wavelength.

[**Precautions**]

(1) Beware of burns.

(2) Lamp B must not be connected to any potential difference exceeding 2.0 V.

(3) The resistance range of the multimeter must not be used on a circuit that is switched on.

[**Procedure**]

(a) The apparatus shown in Fig. 31.7 is already set up, so that the light from lamp A is incident normally on the grating and the photoresistor is placed in the focal plane of the first-order spectrum. Move the photoresistor along the first-order spectrum and observe how its resistance (measured with the multimeter on the × range) varies with position.

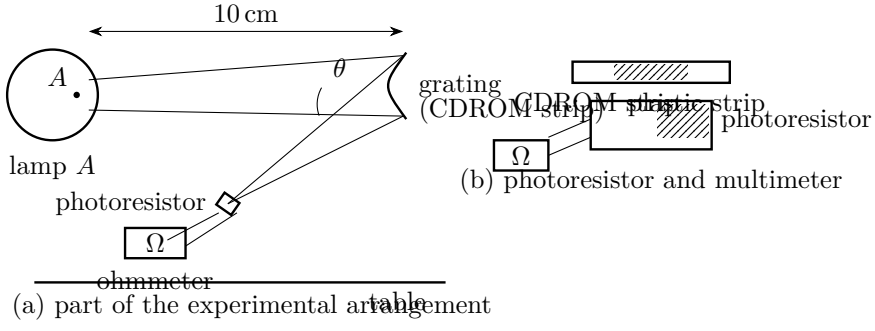


Fig. 31.7

(b) (1) Measure and record the resistance R of the photoresistor at different positions in the first-order spectrum. Record the data in the blank table provided.

(2) Using the graph paper provided, plot the conductance G of the photoresistor against the wavelength λ.

[Note] The angle θ between the direction of light of wavelength λ in the first-order spectrum and the direction in which white light is reflected from the grating is given by

$$\sin\theta = \frac{\lambda}{d},$$

where d is the grating spacing. This grating has 620 lines per millimetre.

Since the emission characteristics of the lamp have not been taken into account, the curve drawn in (b)(2) cannot correctly represent the relation between the conductance of the photoresistor and wavelength. These characteristics will be studied in parts (c) and (d), so that in part (e) a corrected curve is obtained.

[Note] In part (c) three multimeters are used as ammeters. Do not adjust or move these meters. Use the fourth multimeter (marked ×) to measure all voltages.

(c) If the filament of the 50 W lamp radiates as a black body, it can be shown that the relation between the voltage V applied across it and the current I through it may be written as

$$V^3 = CI^5,$$

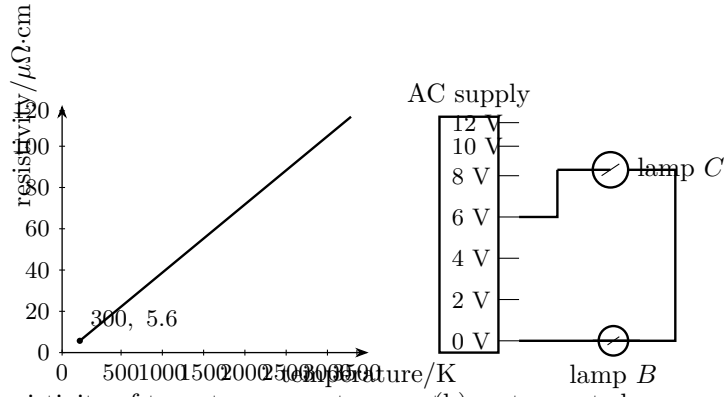
where C is a constant. Measure the corresponding values of V and I for lamp A (in its housing). The ammeter has already been connected; do not adjust it.

(1) Record the data and calculated values in the table on the answer sheet.

(2) Draw a suitable graph on the graph paper provided to show that the filament radiates as a black body.

(d) In order to correct the curve in (b)(2) we need to know the working temperature of the tungsten filament of lamp A. This can be found from the variation of the resistance of the filament with temperature.

[Curve provided] You are provided with a curve of the resistivity of tungsten (in $\mu\Omega \cdot \mathrm{cm}$) against temperature (in K) (Fig. 31.8).



(a) resistivity of tungsten versus temperature (b) meters not shown

Fig. 31.8

If the resistance of lamp A at some known temperature can be found, its temperature when working from a 12 V supply can be found from its resistance at that working voltage. Unfortunately its resistance at room temperature is too small to be measured accurately with this apparatus. However, you are provided with another, smaller lamp C, which has a larger resistance at room temperature that can be measured. According to the following procedure, lamp C can be used as an intermediary. You are also provided with another 12 V, 50 W lamp B identical to lamp A. Lamps B and C are mounted on a board and connected as shown in Fig. 31.8.

(1) With lamp C unlit, measure its resistance at room temperature (using multimeter ×, and taking room temperature to be 300 K). Record this resistance R_{C1} on the answer sheet.

(2) Compare the filaments of lamps B and C using the circuit shown in Fig. 31.8. Vary the current through lamp C with the variable resistor until you see the two superimposed filaments at the same temperature. If the small filament is at a lower temperature than the large one, it appears as a thin dark ring. When equal temperatures are reached, measure the resistances of lamps B and C and record their values R_{C2} and R_B on the answer sheet. Remember that the ammeter has already been connected.

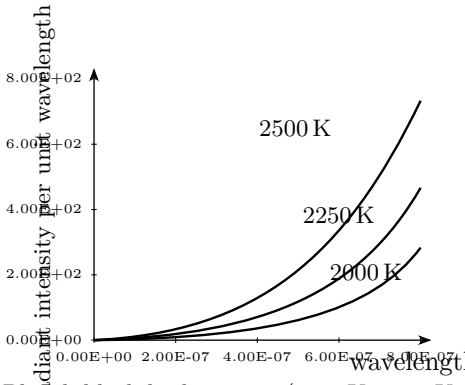
(3) Use the resistance–temperature curve provided to find the temperature at which lamps B and C have the same brightness. Record this temperature T_{2V} on the answer sheet.

(4) Measure the resistance of lamp A (in its housing) when it is connected to the 12 V AC supply. Again, the ammeter has already been connected; do not adjust it. Record this R_{12V} in the table on the answer sheet.

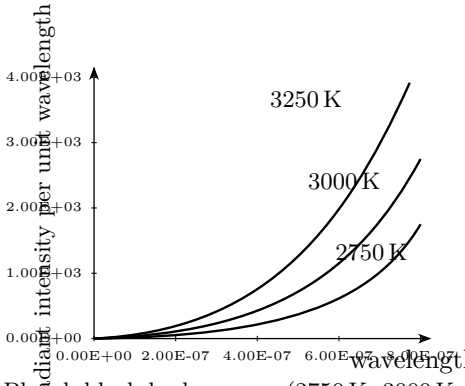
(5) Using the resistances of lamp A (in its housing) at 2 V and at 12 V, and its temperature at 2 V, find the temperature at which it works when connected to the 12 V supply. Record this temperature T_{12V} in the table on the answer sheet.

[Curve provided] You are provided with curves of the relative radiant intensity of a black body (Planck curves) at the temperatures 2000 K, 2250 K, 2500 K, 2750 K, 3000 K and 3250 K (Fig. 31.9).

(e) Using these graphs and the result of (d)(5), draw on the graph paper provided a corrected curve of the conductance of the photoresistor (in arbitrary units) against wavelength. Assume that the conductance of the photoresistor at any wavelength is proportional to the radiant intensity at that wavelength (this assumption is reasonable in this experiment, in which the radiant intensity falling on the photoresistor is low). Also assume that the grating diffracts uniformly over the first-order spectrum.



(a) Planck black-body curves (2000 K, 2250 K, 2500 K)



(b) Planck black-body curves (2750 K, 3000 K, 3250 K)

Fig. 31.9

Solutions to the Experimental Problems

Problem 4

(a) Connect up the apparatus and instruments as required in the figure. Let the white light produced by lamp A fall normally on the concave reflection grating, and observe the diffracted light of different wavelengths.

(b) Move the photoresistor from position to position, measure the relation between the wavelength λ (in nm) of the light in the first-order spectrum diffracted by the grating and the resistance R of the photoresistor, and convert the resistance R into the conductance $G = 1/R$, entering it in the table.

The wavelength of the light is found from the formula $\sin\theta = \lambda/d$; the angle θ can be obtained by measuring lengths on the base plate with the paper and ruler provided, and then calculating. The resistance of the photoresistor is measured with the ohmmeter range of the digital multimeter marked $\times$. The measured range of wavelengths is the visible light from 400 to 700 nm. There should be more than 9 data points; note that near the peak of the curve the data points should be closer together. Plot the curve of the conductance G of the photoresistor against the wavelength λ on the graph paper; the peak of this curve lies in the range (580 ± 20) nm. (Figure omitted.)

(c) Connect the current range of the digital multimeter in series in the circuit of lamp A, and use the voltage range of the multimeter marked $\times$ to measure the voltage across lamp A.

Measure the relation between the voltage V across the lamp and the current I when lamp A is lit. Plotting V^3 against I^5 gives a straight line, which shows that the filament of lamp A radiates as a black body.

(d) Measuring the filament resistance R_C of lamp C, not connected to the supply, at room temperature 300 K gives $R_{C1} = 13.5\,\Omega$; recording next the resistance R_{C2} of lamp C when it is as bright as lamp B gives $R_{C2} = 75.0\,\Omega$ (measured by the volt–ampere method). Hence, while measuring the voltage across lamp B, the resistance $R_B = 1.20\,\Omega$ of lamp B when it emits the same intensity of light can be found. Consulting curve 1, namely the data for the resistivity of tungsten against temperature T, calculation gives the filament temperature of lamps C and B when they have the same brightness as $T_{2V} = 2130\,\mathrm{K}$.

Using the volt–ampere method, the resistance of lamp A when connected to the 12 V AC supply is found to be $R_{12V} = 2.85\,\Omega$. Similarly, consulting the curve of the resistivity of tungsten against temperature T, from R_B, T_{2V} and R_{12V} one obtains the filament temperature of lamp A when connected to the 12 V AC supply, $T_{12V} = 2940\,\mathrm{K}$, close to 3000 K.

(e) According to the assumption, the conductance of the photoresistor at any wavelength is proportional to the radiant intensity at that wavelength. Consult the 3000 K curve of Fig. 31.9(b), the Planck black-body radiation curve (radiant intensity per unit wavelength against wavelength). Correct the measured conductance G of the photoresistor to obtain the corrected conductance G', and plot the G'–λ relation. (Figure omitted.)

Problem 5

The magnetic slider

In this experiment, the uncertainties must be stated for all measured data and experimental results, and the data points in the graphs must also be marked with their uncertainties.

[Aim]

To study the forces acting on a slider as it slides down an inclined plane.

[Precautions]

Do not touch the round face of the slider or the paper surface on the inclined track directly with your hands. Use the gloves provided. For convenience, the two faces of the round slider have been covered with paper of different colours.

[Timing]

Two magnetic sensors are mounted below the inclined plane of the sliding track and are used to trigger the circuit of the "time-measuring box" on the table. When the slider moves between the two magnetic sensors, the green light-emitting diode on the box lights up. The multimeter is used to measure the potential difference across a capacitor; when the green light-emitting diode is lit, the capacitor is connected to a constant-current source and charged (its charging current is constant and proportional to the potential difference across the battery). Hence the reading of the charged voltage displayed by the multimeter can be used to measure the time taken by the slider to travel between the two magnetic sensors. This reading can already represent, in some unit, the speed of the slider between the two sensors.

[Operating the timer]

(1) While the black button on the side of the time-measuring box is pressed, the circuit of the box is switched on and can be used for time measurement.

(2) At this time, if the green light is on, place the slider with its light-coloured face upwards (in this experiment the light-coloured face of the slider is always kept upwards) and put it between the two sensors; let it slide down the incline past the lower sensor, whereupon the green light should go out.

(3) Before releasing the slider, the potential difference across the capacitor can be discharged to zero by pressing and holding the red button on the timer box (for at least 10 seconds).

(4) The battery voltage can be measured by connecting the multimeter to the two terminals marked with the battery symbol.

[Definitions]

(1) A body sliding down an inclined plane is acted on by a resistive force F along the plane and a normal force N. Define

$$\xi = \frac{F}{N}.$$

(2) When the resistance arises only from friction, ξ is equal to μ_k, which is called the coefficient of kinetic friction of the surface; it is independent of speed.

(3) When the blue (dark) face is in contact with the inclined plane, define

$$\xi_d = \frac{F_d}{N}.$$

Here the force F_d along the plane arises partly from surface friction and partly from magnetic effects.

(4) Define the variable corresponding to magnetic effects alone:

$$\xi_{d'} = \xi_d - \mu_k.$$

[Important hints and suggestions]

(1) It is very useful at the beginning to study the sliding behaviour of the slider qualitatively.

(2) Before making quantitative measurements, think about the underlying physics. Remember to present the results by means of graphs as far as possible.

(3) Unless you have enough time, do not record too much experimental data.

(4) In the experiment you measure the potential difference across an electrolytic capacitor; unlike a simple air capacitor, a slow leakage is normal, and the potential difference will not remain perfectly stable.

(5) You are provided with a slider and a 9.0 V battery; please economise on electricity. The constant current that charges the capacitor is proportional to the potential difference across the battery; note the potential difference across the battery and record the result. If the potential difference across the battery falls below 8.4 V, the sensors may not work reliably, and the battery should then be replaced.

(6) Your answer packet contains only 4 sheets of graph paper; no further graph paper will be provided. After the experiment you may take the slider away.

(7) If the timer cannot be used, ask the invigilator for help.

[Data]

Weight of the slider $= 5.84 \times 10^{-2}$ N

The reading of the voltmeter represents the time for which the slider slides between the two sensors; 1 V corresponds to 0.213 s (when the potential difference of the battery is 9 V)

Distance between the two sensors = 0.294 m

[Contents of the experiment]

(1) Using only the equipment provided, investigate how $\xi_{d'}$ depends on the speed v_b of the slider on an inclined track making an angle θ with the horizontal.

(2) On the answer sheet, state the algebraic equations or relations used to analyse the experimental results and to plot the graphs.

(3) Propose a quantitative model to explain the experimental results. Use the collected data to justify this model.

Problem 5

When the magnetic slider slides down an aluminium inclined plane from top to bottom with uniform speed v, the induced emf (and hence the induced current) is proportional to the speed v, so the magnetic damping force may be taken as $F = Kv$, where K is a constant. This magnetic damping force may also be written as $\xi_d W \cos\theta$, where W is the weight of the slider and ξ_d may be regarded as a "magnetic" friction coefficient; θ is the angle between the inclined plane and the horizontal.

When the slider slides down the incline with uniform speed, the resultant of the forces parallel to the incline is zero, so that

$$W \sin\theta = Kv + \mu W \cos\theta,$$

where μ is the coefficient of kinetic friction between the slider and the incline. Dividing both sides of this equation by $W \cos\theta$ gives the equation

$$\tan\theta = \frac{Kv}{W\cos\theta} + \mu.$$

Plotting a straight-line graph of $\tan\theta$ against $v/\cos\theta$ gives the slope K/W and the intercept μ; hence the magnetic friction coefficient ξ_d can be found. Its relation to the speed v and the angle θ is

$$\xi_d = \text{slope} \cdot \frac{v}{\cos\theta},$$

where the value of the slope is obtained from the graph.

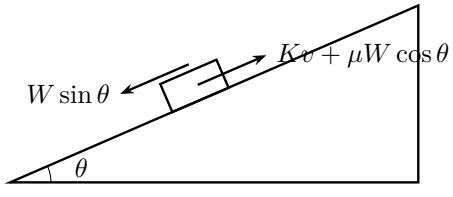


Fig. 31.10

[Note] The raw reading of the timer is a voltage V (the voltage read by the digital multimeter). V can be converted into a time t by the relation $t = 0.213V$; hence the speed of the slider can be found as $v = 0.294/t$, that is, the speed of the slider is $v = 1.380/V$ (where V is the voltage across the capacitor).

This problem is a design-and-investigation competition problem; the manipulation, measurement and analysis should include the following.

(1) Measure the voltage across the laminated battery with the voltage range of the digital multimeter, and then calibrate the voltage across the capacitor in the timer:

$$V_{\text{correct}} = V_{\text{reading}} \cdot (9.00/\text{battery voltage}).$$

(2) The experiment uses angles of inclination θ in the range $\theta_{\max} > \theta > \theta_{\min}$. Show that within this range the slider moves with uniform speed between the two sensors; this can be shown by recording the same time measured by the timer when the slider starts from different heights. Record these data.

(3) At different angles of inclination θ, let the slider slide down with uniform speed, record the time t measured by the timer, and find the speed v of the slider as it passes between the two integrated Hall sensors. Change the angle θ and take measurements at at least 5 points; plot a

straight-line graph of $\tan\theta$ against $v/\cos\theta$. Multiplying the slope of the line by $v/\cos\theta$ gives ξ_d, and the intercept is the coefficient of kinetic friction μ; the result is that μ should lie in the range 0.2–0.4. A layer of paper is stuck to both the aluminium incline and the surface of the slider; do not touch them with your hands.

The 32nd International Physics Olympiad

Translated and compiled by Zheng Yongling, Jiang Zuimin and Lu Shenlong, Fudan University

Theoretical Problems

Problem 1

Part A. The klystron

A klystron is used to amplify high-frequency signals. A klystron consists essentially of two cavities separated by a distance b, each cavity containing a pair of parallel plates, as shown in Fig. 32.1.

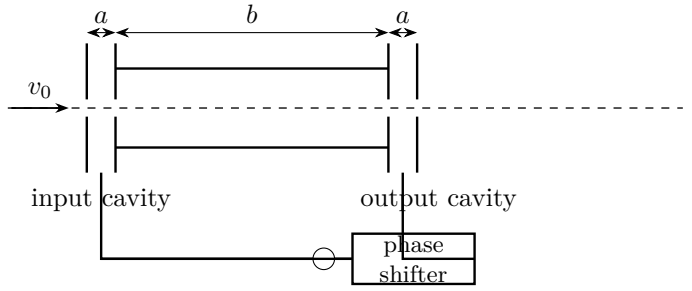


Fig. 32.1

A beam of electrons with initial velocity v_0 passes through small holes in the plates and traverses the whole system. The high-frequency signal to be amplified is applied to the two pairs of electrode plates with a certain phase difference (one period corresponds to a phase of 2π), thereby producing an alternating horizontal electric field in each cavity. When the electric field in the input cavity points to the right, the electrons entering the cavity are decelerated; conversely, when the electric field points to the left, the electrons are accelerated. In this way, the electrons emerging from the input cavity will, after travelling a certain distance, bunch together into short electron bunches. If the output cavity is located at the place where the short electron bunches are formed, then, provided the phase of the voltage applied to it is chosen appropriately, the electric field in the output cavity will absorb energy from the electron beam. Suppose that the voltage signal is a square wave of period $T = 1.0 \times 10^{-9}\,\mathrm{s}$ and voltage $V = \pm 0.5\,\mathrm{V}$. The initial velocity of the electron beam is $v_0 = 2.0 \times 10^6\,\mathrm{m/s}$, and the specific charge of the electron is $e/m = 1.76 \times 10^{11}\,\mathrm{C/kg}$. Assume that the spacing a is very small, so that the transit time of an electron through a cavity can be neglected. Keeping 4 significant figures, calculate:

(a) the distance b over which the electrons can bunch into short electron bunches. Write your result on the answer sheet. (1.5 points)

(b) the required phase difference between the output cavity and the input cavity, provided by the phase shifter. Write your result on the answer sheet. (1.0 point)

Part B. Molecular spacing

Let d_L and d_V denote respectively the average spacing of water molecules in the liquid phase and in the vapour phase. Assume that both phases are at 100°C and at a pressure of one atmosphere, and that the water vapour may be treated as an ideal gas. Using the following data, calculate the ratio d_V/d_L. Write your data on the answer sheet. (2.5 points)

- Density of water in the liquid phase: $\rho_L = 1.0 \times 10^3\,\mathrm{kg/m^3}$
- Molar mass of water: $M = 1.8 \times 10^{-2}\,\mathrm{kg/mol}$
- One atmosphere: $p_a = 1.0 \times 10^5\,\mathrm{N/m^2}$
- Gas constant: $R = 8.3\,\mathrm{J/(mol \cdot K)}$
- Avogadro constant: $N_A = 6.0 \times 10^{23}\,\mathrm{mol^{-1}}$

Part C. A simple sawtooth signal generator

A sawtooth voltage waveform V_0 can be obtained at the terminals of the capacitor C in Fig. 32.2. In the figure, R is a variable resistor, V_s is an ideal battery, and SG is a spark gap consisting of two electrodes whose separation is adjustable. When the voltage across SG exceeds the breakdown voltage V_f, the air between the two electrodes is broken down; the spark gap then becomes a short circuit, and remains short-circuited until the voltage between its electrodes becomes very small.

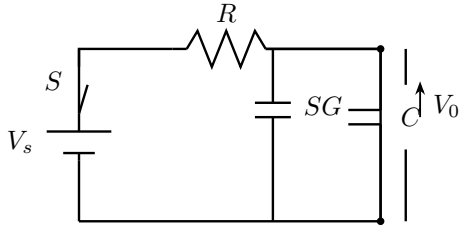


Fig. 32.2

(a) After the switch is closed, plot the waveform of the voltage V_0 as a function of time. (0.5 points)

(b) Under what condition can an almost linear sawtooth voltage waveform V_0 be obtained? Write the result on the answer sheet. (0.2 points)

(c) Assuming that the above condition is satisfied, derive a simple expression for the period T of this waveform. Write the result on the answer sheet. (0.4 points)

(d) If only the period is to be changed, what should be changed (only R, or only SG, or both R and SG)? Write the result on the answer sheet. (0.2 points)

(e) If only the amplitude is to be changed, what should be changed (only R, or only SG, or both R and SG)? Write the result on the answer sheet. (0.2 points)

(f) Given another variable power supply, design and draw a new circuit, and indicate the two terminals from which the output voltage waveform V_0' shown in Fig. 32.3 can be obtained. (1.0 point)

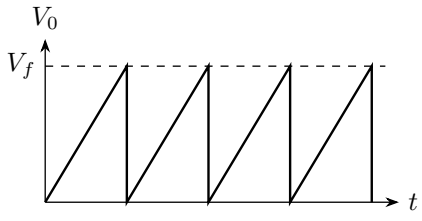


Fig. 32.3

Part D. Atomic beam

In Fig. 32.4, a group of atoms in an oven is heated to a certain temperature T, and these atoms emerge in the horizontal direction through a small hole of diameter D in the side of the oven (the size of D is comparable with atomic dimensions), thus forming an atomic beam. When this beam of atoms has travelled a horizontal distance L, estimate the diameter of the beam. Let the mass of an atom be M. Write your result on the answer sheet. (2.5 points)

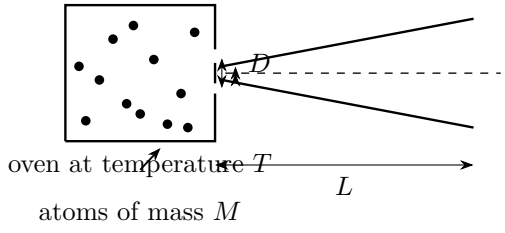


Fig. 32.4

Solutions

Problem 1

Part A

(a) The electrons are decelerated during the first half period and accelerated during the second half. Let the velocity after deceleration be $v_{\rm ret}$ and that after acceleration be $v_{\rm acc}$; then

$$v_{\rm ret} = \sqrt{v_0^2 - 2(e/m)V} = 1.956 \times 10^6\,{\rm m/s},$$
$$v_{\rm acc} = \sqrt{v_0^2 + 2(e/m)V} = 2.044 \times 10^6\,{\rm m/s}.$$

After a time t the two kinds of electrons have travelled the distances

$$x_{\rm ret} = v_{\rm ret}t, \qquad x_{\rm acc} = v_{\rm acc}(t - T/2).$$

Setting $x_{\rm ret} = x_{\rm acc}$, we obtain the time at which the electrons bunch into a short beam:

$$t_b = \frac{v_{\rm acc}T}{2(v_{\rm acc} - v_{\rm ret})} = 11.61T.$$

Hence

$$b = v_{\rm ret}t_b = 2.272 \times 10^{-2}\,{\rm m}.$$

(b) Phase shift:

$$\Delta\varphi = \pm\left(\frac{t_b}{T} - n\right) 2\pi = \pm 0.61 \times 2\pi = \pm 220^\circ,$$

or

$$\Delta\varphi = \pm 140^\circ.$$

Part B

Treating the water vapour as an ideal gas, its density is

$$\rho_V = \frac{Mp_a}{RT}.$$

Hence

$$\frac{d_V}{d_L} = \left(\frac{\rho_L}{\rho_V}\right)^{1/3} = \left(\frac{RT\rho_L}{Mp_a}\right)^{1/3} \approx 12.$$

Part C

(a) As shown in Fig. 32.5.

(b) $V_s \gg V_f$.

(c) After the switch is closed, the output voltage as a function of time is

$$V = V_s\left(1 - \mathrm{e}^{-t/RC}\right).$$

Setting $V = V_f$, and since $V_f \ll V_s$, we must have

$$t = T \ll RC, \qquad \mathrm{e}^{-T/RC} \approx 1 - \frac{T}{RC},$$

$$V_f \approx V_s\left[1 - \left(1 - \frac{T}{RC}\right)\right] = V_s\frac{T}{RC},$$

$$T = RC\frac{V_f}{V_s}.$$

(d) R.

(e) Change R and SG simultaneously so that $RV_f =$ constant.

(f) Either of the two circuits below is acceptable.

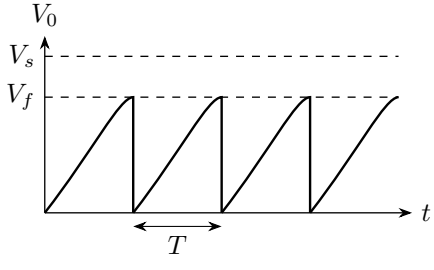


Fig. 32.5

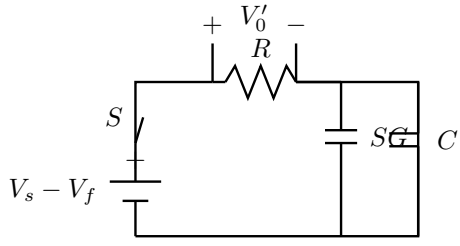


Fig. 32.6(a)

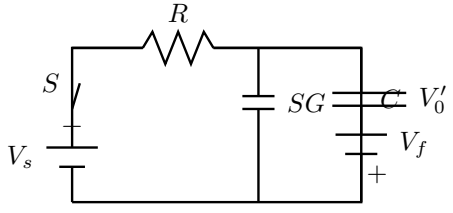


Fig. 32.6(b)

Part D

When the atomic beam passes through the hole of diameter D, the quantum-mechanical uncertainty relation gives for the y-component of its momentum

$$\Delta p_y \approx \frac{\hbar}{D}, \quad \frac{h}{D}, \quad \frac{h}{2D}.$$

The corresponding velocity components are

$$\Delta v_y \approx \frac{\hbar}{MD}, \quad \frac{h}{MD}, \quad \frac{h}{2MD}.$$

The diameter of the atomic beam increases by the amount $\Delta D = 2\Delta v_y t$, where t is the time of flight.

When the oven temperature is T, the kinetic energy of an atom as it leaves the hole is

$$E_k = \frac{1}{2}Mv^2 = \frac{3}{2}kT, \quad kT, \quad \frac{1}{2}kT.$$

Hence

$$v = \sqrt{\frac{3kT}{M}}, \quad \sqrt{\frac{2kT}{M}}, \quad \sqrt{\frac{kT}{M}}.$$

The atomic beam travels the distance L in the time $t = L/v$; therefore

$$\Delta D = 2\Delta v_y t \approx 2\frac{L}{v}\frac{\hbar}{MD} = \frac{2L\hbar}{MD}\sqrt{\frac{M}{3kT}}$$
$$= \frac{2L\hbar}{D\sqrt{3MkT}},$$

or by substituting the other corresponding quantities above. Thus the new diameter of the atomic beam after travelling the distance L is

$$D_{\text{new}} = D + \frac{2L\hbar}{D\sqrt{3MkT}},$$

or the expression obtained by substituting the other corresponding quantities above.

Problem 2

Binary star system

(a) As is well known, most stars form binary systems. One kind of binary system consists of an ordinary star of mass m_0 and radius R, and a compact neutron star of larger mass M, revolving around each other. In all that follows, ignore the motion of the Earth. The following information is obtained from observations of this binary system:

(1) The maximum angular displacement of the ordinary star is $\Delta\theta$, while the maximum angular displacement of the neutron star is $\Delta\varphi$ (Fig. 32.7).

(2) The time required to change from one state of maximum displacement (I) to the other state of maximum displacement (II) is τ.

(3) The radiation characteristics of the ordinary star show that its surface temperature is T, and that the energy radiated per unit time per unit area at the Earth's surface is P.

(4) Because of the gravitational field of the ordinary star, the calcium lines in this radiation differ from the normal wavelength λ_0 by $\Delta\lambda$ [in this calculation the mass of a photon of wavelength λ may be taken as $h/(c\lambda)$].

Find an expression for the distance l from the Earth to this binary system, using only the observed quantities and universal constants. Write your result on the answer sheet. (7.0 points)

(b) Assume that $M \gg m_0$, and that the ordinary star moves essentially in a circular orbit of radius r_0 around the neutron star. Assume that the ordinary star begins to emit gas toward the neutron star with velocity v_0 (relative to the ordinary star) (Fig. 32.8). In this problem consider only the gravitational action of the neutron star, and neglect the change in the orbit of the ordinary star. Find the distance of closest approach r_f of the gas to the neutron star (Fig. 32.8). Write your result on the answer sheet. (3.0 points)

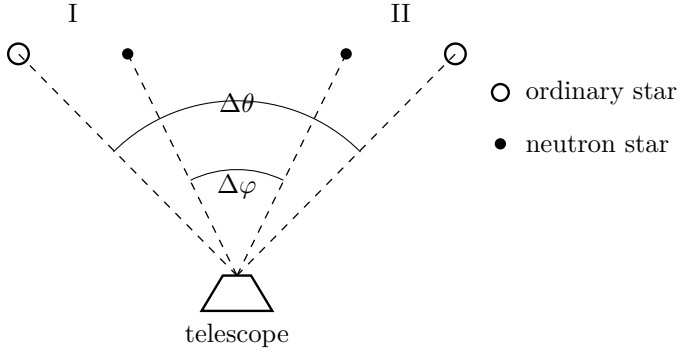


Fig. 32.7

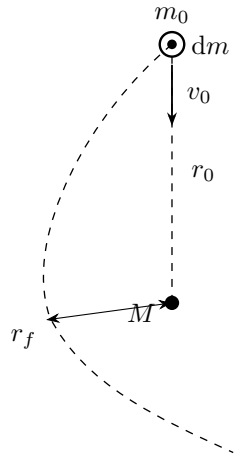


Fig. 32.8

Solutions

Problem 2

(a) The centre of mass of the binary system may be regarded as fixed. Let r_1 be the distance of the ordinary star from the centre of mass and r_2 that of the neutron star; from Fig. 32.7,

$$r_1 = l(\Delta\theta/2), \qquad r_2 = l(\Delta\varphi/2). \tag{1}$$

Hence

$$l = \frac{2(r_1 + r_2)}{\Delta\theta + \Delta\varphi}. \tag{2}$$

It is seen that the value of l depends on $r_1 + r_2$. From Newton's law of motion, and noting the relation between the angular velocity ω of the binary system and the observed quantity τ,

$$\omega = \frac{2\pi}{2\tau} = \frac{\pi}{\tau},$$

we have

$$\frac{GMm_0}{(r_1 + r_2)^2} = m_0 \left(\frac{\pi}{\tau}\right)^2 r_1,$$
$$\frac{GMm_0}{(r_1 + r_2)^2} = M \left(\frac{\pi}{\tau}\right)^2 r_2.$$

From these two equations,

$$\frac{r_1}{r_2} = \frac{M}{m_0},$$

and

$$Gm_0 = \left(\frac{\pi}{\tau}\right)^2 r_2(r_1 + r_2)^2. \tag{3}$$

It is seen that the value of $r_1 + r_2$ depends on m_0, while m_0 can be found from the gravitational redshift of the spectrum. From energy conservation, and noting the relation between the photon mass and the wavelength, we have

$$\frac{hc}{\lambda_0} - \frac{Gm_0}{R}\frac{h}{c\lambda_0} = \frac{hc}{\lambda_0 + \Delta\lambda}.$$

Hence

$$R = \frac{Gm_0(\lambda_0 + \Delta\lambda)}{c^2 \Delta\lambda}. \tag{4}$$

Thus m_0 is related to R. But R can be related to the observed quantity P:

$$P = \sigma T^4 \frac{4\pi R^2}{4\pi l^2} = \sigma T^4 \frac{R^2}{l^2}. \tag{5}$$

From equations (1), (2), (3), (4) and (5) the value of l expressed in terms of the observed quantities is obtained:

$$l = \left[\frac{2\tau c}{\pi(\Delta\theta + \Delta\varphi)}\right] \left[\frac{2\Delta\lambda (P/\sigma T^4)^{1/2}}{\Delta\varphi(\lambda_0 + \Delta\lambda)}\right]^{1/2}. \tag{6}$$

(b) The angular momentum of the gas element $\mathrm{d}m$ is conserved:

$$r_0^2 \omega_0 \, \mathrm{d}m = r_f^2 \omega_f \, \mathrm{d}m, \tag{7}$$

$$\omega_f = \frac{\omega_0 r_0^2}{r_f^2},$$

where ω_f is the angular velocity of the element when it is closest to the neutron star. The quantity ω_0 is determined by the dynamical relation of the original state:

$$\omega_0^2 = \frac{GM}{r_0^3}. \tag{8}$$

The energy of the element is conserved, that is,

$$\frac{1}{2}\mathrm{d}m \left(v_0^2 + r_0^2 \omega_0^2\right) - \frac{GM \, \mathrm{d}m}{r_0} = \frac{1}{2}\mathrm{d}m \, r_f^2 \omega_f^2 - \frac{GM \, \mathrm{d}m}{r_f}. \tag{9}$$

Combining equations (7), (8) and (9), we obtain

$$v_0^2 - \frac{GM}{r_0} = \frac{GMr_0}{r_f^2} - \frac{2GM}{r_f}.$$

That is,

$$\left(v_0^2 - \frac{GM}{r_0}\right) r_f^2 + 2GMr_f - GMr_0 = 0.$$

Solving this quadratic equation for r_f, we obtain

$$r_f = \frac{v_0 \sqrt{GMr_0} - GM}{v_0^2 r_0 - GM} \, r_0. \tag{10}$$

Problem 3

Magnetohydrodynamic generator

A horizontal rectangular plastic pipe, closed at the top, bottom, front and back, has width w and height h, and is filled with mercury of resistivity ρ. A pressure difference p produced by a turbine gives the fluid a constant flow velocity v_0. On each of the front and rear side faces of the pipe there is a copper plate of length L, as shown in the figure. The actual motion of the fluid is very complicated; for simplicity the following assumptions are made:

(1) In spite of the viscosity of the fluid, the velocity is uniform over the whole cross-section.

(2) The velocity of the fluid is always proportional to the net external force acting on it.

(3) The fluid is incompressible.

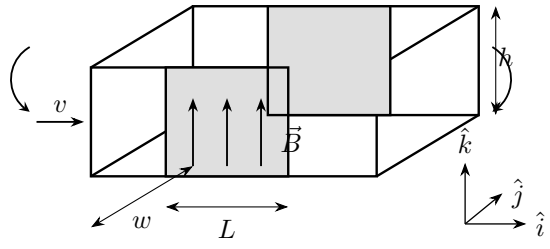


Fig. 32.9

The two front and rear side faces made of copper are short-circuited externally. A uniform magnetic field $\vec{B}$ directed vertically upward is applied only in the region between the two copper side faces. The arrangement is shown in the figure, which also shows the unit vectors $\hat{i}$, $\hat{j}$, $\hat{k}$ used in the answers.

(a) Find the force exerted by the magnetic field on the fluid (express it in terms of L, B, h, w, ρ and the new steady velocity v after the magnetic field is applied). Write the result on the answer sheet. (2.0 points)

(b) Derive an expression for the new steady velocity v of the fluid after the magnetic field is applied (in terms of v_0, p, L, B and ρ). Write the result on the answer sheet. (3.0 points)

(c) To increase the velocity to its original value v_0, the power of the turbine must be increased. Derive an expression for the increased power. Write the result on the answer sheet. (2.0 points)

(d) The magnetic field is now removed, and water flowing with velocity v_0 replaces the mercury. An electromagnetic wave of a single frequency f is sent along the direction of the water flow. Let the refractive index of water be n and $v_0 \ll c$. Derive an expression for the additional phase difference, caused by the motion of the fluid, of the electromagnetic wave between its entry into and its exit from the region between the two side faces of length L. Write the result on the answer sheet. (3.0 points)

Solutions

Problem 3

(a)

$$\mathcal{E} = vBw, \qquad I = \frac{vBw}{(\rho w/Lh)} = \frac{vBLh}{\rho}.$$

Therefore

$$F = IBw = \frac{vB^2Lhw}{\rho}.$$

The direction of this force is along the negative x-axis, that is, opposite to the flow velocity.

(b)

$$\frac{v}{v_0} = \frac{phw - F}{phw} = 1 - \frac{F}{phw}.$$

Substituting the expression for F, we obtain

$$v = v_0 \left[1 + \frac{v_0 B^2 L}{p\rho}\right]^{-1},$$

or

$$v = \frac{v_0 p\rho}{p\rho + v_0 B^2 L}.$$

(c) To increase the velocity to its original value v_0, the pump must increase the pressure by Δp in order to overcome the force exerted by the magnetic field on the current (which is opposite to the pressure force):

$$\Delta p\, hw = F,$$

where F is the value of F when the velocity is $v = v_0$. The increase in power is $\Delta P = \Delta p\, hw\, v_0 = Fv_0$. Substituting the value of F at $v = v_0$ into this expression, we obtain

$$\Delta P = \frac{v_0^2 B^2 Lhw}{\rho}.$$

(d) The speed of light in stationary water is $u = c/n$. The speed in water flowing with velocity v, according to the relativistic velocity addition formula, is

$$u' = \frac{c/n + v}{1 + (c/n)v/c^2} = \frac{c/n + v}{1 + v/(nc)}.$$

For low flow velocities ($v \ll c$), neglecting quantities of order v^2/c^2 and higher in the factor $(1 + v/nc)^{-1}$, we obtain

$$\begin{aligned} u' = \frac{c/n + v}{1 + v/(nc)} &\approx \left(\frac{c}{n} + v\right)\left(1 - \frac{v}{nc}\right) \\ &\approx \frac{c}{n} + v\left(1 - \frac{1}{n^2}\right). \end{aligned}$$

Thus

$$\Delta u = u' - u = v\left(1 - \frac{1}{n^2}\right),$$

and

$$\Delta\varphi = 2\pi f\, \Delta T.$$

Since $T = L/u$,

$$\Delta T = \frac{\Delta u\, L}{u^2} \approx \frac{Lv}{c^2}(n^2 - 1).$$

With $v = v_0$, therefore,

$$\Delta\varphi = 2\pi f \frac{L}{c^2}(n^2 - 1)v_0.$$

Experimental Problems

Problem 4

Rotating liquid

This experiment consists of three main parts:

(1) Study the shape of the surface of a rotating liquid and determine the acceleration due to gravity.

(2) Study the rotating liquid as an optical system.

(3) Determine the refractive index of the liquid.

When a cylindrical container filled with liquid rotates uniformly with angular velocity ω about the vertical axis through its centre, the surface of the liquid becomes a paraboloid (Fig. 32.10). In equilibrium, let the tangent plane at a point $P(x, y)$ on the liquid surface make an angle θ with the horizontal; then

$$\tan\theta = \omega^2 x/g \qquad \text{for } |x| \le R, \tag{1}$$

where R is the radius of the container and g the acceleration due to gravity.

It can further be shown that, for $\omega < \omega_{\max}$ (where $\omega_{\max}$ is the angular velocity at which the centre of the rotating liquid surface just touches the bottom of the container), when $x = x_0 = R/\sqrt{2}$,

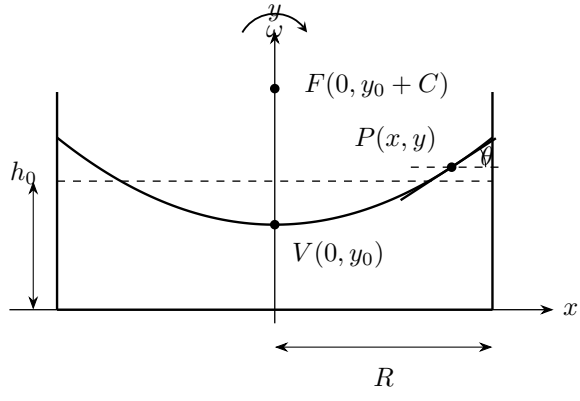
$$y(x_0) = h_0, \tag{2}$$

that is, the height of the rotating liquid there is the same as its height when the liquid is not rotating.

The shape of the surface of the rotating liquid is the paraboloid given by the equation

$$y = y_0 + \frac{x^2}{4C}. \tag{3}$$

The vertex of the paraboloid is at $V(0, y_0)$ and its focus at $F(0, y_0 + C)$. When rays parallel to the axis of symmetry (the optical axis) are reflected from this paraboloid, they all meet at the focus F (Fig. 32.10).



Original depth h_0; the parabolic surface produced when a vessel of radius R rotates about the y-axis with constant angular velocity ω, showing the inclination θ at a point $P(x, y)$, the vertex V and the focus F.

Fig. 32.10

Apparatus

(1) A cylindrical rigid plastic cup filled with liquid glycerine, with millimetre scales pasted on the bottom and on the side wall.

(2) A turntable driven by a small DC motor; the motor is supplied by a DC power supply with adjustable voltage, by means of which the angular velocity of the turntable can be controlled.

(3) A horizontally placed transparent screen, on which transparent or translucent coordinate paper can be placed. The position of the screen can be adjusted in the vertical or horizontal direction.

(4) A laser pen fixed on a support. The position of the laser pen can be adjusted. The cap on its head is interchangeable.

(5) One replacement cap for the laser pen.

(6) A ruler.

(7) A marker pen.

(8) A stopwatch. Its left button resets, its middle button selects the working mode, and its right button starts or stops the timing.

(9) A grating marked with 500 or 1000 lines/mm.

(10) A spirit level.

(11) Protective goggles.

Precautions

(1) Do not look directly into the laser beam. Note that the laser beam reflected from a mirror-like surface is also very dangerous; for safety, always use protective goggles.

(2) Throughout the experiment, handle the cup containing glycerine with care when moving it.

(3) The turntable has been levelled in advance; the spirit level is used only to adjust the horizontal position of the transparent screen.

(4) During the whole experiment you will observe several spots on the screen; they are formed by beams reflected or refracted at the interfaces between the air, the liquid, the screen and the cup. Make sure that the beam you measure is the one required by the experiment.

(5) The rotation speed must be changed slowly, and you must wait long enough for the liquid to reach equilibrium before making measurements.

Experimental procedure

Part 1. Determination of g with a rotating liquid (7.5 points)

(1) Derive equation (1).

(2) Measure the height h_0 of the liquid in the container and the inner diameter $2R$ of the container.

(3) Insert a screen between the light source and the container; measure the distance H between the turntable and the screen (Fig. 32.11).

(4) Adjust the laser pen so that the beam is vertically downward and passes through the liquid surface at a horizontal distance $x_0 = R/\sqrt{2}$ from the centre of the container.

(5) Slowly rotate the turntable. Take care not to let the centre of the rotating liquid surface touch the bottom of the container.

(6) It is known that at $x_0 = R/\sqrt{2}$, whatever the angular velocity ω, the height of the liquid surface there is the same as when the liquid is stationary. Use this fact to measure the inclination θ of the tangent plane to the liquid surface at x_0 for different angular velocities ω, and thereby determine the acceleration due to gravity g experimentally.

(7) Tabulate the measured and calculated values for each ω.

(8) Draw a graph in order to calculate g.

(9) Calculate the value of g and its experimental error.

(10) Write $2R$, x_0, h_0, H and the experimentally determined value of g and its error on the answer sheet.

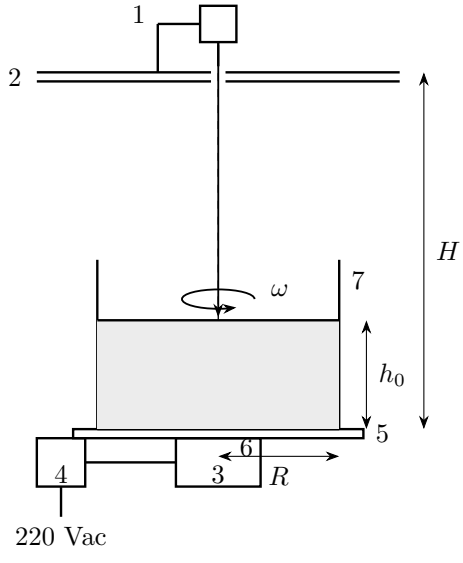


Fig. 32.11

1. laser pen on a stand; 2. transparent screen; 3. motor; 4. motor controller; 5. turntable; 6. rotating shaft; 7. cylindrical container.

Part 2. Optical system

In this part of the experiment the rotating liquid is regarded as an imaging optical system. Since the curvature of the liquid surface varies with the angular velocity of rotation, the focal length of this optical system depends on ω.

(a) Study of the focal length (5.5 points)

(1) Adjust the laser pen so that the laser beam is directed vertically downward onto the centre of the bottom of the cup. Mark the point where the beam intersects the screen as P. The line joining the centre of the bottom of the cup to this point is the optical axis of the system (Fig. 32.11).

(2) Since the liquid surface acts like a parabolic mirror, any beam parallel to the optical axis will pass through the focus F on the optical axis after reflection from this paraboloid.

(3) Adjust the rotation speed so that the focus falls on the screen. Measure the angular velocity and the distance H between the screen and the turntable.

(4) Repeat the above steps for different values of H.

(5) Enter the measured values of $2R$ and h_0, and the measured value of ω corresponding to each H, on the answer sheet.

(6) Using your measured data, draw a graph and find the relation between the focal length and the angular velocity. For simplicity you may take approximately $H - h_0$ as the focal length. Write your result on the answer sheet.

(b) Analysis of the "image" (that is, what you see on the screen) (3.5 points)

In this part of the experiment you will analyse the "image" produced by this optical system. To this end, proceed as follows.

(1) Unscrew the cap of the laser pen counterclockwise.

(2) Screw on clockwise the new cap of the laser pen (placed in the sealed bag). Your laser pen can now form a clear pattern rather than a thin beam.

(3) Adjust the position of the laser pen so that the laser beam falls almost vertically near the centre of the cup.

(4) Place a sheet of translucent coordinate paper on the horizontal screen near the cup, so that the incident laser beam does not hit the translucent paper while the reflected laser beam can fall on it.

(5) With the liquid not rotating, observe the size and orientation of the "image" formed by the emerging laser beam, and the size and orientation of the "image" formed by the reflected laser beam.

(6) Start rotating the liquid, gradually increasing the rotation speed to the maximum that the apparatus can reach, while observing the changes of the "image" on the screen. As ω increases, you will see several different frequency ranges, in which the properties of the "image" are completely different. In order to describe these observations, add rows to the table on the answer sheet, fill in each row the frequency range observed above, and complete the table using the symbols provided. The meanings of these symbols are explained on the answer sheet.

Part 3. Refractive index (3.5 points)

In this part of the experiment we shall use a grating to determine the refractive index of the given liquid. When a monochromatic beam of wavelength λ is incident normally on a diffraction grating, the maximum of the diffraction pattern is observed at α_m, where α_m satisfies the equation

$$m\lambda = d\sin\alpha_m, \tag{4}$$

where m is the order of diffraction and d is the grating constant. In this part of the experiment a diffraction grating is used to determine the wavelength of the laser and the refractive index of the liquid (Fig. 32.12).

(1) Use the grating to measure the wavelength of the laser pen, and write your result on the answer sheet.

(2) Immerse the grating vertically in the liquid and place it at the centre of the cup.

(3) Adjust the laser beam so that it enters the liquid through the wall of the cup and is incident normally on the grating.

(4) Observe the diffraction pattern on the millimetre scale pasted on the opposite side wall of the cup, and make the necessary measurements.

(5) Use your measurements to calculate the refractive index of the liquid (ignore the effect of the plastic cup on the light path).

(6) Write your experimental results on the answer sheet.

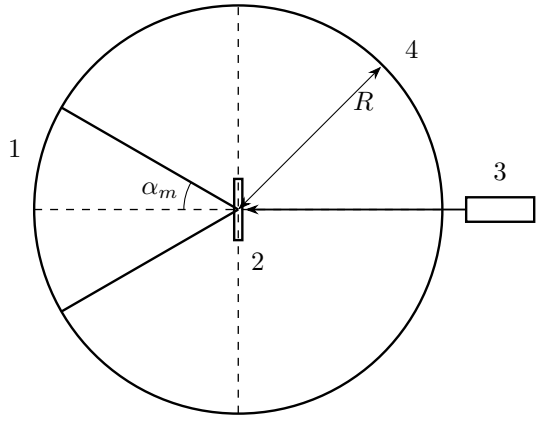


Fig. 32.12

1. side wall with scale; 2. grating on a support; 3. laser pen; 4. cylindrical container.

Solutions to the Experimental Problems

Part 1. Determination of the acceleration due to gravity

Theoretical derivation

Consider a small element of mass m on the liquid surface. When it is in equilibrium (Fig. 32.13),

$$N\cos\theta = mg,$$

$$N \sin\theta = m\omega^2 x.$$

The shape of the liquid surface can be derived as follows:

$$\tan\theta = \frac{\mathrm{d}y}{\mathrm{d}x}, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\omega^2 x}{g},$$

hence

$$y = \frac{\omega^2 x^2}{2g} + y_0$$

(y_0 being the height at $x = 0$). For a certain point $x = x_0$, the height h_0 of the liquid surface is the same as when the liquid is not rotating. In this case,

$$h_0 = y_0 + \frac{\omega^2 x_0^2}{2g}, \tag{1}$$

and

$$x_0^2 = \frac{2g(h_0 - y_0)}{\omega^2}.$$

Since the volume of the liquid is unchanged,

$$\pi R^2 h_0 = \int_0^R y(2\pi x\,\mathrm{d}x) = 2\pi \int_0^R \left(y_0 + \frac{\omega^2 x^2}{2g}\right) x\,\mathrm{d}x,$$

$$y_0 = h_0 - \frac{\omega^2 R^2}{4g}. \tag{2}$$

From equations (1) and (2) we obtain

$$x_0 = \frac{R}{\sqrt{2}}.$$

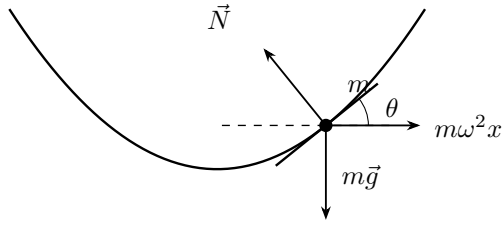


Fig. 32.13

Experimental part

The measured values of h_0, H, x_0 and other quantities are given in Table 32.1.

Table 32.1

$2R/$ mm	$x_0/$ mm	$h_0/$ mm	$H/$ mm
145.0	51.0	30.0	160.0

$H - h_0 = 130.0\,\mathrm{mm}$.

For rotations with small angular velocity, the time $10T$ taken by the disc to complete 10 periods is measured; for rotations with larger angular velocity, the time $15 \sim 20T$ taken to complete 15 to 20 periods is measured. Using

$$\tan(2\theta) = \frac{x}{H - h_0}, \qquad \omega = \frac{2\pi}{T}$$

(where x is the horizontal distance between the intersection of the reflected beam with the transparent screen and the incident beam), the experimental data are tabulated as follows. Plotting $\tan\theta$ against ω^2 gives Fig. 32.14.

Table 32.2

r mm	$10T$ s	ω rad s^{-1}	$\tan 2\theta$	θ rad	θ deg	$\tan\theta$	ω^2 (rad s^{-1})^2
11	21.34	2.94	0.08	0.04	2.4	0.04	8.67
20	15.80	3.98	0.15	0.08	4.4	0.08	15.81
26	14.22	4.42	0.20	0.10	5.7	0.10	19.52
30	12.99	4.84	0.23	0.11	6.5	0.11	23.40
40	11.74	5.35	0.31	0.15	8.6	0.15	28.64
51	10.43	6.01	0.39	0.19	10.7	0.19	36.15
56	9.90	6.35	0.43	0.20	11.7	0.21	40.28
65	9.40	6.68	0.50	0.23	13.3	0.24	44.68
70	9.08	6.92	0.54	0.25	14.2	0.25	47.88
85	8.39	7.49	0.65	0.29	16.6	0.30	56.08
100	7.71	8.15	0.77	0.33	18.8	0.34	66.41
112	7.43	8.46	0.86	0.36	20.4	0.37	71.51
132	7.00	8.98	1.02	0.40	22.7	0.42	80.57
61.4	11.19	6.20	0.47	0.21	11.98	0.21	41.51

(The mean values in the last row of the table are used only for calculating the error.)

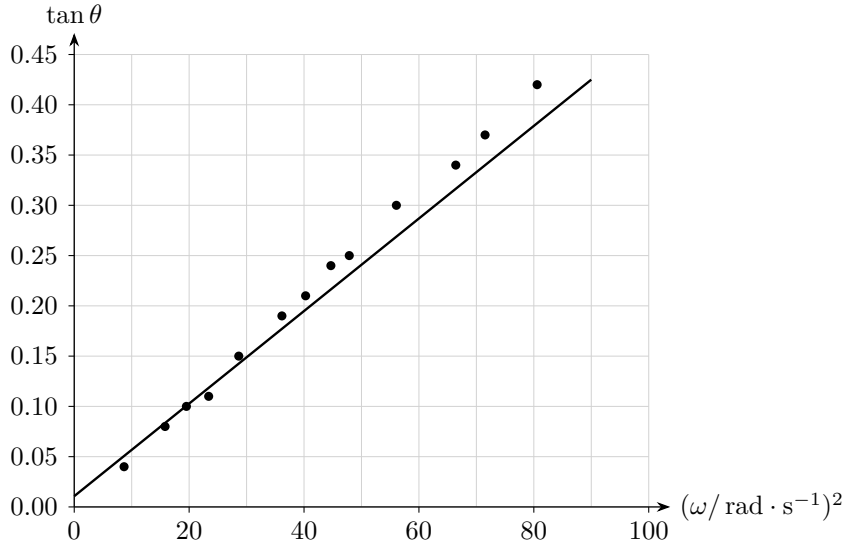


Fig. 32.14

The slope of the straight line in Fig. 32.14 is $0.00520\,(\mathrm{s/rad})^2$; therefore

$$g = \frac{x_0}{\text{slope}} = \frac{5.10}{0.00520}\,\mathrm{cm/s^2} = 981\,\mathrm{cm/s^2}.$$

Error calculation

$$g = \frac{\omega^2 x_0}{\tan\theta},$$

$$\frac{\Delta g}{g} = \sqrt{2\left(\frac{\Delta\omega}{\omega}\right)^2 + \left(\frac{\Delta x_0}{x_0}\right)^2 + \left(\frac{\Delta\tan\theta}{\tan\theta}\right)^2}$$

(here the uncertainty propagation formula is used),

$$\frac{\Delta\omega}{\omega} = \frac{\Delta T}{T},$$

$$\frac{\Delta\tan\theta}{\tan\theta} \approx \frac{\Delta\theta}{\theta} \qquad (\text{from the graph, } \tan\theta \approx \theta),$$

$$\theta \approx \frac{x}{H - h_0}, \qquad \frac{\Delta\theta}{\theta} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta H}{H}\right)^2 + \left(\frac{\Delta h_0}{h_0}\right)^2},$$

$$\frac{\Delta g}{g} = \sqrt{2\left(\frac{\Delta T}{T}\right)^2 + \left(\frac{\Delta x_0}{x_0}\right)^2 + \left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta H}{H}\right)^2 + \left(\frac{\Delta h_0}{h_0}\right)^2}.$$

Since $H = 160.0\,\text{mm}$, $\Delta H = 1\,\text{mm}$, $h_0 = 30.0\,\text{mm}$, $\Delta h_0 = 1\,\text{mm}$, $\bar{x} = 61.4\,\text{mm}$, $\Delta\bar{x} = 1\,\text{mm}$, $\bar{T} = 1.1\,\text{s}$, $\Delta T = 0.01\,\text{s}$, $x_0 = 51.0\,\text{mm}$ and $\Delta x_0 = 1\,\text{mm}$, we obtain

$$g = (981 \pm 42)\,\text{cm/s}^2.$$

$*$ Note: by the method of least squares,

$$g = 982\,\text{cm/s}^2,$$

with standard deviation $\sigma = 33\,\text{cm/s}^2$.

$**$ Using linear approximation, the slope of $\tan\theta$ against ω^2 is 0.00518, with a standard deviation of 5.14×10^{-5}; therefore

$$\frac{\Delta g}{g} = \sqrt{\left[\frac{\Delta(\text{slope})}{\text{slope}}\right]^2 + \left(\frac{\Delta x_0}{x_0}\right)^2} = 0.023,$$

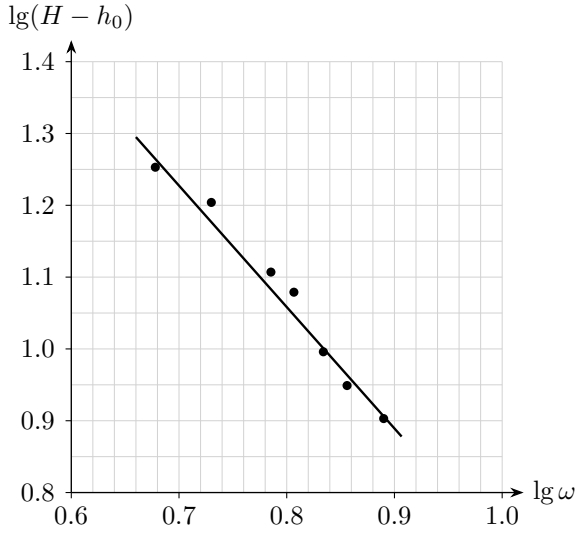
$$g = (981 \pm 23)\,\text{cm/s}^2.$$

Part 2. Measurement of the focal length and analysis of the image

(a) The experimental data are tabulated as follows. Plotting $\lg(H - h_0)$ against $\lg\omega$ gives Fig. 32.15.

Table 32.3

$H/\,\mathrm{mm}$	$10T/\,\mathrm{s}$	$\omega/\,\mathrm{rad}\cdot\mathrm{s}^{-1}$	$\lg\omega$	$(H-h_0)/\,\mathrm{mm}$	$\lg(H-h_0)$
158	10.31	6.09	0.784921	128	2.107
209	13.19	4.76	0.677935	179	2.253
190	11.70	5.37	0.729994	160	2.204
150	9.80	6.41	0.806954	120	2.079
129	9.21	6.82	0.83392	99	1.996
119	8.75	7.18	0.856172	89	1.949
110	8.10	7.76	0.889695	80	1.903



Relation between $\tan\theta$ and ω^2 at x_0.

Fig. 32.15

From Fig. 32.15, the focal length f is a power function of ω: $f = A\omega^n$ with $n \approx -1.7$. (One may also plot $(H-h_0)$ against $1/\omega^2$.)

Noting that for the paraxial region of a spherical mirror we have

$$\tan\theta = \frac{\omega^2 x}{g}, \qquad \theta \approx \frac{x}{R} \approx \tan\theta,$$

this gives

$$R = \frac{g}{\omega^2}, \qquad \text{or} \qquad f = \frac{R}{2} = \frac{g}{2\omega^2}.$$

If the measurement is made in the paraxial region, then $n \approx -2$.

(b) Use the appropriate symbols to describe the pattern formed by reflection that you observe on the screen.

* For $H = 110\,\mathrm{mm}$. ** For $H = 240\,\mathrm{mm}$.

The value of ω is determined by H, h_0 and other quantities.

Notes

(1) Orientation (compared with the object beam observed on the transparent screen): INV—inverted; ER—erect.

Table 32.4

ω range $\mathrm{rad\,s^{-1}}$	Direction	Size trend	Image property
$\omega = 0$	ER		V
$0 < \omega < 8.2^{*}$	ER	D	V
$0 < \omega < 6.3^{**}$			
$8.2 < \omega < 14.6^{**}$	INV	I	R
$6.3 < \omega < 14.0^{**}$			
$14.6 < \omega < \omega^{*}_{\max}$	ER	NC	V
$14.0 < \omega < \omega^{**}_{\max}$			

(2) Change of the size of the pattern as ω increases: I—increases; D—decreases; NC—unchanged.

(3) Property of the image for different frequency ranges: if the screen is above the focus, write R; if the screen is below the focus, write V.

Part 3. Measurement of the refractive index

Measurement of the wavelength

When both the grating and the screen are in air, the light is incident normally. Let L be the distance between the screen and the grating, x the distance of a diffraction spot from the central spot on the screen, and m the order of diffraction.

$$L = 225\,\mathrm{mm}, \quad \bar{x} = 77\,\mathrm{mm}, \quad \text{for } m = \pm 1, \quad d = \frac{1}{500}\,\mathrm{mm},$$
$$\tan\alpha = \frac{\bar{x}}{L} = \frac{77}{225} = 0.342, \qquad \alpha = 18.89^{\circ},$$
$$\lambda = \frac{1}{500}\sin\alpha = 647.6\,\mathrm{nm}.$$

$$L = 128.0\,\mathrm{mm}, \quad \bar{x} = 44.0\,\mathrm{mm}, \quad \text{for } m = \pm 1, \quad d = \frac{1}{500}\,\mathrm{mm},$$
$$\tan\alpha = \frac{44.0}{128.0} = 0.344, \qquad \alpha = 18.97^{\circ},$$
$$\lambda = \frac{1}{500}\sin\alpha = 650.2\,\mathrm{nm}.$$

$$L = 128.0\,\mathrm{mm}, \quad \bar{x} = 111.0\,\mathrm{mm}, \quad \text{for } m = \pm 2, \quad d = \frac{1}{500}\,\mathrm{mm},$$
$$\tan\alpha = \frac{111}{128} = 0.867, \qquad \alpha = 40.9^{\circ},$$
$$\lambda = \frac{1}{2\times 500}\sin\alpha = 655.1\,\mathrm{nm}.$$

The mean value of λ is

$$\bar{\lambda} = 651.0\,\mathrm{nm}.$$

Measurement of the refractive index

When both the grating and the screen are in the liquid, the light is incident normally. From the measurements above,

$$2R = 145.0\,\text{mm}.$$

The distance of a diffraction spot from the central spot, measured along the curved surface, is $R\alpha$. It is found that

$$R\bar{\alpha} = 17.0\,\text{mm}, \qquad \text{for } m = \pm 1,$$

$$\bar{\alpha} = 0.234\,\text{rad}.$$

From

$$n = \frac{m\lambda}{d\sin\alpha},$$

we obtain

$$n = 1.40.$$

If the curvature of the surface is ignored, then

$$\tan\alpha = \frac{17.0}{72.5} = 0.234, \qquad \alpha = 13.20^\circ,$$

$$n = \frac{\lambda}{d\sin\alpha} = \frac{650.9\,\text{nm}}{\frac{1}{500}\,\text{mm}\times 10^6\sin\alpha} = 1.43.$$

The 33rd International Physics Olympiad

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Theoretical Problems

Problem 1. Ground-penetrating radar

Ground-penetrating radar (GPR) detects and locates objects near the Earth's surface by transmitting electromagnetic waves into the ground and receiving the waves reflected back from underground objects. The antenna and the detector are placed directly on the ground, at the same location.

The electric field of a linearly polarised plane electromagnetic wave of angular frequency ω propagating in the z direction is given by

$$E = E_0 \mathrm{e}^{-\alpha z}\cos(\omega t - \beta z), \tag{1}$$

where E_0 is a constant, α is the attenuation constant and β is the wave number, given respectively by

$$\begin{aligned}
\alpha &= \omega\left\{\frac{\mu\varepsilon}{2}\left[\left(1+\frac{\sigma^2}{\varepsilon^2\omega^2}\right)^{1/2} - 1\right]\right\}^{1/2}, \\
\beta &= \omega\left\{\frac{\mu\varepsilon}{2}\left[\left(1+\frac{\sigma^2}{\varepsilon^2\omega^2}\right)^{1/2} + 1\right]\right\}^{1/2},
\end{aligned} \tag{2}$$

where μ, ε and σ are the magnetic permeability, the permittivity and the conductivity respectively.

When the amplitude of the radar signal on reaching an object falls to 1/e (about 37%) of its initial value, the object can no longer be detected. Electromagnetic waves with frequencies commonly between 10 MHz and 1000 MHz are used, so as to adjust the detection range and the resolution.

The performance of GPR depends on its resolution. The resolution is determined by the minimum separation between two adjacent reflecting bodies to be detected, this minimum separation corresponding to a minimum phase difference of 180° between the two reflected waves at the detector.

Problems (Known: $\mu = 4\pi \times 10^{-7}\,\mathrm{H/m}$, $\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F/m}$.)

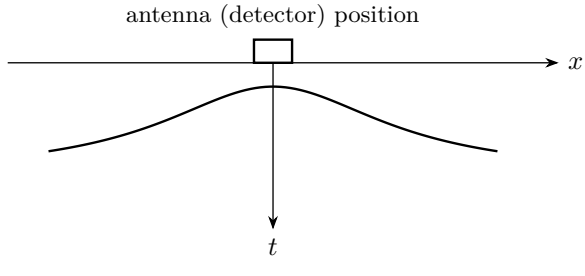
(1) Assuming that the ground is a non-magnetic substance ($\mu = \mu_0$) and satisfies the condition $\left(\frac{\sigma}{\omega\varepsilon}\right)^2 \ll 1$, use equations (1) and (2) to derive the expression for the propagation speed v (in terms of μ and ε). (1.0 point)

(2) Determine the maximum depth of an object that can be detected underground, given that the conductivity of the ground is 1.0 mS/m, its permittivity is $9\varepsilon_0$, and the condition $\left(\frac{\sigma}{\omega\varepsilon}\right)^2 \ll 1$ is satisfied. (2.0 points)

($1\,\mathrm{S} = \Omega^{-1}$, $\mu = \mu_0$.)

(3) Consider two conducting rods buried horizontally and parallel to each other underground, both at a depth of 4 m. The conductivity of the ground is known to be 1.0 mS/m and its permittivity $9\varepsilon_0$. Suppose that GPR measurements are made just above the position of one of the conducting rods, and that the detector is a point. Determine the lowest frequency required for the transverse resolution to reach 50 cm. (3.5 points)

(4) In order to determine the depth d of a conducting rod buried in a stratum with the same conditions as in question (2), measurements are made along the direction perpendicular to the conducting rod. The results can be represented by Fig. 33.1 as the relation between the propagation time t of the electromagnetic wave and the detection position x. Taking $t_{\min} = 100\,\mathrm{ns}$, derive the function $t(x)$ and determine d. (3.5 points)



Propagation time t versus detection position x; $t_{\min} = 100\,\mathrm{ns}$.

Fig. 33.1

Solutions

Solution 1

(1) Propagation speed of the radar wave in the medium From the phase $\omega t - \beta z$ being constant,

$$\omega t - \beta z = \text{constant} \quad \Rightarrow \quad \beta z = -\text{constant} + \omega t,$$

hence

$$v = \frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\omega}{\beta}.$$

Therefore

$$v = \frac{\omega}{\beta} = \frac{\omega}{\omega\left\{\frac{\mu\varepsilon}{2}\left[\left(1+\frac{\sigma^2}{\varepsilon^2\omega^2}\right)^{1/2}+1\right]\right\}^{1/2}} = \frac{1}{\left\{\frac{\mu\varepsilon}{2}\left[\left(1+\frac{\sigma^2}{\varepsilon^2\omega^2}\right)^{1/2}+1\right]\right\}^{1/2}}.$$

Since $\sigma^2/\varepsilon^2\omega^2 \ll 1$,

$$v = \frac{1}{\left\{\frac{\mu\varepsilon}{2}(1+1)\right\}^{1/2}} = \frac{1}{\sqrt{\mu\varepsilon}}.$$

(2) Maximum detectable depth of an object underground The maximum detectable depth (the skin depth) δ is inversely proportional to the attenuation constant:

$$\begin{aligned}\delta = \frac{1}{\alpha} &= \frac{1}{\omega\left\{\frac{\mu\varepsilon}{2}\left[\left(1+\frac{\sigma^2}{\varepsilon^2\omega^2}\right)^{1/2}-1\right]\right\}^{1/2}} \\ &= \frac{1}{\omega\left\{\frac{\mu\varepsilon}{2}\left[\left(1+\frac{1}{2}\frac{\sigma^2}{\varepsilon^2\omega^2}\right)-1\right]\right\}^{1/2}} = \frac{1}{\omega\left\{\frac{\mu\varepsilon}{2}\cdot\frac{1}{2}\frac{\sigma^2}{\varepsilon^2\omega^2}\right\}^{1/2}} \\ &= \left(\frac{2}{\sigma}\right)\left(\frac{\varepsilon}{\mu}\right)^{1/2}.\end{aligned}$$

Numerically,

$$\delta = \frac{5.31\sqrt{\varepsilon_r}}{\sigma}\,\mathrm{m},$$

where σ is measured in mS/m.

For a medium with conductivity 1.0 mS/m and relative permittivity 9, the skin depth is

$$\delta = \frac{5.31\sqrt{9}}{1.0} = 15.93\ (\mathrm{m}).$$

(3) Transverse resolution As shown in Fig. 33.2, for the two reflected waves to have a phase difference of 180°, the difference between their path lengths must be $\lambda/2$, so the extra one-way path to the second rod is $\lambda/4$. Hence

$$r^2 + d^2 = \left(d + \frac{\lambda}{4}\right)^2,$$

and therefore

$$r = \left(\frac{\lambda d}{2} + \frac{\lambda^2}{16}\right)^{1/2}.$$

With $r = 0.5\,\mathrm{m}$ and $d = 4\,\mathrm{m}$,

$$\frac{1}{2} = \left(\frac{4\lambda}{2} + \frac{\lambda^2}{16}\right)^{1/2}, \qquad \lambda^2 + 32\lambda - 4 = 0.$$

Thus the wavelength is

$$\lambda = 0.125\,\mathrm{m}.$$

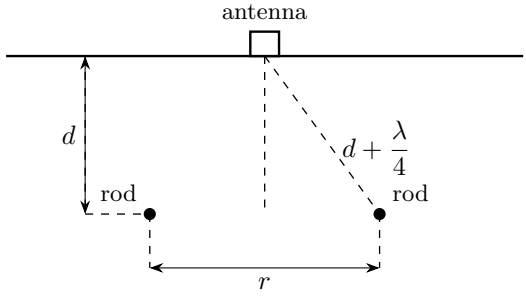


Fig. 33.2

The propagation speed of the signal in the medium is

$$\begin{aligned} v &= \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_0\mu_r\varepsilon_0\varepsilon_r}} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \cdot \frac{1}{\sqrt{\mu_r\varepsilon_r}} \\ &= \frac{c}{\sqrt{\mu_r\varepsilon_r}} = \frac{0.3}{\sqrt{\varepsilon_r}}\,\mathrm{m/ns} = 0.1\,\mathrm{m/ns} = 10^8\,\mathrm{m/s}, \end{aligned}$$

where

$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}, \qquad \mu_r = 1.$$

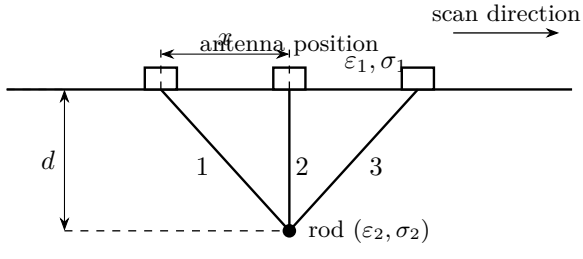
The lowest frequency required to resolve the two separate rods is

$$f_{\min} = \frac{v}{\lambda}.$$

Thus

$$f_{\min} = \frac{0.3/\sqrt{9}}{0.125} \times 10^9\ (\mathrm{Hz}) = 800\ (\mathrm{MHz}).$$

(4) Dependence of the propagation time on position When the antenna and the detector are at a certain position on the ground, the path of the electromagnetic wave is as shown in Fig. 33.3.



Rays 1, 2 and 3 from three antenna positions to the buried rod.

Fig. 33.3

The propagation time t as a function of x satisfies

$$\left(\frac{tv}{2}\right)^2 = d^2 + x^2,$$

so

$$t(x) = \sqrt{\frac{4d^2 + 4x^2}{v^2}}, \qquad t(x) = \frac{2\sqrt{\varepsilon_r}}{0.3}\sqrt{d^2 + x^2}.$$

For $x = 0$,

$$100 = 2 \times \frac{3}{0.3} d,$$

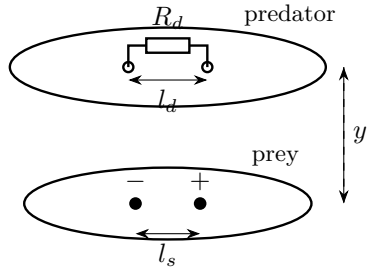
and hence

$$d = 5\,\mathrm{m}.$$

Problem 2. Sensing electric signals

Some marine animals can detect other animals at a certain distance from them, because respiration or other processes, including muscle contraction, produce electric currents. Some carnivorous animals (predators) use such electric signals to locate their prey (even if the prey is buried in the seabed sand).

The electricity-generating mechanism of the prey and the detection mechanism of the predator can be represented by the model of Fig. 33.4. The current in the prey's body flows between two spheres carrying positive and negative potentials respectively. Assume that the distance between the two spheres is l_s, that the radius of each sphere is r_s, that r_s is much smaller than l_s, and that the resistivity of seawater is ρ.



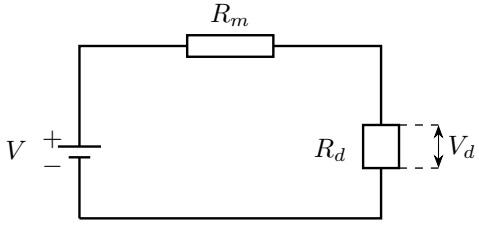
Model of the predator detecting the electric signal from the prey.

Fig. 33.4

Assume that the resistivity of the prey's body is the same as that of the surrounding seawater, so that the boundary of the prey's body in the figure may be ignored.

In order to describe the detection of the electric signal generated by the prey, the detector may similarly be simplified as two spheres on the predator's body, in contact with the surrounding seawater, parallel to the pair of electrodes of the prey, the two spheres being separated by l_d, each sphere having radius r_d, with r_d much smaller than l_d. In the case under consideration, the centre of the detector is at a distance y directly above the signal source, and the line joining the two spheres is parallel to the electric field, as shown in Fig. 33.4. The distances l_s and l_d between the two spheres of the source and of the detector are both much smaller than y. Assume that the electric field strength is constant along the line joining the two detector spheres; then the detector, the prey, the surrounding seawater and the predator form a closed circuit, as shown in

Fig. 33.5. In the figure, V is the potential difference between the two detector spheres produced by the electric field induced by the prey (because of the biological effects of the predator, V here is equivalent to an electromotive force — translator's note), and R_m is the resistance produced when the two spheres of radius r_d are surrounded by the seawater (the internal resistance). In addition, V_d and R_d are respectively the potential difference and the resistance between the two detecting spheres in the predator's body.



Equivalent closed circuit including the predator, the prey and the surrounding seawater.

Fig. 33.5

Problems (1) Determine the current density vector $\vec{j}$ at a distance r from a point current source I_s placed in an infinite conducting medium (the current through unit area). (1.5 points)

(2) Using the law $\vec{E} = \rho\vec{j}$, determine the field strength $\vec{E}_P$ at the midpoint P of the line joining the two detector spheres, assuming that the current between the two spheres in the prey's body is I_s. (2.0 points)

(3) Determine the potential difference V_s between the two source spheres in the prey's body (1.5 points), the resistance R_s between the two source spheres (0.5 points) and the electric power P_s generated by the source (0.5 points).

(4) Determine the resistance R_m between the two detecting spheres in Fig. 33.5 (0.5 points) and the potential difference V_d between the two detecting spheres (1.0 point). Calculate the electric power of the detector (0.5 points). (The original text reads "the electric power transmitted from the source to the detector" — translator's note.)

(5) Determine the optimum value of R_d for which the detected electric power is maximum (1.5 points), and determine this maximum power (0.5 points).

Solution 2

(1) Current density of a point source When a point current source I_s is located in an infinite isotropic medium, the current density vector at a distance r from the point source is

$$\vec{j} = \frac{I_s}{4\pi r^2}\hat{r}.$$

(2) Field strength at the midpoint of the detector The problem assumes that the resistivity of the prey's body is the same as that of the surrounding seawater; this means that the boundary of the prey disappears, and the two spheres are as if located in an infinite isotropic medium of resistivity ρ. When the current produced by one small sphere is I_s, the current density at a distance r from the centre of the sphere is again

$$\vec{j} = \frac{I_s}{4\pi r^2}\hat{r}.$$

Since the resistivity of seawater is ρ, the electric field at r is

$$\vec{E}(r) = \rho\vec{j} = \frac{\rho I_s}{4\pi r^3}\vec{r}.$$

In our model there are two small spheres, one at a positive voltage relative to the other, so that the current I_s flows from the positively charged sphere to the negatively charged sphere, see Fig. 33.6. If the two spheres are separated by l_s, the field strength at $P(0, y)$ is

$$\begin{aligned}\vec{E}_P &= \vec{E}_+ + \vec{E}_- \\ &= \frac{\rho I_s}{4\pi}\left\{\frac{1}{[(l_s/2)^2 + y^2]^{3/2}}\left(-\frac{l_s}{2}\hat{i} + y\hat{j}\right) + \frac{1}{[(l_s/2)^2 + y^2]^{3/2}}\left(-\frac{l_s}{2}\hat{i} - y\hat{j}\right)\right\} \\ &= \frac{\rho I_s}{4\pi}\left\{\frac{l_s(-\hat{i})}{[(l_s/2)^2 + y^2]^{3/2}}\right\}.\end{aligned}$$

For $l_s \ll y$,

$$\vec{E}_P \approx \frac{\rho I_s l_s}{4\pi y^3}(-\hat{i}).$$

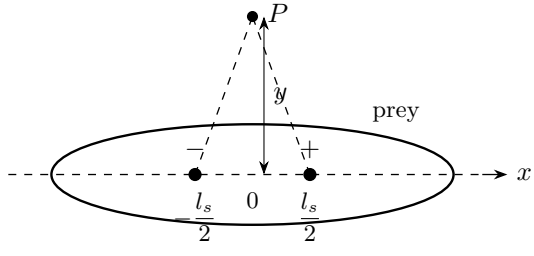


Fig. 33.6

(3) Potential difference, resistance and power of the source The field strength on the x axis along the line joining the two spheres is

$$\vec{E}(x) = \frac{\rho I_s}{4\pi}\left[\frac{1}{(x - l_s/2)^2} + \frac{1}{(x + l_s/2)^2}\right](-\hat{i}).$$

The potential difference producing the given current I_s is

$$\begin{aligned}V_s = \Delta V = V_+ - V_- &= \int_{l_s/2 - r_s}^{-l_s/2 + r_s} \vec{E}(x)\cdot \mathrm{d}\vec{x} \\ &= \frac{\rho I_s}{4\pi}\left\{\left[\frac{1}{r_s} - \frac{1}{l_s - r_s}\right] - \left[\frac{1}{l_s - r_s} - \frac{1}{r_s}\right]\right\} \\ &= \frac{\rho I_s}{4\pi}\left(\frac{2}{r_s} - \frac{2}{l_s - r_s}\right) = \frac{2\rho I_s}{4\pi}\cdot\frac{l_s - 2r_s}{(l_s - r_s)r_s} \\ &= \frac{\rho I_s}{2\pi r_s}\cdot\frac{l_s - 2r_s}{l_s - r_s}.\end{aligned}$$

For $l_s \gg r_s$,

$$V_s = \Delta V \approx \frac{\rho I_s}{2\pi r_s}.$$

The resistance between the two source spheres is

$$R_s = \frac{V_s}{I_s} = \frac{\rho}{2\pi r_s}.$$

The power generated by the source is

$$P = I_s V_s = \frac{\rho I_s^2}{2\pi r_s}.$$

(4) Resistance, potential difference and power of the detector As shown in Fig. 33.7, V is the electromotive force produced between the two detector spheres by the electric field induced by the prey, R_m is the internal resistance produced by the surrounding seawater, V_d is the potential difference between the two detector spheres, R_d is the resistance between the two detector spheres in the predator's body, and i_d is the current flowing through the closed circuit.

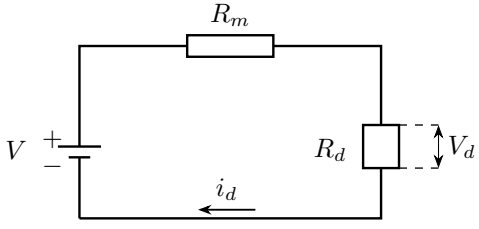


Fig. 33.7

By analogy with the resistance between the two source spheres, the resistance of the medium of resistivity ρ between the two detector spheres (each of radius r_d) is

$$R_m = \frac{\rho}{2\pi r_d}.$$

Since l_d is much smaller than y (see Fig. 33.4), the electric field between the two detector spheres may be regarded as constant, and is

$$E = \frac{\rho I_s l_s}{4\pi y^3}.$$

Hence the electromotive force existing between the two detector spheres in the medium is

$$V = E l_d = \frac{\rho I_s l_s l_d}{4\pi y^3}.$$

The potential difference between the two detector spheres is then

$$V_d = V \frac{R_d}{R_d + R_m} = \frac{\rho I_s l_s l_d}{4\pi y^3} \cdot \frac{R_d}{R_d + \dfrac{\rho}{2\pi r_d}}.$$

The power of the detector is

$$P_d = i_d V_d = \frac{V}{R_d + R_m} V_d = \left(\frac{\rho I_s l_s l_d}{4\pi y^3}\right)^2 \frac{R_d}{\left(R_d + \dfrac{\rho}{2\pi r_d}\right)^2}.$$

(5) Optimum resistance and maximum power When

$$\frac{R_d}{\left(R_d + \dfrac{\rho}{2\pi r_d}\right)^2} = \frac{R_d}{(R_d + R_m)^2}$$

reaches its maximum, the power P_d reaches its maximum. Hence

$$\frac{\mathrm{d}}{\mathrm{d}R_d}\left[\frac{R_d}{(R_d + R_m)^2}\right] = \frac{1 \cdot (R_d + R_m)^2 - R_d \cdot 2(R_d + R_m)}{(R_d + R_m)^4} = 0,$$

so

$$(R_d + R_m) - 2R_d = 0,$$

and therefore

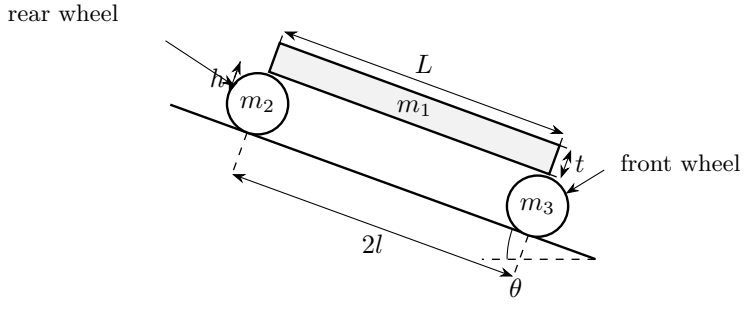
$$R_{d,\mathrm{opt}} = R_m = \frac{\rho}{2\pi r_d}.$$

The maximum power is

$$P_{d,\max} = \left(\frac{\rho I_s l_s l_d}{4\pi y^3}\right)^2 \frac{\pi r_d}{2\rho} = \frac{\rho (I_s l_s l_d)^2 r_d}{32\pi y^6}.$$

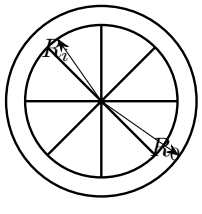
Problem 3. Motion of a heavy vehicle on an inclined road

Fig. 33.8 is a simplified model of a heavy vehicle (a road roller). The vehicle has a cylinder at the front and another at the rear as wheels, and moves on a road inclined at an angle θ. Each cylindrical wheel has total mass $m_2 = m_3 = M$ and consists of a cylindrical shell of outer radius R_0 and inner radius $R_i = 0.8R_0$ together with 8 spokes; the total mass of the 8 spokes is $0.2M$, and the mass of the lower frame supporting the vehicle body may be neglected. The model of the cylindrical wheel is shown in Fig. 33.9. The heavy vehicle now moves along the road under the action of gravity and friction.



Simplified model of the heavy vehicle moving on an inclined road.

Fig. 33.8



Simplified model of the cylindrical wheel.

Fig. 33.9

The static and kinetic friction coefficients between the cylindrical wheels and the ground are μ_s and μ_k respectively; the mass of the vehicle body is $m_1 = 5M$, its length is L, its thickness is t, and the distance between the front and rear wheels is $2l$. The distance from the centre of a cylindrical wheel to the bottom of the vehicle body is h. The friction between the cylindrical wheels and their axles may be neglected.

Problems (1) Calculate the moment of inertia of each cylindrical wheel. (1.5)

(2) Draw the forces acting on the vehicle body, the front wheel and the rear wheel. Write down the equations of motion for each part. (2.5 points)

(3) Suppose that the vehicle starts from rest and moves freely under the action of gravity and friction. Describe all possible forms of motion of the system, and derive the corresponding acceleration of the vehicle in terms of the physical quantities given in the problem. (4.0 points)

(4) Suppose that the heavy vehicle starts from rest and, after travelling a distance d metres in pure rolling, enters another part of the road where all the friction coefficients drop to smaller constants μ_s' and μ_k', so that the two cylindrical wheels begin to slip. Calculate the linear velocity and the angular velocity of each cylindrical wheel after the vehicle has travelled a total distance s. Assume that d and s are both much larger than the linear dimensions of the vehicle. (2.0 points)

Solution 3

To simplify the model, use the simplified drawing of Fig. 33.10 and set $h_1 = h + 0.5t$, $R_0 = R$.

(1) Moment of inertia of a cylindrical wheel It is known that

$$R_i = 0.8R_0,$$

the mass of the cylindrical part is

$$m_{\text{cyl}} = 0.8M,$$

and the mass of each spoke is

$$m_{\text{spoke}} = 0.025M.$$

The moment of inertia is

$$I = \int r^2 \,\mathrm{d}m = \int_{\text{cyl}} r^2 \,\mathrm{d}m + \int_{\text{spoke 1}} r^2 \,\mathrm{d}m + \cdots + \int_{\text{spoke 8}} r^2 \,\mathrm{d}m.$$

For the cylindrical part, with an areal density σ,

$$\begin{aligned}\int_{\text{cyl}} r^2 \,\mathrm{d}m &= 2\pi\sigma \int_{R_i}^{R_0} r^3 \,\mathrm{d}r = 0.5\pi\sigma \left(R_0^4 - R_i^4\right) \\ &= 0.5 m_{\text{cyl}} \left(R_0^2 + R_i^2\right) = 0.656MR^2.\end{aligned}$$

For each spoke, with a linear density λ,

$$\begin{aligned}\int_{\text{spoke}} r^2 \,\mathrm{d}m &= \lambda \int_0^{R_i} r^2 \,\mathrm{d}r = \frac{1}{3}\lambda R_i^3 = \frac{1}{3} m_{\text{spoke}} R_i^2 \\ &= \frac{1}{3}(0.025M)(0.64R^2) = 0.00533MR^2.\end{aligned}$$

Hence the moment of inertia of each cylindrical wheel is

$$I = 0.656MR^2 + 8 \times 0.00533MR^2 = 0.7MR^2.$$

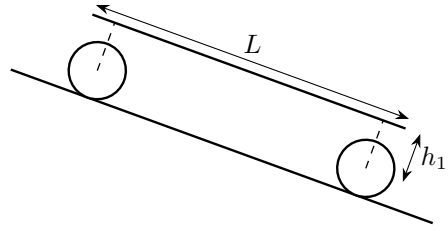


Fig. 33.10

(2) Force diagrams and equations of motion For simplicity of analysis the system is divided into three parts: the vehicle body (part 1), which may be regarded as a uniform thin plate; the rear cylindrical wheels (the two rear cylindrical wheels taken together as part 2); and the front cylindrical wheels (the two front cylindrical wheels taken together as part 3).

Part 1: vehicle body.

As shown in Fig. 33.11, the equations of motion for this part are as follows. Along the incline,

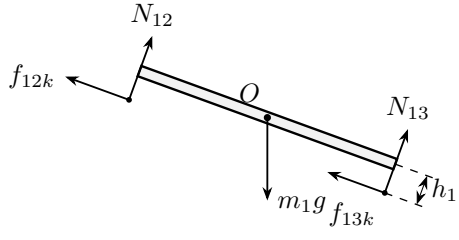
$$m_1 g \sin\theta - f_{12k} - f_{13k} = m_1 a. \tag{1}$$

Perpendicular to the incline,

$$m_1 g \cos\theta = N_{12} + N_{13}. \tag{2}$$

The total moment about O is zero:

$$N_{12} l - N_{13} l + f_{12k} h_1 + f_{13k} h_1 = 0. \tag{3}$$



Forces acting on the vehicle body.

Fig. 33.11

Part 2: rear cylindrical wheels.

As shown in Fig. 33.12, the equations of motion of the rear wheel are

$$f_{21h} - f_2 + Mg \sin\theta = Ma, \tag{4}$$

$$N_2 - N_{21} - Mg \cos\theta = 0. \tag{5}$$

For pure rolling,

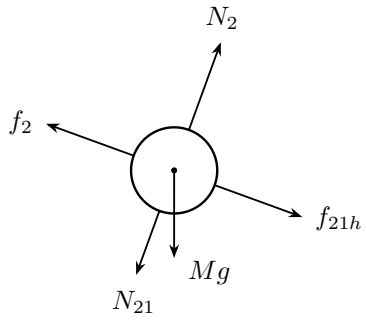
$$f_2 R = I\alpha_2 = I\frac{a_2}{R},$$

or

$$f_2 = \frac{I}{R^2} a. \tag{6}$$

For rolling with slipping,

$$f_2 = \mu_k N_2. \tag{7}$$



Forces acting on the rear cylindrical wheel.

Fig. 33.12

Part 3: front cylindrical wheels.
As shown in Fig. 33.13, the equations of motion of the front wheel are

$$f_{31h} - f_3 + Mg\sin\theta = Ma, \tag{8}$$

$$N_3 - N_{31} - Mg\cos\theta = 0. \tag{9}$$

For pure rolling,

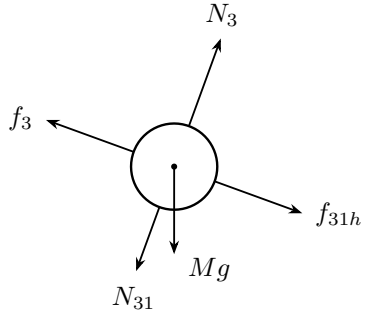
$$f_3 R = I\alpha_3 = I\frac{a_3}{R},$$

or

$$f_3 = \frac{I}{R^2}a. \tag{10}$$

For rolling with slipping,

$$f_3 = \mu_k N_3. \tag{11}$$



Forces acting on the front cylindrical wheel.

Fig. 33.13

(3) Possible forms of motion and the corresponding accelerations From equations (2), (5) and (9),

$$\begin{aligned} m_1 g\cos\theta &= N_2 - m_2 g\cos\theta + N_3 - m_3 g\cos\theta, \\ N_2 + N_3 &= (m_1 + m_2 + m_3)g\cos\theta = 7Mg\cos\theta. \end{aligned} \tag{12}$$

From equations (3), (5) and (8),

$$(N_3 - Mg\cos\theta)l - (N_2 - Mg\cos\theta)l = h_1(f_2 + Ma - Mg\sin\theta + f_3 + Ma - Mg\sin\theta),$$
$$N_3 - N_2 = \frac{h_1}{l}(f_2 + 2Ma - 2Mg\sin\theta + f_3). \tag{13}$$

Both front and rear wheels in pure rolling.
From equations (4) and (6),

$$f_{21h} = \left(\frac{I}{R^2}\right)a + Ma - Mg\sin\theta. \tag{14}$$

From equations (8) and (10),

$$f_{31h} = \left(\frac{I}{R^2}\right)a + Ma - Mg\sin\theta. \tag{15}$$

From equations (1), (14) and (15),

$$5Mg\sin\theta - \left[\left(\frac{I}{R^2}\right)a + Ma - Mg\sin\theta\right] - \left[\left(\frac{I}{R^2}\right)a + Ma - Mg\sin\theta\right] = m_1 a,$$
$$7Mg\sin\theta = \left(\frac{2I}{R^2} + 7M\right)a.$$

Hence

$$a = \frac{7Mg\sin\theta}{7M + 2\dfrac{I}{R^2}} = \frac{7Mg\sin\theta}{7M + 2\cdot\dfrac{0.7MR^2}{R^2}} = 0.833g\sin\theta. \tag{16}$$

The normal forces are

$$\begin{aligned} N_3 &= \frac{7M}{2}g\cos\theta + \frac{h_1}{l}\left[\left(M + \frac{I}{R^2}\right)\times 0.833g\sin\theta - Mg\sin\theta\right] \\ &= 3.5Mg\cos\theta + \frac{h_1}{l}\left[(M + 0.7M)\times 0.833g\sin\theta - Mg\sin\theta\right] \\ &= 3.5Mg\cos\theta + 0.41\frac{h_1}{l}Mg\sin\theta, \\ N_2 &= \frac{7M}{2}g\cos\theta - \frac{h_1}{l}\left[\left(\frac{I}{R^2} + M\right)\times 0.833g\sin\theta - Mg\sin\theta\right] \\ &= 3.5Mg\cos\theta - \frac{h_1}{l}\left[(0.7M + M)\frac{7Mg\sin\theta}{2\times 0.7M + 7M} - Mg\sin\theta\right] \\ &= 3.5Mg\cos\theta - 0.41\frac{h_1}{l}Mg\sin\theta. \end{aligned}$$

The condition for pure rolling is

$$f_2 \le \mu_s N_2 \quad \text{and} \quad f_3 \le \mu_s N_3, \qquad \frac{I}{R^2}a \le \mu_s N_2 \quad \text{and} \quad \frac{I}{R^2}a \le \mu_s N_3. \tag{17}$$

From the left-hand inequality of (17),

$$0.7M \times 0.833g\sin\theta \le \mu_s\left(3.5Mg\cos\theta - 0.41\frac{h_1}{l}Mg\sin\theta\right),$$

so

$$\tan\theta \le \frac{3.5\mu_s}{0.5831 + 0.41\mu_s \dfrac{h_1}{l}}.$$

From the right-hand inequality of (17),

$$0.7M \times 0.833g\sin\theta \le \mu_s \left(3.5Mg\cos\theta + 0.41\frac{h_1}{l}Mg\sin\theta\right),$$

so

$$\tan\theta \le \frac{3.5\mu_s}{0.5831 - 0.41\mu_s \dfrac{h_1}{l}}.$$

The former inequality should be taken.

Both front and rear wheels rolling with slipping.

From equation (4),

$$f_{21h} = Ma + \mu_k N_2 - Mg\sin\theta. \tag{18}$$

From equation (8),

$$f_{31h} = Ma + \mu_k N_3 - Mg\sin\theta. \tag{19}$$

Substituting (18) and (19) into (1),

$$5Mg\sin\theta - (Ma + \mu_k N_2 - Mg\sin\theta) - (Ma + \mu_k N_3 - Mg\sin\theta) = m_1 a,$$

$$a = \frac{7Mg\sin\theta - \mu_k N_2 - \mu_k N_3}{7M} = g\sin\theta - \frac{\mu_k(N_2 + N_3)}{7M}. \tag{20}$$

Since

$$N_3 + N_2 = 7Mg\cos\theta,$$

the above two equations give

$$a = g\sin\theta - \mu_k g\cos\theta.$$

The condition for both front and rear wheels to slip simultaneously is the opposite of the pure-rolling condition, namely

$$f_2 > \mu_s N_2' \quad \text{and} \quad f_3 > \mu_s N_3', \qquad \frac{I}{R^2}a > \mu_s N_2' \quad \text{and} \quad \frac{I}{R^2}a > \mu_s N_3', \tag{21}$$

where N_2' and N_3' are the values calculated for pure rolling of both wheels.

Finally,

$$\tan\theta > \frac{3.5\mu_s}{0.5831 + 0.41\mu_s \dfrac{h_1}{l}},$$

and

$$\tan\theta > \frac{3.5\mu_s}{0.5831 - 0.41\mu_s \dfrac{h_1}{l}}.$$

The latter inequality should be taken.

One wheel in pure rolling and the other rolling with slipping.

From (17) it is seen that it must be the front wheel that rolls purely and the rear wheel that slips.

From equation (4),

$$f_{21h} = m_2 a + \mu_k N_2 - m_2 g \sin\theta. \tag{22}$$

From equation (10),

$$f_{31h} = m_3 a + \left(\frac{I}{R^2}\right) a - m_3 g \sin\theta. \tag{23}$$

From equations (1), (22) and (23),

$$\begin{aligned} m_1 g \sin\theta - (m_2 a + \mu_k N_2 - m_2 g \sin\theta) \\ - \left[m_3 a + \left(\frac{I}{R^2}\right) a - m_3 g \sin\theta\right] = m_1 a, \\ m_1 g \sin\theta + m_2 g \sin\theta + m_3 g \sin\theta - \mu_k N_2 = \left[\left(\frac{I}{R^2}\right) + m_3\right] a + m_2 a + m_1 a, \\ 5Mg \sin\theta + Mg \sin\theta + Mg \sin\theta - \mu_k N_2 = (0.7M + M)a + Ma + 5Ma. \end{aligned}$$

Hence

$$a = \frac{7Mg \sin\theta - \mu_k N_2}{7.7M} = 0.9091 g \sin\theta - \frac{\mu_k N_2}{7.7M}. \tag{24}$$

Also,

$$\begin{aligned} N_3 - N_2 &= \frac{h_1}{l}\left(\mu_k N_2 + \frac{I}{R^2} a + 2Ma - 2Mg \sin\theta\right), \\ N_3 - N_2 &= \frac{h_1}{l}\left(\mu_k N_2 + 2.7 \times 0.9091 Mg \sin\theta - \frac{2.7\mu_k N_2}{7.7} - 2Mg \sin\theta\right), \\ N_3 - N_2 &\left(1 + 0.65\mu_k \frac{h_1}{l}\right) = 0.4546 \frac{h_1}{l} Mg \sin\theta. \end{aligned}$$

Together with

$$N_3 + N_2 = 7Mg \cos\theta,$$

this gives

$$\begin{aligned} N_2 &= \frac{7Mg \cos\theta - 0.4546 \dfrac{h_1}{l} Mg \sin\theta}{2 + 0.65\mu_k \dfrac{h_1}{l}}, \\ N_3 &= 7Mg \cos\theta - \frac{7Mg \cos\theta - 0.4546 \dfrac{h_1}{l} Mg \sin\theta}{2 + 0.65\mu_k \dfrac{h_1}{l}}. \end{aligned} \tag{25}$$

Substituting the above results into (24) gives

$$\begin{aligned} a &= 0.9091 g \sin\theta - \frac{\mu_k N_2}{7.7M} \\ &= 0.9091 g \sin\theta - \frac{\mu_k}{7.7} \cdot \frac{7g \cos\theta - 0.4546 \dfrac{h_1}{l} g \sin\theta}{2 + 0.65\mu_k \dfrac{h_1}{l}}. \end{aligned} \tag{26}$$

The condition for this rolling-and-slipping case is

$$f_2 > \mu_s N_2' \quad \text{and} \quad f_3 \le \mu_s N_3', \qquad \frac{I}{R^2}a > \mu_s N_2' \quad \text{and} \quad \frac{I}{R^2}a \le \mu_s N_3', \tag{27}$$

where N_2' and N_3' are the normal forces in the pure-rolling case.

(4) Motion after the friction coefficients decrease Suppose that, after rolling through a distance of d metres, both the front and rear wheels begin to slip, and that the total distance travelled is s metres. Let t_1 be the time taken to cover the distance d. Then

$$v_1 = v_0 + a_1 t_1 = 0 + a_1 t_1 = a_1 t_1,$$
$$d = v_0 t_1 + \frac{1}{2}a_1 t_1^2 = \frac{1}{2}a_1 t_1^2.$$

$$\begin{aligned} t_1 &= \sqrt{\frac{2d}{a_1}}, \\ v_{t_1} &= a_1\sqrt{\frac{2d}{a_1}} = \sqrt{2da_1} \\ &= \sqrt{2d \cdot 0.833g\sin\theta} = \sqrt{1.666dg\sin\theta}. \end{aligned} \tag{28}$$

After the car has rolled through a distance d, the front and rear wheels have the same angular velocity, namely

$$\omega_{t_1} = \frac{v_{t_1}}{R} = \frac{1}{R}\sqrt{1.666dg\sin\theta}. \tag{29}$$

The car then slides. Let t_2 be the time the car takes to slide from the distance d to the distance s; then

$$\begin{aligned} v_{t_2} &= v_{t_1} + a_2 t_2 = \sqrt{1.666dg\sin\theta} + a_2 t_2, \\ s - d &= v_{t_1} t_2 + \frac{1}{2}a_2 t_2^2, \\ t_2 &= \frac{-v_{t_1} + \sqrt{v_{t_1}^2 + 2a_2(s-d)}}{a_2}, \\ v_{t_2} &= \sqrt{1.666dg\sin\theta} - v_{t_1} + \sqrt{v_{t_1}^2 + 2a_2(s-d)}. \end{aligned} \tag{30}$$

Substituting the results obtained above for v_{t_1} and a_2 into this expression gives the final result.

As for the angular velocity: while the wheel slides, the torque acting on it is

$$\begin{aligned} \tau &= \mu_k N R, \\ \alpha &= \frac{\tau}{I} = \frac{\mu_k N R}{I}. \end{aligned} \tag{31}$$

$$\begin{aligned} \omega_{t_2} &= \omega_{t_1} + \alpha t_2 \\ &= \frac{1}{R}\sqrt{1.666dg\sin\theta} + \frac{\mu_k N R}{I} \cdot \frac{-v_{t_1} + \sqrt{v_{t_1}^2 + 2a_2(s-d)}}{a_2}. \end{aligned}$$

Substituting the normal force N of each cylindrical wheel separately into the above expression, one obtains the angular velocity of that wheel.

Experimental Problems and Solutions

Problem 4. Determination of e/k_B by electrolysis

Background theory The electrolysis of water is described by the following reactions:

$$\mathrm{H_2O} \longrightarrow 2\mathrm{H^+} + \mathrm{O^{2-}},$$
$$2\mathrm{H^+} + 2\mathrm{e^-} \longrightarrow \mathrm{H_2}; \qquad \mathrm{O^{2-}} \longrightarrow \tfrac{1}{2}\mathrm{O_2} + 2\mathrm{e^-}.$$

When a current passes between a pair of electrodes immersed in water, the above reactions take place. Assume that both gases produced in the reaction are ideal gases.

One of the gases produced by the reaction is collected in a test tube bearing an arbitrary scale. If the total charge transferred and the volume of the gas in the test tube are known, e/k_B can be determined, where e is the charge of the electron and k_B is the Boltzmann constant.

For this purpose the experiment is divided into two parts.

Part A: a dynamical method is used to calibrate the arbitrary scale on the test tube. The result is then used in Part B.

Part B: the physical quantity e/k_B is determined by the electrolysis of water.

The two parts, A and B, may be performed in either order.

The following physical quantities are assumed known:

acceleration due to gravity $g = (9.78 \pm 0.01)\,\mathrm{m/s^2}$; the ratio of the inner to the outer radius of the test tube is $\alpha = 0.82 \pm 0.01$.

The temperature T and the pressure p of the laboratory are supplied by the examination site.

Apparatus

(1) A test tube with an arbitrary scale.

(2) Three insulated copper wires of different diameters: a thick brown wire, a thin brown wire and a blue wire.

(3) An adjustable power supply (0–60 V, maximum output current 1 A).

(4) One plastic container and one bottle of water.

(5) A copper block with a plastic screw clamp, used to hold the electrode in position so that the insulation of the wire is not damaged.

(6) A digital stopwatch.

(7) A multimeter (observe the correct procedure for its use).

(8) A wooden test-tube clamp, to keep the test tube vertical.

(9) A multipurpose dropper.

(10) A vertical support.

(11) One bottle of white correction fluid for making marks.

(12) A knife.

(13) A pair of scissors.

(14) A roll of transparent adhesive tape.

(15) One steel ball.

(16) One piece of stainless steel sheet, to serve as an electrode.

(17) Graph paper.

Note All the scales on the graph paper and on the instruments (for example the test tube) are marked in the same unit, but the unit is not the millimetre.

Experimental tasks Part A. Calibration of the arbitrary scale on the test tube

(1) Determine a dynamical method by which an arbitrary length scale can be converted into a conventional scale.

(2) Write down the relation between your measured quantities and the readings of the arbitrary scale of the test tube, and draw a sketch of the experimental arrangement.

(3) Collect and analyse the experimental data, calculate and calibrate the scale of the unknown length.

Part B. Determination of the physical quantity e/k_B

(1) Set up an electrolysis experiment with the test tube suitably mounted, so as to collect one of the gases produced during the reaction.

(2) Derive the relation between the time t to be measured in the experiment, the current I, and the height difference Δh of the water level in the test tube.

(3) Collect and analyse the experimental data. As a simplification, you may assume that the pressure of the gas inside the test tube remains constant throughout the experiment.

(4) Determine the value of e/k_B.

Questions Part A. Calibration of the arbitrary scale on the test tube

(1) Explain the method you have chosen, and draw a sketch of the experimental arrangement for this method. (1.0 point)

(2) Write down the relation between the quantities measured in your chosen method, and state all the approximations used in obtaining this expression. (1.0 point)

(3) Collect and tabulate the data in the following order: physical quantity, value, unit. (1.0 point)

(4) Use a graph of the relation between two independently measured quantities to demonstrate the quality of the calibration, and state the range over which this calibration is usable. (0.5 point)

(5) Determine, in millimetres, the smallest division of the arbitrary scale, and estimate the error caused by the measurement. (1.5 point)

Part B. Determination of the physical quantity e/k_B

(1) Draw a sketch of the experimental arrangement. (1.0 point)

(2) Derive the following expression. (1.5 point)

$$I\Delta t = \frac{e}{k_B}\,\frac{2p(\pi r^2)}{T}\,\Delta h.$$

(3) Collect and tabulate the data in the following format: (1.0 point)
physical quantity (value, unit).

(4) Determine the value of e/k_B and estimate its error. (1.5 point)

Solution to Problem 4

Part A. Calibration of the arbitrary scale on the test tube

(1) In the experiment a simple pendulum system is chosen to calibrate the arbitrary scale; the sketch is shown in Fig. 33.1. (The graduations of the test tube used in the experiment do not have a division value of 1 mm; such a scale is defined as an arbitrary scale.)

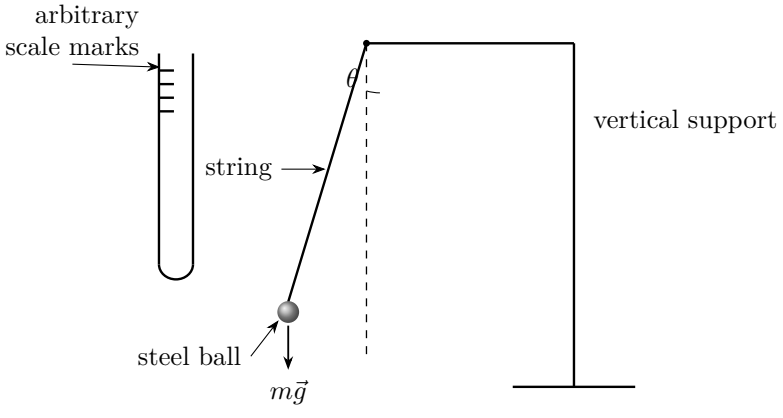


Fig. 33.1

(2) From Newton's second law,

$$F = m\,\frac{\mathrm{d}^2(l\theta)}{\mathrm{d}t^2} = ml\,\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2},$$

where F is the force, m the mass of the ball, θ the angle of the pendulum from the vertical direction, l the length of the pendulum, and t the time.

$$F = -mg\sin\theta,$$

$$m\,\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{mg\sin\theta}{l} = 0,$$

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g\sin\theta}{l} = 0.$$

Making the approximation $\sin\theta \approx \theta$,

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{l}\theta = 0.$$

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \omega^2\theta = 0 \qquad (\omega^2 = g/l). \tag{1}$$

Equation (1) is the equation of simple harmonic motion; its angular frequency ω is

$$\omega = \frac{2\pi}{T_{\mathrm{osc}}} = \sqrt{\frac{g}{l}}, \qquad T^2 = 4\pi^2\frac{l}{g}.$$

Expression for the measured quantity: period $T = 2\pi\sqrt{\dfrac{l}{g}}$.

Approximations: an ideal simple pendulum satisfies the conditions that the mass of the string is much smaller than the mass of the steel ball, the radius of the steel ball is much smaller than the length of the string, and the stretching of the string and the air resistance are neglected.

(3) A sample of the data obtained from the simple-pendulum experiment. The number of periods is ≥ 20, the difference between successive periods is ≥ 0.01 s, and the number of data sets is ≥ 4.

Table 33.1

No.	Time for 50 periods (s)	Period T (s)	Scale marked on the string (arbitrary scale)
1	91.47	1.829	200
2	89.09	1.782	150
3	86.45	1.729	100
4	83.80	1.676	50

(4)

Table 33.2

No.	Period T (s)	Scale marked on the string (arbitrary scale)	Square of period T^2 (s^2)
1	1.829	200	3.347
2	1.782	150	3.176
3	1.729	100	2.989
4	1.676	50	2.809

Graph of T^2 against the arbitrary scale on the string. (Omitted.)

(5) Determination of the smallest division of the arbitrary scale (unit: mm).

$$T_{\mathrm{osc}_1}^2 = \frac{4\pi^2}{g} l_1, \qquad T_{\mathrm{osc}_2}^2 = \frac{4\pi^2}{g} l_2,$$

$$T_{\mathrm{osc}_1}^2 - T_{\mathrm{osc}_2}^2 = \frac{4\pi^2}{g}(l_1 - l_2) = \frac{4\pi^2}{g}\Delta l,$$

$$\Delta l = \frac{g}{4\pi^2}\left(T_{\mathrm{osc}_1}^2 - T_{\mathrm{osc}_2}^2\right),$$

or other equivalent expressions, graphical methods, and so on.

Table 33.3

No.	Difference of the squares of the periods (s^2)	Calculated Δl (m)	Δl (arbitrary scale)	Smallest unit of the arbitrary scale (mm)
1	$T_1^2 - T_2^2 = 0.171$	0.0426	50	0.85
2	$T_1^2 - T_3^2 = 0.357$	0.0886	100	0.89
3	$T_1^2 - T_4^2 = 0.537$	0.1330	150	0.89
4	$T_2^2 - T_3^2 = 0.185$	0.0460	50	0.92
5	$T_2^2 - T_4^2 = 0.365$	0.0907	100	0.91
6	$T_3^2 - T_4^2 = 0.180$	0.0448	50	0.90

Mean value of the smallest unit of the arbitrary scale: $\bar{l} = 0.89\,\mathrm{mm}$.

Estimation of the error caused by the measurement.

Table 33.4

No.	Smallest unit of the arbitrary scale (mm)	$(l - \bar{l})$ (mm)	$(l - \bar{l})^2$ (mm^2)
1	0.85	−0.04	0.0016
2	0.89	0	0
3	0.89	0	0
4	0.92	0.03	0.0009
5	0.91	0.02	0.0004
6	0.90	0.01	0.0001

Hence the standard deviation is

$$\Delta l = \sqrt{\frac{\sum_{i=1}^{N}(l_i - \bar{l})^2}{N-1}} = \sqrt{\frac{0.003}{5}}\,(\mathrm{mm}) = 0.02\,(\mathrm{mm}).$$

Other reasonable methods may also be used.

Part B. Determination of the physical quantity e/k_B by electrolysis of water

(1) Experimental arrangement

The experimental arrangement is shown in Fig. 33.2.

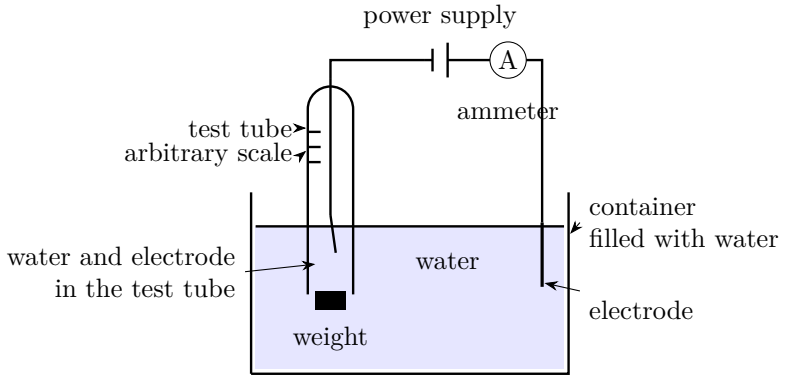


Fig. 33.2

(2) Derivation of the relation between the time t, the current I and the water-level difference Δh.

$$I = \frac{\Delta Q}{\Delta t}.$$

In the reaction $2H^+ + 2e^- \longrightarrow H_2$, the charge transferred in producing a definite number of molecules (ΔN) is

$$\begin{aligned}
\Delta Q &= 2e\Delta N, \\
I &= \frac{\Delta N \cdot 2e}{\Delta t}, \\
p\Delta V = \Delta N k_B T &= \frac{I\Delta t}{2e} k_B T, \\
p\,\Delta h(\pi r^2) &= \frac{I\Delta t}{2}\,\frac{k_B T}{e}, \\
I\Delta t &= \frac{e}{k_B}\,\frac{2p(\pi r^2)}{T}\,\Delta h.
\end{aligned}$$

(3) Experimental data

Table 33.5

No.	Δh (arbitrary scale)	I (mA)	Δt (s)
1	12	4.00	1 560
2	16	4.00	2 281
3	20	4.00	2 940
4	24	4.00	3 600

- The circumference of the test tube is 46 arbitrary scale divisions.

- In view of the error range permitted by the uncertainty in reading the water surface, choose $\Delta h \geq 4$ scale divisions; in view of the disturbance of the current, choose $I \leq 4\,\mathrm{mA}$.
- Number of data sets ≥ 4.

External conditions under which the experimental data were obtained:

$$T = 300\,\mathrm{K},$$
$$p = 1.00 \times 10^5\,\mathrm{Pa}.$$

(4) Measurement of the value of e/k_B

Table 33.6

No.	Δh (arbitrary scale)	Δh (mm)	I (mA)	Δt (s)	$I\Delta t$ (C)
1	12	10.68	4.00	1 560	6.24
2	16	14.24	4.00	2 281	9.12
3	20	17.80	4.00	2 940	11.8
4	24	21.36	4.00	3 600	14.4

The graph of the data $I\Delta t$ against Δh listed above is omitted.

The slope obtained from the graph is $764\,\mathrm{C}\cdot\mathrm{m}^{-1}$.

$$\frac{e}{k_B} = \frac{764 \times 300 \times \pi}{2 \times 10^5 \times (23 \times 0.89 \times 10^{-3} \times 0.82)^2}\ (\mathrm{C}\cdot\mathrm{K/J})$$
$$= 1.28 \times 10^4\ (\mathrm{C}\cdot\mathrm{K/J}) \qquad (1.0\text{ point})$$

Alternative solution (equal marks)

Table 33.7

No.	Δh (mm)	$I\Delta t/10^3$ C	Slope	$(e/k_B)/10^4$ $(\mathrm{C{\cdot}K{\cdot}J^{-1}})$
1	10.68	6.24	584	0.977
2	14.24	9.12	640	1.07
3	17.80	11.8	663	1.11
4	21.36	14.4	674	1.13

The mean value of e/k_B is $1.07 \times 10^4\,\mathrm{C}\cdot\mathrm{K/J}$.

The standard deviation obtained by calculation is about $0.03 \times 10^4\,\mathrm{C}\cdot\mathrm{K/J}$.

Problem 5. Optical black box

Description In this problem the student must determine the unknown optical elements inside a box. The box is sealed and has only two narrow slits, which are covered with red plastic sheet. The optical elements in the box are to be identified from the optical phenomena observed in the experiment; the small thickness effect of the plastic covering may be neglected.

The line through the centres of the two slits defines the axis of the box. Apart from the red plastic covering, the box contains 3 of the following optical elements (they may be of the same kind or of different kinds):

(1) mirrors, which may be plane or spherical;

(2) lenses, which may be convex or concave;

(3) transparent plates with two parallel surfaces (so-called plane-parallel plates);

(4) prisms;

(5) diffraction gratings.

The material of all the transparent elements has a refractive index $n = 1.47$ at the wavelength used.

Apparatus

(1) A laser pointer of wavelength 670 nm. Take care not to look directly into the laser beam.

(2) An optical bench.

(3) A platform, movable along the optical bench, on which the black box is mounted.

(4) A screen, which can be mounted at one end of the optical bench; it can be removed when other measurements are made.

(5) Graph paper, which can be stuck to the screen with adhesive tape.

(6) A vertical support with front and back clamps.

Note All the scales marked on the graph paper and on the experimental instruments are graduated in the same unit, but the unit is not the millimetre.

Questions

(1) Determine what each of the 3 elements is, and give the characteristics of each of them.

Table 33.8

Element type	Element characteristics
Reflecting mirror	radius of curvature; angle between the mirror axis and the axis of the box
Lens*	focal length; convex or concave; and their positions in the box
Plane-parallel plate	thickness; angle between the plate and the axis of the box
Prism	apex angle; angle between one refracting face and the axis of the box
Diffraction grating*	spacing and direction of the ruled lines, and its position in the box

* Assume that the principal plane of the lens and the plane of the grating are perpendicular to the axis of the box.

The final answers for the characteristic parameters of each element (such as the focal length and the radius of curvature) should be expressed in millimetres, micrometres, or divisions of the graph paper. You need not determine the accuracy of the results.

(2) Write down the names of the optical elements in the box.

No. 1 ..... (0.5 point)

No. 2 ..... (0.5 point)

No. 3 ..... (0.5 point)

(3) Below is a cross-sectional diagram of the box. On the diagram, sketch the positions of the 3 elements inside the box, and label each element with the number used in answer 1. (0.5 point for each correct position)

(4) Redraw the diagram of answer 2 and mark on it detailed information, such as the angles of the elements, the distances of the elements from the slits, and the orientation or direction of the elements. (1.0 point)

(5) Synthesise the observed data (0.5 point); with the help of the diagram, derive an appropriate formula and determine the characteristics of optical element No. 1 (1.0 point); calculate the required characteristics and enter the answers in the table below (0.5 point).

Table 33.9

Name of element No. 1	Characteristics

(6) Synthesise the observed data (0.5 point); with the help of the diagram, derive an appropriate formula and determine the characteristics of optical element No. 2 (1.0 point); calculate the required characteristics and enter the answers in the table below (0.5 point).

Table 33.10

Name of element No. 2	Characteristics

(7) Synthesise the observed data (0.5 point); with the help of the diagram, derive an appropriate formula and determine the characteristics of optical element No. 3 (1.0 point); calculate the required characteristics and enter the answers in the table below (0.5 point).

Table 33.11

Name of element No. 3	Characteristics

Solution to Problem 5

(1) The optical elements are, respectively,

- No. 1: diffraction grating;
- No. 2: diffraction grating;
- No. 3: plane-parallel transparent plate.

(2) Cross-section of the box (Fig. 33.3).

The laser beam enters the box as shown in Figs. 33.4 and 33.5. The positions of the light spots on the screen are the points indicated (marked with crosses).

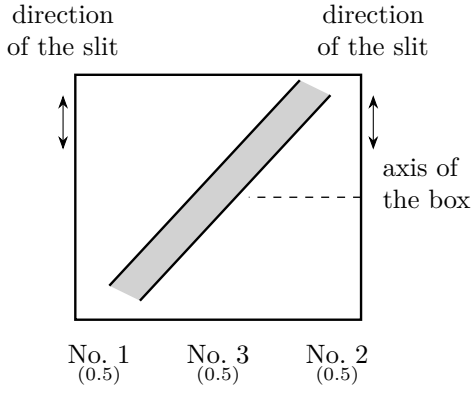


Fig. 33.3

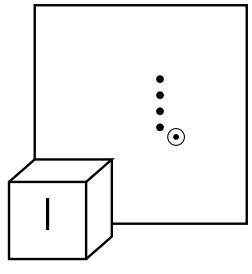


Fig. 33.4

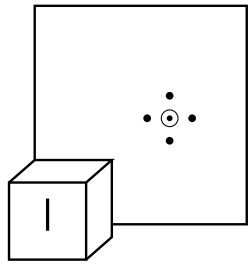


Fig. 33.5

(3) Supplementary explanations:

The distance between grating No. 1 and the left wall of the box is almost zero (Fig. 33.6).

The ruled lines of grating No. 1 make a right angle with the slit of the box.

The distance between grating No. 2 and the right wall of the box is almost zero.

The ruled lines of grating No. 2 are parallel to the slit of the box.

Schematic diagram of the box to be analysed (Fig. 33.7).

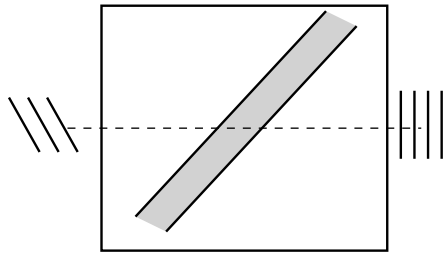


Fig. 33.6

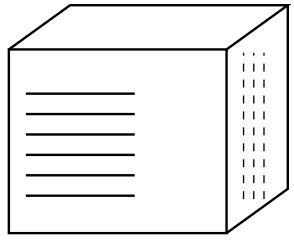


Fig. 33.7

(4) Diffraction grating (Fig. 33.8)

The optical path difference is

$$\Delta = d \sin\theta,$$

where d is the grating constant, θ the angle of diffraction and m the order of diffraction.

The condition for diffraction is

$$\Delta = m\lambda.$$

Hence, for first-order diffraction ($m = 1$),

$$\sin\theta = \lambda/d.$$

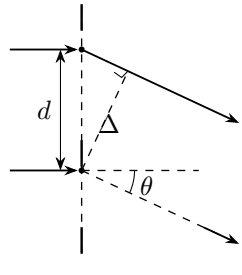


Fig. 33.8

Observed data

Table 33.12

$\tan\theta$	θ (°)	$\sin\theta$
0.34	18.8	0.322
0.32	17.7	0.305
0.32	17.7	0.305

Table 33.13

Name of element No. 1	Specific specification
Diffraction grating	grating constant = 2.16 μm grating lines at right angles to the slit

(5) Diffraction grating

Observed data

Table 33.14

$\tan\theta$	θ (°)	$\sin\theta$
1.04	46.1	0.721
0.96	43.8	0.693
1.08	47.2	0.734

Table 33.15

Name of element No. 2	Specific specification
Diffraction grating	grating constant = 0.936 μm grating lines parallel to the slit

(6) Plane-parallel smooth transparent glass plate (Fig. 33.9)

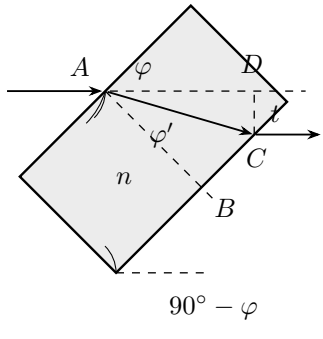


Fig. 33.9

- Snell's law: $\sin\varphi = n\sin\varphi', \quad \varphi' = \angle BAC.$
- Optical path inside the transparent glass plate: $AC = AB/\cos\varphi'.$
- $AB = h =$ thickness of the transparent glass plate.
- Displacement of the beam:

$$CD = t = AC\sin\angle CAD, \qquad \angle CAD = \varphi - \varphi'.$$

Hence

$$t = h\sin\varphi\left[1 - \frac{\cos\varphi}{(n^2 - \sin^2\varphi)^{1/2}}\right].$$

Observed data

Table 33.16

φ	t
0	0; the beam makes an angle of 49° with the central axis
49°	7.3 arbitrary scale

Table 33.17

Name of element No. 3	Specific specification
Parallel transparent glass plate	thickness = 17.9 mm angle with the central axis of the box = 49°

The 34th International Physics Olympiad

Translated and edited by Zheng Yongling and Jiang Zuimin, Fudan University

Theoretical Problems

Problem 1. *A pendulum with a falling bob.*

A rigid cylinder of radius R is mounted horizontally above the ground. A bob of mass m is suspended by a string of length L ($L > 2\pi R$) and negligible mass from the point A at the top of the cylinder, as shown in Fig. 34.1. The bob is pulled aside until it is on the same horizontal level as A, with the string taut, and is then released from rest. Any stretching of the string may be neglected. The bob may be regarded as a particle and moves only in the vertical plane perpendicular to the axis of the cylinder. For this reason the bob is sometimes also called a "particle". The acceleration due to gravity is g.

As shown in Fig. 34.1, take O as the origin of coordinates. When the particle has descended to the point P, the string is tangent to the surface of the cylinder at Q. The length of the segment QP is denoted by s. At Q the unit vector in the tangential direction is $\hat{t}$, and the unit vector in

the radial direction is $\hat{r}$. Take the vertical line along OA, pointing upwards, as the x axis; when OA rotates counterclockwise to OQ, its angular displacement θ is taken as positive.

When $\theta = 0$, the length of the segment s equals L, and the gravitational potential energy U of the particle is set equal to zero. Because of the motion of the particle, the instantaneous rates of change of θ and s with time are denoted by $\dfrac{d\theta}{dt}$ and $\dfrac{ds}{dt}$, respectively.

Unless otherwise stated, all speeds and velocities are relative to the fixed point O.

Part A

In part A only the motion of the particle while the string is taut is considered. Using the quantities given above (namely s, θ, $\dfrac{ds}{dt}$, $\dfrac{d\theta}{dt}$, R, L, g, $\hat{t}$, $\hat{r}$), find:

(a) the relation between $\dfrac{d\theta}{dt}$ and $\dfrac{ds}{dt}$;

(b) the velocity $\vec{v}_Q$ of the moving point Q relative to O;

(c) when the particle is at P, its velocity $\vec{v}'$ relative to the moving point Q;

(d) when the particle is at P, its velocity $\vec{v}$ relative to O;

(e) when the particle is at P, the component in the direction $\hat{t}$ of its acceleration relative to O;

(f) when the particle is at P, its gravitational potential energy U;

(g) the speed v_m of the particle at the lowest point of its trajectory.

Part B

In part B the ratio of L to R is

$$\frac{L}{R} = \frac{9\pi}{8} + \frac{2}{3}\cot\frac{\pi}{16} = 3.534 + 3.352 = 6.886.$$

(h) When the string from Q to P is straight and its length is shortest, what is the speed v_t of the particle? (Express it in terms of g and R.)

(i) When the particle has swung to the other side of the cylinder and reached the highest point, what is its speed v_H? (Express it in terms of g and R.)

Part C

In part C the bob of mass m is not suspended from A; instead, a string passes over the top of the cylinder and is joined on the other side to a heavier object of mass M, as shown in Fig. 34.2. This heavy object may also be regarded as a particle.

Initially the bob is pulled to the same horizontal level as A, while the heavy object at the other end hangs below the point O; the string is taut and the length of its horizontal part is L. The bob is then released from rest, and the heavy object begins to fall. Assume that the bob remains in the vertical plane and can swing past the vertically hanging heavy object and string without colliding with them.

The kinetic friction between the string and the surface of the cylinder is negligible, but the static friction is large enough that, once the speed of the falling heavy object has become zero, the object can be held at rest.

(j) Suppose that after the heavy object has fallen a distance D its speed becomes zero, and that $L - D \gg R$. If the bob can thereafter swing around the cylinder to $\theta = 2\pi$ while the two free segments of the string remain straight, then the ratio $\alpha = D/L$ must be not less than a certain value α_c. Neglecting terms of order R/L and higher, estimate the critical value α_c (express it in terms of M/m).

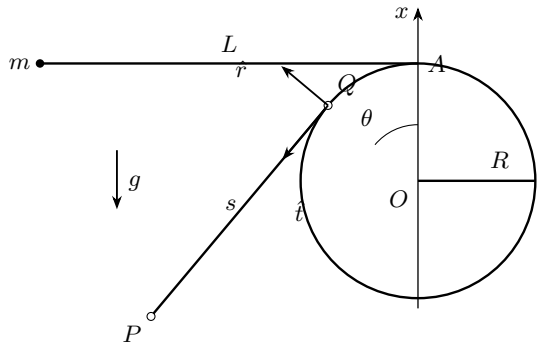


Fig. 34.1

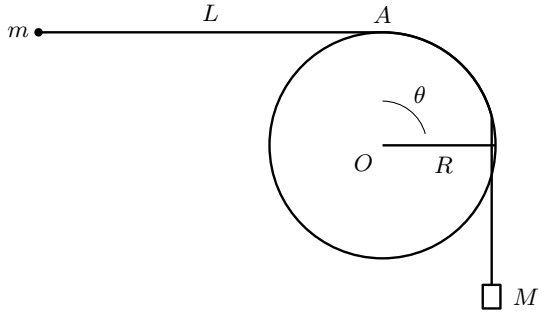


Fig. 34.2

Solutions

Solution 1.

Part A

(a) Since the length of the string, $L = s + R\theta$, is constant,[1] we have

$$\frac{ds}{dt} + R\frac{d\theta}{dt} = 0. \tag{A1*}$$

(b) Relative to O, the point Q moves on a circle of radius R with angular velocity $\dfrac{d\theta}{dt}$; hence

$$\vec{v}_Q = R\frac{d\theta}{dt}\,\hat{t} = -\frac{ds}{dt}\,\hat{t}. \tag{A2*}$$

(c) Relative to Q, the displacement of the particle during the time interval Δt is

$$\Delta\vec{r}' = (s\,\Delta\theta)(-\hat{r}) + \Delta s\,\hat{t}.$$

See Fig. 34.3. Hence

$$\vec{v}' = \frac{\Delta\vec{r}'}{\Delta t} = -s\frac{d\theta}{dt}\,\hat{r} + \frac{ds}{dt}\,\hat{t}. \tag{A3*}$$

[1] Equation numbers marked with an asterisk (*) denote answers to the problem; the same convention applies below.

(d) The velocity of the particle relative to O is the sum of the two velocities given by (A2) and (A3); therefore

$$\vec{v} = \vec{v}' + \vec{v}_Q = -s\frac{d\theta}{dt}\,\hat{r} + \frac{ds}{dt}\,\hat{t} - \frac{ds}{dt}\,\hat{t} = -s\frac{d\theta}{dt}\,\hat{r}. \qquad (A4^*)$$

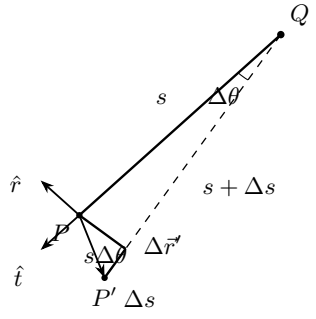


Fig. 34.3

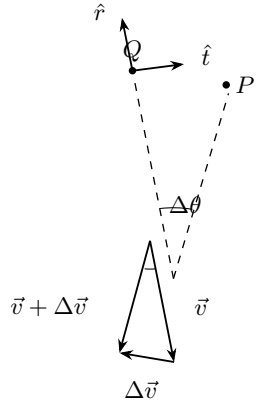


Fig. 34.4

(e) Referring to Fig. 34.4, the component of the velocity change $\Delta\vec{v}$ in the $(-\hat{t})$ direction is given by

$$(-\hat{t}) \cdot \Delta\vec{v} = v\,\Delta\theta = v\frac{d\theta}{dt}\,\Delta t.$$

Hence the $\hat{t}$ component of the acceleration $\vec{a} = \Delta\vec{v}/\Delta t$ is $\hat{t} \cdot \vec{a} = -v\dfrac{d\theta}{dt}$. From (A4) the speed of the particle is $s\dfrac{d\theta}{dt}$; therefore the $\hat{t}$ component of the acceleration of the particle at P is

$$a_t = \vec{a} \cdot \hat{t} = -v\frac{d\theta}{dt} = -\left(s\frac{d\theta}{dt}\right)\frac{d\theta}{dt} = -s\left(\frac{d\theta}{dt}\right)^2. \qquad (A5^*)$$

Note that, from Fig. 34.4, the radial component of the acceleration of the particle can also be obtained in a similar way:

$$a_r = \vec{a} \cdot \hat{r} = -\frac{dv}{dt} = -\frac{d}{dt}\left(s\frac{d\theta}{dt}\right).$$

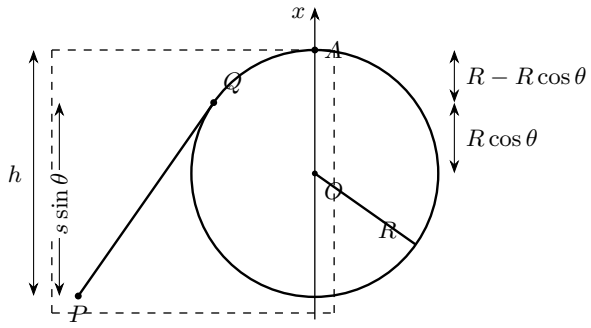


Fig. 34.5

(f) From Fig. 34.5, the gravitational potential energy of the particle can be expressed in terms of s and θ as

$$U(\theta) = -mgh = -mg\left[R(1-\cos\theta) + s\sin\theta\right]. \qquad (A6^*)$$

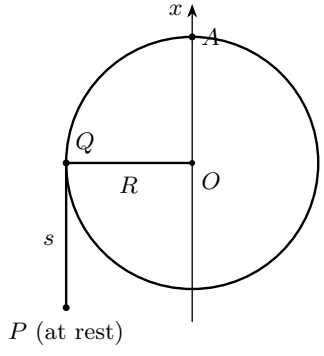


Fig. 34.6

(g) At the lowest point of its trajectory the gravitational potential energy U of the particle must take its minimum value U_m. This position of minimum potential energy can be found from the equilibrium position of the particle. It is not difficult to see that it is the position $\theta = \dfrac{\pi}{2}$, i.e. $s = L - \dfrac{\pi R}{2}$ (Fig. 34.6). Hence

$$U_m = U\left(\frac{\pi}{2}\right) = -mg\left(R + L - \frac{\pi R}{2}\right). \qquad (A7)$$

Initially the mechanical energy of the particle is zero; by conservation of mechanical energy, the speed v_m of the particle at the lowest point of its trajectory must satisfy

$$E = 0 = \frac{1}{2}mv_m^2 + U_m. \qquad (A8)$$

From (A7) and (A8) we obtain

$$v_m = \sqrt{-\frac{2U_m}{m}} = \sqrt{2g\left[R + \left(L - \frac{\pi R}{2}\right)\right]}. \qquad (A9^*)$$

Part B

(h) From (A6), the total mechanical energy of the particle can be written as

$$E = 0 = \frac{1}{2}mv^2 + U(\theta) = \frac{1}{2}mv^2 - mg\left[R(1-\cos\theta) + s\sin\theta\right]. \tag{B1}$$

From (A4) the speed of the particle is $s\dfrac{d\theta}{dt}$; hence from (B1) we obtain

$$v^2 = \left(s\frac{d\theta}{dt}\right)^2 = 2g\left[R(1-\cos\theta) + s\sin\theta\right]. \tag{B2}$$

Let T be the tension in the string. As shown in Fig. 34.7, the $\hat{t}$ component of the net force acting on the particle is $-T + mg\sin\theta$; therefore, by Newton's second law,

$$m\left[-s\left(\frac{d\theta}{dt}\right)^2\right] = -T + mg\sin\theta. \tag{B3}$$

From (B2) and (B3) the tension can be expressed as

$$\begin{aligned} T &= m\left[g\sin\theta + s\left(\frac{d\theta}{dt}\right)^2\right] = \frac{mg}{s}\left[2R(1-\cos\theta) + 3s\sin\theta\right] \\ &= \frac{2mgR}{s}\left[\tan\frac{\theta}{2} - \frac{3}{2}\left(\theta - \frac{L}{R}\right)\right]\sin\theta = \frac{2mgR}{s}(y_1 - y_2)\sin\theta \end{aligned} \tag{B4}$$

The curves of the functions $y_1 = \tan\dfrac{\theta}{2}$ and $y_2 = \dfrac{3}{2}\left(\theta - \dfrac{L}{R}\right)$ are drawn in Fig. 34.8.

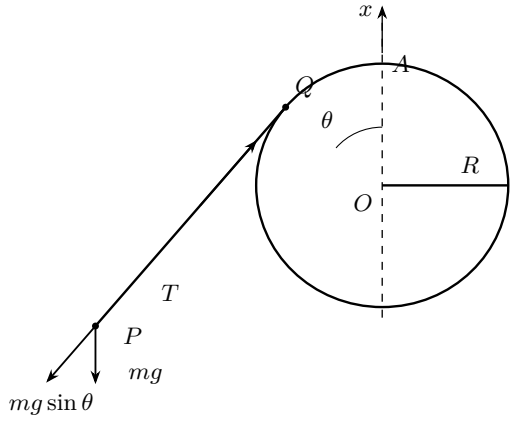


Fig. 34.7

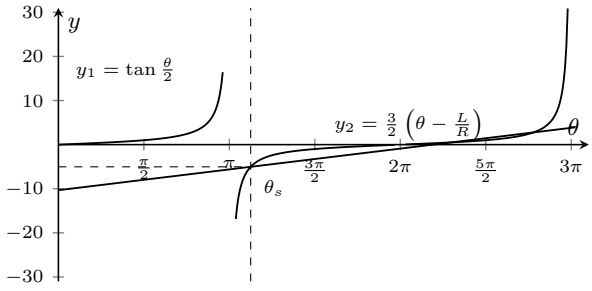


Fig. 34.8

Table 34.1

	$y_1 - y_2$	$\sin\theta$	Tension T
$0 < \theta < \pi$	positive	positive	positive
$\theta = \pi$	$+\infty$	0	positive
$\pi < \theta < \theta_s$	negative	negative	positive
$\theta = \theta_s$	zero	negative	zero
$\theta_s < \theta < 2\pi$	positive	negative	negative

From (B4) and Fig. 34.8 the results listed in Table 34.1 are obtained. Let θ_s ($\pi < \theta_s < 2\pi$) denote the angle at which $y_2 = y_1$; it is given by

$$\frac{3}{2}\left(\theta_s - \frac{L}{R}\right) = \tan\frac{\theta_s}{2}, \tag{B5}$$

or

$$\frac{L}{R} = \theta_s - \frac{2}{3}\tan\frac{\theta_s}{2}. \tag{B6}$$

Since the ratio L/R is given by

$$\frac{L}{R} = \frac{9\pi}{8} + \frac{2}{3}\cot\frac{\pi}{16} = \left(\pi + \frac{\pi}{8}\right) - \frac{2}{3}\tan\frac{1}{2}\left(\pi + \frac{\pi}{8}\right), \tag{B7}$$

equations (B6) and (B7) give

$$\theta_s = \frac{9\pi}{8}.$$

From Table 34.1 it is seen that the tension T is always positive in the range $0 < \theta < \theta_s$ (that is, the string must remain taut). As soon as θ reaches θ_s the tension becomes zero, and thereafter the free part of the string will no longer be taut. The smallest possible value $s_{\min}$ of s occurs at $\theta = \theta_s$ and is given by

$$s_{\min} = L - R\theta_s = R\left(\frac{9\pi}{8} + \frac{2}{3}\cot\frac{\pi}{16} - \frac{9\pi}{8}\right) = \frac{2R}{3}\cot\frac{\pi}{16} = 3.352R. \tag{B8}$$

When $\theta = \theta_s$, $T = 0$; from (B2) and (B3) we obtain $v^2 = -gs\sin\theta$, so the speed v_t is

$$v_t = \sqrt{-gs_{\min}\sin\theta_s} = \sqrt{\frac{2gR}{3}\cot\frac{\pi}{16}\sin\frac{\pi}{8}} = \sqrt{\frac{4gR}{3}}\cos\frac{\pi}{16} = 1.133\sqrt{gR}. \tag{B9*}$$

(i) When $\theta \geq \theta_s$ the particle moves like a projectile in the gravitational field, as shown in Fig. 34.9. The particle is projected from the position $P(x_s, y_s)$ with initial speed v_t at an elevation angle $\varphi = \left(\dfrac{3\pi}{2} - \theta_s\right)$ to the y axis.

The speed v_H of the particle at the highest point of its parabolic trajectory is equal to the y component of its initial velocity at the moment of projection. Hence

$$v_H = v_t \sin(\theta_s - \pi) = \sqrt{\frac{4gR}{3}} \cos\frac{\pi}{16} \sin\frac{\pi}{8} = 0.4334\sqrt{gR}. \tag{B10*}$$

The horizontal distance H travelled by the particle from the point of projection P to the highest point is

$$H = \frac{v_t^2 \sin 2(\theta_s - \pi)}{2g} = \frac{v_t^2}{2g} \sin\frac{\pi}{4} = 0.4535R. \tag{B11}$$

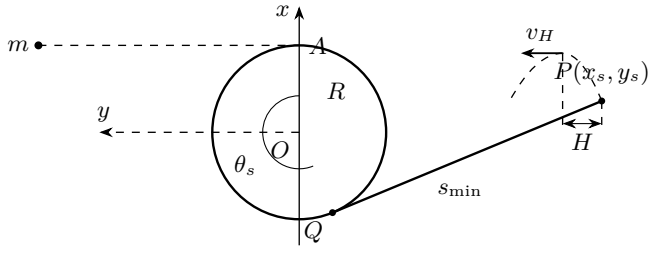


Fig. 34.9

When $\theta = \theta_s$, the coordinates of the particle are given by

$$\begin{aligned} x_s &= R\cos\theta_s - s_{\min}\sin\theta_s = -R\cos\frac{\pi}{8} + s_{\min}\sin\frac{\pi}{8} = 0.3587R, \\ y_s &= R\sin\theta_s + s_{\min}\cos\theta_s = -R\sin\frac{\pi}{8} - s_{\min}\cos\frac{\pi}{8} = -3.479R. \end{aligned} \tag{B12}$$

Obviously $|y_s| > (R + H)$, so the particle can reach its highest position without colliding with the surface of the cylinder.

Part C

(j) Suppose that the heavy object is initially at a distance h below the point O, as shown in Fig. 34.10. When the heavy object has fallen a distance D and then stopped, application of the law of conservation of mechanical energy to the particle–heavy object system gives

$$-Mgh = E' - Mg(h + D), \tag{C1}$$

where E' is the total mechanical energy of the particle after the heavy object has stopped. Hence

$$E' = MgD. \tag{C2}$$

Let the total length of the string be Λ. Its value at an arbitrary angular displacement θ must be the same as its value at $\theta = 0$, i.e.

$$\Lambda = L + \frac{\pi}{2}R + h = s + R\left(\theta + \frac{\pi}{2}\right) + (h + D). \tag{C3}$$

Noting that $D = \alpha L$ and introducing $l = L - D$, we have

$$l = L - D = (1 - \alpha)L. \tag{C4}$$

From (C3) and (C4) we obtain

$$s = L - D - R\theta = l - R\theta. \tag{C5}$$

After the heavy object has stopped, the total mechanical energy of the particle is conserved. According to (C2), equation (B1) now becomes

$$E' = MgD = \frac{1}{2}mv^2 - mg\left[R(1-\cos\theta) + s\sin\theta\right]. \tag{C6}$$

Hence the square of the speed of the particle is

$$v^2 = \left(s\frac{d\theta}{dt}\right)^2 = \frac{2MgD}{m} + 2gR\left[1 - \cos\theta + \frac{s}{R}\sin\theta\right]. \tag{C7}$$

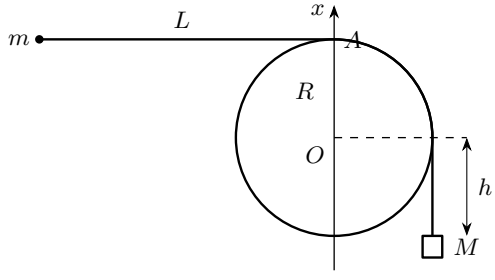


Fig. 34.10

Since (B3) still holds, the tension T is given by

$$-T + mg\sin\theta = m\left[-s\left(\frac{d\theta}{dt}\right)^2\right]. \tag{C8}$$

From the last two equations we obtain

$$\begin{aligned} T &= m\left[s\left(\frac{d\theta}{dt}\right)^2 + g\sin\theta\right] = \frac{mg}{s}\left[\frac{2MD}{m} + 2R(1-\cos\theta) + 3s\sin\theta\right] \\ &= \frac{2mgR}{s}\left[\frac{MD}{mR} + (1-\cos\theta) + \frac{3}{2}\left(\frac{l}{R} - \theta\right)\sin\theta\right], \end{aligned} \tag{C9}$$

where equation (C5) has been used in obtaining the last expression.

Now introduce the function

$$f(\theta) = 1 - \cos\theta + \frac{3}{2}\left(\frac{l}{R} - \theta\right)\sin\theta. \tag{C10}$$

By the condition $l = L - D \gg R$, the above expression can be written as

$$f(\theta) \approx 1 + \frac{3}{2}\frac{l}{R}\sin\theta - \cos\theta = 1 + A\sin(\theta - \varphi), \tag{C11}$$

where

$$A = \sqrt{1 + \left(\frac{3l}{2R}\right)^2}, \qquad \varphi = \arctan\frac{2R}{3l}. \tag{C12}$$

From (C11) it is seen that the minimum value of $f(\theta)$ is given by

$$f_{\min} = 1 - A = 1 - \sqrt{1 + \left(\frac{3l}{2R}\right)^2}. \tag{C13}$$

According to the statement of the problem, the tension T must remain positive while the particle swings around the cylinder (i.e. the string remains straight); from (C9) we obtain the inequality

$$\frac{MD}{mR} + f_{\min} = \frac{M(L-l)}{mR} + 1 - \sqrt{1 + \left(\frac{3l}{2R}\right)^2} \geq 0. \tag{C14}$$

That is,

$$\left(\frac{ML}{mR}\right) + 1 \geq \left(\frac{Ml}{mR}\right) + \sqrt{1 + \left(\frac{3l}{2R}\right)^2} \approx \left(\frac{Ml}{mR}\right) + \left(\frac{3l}{2R}\right). \tag{C15}$$

By (C4), equation (C15) can be rewritten as

$$\left(\frac{ML}{mR}\right) + 1 \geq \left(\frac{ML}{mR} + \frac{3L}{2R}\right)(1 - \alpha). \tag{C16}$$

Neglecting terms of order R/L and higher, the above inequality leads to

$$\alpha \geq 1 - \frac{\dfrac{ML}{mR} + 1}{\dfrac{ML}{mR} + \dfrac{3L}{2R}} = \frac{\dfrac{3L}{2R} - 1}{\dfrac{ML}{mR} + \dfrac{3L}{2R}} = \frac{1 - \dfrac{2R}{3L}}{\dfrac{2M}{3m} + 1} \approx \frac{1}{1 + \dfrac{2M}{3m}}. \tag{C17}$$

Thus the critical value of the ratio D/L is

$$\alpha_c = \frac{1}{1 + \dfrac{2M}{3m}}. \tag{C18*}$$

Problem 2. *A piezoelectric crystal resonator under an alternating voltage.*

Consider a uniform long rod of natural length l and cross-sectional area A (Fig. 34.11). When its two end faces are acted on by normal forces F of equal magnitude and opposite directions, the change in its length is Δl. The stress T is defined as F/A, and the relative change in length, $\Delta l/l$, is called the strain S. Hooke's law expressed in terms of stress and strain can be written as

$$T = YS$$

or

$$\frac{F}{A} = Y\frac{\Delta l}{l}. \tag{1}$$

The proportionality constant Y in the above expressions is called Young's modulus of the rod material. Note that a compressive stress T shortens the rod; the corresponding longitudinal force is negative (i.e. $F < 0$), and the change in length is also negative (i.e. $\Delta l < 0$). Hence the relation between the compressive stress and the pressure p is $p = -T$.

For a uniform long rod of density ρ, the speed u of a longitudinal wave (i.e. a sound wave) propagating along the rod is given by

$$u = \sqrt{\frac{Y}{\rho}}. \tag{2}$$

In answering the following questions, assume that damping and loss effects may be neglected.

Part A Mechanical properties

A uniform semi-infinite rod of density ρ extends from $x = 0$ to $x = \infty$. It is initially at rest and free of stress. During a very short time interval Δt, a piston applies a pressure of constant magnitude to the left end face of the rod at $x = 0$ (Fig. 34.12), producing a pressure wave which propagates to the right with speed u.

(a) If the left end face of the rod moves with constant velocity v (Fig. 34.12), what are the strain S and the pressure p at the left end of the rod during the time interval Δt? (The answers may contain only ρ, u and v.)

(b) Consider a longitudinal wave travelling in the rod along the x direction. Let x denote the equilibrium position of a cross-section of the rod when it is stress-free (Fig. 34.13), and let $\xi(x, t)$ denote the displacement of this cross-section at time t. Suppose that

$$\xi(x, t) = \xi_0 \sin k(x - ut), \tag{3}$$

where ξ_0 and k are constants. Find the velocity $v(x, t)$, the strain $S(x, t)$ and the pressure $p(x, t)$ as functions of x and t.

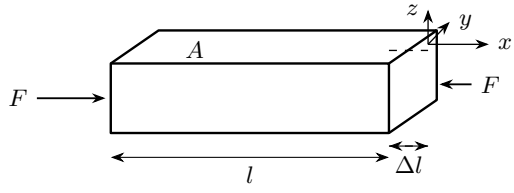


Fig. 34.11

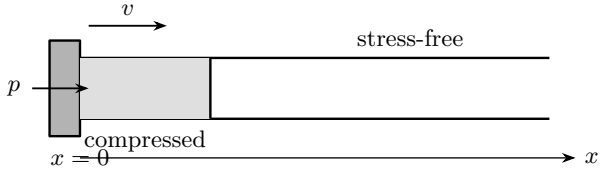


Fig. 34.12

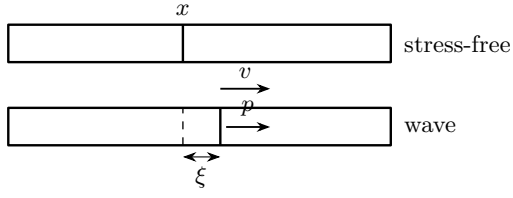


Fig. 34.13

Part B Electromechanical properties (including the piezoelectric effect)

Consider a uniform quartz plate of length b, width w and thickness h (Fig. 34.14). Its length is along the x axis and its thickness along the z axis. The upper and lower surfaces of the plate are each coated with a thin metal film serving as an electrode; two leads soldered to the centres of the electrodes also serve as supports (Fig. 34.15). For longitudinal vibrations along the x axis these two electrode centres may be regarded as fixed.

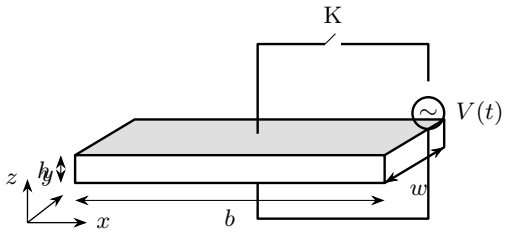


Fig. 34.14

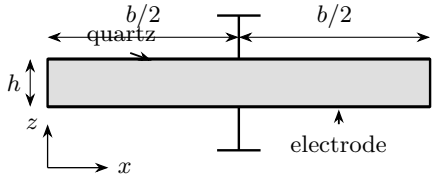


Fig. 34.15

The density ρ of this quartz crystal is $2.65 \times 10^3\,\mathrm{kg/m^3}$ and its Young's modulus Y is $7.87 \times 10^{10}\,\mathrm{N/m^2}$. The length b of the plate is 1.00 cm, while its width w and thickness h satisfy $w \ll b$ and $h \ll w$. When the switch K is not closed, only standing waves of the longitudinal mode along the x axis are excited in the quartz plate.

For a standing wave of frequency $f = \omega/2\pi$, the displacement $\xi(x,t)$ at time t of the cross-section whose equilibrium position is at x can be written as

$$\xi(x,t) = 2\xi_0\, g(x) \cos \omega t \qquad (0 \le x \le b), \tag{4a}$$

where ξ_0 is a positive constant and the position function $g(x)$ can be written in the form

$$g(x) = B_1 \sin k\left(x - \frac{b}{2}\right) + B_2 \cos k\left(x - \frac{b}{2}\right). \tag{4b}$$

The maximum value of $g(x)$ is 1, and $k = \omega/u$. Remember that the centre of the electrode is at rest, while the left and right end faces of the plate are free, so that the stress (or pressure) there is zero.

(c) For the longitudinal standing wave of this quartz plate, determine the values of B_1 and B_2 in equation (4b).

(d) What are the two lowest frequencies of the longitudinal standing waves excited in this quartz plate?

The piezoelectric effect is a property of quartz crystals. When a crystal is compressed or stretched, a voltage appears across its two ends; conversely, if a voltage is applied across the crystal, the crystal expands or contracts according to the polarity of the voltage. Hence mechanical and electrical oscillations are coupled to each other through the quartz crystal and give rise to resonance.

To describe the piezoelectric effect, suppose that when an electric field E along the z direction exists in the quartz plate, the surface charge densities on its upper and lower electrodes are $+\sigma$ and $-\sigma$ respectively. Let S and T denote the strain and the stress of the plate along the x

direction. Then the piezoelectric effect of the quartz plate can be described by the following set of relations:

$$S = \frac{1}{Y}T + d_p E, \tag{5a}$$

$$\sigma = d_p T + \varepsilon_T E, \tag{5b}$$

where $1/Y = 1.27 \times 10^{-11}\,\mathrm{m^2/N}$ is the elastic compliance at constant electric field (i.e. the reciprocal of Young's modulus Y), $\varepsilon_T = 4.06 \times 10^{-11}\,\mathrm{F/m}$ is the permittivity at constant stress, and $d_p = 2.25 \times 10^{-12}\,\mathrm{m/V}$ is the piezoelectric constant.

When the switch K in Fig. 34.14 is closed, an alternating voltage $V(t) = V_m \cos \omega t$ is applied between the two electrodes, so that a uniform electric field $E(t) = V(t)/h$ exists in the plate. A steady state is eventually reached, in which a longitudinal standing-wave oscillation along the x direction with angular frequency ω appears in the plate.

When the field E is uniform, the relation $\lambda = u/f$ between the wavelength λ and the frequency f of the longitudinal standing wave is still satisfied, with u given by (2). However, as (5a) shows, $T = YS$ no longer holds, although the definitions of stress and strain remain unchanged and the two end faces of the plate remain free and stress-free.

(e) Considering equations (5a) and (5b), the surface charge density σ on the lower electrode, as a function of x and t, is

$$\sigma(x,t) = \left[D_1 \cos k\left(x - \frac{b}{2}\right) + D_2\right] \frac{V(t)}{h},$$

where $k = \omega/u$. Find expressions for D_1 and D_2.

(f) The total charge $Q(t)$ on the lower electrode is related to the voltage $V(t)$ by

$$Q(t) = \left[1 + a^2\left(\frac{2}{kb}\tan\frac{kb}{2} - 1\right)\right] C_0 V(t). \tag{6}$$

Find the expression for the constant C_0, and the expression and numerical value of a^2.

Solution 2.

Part A

(a) Referring to Fig. 34.16, the left end face of the rod moves a distance $v\Delta t$, while at the same time the pressure wave propagates a distance $u\Delta t$, with $u = \sqrt{Y/\rho}$. Hence the strain at the left end face is

$$S = \frac{\Delta l}{l} = \frac{-v\Delta t}{u\Delta t} = -\frac{v}{u}. \tag{A1a*}$$

According to Hooke's law, the pressure at the left end face is

$$p = -YS = Y\frac{v}{u} = \rho u v. \tag{A1b*}$$

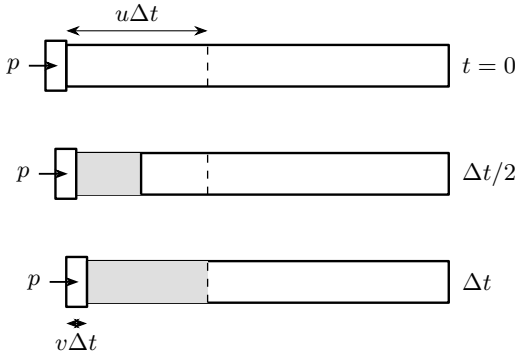


Fig. 34.16

(b) The velocity v is related to the displacement ξ, which executes simple harmonic motion of angular frequency $\omega = ku$ (the displacement ξ may also be regarded as the projection of a uniform circular motion, as shown in Fig. 34.17). Hence, if

$$\xi(x,t) = \xi_0 \sin k(x - ut),$$

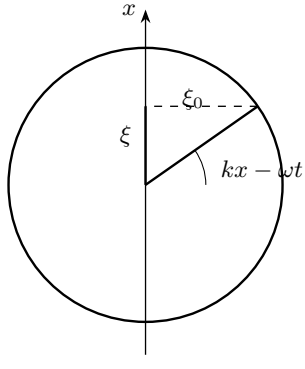


Fig. 34.17

then

$$v(x,t) = -ku\xi_0 \cos k(x - ut). \tag{A2*}$$

The relations of the strain and the pressure to the velocity are the same as in part (a); hence

$$S(x,t) = -\frac{v(x,t)}{u} = k\xi_0 \cos k(x - ut), \tag{A3*}$$

$$\begin{aligned} p(x,t) &= \rho u v(x,t) = -k\rho u^2 \xi_0 \cos k(x - ut) \\ &= -YS(x,t) = -kY\xi_0 \cos k(x - ut). \end{aligned} \tag{A4*}$$

Alternatively, the answers can also be obtained by differentiation:

$$\begin{aligned} v(x,t) &= \frac{d\xi}{dt} = -ku\xi_0 \cos k(x - ut), \\ S(x,t) &= \frac{d\xi}{dx} = k\xi_0 \cos k(x - ut), \\ p(x,t) &= -Y\frac{d\xi}{dx} = -kY\xi_0 \cos k(x - ut). \end{aligned}$$

Part B

(c) Since the angular frequency ω and the propagation speed u are known, the wavelength can be obtained from $\lambda = 2\pi/k$, where $k = \omega/u$. According to the statement of the problem, the position-dependent part of the displacement is described by

$$g(x) = B_1 \sin k\left(x - \frac{b}{2}\right) + B_2 \cos k\left(x - \frac{b}{2}\right). \tag{B1}$$

Since the centre of the electrode has been assumed fixed, $g\left(\frac{b}{2}\right) = 0$, which leads to $B_2 = 0$. Since the maximum value of $g(x)$ is 1, we obtain $B_1 = \pm 1$; hence

$$g(x) = \pm \sin \frac{\omega}{u}\left(x - \frac{b}{2}\right). \tag{B2*}$$

Thus the displacement is

$$\xi(x,t) = \pm 2\xi_0 \sin \frac{\omega}{u}\left(x - \frac{b}{2}\right)\cos \omega t. \tag{B3}$$

(d) Since the pressure p (or the stress T) vanishes at the two end faces of the plate (i.e. at $x = 0$ and $x = b$), the answer to this question can be obtained by analogy with the frequencies of the acoustic standing waves in a pipe of length b open at both ends. However, because the centre of the electrode is fixed, all even harmonics of the fundamental must be excluded, since at the central cross-section of the plate they have a displacement antinode rather than a node.

Since for the fundamental the wavelength is $\lambda = 2b$, the fundamental frequency is given by $f_1 = u/(2b)$. The propagation speed u of the wave is given by

$$u = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{7.87 \times 10^{10}}{2.65 \times 10^3}}\,\mathrm{m/s} = 5.45 \times 10^3\,\mathrm{m/s}. \tag{B4}$$

Given $b = 1.00 \times 10^{-2}\,\mathrm{m}$, the two lowest standing-wave frequencies are

$$f_1 = \frac{u}{2b} = 273\,\mathrm{kHz}, \qquad f_3 = 3f_1 = \frac{3u}{2b} = 818\,\mathrm{kHz}. \tag{B5*}$$

Another solution of parts (c) and (d):

The longitudinal standing wave in the crystal wafer has a displacement node at $x = b/2$. This standing wave may be regarded as the superposition of two waves travelling in opposite directions; hence its displacement and velocity must have the following form:

$$\begin{aligned}\xi(x,t) &= 2\xi_m \sin k\left(x - \frac{b}{2}\right)\cos \omega t \\ &= \xi_m \left[\sin k\left(x - \frac{b}{2} - ut\right) + \sin k\left(x - \frac{b}{2} + ut\right)\right]\end{aligned} \tag{B6}$$

$$\begin{aligned}v(x,t) &= -ku\xi_m \left[\cos k\left(x - \frac{b}{2} - ut\right) - \cos k\left(x - \frac{b}{2} + ut\right)\right] \\ &= -2\omega\xi_m \sin k\left(x - \frac{b}{2}\right)\sin \omega t\end{aligned} \tag{B7}$$

where $\omega = ku$; the first and the second term in the square brackets represent respectively the wave travelling in the $+x$ direction and the wave travelling in the $-x$ direction. Note that if $\xi_m = \pm\xi_0$, equation (B6) is the same as equation (B3).

For a wave travelling in the $-x$ direction, the velocity v in equations (A1a) and (A1b) must be replaced by $-v$; hence we have

$$S = -\frac{v}{u} \quad \text{and} \quad p = \rho u v \qquad \text{(wave travelling in the } +x \text{ direction)} \tag{B8}$$

$$S = \frac{v}{u} \quad \text{and} \quad p = -\rho u v \qquad \text{(wave travelling in the } -x \text{ direction)} \tag{B9}$$

The strain and the pressure of part (b) are then

$$\begin{aligned} S(x,t) &= -k\xi_m \left[-\cos k\left(x - \frac{b}{2} - ut\right) - \cos k\left(x - \frac{b}{2} + ut\right)\right] \\ &= 2k\xi_m \cos k\left(x - \frac{b}{2}\right)\cos \omega t \end{aligned} \tag{B10}$$

$$\begin{aligned} p(x,t) &= -\rho u\omega\xi_m \left[\cos k\left(x - \frac{b}{2} - ut\right) + \cos k\left(x - \frac{b}{2} + ut\right)\right] \\ &= -2\rho u\omega\xi_m \cos k\left(x - \frac{b}{2}\right)\cos \omega t \end{aligned} \tag{B11}$$

Note that v, S and p can also be obtained, as in part (b), by differentiating ξ.

Since the two end faces of the wafer ($x = 0$ and $x = b$) are free, the stress or pressure at the two end faces must vanish at every instant. By equation (B11) this necessarily requires $\cos(kb/2) = 0$, that is,

$$kb = \frac{\omega b}{u} = \frac{2\pi f b}{\lambda f} = n\pi, \qquad n = 1, 3, 5, \ldots \tag{B12}$$

Expressed in terms of the wavelength λ, equation (B12) becomes

$$\lambda = \frac{2b}{n}, \qquad n = 1, 3, 5, \ldots \tag{B13}$$

The frequencies are then

$$f = \frac{u}{\lambda} = \frac{nu}{2b} = \frac{n}{2b}\sqrt{\frac{Y}{\rho}}, \qquad n = 1, 3, 5, \ldots \tag{B14}$$

This result is the same as that given by equations (B4) and (B5).

(e) From equations (5a) and (5b) of the problem, the piezoelectric effect leads to the following equations:

$$T = Y(S - d_p E) \tag{B15}$$

$$\sigma = Y d_p S + \varepsilon_T \left(1 - Y\frac{d_p^2}{\varepsilon_T}\right) E \tag{B16}$$

For any longitudinal standing wave in the wafer, $x = b/2$ is a displacement node; therefore the displacement ξ and the strain S must have the forms of equations (B6) and (B10):

$$\xi(x,t) = \xi_m \sin k\left(x - \frac{b}{2}\right)\cos(\omega t + \varphi) \tag{B17}$$

$$S(x,t) = k\xi_m \cos k\left(x - \frac{b}{2}\right)\cos(\omega t + \varphi) \tag{B18}$$

Here a phase constant φ has been added to the time-dependent factor.

It has been assumed that the electric field between the electrodes is uniform and depends only on time:

$$E(x,t) = \frac{V(t)}{h} = \frac{V_m \cos\omega t}{h} \tag{B19}$$

Substituting equations (B18) and (B19) into equation (B15) gives

$$T = Y\left[k\xi_m \cos k\left(x - \frac{b}{2}\right)\cos(\omega t + \varphi) - \frac{d_p V_m}{h}\cos\omega t\right] \tag{B20}$$

Since the two end faces of the wafer ($x = 0$ and $x = b$) are free, the stress T at the two end faces must vanish at every instant; this necessarily requires $\varphi = 0$ and

$$k\xi_m \cos\frac{kb}{2} = d_p\frac{V_m}{h} \tag{B21}$$

Since $\varphi = 0$, it follows from equations (B16), (B18) and (B19) that the surface charge density has the same time dependence and may be written as

$$\sigma(x,t) = \sigma(x)\cos\omega t \tag{B22}$$

The factor $\sigma(x)$ depending on x may be written as

$$\begin{aligned}\sigma(x) &= Y d_p k\xi_m \cos k\left(x - \frac{b}{2}\right) + \varepsilon_T\left(1 - Y\frac{d_p^2}{\varepsilon_T}\right)\frac{V_m}{h} \\ &= \left[Y\frac{d_p^2}{\cos\dfrac{kb}{2}}\cos k\left(x - \frac{b}{2}\right) + \varepsilon_T\left(1 - Y\frac{d_p^2}{\varepsilon_T}\right)\right]\frac{V_m}{h}\end{aligned} \tag{B23}$$

(f) At time t the total surface charge $Q(t)$ on the lower electrode is obtained by integrating $\sigma(x,t)$ of equation (B22) over the surface of the electrode. The result is

$$\begin{aligned}\frac{Q(t)}{V(t)} &= \frac{1}{V(t)}\int_0^b \sigma(x,t)w\,\mathrm{d}x = \frac{1}{V_m}\int_0^b \sigma(x)w\,\mathrm{d}x \\ &= \frac{w}{h}\int_0^b \left[Y\frac{d_p^2}{\cos\dfrac{kb}{2}}\cos k\left(x - \frac{b}{2}\right) + \varepsilon_T\left(1 - Y\frac{d_p^2}{\varepsilon_T}\right)\right]\mathrm{d}x \\ &= \left(\varepsilon_T\frac{bw}{h}\right)\left[Y\frac{d_p^2}{\varepsilon_T}\left(\frac{2}{kb}\tan\frac{kb}{2}\right) + \left(1 - Y\frac{d_p^2}{\varepsilon_T}\right)\right] \\ &= C_0\left[\alpha^2\left(\frac{2}{kb}\tan\frac{kb}{2}\right) + (1 - \alpha^2)\right]\end{aligned}$$

where

$$C_0 = \varepsilon_T\frac{bw}{h}$$

$$\alpha^2 = Y\frac{d_p^2}{\varepsilon_T} = \frac{(2.25\times10^{12})^2}{1.27\times10^{-11}\times4.06\times10^{-11}} = 9.82\times10^{-3} \tag{B25}$$

(the constant α is called the electromechanical coupling coefficient).

Note. The result $C_0 = \varepsilon_T bw/h$ may be seen from equation (5) of the problem by considering the static limit $k = 0$: when $x \ll 1$, $\tan x \approx x$, and hence

$$\lim_{k\to0}\frac{Q(t)}{V(t)} \approx C_0\left[\alpha^2 + (1-\alpha^2)\right] = C_0 \tag{B26}$$

Obviously, the constant C_0 is just the capacitance of the parallel-plate capacitor formed by the two electrodes (of area bw) with the quartz wafer (of thickness h and permittivity ε_T) as dielectric, and so it equals $\varepsilon_T bw/h$.

Problem 3

Part A: Neutrino mass and neutron decay

A free neutron of mass m_n, at rest in the laboratory frame, decays into three non-interacting particles: a proton, an electron and an antineutrino. The rest mass of the proton is m_p, and the rest mass m_ν of the antineutrino is assumed to be non-zero but much smaller than the rest mass m_e of the electron. The speed of light in vacuum is denoted by c. The measured values are as follows:

$$m_n = 939.565\,63\,\mathrm{MeV/c^2}$$
$$m_p = 938.272\,31\,\mathrm{MeV/c^2}$$
$$m_e = 0.510\,990\,7\,\mathrm{MeV/c^2}$$

In what follows all energies and velocities are referred to the laboratory frame. Let E be the total energy possessed by the electron produced in the decay.

(a) Find the maximum possible value $E_{\max}$ of E, and the speed v_m of the antineutrino when $E = E_{\max}$. Both answers must be expressed in terms of the rest masses of the particles and the speed of light. Given that $m_\nu < 7.3\,\mathrm{eV/c^2}$, calculate $E_{\max}$ and v_m/c to 3 significant figures.

Part B: Polishing

A transparent glass hemisphere has radius R, mass m and refractive index n. The medium outside the hemisphere has refractive index 1. A monochromatic parallel laser beam is incident along the normal on the central part of the flat surface of the hemisphere, as shown in Fig. 3.1. In the figure the acceleration due to gravity g points vertically downwards. The circular cross-section of the laser beam has radius δ, which is much smaller than R. Both the glass hemisphere and the laser beam are symmetric about the z axis.

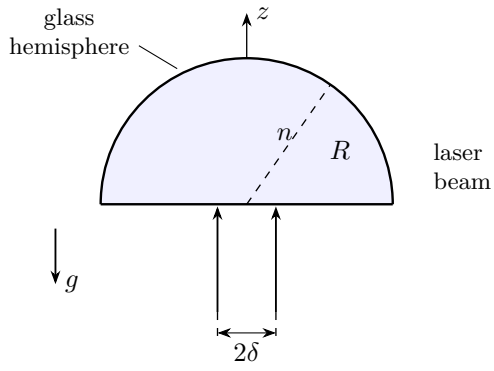


Fig. 3.1

The glass hemisphere does not absorb any of the laser light. The surface of the glass has been treated with an optical coating, so that the reflection of the incident and emergent light at the flat surface and at the spherical surface may be neglected, and the optical path of the laser light in the coating may also be neglected.

(b) Neglecting terms of order $(\delta/R)^3$ and higher, find the laser power P required to balance the weight of the glass hemisphere.

Hint: when the angle θ is much smaller than 1,

$$\cos\theta \approx 1 - \theta^2/2.$$

Solution of Problem 3

Part A: Neutrino mass and neutron decay

In the laboratory frame (that is, the rest frame of the neutron), let the electron produced in the neutron decay have energy $E_e\,(= E)$ and momentum $\vec{p}_e$, the proton have energy E_p and momentum $\vec{p}_p$, and the antineutrino have energy E_ν and momentum $\vec{p}_\nu$. Denoting the magnitude of the momentum $\vec{p}_e$ by p_e, we have

$$\left.\begin{aligned} E_e^2 &= m_e^2c^4 + p_e^2c^2 \\ E_p^2 &= m_p^2c^4 + p_p^2c^2 \\ E_\nu^2 &= m_\nu^2c^4 + p_\nu^2c^2 \end{aligned}\right\} \tag{A1}$$

From the conservation of energy and momentum,

$$E_p + E_\nu = E_n - E_e \tag{A2}$$

$$\vec{p}_p + \vec{p}_\nu = -\vec{p}_e \tag{A3}$$

where E_n is the rest energy of the neutron. Squaring both sides of equations (A2) and (A3) gives

$$E_p^2 + E_\nu^2 + 2E_pE_\nu = E_n^2 + E_e^2 - 2E_nE_e \tag{A4}$$

$$p_p^2 + p_\nu^2 + 2\vec{p}_p \cdot \vec{p}_\nu = p_e^2 \tag{A5}$$

Multiplying both sides of the above equation by c^2,

$$p_p^2c^2 + p_\nu^2c^2 + 2\vec{p}_p \cdot \vec{p}_\nu c^2 = p_e^2c^2 \tag{A5a}$$

Subtracting equation (A5a) from equation (A4) and using equation (A1), we obtain

$$m_p^2c^4 + m_\nu^2c^4 + 2\left(E_pE_\nu - \vec{p}_p\cdot\vec{p}_\nu c^2\right) = E_n^2 + m_e^2c^4 - 2E_nE_e \tag{A6}$$

that is,

$$2E_nE_e = E_n^2 + m_e^2c^4 - m_p^2c^4 - m_\nu^2c^4 - 2\left(E_pE_\nu - \vec{p}_p\cdot\vec{p}_\nu c^2\right) \tag{A7}$$

If θ denotes the angle between $\vec{p}_p$ and $\vec{p}_\nu$, then $\vec{p}_p\cdot\vec{p}_\nu = p_pp_\nu\cos\theta \le p_pp_\nu$, and from the above equation we obtain

$$2E_nE_e \le E_n^2 + m_e^2c^4 - m_p^2c^4 - m_\nu^2c^4 - 2\left(E_pE_\nu - p_pp_\nu c^2\right) \tag{A8}$$

It is seen from this equation that the electron energy reaches its maximum when $\theta = 0$, that is, when the antineutrino and the proton move in the same direction.

The quantities E_p, p_p, E_ν and p_ν may be related respectively to m_p and m_ν by means of the energy-momentum triangle (Fig. 3.2), which gives

$$\left.\begin{aligned} E_p &= \frac{m_pc^2}{\cos\varphi_p} \\ E_\nu &= \frac{m_\nu c^2}{\cos\varphi_\nu} \end{aligned}\right\} \tag{A9}$$

$$\left.\begin{aligned} p_pc &= m_pc^2\tan\varphi_p \\ p_\nu c &= m_\nu c^2\tan\varphi_\nu \end{aligned}\right\} \tag{A10}$$

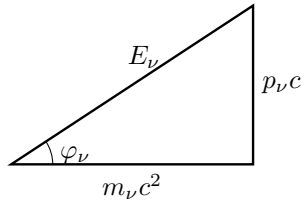


Fig. 3.2

Equation (A8) then becomes

$$2E_nE_e \le E_n^2 + m_e^2c^4 - m_p^2c^4 - m_\nu^2c^4 - 2m_pm_\nu c^4\left[\frac{1-\sin\varphi_p\sin\varphi_\nu}{\cos\varphi_p\cos\varphi_\nu}\right] \tag{A11}$$

The factor in the brackets in the last term of the above equation may be written as

$$\begin{aligned} \frac{1-\sin\varphi_p\sin\varphi_\nu}{\cos\varphi_p\cos\varphi_\nu} &= \frac{1-\sin\varphi_p\sin\varphi_\nu-\cos\varphi_p\cos\varphi_\nu}{\cos\varphi_p\cos\varphi_\nu} + 1 \\ &= \frac{1-\cos(\varphi_p-\varphi_\nu)}{\cos\varphi_p\cos\varphi_\nu} + 1 \ge 1 \end{aligned} \tag{A12}$$

Obviously the minimum of this factor occurs when $\varphi_p = \varphi_\nu$, that is, when the antineutrino and the proton move with the same velocity. Hence from equation (A11) the maximum value of E_e is

$$\begin{aligned} E_{\max} &= \frac{1}{2E_n}\left[E_n^2 + m_e^2c^4 - m_p^2c^4 - m_\nu^2c^4 - 2m_pm_\nu c^4\right] \\ &= \frac{c^2}{2m_n}\left[m_n^2 + m_e^2 - (m_p+m_\nu)^2\right] \end{aligned} \tag{A13}$$

Substituting the corresponding numerical values and using $m_\nu \ll m_p$, so that m_ν is neglected, we obtain

$$E_{\max} \approx 1.292\,569\,\text{MeV} \approx 1.29\,\text{MeV} \tag{A14}$$

When the antineutrino and the proton move with the same velocity, the speed of the antineutrino is

$$v_\nu = \frac{p_\nu}{m_\nu} = \frac{p_\nu}{E_\nu/c^2} = \frac{c^2 p_\nu}{E_\nu}.$$

Similarly, the speed of the proton is

$$v_p = \frac{c^2 p_p}{E_p}.$$

From these two equations,

$$v_\nu = v_p = c^2 \frac{p_\nu + p_p}{E_\nu + E_p} = c\,\frac{p_e c}{E_\nu + E_p} = c\,\frac{\sqrt{E_e^2 - m_e^2 c^4}}{E_n - E_e} \tag{A15}$$

Here E_e must be taken as $E_{\max}$. Substituting $E_{\max}$ of equation (A13) into the above equation gives the corresponding value v_m of v_ν:

$$v_m = c\,\frac{\sqrt{E_{\max}^2 - m_e^2 c^4}}{E_n - E_{\max}} = c\,\frac{\dfrac{1}{2m_n}\sqrt{\left[m_n^2 + m_e^2 - (m_p + m_\nu)^2\right]^2 - 4m_n^2 m_e^2}}{m_n - \dfrac{1}{2m_n}\left[m_n^2 + m_e^2 - (m_p + m_\nu)^2\right]}$$

$$= c\,\frac{\sqrt{\left[m_n^2 + m_e^2 - (m_p + m_\nu)^2\right]^2 - 4m_n^2 m_e^2}}{2m_n^2 - \left[m_n^2 + m_e^2 - (m_p + m_\nu)^2\right]} \approx 0.00127c \tag{A16}$$

Part B: Polishing

(b) Referring to Fig. 3.3, the light incident on the spherical surface obeys Snell's law:

$$n \sin\theta_i = \sin\theta_t \tag{B1}$$

Neglecting terms of order $(\delta/R)^3$ and higher in the sine functions, the above equation becomes

$$n\theta_i = \theta_t \tag{B2}$$

For the triangle FAC in Fig. 3.3,

$$\beta = \theta_t - \theta_i \approx n\theta_i - \theta_i = (n-1)\theta_i \tag{B3}$$

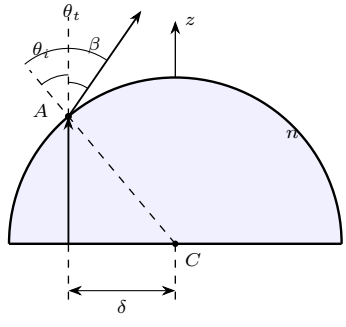


Fig. 3.3

Let the frequency of the incident light be f_0. If the number of photons incident per unit time on unit area of the flat surface is n_p, then the total number of photons incident per unit time on the flat surface is $n_p\pi\delta^2$. The total power of the photons incident on the flat surface is $P = (n_p\pi\delta^2)(hf_0)$, where h is Planck's constant; hence

$$n_p = \frac{P}{\pi\delta^2 hf_0} \tag{B4}$$

The number of photons incident per unit time on an annulus of inner radius r and outer radius $r + \mathrm{d}r$ of the flat surface is $n_p(2\pi r\,\mathrm{d}r)$, where $r = R\sin\theta_i \approx R\theta_i$; hence the number of photons incident per unit time on the annulus is

$$n_p(2\pi r\,\mathrm{d}r) \approx n_p(2\pi R^2)\theta_i\,\mathrm{d}\theta_i \tag{B5}$$

When these photons are refracted at the spherical surface, the z component of the momentum of the light carried away per unit time is

$$\begin{aligned}\mathrm{d}F_z &= n_p(2\pi r\,\mathrm{d}r)\frac{hf_0}{c}\cos\beta \approx n_p\frac{hf_0}{c}(2\pi R^2)\left(1-\frac{\beta^2}{2}\right)\theta_i\,\mathrm{d}\theta_i \\ &\approx n_p\frac{hf_0}{c}(2\pi R^2)\left[\theta_i - \frac{(n-1)^2}{2}\theta_i^3\right]\mathrm{d}\theta_i\end{aligned} \tag{B6}$$

Therefore the total momentum carried away by the photons per unit time is

$$F_z = 2\pi R^2 n_p\frac{hf_0}{c}\int_0^{\theta_m}\left[\theta_i - \frac{(n-1)^2}{2}\theta_i^3\right]\mathrm{d}\theta_i = \pi R^2 n_p\frac{hf_0}{c}\theta_m^2\left[1-\frac{(n-1)^2}{4}\theta_m^2\right] \tag{B7}$$

where $\theta_m = \arcsin\dfrac{\delta}{R} \approx \dfrac{\delta}{R}$. Then, using equation (B4), the above equation becomes

$$F_z = \frac{\pi R^2 P}{\pi\delta^2 hf_0}\frac{hf_0}{c}\frac{\delta^2}{R^2}\left[1-\frac{(n-1)^2\delta^2}{4R^2}\right] = \frac{P}{c}\left[1-\frac{(n-1)^2\delta^2}{4R^2}\right] \tag{B8}$$

The optical levitation force is equal to the sum of the z components of the forces exerted on the glass hemisphere by the incident light and by the refracted light, and may be written as

$$\frac{P}{c} + (-F_z) = \frac{P}{c} - \frac{P}{c}\left[1-\frac{(n-1)^2\delta^2}{4R^2}\right] = \frac{(n-1)^2\delta^2}{4R^2}\frac{P}{c} \tag{B9}$$

This force must equal the weight of the glass hemisphere, from which the laser power required to suspend the hemisphere is obtained:

$$P = \frac{4mgcR^2}{(n-1)^2\delta^2} \tag{B10}$$

Experimental Problems and Solutions

Problem 4

Experiment on the electro-optical properties of liquid crystals

[Experimental apparatus]

(1) Available instruments

They are listed in Table 4.1.

Table 4.1

Code	Apparatus	Qty	Code	Apparatus	Qty
A	Photodetector (PD)	1	I	Battery	2
B	Polarizer with rotating mount	2	J	Battery holder	1
C	90° TN-LC liquid-crystal cell with rotating holder	1	K	Optical bench	1
D	Function generator	1	L	Translucent paper	2
E	Laser diode (LD)	1	M	Ruler	1
F	Multimeter	2	N	White adhesive tape* (for marking the apparatus)	1
G	Parallel liquid-crystal cell	1	O	Scissors	1
H	Variable resistor	1	P	Graph paper	10

*Do not make marks on the apparatus; if necessary, stick a piece of white adhesive tape on it and make the marks on the tape.

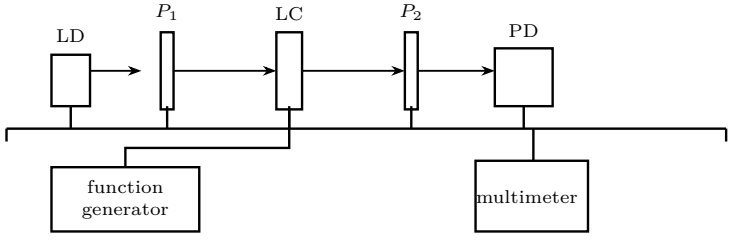


Fig. 4.1

(2) Instructions for the multimeter (Fig. 2)

- The “DC/AC” switch selects direct or alternating current.
- Use the “VΩ” and “COM” terminals to measure voltage or resistance.
- Use the “mA” and “COM” terminals to measure small currents; the digital display then gives the current in mA.
- Use the function knob to select the appropriate function and measuring range. “V” denotes voltage measurement, “A” current measurement, and “Ω” resistance measurement.

(3) Instructions for the function generator (Fig. 3)

- The digital display shows the frequency.
- Use the coarse and the fine adjustment knobs to select the appropriate frequency.
- Use the waveform selection buttons, the leftmost one for a square wave.
- Use the output amplitude knob to change the output voltage.

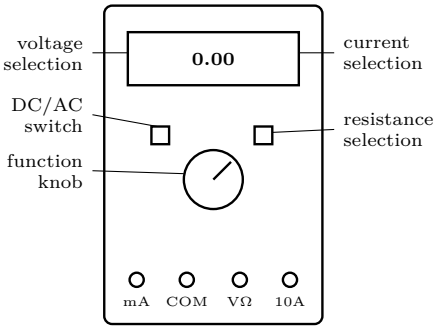


Fig. 4.2

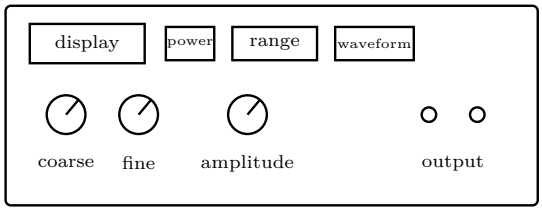


Fig. 4.3

[Experimental Problems]

Part A: Optical characteristics of the laser diode

I. Introduction

(1) Laser diode

The light source used in this experiment is a laser diode, whose laser light has a wavelength of 650 nm. When the diode current exceeds a certain threshold value, the laser diode emits monochromatic, partially polarised coherent light, which is the so-called laser light. When the current through the laser diode is below the threshold value, the luminous intensity is very small; above the threshold value the luminous intensity increases rapidly with increasing current, and the luminous intensity varies linearly with the current. If the current continues to increase, the rate of increase of the luminous intensity slows down instead, because the temperature of the laser diode itself rises. Therefore the optimum operating range of a laser diode is the linear region. Usually the threshold current I_{th} is defined as the value at which the extrapolated straight line of this linear region meets the current axis.

Note. Under no circumstances should you look directly into the laser beam; doing so will damage your eyes!

(2) Photodetector

The photodetector used in this experiment is made of a photodiode in series with a current amplifier. When a bias voltage is applied to the photodiode, the diode produces a photocurrent when illuminated. At constant temperature and for monochromatic incident light, the photocurrent is proportional to the incident light intensity. The current amplifier converts the photocurrent into an output voltage in a fixed proportion. The conversion ratio of this photodetector has two settings: a high-gain setting and a low-gain setting. In this experiment the low-gain setting is used throughout. If the incident light intensity is too high, because of the limitations of the current amplifier itself, the output voltage no longer increases in proportion to the photocurrent but gradually approaches a saturation value of about 8 V. Hence the normal working range of

the whole photodetector is the region in which the light intensity is genuinely proportional to the output voltage; if the light intensity is too high and the output voltage saturates, the reading of the photodetector will no longer represent the incident light intensity correctly.

II. Experimental procedure

Measurement of the electro-optical characteristics of the laser diode

For the experiment to succeed, the alignment of the light beam through the various parts of the apparatus is crucial. In addition, the light source and the detector must be operated under suitable conditions. Part A is concerned with these questions and with the degree of polarisation.

(1) As shown in Fig. 4.4, place the laser diode and the photodetector on the same horizontal line in the optical bench, and connect the variable resistor, the battery, the ammeter, the voltmeter, the laser diode and the photodetector as shown in Fig. 4.5. Adjust the value of the variable resistor so that the current through the laser diode is about 25 mA; the diode then emits a suitable laser beam. Set the photodetector to low-gain amplification and adjust suitably the heights of the laser diode and of the photodetector and their orientation relative to each other, so that the laser beam enters exactly the small hole in the box of the photodetector and the output voltage of the photodetector reaches its maximum value, thus ensuring that the two are aligned with each other.

Note. Do not let the black and red terminals of the battery touch each other, lest a short circuit be caused!

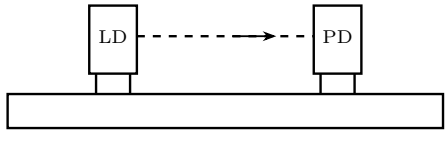


Fig. 4.4

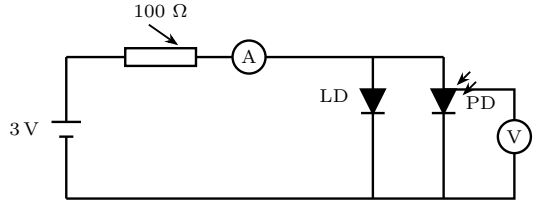


Fig. 4.5

(2) Use the output voltage of the photodetector to represent the intensity J of the laser light, and adjust the variable resistor suitably to change the current I through the laser diode from zero up to the maximum current, measuring at the same time the relation between J and I. Take care to choose the intervals of the current increments appropriately.

Problem A-(1)

Measure, tabulate and plot the J–I curve.

Problem A-(2)

Estimate the maximum current I_m within the linear region of the J–I curve, together with its uncertainty. Indicate the linear range with an arrow (↓), determine the threshold current I_{th}, and carry out a detailed error analysis.

(3) Choose the current of the laser diode to be $I_{\mathrm{th}} + 2(I_m - I_{\mathrm{th}})/3$ as the working point, so as to ensure that both the laser diode and the photodetector work normally.

(4) In preparation for the experiments of Part B: place a polarizer near the laser diode on the optical bench as shown in Fig. 4.6, taking care that the laser beam strikes the central part of the polarizer. Adjust the polarizer so that the incident laser beam is perpendicular to the plane of the polarizer. (Hint: you may insert a sheet of translucent paper as a screen and check whether the spots produced by the incident and the reflected light coincide, so as to ensure that the incident light is perpendicular to the polarizer.)

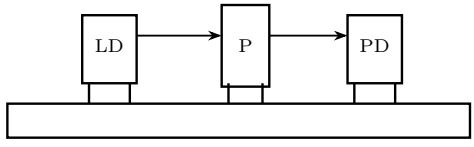


Fig. 4.6

(5) Keeping the current of the laser diode fixed, place a second polarizer on the optical bench, taking care that the optical alignment is suitable; that is, place the light source, the detector and the two polarizers on one straight line, and make sure that the plane of each polarizer is perpendicular to the beam.

Part B: Optical properties of nematic liquid crystals and the electro-optical switching characteristics of a 90° TN-LC cell

I. Introduction

(1) Liquid crystals

A liquid crystal (abbreviated LC) is a substance intermediate between a solid crystal and an amorphous liquid. A nematic liquid crystal consists of long rod-shaped organic molecules; the average orientation of these molecules can easily be arranged in a regular order, and the direction of the long axis of the rod-shaped liquid-crystal molecules can also be controlled by applying an electric field. In most liquid-crystal devices the orientation of the liquid-crystal molecules is required to be uniform and to point in a preset direction. The structure of the liquid-crystal cell (LC cell) used in this experiment is shown in Fig. 4.7. Rubbing the PI alignment film shown in Fig. 4.7 in a single direction with lens paper produces regularly arranged grooves on the surface of the PI film; these make the adjacent long liquid-crystal molecules align along the direction of the grooves, and then, through the interaction between molecules, the other liquid-crystal molecules also become regularly arranged, so that a uniformly oriented nematic liquid crystal is obtained. The local molecular orientation is called the director of the liquid crystal at that point.

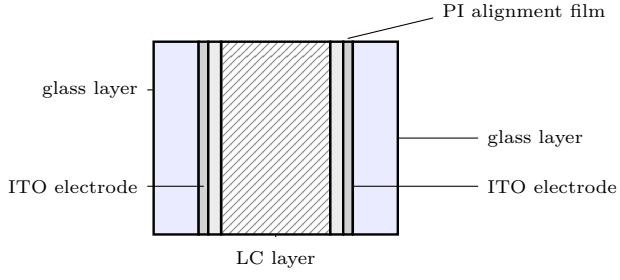


Fig. 4.7

A liquid-crystal cell exhibits double refraction, that is, it has two refractive indices. When light propagates along the director, all polarisation components propagate with the same speed $v_o = c/n_o$, where n_o is called the "ordinary refractive index"; this direction of propagation (the director) is called the optic axis of the liquid-crystal cell. When light propagates perpendicular to

the optic axis, there are two speeds of propagation: when the electric field of the polarised light is perpendicular to the optic axis, the speed of light is still $v_o = c/n_o$; but when the electric field of the polarised light is parallel to the optic axis, the speed of light is $v_e = c/n_e$, where n_e is called the "extraordinary refractive index". The birefringence (that is, the optical anisotropy) is defined as the difference between the extraordinary and the ordinary refractive indices, $\Delta n = n_e - n_o$.

(2) The 90° twisted nematic liquid-crystal cell (90° TN-LC)

In the 90° twisted nematic liquid-crystal cell shown in Fig. 4.8, the LC director of the rear PI alignment film is twisted by 90° relative to the LC director of the front PI alignment film. Hence the liquid-crystal molecules also gradually twist through 90° in the liquid-crystal layer. Let the director of the front PI alignment film be parallel to the transmission axis of the polarizer. The incident unpolarised light becomes linearly polarised after passing through the polarizer. When this linearly polarised light passes through the 90° TN-LC, its direction of polarisation twists together with the twisting of the long axes of the liquid-crystal molecules, so that the light transmitted through the TN-LC remains linearly polarised, but its direction of polarisation has been rotated by 90°. Since the electric field of the incident polarised light always remains parallel to the long axis of the liquid-crystal molecules, the light behaves as an extraordinary ray, with refractive index n_e. (Similarly, if the direction of polarisation of the incident polarised light always remains perpendicular to the long axis of the liquid-crystal molecules, it is also subject to the 90° rotation, but it behaves as an ordinary ray, with refractive index n_o.) If the direction of polarisation of the analyser (the second polarizer) is set parallel to that of the polarizer (Fig. 4.9), the light cannot pass through the analyser. This setting is called the "normally black" mode (NB). But if a voltage V is applied across the TN cell and V exceeds the critical value V_c, the long axes of the liquid-crystal molecules become aligned parallel to the applied electric field under its action; this gradually destroys the guiding and twisting action on the direction of polarisation of the polarised light, and the light gradually passes through the analyser. Let V_{10} and V_{90} denote the applied voltages at which the transmitted light intensity reaches 10% and 90% of the maximum transmitted light intensity respectively; then the "electro-optical switching slope" γ may be defined as

$$\gamma = \frac{V_{90} - V_{10}}{V_{10}}.$$

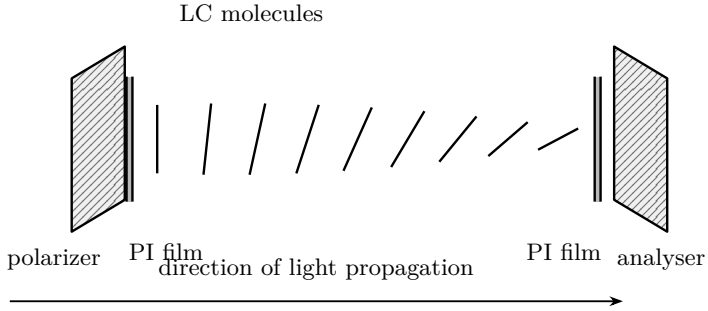


Fig. 34.8

Fig. 34.8 shows schematically the twisted alignment of the liquid-crystal molecules in a 90° TN-LC cell: the light passes successively through the polarizer, the two PI alignment films and the analyser in the direction indicated.

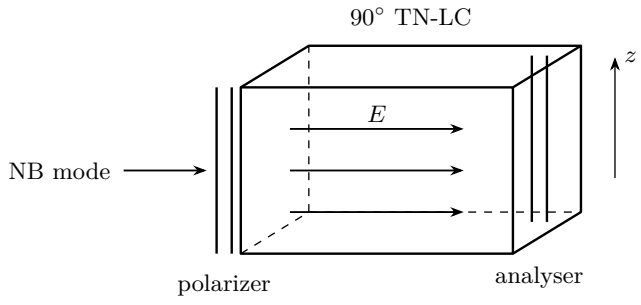


Fig. 34.9

Fig. 34.9: the NB-mode setting of the 90° TN cell.

II. Experimental procedure Set the "90° TN-LC" cell to the "normally black" mode and place it between two polarizers whose polarization directions are parallel to each other. Using a function generator, apply a 100 Hz square-wave voltage to the ITO electrodes on the two sides of the liquid-crystal cell. Increase the voltage $V_{\rm rms}$ gradually from 0 to 7.2 V.

- Near the critical points, take several additional sets of data if necessary.

Problem B-(1) Measure the NB 90° TN-LC cell, tabulate the data and plot the electro-optic characteristic curve (I versus $V_{\rm rms}$); determine the switching slope γ, $\left(\gamma = \dfrac{V_{90} - V_{10}}{V_{10}}\right)$.

Problem B-(2) Determine the critical voltage V_c of the NB 90° TN-LC cell. Show clearly on your graph the method by which you determine the value of V_c. Hint: when the applied voltage exceeds the critical voltage, the transmittance of the light rises steeply.

Part C Optical properties of nematic liquid crystals

Electro-optic switching characteristics of a parallel-aligned liquid-crystal cell

I. Introduction *Uniformly parallel-aligned liquid-crystal cell (Fig. 34.10)*

A uniformly parallel-aligned liquid-crystal cell is a cell in which the dominant directions of the front PI alignment film and of the rear PI alignment film are parallel to each other (Fig. 34.10). If the linear polarization direction of the light incident through the polarizer is parallel to the dominant LC direction, then in the liquid crystal the light propagates as an extraordinary wave, and the light undergoes only a single (pure) phase change.

If the incident polarization direction makes an angle of 45° with the dominant LC direction (see Fig. 34.11), then, because the ordinary and the extraordinary waves have different velocities in the LC, a "phase retardation" occurs between these two waves. In this case, with $\theta = 45°$ and with the polarization directions of the two polarizers parallel to each other, the transmittance of the light transmitted through the analyser (that is, the ratio of the transmitted light intensity to the intensity of the light incident on the cell) is

$$T_{\parallel} = \cos^2 \frac{\delta}{2}.$$

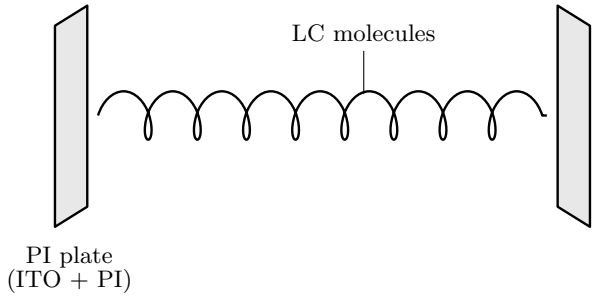


Fig. 34.10

Fig. 34.10: a uniformly parallel-aligned liquid-crystal cell, whose two PI plates (ITO + PI) are parallel to each other.

The "phase retardation" δ in the above expression can be written as

$$\delta = 2\pi d\,\Delta n(V,\lambda)/\lambda,$$

where d is the thickness of the liquid-crystal layer, λ is the wavelength of the light in air, V is the root-mean-square value of the alternating voltage, and $\Delta n \equiv n_e - n_o$ is the birefringence of the liquid crystal; Δn is a function of λ and V. Both Δn and δ attain their maximum values when V is equal to zero; moreover, Δn decreases as V increases.

In the general case the transmittance can be expressed as follows:

$$T_{\parallel} = 1 - \sin^2 2\theta \, \sin^2 \frac{\delta}{2},$$

$$T_{\perp} = \sin^2 2\theta \, \sin^2 \frac{\delta}{2},$$

where $\parallel$ and $\perp$ denote respectively the two cases in which the polarization direction of the analyser is parallel and perpendicular to the polarization direction of the polarizer.

II. Experimental procedure

(1) Replace the 90° TN liquid-crystal cell by the uniformly parallel-aligned liquid-crystal cell.

(2) With $V = 0$, set $\theta = 45°$ (Fig. 34.11). In the experiment, with the transmission axes of the polarizer and of the analyser perpendicular to each other, rotate the liquid-crystal specimen until the transmittance reaches its maximum value ($T_{\perp}$); this determines $\theta = 45°$. Record $T_{\perp}$. Then set the transmission axes of the polarizer and of the analyser parallel to each other and measure $T_{\parallel}$ (again with the applied voltage $V = 0$), so as to obtain the phase retardation δ and the birefringence Δn.

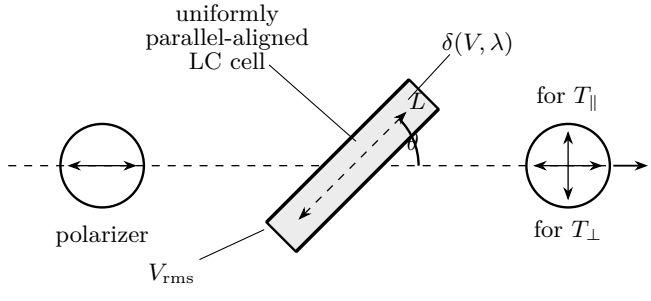


Fig. 34.11

Fig. 34.11: schematic diagram of the experimental setup (the arrow L indicates the dominant direction).

Problem C-(1) Given that the wavelength of the laser is $\lambda = 650\,\text{nm}$, that the thickness of the liquid-crystal layer is $d = 7.7\,\mu\text{m}$ and that $\Delta n \approx 0.25$, use the experimental data for $T_\perp$ and $T_\parallel$ obtained above to calculate the precise values of δ and Δn of this cell at $V = 0$.

(3) As in experiment (1) above, with $\theta = 45°$ fixed, use a function generator to apply a 100 Hz square wave to the ITO electrodes at the two sides of the uniformly parallel-aligned liquid-crystal specimen. With the transmission directions of the polarizer and of the analyser set parallel to each other, measure the electro-optic switching curve of $T_\parallel$, that is, $T_\parallel$ versus the applied voltage V_{rms}, for V_{rms} between 0 and 7 V.

(Hint: the result of measuring the switching curve of the transmittance $T_\perp$ may help to improve the accuracy of the measurement of $T_\parallel$; the measured result for $T_\perp$ is not needed in answering the following questions.)

- Remember to take several additional sets of data near the critical points when necessary (especially between 0.5 and 4.0 V).

Problem C-(2) In the $\theta = 45°$ configuration, measure the parallel-aligned liquid-crystal cell, tabulate the data and plot the electro-optic switching curve of $T_\parallel$.

Problem C-(3) From the data of the electro-optic switching, find the value of the applied voltage V_π at which $\delta = \pi$. Hint: V_π is the applied voltage for which the phase retardation is $\delta = \pi$ (180°). Remember that Δn is a function of the applied voltage and that Δn decreases as V increases. In fixing the accurate value of V_π you may consider using extrapolation.

Solution to Problem 4

Part A Measurement of the electro-optic characteristics of the laser diode

Problem A-(1) (a) Data: a table of data with the variables and units indicated.

Table 4.2 J–I data

I/mA	9.2	15.2	19.5	21.6	22.2	22.7	23.0	23.4	23.8
J/V	0.00	0.01	0.02	0.03	0.05	0.06	0.09	0.12	0.30
I/mA	24.2	24.6	25.0	25.4	25.8	26.2	26.6	27.0	27.4
J/V	0.66	1.02	1.41	1.88	2.23	2.64	3.04	3.36	3.78
I/mA	27.8	28.2	28.6	29.0	29.4	29.8	30.2	30.5	31.0
J/V	4.12	4.48	4.79	5.13	5.44	5.72	6.05	6.25	6.55
I/mA	31.4	31.8	32.2	32.6	33.0	33.4	33.8	34.2	34.6
J/V	6.75	6.99	7.22	7.40	7.60	7.78	7.93	8.07	8.14
I/mA	35.0	35.5	36.0	36.5	37.0	37.6	38.0	38.6	
J/V	8.18	8.20	8.22	8.24	8.24	8.25	8.26	8.27	

Current error: $\pm 0.1\,\mathrm{mA}$; voltage error: $\pm 0.01\,\mathrm{V}$.

(b) Plotting: choose the abscissa and ordinate scales and units that match the range and the precision of the experimental data.

(c) Drawing the curve: use the appropriate data and a suitable shape of line.

- As shown in Fig. 34.12: onset → 0 → threshold → linear → saturation.

Problem A-(2) (a) Mark the linear region of Fig. 34.12; see Fig. 34.13.

(b) Carry out the error analysis by the method of least squares or visually with a ruler.

(c) $I_m \pm \Delta I_m$: obtain suitable data for I_m and the error ($\pm\Delta I_m$) from the linear region of the J–I curve.

(d) A suitable value of I_{th} and its error:

$$I_{\mathrm{th}} = [(21 \sim 26) \pm (0.01 \text{ or } 0.2)]\,\mathrm{mA}.$$

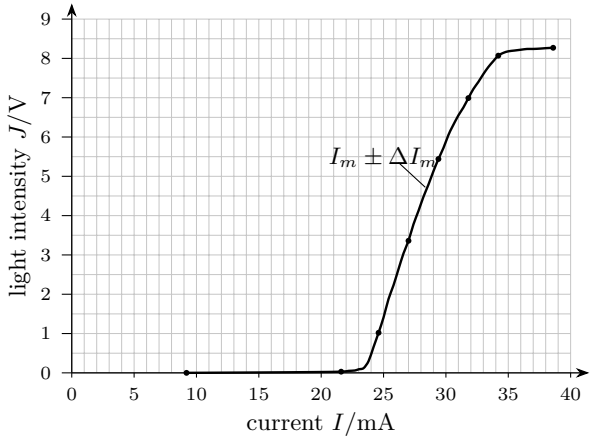


Fig. 34.12

Fig. 34.12: the J–I relation.

Table 4.3

Least-squares fit	Visual estimation with a ruler
error in the graph $0.0x$ mA	error in the graph $0.0x$ mA
least-squares method	enlarged scale graph
error analysis	draw 3 curves for the error analysis

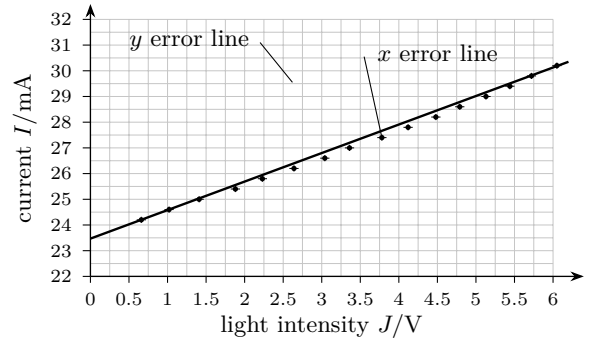


Fig. 34.13

Fig. 34.13: the straight line and the extrapolation.

A1-1

- Least-squares method

$$I = mJ + b \quad \longrightarrow \quad b = I_{\text{th}},$$

for $y = mx + b$.

Table 4.4

n	y: I/mA	x: J	xy	x^2	$y(x) = mx + b$	$[y - y(x)]^2$
1	23.8	0.30	7.14	0.0900	23.7937	3.969E-05
2	24.2	0.66	15.972	0.4356	24.17134	0.000821
3	24.6	1.02	25.092	1.0404	24.54898	0.00260
4	25.0	1.41	35.25	1.9891	24.95809	0.00176
5	25.4	1.88	47.752	3.5344	25.45112	0.00261
6	25.8	2.23	57.534	4.9729	25.81827	0.000334
7	26.2	2.64	69.168	6.9696	26.24836	0.00234
8	26.6	3.04	80.864	9.2416	26.66796	0.00462
9	27.0	3.36	90.72	11.2896	27.00364	1.325E-05
10	27.4	3.78	103.572	14.2884	27.44422	0.00196
11	27.8	4.12	114.536	16.9744	27.80088	7.733E-07
12	28.2	4.48	126.336	20.0704	28.17852	0.000461
13	28.6	4.79	136.994	22.9441	28.50371	0.00927
	$\sum y =$ 340.6	$\sum x =$ 33.71	$\sum xy =$ 910.93	$\sum x^2 =$ 113.840		$\sum [y - y(x)]^2$ = 0.0268

$$\Delta = N\sum x^2 - \Big(\sum x\Big)^2 = 13(113.840) - (33.71)^2 = 343.6$$

$$m = \frac{1}{\Delta}\Big(N\sum xy - \sum x\sum y\Big) = \frac{13(910.93) - (33.71)(340.6)}{343.556} = 1.049$$

$$b = \frac{1}{\Delta}\Big(\sum x^2\sum y - \sum x\sum xy\Big) = \frac{(113.840)(340.6) - (33.71)(910.93)}{343.556} = 23.48$$

$$\sigma_y = \frac{1}{N-2}\sqrt{\sum [y - y(x)]^2} = \frac{1}{13-2}\sqrt{0.0268} = 0.015$$

$$\sigma = \sqrt{(\sigma_y)^2 + \Big(\frac{\mathrm{d}y}{\mathrm{d}x}\sigma_x\Big)^2} = \sqrt{(0.015)^2 + (1.049 \times 0.005)^2} = 0.016$$

$$\sigma_m = \sqrt{\frac{N\sigma^2}{\Delta}} = \sqrt{\frac{13 \times 0.016^2}{343.556}} = 0.0031$$

$$\sigma_b = \sqrt{\frac{\sum x^2}{\Delta}}\,\sigma = 0.016 \times \sqrt{\frac{113.840}{343.556}} = 0.0092$$

$$I_{\text{th}} = (23.48 \pm 0.01)\,\text{mA}.$$

A1-2

- Visual estimation:

$$I = mJ + b \quad \longrightarrow \quad b = I_{\text{th}},$$

for $y = mx + b$. Straight line 1: $y = 1.00x + 23.66$; straight line 2: $y = 1.05x + 23.48$; straight line 3: $y = 1.13x + 23.31$.

$$I_{\text{th}}(\text{mean}) = 23.48\,\text{mA}, \qquad \Delta I_{\text{th}} = 0.18\,\text{mA},$$

$$I_{\text{th}} = (23.5 \pm 0.2)\,\text{mA}.$$

Part B Optical properties of the nematic liquid-crystal cell Electro-optic switching characteristics of the 90° TN-LC cell

Problem B-(1) (a) A table of data with the variables and units indicated.

Table 4.5

applied voltage/V	light intensity/V	applied voltage/V	light intensity/V	applied voltage/V	light intensity/V
0.00	0.00	1.78	0.21	3.41	1.39
0.10	0.00	1.81	0.26	3.50	1.40
0.20	0.00	1.85	0.33	3.60	1.39
0.30	0.00	1.90	0.44	3.70	1.40
0.40	0.00	1.96	0.57	3.80	1.40
0.50	0.00	2.03	0.70	4.03	1.40
0.60	0.00	2.08	0.80	4.22	1.40
0.70	0.00	2.15	0.92	4.40	1.39
0.80	0.00	2.21	1.02	4.61	1.39
0.90	0.00	2.28	1.10	4.78	1.40
1.00	0.00	2.33	1.14	5.03	1.39
1.10	0.02	2.39	1.19	5.20	1.39
1.20	0.04	2.44	1.22	5.39	1.38
1.24	0.04	2.50	1.26	5.61	1.39
1.30	0.04	2.55	1.27	5.81	1.38
1.34	0.03	2.60	1.29	6.02	1.38
1.38	0.02	2.67	1.32	6.21	1.38
1.45	0.01	2.72	1.33	6.40	1.38
1.48	0.01	2.85	1.36	6.63	1.38
1.55	0.02	2.97	1.37	6.80	1.38
1.59	0.03	3.11	1.38	7.02	1.38
1.64	0.05	3.20	1.39	7.20	1.38
1.71	0.11	3.32	1.39		

(b) Choose suitable scales and units for the abscissa and the ordinate according to the range and the precision of the experiment.

(c) Measure and plot the J–V_{rms} curve of the light intensity against the applied voltage (see Fig. 34.14).

- In the normally black mode the transmitted light intensity is less than 0.05 V.
- Before the applied voltage reaches the critical value there is a small light bounce.

- Once the applied voltage exceeds the critical value, the transmitted light intensity increases steeply.
- Once the applied voltage exceeds 3 V, a plateau region appears in the transmitted light intensity.

(d) The value of γ and its error.

- Find the maximum value of the light intensity for applied voltages between 3.0 and 7.2 V.
- Determine the value corresponding to 90% of the maximum light intensity and obtain, by interpolation, the applied voltage V_{90}.
- Determine the value corresponding to 10% of the maximum light intensity and obtain, by interpolation, the applied voltage V_{10}.
- The correct value of $\gamma \pm \Delta\gamma$ is $(0.42 \sim 0.44) \pm 0.02$.

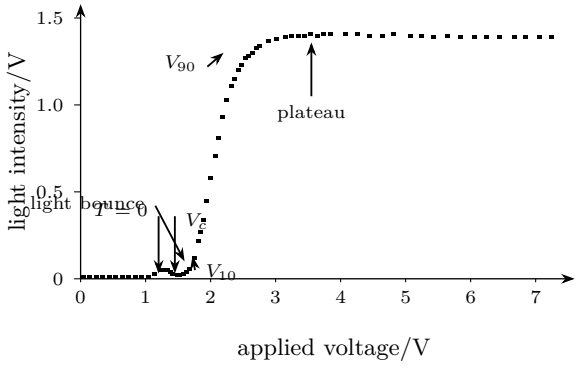


Fig. 34.14

Fig. 34.14: the J–$V_{\rm rms}$ curve of the 90° TN-LC cell.

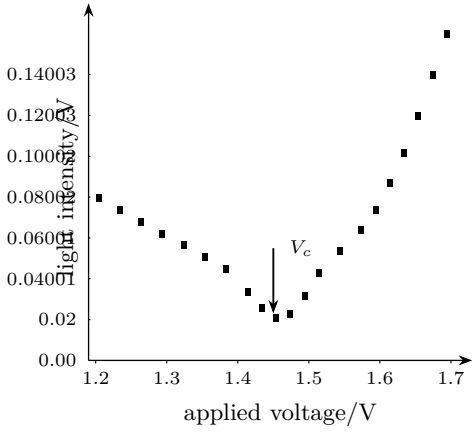


Fig. 34.15

Fig. 34.15: the region around V_c on an enlarged scale.

Problem B-(2) (a) The value of V_c and its error, $V_c \pm \Delta V_c$.

- Enlarge the scale of the graph and take more measurement points in the neighbourhood of V_c (see Fig. 34.15).
- Determine the value of V_c at which the transmitted light intensity increases steeply.
- The correct data for $V_c \pm \Delta V_c$ are $[(1.20 \sim 1.50) \pm 0.01]\,\mathrm{V}$.

The data of this graph do not correspond one-to-one to the data of Table 4.5; this graph merely shows how V_c is obtained.

Part C Optical properties of the nematic liquid-crystal cell Electro-optic switching characteristics of the uniformly parallel-aligned liquid-crystal cell

Problem C-(1) (a) The values of δ and Δn and their errors.

- Obtain the value of $T_\parallel$ and take the average.
- Determine the order m.
- The correct values of δ are $15.7 \sim 18.2$.
- The correct values of Δn are $0.20 \sim 0.24$.

$$T_\parallel = \frac{0.31 + 0.31 + 0.31}{3}\,\mathrm{V} = (0.31 \pm 0.01)\,\mathrm{V},$$
$$T_\perp = \frac{1.04 + 1.03 + 1.04}{3}\,\mathrm{V} = (1.04 \pm 0.01)\,\mathrm{V},$$

$$\tan\frac{\delta}{2} = \pm\sqrt{\frac{T_\perp}{T_\parallel}} = 1.83, \quad \text{hence} \quad \delta = 4.14 + 2m\pi \ \ (\text{or} \ -2.14 + 2m\pi).$$

$$\delta = \frac{2\pi d\,\Delta n}{\lambda} = \frac{2\pi \times 7.7 \times 0.25}{0.65} = 18.6.$$

Taking $m = 2$ (or 3) gives $\delta = 16.7$ (5.32π).

$$\text{From} \ \ \delta = \frac{2\pi d\,\Delta n}{\lambda} \ \ \text{we obtain} \ \ \Delta n = \frac{\delta\lambda}{2\pi d} = 0.22.$$

Permitted values: $\Delta n = (0.20 \sim 0.24)$.

- If $\tan\dfrac{\delta}{2} = 1.83$, then δ may be 4.68π or 6.68π, which is inconsistent with the data graph of Problem C-(2).

Problem C-(2) (a) A table of data with the variables and units indicated.

Table 4.6

applied voltage/V	light intensity/V	applied voltage/V	light intensity/V	applied voltage/V	light intensity/V
0.00	0.30	1.08	0.25	1.50	0.13
0.10	0.30	1.11	0.40	1.53	0.06
0.20	0.29	1.14	0.67	1.59	0.03
0.30	0.29	1.17	0.93	1.62	0.05
0.40	0.29	1.20	1.25	1.65	0.15
0.50	0.28	1.26	1.31	1.68	0.24
0.60	0.26	1.29	1.36	1.71	0.34
0.70	0.23	1.32	1.32	1.74	0.49
0.80	0.19	1.35	1.09	1.77	0.63
0.90	0.09	1.38	0.85	1.80	0.78
0.99	0.00	1.41	0.62	1.83	0.92
1.02	0.06	1.44	0.46	1.86	1.05
1.05	0.16	1.47	0.29	1.89	1.19

Table 4.6 (continued)

applied voltage/V	light intensity/V	applied voltage/V	light intensity/V	applied voltage/V	light intensity/V
1.92	1.27	2.79	0.26	4.20	0.11
1.95	1.34	2.82	0.23	4.30	0.14
1.98	1.40	2.85	0.21	4.40	0.16
2.01	1.47	2.88	0.18	4.50	0.19
2.04	1.48	2.91	0.16	4.60	0.22
2.07	1.48	2.94	0.14	4.70	0.25
2.10	1.48	2.97	0.12	4.80	0.28
2.13	1.45	3.00	0.09	4.90	0.31
2.16	1.42	3.06	0.08	5.01	0.34
2.19	1.38	3.09	0.06	5.11	0.37
2.22	1.33	3.12	0.05	5.21	0.39
2.25	1.27	3.18	0.04	5.29	0.42
2.28	1.20	3.21	0.03	5.39	0.44
2.31	1.14	3.24	0.02	5.51	0.48
2.34	1.07	3.27	0.02	5.57	0.49
2.37	1.00	3.30	0.01	5.70	0.52
2.40	0.94	3.33	0.00	5.80	0.55
2.43	0.87	3.36	0.00	5.90	0.57
2.46	0.79	3.39	0.00	6.01	0.60
2.49	0.72	3.42	0.00	6.10	0.62
2.52	0.66	3.45	0.00	6.19	0.64
2.55	0.61	3.48	0.00	6.30	0.66
2.58	0.56	3.51	0.00	6.40	0.69
2.61	0.51	3.60	0.01	6.60	0.73
2.64	0.46	3.70	0.02	6.70	0.74
2.67	0.42	3.80	0.03	6.80	0.76
2.70	0.37	3.90	0.04	7.00	0.80
2.73	0.33	4.00	0.07	7.20	0.83
2.76	0.30	4.10	0.09		

(b) Choose suitable scales and units for the abscissa and the ordinate according to the range and the precision of the experiment.

(c) Measure $T_{\parallel}$ as the applied voltage varies and plot the $T_{\parallel}$–V_{rms} curve (see Fig. 34.16).

- There are 3 minima and 2 sharp maxima of the light.
- The difference between the two maxima is less than 15%.
- The values at the minima are less than 0.1 V.

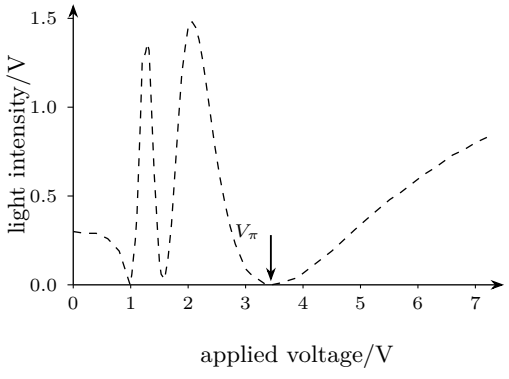


Fig. 34.16

Fig. 34.16: the $T_{\parallel}$–$V_{\rm rms}$ curve.

Problem C-(3) The value of V_π and its error.

- Enlarge the scale of the graph and take more data in the neighbourhood of V_π (see Fig. 34.17).
- Mark the correct value of V_π.
- Obtain the value of V_π by interpolation or by rounding off.
- The correct value of V_π is $[(3.2 \sim 3.5) \pm 0.01]$ V.

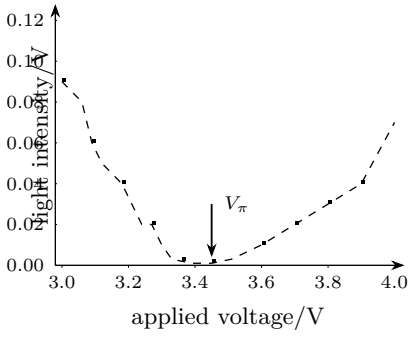


Fig. 34.17

Fig. 34.17: the region around V_π on an enlarged scale.

Appendix B

Physical Data and Formulas

The 1986 recommended values of the basic physical constants are listed in Table B.1.

Table B.1 Basic Physical Constants

Quantity	Symbol	Value	Unit	Relative uncertainty (ppm)
Speed of light in vacuum	c	299 792 458	$\mathrm{m\,s^{-1}}$	(exact)
Triple-point temperature	T_0	273.16	K	(exact)
Vacuum permeability	μ_0	$4\pi \times 10^{-7}$ $= 12.566\,370\,614\ldots \times 10^{-7}$	$\mathrm{N\,A^{-2}}$	(exact)
Vacuum permittivity, $1/\mu_0 c^2$	ε_0	$8.854\,187\,817\ldots$	$10^{-12}\,\mathrm{F\,m^{-1}}$	(exact)
Newtonian gravitational constant	G	6.672 59(85)	$10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$	128
Planck constant	h	6.626 075 5(40)	$10^{-34}\,\mathrm{J\,s}$	0.60
$h/2\pi$	$\hbar$	1.054 572 66(63)	$10^{-34}\,\mathrm{J\,s}$	0.60
Elementary charge	e	1.602 177 33(49)	$10^{-19}\,\mathrm{C}$	0.30
Magnetic flux quantum, $h/2e$	Φ_0	2.067 834 61(61)	$10^{-15}\,\mathrm{Wb}$	0.30
Electron mass	m_e	9.109 389 7(54)	$10^{-31}\,\mathrm{kg}$	0.59
Proton mass	m_p	1.672 623 1(10)	$10^{-27}\,\mathrm{kg}$	0.59
Proton–electron mass ratio	m_p/m_e	1 836.152 701(37)	—	0.020
Neutron mass	m_n	1.674 928 6(10)	$10^{-27}\,\mathrm{kg}$	0.59
Compton wavelength, $h/m_e c$	λ_C	2.426 310 58(22)	$10^{-12}\,\mathrm{m}$	0.089
Fine-structure constant, $\mu_0 c e^2/2h$	α	7.297 353 08(33)	10^{-3}	0.045
Inverse fine-structure constant	α^{-1}	137.035 989 5(61)	—	0.045
Rydberg constant, $m_e c\alpha^2/2h$	R_∞	10 973 731.534(13)	$\mathrm{m^{-1}}$	0.0012
Avogadro constant	N_A, L	6.022 136 7(36)	$10^{23}\,\mathrm{mol^{-1}}$	0.59
Faraday constant, $N_A e$	F	96 485.309(29)	$\mathrm{C\,mol^{-1}}$	0.30
Molar gas constant	R	8.314 510(70)	$\mathrm{J\,mol^{-1}\,K^{-1}}$	8.4
Boltzmann constant, R/N_A	k	1.380 658(12)	$10^{-23}\,\mathrm{J\,K^{-1}}$	8.5
Stefan–Boltzmann constant, $(\pi^2/60)k^4/\hbar^3 c^2$	σ	5.670 51(19)	$10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$	34
		Non-SI units accepted for use with the SI		
Electron volt, $(e/C)\,\mathrm{J} = \lvert e\rvert\,\mathrm{J}$	eV	1.602 177 33(49)	$10^{-19}\,\mathrm{J}$	0.30
Atomic mass unit (unified), $1\,\mathrm{u} = m_u = \frac{1}{12}m(^{12}\mathrm{C})$	u	1.660 540 2(10)	$10^{-27}\,\mathrm{kg}$	0.59

From E. Richard Cohen and B. N. Taylor, *Reviews of Modern Physics*, Vol. 59, No. 4, October 1987, p. 1139.

Solar-System Data

The astronomical data of the solar system are listed in Table B.2.

Table B.2 Solar-System Data

Body	Orbiting body	Mean orbital radius (m)	Body radius (m)	Period (s)	Mass (kg)	Number of satellites
Sun	Milky Way	5.6×10^{20}	6.96×10^{8}	8×10^{15}	1.99×10^{30}	
Mercury	Sun	5.79×10^{10}	2.42×10^{6}	7.60×10^{6}	3.35×10^{23}	
Venus	Sun	1.08×10^{11}	6.10×10^{6}	1.94×10^{7}	4.89×10^{24}	
Earth	Sun	1.50×10^{11}	6.37×10^{6}	3.16×10^{7}	5.98×10^{24}	1
Mars	Sun	2.28×10^{11}	3.38×10^{6}	5.94×10^{7}	6.45×10^{23}	2
Jupiter	Sun	7.78×10^{11}	7.13×10^{7}	3.74×10^{8}	1.90×10^{27}	16
Saturn	Sun	1.43×10^{12}	6.04×10^{7}	9.35×10^{8}	5.69×10^{26}	17
Uranus	Sun	2.87×10^{12}	2.38×10^{7}	2.64×10^{9}	8.73×10^{25}	5
Neptune	Sun	4.50×10^{12}	2.22×10^{7}	5.22×10^{9}	1.03×10^{26}	2
Pluto	Sun	5.91×10^{12}	3×10^{6}	7.82×10^{9}	5.4×10^{24}	1
Moon	Earth	3.84×10^{8}	1.74×10^{6}	2.36×10^{6}	7.35×10^{22}	

Common Physical Data

Pi

$$\pi = 3.141\,592\,653\,5 \approx 3.14$$

Base of natural logarithms

$$\mathrm{e} = 2.718\,281\,83 \approx 2.72$$

Conversion factor for logarithms

$$\log_{\mathrm{e}} 10 = 2.302\,585\,09 \approx 2.3026$$

$$(\ln N \approx 2.3026 \lg N)$$

Radius of the Earth (radius of a sphere of the same volume as the Earth)

$$R = 6.371 \times 10^{6}\ \mathrm{m} \approx 6400\ \mathrm{km}$$

Mass of the Earth

$$M = 5.977 \times 10^{24}\ \mathrm{kg}$$

Acceleration of gravity at the Earth's surface

$$g = 9.806\,65\ \mathrm{m/s^2}$$

Standard atmospheric pressure

$$\begin{aligned} p_0 &= 76\ \mathrm{cmHg} = 1.013\,25 \times 10^{5}\ \mathrm{Pa} \\ &= 1.033\,6\ \mathrm{kgf/cm^2} \\ &= 1\,013.25\ \mathrm{mbar} \end{aligned}$$

Speed of sound in air
at 0°C and standard atmospheric pressure: 331.4 m/s;
usual value at room temperature: 340 m/s.
Densities of gases (under standard conditions)
air 1.293 $\mathrm{kg/m^3} \approx 0.001\,29\ \mathrm{g/cm^3}$;
hydrogen $0.089\,9\ \mathrm{kg/m^3}$;
helium $0.178\,6\ \mathrm{kg/m^3}$.
Density of mercury (at 0° C)

$$13.595\,0 \times 10^{3}\ \mathrm{kg/m^3} \approx 13.6\ \mathrm{g/cm^3}$$

Density of water (pure water at 4° C)

$$1.0 \times 10^{3}\ \mathrm{kg/m^3} = 1\ \mathrm{g/cm^3}$$

Density of titanium

$$4.510 \times 10^{3}\ \mathrm{kg/m^3}$$

Density of copper

$$8.93 \times 10^{3}\ \mathrm{kg/m^3}$$

Density of iron

$$7.860 \times 10^{3}\ \mathrm{kg/m^3}$$

Specific heat of air at constant pressure

$$\begin{aligned} 1007\ \mathrm{J/(kg \cdot K)} &\approx 1\ \mathrm{J/(g \cdot {}^{\circ}\,C)} \\ &\approx 0.24\ \mathrm{cal/(g \cdot {}^{\circ}\,C)} \end{aligned}$$

Surface-tension coefficient of water

$$0.073\ \mathrm{N/m}$$

Specific heat of water

$$1\ \mathrm{cal/(g \cdot {}^{\circ}\,C)} = 4.18 \times 10^{3}\ \mathrm{J/(kg \cdot K)}$$

Boiling point of water (at 1 atm)

$$100^{\circ}\mathrm{C} = 373.15\ \mathrm{K}$$

Heat of vaporisation of water (at 1 atm)

$$539\ \mathrm{cal/g} = 2.26 \times 10^{6}\ \mathrm{J/kg}$$

Freezing point (ice point) of water (at 1 atm)

$$0^{\circ}\mathrm{C} = 273.15\ \mathrm{K}$$

Heat of fusion of ice (at 1 atm)

$$80\ \mathrm{cal/g} = 3.35 \times 10^{5}\ \mathrm{J/kg}$$

Triple point of water

$$273.16\ \mathrm{K} \pm 0.01\ \mathrm{K}$$

Absolute zero

$$0\ \mathrm{K} = -273.15\ {}^{\circ}\mathrm{C} \approx -273\ {}^{\circ}\mathrm{C}$$

Volume expansion coefficient of gases

$$\frac{1}{273}\ \text{deg}^{-1} = 0.003\,66\ \text{deg}^{-1}$$

Pressure coefficient of gases

$$\frac{1}{273}\ \text{deg}^{-1} \approx 0.003\,66\ \text{deg}^{-1}$$

Tensile strength of steel

$$500 \sim 2000\ \text{MPa}$$

Mechanical equivalent of heat

$$J = 4.186\,8\ \text{J/cal} \approx 427\ \text{kgf/kcal}$$

Thermal equivalent of work

$$0.238\,85\ \text{cal/J} \approx 0.24\ \text{cal/J}$$

Horsepower

1 international horsepower = 75 kgf m/s = 735 W.

Faraday constant

$$F = 9.648\,456 \times 10^4\ \text{C/mol} \approx 9.65 \times 10^4\ \text{C/mol}$$

Refractive index of water

1.33

Refractive index of air

1.000 293

Refractive index of glass

$1.5 \sim 1.8$

Refractive index of diamond

2.42

Trigonometric Formulas

Addition formulas

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$
$$\cot(\alpha \pm \beta) = \frac{\cot\alpha\cot\beta \mp 1}{\cot\beta \pm \cot\alpha}$$

Formulas converting sums and differences into products and vice versa

$$\sin\alpha + \sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$
$$\sin\alpha - \sin\beta = 2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$$
$$\cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$
$$\cos\alpha - \cos\beta = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$$
$$\tan\alpha \pm \tan\beta = \frac{\sin(\alpha \pm \beta)}{\cos\alpha\cos\beta}$$
$$\cot\alpha \pm \cot\beta = \pm\frac{\sin(\alpha \pm \beta)}{\sin\alpha\sin\beta}$$
$$\sin\alpha\sin\beta = -\frac{1}{2}\big[\cos(\alpha+\beta) - \cos(\alpha-\beta)\big]$$
$$\cos\alpha\cos\beta = \frac{1}{2}\big[\cos(\alpha+\beta) + \cos(\alpha-\beta)\big]$$
$$\sin\alpha\cos\beta = \frac{1}{2}\big[\sin(\alpha+\beta) + \sin(\alpha-\beta)\big]$$

Double-angle formulas

$$\sin 2\alpha = 2\sin\alpha\cos\alpha = \frac{2\tan\alpha}{1+\tan^2\alpha}$$
$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha = \frac{1-\tan^2\alpha}{1+\tan^2\alpha}$$
$$\tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}$$
$$\cot 2\alpha = \frac{\cot^2\alpha - 1}{2\cot\alpha}$$

Half-angle formulas

$$\sin\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{2}}$$
$$\cos\frac{\alpha}{2} = \pm\sqrt{\frac{1+\cos\alpha}{2}}$$
$$\tan\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}} = \frac{1-\cos\alpha}{\sin\alpha} = \frac{\sin\alpha}{1+\cos\alpha}$$
$$\cot\frac{\alpha}{2} = \pm\sqrt{\frac{1+\cos\alpha}{1-\cos\alpha}} = \frac{1+\cos\alpha}{\sin\alpha} = \frac{\sin\alpha}{1-\cos\alpha}$$

Calculus Formulas

Differential Formulas

The derivative of $y = f(x)$ is defined as the limit of the slope $\Delta y/\Delta x$ of the y–x curve:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

General relations for differentials

$$\frac{\mathrm{d}}{\mathrm{d}x}\big[f(x) \pm g(x)\big] = \frac{\mathrm{d}f}{\mathrm{d}x} \pm \frac{\mathrm{d}g}{\mathrm{d}x}$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\big[f(x)g(x)\big] = f\,\frac{\mathrm{d}g}{\mathrm{d}x} + g\,\frac{\mathrm{d}f}{\mathrm{d}x}$$
$$\frac{\mathrm{d}(f/g)}{\mathrm{d}x} = \frac{g\,\mathrm{d}f/\mathrm{d}x - f\,\mathrm{d}g/\mathrm{d}x}{g^2}$$

Chain rule. If $y = f(x)$ and $x = g(z)$, then

$$\frac{\mathrm{d}f(x)}{\mathrm{d}z} = \frac{\mathrm{d}f(x)}{\mathrm{d}x}\,\frac{\mathrm{d}x}{\mathrm{d}z}.$$

Derivatives of some special functions (a and n are constants)

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x \qquad \frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\arcsin x = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\cos x = -\sin x \qquad \frac{\mathrm{d}}{\mathrm{d}x}a^x = a^x \ln a \qquad \frac{\mathrm{d}}{\mathrm{d}x}\cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\tan x = \sec^2 x \qquad \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^x = \mathrm{e}^x \qquad \frac{\mathrm{d}}{\mathrm{d}x}\tan^{-1} x = \frac{1}{1+x^2}$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\cot x = -\csc^2 x \qquad \frac{\mathrm{d}}{\mathrm{d}x}\ln x = \frac{1}{x} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\cot^{-1} x = -\frac{1}{1+x^2}$$

Integral Formulas

The integral of a function $f(x)$ over the interval from a to b is written as

$$I = \int_a^b f(x)\,\mathrm{d}x.$$

It equals the area of the region bounded by the lines $x = a$, $x = b$, the curve $f(x)$ and the x-axis. The basic rule of calculation is: if the upper limit is a variable w, then

$$I(w) = \int_a^w f(x)\,\mathrm{d}x,$$

$$\frac{\mathrm{d}}{\mathrm{d}w}I(w) = \frac{\mathrm{d}}{\mathrm{d}w}\int_a^w f(x)\,\mathrm{d}x = f(w).$$

Therefore we regard integration as the inverse operation of differentiation. The indefinite integral $I(x)$ of $f(x)$ is a function whose derivative is $f(x)$; for example, the indefinite integral of ax^2+bx+c is

$$\frac{1}{3}ax^3 + \frac{1}{2}bx^2 + cx + d.$$

Some common formulas for indefinite integrals are as follows (f, g, u and v are functions; a, b and c are constants).

$$\int \mathrm{d}x = x + c$$

$$\int \frac{\mathrm{d}f(x)}{\mathrm{d}x}\,\mathrm{d}x = f(x) + c$$

$$\int af(x)\,\mathrm{d}x = a\int f(x)\,\mathrm{d}x$$

$$\int \big[af(x) + bg(x)\big]\,\mathrm{d}x = a\int f(x)\,\mathrm{d}x + b\int g(x)\,\mathrm{d}x$$

$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u$$

Below is a table of common definite and indefinite integrals.

Indefinite integrals (an arbitrary constant should be added after each integral; a, b and n denote constants)

$$\int x^n\,\mathrm{d}x = \frac{x^{n+1}}{n+1}$$

$$\int (a+bx)^n\,\mathrm{d}x = \frac{(a+bx)^{n+1}}{b(n+1)} \qquad (n \neq -1)$$

$$\int \frac{\mathrm{d}x}{x} = \ln x$$

$$\int \frac{\mathrm{d}x}{a+bx} = \frac{1}{b}\ln(a+bx)$$

$$\int \frac{\mathrm{d}x}{a+bx^2} = \frac{1}{\sqrt{ab}}\tan^{-1}\left(\sqrt{\frac{b}{a}}\,x\right) \qquad (ab > 0)$$

$$\int \frac{\mathrm{d}x}{a+bx^2} = \frac{1}{2\sqrt{|ab|}}\ln\left(\frac{a - x\sqrt{|ab|}}{a + x\sqrt{|ab|}}\right) \qquad (ab < 0)$$

$$\int \frac{x\,\mathrm{d}x}{(a+bx^2)^n} = -\frac{1}{2b(n-1)(a+bx^2)^{n-1}} \qquad (n \neq 1)$$

$$\int \frac{x\,\mathrm{d}x}{a+bx^2} = \frac{1}{2b}\ln(a+bx^2)$$

Let $u = \sqrt{a+cx^2}$ and $I = \displaystyle\int \frac{\mathrm{d}x}{u}$; then

$$I = \frac{1}{\sqrt{c}}\ln(x\sqrt{c} + u) \qquad (c > 0)$$

$$= \frac{1}{\sqrt{-c}}\arcsin\left(x\sqrt{-\frac{c}{a}}\right) \qquad (c < 0 \text{ and } a > 0)$$

and hence

$$\int u\,\mathrm{d}x = \frac{1}{2}(xu + aI).$$

$$\int \frac{\mathrm{d}x}{u} = I$$

$$\int xu\,\mathrm{d}x = \frac{u^3}{3c}$$

$$\int \frac{x\,\mathrm{d}x}{u} = \frac{u}{c}$$

$$\int \mathrm{e}^{ax}\,\mathrm{d}x = \frac{\mathrm{e}^{ax}}{a}$$

$$\int x\mathrm{e}^{ax}\,\mathrm{d}x = \frac{\mathrm{e}^{ax}}{a^2}(ax-1)$$

$$\int x^2\mathrm{e}^{ax}\,\mathrm{d}x = \frac{\mathrm{e}^{ax}}{a^3}(a^2x^2 - 2ax + 2)$$

$$\int \frac{\mathrm{d}x}{a + b\mathrm{e}^{mx}} = \frac{x}{a} - \frac{\ln(a + b\mathrm{e}^{mx})}{am}$$

$$\int \ln ax\,\mathrm{d}x = (x\ln ax) - x$$

$$\int \sin ax\,\mathrm{d}x = -\frac{\cos ax}{a}$$

$$\int \cos ax\,\mathrm{d}x = \frac{\sin ax}{a}$$

$$\int \tan ax\,\mathrm{d}x = -\frac{\ln(\cos ax)}{a}$$

$$\int \sin^2 ax\,\mathrm{d}x = \frac{x}{2} - \frac{\sin 2ax}{4a}$$

$$\int \cos^2 ax\,\mathrm{d}x = \frac{x}{2} + \frac{\sin 2ax}{4a}$$

$$\int \tan^2 ax\,\mathrm{d}x = \frac{\tan ax}{a} - x$$

$$\int \arcsin\left(\frac{x}{a}\right)\mathrm{d}x = x\arcsin\left(\frac{x}{a}\right) + \sqrt{a^2 - x^2}$$

$$\int \arccos\left(\frac{x}{a}\right)\mathrm{d}x = x\arccos\left(\frac{x}{a}\right) - \sqrt{a^2 - x^2}$$

$$\int \arctan\left(\frac{x}{a}\right)\mathrm{d}x = x\arctan\left(\frac{x}{a}\right) - \left(\frac{a}{2}\right)\ln(a^2 + x^2)$$

Definite integrals $(a > 0)$

$$\int_0^\infty \mathrm{e}^{-ax}\,\mathrm{d}x = \frac{1}{a}$$

$$\int_0^\infty x^n \mathrm{e}^{-ax}\,\mathrm{d}x = \frac{n!}{a^{n+1}}$$

$$\int_0^\infty \frac{\mathrm{d}x}{1+\mathrm{e}^{ax}} = \frac{\ln 2}{a}$$

$$\int_0^\infty \mathrm{e}^{-a^2x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2a}$$

$$\int_0^\infty x\mathrm{e}^{-a^2x^2}\,\mathrm{d}x = \frac{1}{2a^2}$$

$$\int_0^\infty x^2\mathrm{e}^{-a^2x^2}\,\mathrm{d}x = \frac{1}{4}\sqrt{\frac{\pi}{a^3}}$$

$$\int_0^\infty x^3\mathrm{e}^{-a^2x^2}\,\mathrm{d}x = \frac{1}{2a^4}$$

$$\int_0^\infty x^4\mathrm{e}^{-a^2x^2}\,\mathrm{d}x = \frac{3}{8}\sqrt{\frac{\pi}{a^5}}$$

$$\int_0^\infty \frac{\sin ax}{x}\,\mathrm{d}x = \frac{\pi}{2}$$

Approximation Formulas

In the approximation formulas, the angles of trigonometric functions are measured in radians.

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots \qquad (|x| < 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \qquad (x^2 < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \qquad (x^2 < \infty)$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots \qquad \left(x^2 < \frac{\pi^2}{4}\right)$$

$$\cot x = \frac{1}{x} - \frac{x}{3} - \frac{x^3}{45} - \frac{2x^5}{945} - \cdots \qquad (x^2 < \pi^2)$$

$$\arcsin x = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \cdots \qquad (x^2 < 1)$$

$$\arccos x = \frac{\pi}{2} - \arcsin x \qquad (x^2 < 1)$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots = \frac{\pi}{2} - \operatorname{arccot} x \qquad (x^2 < 1)$$

$$= \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^3} - \frac{1}{5x^5} + \cdots \qquad (x^2 > 1)$$

$$\mathrm{e}^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \qquad (x^2 < \infty)$$

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots \qquad (x^2 < 1)$$

$$\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{1}{3}x^3 + \frac{1}{5}x^5 + \frac{1}{7}x^7 + \cdots\right) \qquad (x^2 < 1)$$

$$\ln\left(\frac{x+1}{x-1}\right) = 2\left[\frac{1}{x} + \frac{1}{3}\left(\frac{1}{x}\right)^3 + \frac{1}{5}\left(\frac{1}{x}\right)^5 + \cdots\right] \qquad (x^2 > 1)$$